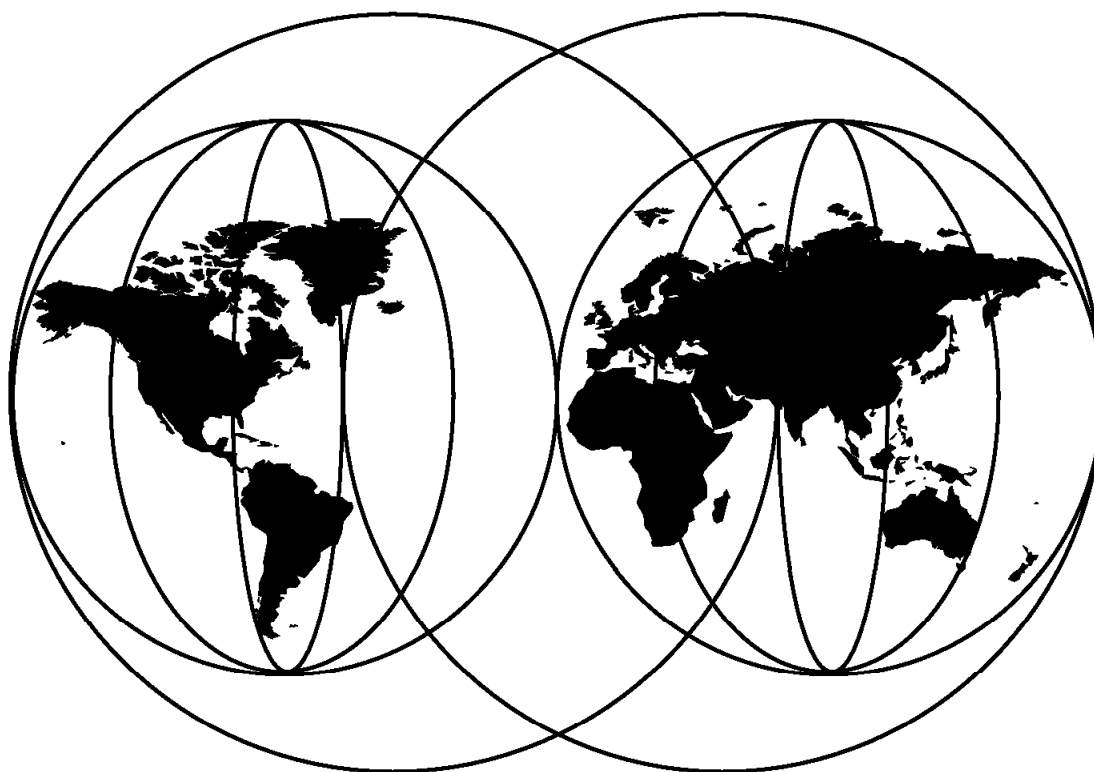


Converting to Removable Media Manager: A Practical Guide

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**Converting to Removable Media Manager:
A Practical Guide**

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Take Note!

Before using this information and the product it supports, be sure to read the general information in Appendix F, "Special Notices" on page 679.

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This edition applies to Version 1, Release 4, of DFSMS/MVS, Program Number 5695-DF1 for use with the MVS/ESA product for the OS/390 system.

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Preface

This redbook is written for people who are planning to convert from a vendor tape management system to DFSMSrmm. We have designed this book to help you with all aspects of the conversion, from the early planning stage through customization and production implementation of DFSMSrmm. Use it to decide whether to convert to DFSMSrmm, and then use it to help plan and execute that conversion. It provides you with enough information to prepare a plan for conversion and then takes you step by step through that plan. We provide details on the differences between DFSMSrmm and vendor tape management systems and compare the terminology, data, and functions. We explain how to use the IBM-supplied sample conversion programs and get the converted data validated and ready for use in a production environment. Working samples that are ready for use both during and after conversion are included.

Note: In February, 2000 a chapter describing the conversion from Control-T to DFSMSrmm was added to this book.

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Comments welcome

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Chapter 1. Introduction

DFSMS/MVS Removable Media Manager is a full function tape management system and is available as DFSMSrmm, a functional component of DFSMS/MVS.

DFSMSrmm enables you to manage your removable media as an enterprise library across systems that can share DASD. It can manage all of your tape volumes and the data sets on those volumes. It protects tape data sets from being accidentally overwritten, manages the movement of tape volumes between libraries and vaults over the life of the tape data sets, and expire and scratch tapes, all according to policies that you define. DFSMSrmm also manages other removable media that you define to it; for example, it can record the shelf location for optical disks and track their vital record status. DFSMSrmm does not yet automatically record information for optical volumes.

Objectives of Chapter

- Introduce the DFSMSrmm product, the conversion process, and the factors that you should consider in preparing and planning for conversion.

Audience

- Decision makers (data center managers and storage management consultants)
- Project leaders
- Project participants.

At the end of this chapter you will have a basic understanding of the conversion process as well as the functional capabilities and potential of DFSMSrmm.

1.1 Before You Begin

This redbook contains information on the DFSMSrmm Application Programming Interface (API) which became available in April, 1998, and the vital record specification (VRS) enhancements which became available in June, 1998. The information in this book applies to systems at the maintenance levels in Table 1.

Table 1. DFSMSrmm Maintenance Levels for this Document	
DFSMSrmm 1.4	DFSMSrmm 1.3
OW30369	OW30369
OW30969	OW33400

For the latest information on DFSMSrmm functions refer to the following sources of information:

- IBM announcement letters
- The DFSMSrmm publications
- The documentation provided in softcopy with the product in the following members of SAMPLIB:
 - EDGCMM01 - Contains the *Data Extract Program User Guide for Conversion to DFRMM and DFSMSrmm*, which includes the latest information about tools and samples available to help you convert to DFSMSrmm.

- EDGDOC - Contains information about new functions shipped since the general availability of the latest release of DFSMSrmm. It is ready for online viewing and can be printed.
- EDGDOCS - The source documentation used to generate the contents of EDGDOC.
- EDGDOCX - Contains information about the API. It is ready for online viewing and can be printed.
- EDGDOCXS - The source documentation used to generate the contents of EDGDOC. You must have BookMaster and SCRIPT/VS to format this information.
- APARs II08155 and II10888 - Contain information about new functions available in DFSMSrmm.

1.2 Volume Management

Most of the volumes that DFSMSrmm manages in your installation are probably scratch volumes—volumes that are used again and again by different users. Each time a volume is used it is retained and managed by policies that you define to DFSMSrmm. When the data is no longer required, the volume is returned to scratch status and is ready for use by another user. Figure 1 on page 3 shows the life cycle of a volume and illustrates how the use of DFSMSrmm commands and actions affect that cycle.

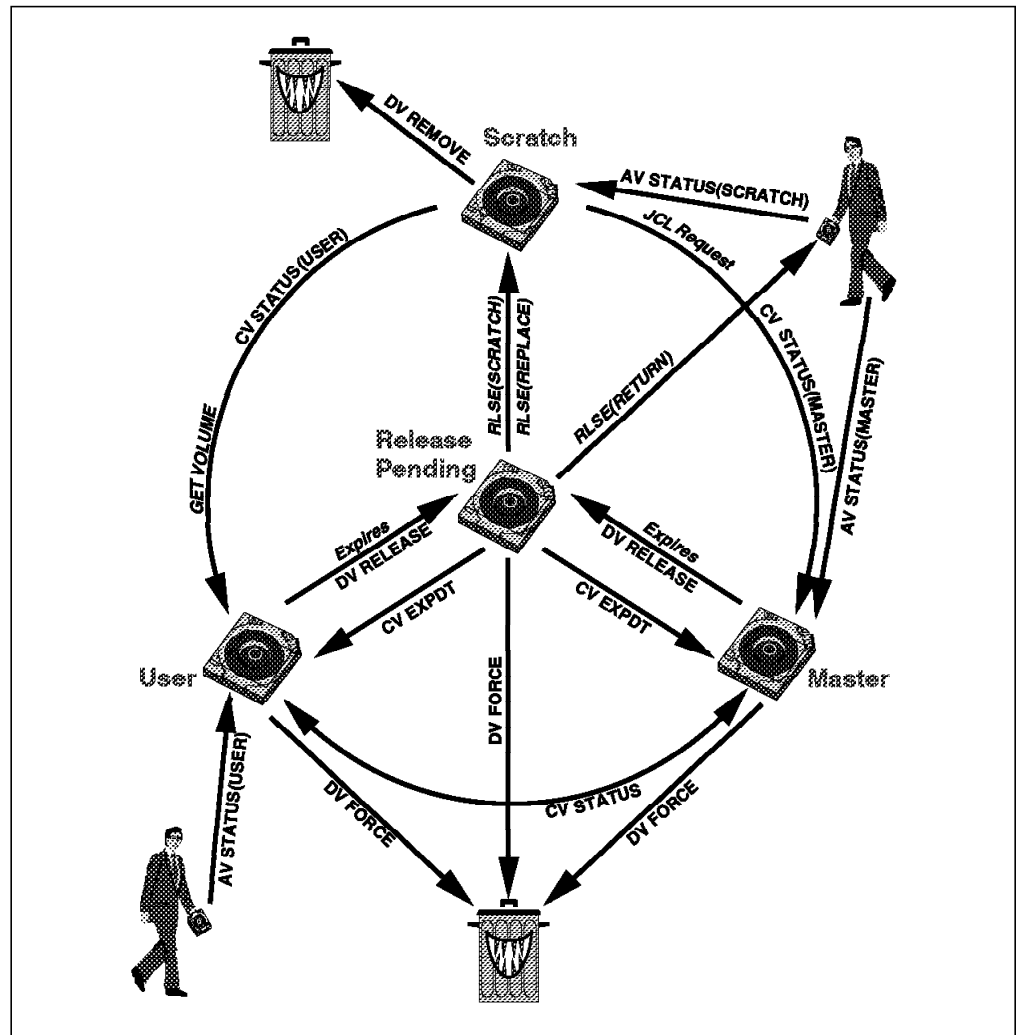


Figure 1. Life Cycle of a Volume

DFSMSrmm records information for data sets on all files of a tape volume. It also can manage the RACF TAPEVOL profiles for tape volumes and uncatalog data sets. Figure 2 on page 4 shows how DFSMSrmm processes records and catalogs related to tape data sets.

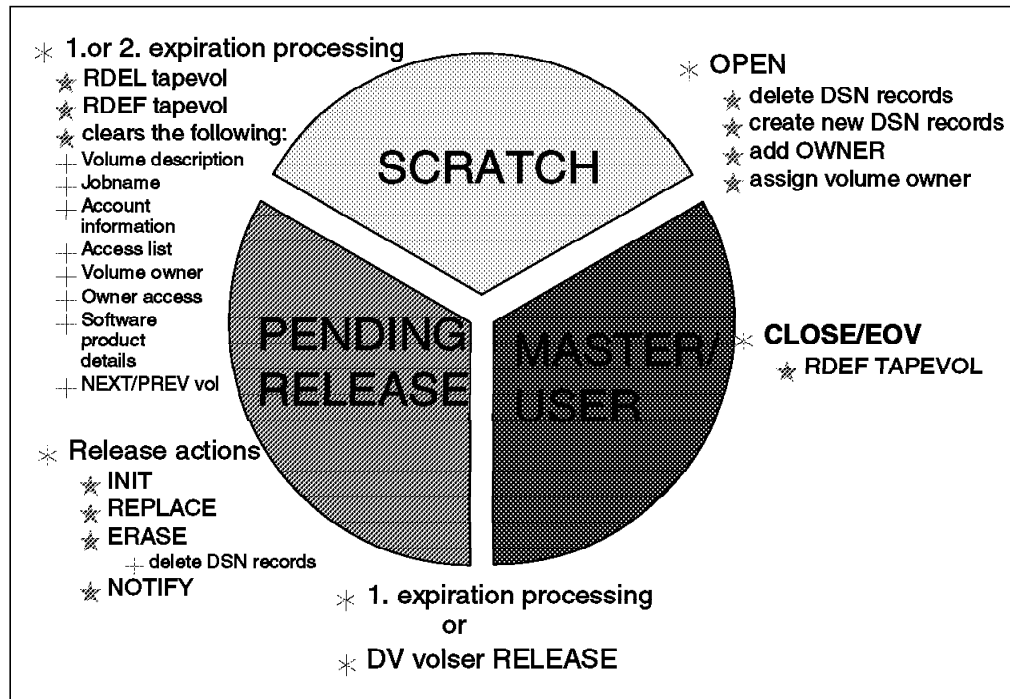


Figure 2. Timing of Related Events for a Volume

In addition to managing scratch volumes, DFSMSrmm can manage any volume you define to it, including those that you treat as foreign volumes under your existing system. DFSMSrmm supports any volume serial (VOLSER) and allows you to use a rack number that is different from the VOLSER. Thus you can manage all volumes except those that are duplicates of volumes already defined to DFSMSrmm. You can still process volumes as foreign to DFSMSrmm by either not defining them to DFSMSrmm or using the DFSMSrmm ignore support through the EDGUX100 exit.

1.2.1 Validation

DFSMSrmm automatically validates volumes, ensuring that only valid scratch volumes are mounted for nonspecific mount requests and that the right volume is mounted for a specific mount request. This validation eliminates the unintentional overwriting of a valid master volume or a volume retained for disaster recovery or vital record management.

When a data set on a volume is opened and closed, DFSMSrmm automatically:

- Changes the volume status from scratch to master for nonspecific mount requests
- Sets an expiration date for the volume and ensures that the maximum expiration date is not exceeded (repeated at close time)
- Records information about data sets on the volume (the data set name is recorded at open time; all other information is recorded at close time)
- Counts the number of times a volume is used
- Counts the number of temporary and permanent errors encountered
- Sets a security classification based on the data sets that reside on the volume
- Prevents reading of data on a volume in scratch status when DFSMSrmm is running in protect mode.

All of the above actions occur at open time, except where noted.

1.2.2 Library and Storage Location Management

You decide where to store your removable media according to how often the media are accessed and for what purpose they are retained. For example, you might keep volumes that are frequently accessed in an automated tape library dataserwer, and you probably use at least one storage location or vault to retain volumes for disaster recovery and audit purposes. You might also have locations where volumes are sent for further processing. Such locations would be data centers within your company or customer and vendor sites.

DFSMSrmm manages:

- **Removable media library**, which contains all tape and optical volumes that are available for immediate use and includes the shelves where they reside. A removable media library usually includes other libraries, such as:
 - System-managed tape libraries; for example, the IBM 3494 and 3495 Automated Tape Library Dataserver models, and the manual IBM 3495 Tape Library Dataserver.
 - Non-system-managed tape libraries, or traditional tape libraries.
- **Storage locations** that are not part of the removable media library because their volumes are not generally available for immediate use. Storage locations are typically used to store removable media that are kept for disaster recovery or vital records.

Figure 3 shows the relationship of the libraries and storage locations.

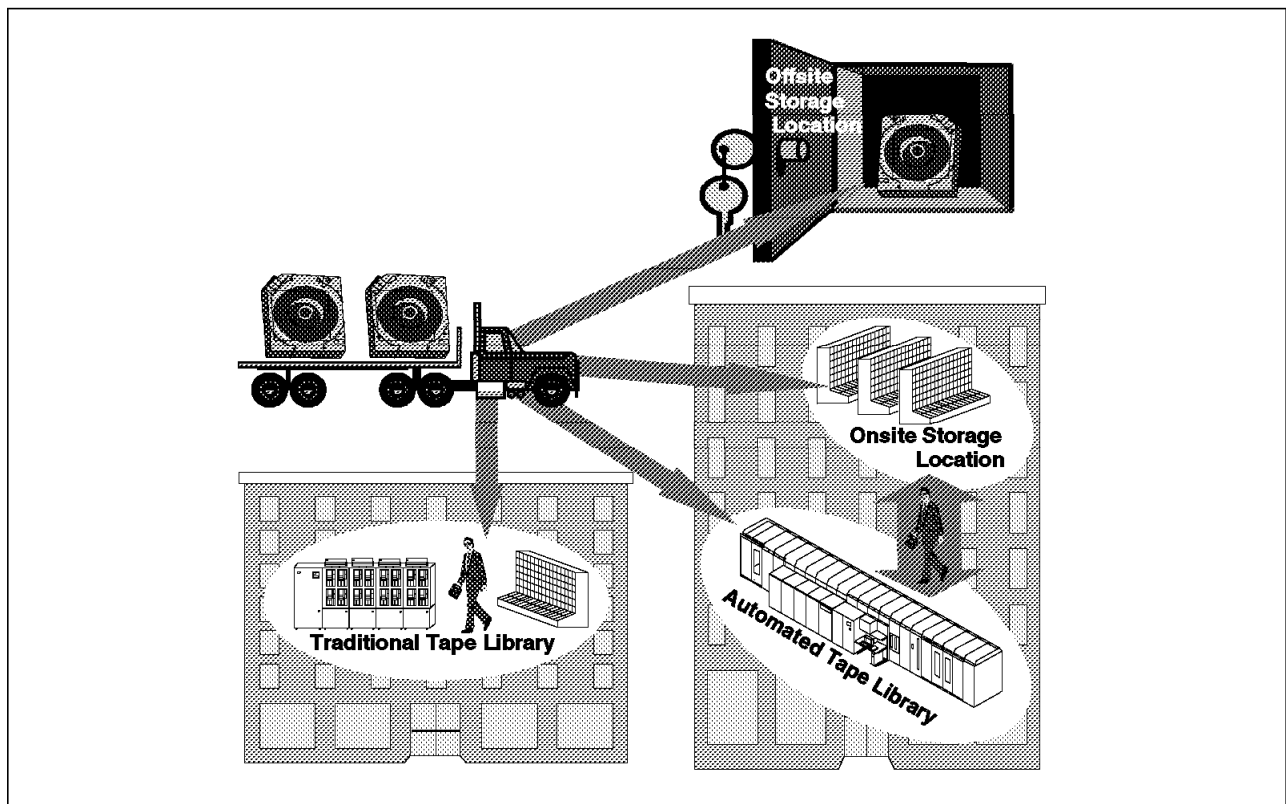


Figure 3. Removable Media Libraries and Storage Locations

1.2.2.1 Removable Media Library

In the removable media library, you store your volumes in shelves, where each volume occupies a single *shelf location*. In the RMM TSO subcommands and DFSMSrmm Interactive System Productivity Facility (ISPF) dialog, a shelf location in the library is called a *rack number*. A rack number matches the volume's external label. DFSMSrmm uses the external VOLSER to assign a rack number when adding a volume, unless you specify otherwise. The rack number is the external VOLSER. Thus both must be six alphanumeric characters. The internal VOLSER can be one to six alphanumeric characters. There are no restrictions on the format of the six character volume serial numbers you can define to DFSMSrmm.

System-Managed Tape Library: A system-managed tape library is a collection of tape volumes and tape devices, defined in the tape configuration database (TCDB).

The TCDB is an ICF catalog of type VOLCAT. It is a new SMS control data set containing tape library and tape volume records.

A system-managed tape library can be either automated or manual. You can have several automated or manual tape library dataservers; you use an installation-defined library name to define each automated tape library datserv or manual tape library datserv to the system, and DFSMSrmm treats each system-managed tape library as a separate location or destination.

An automated tape library datserv is a device consisting of robotics components, cartridge storage areas (or shelves), tape subsystems, and controlling hardware and software, together with the set of tape volumes that reside in the library and can be mounted on the library tape drives. IBM's automated tape libraries are the IBM 3494 and 3495 Automated Tape Library Dataservers.

A manual tape library datserv is a set of tape drives and the set of system-managed volumes the operator can mount on those drives. IBM's manual tape library is the IBM 3495 Tape Library Datserv Model M10.

Non-System-Managed Tape Library: A non-system-managed tape library is all the volumes, shelves, and drives not in an automated or manual tape library datserv. You might know this library as the traditional "tape library." DFSMSrmm provides complete tape management functions for the volumes and shelves in this traditional tape library. DFSMSrmm defines volumes in a non-system-managed library as being *shelf resident*.

All tape media and drives that the operating system supports are supported in a non-system-managed tape library environment. Using DFSMSrmm, you can fully manage all types of tapes in a non-system-managed tape library, including 3420 reels and 3480 and 3590 cartridge system tapes.

1.2.2.2 Storage Location

A storage location comprises shelf locations that you define to DFSMSrmm. A shelf location in a storage location is identified by a *bin number*. DFSMSrmm manages two types of storage locations: installation-defined storage locations and DFSMSrmm built-in storage locations.

You can define an unlimited number of installation-defined storage locations. You can use any eight-character alphanumeric name for a storage location and

define the type or shape of the media in the location and the bin numbers that DFSMSrmm assigns to the shelf locations in the location. You can request DFSMSrmm shelf management when you want DFSMSrmm to assign a specific shelf location to a volume in the location.

You can also use the DFSMSrmm built-in storage locations, LOCAL, DISTANT, and REMOTE. Although the names of these locations imply their purpose, they do not mandate their actual location. All volumes can be in the same or separate physical locations. For example, an installation could have the LOCAL storage location onsite, as a vault in the computer room, the DISTANT storage location could be a vault in an adjacent building, and the REMOTE storage location could be a secure facility across town. An installation defined storage location of SHELF could be onsite, and used for volume movement within a library to make better use of the library. Because DFSMSrmm provides shelf management for storage locations, storage locations can be managed at the shelf location level. Optionally you can define storage locations that are not shelf-managed.

Although DFSMSrmm automatically shelf-manages built-in storage locations, you must first define the bins you want to use to DFSMSrmm. For bin numbers in built-in storage locations, the numbers are fixed in range, starting at bin number 000001.

1.2.3 Policies for Retention and Movement

DFSMSrmm provides policy management for movement and retention at the data set level. Every tape data set can have a policy, and each policy can specify movement as well as retention. The retention and movement policies you define to DFSMSrmm are known as *vital record specifications* (VRSSs). You use them to indicate how long and where you want to keep data sets or volumes. You also use them to define how volumes are to be moved among the libraries that DFSMSrmm supports and the storage locations defined for vital records and disaster recovery purposes.

You can create a vital record specification *chain* to cause a sequence of volume moves. The first vital record specification in the chain is a data set or volume vital record specification. These specify one location in which the data set or volume being retained is to be stored. Add one name vital record specification for each additional location or retention value that applies to the volume or data set.

DFSMSrmm records the starting locations for volume when the volume is initially defined to DFSMSrmm, or when volume information is changed. This starting location is known to DFSMSrmm as a *home* location. Home is where volumes start from and are returned to when the identified retention and movement actions have been completed.

1.2.3.1 Retention

Use data set names and data set name masks to define retention policies for data sets. Use job names and job name masks to define retention policies to further qualify the criteria for applying retention and movement policies. For data sets, you can request the following types of retention:

Retention by PARMLIB option - VRSEL (NEW|OLD)

You can control the extent to which the data set name and management value VRSSs are used.

Retention by cycles

You can retain a minimum number of cycles or copies of a data set. This type of retention applies to generation data groups (GDGs) or like-named data sets identified by pseudo-GDG data set names. For non-GDG data sets, DFSMSrmm considers each occurrence of a data set to be a cycle.

Retention by BYDAYSCYCLE

You can specify to retain all instances of a data set created on a single day as a single cycle.

Retention by data set name until expired

You can retain a data set until the volume expiration date has been reached or the retention amount has elapsed.

Retention by number of elapsed days

You can retain each copy of a data set a given number of days since it was created.

Retention by days since last referenced

You can retain each copy of a data set for a set number of days since the data set was read or written.

Retention while data set is cataloged

You can retain any data set as long as it remains cataloged. Catalog status can also be used in combination with management by cycles, elapsed days, days since last referenced, and volume expiration date.

Retention to a specific date

You can set a deletion date for a VRS. When that date is reached, the VRS is deleted. All data sets and volumes that would match the VRS become eligible for release processing, or they might match a less specific VRS that specifies different retention and movement information.

Retention by expiration date

You can retain the data set on a volume as long as the volume expiration date has not yet been reached. You can also use a combination of catalog status, volume expiration date, and other retention options to retain data sets.

Retention of open data sets

You can specify a separate policy to apply to all data sets that are currently open.

Retention of data sets closed by abend processing

You can specify a separate policy to apply to all data sets that were open at the time of an application or system abend.

Your installation can also define a *VRS management value* to define retention policies. If you have specified VRSEL(NEW) in the EDGRMMxx PARMLIB member you can have up to two policies per data set: one policy using a data set name mask, and one using VRS management class or VRS management value. The definition of VRSEL(OLD) allows only one policy for each data set. For information refer to 12.1.1, "The Best Match Fit Sequence" on page 474. A VRS management value assigns management and retention values to tape data sets. You can define data set VRSS for VRS management values to provide support for special JCL-specified expiration dates and to allow DFSMSrmm to manage those data sets with special dates. You can use the sample installation

exit, EDGUX100, in SAMPLIB to assign VRS management values. You can use management class names when you are managing system-managed volumes.

To define retention and movement policies for volumes, you can use a specific or generic VOLSER. If you use a specific VOLSER, the volume matching that number is retained by days since creation. If you use a generic VOLSER, any volumes matching the generic VOLSER are retained.

1.2.3.2 Movement

DFSMSrmm records the starting location for a volume when the volume is initially defined to DFSMSrmm, or through a change to the volume information. This starting location is known to DFSMSrmm as a *home* location. Volumes are returned to the home location when the identified retention and movement actions have been completed.

You can give any system-managed library or storage location as a target destination for a volume move.

You can also define policies to provide retention information for data sets and volumes that must be moved through multiple locations before they expire. You define such policies by creating a VRS chain.

When multiple data sets on the same volume are retained by VRSs, and each VRS contains a different destination, DFSMSrmm decides where to move the volume according to priority number.

Unlike some other systems, DFSMSrmm does not do storage location management on the basis of the first file on a volume. Storage location management is based on the priority of the storage location management requirements for all data sets on a volume. Thus DFSMSrmm must sometimes do conflict resolution and only retains and moves volumes with a data set retained by a VRS.

Storage location management with DFSMSrmm is essentially similar to vault management in other tape management systems. What seems to be unique to DFSMSrmm is the concept of volume-in-transit. In other words, volumes have been scheduled for a move or are on their way, but they have not yet been received at the storage location. Operators can thus tell which volumes are still in transit. A special command moves volumes from in-transit to storage-location-resident status. Installations that have established special storage location names to cover transport time will need fewer storage locations. Installations that do not want librarians issuing this command can automate the confirmation of movement by issuing TSO commands or through REXX procedures.

If you have VRSEL(NEW) specified in the EDGRMMxx member of PARMLIB, you can now have separate movement and retention policies. Use a VRS based on management value or management class name to define retention criteria; use a VRS based on data set name mask for defining movement.

1.2.4 Environment

All data set, volume, and policy information is kept in the control data set (CDS). The CDS is a VSAM key-sequenced data set (KSDS) that contains all inventory information.

All DFSMSrmm interactions take place under control of the subsystem. Utilities are provided to start housekeeping functions so that you can schedule DFSMSrmm work through your production scheduling systems. One benefit of running as a subsystem is evident in the support for system-managed tape: DFSMSrmm uses a subsystem-to-subsystem interface to provide as much support as possible without involving the operator.

DFSMSrmm supplies ISPF panels as well as TSO commands. Thus almost anything you can do online you can also do in batch through TSO command execution.

DFSMSrmm provides utilities to manage your inventory, create reports, maintain the DFSMSrmm CDS, and erase and initialize volumes.

Use the EDGHSKP utility, with the DFSMSrmm subsystem active, to run inventory management activities, which include:

- Vital record processing, to determine which data sets to retain and which volume moves are required, based on VRSs
- A trial-run of inventory management vital record processing without making any changes to the control data set or journal
- Expiration processing, to identify volumes ready to be released and returned to scratch
- Storage location management processing, to assign shelf locations to volumes being moved to storage locations
- Backing up the CDS and the journal, and clearing the journal
- Creating an extract data set for report generation

Use the EDGUTIL utility to create, update, and verify the CDS. Use the EDGBKUP utility to back up and recover the CDS and journal. EDGUTIL and EDGBKUP can be executed independently of the subsystem.

1.2.4.1 Backup and Recovery

Using the backups of the CDS and journal taken by the EDGHSKP utility during inventory management you can recover the DFSMSrmm CDS at any time.

Figure 4 on page 11 shows you how you can use the latest backup of the CDS and the current journal file to restore and forward recover to the point of failure.

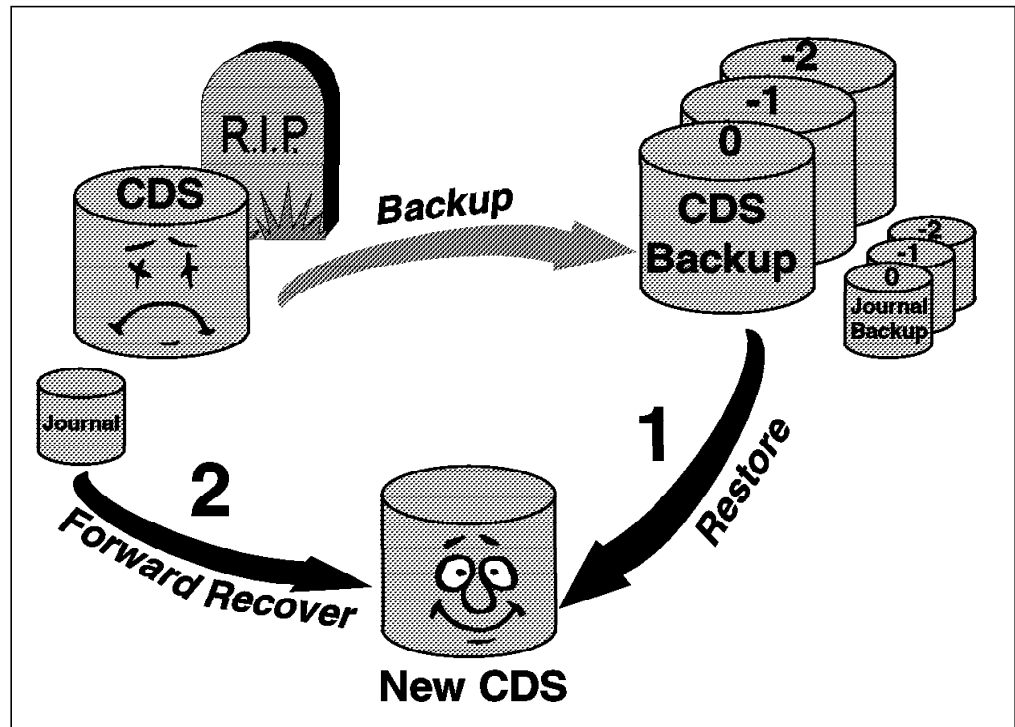


Figure 4. Recovering the CDS from Latest Backup

If the latest backup of the CDS is not available, you can use an older backup and the backups of the journal to recover. Figure 5 shows you how you can use the previous backup generation of the CDS, the latest journal backup, and the current journal file to restore and forward recover to the point of failure.

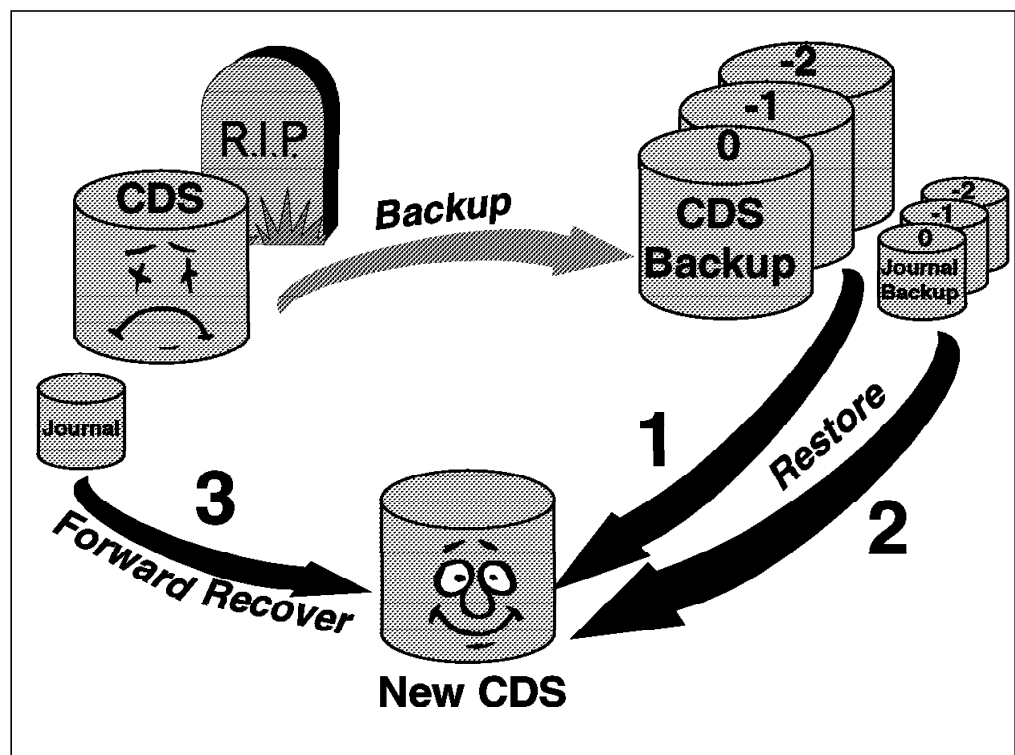


Figure 5. Recovering the CDS from Backup -1

Using this process you can recover from any CDS backup as long as you have the corresponding journal backups.

When backup is performed using the DFSMSDss DUMP command, DFSMSrmm backs up the CDS and then the journal. DFSMSrmm resets the journal data set and discards journal records if the back up completes successfully. To restore the CDS using the latest CDS backup, both the latest journal backup and the current journal must be used for forward recovery. DFSMSrmm clears the journal when the CDS and journal are backed up. Clearing the journal data set prevents the journal from becoming full, and reduces the risk of losing updates to the CDS. Backing up the journal using DFSMSDss allows you to write the data directly to tape. Using DFSMSDss concurrent copy allows simultaneous updates to the DFSMSrmm CDS so that other tape activity need not be interrupted.

1.2.4.2 Reports

You can obtain information and create reports, using:

- DFSMSrmm ISPF dialog or RMM TSO commands
- EDGAUD, EDGJRPT JCL, and EDGRPTD report utilities
- DFSORT's ICETOOL utility

Searches: You can do an online search using either the DFSMSrmm ISPF dialog or RMM TSO subcommands to create lists of resources and display information recorded in the DFSMSrmm CDS. Here are some examples:

- Operators can create lists of scratch volumes to be pulled for use.
- Tape librarians and system programmers can create lists of software products and the volumes on which they reside.
- General users can create lists of volumes they own, such as the example in Figure 6.

Volume	Owner	Rack	Assigned date	Expiration date	Location	Dsets	St	Act	Dest.
VOL600	AMYW01	RAC500	06/11/1998	11/11/1998	SHELF	0	UR	SI	
VOL601	AMYW01	RAC501	06/11/1998	11/11/1998	SHELF	0	UR	SI	
VOL603	AMYW01	RAC502	06/11/1998	11/11/1998	SHELF	0	UR	SI	
EDG3011I 3 ENTRIES LISTED									

Figure 6. List of Volumes Owned by a Single User

With DFSMSrmm, you can use the RMM TSO SEARCH subcommands with the CLIST operand to create a data set of executable subcommands. For example, you can create subcommands to confirm volume movement for volumes identified during a SEARCHVOLUME request.

Report Utilities: You can create several types of reports using two DFSMSrmm report utilities. Use EDGRPTD or EDGJRPT JCL to create movement and inventory reports and EDGAUD to create security and audit reports. EDGRPTD and EDGJRPT use the DFSMSrmm extract data set as input. EDGAUD uses system management facility (SMF) records as input.

You can use the reports to:

- Identify volumes that should be moved between the removable media library and storage locations
- Determine your volume inventory in the removable media library and storage locations
- Identify volumes that are in transit or should be marked as moved

- Identify all accesses to volumes and changes to information recorded in the DFSMSrmm CDS
- Separate volumes waiting to return to scratch from those that are private or have other release actions pending.

You can use DFSORT or a similar program to generate a formatted report using the information in the EXTRACT data set created by the EDGHSKP utility. Also, the ACTIVITY file and the EDGJACTP sample JCL can be used to create a report from it. For example, you could produce an extract data set listing all volumes to be used on VM with information about volume owners. Then use DFSORT's ICETOOL utility to sort the information by volume and produce a report, complete with title and header information.

1.2.4.3 Security

You can choose the authorization levels of users for all DFSMSrmm functions. DFSMSrmm uses the MVS system authorization facility (SAF) for its authorization checking. You define DFSMSrmm resources to Resource Access Control Facility (RACF) for use during authorization checking. DFSMSrmm can create volume profiles, change them, and delete them on registration, expiration, or release of volumes. You can use the access list that DFSMSrmm provides to set the access list in RACF. You can use this same list for authorization checking on non-RACF systems. To display the DFSMSrmm access list, use the RMM TSO LISTVOLUME subcommand or the DFSMSrmm ISPF dialog volume list function. You can also find the access list in the volume records in the report extract data set.

DFSMSrmm provides automatic security classification through installation-specified criteria based on data set names. It provides the following control of classified volumes:

- Audit trail of access and change of status through SMF. This audit trail produces information about RACF user ID, group, and job name.
- Operator confirmation of use.
- Erasure of data when a volume is released before it is returned to scratch status.

DFSMSrmm provides the following ways of optionally keeping an audit trail for volumes defined to it:

- Control data set information
- SMF audit records
- RACF audit information

1.2.5 Benefits of DFSMSrmm

DFSMSrmm has some benefits compared with other vendor tape management systems:

1. DFSMSrmm does not need "special dates" to implement expiration by last reference, cycle, or uncataloged. DFSMSrmm has easy-to-understand retention policies:
 - JCL controlled
 - LABEL=RETPD=nnnn
 - LABEL=EXPDT=yyddd
 - VRS controlled
 - While data set is cataloged
 - By number of elapsed days

- By day last referenced
 - By cycles and BYDAYSCYCLES
 - To a specific date
 - By expiration date
 - Using specific or generic filter criteria (like ISMF standard)
 - Data set name
 - VOLSER
 - Job name
2. DFSMSrmm has no problems with different retention definitions; you can mix VRS and JCL parameters at any time.
 3. DFSMSrmm has no limit for defining racks, volumes, and other resources to it.
 4. DFSMSrmm VOLSERS have no limitations:
 - They can have a length of 1 through 6 characters.
 - Alphanumeric and national characters can be used.
 5. DFSMSrmm controls the initialization and erasure of volumes.
 6. DFSMSrmm CDSs can be dynamically extended at any time by adding, for example, volumes, racks, program products, owners, and data sets.
 7. DFSMSrmm parameters can be changed without interrupting tape activities, for instance:
F DFRMM,M=xx (restarting the DFSMSrmm subsystem)
 8. DFSMSrmm can manage scratch, master, and user tapes. These can include transient volumes such as: foreign or external, PTF and/or APAR, and program products.
 9. DFSMSrmm supports RACF through TAPEVOL, TAPEDSN, TAPEVOL, and TAPEDSN.
 10. DFSMSrmm resources are protected with RACF profiles in the FACILITY class (for example, the use of DFSMSrmm utilities, DFSMS/MVS Removable Media Manager TSO subcommands, and ISPF dialog).
 11. DFSMSrmm can reply to the WTOR IEC507D (when you have modified the expiration date or released a volume in the DFSMSrmm CDS).
 12. DFSMSrmm writes SMF audit and security records.
 13. DFSMSrmm supports system-managed tape libraries, such as the automated IBM 3494 and IBM 3495 Tape Library Dataserver, the IBM 3495 Tape Library Dataserver Model M10, and the IBM 3590 High Performance Tape Subsystem, and is an integrated part of the IBM system-managed tape solution. It supports non-system-managed tape libraries, including all other robots.
 14. DFSMSrmm has an Application Programming Interface (API) through which you can issue any of the DFSMS/MVS Removable Media Manager TSO subcommands from most programming environments. The data is returned in either line mode or structured fields.
 15. DFSMSrmm has a TSO command interface, which includes an interface to REXX so that you can easily produce your own procedures that interface to RMM.
 16. DFSMSrmm uses ISPF as a user shell, which with the REXX interface provides a very flexible and extendible user interface.
 17. DFSMSrmm runs in parallel with other tape management systems.
 18. The DFSMSrmm report extract data set can be used to customize any type of reports.

1.3 Conversion Process Overview

The conversion process we describe in this book transfers your tape data information from your current tape management system database to the DFSMSrmm database, while continuing to meet your tape management requirements. Figure 7 shows how detailed data about tape volumes is converted to DFSMSrmm format.

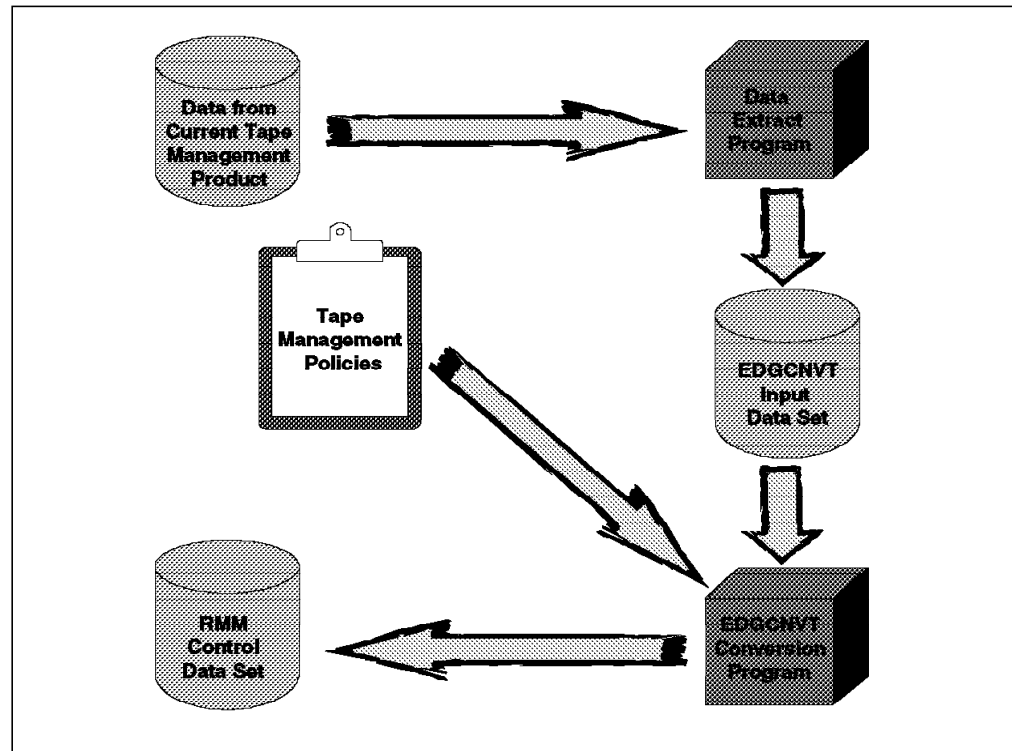


Figure 7. Vendor Product Data Conversion to DFSMSrmm

The data transfer is really a three-phase method. The first phase is the *data extraction phase*, where the data that DFSMSrmm needs is derived from your current tape management system database. The second phase, the *convert phase*, is the manipulation of the extracted data into a format that is, when loaded into a VSAM KSDS, useful to DFSMSrmm. The EDGCNVT and EDGCVOVL programs accomplish the second phase. The third phase, referred to as the *postprocessing phase*, is the loading of the VSAM KSDS, the addition of the DFSMSrmm CDS control record, and the utility run that cross checks the records in the DFSMSrmm database, one against the other, to verify completeness and accuracy.

The conversion process involves more than simple data transfer. It includes testing required functions, validating requirements, and educating users on the changes involved. In this book we describe the three phases of transferring data and help you plan for execution of the complete conversion process.

You can convert to DFSMSrmm in several different ways. For example, you could install DFSMSrmm to manage a set of volumes that your existing tape management system does not manage. Over time you can increase the number of volumes that DFSMSrmm manages and reduce the number of volumes that your existing system manages.

Another way of converting to DFSMSrmm—the conversion process that we recommend, and that the rest of this book describes—is to convert all existing volumes from your existing tape management system to DFSMSrmm. You convert your current management requirements into DFSMSrmm policies, and, after a period during which you validate that those requirements are being met, you remove your existing tape management system.

If you do not have a vendor tape management system, you can still convert to DFSMSrmm as long as you have some information about the volumes in your installation. For example, you may have manual records, or all tape data sets may be cataloged in an ICF catalog.

Included in the conversion process is the requirement to expand or consolidate the number of systems and tape databases. For example, you may have two tape databases on two systems today, but you may want to convert to a single database that is shared by the two systems or used by a single image produced from merging the two systems.

1.3.1 Structure

The structure of this book is geared to the conversion process that you should follow.

The process consists of eight stages:

1. **Learn about DFSMSrmm.**

In stage 1 you ensure that you have enough knowledge about DFSMSrmm to prepare for and plan conversion.

2. **Plan conversion.**

In stage 2 you use your knowledge of your current tape management system and the conversion process to build a plan.

3. **Start DFSMSrmm.**

In stage 3 you implement DFSMSrmm on your system. You install and test basic DFSMSrmm functions without impacting your current tape management.

4. **Extract data.**

In stage 4 you use the supplied data extract tools to pull details of tape data from your existing system. You analyze your tape management setup to understand which data to convert.

5. **Convert to DFSMSrmm.**

In stage 5 you convert the data extracted from your current system to DFSMSrmm format and load it into an DFSMSrmm CDS ready for use by DFSMSrmm.

6. **Validation.**

In stage 6 you run DFSMSrmm in parallel with your existing tape management system and ensure that DFSMSrmm provides the management functions you require.

7. **Production cut over.**

In stage 7 you remove the old tape management system and continue with DFSMSrmm in control.

8. **Full exploitation.**

In stage 8 you take advantage of some DFSMSrmm functions that will help improve the management of removable media in your installation.

1.3.2 Planning Considerations

As you prepare for conversion there are many factors to consider. Among them are:

- Software maintenance levels
- Timing of events in your installation
- Change management
- Limiting business risk

1.3.2.1 Software Maintenance Levels

You should know the exact release levels of your existing tape management software and the proposed level of DFSMSrmm. The level of DFSMSrmm can alter the functions that are available. The higher the maintenance level, the more new functions are available for you to use. DFSMSrmm ships many new functions through PTFs, so we recommend that you ensure that the level to which you plan to convert is the highest available.

Once you have started down the conversion path and have performed testing, we recommend that you install only the maintenance required to fix problems you have experienced. In this way you avoid having to repeat tests just to validate that a new software level has not regressed tests you have already completed. Once DFSMSrmm is in production you can revert to your usual software maintenance strategy.

1.3.2.2 Timing of Events in Your Installation

Be aware of the key events and changes planned for your installation over the potential conversion period. It is possible that any of these events could have an impact on your plan, or that changes you plan to make could impact events that are already in plan.

1.3.2.3 Change Management

As you develop your plan, use your existing change management process to add events to your installation's change schedule. This approach will ensure that any conflicts in what you plan can be resolved as early as possible and enable you to identify where impacts may arise if the plan does not execute as expected.

1.3.2.4 Limiting Business Risk

A key objective of the conversion process is to minimize or eliminate any risk to your business that could occur through the introduction of change into your systems. At each stage we identify what you can do to prepare and test each change so that there are no surprises as DFSMSrmm is introduced. Because you can run DFSMSrmm in parallel with your existing tape management system, you do not need to make a commitment to DFSMSrmm before you are actually ready to commit. You can take time to ensure that DFSMSrmm is meeting your business needs.

If you follow the conversion process described in this book and make decisions appropriate for your business, you will minimize your business risks.

1.3.3 Samples and Programs

In addition to this book and the DFSMSrmm publications listed in Appendix G, “Related Publications” on page 681, sample Jobs, CLISTs, REXX execs and programs are available to help you convert to DFSMSrmm in SAMPLIB.

The sample JCL is set up to work in a typical installation. Some of the sample JCL is commented out so that you can choose to use it or not as is your preference. Procedures, which are normally part of other IBM products, are used to assemble and compile the sample code. You do not have to assemble, link, and compile code before you can use it. If these procedures are not available to you, you can edit the JCL to uncomment the statements that execute the programs from a STEPLIB. If you choose to link-edit the programs yourself, you can use the standard link-edit options. None of the programs are reentrant.

1.3.3.1 Programs Provided with DFSMSrmm

Table 2 lists the programs provided with DFSMSrmm.

Table 2. Programs Provided with DFSMSrmm		
Name	Location	Function
EDGCNXIT	SAMPLIB	Source for EDGCNVT user exit EDGCNXIT
EDGCNVT	LINKLIB	Program to convert data to DFSMSrmm format
EDGCDREC	MODGEN	Macro that maps the input data set record to EDGCNVT
EDGCEREC	MODGEN	Macro that maps the input empty shelf record to EDGCNVT
EDGCKREC	MODGEN	Macro that maps the input VRS policy record to EDGCNVT
EDGCLREC	MODGEN	Macro that maps the input volume and first data set record to EDGCNVT
EDGCOREC	MODGEN	Macro that maps the input owner record to EDGCNVT
EDGCVOVL	LINKLIB	Program to complete owner records
EDGUTIL	LINKLIB	Program to initialize and update the CDS control record and validate and correct the contents of the CDS

1.3.3.2 Samples Described in This Book

Refer to Appendix D, “Sample Exits and Reports” on page 603 for a list of all samples described in this book.

Chapter 2. Planning for Conversion

In this chapter we introduce you to the conversion process and help you prepare a plan for conversion to DFSMSrmm.

Objectives of Chapter

- Outline the key steps in converting to DFSMSrmm
- Help formulate a plan and schedule
- Identify the milestones and objectives
- Introduce the stages that are covered in subsequent chapters
- Identify potential impacts on the production environment, such as IPLs, and the deactivation and removal of your existing tape management system.

Audience

- Data center managers
- System programmers
- Anyone who plays an active role in either the conversion or tape management.

At the end of this chapter you will understand the process to follow for conversion to DFSMSrmm and be able to develop a plan to execute that process.

2.1 Conversion Process

The objective of conversion is to use DFSMSrmm rather than your current tape management system to meet your tape management requirements. You can convert to DFSMSrmm in different ways. For example, you can:

- Define your volumes to DFSMSrmm by using the DFSMSrmm ADDVOLUME subcommand or the DFSMSrmm API.
- Use DFSMSrmm to manage a set of volumes that are not managed by your existing tape management system. Gradually increase the number of DFSMSrmm-managed volumes and remove volumes from your existing system.
- Define all known volumes to DFSMSrmm and run DFSMSrmm in parallel with your existing system until DFSMSrmm knows all details about the volumes.
- Use the conversion program and sample extract programs provided with DFSMSrmm to directly convert your tape management data and policies to DFSMSrmm format. Once data is converted, disable and remove your existing tape management system.
- Analyze your existing tape management setup and review your tape management policies for current applicability. Build a customized environment for DFSMSrmm that meets your installation's needs. Use the conversion program and sample extract programs provided with DFSMSrmm to directly convert your tape management data and policies to DFSMSrmm format. Run DFSMSrmm in parallel for a period of time to validate that the conversion is successful and that DFSMSrmm continues to meet your needs.

You could use any of the above processes to convert to DFSMSrmm, but only one process, the last process, includes the activities required to prepare for and validate conversion. This is the process that we recommend. In the rest of this book, we describe it in detail.

If you plan to use one of the other processes, you will still find that the information we discuss and present is useful. Therefore, we suggest that you review and use the information and learn from the experiences and knowledge that we draw on to put the process together.

DFSMSrmm provides functions that help with the conversion process. For example, you can use the DFSMSrmm running modes—manual, record, warning, and protect—to help you start DFSMSrmm on the same system on which you run your existing tape management system. You can start DFSMSrmm in manual mode and use the DFSMSrmm commands and utilities without DFSMSrmm recording tape activity. Any mode other than manual activates the DFSMSrmm tape recording function.

Using the different modes you can use DFSMSrmm to gain experience and to educate those that are involved in the conversion.

Note: Before starting DFSMSrmm ensure that you have read and understand the information provided in Chapter 3, “Starting DFSMSrmm” on page 27. If you start DFSMSrmm with unsuitable options you can potentially have an impact on your current tape management environment.

As we describe and take you through the conversion process we identify where you should perform testing and what type of testing to perform. You do not test the DFSMSrmm product, but test your use of the product to ensure that DFSMSrmm can meet your needs. Testing is performed and described in:

- Chapter 3, “Starting DFSMSrmm”
- Chapter 4, “Converting from CA-1 to DFSMSrmm,” Chapter 5, “Converting from TLMS to DFSMSrmm,” Chapter 7, “Converting from EPIC to DFSMSrmm,” and Chapter 8, “Converting from Manual Management to DFSMSrmm”¹
- Chapter 9, “Building the DFSMSrmm CDS”
- Chapter 11, “Cutover to Production”

Once you have converted to DFSMSrmm we want you to take advantage of the functions provided in the DFSMSrmm product. Chapter 13, “Additional DFSMSrmm Function” on page 495 provides details of some DFSMSrmm functions that you may want to review and consider implementing once the conversion is completed.

2.2 Conversion Stages

In this book we present information sequentially, in the order of the stages in the conversion process. Your route through the process most likely will not follow the sequential stages exactly, however. At times you will have to repeat a stage or step after modifying DFSMSrmm or the data. Such repetition will help you get as close as possible to the DFSMSrmm setup that will meet your requirements.

Figure 8 on page 21 illustrates the process that you should follow to successfully convert to DFSMSrmm.

¹ You use one of these chapters, depending on your current tape management system.

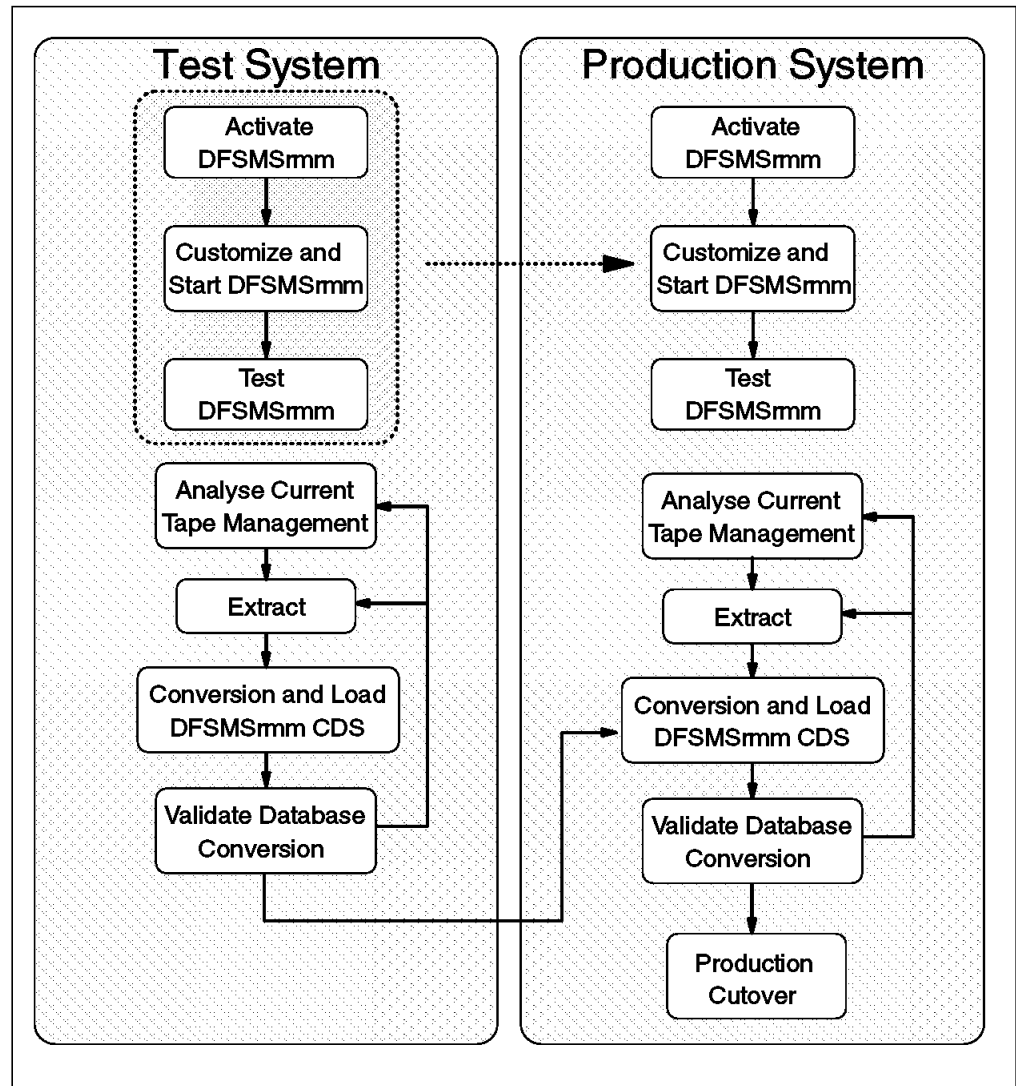


Figure 8. Conversion Process Flow

You can see that you must follow the same basic process in both a test and a production environment. In this book we guide you through the conversion process in a test environment. When you actually perform the conversion in production, you will be able to execute the steps easily with minimum risk.

If you have multiple MVS images or a shared DASD complex, you must repeat several of the conversion stages for each image or complex. We recommend which stages to repeat at the time we describe the stage.

The conversion stages are:

- Learning about DFSMSrmm
- Planning for conversion
- Starting DFSMSrmm
- Analyzing current environment
- Extracting data for conversion
- Completing conversion
- Parallel running and conversion
- Cutover to production
- Measuring success

- Implementing additional DFSMSrmm functions

Each stage has a reference to the chapter that covers it in detail.

2.2.1 Learning About DFSMSrmm

Before converting data to DFSMSrmm you must have some knowledge of the DFSMSrmm product. Chapter 1, “Introduction” on page 1 provides a quick introduction to DFSMSrmm. You can also gather more information by reading the *DFSMSrmm Guide and Reference*, SC26-4931 and *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

You can use the DFSMSrmm product itself to help you learn about DFSMSrmm. Use the information in Chapter 3, “Starting DFSMSrmm” on page 27 to get DFSMSrmm started in a test environment and then use the DFSMSrmm commands and ISPF dialog to gain experience. Ensure that the volumes you use are not the volumes that your existing tape management system requires or manages.

All system users and support people (for example, administrators, librarians, system programmers, and operators) who have any tape management responsibilities or use tape volumes require some education and knowledge about DFSMSrmm. Use the DFSMSrmm knowledge and experience that you gain to demonstrate the product to your users and colleagues.

2.2.2 Planning for Conversion

Once you have some basic knowledge about DFSMSrmm and the conversion process you can begin to plan how best to convert to DFSMSrmm. See 2.3, “Preparing a Conversion Plan” on page 25, where we describe how to build a plan and explain how to ensure that your conversion has the greatest chance of success.

2.2.3 Starting DFSMSrmm

At this stage you execute the steps necessary to get the DFSMSrmm product installed and started on one of your systems. This stage includes basic product setup and verifying that DFSMSrmm is installed correctly. See Chapter 3, “Starting DFSMSrmm” on page 27 for a detailed explanation of this stage.

If you are using the DFSMS/MVS product, you already have the DFSMSrmm component installed. Parts of the DFSMSrmm product are in the system link pack area (LPA), so you must IPL your system to ensure that the parts are installed into the LPA. Some maintenance to DFSMSrmm may also update the parts of DFSMSrmm that reside in the LPA. Remember that you should plan an IPL before starting DFSMSrmm after the initial SMP/E install, or after using SMP/E to apply maintenance to DFSMSrmm code that resides in LPA.

At this stage we recommend that you plan to test:

- Basic DFSMSrmm functions, such as:
 - Inventory management
 - Creating reports
 - Maintaining the CDS
 - Verifying the CDS
 - Initializing and erasing volumes

- Specific functions that you know are used in your existing tape management system, such as:
 - Scratch pooling
 - Keyword expiration dates
- Exits that must be changed to support parallel running, such as:
 - ARCTVEXT
 - CBRUXENT
 - CBRUXVNL

Note: With DFSMS/MVS Version 1 Release 4.0, DFSMSHsm and DFSMSRmm communicate directly without calling the ARCTVEXT exit. The DFSMSRmm EDGTVEXT programming interface will be used. Prior to DFSMS/MVS Version 1 Release 4.0 a sample ARCTVEXT exit was shipped with DFSMSRmm.

You can test DFSMSRmm by using test tape volumes and data or the data converted from your existing tape management system. Care must be taken when you use converted data for testing, so we recommend using test volumes and a test CDS.

2.2.4 Analyzing Current Environment

Before you can proceed with the conversion, you must understand which tape management functions and policies you are using today. You gain this understanding by analyzing your current tape management system implementation.

Analysis covers:

1. Inventory data gathering
 - Volumes
 - Media
 - Policies
 - Locations
 - Reports
2. Functions used
3. Interfaces to other products

To help you analyze your current environment, we have written a chapter for each vendor product for which we provide conversion support samples. Use the information in the following chapters to perform your analysis:

- For CA-1, use Chapter 4, “Converting from CA-1 to DFSMSRmm” on page 65.
- For TLMS, use Chapter 5, “Converting from TLMS to DFSMSRmm” on page 169.
- For EPIC, use Chapter 7, “Converting from EPIC to DFSMSRmm” on page 323.
- If you do not have a tape management system or have a system for which IBM does not yet provide sample extract programs, use Chapter 8, “Converting from Manual Management to DFSMSRmm” on page 367.

2.2.5 Extracting Data for Conversion

Before you can complete the conversion, you must extract the data from your current tape management system database.

Each of the product-specific chapters (Chapter 4, “Converting from CA-1 to DFSMSrmm” on page 65 to Chapter 8, “Converting from Manual Management to DFSMSrmm” on page 367) includes a section on how to execute and prepare for running the sample extract programs.

2.2.6 Completing Conversion

During this stage you build an DFSMSrmm CDS using the volume and data set details and the policy details extracted from your existing tape management system. Typically, you would convert one database to a single DFSMSrmm CDS. However, you may have a requirement to consolidate systems or databases and may want to convert the consolidated database to an existing DFSMSrmm CDS. Chapter 9, “Building the DFSMSrmm CDS” on page 403 explains what you must do to complete the conversion and refers you to 13.4, “Merging and Splitting the DFSMSrmm CDS” on page 512.

2.2.7 Parallel Running and Validation

This is a key stage in the conversion process. You use it to ensure that the data converted to DFSMSrmm is correct and that the converted policies provide the same retention that your existing tape management system provides.

Chapter 10, “Parallel Running and Validation” on page 427 provides detailed information on preparing DFSMSrmm and the system for parallel running. During this stage you may need to repeat the data conversion, make minor adjustments to the policies used with DFSMSrmm, or change the control statements for running the sample extract programs or the EDGCNVT convert program.

During this stage it is important that you minimize the manual changes made to your existing tape management system to avoid discrepancies with DFSMSrmm. However, the conversion is easily repeated, so you can repeat it whenever necessary. Each time you repeat the data conversion the tape management systems should be inactive to ensure that both products have the same point-in-time information.

The parallel running and validation stage can be as long or as short as you want it to be. For example, you may want to run in parallel for a month so that the different retention policies used during the month can be validated.

Once you have confirmed that the conversion is successful, and validation confirms that DFSMSrmm is providing the functions you require, you can proceed to the next stage.

2.2.8 Cutover to Production

Chapter 11, “Cutover to Production” on page 455 provides details of how to switch your tape management system to DFSMSrmm. As part of the cutover, you may need to change system exits, startup options, and inventory management. Also, you must now disable or remove your existing tape management system. You may need to do an IPL to implement some of the changes.

We recommend that you have a fallback or recovery plan in place in case any of the cutover actions cause problems.

2.2.9 Measuring Success

Appendix B, “Conversion Checklist” on page 585 provides a sample checklist for you to use to track the conversion tasks and verify that production cutover is successful. After you have prepared your plan, ensure that any additional key tasks are identified and added to the checklist. For each task include target dates and who is responsible so that you can manage and measure your plan.

2.2.10 Implementing Additional DFSMSrmm Functions

Once the conversion is successful you should consider implementing additional DFSMSrmm functions. We recommend that you consider doing this soon after cutover to production, as your knowledge of DFSMSrmm is fresh, and you may not have an opportunity to come back to add these functions because of other activities in your installation.

Chapter 13, “Additional DFSMSrmm Function” on page 495 details some of the additional DFSMSrmm functions that you can implement.

2.3 Preparing a Conversion Plan

You can increase the chances of a successful conversion by managing the conversion as a project. This approach ensures that the required tasks are performed and that the activities are tracked and managed.

It is likely that many other activities will be occurring in your installation at the same time. With so much going on it is all too easy to forget about a task or assume that it has been completed. Although managing the conversion as a project may appear to be unnecessary, changing your tape management system will affect many different groups of people in your installation, and your project will be able to keep these groups apprised of the conversion activities.

To ensure a successful conversion we recommend that you:

- Appoint a project manager.
- Appoint a project leader.
- Set objectives.
- Create a plan and schedule.
- Customize and use the checklist provided in Appendix B, “Conversion Checklist” on page 585 to track tasks and measure success.

Use this book to help build a plan for conversion. Each chapter identifies tasks that are recommended and tasks that are mandatory.

As you customize the checklist you will begin to understand which activities are required and how they must be managed in your installation. Based on this information you can construct a schedule and add the key steps to your change management system.

Some stages may require that you do an IPL or stop using tapes for short periods of time. These activities may have an impact on your production environment, so you should schedule them around other activities.

We recommend that you test the success of your DFSMSrmm installation, your planned use of DFSMSrmm, and any customization of system options, parameters, and exits. You will need a test system, or you will have to set up a test environment where you can run DFSMSrmm without impacting your production systems.

The sample schedule shown below is representative of a typical conversion project. Within the conversion plan there are specific points at which data is converted, and allowance has been made for a period of parallel running.

Activities	1999			
	Aug	Sep	Oct	Nov
Kick-off meeting	△ Establish project			
Learning about DFSMSrmm	■			
IPL	△			
Starting DFSMSrmm	■			
Analyzing current environment		■		
Extracting data		△ parallel	△ cutover	
Parallel running and validation		■		
Cutover to production			△ Measure success	
Remove old TMS			■	
Implementing additional RMM functions			■	
Education		△ operators	△ all users	

We do not want to dictate how long it should take to execute a conversion plan. We have experience with schedules that are much shorter and much longer than indicated in the above schedule, but the sample schedule is realistic and allows sufficient time for learning a new product and gaining experience with its use. Use the schedule as a model for your own schedule and as input to your existing change management and activity schedules.

Once you have customized the checklist in Appendix B, "Conversion Checklist" on page 585 and have prepared a schedule you are ready to proceed with the conversion process. Establish checkpoints along the way and review progress with the implementation team and your management.

Chapter 3. Starting DFSMSrmm

In this chapter we describe how to start DFSMSrmm on your system and test DFSMSrmm functions without affecting your current tape management system.

Objectives of Chapter

- Install and start DFSMSrmm on a test system or in a test environment
- Provide a base for learning about DFSMSrmm.

Audience

- System programmers
- Storage administrators and librarians

Considerations: If you have a current tape management system, it is essential that you introduce DFSMSrmm in a way that avoids any impact to your current processing. It is possible to do that, but you must select the correct DFSMSrmm PARMLIB options and ensure that the correct installationwide exits are in use. When you start DFSMSrmm, your existing tape management system can continue to run and be unaffected.

Although we recommend using test data and test volumes initially with DFSMSrmm, you can start DFSMSrmm by using a CDS that you created during the extract and conversion steps. If you start DFSMSrmm with a converted CDS, you should consider the environment within which you plan to run or test. Consider whether the test environment shares any data, catalogs, or volumes with your production environment. Any changes that DFSMSrmm makes as a result of processing could possibly update shared data. Use the correct DFSMSrmm PARMLIB options to prevent updates to RACF, the TCDB, and catalogs.

No matter how you plan to use DFSMSrmm, when you introduce it into any of your systems, it becomes visible to operators and perhaps some users. Those who will come into contact with DFSMSrmm should be provided with at least basic information about the product and how it affects them.

At the end of this chapter you will have started DFSMSrmm in a test environment and set it up for basic operations in your environment. You will be ready to create a CDS from your current tape management system and start customization.

3.1 DFSMSrmm Implementation Steps

In this section we guide you through the process of getting DFSMSrmm started on your system with minimal effort and without impacting any existing tape management system you might have. The intention is to do only essential customization and setup in order to get DFSMSrmm started quickly. In a later stage, when you have more knowledge of DFSMSrmm and your current tape management environment is ready for conversion, you will tailor and customize the DFSMSrmm installation to more specifically meet your needs.

Follow the steps to implement DFSMSrmm as documented in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, in conjunction with the modifications explained in this chapter.

3.1.1 Update Installationwide Exits

The installationwide exits that you install with DFSMSrmm replace any previously installed exits. The DFSMSrmm supplied programming interface and exits are:

- EDGTVEXT Programming Interface
- IGXMSGEX
- OAM exits
 - CBRUXENT (OAM cartridge entry exit)
 - CBRUXEJC (OAM cartridge eject exit)
 - CBRUXCUA (OAM cartridge change use attributes exit)
 - CBRUXVNL (OAM volume not in library exit)

Before you install DFSMSrmm you must determine if any of the DFSMSrmm supplied exits are used on your system. If they are used, and you still require the functions during parallel running or after conversion to DFSMSrmm, you must merge your existing exits with the DFSMSrmm supplied exits.

The EDGTVEXT programming interface is used by DFSMSHsm to tell DFSMSrmm which tapes are no longer in use by DFSMSHsm.

Prior to DFSMS/MVS Version 1 Release 4, the ARCTVEXT exit was used by DFSMSrmm to manage tapes that DFSMSHsm uses. With DFSMS/MVS Version 1 Release 4, DFSMSHsm and DFSMSrmm communicate directly without calling the ARCTVEXT exit. You do not need to activate the ARCTVEXT exit. The EDGTVEXT programming interface frees ARCTVEXT from DFSMSrmm usage, so you can use ARCTVEXT for other tape management systems.

If you will share one or more exits between your existing tape management system and DFSMSrmm, you must check if updates are needed in any of the exits to achieve coexistence when running in parallel.

There are two ways to do this. You can merge your existing exits or use a router exit which invokes your existing exit and the DFSMSrmm exit. There are detailed samples of the DFSMSHsm ARCTVEXT exits in Appendix D, "Sample Exits and Reports" on page 603 .

3.1.2 Update and Validate PARMLIB Members

Update the following SYS1.PARMLIB members:

- IEFSSNxx
- IKJTSOxx
- IFAPRDxx (has replaced IGDDFPKG)
- EDGRMMxx.
- SMFPRMxx (Optional)
- GRSRNLxx (Optional)
- SCHEDxx (Optional)

Perform this step once for each MVS image.

3.1.2.1 IEFSSNxx

There are considerations when defining the DFSMSrmm subsystem name and started procedure name described in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, under the topic "Updating IEFSSNxx". You should use that information along with the information provided here before updating the IEFSSNxx PARMLIB member.

You need to make two changes in IEFSSNxx:

- Define a DFSMSrmm subsystem name to MVS
- Add the name of the subsystem interface initialization program, EDGSSSI, to fully enable DFSMSrmm.

We recommend that you implement the changes in two steps. Define the DFSMSrmm system at this time as described in the above-mentioned manual. Then you can add the name of the subsystem interface later as described in "Initializing the DFSMSrmm Subsystem and Tape Recording". If you make only the first change at this time, you can choose whether to start the DFSMSrmm procedure and what time best suits your particular situation. If you implement both changes at once, DFSMSrmm rejects all tape mounts until the subsystem procedure is active or you use the EDGRESET utility to disable the interface.

Enabling the DFSMSrmm subsystem interface ensures that the interface will start every time you IPL the system. This step ensures that users cannot use tapes before the DFSMSrmm subsystem starts. To enable DFSMSrmm, update the IEFSSNxx member of SYS1.PARMLIB. Add EDGSSSI, as the DFSMSrmm subsystem initialization program, as shown in Figure 9.

SUBSYS SUBNAME(SMS)	/* SMS */	
INITRTN(IGDSSIIN)		
SUBSYS SUBNAME(JES2)	/* JES2 PRIMARY SUBSYSTEM START	*/
PRIMARY(YES) START(YES)		
SUBSYS SUBNAME(DFRM)	/* Name of the DFSMSrmm subsystem	*/
INITRTN(EDGSSSI)	/* DFSMSrmm initialization routine	*/
SUBSYS SUBNAME(AOPA)	/* Netview	*/

Figure 9. Defining DFSMSrmm to MVS through IEFSSNxx

Figure 9 shows the correct relative position of the DFSMSrmm subsystem, updated to include the initialization program.

Use the MVS system command SETSSI to define the DFRM subsystem dynamically. You can issue the SETSSI command from one of the following:

- A console that has master authority
- A console to which an operator with sufficient RACF authority has logged on.

To define the DFSMSrmm subsystem, use the following MVS system command:

```
SETSSI ADD,SUBNAME=DFRM
```

where

ADD specifies that a subsystem is to be dynamically added.

SUBNAME= specifies the subsystem name to be dynamically added.

3.1.2.2 IKJTSOxx

Update IKJTSOxx to authorize DFSMSrmm commands and utilities. If you use the TSO CSECT facility rather than IKJTSOxx, refer to *TSO/E Customization*, SC28-1307, for information on updating.

To authorize the TSO RMM command, add the statement shown in Figure 10.

```
AUTHCMD NAMES(RMM)
```

Figure 10. Updating IKJTSOxx to Authorize RMM Commands

To authorize the DFSMSrmm utilities that you can use within TSO or call through the TSO Service Facility, add the statements shown in Figure 11.

```
AUTHTSF NAMES(...  
                EDGHSKP  
                EDGUTIL  
                EDGRPTD  
                EDGAUD  
                ...)  
AUTHPGM NAMES(...  
                EDGHSKP  
                EDGUTIL  
                EDGRPTD  
                EDGAUD  
                ... )
```

Figure 11. Updating IKJTSO to Call DFSMSrmm through TSO

To use the updated version of the IKJEFTxx member dynamically use the following commands in your TSO session:

```
TSO PARMLIB LIST  
TSO PARMLIB UPDATE(**)
```

3.1.2.3 IFAPRDxx

When an installation orders an IBM product, like OS/390, that provides product enablement, IBM supplies a tailored IFAPRD00 member of SYS1.PARMLIB. This tailored member enables the product and any optional features ordered with the product. IFAPRDxx is the PARMLIB member that ensures you are licensed to use the components of DFSMS/MVS you want. IFAPRDxx replaced IGDDFPKG in OS/390 Version 1 Release 1.

Review your IFAPRDxx PARMLIB member to verify IFAPRDxx was updated during installation to reflect that DFSMSrmm is licensed for your use (see Figure 12 on page 31). For more information on IGDDFPKG and IFAPRDxx, refer to *OS/390 MVS Initialization and Tuning Reference*, SC28-1751.

```

PRODUCT OWNER(' IBM CORP')
NAME(OS/390)
ID(5647-A01)
VERSION(*) RELEASE(*) MOD(*)
FEATURENAME(DFSMSRMM)
STATE(ENABLED)

```

Figure 12. Updating IFAPRDxx to License DFSMSrmm Functions

You can use the SET PROD operator command to modify the enablement policy dynamically by specifying which IFAPRDxx member the system is to use. Statements in the IFAPRDxx member modify, not replace, an existing policy. The change to the policy takes place immediately but does not affect any product instances that are already running. To update the IFAPRDxx member dynamically use the following MVS system command:

```
SET PROD=xx
```

3.1.2.4 IGDDFPKG

In a non-OS/390 system, PARMLIB member IGDDFPKG is used to specify which DFSMS/MVS functional components are licensed for use on a particular MVS/ESA system, or across a sysplex; see Figure 13.

If DFSMS/MVS is installed on OS/390 Version 1 Release 1, only the IGDDFPKG parameters for Optimizer, ADSM, and DFSORT are processed. For all other DFSMS/MVS products, IFAPRDxx parameters are processed. It is recommended that you only use IFAPRDxx on an OS/390 system.

If DFSMS/MVS is installed on OS/390 Release 2 or later, IGDDFPKG is not processed. All product enablement policy is controlled by IFAPRDxx parameters.

```

/*****
/*  Valid Offerings      Components Enabled
/*  -----
/*      1                DFSMSdfp
/*      2                DFSMSdfp + DFSMSdss
/*      3                DFSMSdfp + DFSMSdss + DFSMShsm
/*      4                DFSMSdfp + DFSMSrmm
/*      Full             DFSMSdfp + DFSMSdss + DFSMShsm + DFSMSrmm
/*****
DFSMS_OFFERING=FULL

```

Figure 13. Updating IGDDFPKG to License DFSMSrmm Functions

3.1.2.5 EDGRMMxx

Use the following information together with the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, which describes in detail the options you can use with DFSMSrmm. In this section we describe the options that you can specify in EDGRMMxx if DFSMSrmm is running in parallel mode.

Create an EDGRMMxx PARMLIB member for DFSMSrmm (see Figure 14 on page 35). The member name is in the form EDGRMMxx, where xx is a two-character alphanumeric suffix of your choice. The default is EDGRMM00. A sample EDGRMMxx member is provided in SAMPLIB member EDGIVPPM; you

can copy it to PARMLIB and modify it according to your installation requirements.

For a description of the complete set of PARMLIB options that can be set in EDGRMMxx, refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

At this stage of your DFSMSrmm implementation we recommend that you specify the following in EDGRMMxx:

- **OPTION OPMODE** - Use either OPMODE(M) or OPMODE(R) at this stage. Use of the other modes can either cause additional messages to the user or impact tape processing as DFSMSrmm takes control.
If you have any system managed tapes, you should use OPMODE(R) to avoid updates to the TCDB made by DFSMSrmm.
- **OPTION TPRACF** - Use TPRACF(N) at this stage. Use of the other values will result in DFSMSrmm making updates to the RACF security profiles for tape data.
- **OPTION UNCATALOG** - Use UNCATALOG(N) at this stage. Use of the other values will result in DFSMSrmm uncataloging tape data that could be under the control of your existing product.

Following is a list of the most important options in the EDGRMMxx PARMLIB member if DFSMSrmm is running in parallel to another tape management system:

OPMODE	Specifies the mode of DFSMSrmm
R	Record-only mode. You can use DFSMSrmm TSO commands, ISPF dialog, and inventory management functions. In addition, DFSMSrmm records information about magnetic tape volumes used on the system, including details about volume owners and data set names. For volumes in an automated tape library dataserber, DFSMSrmm records information about entry and eject activities. During entry processing, DFSMSrmm provides volume status information to OAM. If a volume is not already defined to DFSMSrmm, DFSMSrmm automatically defines it as long as there are no conflicts with either the VOLSER or corresponding rack number. DFSMSrmm does not validate or reject volumes when in record-only mode. When DFSMSrmm is running in record-only mode, the volume status in the TCDB is not updated. When you issue DFSMSrmm commands, the TCDB is updated if appropriate and the volume is system-managed. When volumes are returned to scratch status, no TCDB updates are made.
W	Warning mode. In addition to performing the actions described for record-only mode, DFSMSrmm validates magnetic tape volumes as you use them. If DFSMSrmm discovers errors, it issues error messages instead of rejecting tapes.
DATEFORM	Specifies the date format for messages and reports (use the date format preferred in your computing center).

DSNAME	Specifies the name of the DFSMSrmm CDS (use a name up to 44 characters long). Default: If you do not specify DSNAME, you must specify the data set name in the MASTER DD statement in the DFSMSrmm started procedure. If you specify a name for both DSNAME and MASTER DD, DFSMSrmm ignores the MASTER DD statement.
JRNLMAME	Specifies the name of the journal. Specify a name up to 44 characters long. Default: If you do not specify JRNLMAME in EDGRMMxx, you can specify a name in the JOURNAL DD statement in the DFSMSrmm started procedure. If you do not specify a journal name in either EDGRMMxx or the started procedure, DFSMSrmm does not provide journaling. If you specify a name in both, DFSMSrmm uses the JRNLMAME value and ignores the JOURNAL DD statement.
MAXHOLD	Specifies the maximum number of I/O operations to the CDS before the reserve is released and reacquired. Specify MAXHOLD to minimize the impact that long operations, such as RMM TSO ADD subcommands with large count values, or large searches, have on other users.
RETPD	Specifies the default retention period for all new data sets on tape volumes (use a value between 0 and 9999 days). The specified value is added to the current date to determine the expiration date. Note: The value you specify for the RETPD operand should be the same as the value you use in your current tape management system.
TPRACF	Do not create RACF tape profiles.
UNCATALOG	Do not use DFSMSrmm uncatalog processing. If UNCATALOG(N) is set DFSMSrmm does not uncatalog data sets under any circumstances. Attention: DFSMSrmm will uncatalog data sets in each operating mode if the UNCATALOG function is set to Y or S.
VRSJOBNAME	Specify this operand to select whether DFSMSrmm uses the job name from VRS as the primary or secondary match. Specify VRSJOBNAME(1) so that the job name takes precedence over the data set name during VRS match processing. Specify VRSJOBNAME(2) so that the data set name takes precedence over the job name during VRS match processing. This is the option that most closely matches the use of job names in vendor tape management systems.
REJECT	DFSMSrmm can reject a range of tapes only if you first define them to DFSMSrmm. It can reject all tapes not defined to it, which you specify with a single asterisk for the prefix, but it cannot reject a range of tapes not defined to it. Note: If you specify REJECT ANYUSE(*), all tapes not defined to the DFSMSrmm CDS cause an error message to be issued by DFSMSrmm when the tapes are used.

SECCLS	Use the SECCLS command to define security classes for data sets and volumes. These security classes appear in reports and RMM TSO subcommand output. DFSMSrmm records these security classes in the DFSMSrmm CDS only, it does not make RACF aware of them. There is no connection between these definitions and any similar definitions or function provided in RACF, but you can use similar values for overall consistency.
MNTMSG	Use the MNTMSG command to tailor mount and fetch messages so they display the VOLSER and pool name or rack number. This information makes it easier for the operator to pull and mount volumes.
VLPOOL	Use the VLPOOL command to define pools. When you add a new volume to the library, DFSMSrmm assigns it a shelf location from the specified pool. In DFSMSrmm you can specify two categories of pools. The TYPE parameter specifies the type of pool: R A rack pool is shelf space assigned to hold volumes that are generally read-only, and that enter and leave your installation on an ad hoc basis. These volumes are typically software product volumes and customer volumes and do not adhere to your installation's naming conventions. S A scratch pool is assigned to shelf space to hold all other volumes in the removable media library. The volumes assigned to this shelf space can be used to satisfy scratch requests as long as the volumes are in scratch status. Once the volume has been written to, it becomes a master volume until the data is no longer required by the installation. The volume remains in the same DFSMSrmm scratch pool in that it occupies the same shelf space regardless of status. We recommend that you specify all of the pools that are defined in your current tape management system when running in parallel.
VRSEL	Use VRSEL to control how DFSMSrmm inventory management vital record processing uses retention and movement information defined in VRSSs. OLD Specify old if you want to: • Ignore retention information defined in NAME VRSSs • Ignore release options defined in VRSSs • Treat the ANDVRS in the RMM ADDVRS command as if it is a NEXTVRS NEW Specify NEW if you want: • DFSMSrmm to process retention information in NAME VRSSs, release options defined in VRSSs, and VRSSs chained using the RMM ADDVRS operand. • If you are converting using the conversion tools for CA-1 or TLMS for functions close to your current function.

Figure 14 on page 35 is a sample EDGRMMxx PARMLIB member you can use as a sample when preparing for DFSMSrmm recording mode.

OPTION	OPMODE(R)	/* Recording Mode */ -
	ACCOUNTING(J)	/* Accounting from JOB */ -
	BACKUPPROC(EDGBKUP)	/* Name of BACKUP-proc */ -
	BLP(RMM)	/* RMM controls BLP */ -
	CATRETPD(0012)	/* catalog retention */ -
	CDSID(PROD)	/* control data set id */ -
	DATEFORM(J)	/* Date format */ -
	DSNAME(RMM.PROD.CDS)	/* CDS data set name */ -
	IPLDATE(N)	/* IPL date checking */ -
	JRNLCNAME(RMM.PROD.JRNL)	/* JRNL data set name */ -
	JOURNALFULL(75)	/* Percentage JRNL full */ -
	LINECOUNT(054)	/* Lines per page */ -
	MASTEROVERWRITE(LAST)	/* Overwriting of a vol */ -
	MAXHOLD(100)	/* Number of I/O oper. */ -
	MAXRETPD(9999)	/* Maximum retention */ -
	MSG(M)	/* case for message txt */ -
	NOTIFY(Y)	/* Notify volume owners */ -
	PDA(OFF)	/* PDA is disabled */ -
	PDABLKCT(255)	/* number of blocks */ -
	PDABLKCT(31)	/* blocksize in K */ -
	PDALOG(ON)	/* PDA output disabled */ -
	RETPD(0005)	/* Default retention */ -
	SCRATCHPROC(EDGXPROC)	/* ATL/MTL procedure */ -
	SMFAUD(248)	/* SMF audit records */ -
	SMFSEC(249)	/* SMF security records */ -
	SYSID(EGZB)	/* Name of the system */ -
	TPRACF(N)	/* RACF tape support */ -
	UNCATALOG(N)	/* Catalog support */ -
	VRSCCHANGE(INFO)	/* No additional action */ -
	VRSJOBNAME(2)	/* DATASETNAME/JOBNAME */ -
	VRSMIN(0000000100,WARN)	/* DATASETNAME/JOBNAME */ -
	VRSEL(NEW)	/* Use VRS enhancements */ -
	TLCSV1(N)	/* Call TLCS O/C/EOV */ -

Figure 14 (Part 1 of 4). Sample EDGRMMxx PARMLIB Member

```

/* ***** */
MNTMSG MSGID(IEF233A) ID(001) VOLUME(016) RACK(016)
MNTMSG MSGID(IEF233D) ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEF234E K') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEF234E R') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEF234E D') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(IEF455D) ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(IEC501A) ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEC502E K') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEC502E D') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEC502E R') ID(001) VOLUME(016) RACK(999)
MNTMSG MSGID(' IEC502E RD') ID(001) VOLUME(017) RACK(999)
MNTMSG MSGID(' IEC502E RK') ID(001) VOLUME(017) RACK(999)
MNTMSG MSGID(IAT5110) ID(001) VOLUME(044) RACK(999)
MNTMSG MSGID(IAT5210) ID(001) VOLUME(050) RACK(999)
MNTMSG MSGID(IAT5410) ID(001) VOLUME(020) RACK(999)
/* ***** */
REJECT ANYUSE(*)
/* ***** */
VLPOOL PREFIX(SC*) -
      TYPE(S) -
      RACF(N) -
      MEDIANAME(3590) -
      EXPDTCHECK(N) -
      DESCRIPTION('3590 DEFAULT SCRATCH POOL')
VLPOOL PREFIX(N*) -
      TYPE(R) -
      RACF(N) -
      EXPDTCHECK(O) -
      DESCRIPTION(' TEST POOL FOR NORBERT')
VLPOOL PREFIX(F*) -
      TYPE(R) -
      RACF(Y) -
      EXPDTCHECK(Y) -
      DESCRIPTION(' FOREIGN CARTRIDGES')
VLPOOL PREFIX(PP*) -
      TYPE(R) -
      RACF(Y) -
      EXPDTCHECK(Y) -
      DESCRIPTION(' SOFTWARE CARTRIDGES')

```

Figure 14 (Part 2 of 4). Sample EDGRMMxx PARMLIB Member

```

/* ***** */
LOCDEF    LOCATION(SAFE)                -
          TYPE(STORAGE)                 -
          PRIORITY(2000)                 -
          MEDIANAME(*)                   -
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(SANJOSE)             -
          TYPE(STORAGE)                 -
          PRIORITY(2000)                 -
          MEDIANAME(*)                   -
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(NEWYORK)             -
          TYPE(STORAGE)                 -
          PRIORITY(2000)                 -
          MEDIANAME(*)                   -
          MANAGEMENTTYPE(NOBS)
LOCDEF    LOCATION(IBMATL1)             -
          TYPE(LIBRARY)                 -
          PRIORITY(4800)
/* ***** */
SECCLS    NUMBER(010)                   -
          NAME(UCL)                     -
          SMF(N)                        -
          MESSAGE(N)                    -
          ERASE(N)                      -
          DESCRIPTION(' UNCLASSIFIED')  -
          MASK(**)
SECCLS    NUMBER(050)                   -
          NAME(IUO)                     -
          SMF(Y)                        -
          MESSAGE(N)                    -
          ERASE(N)                      -
          DESCRIPTION(' INTERNAL USE ONLY')
          MASK(**.IUO.**,**.INTERNAL.**,**.INT.***)
SECCLS    NUMBER(100)                   -
          NAME(IC)                      -
          SMF(Y)                        -
          MESSAGE(N)                    -
          ERASE(N)                      -
          DESCRIPTION(' CONFIDENTIAL')  -
          MASK(**.IC.***)

```

Figure 14 (Part 3 of 4). Sample EDGRMMxx PARMLIB Member

SECCLS	NUMBER(125)	-
	NAME(ICR)	-
	SMF(Y)	-
	MESSAGE(N)	-
	ERASE(Y)	-
	DESCRIPTION(' CONFIDENTIAL RESTRICTED')	-
	MASK(**.ICR.**)	-
SECCLS	NUMBER(150)	-
	NAME(RIC)	-
	SMF(Y)	-
	MESSAGE(Y)	-
	ERASE(Y)	-
	DESCRIPTION(' REGISTERED CONFIDENTIAL')	-
	MASK(**.RIC.**)	-

Figure 14 (Part 4 of 4). Sample EDGRMMxx PARMLIB Member

3.1.2.6 SMFPRMxx (Optional)

If you want security or audit records, update SMFPRMxx to define the SMF record numbers that DFSMSrmm generates. In DFSMSrmm, you define the SMF record numbers in the EDGRMMxx PARMLIB member using two operands: OPTION SMFAUD for auditing records and OPTION SMFSEC for security records. For more information on these commands, see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

Select two SMF record numbers in the 128 to 255 range that your installation is not using. Add your SMFAUD and SMFSEC record numbers to SMFPRMxx as specified in *OS/390 MVS System Management Facilities (SMF)*, GC28-1783. By default, there is no SMF recording.

3.1.2.7 GRSRNLxx (Optional)

If you are not using global resource serialization (GRS) to control enqueues among sharing systems, you do not have to update GRSRNLxx. If you have multiple systems connected in a GRS ring, decide how you want GRS to handle the reserve on the CDS:

1. If the volume where the CDS resides does not contain other critical data, and real reserves will not cause any shared DASD contention problems, add the reserve name to the GRS exclusion list. This will leave the real hardware reserve in effect and cause the associated SYSTEMS enqueue to be converted to a local SYSTEM enqueue. The delay associated with sending the additional unnecessary enqueue around the GRS ring will then be avoided.

Figure 15 shows the required statements in GRSRNLxx to convert the SYSTEMS enqueue to a local SYSTEM enqueue.

```
RNLDEF RNL(EXCL) TYPE(SPECIFIC)
QNAME(SYSZRMM)
RNAME(MASTER.RESERVE)
```

Figure 15. Converting the SYSTEMS Enqueue to a Local SYSTEM Enqueue

2. If the volume where the CDS resides contains other critical data, and real reserves could impact other systems or cause DASD lockouts, add the

reserve name to the GRS reserve conversion list. This will leave the global SYSTEMS ENQUEUE in effect and remove the real hardware reserve.

Figure 16 shows the required statements in GRSRNLxx to convert the RESERVE to a SYSTEMS enqueue.

```
RNLDEF RNL(CON) TYPE(SPECIFIC)
QNAME(SYSZRMM)
RNAME(MASTER.RESERVE)
```

Figure 16. Converting the RESERVE to a SYSTEMS Enqueue

Note: If you are not using GRS, and use an equivalent product instead, use the ENQ resource name information provided in the DFSMSrmm publications to define the resource correctly to the other product. Do not use generic or global options for SYSZRMM. You must ensure that each resource with SYSZRMM major name is processed correctly.

3.1.2.8 SCHEDxx (Optional)

Update SCHEDxx if you want to ensure that the PARMLIB data set named in the DFSMSrmm started procedure does not cause enqueue contention while DFSMSrmm is running. If the DFSMSrmm procedure JCL allocates the PARMLIB data set, we recommend that you avoid the resulting enqueue. If you use the default of SYS1.PARMLIB, or a data set in the system PARMLIB concatenation, you do not need to update SCHEDxx.

Update the program properties table (PPT) in SCHEDxx as shown in Figure 17.

```
PPT  PGMNAME(EDGMAIN)
      NODSI
PPT  PGMNAME(EDGRESET)
      NODSI
```

Figure 17. Updating SCHEDxx

If you update SCHEDxx, the DFSMSrmm started procedure will not enqueue any of the data sets it allocates. Someone who is authorized to use these data sets could scratch them while DFSMSrmm is running without DFSMSrmm knowing about it.

To update the SCHEDxx member dynamically, enter the following command from an MVS system console:

```
SET SCH=xx
```

3.1.3 Update the Procedure Library

Create a procedure in SYS1.PROCLIB to start the DFSMSrmm subsystem address space. Figure 18 on page 40 shows sample JCL where:

- DFRMM** Specifies the procedure name. You can use any procedure name, up to eight characters long, subject to the restriction documented under Updating IEFSSNxx in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.
- M** Allows you to specify a PARMLIB member suffix. M=00 tells DFSMSrmm to use EDGRMM00.

OPT Allows you to specify:

OPT=RESET to disable the subsystem interface
OPT=MAIN to execute the DFSMSrmm subsystem

Before DFSMSrmm disables the subsystem interface, it ensures that the user who made the request is authorized.

PARMLIB, IEFORDER, MASTER, and JOURNAL DD

Are optional statements. If not coded in the JCL, you must supply the CDS name in EDGRMMxx, as well as the journal data set name if you require journaling. You must catalog the journal. If you do not code PARMLIB, IEFORDER is redundant.

EDGPDOX and EDGPDOY DD Are optional statements for external trace output recording. If you do not specify these DD statements, no logging of the PDA trace output is performed. *hlq* is the high-level qualifier for the PDA trace data sets (for example, the DFSMSrmm RACF user ID), and *hostid* is the MVS system image id.

The EDGPDOX and EDGPDOY data sets must be preallocated and cataloged to be used by DFSMSrmm. Both data sets must be allocated on the same volume. The DFSMSrmm RACF user ID requires ALTER access to the EDGPDOX and EDGPDOY data sets.

```
//DFRMM PROC M=00,OPT=MAIN
//IEFPROC EXEC PGM=EDG&OPT.,PARM='&M',TIME=1440,REGION=4096K
//PARMLIB DD DDNAME=IEFORDER
//IEFORDER DD DISP=SHR,DSN=SYS1.PARMLIB
//*MASTER DD DISP=SHR,DSN=RMM.CONTROL.DSET
//*JOURNAL DD DISP=SHR,DSN=RMM.JOURNAL.DSET
//EDGPDOX DD DISP=SHR,DSN=hlq.hostid.RMMPDOX
//EDGPDOY DD DISP=SHR,DSN=hlq.hostid.RMMPDOY
```

Figure 18. Creating a DFSMSrmm Procedure in SYS1.PROCLIB

3.1.4 Create the DFSMSrmm CDS

The DFSMSrmm CDS is a VSAM KSDS that contains the complete inventory of the removable media library. DFSMSrmm records all changes made to the inventory, such as adding or deleting volumes, in the CDS.

To create the CDS on the system, perform the following steps:

1. Assess DASD space and placement.
2. Allocate space for the CDS.
3. Protect the CDS.
4. Initialize the CDS for DFSMSrmm subsystem use by running the EDGUTIL utility.
5. Back up the CDS.

The JCL in Figure 19 has been modified to allocate a bigger INDEX than that calculated by standard VSAM. Note, however, that we recommend using cached DASD rather than IMBED and REPLICATE.

There is sample JCL to allocate the CDS in SYS1.SAMPLIB member EDGJMFAL which you can modify to meet your installation standards.


```

//LCLCDSAL JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* ***** *
//*          STEP 1 Delete old DFSMSrmm control data set *
//*          2 Allocate a new DFSMSrmm control data set *
//* ***** *
//STEP01 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE RMM.PROD.CDS CL
SET MAXCC = 0
/*
//STEP02 EXEC PGM=IDCAMS                      ALLOCATION MASTER FILE
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR,UNIT=3390,VOL=SER=DFRMM4
//SYSIN DD *
DEFINE CLUSTER (NAME(RMM.PROD.CDS)             -
                FILE(MASTER)                   -
                FREESPACE(35 20)                -
                KEY(56 0)                      -
                REUSE                           -
                RECSZ(512 9216)                 -
                SHR(3 3)                       -
                STORCLAS(storage_class)        -
                VOLUMES(DFRMM4))               -
DATA          (NAME(RMM.PROD.CDS.DATA)         -
                KILOBYTES(35532 3553)          -
                CISZ(10240)                    -
                FILE(MASTER))                  -
INDEX         (NAME(RMM.PROD.CDS.INDEX)        -
                CISZ(2048)                     -
                KILOBYTES(3815 381)            -
                FILE(MASTER)                   -
                IMBED                           -
                REPLICATE)
/*

```

Figure 19. Sample JCL to Create the DFSMSrmm CDS

Before using the DFSMSrmm CDS, a control record must be written in the CDS using the EDGUTIL utility. The control record contains information about the number of shelf locations in the library and storage locations.

Sample JCL is provided in SYS1.SAMPLIB member EDGJUTIL which you can modify to meet your installation standards.

Figure 20 on page 42 shows sample JCL that can be used to initialize the CDS.

```

//EDGUTIL JOB ,140.SCHLUMBERGER,MSGCLASS=H,REGION=6M,
//      MSGLEVEL=(1,1),NOTIFY=SCHLUM
//*****
//*          STEP 1  Initializing the Control Data Set          *
//*****
//EDGUTIL EXEC PGM=EDGUTIL,PARM='CREATE'
//SYSPRINT DD  SYSOUT=*
//MESSAGE DD  SYSOUT=*
//MASTER DD  DISP=OLD,DSN=RMM.PROD.CDS
//SYSIN DD  *
CONTROL
/*

```

Figure 20. Sample JCL to Initialize the DFSMSrmm CDS

For initial testing with DFSMSrmm, a small CDS will do. When you convert your existing data to DFSMSrmm, you will need a CDS sized to accommodate your existing and planned requirements. See 9.3, “DFSMSrmm CDS Creation” on page 420 for more information about how to create a CDS for conversion.

3.1.5 Create the DFSMSrmm Journal

The DFSMSrmm journal contains a record of all changes made to the CDS since the last backup. We recommend that you create the journal and use it to forward recover changes made since the last backup. Each time you successfully back up the CDS by using the EDGHSKP utility, the journal data set is cleared.

We recommend that you allocate the journal data set on a different volume than the CDS, back up the journal data set, and maintain multiple generations of it.

To create the DFSMSrmm journal on the system, perform the following steps:

1. Assess DASD space and placement
2. Allocate space for the journal
3. Protect the journal.

Sample JCL to create the DFSMSrmm journal is provided in SYS1.SAMPLIB member EDGJNLAL. You can modify the sample JCL to meet your installation standards.

Figure 21 on page 43 shows sample JCL to create the DFSMSrmm journal.

```

//LCLJRNLA JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
//*          STEP 1  DELETE OLD DFSMSRMM JOURNAL                      *
//*          2  ALLOCATE A NEW DFSMSRMM JOURNAL                      *
//* *****
//STEP01 EXEC PGM=IDCAMS
//SYSPRINT DD  SYSOUT=*
//SYSIN DD  *
DEL RMM.PROD.JRNL
SET MAXCC=0
/*
//STEP02 EXEC PGM=IEFBR14
//SYSPRINT DD  SYSOUT=*
//JOURNAL DD  DISP=(,CATLG,DELETE),
//          DSN=RMM.PROD.JRNL,
//          UNIT=3390,VOL=SER=DFRMM3,
//          AVGREC=U,SPACE=(4096,(6957),,CONTIG)

```

Figure 21. Sample JCL to Create the DFSMSrmm Journal

For initial testing with DFSMSrmm, either no journal or a small journal will do. Later, when you use converted data, you will need to use a journal that is large enough for your daily needs.

3.1.6 Protecting DFSMSrmm Resources

To protect DFSMSrmm functions you need to use an external security product, such as RACF. Invocation to this product should be through the security authorization facility (SAF). If RACF is not installed you must provide equivalent function through the SAF interface.

Appendix A, “Security Considerations and Implementation” on page 561 provides a detailed description on how to protect your DFSMSrmm resources using RACF, and sample RACF definitions. The security implementation tasks are as follows:

- Assign a RACF user ID to DFSMSrmm
- Define RACF groups for DFSMSrmm
- Define DFSMSrmm resources to RACF
 - STGADMIN.EDG.FORCE
 - STGADMIN.EDG.HOUSEKEEP
 - STGADMIN.EDG.IGNORE.TAPE.volser
 - STGADMIN.EDG.LABEL.volser
 - STGADMIN.EDG.LISTCONTROL
 - STGADMIN.EDG.MASTER
 - STGADMIN.EDG.NOLABEL.volser
 - STGADMIN.EDG.OPERATOR
 - STGADMIN.EDG.OWNER.userid
 - STGADMIN.EDG.RELEASE
 - STGADMIN.EDG.RESET.SSI
 - STGADMIN.EDG.VRS
 - STGADMIN.EDG.INERS.WRONG.LABEL
 - Generic DFSMSrmm data set profile
 - Data set profiles for CDS, journal, and message data sets
 - Data set profiles for housekeeping data sets
 - Data set profiles for audit and report data sets

- Define the STGADMIN.ADR.DUMP.CNCURRNT resource

Before you start DFSMSrmm you need to create a RACF user ID for DFSMSrmm and define the DFSMSrmm started task to RACF. To fully implement RACF security for DFSMSrmm, refer to Appendix A, “Security Considerations and Implementation” on page 561.

3.1.6.1 Assigning DFSMSrmm a RACF User ID

You can assign a RACF user ID that matches the name of the DFSMSrmm procedure you created in 3.1.3, “Update the Procedure Library” on page 39, but any installation-selected RACF user ID is acceptable. As data sets are created for use by the DFSMSrmm procedure, add the DFSMSrmm RACF user ID to the access list for the data sets.

For a started procedure to access any of the system’s resources, a user ID must be associated with that task. The user ID should have all the proper authorizations for accessing the system resources, therefore it should be assigned the RACF OPERATIONS attribute. We recommend defining a new user ID for each started procedure rather than using the default started task user ID (for example, STCUSER). In our system we associated userid *DFRMM* with the DFSMSrmm started procedure. Following is the RACF command we used to define the DFSMSrmm user ID:

```
ADDUSER DFRMM OPERATIONS DFLTGRP(SYS1) NAME('DFSMSrmm Userid')
```

Note: If you do not specify the DFLTGRP parameter on the ADDUSER command, the current connect group of the user issuing the ADDUSER command is used as the default.

3.1.6.2 Identifying Started Procedures to RACF

Before RACF 2.1, the only way to associate a started procedure with a RACF user ID was by coding the RACF started procedures table, ICHRIN03. With RACF 2.1, assigning RACF identities to started procedures has been greatly simplified by the introduction of the RACF STARTED class. You can add or modify security definitions for new and existing started procedures by issuing the RDEFINE and RALTER commands.

Even though the use of the RACF STARTED class is the preferred way of identifying started procedures to RACF, it is still mandatory to have the ICHRIN03 module. RACF cannot be initialized if ICHRIN03 is not present in the system. A dummy ICHRIN03 is shipped and installed with RACF.

RACF STARTED Class: The STARTED class allows you to assign RACF identities to started procedures dynamically using the RDEFINE and RALTER commands. Resource names in the STARTED class have the format *membername.jobname*. You assign identities such as the RACF user ID and group ID, using fields in the STDATA segment. You can define the started procedure resource, using either a generic profile name or a discrete profile name. A RACF generic profile describes one or more data sets that have a similar name structure. A RACF discrete profile describes a specific data set on a specific volume. In our system we created generic profiles for started procedures. Issue the following RACF commands to assign RACF identities to the DFSMSrmm started procedure:

```
RDEFINE STARTED (DFRMM.*) UACC(NONE) STDATA(USER(DFRMM) GROUP(SYS1))
SETROPTS RACLIST(STARTED) REFRESH
SETROPTS GENERIC(STARTED) REFRESH
```

After you have added profiles to the RACF STARTED class, refresh the in-storage profiles, using the SETROPTS REFRESH command. The SETROPTS GENERIC command is needed only when you define generic profiles.

Note: If you plan to use the EDGLABEL, EDGXPROC, or the EDGBKUP procedure, you must define the procedures in the RACF STARTED class or in ICHRIN03.

For more information on defining procedures to the RACF STARTED class, refer to the *OS/390 Security Server (RACF) Security Administrator's Guide*, SC28-1915.

Figure 22 shows a batch job that can be used to add the DFSMSrmm user ID in the RACF STARTED class.

```
//RACFRMM JOB ,RMM,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
//* Note
//* ====
//* To execute this job you need the RACF SPECIAL attribute
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* CREATE AN ENTRY IN THE DYNAMIC STARTED PROCEDURE TABLE FOR DFRMM */
/* ***** */
RDEFINE STARTED DFRMM.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
```

Figure 22. JCL to Add DFSMSrmm User ID in RACF STARTED Class

ICHRIN03 Started Procedures Table: We do not recommend using the RACF started procedures table (ICHRIN03). The preferred way of adding started procedure users to the RACF database is by using the STARTED class. Refer to A.3.2.2, "ICHRIN03 Started Procedures Table" on page 567 for information and sample JCL to identify the DFSMSrmm started procedure to RACF by assembling ICHRIN03.

3.1.6.3 STGADMIN.EDG.RESET.SSI Facility Class Profile

At this stage there is one security profile in particular that you should be aware of and create. When you first install and start to use DFSMSrmm, it is likely that you will need to stop or even remove DFSMSrmm from your system. Although you can easily stop DFSMSrmm, if you have EDGSSSI in the subsystem name table DFRM entry, or have been using a running mode that supports tape recording, DFSMSrmm prevents all tape usage until it is restarted. To allow tapes to be used, you must restart DFSMSrmm, or remove it from the system.

The RACF FACILITY class profile STGADMIN.EDG.RESET.SSI controls the use of the RESET facility for removing DFSMSrmm from the system. Be sure to define this security profile and authorize the DFSMSrmm started task to reset the subsystem interface. Grant the DFSMSrmm started task procedure RACF user

ID ALTER access authority to STGADMIN.EDG.RESET.SSI. Refer to A.3.6, “Removing DFSMSrmm from the System” on page 576 for specific details.

Note: You can use the RESET facility without defining this profile when you do not have a security product installed.

3.1.7 Restart MVS with DFSMSrmm Implemented

You are ready to start the system with DFSMSrmm implemented. You may be able to dynamically implement changes to the MVS system PARMLIB members or modified installation exits depending on your MVS level. You must:

- IPL with CLPA to include new levels of the DFSMSrmm code that have LPALIB as the target library
- IPL when you make changes to IFAPRDxx to change the licensing options.

3.1.8 Setting Up DFSMSrmm Utilities

There are several DFSMSrmm utilities that you should set up now. Refer to the DFSMSrmm Implementation and Customization Guide for the following information for each utility:

- A sample schedule of how frequently to run DFSMSrmm utilities.
- Information on using EDGHSKP to run the following inventory management activities:
 - Vital record processing to determine which volumes to retain and which volume moves are required, based on VRSS.
 - Expiration processing to identify volumes ready to be released and returned to scratch.
 - Storage location management processing to assign shelf locations to volumes being moved to storage locations
 - Backup of the CDS and journal.
 - Creation of an extract data set. You can produce movement and inventory reports by producing an extract data set from the control data set and creating a report from it with the report utility.
- Information about EDGAUD and EDGRPTD, two utilities that help you get information about your removable media library and storage locations. You can also get security related information about volumes and data sets defined to DFSMSrmm and audit trail information about volumes, shelf assignments, and user activity.
- Information about using the DFSMSrmm utilities to back up and recover the CDS, back up the journal, and check the integrity of the information contained in the CDS. Use the EDGBKUP utility to back up and recover the control data set and back up the journal, and the EDGUTIL utility to create, update, and verify the control data set.

Use the DFSMSrmm backup utilities rather than other backup utilities, such as access method services EXPORT, because DFSMSrmm provides the necessary serialization and forward recovery functions. DFSMSrmm backup utilities check whether the control data set is in use, tell the DFSMSrmm subsystem that backup or recovery is in process, and provide a way to forward recover.

- Information about EDGINERS, the DFSMSrmm utility that helps you erase and initialize tape volumes either automatically or manually.

You can use EDGINERS to replace the DFSMSdfp utility IEHINITT.

3.1.9 Running the Installation Verification Program

The quickest and simplest way of getting DFSMSrmm started and verifying that the key pieces are installed successfully is to run the installation verification procedure (IVP). The IVP is documented in detail in Appendix A in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932. The appendix includes considerations for running in an environment where another tape management system is in use.

3.2 Verifying Your DFSMSrmm Implementation

To test DFSMSrmm, you can use a small, test CDS or a converted CDS. We recommend that you use simple DFSMSrmm function tests for your first testing. Later, after you have analyzed the data and customized the exits and parameters, you can test those functions that will be used in production.

Verifying DFSMSrmm functions by submitting batch jobs is quite simple. In this section we show you sample batch jobs we used to verify DFSMSrmm functions in our system. Run the sample jobs in the sequence shown to test basic DFSMSrmm functions in your system.

3.2.1 Display PARMLIB Options and Control Information

Use the RMM LISTCONTROL subcommand to display information from the DFSMSrmm CDS control record and options defined in the EDGRMMxx PARMLIB member.

Example: Display your installation's options and rules, restricting the information displayed to the control record information and system options only (see Figure 23). Specify the ALL operand to display all options and rules defined to your installation.

```
//LISTCNTL JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,  
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)  
//* *****  
//*          STEP 1  DISPLAY INFORMATION  
//* *****  
//STEP01   EXEC PGM=IKJEFT01  
//SYSTSPRT DD   SYSOUT=*  
//SYSTSIN  DD   *  
           RMM LISTCONTROL ALL  
/*
```

Figure 23. Sample JCL to Display PARMLIB Options and Control Information

3.2.2 Add Owner Information

Use the ADDOWNER subcommand to define an owner to DFSMSrmm. An owner can be an individual or group defined by a RACF group name, or any value you choose. The owner value contains one to eight alphanumeric characters.

DFSMSrmm automatically creates an owner record if a user who is not defined to DFSMSrmm requests a job that writes to a volume managed by DFSMSrmm. DFSMSrmm uses the user ID that requests the job as an DFSMSrmm owner ID. In all other cases the owner must be defined manually, and must be defined before you can use an RMM TSO subcommand that includes owner information.

Use the LISTOWNER subcommand to display information about a single owner defined to DFSMSrmm.

Example: Add details of a new owner and in STEP02 request information recorded by DFSMSrmm about the owner whose owner ID is SCHLUM (see Figure 24).

```
//ADDOWNER JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* ***** *
//*      STEP 1  ADDING OWNER INFORMATION      *
//*      2  DISPLAY INFORMATION                *
//* ***** *
//STEP01 EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM ADDOWNER SCHLUM -
      FNAME('NORBERT') -
      SNAME('SCHLUMBERGER') -
      DEPARTEMENT('International Projects (6304)') -
      ADD1('IBM Deutschland Informationssysteme GmbH') -
      ADD2('Am Keltenwald 1') -
      ADD3('D-71139 Ehningen') -
      INTEL('919-3579') -
      EXTEL('++49-(0)-7034-15-3579') -
      USER(DEIBMPVS) -
      NODE(IBMAIL)
/*
//STEP02 EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM LISTOWNER SCHLUM
/*
```

Figure 24. Sample JCL to Add Owner Information

3.2.3 Add Racks to DFSMSrmm

The ADDRACK subcommand defines shelf locations in the removable media library and storage locations. DFSMSrmm defines shelf spaces in the removable media library as rack numbers and bin numbers in storage locations. Use the ADDRACK subcommand to define empty, or available, rack and bin numbers to DFSMSrmm. You must supply an initial rack number. You have to add racks before you can add a volume serial number to DFSMSrmm.

The SEARCHRACK subcommand creates a list of shelf locations defined in the removable media library. Shelf locations in the removable media library are called rack numbers.

Example: Add 20 racks to the removable media library, starting with rack number SC0000. In STEP02 create a list of the created rack numbers (see Figure 25 on page 49).


```

//ADDRACK JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* ***** *
//*          STEP 1  ADDING 20 RACKS *
//*          2  DISPLAY INFORMATION *
//* ***** *
//STEP01 EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM ADDRACK SC0000 -
          LOCATION(SHELF) -
          MEDIANAME(3590) -
          COUNT(20)

/*
//STEP02 EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM SEARCHRACK RACK(SC*) LIMIT(*)
/*

```

Figure 25. Sample JCL to Add Racks

3.2.4 Add Volumes to DFSMSrmm

The ADDVOLUME subcommand adds one or more volumes to DFSMSrmm. You have to define all volumes you will use in the DFSMSrmm CDS.

DFSMSrmm monitors all tape volume mounts and automatically updates information about your pre-defined tape volumes when they are used.

The SEARCHVOLUME subcommand creates a list of volumes matching the search criteria you specify. You can display lists of volumes according to ownership, assigned date, status, movement, action, pool, or media name.

Example: Add 10 new scratch volumes in STEP01 to the removable media library using a rack number that is the same as the VOLSER. The volumes will not be initialized. In STEP02 add 10 new scratch volumes to the removable media manager into the racks, but in this case the volumes must be labeled before they can be used, as indicated by INIT(Y). In STEP03 create a list of all added volumes (see Figure 26 on page 50).

```

//ADDVOLS JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
/*          STEP 1  ADDING 10 VOLUMES                                     *
/*          2  ADDING 10 VOLUMES (these volumes must be labeled       *
/*                                     before they can be used, as      *
/*                                     indicated by INIT(Y)              *
/*          3  DISPLAY INFORMATION                                     *
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD  *
RMM ADDVOLUME SC0000 -
                STATUS(SCRATCH) -
                DENSITY(*) -
                INITIALIZE(N) -
                LABEL(SL) -
                LOCATION(SHELF) -
                MEDIANAME(3590) -
                MEDIATYPE(*) -
                POOL(SC*) -
                USE(MVS) -
                COUNT(10)
/*
//STEP02 EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD  *
RMM ADDVOLUME SC0010 -
                STATUS(SCRATCH) -
                DENSITY(*) -
                INITIALIZE(Y) -
                LABEL(SL) -
                LOCATION(SHELF) -
                MEDIANAME(3590) -
                MEDIATYPE(*) -
                POOL(SC*) -
                USE(MVS) -
                COUNT(10)
/*
//STEP03 EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD  *
RMM SEARCHVOLUME VOLUME(SC*) OWNER(*) LIMIT(*)
/*

```

Figure 26. Sample JCL to Add Volume Information

3.2.5 Add Bins to DFSMSrmm

The ADDBIN subcommand defines shelf locations in the storage locations. DFSMSrmm defines shelf spaces in storage locations as bin numbers. Use the ADDBIN subcommand to define empty bin numbers to DFSMSrmm.

The SEARCHBIN subcommand creates a list of shelf locations defined in the storage locations. Shelf locations in the storage locations are called bin numbers.

Example: Add 20 empty shelf locations to the built-in storage location SANJOSE, starting with bin number 000001. The example assumes that you have defined storage location SANJOSE with the LOCDEF command in PARMLIB member EDGRMMxx. In STEP02 create a list of the created bin numbers (see Figure 27 on page 51).

```
//ADDBIN JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
//*      STEP 1  ADDING 10 BINS TO built_in storage location SANJOSE      *
//*      2      DISPLAY INFORMATION                                       *
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD   *
RMM ADDBIN 000001 -
           LOCATION(SANJOSE) -
           MEDIANAME(3590) -
           COUNT(10)
/*
//STEP02 EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD   *
RMM SEARCHBIN BIN(*) LIMIT(*) LOCATION(SANJOSE)
/*
```

Figure 27. Sample JCL to Add Bins

DFSMSrmm distinguishes between built-in storage location names and installation defined storage location names. Built-in storage location names are LOCAL, DISTANT, and REMOTE and defined by DFSMSrmm. A LOCDEF command is required in the DFSMSrmm PARMLIB member EDGRMMxx for each installation defined storage location name.

3.2.6 Add a VRS to DFSMSrmm

The RMM ADDVRS command adds a new VRS to DFSMSrmm. A VRS is used to define retention and movement policies for data set and volumes.

The LISTVRS subcommand displays details about a single VRS. Specify a data set name when requesting information about a data set VRS.

Example: Add a data set VRS to retain all data sets matching the DSNAME mask SCHLUM.RMMTEST.MOVE.** in location REMOTE until the data set is no longer cataloged. In STEP02 request information about the data set VRS defined for the SCHLUM.RMMTEST.MOVE.** data (see Figure 28 on page 52).

```

//ADDVRS   JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
//*          STEP 1  ADDING A VITAL RECORD SPECIFICATION          *
//*          2  DISPLAY INFORMATION                               *
//* *****
//STEP01   EXEC PGM=IKJEFT01
//SYSTSPRT DD   SYSOUT=*
//SYSTSIN  DD   *
RMM ADDVRS DSNAME('SCHLUM.RMMTEST.MOVE.**') -
          CYCLES -
          COUNT(99999) -
          DESCRIPTION('RETAIN AND MOVE DATA SETS') -
          LOCATION(REMOTE) -
          OWNER(SCHLUM) -
          STORENUMBER(1) -
          WHILECATALOG
/*
//STEP02   EXEC PGM=IKJEFT01
//SYSTSPRT DD   SYSOUT=*
//SYSTSIN  DD   *
RMM LISTVRS DSNAME('SCHLUM.RMMTEST.MOVE.**')
/*

```

Figure 28. Sample JCL to Add a VRS

3.2.7 Create Tape Data Sets

Use the IEBGENER utility to create a multi-file, multi-volume chain to understand the differences between DFSMSrmm and your current tape management system (Figure 29).

```

//CREATE1   JOB DE06304,140.SCHLUMBERGER,...
//* *****
//*          * 1. STEP  ALLOC 1. FILE  DSN=SCHLUM.RMMTEST.FILE1.VOL1  *
//*          *          VOLUME 1 / FILE 1 / RETPD 5480 (APPROX. 15 Y.) *
//*          *****
//STEP01   EXEC PGM=IEBGENER
//SYSPRINT DD   SYSOUT=*
//SYSUT1   DD   DSN=DFRMM.TEST.DATA.SMALL,DISP=SHR
//SYSUT2   DD   DSN=SCHLUM.RMMTEST.FILE1.VOL1,
//          DISP=(,CATLG,DELETE),
//          UNIT=IBMATL1,VOL=(,RETAIN),
//          DCB=*.SYSUT1,LABEL=RETPD=5480
//SYSIN    DD   DUMMY
//* *****
//*          * 2. STEP  ALLOC 2. FILE  DSN=SCHLUM.RMMTEST.FILE2.VOL1+2 *
//*          *          VOLUME 1 FILE 2 + VOLUME 2 FILE 1 / RETPD = 5  *
//*          *****

```

Figure 29 (Part 1 of 2). Sample JCL to Create Data Sets on Tape

```

//STEP02 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=DFRMM.TEST.DATA.BIG,DISP=SHR
//SYSUT2 DD DSN=SCHLUM.RMMTEST.FILE2.VOL12,
//      DISP=(,CATLG,DELETE),
//      DCB=*.SYSUT1,LABEL=(2,SL,,RETPD=5),
//      VOL=(,RETAIN,,REF=*.STEP01.SYSUT2)
//SYSIN DD DUMMY *
//* *****
//* * 3. STEP ALLOC 3. FILE DSN=SCHLUM.RMMTEST.FILE3.VOL2 *
//* * VOLUME 2 / FILE 2 / RETPD 3 *
//* *****
//STEP03 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=DFRMM.TEST.DATA.SMALL,DISP=SHR
//SYSUT2 DD DSN=SCHLUM.RMMTEST.FILE3.VOL2,
//      DISP=(,CATLG,DELETE),
//      DCB=*.SYSUT1,LABEL=(3,SL,,RETPD=3),
//      VOL=(,RETAIN,,REF=*.STEP02.SYSUT2)
//SYSIN DD DUMMY
//* *****
//* * 4. STEP ALLOC 4. FILE DSN=SCHLUM.RMMTEST.FILE4.VOL23 *
//* * VOLUME 2 FILE 3 + VOLUME 3 FILE 1 / RETPD = 4 *
//* *****
//STEP04 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=DFRMM.TEST.DATA.BIG,DISP=SHR
//SYSUT2 DD DSN=SCHLUM.RMMTEST.FILE4.VOL23,
//      DISP=(,CATLG,DELETE),
//      DCB=*.SYSUT1,LABEL=(4,SL,,RETPD=4),
//      VOL=(,RETAIN,,REF=*.STEP02.SYSUT2)
//SYSIN DD DUMMY
//* *****
//* * 5. STEP ALLOC 5. FILE DSN=SCHLUM.RMMTEST.FILE5.VOL3 *
//* * VOLUME 3 / FILE 2 / RETPD 10 *
//* *****
//STEP05 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=DFRMM.TEST.DATA.SMALL,DISP=SHR
//SYSUT2 DD DSN=SCHLUM.RMMTEST.FILE5.VOL3,DISP=(,CATLG,DELETE),
//      DISP=(,CATLG,DELETE),
//      DCB=*.SYSUT1,LABEL=(5,SL,,RETPD=10),
//      VOL=(,RETAIN,,REF=*.STEP04.SYSUT2)
//SYSIN DD DUMMY

```

Figure 29 (Part 2 of 2). Sample JCL to Create Data Sets on Tape

Figure 30 on page 54 shows how to create data sets on a tape to be moved to a storage location. In 3.2.6, “Add a VRS to DFSMSrmm” on page 51 we added a DSNAME VRS, and now we will create matching data sets to test VRS processing.

```

//CREATE2 JOB DE03331,140.SCHLUMBERGER,MSGCLASS=H,
//      MSGLEVEL=(1,1),TIME=1439
//STEP01 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSIN DD DUMMY
//SYSUT2 DD DSN=SCHLUM.RMMTEST.MOVE(+1),DISP=(,CATLG,DELETE),
//      DCB=(RECFM=F,LRECL=80,BLKSIZE=80),UNIT=3590
//SYSUT1 DD *
CASE07 RMM TEST 6.FILE
/*
//STEP02 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSIN DD DUMMY
//SYSUT2 DD DSN=SCHLUM.RMMTEST.MOVE(+2),DISP=(,CATLG,DELETE),
//      DCB=(RECFM=F,LRECL=80,BLKSIZE=80),UNIT=3590
//SYSUT1 DD *
CASE07 RMM TEST 7.FILE
/*
//STEP04 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSIN DD DUMMY
//SYSUT2 DD DSN=SCHLUM.RMMTEST.MOVE(+3),DISP=(,CATLG,DELETE),
//      DCB=(RECFM=F,LRECL=80,BLKSIZE=80),UNIT=3590
//SYSUT1 DD *
CASE08 RMM TEST 8.FILE
/*
//STEP04 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSIN DD DUMMY
//SYSUT2 DD DSN=SCHLUM.RMMTEST.MOVE(+4),DISP=(,CATLG,DELETE),
//      DCB=(RECFM=F,LRECL=80,BLKSIZE=80),UNIT=3590
//SYSUT1 DD *
CASE09 RMM TEST 9.FILE
/*

```

Figure 30. Sample JCL to Create Data Sets on Tape to Test VRS Processing

3.2.8 Allocate Inventory Management Data Sets

Before you can use the EDGHSKP utility to run inventory management activities, you must define several data sets, but not all data sets required for all functions. Some data sets used during inventory management must be pre-allocated and cataloged because these data sets are used by both the EDGHSKP utility and the DFSMSrmm subsystem. Figure 31 on page 55 shows sample JCL for allocating these data sets.

```

//ALLOCATE JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
//*          STEP01  pre-allocate MESSAGE data set                      *
//*          pre-allocate REPORT data set                             *
//*          pre-allocate RPTTEXT data set                             *
//*          pre-allocate ACTIVITY data set                             *
//* *****
//STEP01    EXEC PGM=IEFBR14
//MESSAGE   DD DSN=MESSAGE.FILE.NAME,DISP=(NEW,CATLG),
//          UNIT=SYSDA,AVGREC=U,SPACE=(4096,(10,10))
//REPORT    DD DSN=REPORT.FILE.NAME,DISP=(NEW,CATLG),
//          LRECL=137,RECFM=VBA,
//          UNIT=SYSDA,AVGREC=U,SPACE=(4096,(50,30))
//ACTIVITY  DD DSN=ACTIVITY.FILE.NAME,DISP=(NEW,CATLG),
//          UNIT=SYSDA,AVGREC=U,SPACE=(4096,(50,30))
//RPTTEXT   DD DSN=REPORT.EXTRACT.FILE.NAME,DISP=(NEW,CATLG),
//          UNIT=SYSDA,AVGREC=U,SPACE=(4096,(1000,500))

```

Figure 31. Sample JCL to Pre-Allocate Data Sets for EDGHSKP Utility

There is sample JCL in data set SYS1.SAMPLIB member EDGJHKPA that can be used to allocate the files required to run inventory management functions to allocate the ACTIVITY, MESSAGE, REPORT, and RPTTEXT data sets.

3.2.9 Run Inventory Management

After you have created racks, volumes, VRSSs, and data sets, you must run inventory management as follows:

1. Run vital record processing before expiration and storage location management processing to identify which volumes to retain and where volumes should be moved according to their VRSSs.
2. Run storage location management to set a destination location, and optionally assign shelf locations in storage locations to volumes that are being sent out of the removable media library.
3. Run expiration processing to identify those volumes not required for vital records that are ready to expire. During expiration processing, release actions for volumes are noted.
4. Run extract data set processing to a sequential data set that contains information from DFSMSrmm and use it as input to EDGRPTD, the DFSMSrmm reporting utility.
5. Back up the CDS and optionally the journal, and reset the journal data set when the control data set and journal are backed up using the EDGHSKP utility.

Figure 32 on page 56 shows sample JCL to perform inventory management.

```

//LCLHSKP JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
//* Step 01  List all volumes to be returned to scratch status      *
//*          (Availability = PENDING RELEASE)                      *
//*      02  Performing inventory management - EDGHSKP            *
//*          a) VRSEL  (Vital Record Processing)                  *
//*          b) DSTORE (Stores management processing)              *
//*          b) EXPROC (Expiration processing)                     *
//*          d) RPTEXT (Report extract processing)                 *
//*          e) BACKUP (Backing up the control data set, backing up *
//*                the journal data set and clearing out the journal *
//*                data set only if the back up of the control data set *
//*                was successfully)                                *
//*      03  Copy DFSMSrmm message file to sysout                  *
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) ACTION(SCRATCH) LIMIT(*) OWNER(*)
/*
//* *****
//STEP02 EXEC PGM=EDGHSKP,
//      PARM='VRSEL,EXPROC,DSTORE,BACKUP,RPTEXT,DATEFORM(J)'  

//SYSPRINT DD SYSOUT=*
//MESSAGE DD DSN=RMM.PROD.MSGS,DISP=SHR
//REPORT DD DSN=RMM.HSKP.VRSEL,DISP=SHR
//ACTIVITY DD DSN=RMM.HSKP.ACTIFITY,DISP=SHR
//REPTTEXT DD DSN=RMM.HSKP.EXTRACT,DISP=SHR
//BACKUP DD DSN=RMM.HSKP.BACKUP,DISP=(,CATLG,DELETE),
//      SPACE=(TRK,(150,10),RLSE),
//      DCB=(GDG,LRECL=9000,BLKSIZE=23476,RECFM=VB)
//JRNLBKUP DD DSN=RMM.HSKP.JRNLBKUP,DISP=(,CATLG,DELETE),
//      SPACE=(TRK,(10,10),RLSE),
//      DCB=(GDG,LRECL=9000,BLKSIZE=23476,RECFM=VB)
//* *****
//STEP03 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=0
//SYSUT1 DD DSN=RMM.PROD.MSGS,DISP=SHR
//SYSUT2 DD SYSOUT=*
//SYSIN DD DUMMY
//* *****

```

Figure 32. Sample JCL to Run Inventory Management

3.2.10 Confirm Movements

Confirm your outstanding volume moves by using the RMM CHANGEVOLUME command. In DFSMSrmm you have to confirm each volume move, except for volumes moved to an IBM automated tape library dataserer, to confirm that you have moved the volume.

You can confirm a move in two ways:

- To confirm a move for a single volume, use the CHANGEVOLUME subcommand with a VOLSER and the CONFIRMMOVE operand:


```
RMM CHANGEVOLUME vvvvvv CMOVE(SANJOSE)
```

No further actions are required.

- To perform a global confirm of volume movements, specify an asterisk for the VOLSER:

```
RMM CHANGEVOLUME * CMOVE(SANJOSE)
```

In this case you must rerun the storage location management processing to update the CDS.

3.2.11 Reports

You can create several types of reports using two DFSMSrmm report utilities, EDGRPTD and EDGAUD. DFSMSrmm provides standard reports and samples shipped in SAMPLIB. You can use the sample DFSMSrmm EDGJRPT JCL that invokes the EDGRRPTE exec to create the reports. The input to the reporting execs is the extended extract data set. Documentation about the reports is shipped in SYS1.SAMPLIB in the EDGDOC file.

Use the reports created to help you keep track of your DFSMSrmm environment, and compare these reports with those of your old tape management system running in parallel. You can use DFSORTS's ICETOOL to write customized reports, or you can use the DFSMSrmm TSO subcommands to create lists of information defined to DFSMSrmm.

3.2.11.1 Create Reports with EDGRPTD

Use EDGRPTD to create inventory reports for auditing the physical contents of the installation media library and storage locations. You can also use EDGRPTD to create movement reports that list volumes to be moved from one location to another. Use these reports to make an inventory of your volumes and to identify volumes that need to be pulled and moved to other locations.

For detailed information how to create a report, see Appendix D, "Sample Exits and Reports" on page 603.

3.2.11.2 DFSMSrmm Supplied Reports

DFSMSrmm provides REXX execs and JCL in SYS1.SAMPLIB member EDGJRPT that you can use as-is to create the following reports:

- Pull list for scratch tapes sorted by volume serial number
- Pull list for scratch tapes sorted by data set name
- Inventory list by volume serial number
- Inventory list by data set name
- Inventory of data set including number of KB used
- Inventory of volume serial number by location
- Inventory of data set names by location
- Inventory of BIN numbers by location
- List of all data set names at loan location
- List of all volume serial numbers at loan location
- List of all multiple volume, multiple data set
- Movement report including the first data set name on the volume
- Movement report sorted by storage location bin number
- Movement report sorted by volume serial number
- Inventory list sorted by volume serial number including volume count

You can copy these reports and use them to create reports that are tailored for your installation.

3.2.11.3 Create Reports with EDGAUD

DFSMSrmm produces SMF records when you specify the SMFAUD and SMFSEC installation options in the PARMLIB member EDGRMMxx. Before you can create any security or audit reports you must archive your SMF data by using your installation-defined procedures.

Use EDGAUD to create security and audit reports using either previously selected and sorted SMF records or raw SMF data. DFSMSrmm produces SMF records when you specify the SMFAUD or SMFSEC installation options in the PARMLIB member EDGRMMxx. DFSMSrmm uses the current subsystem startup option for the SMF record types and default report option unless you override them with the EDGAUD EXEC parameters.

3.2.12 Run EDGINERS to Initialize Volumes Automatically

With EDGINERS, volumes can be initialized and erased, based on the volume status and actions recorded in the CDS, or you can let your users or operator perform the processing. See the *DFSMSrmm Guide and Reference*, SC26-4931, for more information about operator tasks such as responding to initialization messages, tape mount messages, and using the LABEL procedure to request EDGINERS processing. The *DFSMSrmm Implementation and Customization Guide*, SC26-4932, also describes the EDGLABEL procedure provided by DFSMSrmm.

During automatic processing, the process requests each volume to be mounted, as with any batch job, because an operator reply is not required. When the volume is mounted, processing continues. If the operator cannot mount a volume, DFSMSrmm allows them to skip processing the current requested volume by replying to a WTOR.

As part of volume expiration processing, DFSMSrmm records when volumes need to be initialized or erased as they are released for return to scratch. If you use the EXEC parameters in your JCL to set up automatic processing, DFSMSrmm initializes or erases those volumes without librarian intervention. EDGINERS issues WTOR messages and MSGDISP requests to the operator and drive to get a volume mounted and demounted.

You can run EDGINERS automatic processing to initialize the volumes you are adding to DFSMSrmm. When you add scratch volumes to DFSMSrmm, use the RMM TSO subcommand operand, INIT(Y), on the ADDVOLUME command to ensure that volumes are initialized before they are available as scratch. Specify the initialization parameters on the PARM statement in your EDGINERS JCL. DFSMSrmm selects all volumes marked as requiring initialization and performs automatic processing. Figure 33 on page 59 shows sample JCL for automatic processing.

```

//LCLHSP JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
/* Step 01   Initializing volumes
/* *****
//STEP01 EXEC PGM=EDGINERS,PARM=' MEDIANAME(3590),VERIFY'
//SYSPRINT DD  SYSOUT=*
//TAPE      DD  UNIT=(TAPE,,DEFER)

```

Figure 33. Sample JCL for Automatic Processing

This job initializes the volumes SC0010 through SC0019 added in Figure 26 on page 50. The VERIFY parameter requests that DFSMSrmm ask the operator to remount each volume that has been successfully erased or labeled. The volumes are requested in reverse order, and the volume labels read to ensure that operator errors have not occurred (for example, a mismatch between the internal label and the external label).

3.2.13 Restore the CDS

To restore the CDS, you first delete the existing CDS and then redefine it using Access Method Services (AMS) commands. The EDGBKUP utility is used to forward recover the CDS. During the recovery you must stop all DFSMSrmm activities using the MODIFY DFRMM,QUIESCE command. When the CDS recovery is complete, you must restart DFSMSrmm using the MODIFY DFRMM,M=xx operator command.

You can restore the DFSMSrmm CDS and forward recover the restored CDS with the journal. Use the JOURNAL DD statement to concatenate multiple journal data sets. If you forward recover using multiple journal data sets, ensure that the journal data sets are concatenated in the correct order in which changes were originally made.

When backup is taken using DFSMSdss, by specifying BACKUP(DSS) for any backup being restored, a journal backup is also required for forward recovery. For the latest control data set backup, both the latest journal backup and the active journal must be used for forward recovery. In a remote recovery site, the journal backup is all that is required for successful forward recovery of the restored control data set. Figure 34 and Figure 35 on page 60 show sample JCL to forward recover the CDS.

```

//RESTORE JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)
//* *****
/* Step 01   DELETE THE OLD CONTROL DATA SET
/*          02   ALLOCATE A NEW CONTROL DATA SET
/*          03   RESTORE THE CONTROL DATA SET
/* *****

```

Figure 34 (Part 1 of 2). Sample JCL to Delete and Define the CDS Using AMS

```

//STEP01 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE RMM.PROD.CDS CL
SET MAXCC=0
/*
//STEP02 EXEC PGM=IDCAMS ALLOCATION MASTER FILE
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR,UNIT=3390,VOL=SER=DFRMM4
//SYSIN DD *
DEFINE CLUSTER (NAME(RMM.PROD.CDS) -
FILE(MASTER) -
FREESPACE(35 20) -
KEY(56 0) -
REUSE -
RECSZ(512 9216) -
SHR(3 3) -
STORCLAS(NONSMS) -
VOLUMES(DFRMM4)) -
DATA (NAME(RMM.PROD.CDS.DATA) -
KILOBYTES(35532 3553) -
CISZ(10240) -
FILE(MASTER)) -
INDEX (NAME(RMM.PROD.CDS.INDEX) -
CISZ(2048) -
KILOBYTES(3815 381) -
FILE(MASTER) -
IMBED -
REPLICATE)
/*

```

Figure 34 (Part 2 of 2). Sample JCL to Delete and Define the CDS Using AMS

```

//STEP03 EXEC PGM=EDGBKUP,PARM='RESTORE'
//SYSPRINT DD SYSOUT=*
//MASTER DD DSN=RMM.HSKP.BACKUP,DISP=SHR
//JOURNAL DD DSN=RMM.PROD.JRNL,DISP=SHR

```

Figure 35. Sample JCL to Forward Recover the CDS for a BACKUP(AMS)

Figure 36 shows sample JCL to forward recover the CDS using BACKUP(DSS).

```

//STEP03 EXEC PGM=EDGBKUP,PARM='RESTORE'
//SYSPRINT DD SYSOUT=*
//MASTER DD DSN=RMM.HSKP.BACKUP,DISP=SHR
//JOURNAL DD DSN=RMM.HSKP.JRNLBKUP(0),DISP=SHR
// DD DSN=RMM.PROD.JRNL,DISP=SHR

```

Figure 36. Sample JCL to Forward Recover the CDS for a BACKUP(DSS)

You can check that restore and forward recovery were successful by displaying volume information. All volumes should have been initialized if the job shown in Figure 33 on page 59 ended without any errors.

3.2.14 Verify the CDS

Use the EDGUTIL utility to verify the contents of the CDS. You should perform the verification each time you recover your DFSMSrmm CDS. Figure 37 shows sample JCL to verify the CDS.

```
//VERIFY JOB ,140.SCHLUMBERGER,NOTIFY=SCHLUM,  
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1)  
//* *****  
//* STEP 01 VERIFYING THE CONTENTS OF THE CONTROL DATA SET *  
//* *****  
//STEP01 EXEC PGM=EDGUTIL,PARM='VERIFY(ALL)'  
//SYSPRINT DD SYSOUT=*  
//MASTER DD DISP=SHR,DSN=RMM.PROD.CDS  
//SYSIN DD DUMMY
```

Figure 37. Sample JCL to Verify the Contents of the CDS

3.2.15 Testing Implemented Exits

If you use the following exits, you must test them:

- EDGUX100
 - SCRATCH pooling
 - Special expiration dates
 - 90ddd — retain for a minimum number of days
 - 98000 — bypass DFSMSrmm
 - 98ddd — retain until the specified number of days since last reference has elapsed
 - 99000 — retain for as long as it is cataloged
 - 99ccc — retain for the total number of cycles or generations
 - Special ACCODE keywords
 - xCANORES — bypass DFSMSrmm
 - xCACATLG — retain for as long as it is cataloged
 - xCAPERM — retain data set permanently
 - xCAUSER — retain data set permanently
 - Bypass DFSMSrmm

You can use the supplied sample EDGUX100 exit to support the special expiration dates 98000 and 99000 without modifications to the sample code. If you plan to support other special dates using the exit and have modified it, you also need to test them. The results and the testing of the additional dates are the same as for the 99000 date.

To test use of the 99000 special date, create a new tape data set by specifying EXPDT=99000. Use the DFSMSrmm LISTDATASET command or the list data set dialog to display details of the data set that should include D99000 as the VRS management value.

To test the use of EXPDT=98000, first define a FACILITY class resource to allow the use of the ignore function to bypass DFSMSrmm recording. Whether your test is to check that ignore is allowed or that ignore is not allowed, DFSMSrmm issues a message of explanation. To test the ignore function, DFSMSrmm must be running in either warning or protect mode.

- EDGUX200

The sample EDGUX200 provides no specific function. Test the exit only if you plan to use it and have modified the supplied source code. The exit is intended for communication with software that has an inventory of scratch tapes. To test the exit, you need a tape volume that is in pending release status and is to return to scratch. Run inventory management EXPROC processing and then check that the external inventory has been correctly updated.

- System-managed tape support
 - CBRUXENT (OAM cartridge entry exit)
 - CBRUXEJC (OAM cartridge eject exit)
 - CBRUXCUA (OAM cartridge change use attributes exit)
 - CBRUXVNL (OAM volume not in library exit).

Test these exits only if you have an automated tape library dataset server. DFSMSrmm provides sample exits that are ready to use, installed and active on your system, and do not require modification. However, you can customize the exits to meet your needs. CBRUXVNL is the exit you may want to consider customizing to meet your needs.

A simple set of tests can be used to test these exits:

1. Define a volume to DFSMSrmm for entry to the library:

```
RMM ADDVOLUME volser LOCATION(ATL) STATUS(MASTER)
```

If you list or display the volume, it should be in MASTER status with a destination of ATL.

2. Enter the volume into the library.

- a. To test CBRUXENT, list or or display the volume in DFSMSrmm. It should now show a location of ATL and no destination. If you display the volume using ISMF, the volume should display as private.
- b. To test CBRUXCUA, use ISMF to change the volume status to scratch. If DFSMSrmm is in protect mode, the change should be rejected. If DFSMSrmm is in warning mode, the change should be allowed, but a warning message should be issued that DFSMSrmm only allows the change in warning mode. Messages are not issued in record or manual mode.
- c. To test CBRUXEJC, eject the volume from the library. You can use any method for the ejection. When the eject is completed and the volume displays in DFSMSrmm, the volume should show as being in transit.
- d. To test CBRUXVNL, ensure that the volume is in private or master status, and run a job to create or read a file on the tape.

You should see message EDG8121D on the console requesting that the volume be reentered into the library. Reply RETRY after the volume is entered into the library, and the job should run successfully.

3.2.16 Testing Application Use of Tape

You should be familiar with the ways in which tape is used in your installation. Although we assume that everyone's use is the same, it is surprising how often we find new and devious uses of tape. To avoid discovering problems in production with DFSMSrmm, we suggest that you test the ways in which you use tapes today to ensure that DFSMSrmm supports them as expected. For example, you may use BLP (perhaps to scratch tapes) or nonlabeled tapes, or you may have key business functions that depend on foreign tape processing.

Use tests that you have already in place for your production tape usage under DFSMSrmm to validate that they are supported and that you know which actions may be required to support them. For example, RACF profiles are required to use NL output to scratch and to process tapes that are to be ignored by DFSMSrmm.

DFSMSrmm normally supports only standard label scratch tapes for non-specific NL tapes, and relabels them at time of use to NL and standard label at return to scratch. To use NL scratch tapes you can use the system tape exit IFG019VM. DFSMSrmm now ships a sample IFG019VM exit as member EDG019VM in SAMPLIB. The sample exit checks for a supported request and issued a WTOR to the operator to obtain the volume serial number. Testing should ensure that it does support the installation needs for non-specific NL tape.

You should test the interface between DFSMShsm and DFSMSrmm. Before you start this kind of test, ensure that DFSMShsm has the correct authority for the STGADMIN resources in RACF CLASS FACILITY. If you have a release of DFSMS/MVS earlier than Version 1 Release 4.0 you must define the use of the ARCTVEXT installation exit in the DFSMShsm PARMLIB member. You can test the function of the EDGTVEXT interface, the use of the ARCTVEXT exit EDGTVEXT interface, or the use of the ARCTVEXT exit by deleting a volume from the DFSMShsm tape volume pool using the HSM DELVOL command.

3.3 Education

You should consider which groups of users may be affected by the initial introduction of DFSMSrmm for testing. Your colleagues, the operators, and possibly the tape librarian will need to learn about DFSMSrmm at some stage. At this stage a minimum level of education should be given so that any potential impact on your production system can be minimized by ensuring that affected users are aware of changes.

3.4 New Operator Procedures

When you start to run DFSMSrmm, system operators will most likely be affected. DFSMSrmm can cause the following changes:

- Mount and fetch messages are updated.
- Scratch pool names may be different and are included in the MVS mount messages.
- Tape drive displays are updated with rack number and pool information.
- DFSMSrmm issues WTORS that require operator action and decisions.

In addition, you may want the operator to be involved in starting and stopping DFSMSrmm.

See *DFSMSrmm Guide and Reference*, SC26-4931, for details of messages and activities that affect the operator.

Chapter 4. Converting from CA-1 to DFSMSrmm

In this chapter we describe how to convert from CA-1, a tape management program of Computer Associates, to DFSMSrmm. We compare CA-1 and DFSMSrmm terms and functions and explain the data extraction process.

Objectives of Chapter

- Explain differences between CA-1 and DFSMSrmm
- Describe how to customize DFSMSrmm for equivalent CA-1 functions
- List the functions where there is no equivalence
- Extract data and prepare input for EDGCNVT
- Provide hints and tips

Audience

- System programmers
- Storage administrators and librarians

At the end of this chapter you will have reviewed the implementation of your current tape management system and will understand the detailed changes required to customize DFSMSrmm to meet your tape management needs.

4.1 CA-1 Overview

CA-1 uses your installation's retention management, vault management, and tape pool management policies to manage tape volumes in tape libraries and storage locations.

The CA-1 tape management catalog (TMC) contains two record types with tape volume inventory information:

- TMC volume record
- Data set name block (DSNB)

CA-1 assigns a default retention period defined by your installation to any tape data set that does not have a retention period or an expiration date specified in the JCL EXPDT or RETPD parameter. It also provides unique Julian dates, which have special meanings to manage tape data set retention. Additionally, it manages tape data set retention using a retention data set (RDS). When the condition is met, the retention specified in the retention data set overrides the expiration value in the TMC volume record. Thus you can change the retention period after a tape data set has been created.

CA-1 uses a vaulting policy to manage volume movement from the library to and among the storage locations. The vaulting policy specifies the data set name, the movement criteria, the storage location ID, and the retention criteria. This policy information is stored in the VPDD, which CA-1's batch program uses during batch update processing. When the batch program executes, it matches the data set name entry in the vault pattern description (VPD) with the TMC volume record. If it finds a match, it updates the TMC volume record and generates a pull list report, which you can use to move tape volumes to and among the storage locations.

CA-1 allows you to assign a tape pool to a specific group of users or application. On a scratch mount request, CA-1 issues a mount message with the pool name indicating the tape pool from which the scratch volume should be selected. It matches the VOLSER of the mounted volume with the ranges defined in the tape pool. If the VOLSER is within the defined ranges, it honors the request. Otherwise, it rejects the volume and requests another scratch mount.

4.2 CA-1 and DFSMSrmm

To ensure a successful conversion, it is important that you understand some of the differences between the products and how those differences affect your conversion project.

4.2.1 Terminology

Table 3 compares and defines some CA-1 and DFSMSrmm terms.

<i>Table 3 (Page 1 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
Address of creation unit	Device number	The device number of the drive on which the volume was mounted when DFSMSrmm recorded information about the data set
Address of unit last used	Last use drive	The device number of the drive on which the volume was last mounted when DFSMSrmm recorded information about the data set
Audit file Audit log	Journal	A sequential data set that contains a chronological record of changes made to the DFSMSrmm control data set
Birth date	Creation date	The volume creation date
Control record	Control record	The first CDS record containing key information for DFSMSrmm
Creating DD name	Creating DD name	The DD statement that created the volume
Creating job name	Job name	The name of the job that created the volume
Creating step name	Creating step name	The name of the step that created the volume
Creation date	Assign date	Volume: The date when the volume changed status; either to or from scratch
	Create date	Data set: The date when the data set was first written to tape
Creation time	Assign time	Volume: The time when the volume changed status; either to or from scratch
	Create time	Data set: The time when the data set was first written to tape
CATALOG	WHILECATALOG (VRS)	DFSMSrmm retains the data set only as long as it is cataloged. Issue the RMM TSO subcommand: RMM ADDVRS DSNAME('**') - WHILECATALOG This means to retain all versions of a data set while they are cataloged. The options <i>CYCLES</i> and <i>COUNT(99999)</i> are default.

Table 3 (Page 2 of 9). CA-1 and DFSMSrmm Terminology		
CA-1	DFSMSrmm	Definition
CATLG/ddd	WHILECATALOG and DAYS in VRS combination (VRS)	DFSMSrmm retains the data set for a minimum of days since creation, and after that period DFSMSrmm retains the data set for as long as it is cataloged. Issue the RMM TSO subcommand: RMM ADDVRS DSNAME('**') DAYS COUNT(5) - NEXT(WHILECATALOG) RMM ADDVRS NAME(WHILECATALOG) WHILECATALOG Note: DFSMSrmm PARMLIB option VRSEL(NEW) is required.
CYCLE/ccc Cycle control	CYCLE (VRS)	DFSMSrmm retains data sets based on cycles or copies of a data set. For GDG data set VRS, the retention is based on GDG cycles. For non-GDG data sets, each occurrence of a data set is considered to be a cycle. To retain 20 cycles for each data set issue the TSO RMM subcommand: RMM ADDVRS DSNAME('CYCLE20') - CYCLES COUNT(20)
Data set block count	Block count	The number of data blocks used by the data set
Data set security password	--	Controls access to tape data sets. Password is stored only in the CA-1 TMC.
Data set name	Data set name	The name of the data set
--	Data set size	The size of the data set in kilobytes
Data set sequence	Physical file sequence number Data set sequence number	The physical file sequence number field specifies the relative position of a data set on a volume. The data set sequence number field corresponds to the IBM standard HDR1 Data set sequence number field.
Date last used	Read date Write date	The date when the data set was last read The date when the data set was last written to
Date tape was first used	Create date Create time	The date when a volume was created The time of day when the volume was created
Date tape was last cleaned	--	The date when a volume was last cleaned
Days since last used control LDATE/ddd	LASTREFERENCEDAYS (VRS)	Specifies that all copies of the data set be retained according to the number of elapsed days since the data set was last read or written to. For example, to retain a data set that has not been used for 14 days, issue the TSO RMM subcommand: RMM ADDVRS DSNAME('LDATE014') - LASTREFERENCE COUNT(14)

<i>Table 3 (Page 3 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
DSNB record	Data set record	Data set information. In CA-1, one for each file greater than the first file on volume ¹ . In DFSMSrmm, one for each file on the volume.
Expiration date	Expiration date	The Expiration date field displays the date when the volume will expire.
	Original expiration date	The original Expiration date field displays a volume's original expiration date. This expiration date is set using the JCL EXPDT or RETPD keyword when the volume was used during open, close, EOVS processing.

Table 3 (Page 4 of 9). CA-1 and DFSMSrmm Terminology

CA-1	DFSMSrmm	Definition
EDMID (External data managed)	Data Set Name / Jobname (VRS)	Specifies tapes controlled by an external data manager, such as DFSMSHsm. DFSMSrmm does not recognize EDM's, but: You can retain all DFSMSHsm and/or ADSM tapes that require no movement by specifying the TSO RMM subcommands: <pre> RMM ADDVRS DSNAME('mprefix.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('bprefix.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('authid.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('adsmprefix.**') - COUNT(99999)- DELETEDATE(1999/365) </pre> In addition, you can define VRS to control the retention of data sets that are not created normally by using the TSO RMM subcommand: <pre> RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(hsm_proc) RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(abars_proc) RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(adsm_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(hsm_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(abars_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(adsm_proc) </pre> See "Running DFSMSrmm with DFSMSHsm," in <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, for more information. Note: All ADSM tapes are managed in the same way as DFSMSHsm tapes.
Hooks	Subsystem interface	DFSMSrmm requires no changes to open, close, or EOVS instructions. DFSMSrmm uses the subsystem interface.
Job that last used volume	--	The name of the job that last used the volume

<i>Table 3 (Page 5 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
--	K-Record	The VRS information
Location ID	Loan location	A user-defined location of a volume that is on loan or not in location SHELF, a system-managed library, or any of the storage locations.
Location ID	Storage location	An onsite library or an offsite storage location, also known as vault or store.
Logical record length	Logical record length	The length, in bytes, of the longest record in the data set
Maximum block size	Block size	The block size of the data set
Number of permanent read errors since initialized	Permanent read	The number of permanent errors that occurred while reading the volume since it was created (Permanent errors are not recoverable.)
Number of permanent read errors since cleaned	--	The number of permanent errors that occurred while reading the volume since it was last cleaned
Number of permanent write errors since initialized	Permanent write	The number of permanent errors that occurred while writing to the volume since it was created (Permanent errors are not recoverable.)
Number of permanent write errors since cleaned	--	The number of permanent errors that occurred while writing to the volume since it was last cleaned
Number of temporary read errors since initialized	Temporary read	The number of temporary errors that occurred while reading the volume since it was created (Temporary errors are recoverable.)
Number of temporary read errors since cleaned	--	The number of temporary errors that occurred while reading the volume since it was last cleaned
Number of temporary write errors since initialized	Temporary write	The number of temporary errors that occurred while writing to the volume since it was created (Temporary errors are recoverable.)
Number of temporary write errors since cleaned	--	The number of temporary errors that occurred while writing to the volume since it was last cleaned.
Number of times tape was cleaned	--	The number of times the volume was cleaned
Number of times tape opened since birthday.	Volume use count	The number of times that the volume was opened for either read and write operations In DFSMSrmm the field will be reset to zero if the status changed from SCRATCH to MASTER or USER.

Table 3 (Page 6 of 9). CA-1 and DFSMSrmm Terminology		
CA-1	DFSMSrmm	Definition
--	OWNER	Users are defined to DFSMSrmm through owner IDs. Owners can be individuals, departments, or any logical grouping of people. General users can change information for their own owner ID once they have been defined to DFSMSrmm. DFSMSrmm automatically creates an owner record if a user who is not defined to DFSMSrmm creates a data set on a scratch volume managed by DFSMSrmm. DFSMSrmm uses the RACF user ID that created the data set as the DFSMSrmm owner ID.
Out-of-area	Loan location Storage location	A user-defined location of a volume that is on loan or not in location SHELF, a system-managed library, or any of the storage locations An onsite library or an offsite storage location, also known as vault or store
--	O-Record	Owner information
PERM/PERMANENT permanent hold	1999/365 or 1999/366	Expiration dates of 99365 and 99366 or 1999/365 and 1999/366 are considered "never-scratch" dates.
	A VRS with a COUNT(99999)	Data sets with these expiration dates must be released manually. Use a VRS with COUNT(99999) to automatically manage data sets with permanent retention.
--	Physical file sequence number	The Physical file sequence number field specifies the relative position of a data set on a volume.
PRIVATE	MASTER USER	A MASTER volume contains valid user data and cannot be overwritten when the same data set name is used. A USER volume is assigned to a user and is in use. It can hold any type of data and can be overwritten, rewritten, and used repeatedly until the volume has expired.
--	P-Record	The software product information
--	Rack number	In the removable media library, you store your volumes in shelves, where each volume occupies a single shelf location. This shelf location is called a rack number. A rack number matches the volume's external label. DFSMSrmm uses the external VOLSER to assign a rack number when adding a volume, unless you specify otherwise. The rack number is the external VOLSER. Thus both must be six alphanumeric characters. The format of the VOLSER you define to DFSMSrmm must be one to six alphanumeric characters. The rack number must be six alphanumeric characters.

<i>Table 3 (Page 7 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
Record format	Record format	The format and characteristics of the records in the data set
Recording density	Recording density	The volume's recording density. For a 3420 tape reel, you can use DENSITY as 1600 or 6250. For a 3480 tape cartridge, use a value of 3480.
Recording technique	Recording format	The basic recording format for tape volumes
Retention data set (RDS)	Data Set Name (VRS)	Use VRs to set policies for retaining all removable media. You can also use VRs to move data sets and volumes for disaster recovery and vital records within the removable media library and among your storage locations. You can use EDGUX100 to assign VRS management values or retention periods to data sets based on the converted RDS. See the <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, for more information.
--	R-Record	The library shelf location (rack) information
SCRATCH	SCRATCH	The volume is free and available for use. It contains no data, or the data on the volume is expired or invalid.
SCRATCH POOL	VLPOOL Volume pool	You can use pool IDs to add one or more rack numbers to a specific library in the removable media library. See "Using DFSMSrmm Installation Exits," in the <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, for more information.
--	Security level	You can define security classes for data sets and volumes. These security classes appear in reports and RMM TSO subcommand output. DFSMSrmm records the security classes in the DFSMSrmm CDS; it does not make RACF aware of them. There is no connection between these definitions and any similar definitions or function provided in RACF, but you can use similar values for overall consistency. Each time a data set is opened or closed DFSMSrmm automatically records its security level. DFSMSrmm determines the security classification of a tape volume with multiple data sets by the highest classification found for a single data set.
Slot	Bin number	Displays the shelf location in one of the storage locations assigned to the volume as part of the requirements of vital records processing. If the volume is moving, this is the bin number in the target location.

<i>Table 3 (Page 8 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
--	SMS data class name	A list of data set allocation attributes that SMS assigns to a data set when it is created
--	SMS storage class name	A list of data set storage service attributes that identify performance and availability requirements. SMS uses these attributes to control data placement.
--	SMS storage group name	A list of volumes used for allocation of new system-managed data sets
--	S-Record	The storage location bin information
Tape label type	Label	The volume's label type: AL - ANSI label AUL - ASCII user label BLP - Bypass label processing NL - Non labeled SL - Standard label SUL - Standard user label
--	Tape media type	Specifies the volume's physical media type. Use one of the following: * - The volume is not a cartridge. CST - The volume is a 3480 or 3490 cartridge. ECCST - The volume is an enhanced capacity 3480 or 3490 cartridge. HPCT - High Performance Cartridge Tape (3590) EHPCT - Reserved for Extended High Performance Cartridge Tape
Time last used	--	The time when the data set was last used
TMC (Tape management catalog)	CDS (Control data set)	The DFSMSrmm CDS is a VSAM KSDS that contains the complete inventory of the removable media library. DFSMSrmm records all changes made to the inventory, such as adding or deleting volumes, in the CDS.
TMC record	--	Volume and first data set information
TMSCLEAN	EDGHSKP (Expiration processing)	DFSMSrmm checks the expiration date for volumes not managed by a vital record specification. If the expiration date has been reached, DFSMSrmm changes the volume status to "pending release." DFSMSrmm checks for any actions that should be taken before the volume can be released. DFSMSrmm returns volumes to scratch that are already in "pending release" status and require no action other than being returned to scratch.
TMSGRW (Generalized report writer)	EDGRPDT (Inventory and movement reports)	Create inventory reports for auditing the physical contents of the installation media library and storage locations

<i>Table 3 (Page 9 of 9). CA-1 and DFSMSrmm Terminology</i>		
CA-1	DFSMSrmm	Definition
System option (CA1.PPOTION)	EDGRMMxx	The DFSMSrmm PARMLIB EDGRMMxx member includes many options for setting up DFSMSrmm
Use count at last cleaning	– –	The use count since the volume was last cleaned
Vault	Storage location	A storage location is a place where you send a store volume. Optionally, it is comprised of shelf locations that you define to DFSMSrmm.
– –	V-Record	The volume information
– –	USER	A USER volume is assigned to a user and is in use. It can hold any type of data and can be overwritten, rewritten, and used repeatedly until the volume has expired.
Vault slot number	Bin number	Specifies the shelf location in one of the storage locations assigned to the volume as part of the requirements of vital records processing
Volume serial number	Volume VOLSER	The VOLSER of the volume
Vault pattern description data set (VPDS)	VRS	Use VRSs to set policies for retaining all removable media. You can also use VRSs to move data sets and volumes for disaster recovery and vital records within the removable media library and among your storage locations. See the <i>DFSMSrmm Implementation and Customization Guide</i> , SC26-4932, for more information.
VMS (Vault management system)	EDGHSKP Storage location management	Request storage location management processing when you are using vital record specifications to identify data sets and volumes to be moved between locations.
¹ The information about the first file is stored in the volume record. The information about the second file and up (file sequence numbers from 2 through 9999 of a volume) is stored in DSNB records.		

4.2.2 Functions

Table 4 on page 75 compares the key tape management functions of CA-1 and DFSMSrmm.

Table 4 (Page 1 of 4). CA-1 and DFSMSrmm Functions	
Function	
CA-1	DFSMSrmm
Tape security	
• Tape must be scratch for output scratch request • Data set name on tape must match inventory specific request • Own password protection, as well as a security system interface that uses SAF to interface to the installed security system.	• Tape must be scratch for output scratch request • Data set name on tape must match inventory specific request • 44-character data set name checked against CDS on input processing • RACF interface for tape profile management • Access list for non-RACF system • Security erase on scratch • Prevent read of scratch tapes • Security classification by generic data set name • Master tape overwrite support controlled by PARMLIB option
Retention management	
• Based on RETPD or EXPDT • By cycle or cycles by day • By days since last reference • By nonmanaged • By catalog status • Permanent • Can default retention by data set name • By days and catalog	• Based on RETPD or EXPDT • By cycle or by BYDAYSCYCLE For DFSMSrmm releases 1.2 through 1.4 which do not have OW30969 (VRS enhancements) installed, APAR OW17377 is required, plus a patch available through the IBM service team. For DFSMSrmm releases 1.3 and 1.4 with OW30969 installed there is the BYDAYSCYCLE option for VRSS. • By days since last reference • By days since creation • By catalog status • Permanent • Can specify default and maximum retention installationwide. Default retention can also be specified using generic data set mask in VRS¹ • Retention policies based on full or generic data set name or VOLSER • By cycle while cataloged • By job name • By days since last reference while cataloged • By days since creation while cataloged • By until expired

Table 4 (Page 2 of 4). CA-1 and DFSMSrmm Functions	
Function	
CA-1	DFSMSrmm
EXPDT with Special Meanings	
• 90ddd • 98000 • 98ddd • 99000 • 99ccc • 99365 and 99366 • Supports up to 2155/366	• Dates of 1999/365 and 1999/366 mean never scratch • Full support of dates beyond twentieth century • Installationwide exit can be used to support special dates
ACCODE JCL parameter	
• xCAUSER • xCANORES • xCACATLG • xCAPERM	• The EDGUX100 installationwide exit can be used to support the ACCODE keywords. APAR OW33428 is required.
Security	
• Password data set and SAF interface (see tape security function)	• RACF control (through SAF interface) to authorize different functions
Exit for DFSMSHsm and ADSM	
• Supplied	• Supplied With DFSMSrmm 1.4 and later releases this is a built-in interface with no need for exits. Prior to DFSMSrmm 1.4, ARCTVEXT is required. • Full support for DFSMSHsm use of scratch pools
Interfaces	
• Supplied	• Integrated with DFSMS/MVS • Can test while CA-1 in control

Table 4 (Page 3 of 4). CA-1 and DFSMSrmm Functions

Function	
CA-1	DFSMSrmm
Vault or storage location management	
• Supports up to 48 vaults (no limit in release 5.1 or higher) • Define management by data set name For CA-1 5.2 and later releases, there is now an option to specify volumes in the VPD. – Full or partially qualified For CA-1 5.1 and later there is an option to use generic data set masks. The masks are not compatible with those used in the IBM products DFSMSdss, DFSMSdtp, ISPF, and DFSMSrmm. – Qualified by job name • Manages by vault slot number • Unrestricted movement • Selection for management based on file 1 on the volume	• Supports up to three built-in storage location names LOCAL, DISTANT and REMOTE, unlimited installation-selected storage locations (any eight-character name can be defined), and any number of system-managed libraries • Define management by data set name, VOLSER, or job name – Full or generic (ISMF syntax), GDG or pseudo-GDG • Manages by store bin number • Unrestricted movement • Selection for management based on both a VRS for all data sets on the volume and a hierarchy • Movement hierarchy (PRIORITY) • Supports loan location
Initialization of new tapes	
• Uses “inactive” VOLSERS, updates inventory² • Uses WTOR for integrity	• Interface ensures integrity, updates inventory³ • Full support of system-managed libraries
Label support	
• SL, AL, NL, NSL, BLP (NSL, NL, BLP require user exit routine or SAF)	• SL, AL, NL, BLP, SUL, AUL
Volume pools (scratch request)	
• VOLSER ranges associated with data set name or job name or both	• VOLSER ranges associated with system; can also restrict input • Installationwide exit can be used to support scratch pools based on data set name and job name or any other installation-selected criteria.
Operator messages	
• Scratch mount messages issued with pool name • Uses 3480 display for “NOT SCRATCH” reason codes • Can modify with a usermod the 3480 and 3490 message display with pool name	• Scratch mount messages modified with pool name • Specific requests modified with rack (slot) number in library • 3480, 3490, and 3590 message display modified with pool ID, rack (slot) number, or pool name

Table 4 (Page 4 of 4). CA-1 and DFSMSrmm Functions	
Function	
CA-1	DFSMSrmm
Optional sticky labels	
All, none, or selected (JCL or online request)	Installationwide exit EDGUX100 can be used to create a label and send to dedicated console
Identify tapes out of library	
Out-of-area support	Loan location support
¹DFSMSrmm provides permanent retention using 99365/99366 or 1999/365 and 1999/366. ²In releases prior to CA-1 5.1, the CA-1 utility TMSTPNIT, is a front-end to IEHINITT. An operator reply to WTOR is required. With CA-1 5.1 the TMSTPNIT utility is no longer a front-end to IEHINITT — IEHINITT is not required. ³DFSMSrmm utility EDGINERS includes IEHINITT functions, but DFSMSrmm does not use IEHINITT; DFSMSrmm supports system-managed tape, and requires no operator intervention, that is, no WTOR.	

4.2.3 Retention Methods

To help you understand the way in which CA-1 and DFSMSrmm manage the retention of tape data sets, we compare the management rules of the two products.

Table 5 presents the CA-1 retention methods and the equivalent VRS definitions that you should use in the DFSMSrmm environment.

Table 5 (Page 1 of 3). CA-1 Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
CA-1	DFSMSrmm
Retained by days and catalog	
LABEL=EXPDT=90ddd Specifies that a tape data set will be retained for a minimum number of days (ddd), after which it will be retained for as long as it is cataloged	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSs. For example: <pre>RMM ADDVRS DSN('CATLG005') DAYS COUNT(5) - NEXTVRS(CATALOG) RMM ADDVRS NAME(CATALOG) - WHILECATALOG</pre> The catalog and days control EXPDT=90ddd will be translated to a MV of 'CATLGddd' and a NEXTVRS of 'CATALOG', as in the above example. If VRSMV=YES is specified in the EDGCxLDR data extraction programs, it will convert the 90ddd key word dates to a new special VRS naming format.

Table 5 (Page 2 of 3). CA-1 Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
CA-1	DFSMSrmm
Ignoring duplicate or undefined VOLSERs	
LABEL=EXPDT=98000 LABEL=EXPDT=1998/000 ACCODE=xCANORES Specifies that a volume should be ignored CA-1 also uses TMSUX0A to allow volumes to be ignored	When you use the EDGUX100 installation exit, DFSMSrmm calls the exit each time a volume is opened, allowing you to decide whether you want to ignore the volume. If you choose to use ignore the volume, the installation exit checks the JCL-specified EXPDT value for a special date of 98000, which indicates that the volume should be ignored. If this special date is found, the exit then uses the installation exit parameter list to request that DFSMSrmm ignore the volume. The sample EDGUX100 exit includes this function. You may have to customize the supplied EDGUX100 to get ignore support using a trigger other than EXPDT=98000. APAR OW33428 is required, to support the ACCODE=xCANORES JCL parameter. Note: For this function you need read or update access to the profile: STGADMIN.EDG.IGNORE.TAPE.volser The RACF profile is stored in the FACILITY class.
Days since last used	
LABEL=EXPDT=98ddd Specifies that a tape data set be retained until the specified number of days (ddd) since last reference has elapsed	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSSs. DFSMSrmm uses the D98ddd VRS management value to retain all copies of the data set based on the number of elapsed days (ddd) since the data set was last read or For example, you can use the VRS management value LDATE020 for the special date 98020 and issue the following TSO RMM subcommand to add a new VRS: RMM ADDVRS DSNAME(' LDATE020') - LASTREFERENCEDAYS COUNT(20) If VRSMV=YES is specified in the EDGCxLDR data extraction programs, it will convert the 98ddd key word dates to a new special VRS naming format.

Table 5 (Page 3 of 3). CA-1 Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
CA-1	DFSMSrmm
Catalog control	
LABEL=EXPDT=99000 LABEL=EXPDT=1999/000 ACCODE=xCACATLG Specifies that a tape data set be retained for as long as it is cataloged	For 99000 the sample user exit EDGUX100 clears the expiration date field, so that DFSMSrmm uses the DFSMSrmm PARMLIB default retention period to calculate an expiration date. This prevents DFSMSrmm from treating a special expiration date as an actual expiration date. Also if the exit parameter list indicates that the exit can specify a VRS management value, EDGUX100 returns a VRS management value of D99000. APAR OW33428 is required, to support the ACCODE JCL parameter. The following example defines a VRS for managing the special date, 99000: <pre>RMM ADDVRS DSNAME(' CATALOG') - WHILECATALOG</pre>
Permanent Retention	
LABEL=EXPDT=99365 or 99366 LABEL=EXPDT=1999/365 or 1999/366 ACCODE=xCAUSER ACCODE=xCAPERM These parameters will retain a tape permanently.	Use the sample user exit EDGUX100 to detect the special JCL parameters and return a VRS management value of 'PERM' APAR OW33428 is required, to support the ACCODE JCL parameter. The following example defines a VRS for managing permanently protected tapes: <pre>RMM ADDVRS DSNAME(' PERM')</pre>
Cycle control	
LABEL=EXPDT=99ccc Specifies that a tape data set be retained for the total number of cycles (ccc) or generations.	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSs. Using the VRS management value DFSMSrmm retains all copies, or cycles, of the data set based on the number of elapsed days (ddd) since the data set was last read or written to. For example, you can use the VRS management value CYCLE005 for the special date 99005 and issue the TSO RMM subcommand to add a new VRS: <pre>RMM ADDVRS DSNAME(' CYCLE005') - CYCLES COUNT(5)</pre>

The extract programs provided with DFSMSrmm for CA-1 build K-Records for the keyword dates that you identify. The K-Records are converted to VRS definitions so you do not need to use the ADDVRS command. Also, the keyword dates and the RDS entries are converted into a table that can be changed dynamically and used with EDGUX100 (the EDGCVRSX sample) to assign VRS management values for new data sets. The table is a combination of JCL keyword values and RDS entries.

4.2.4 Volume and Data Set Record Display

When you display information for a volume or a data set, you will find that some CA-1 fields differ from those of DFSMSrmm. Table 6 lists those differences.

<i>Table 6 (Page 1 of 2). CA-1 and DFSMSrmm Volume and Data Set Record Displays</i>	
CA-1	DFSMSrmm
VOLSER	Volume
DSN	Data set name
DSN17	--
EXPDT	Expiration date
	Original expiration date
ACCT	Account number
FLAG1	--
HOOKID	--
FLAG2	--
BATCHID	--
FLAG3	--
FLAG4	--
CDATE	Assign date
CJOB	Job name
CTIME	Assign time
CUNIT	Unit address
LJOB	--
LTIME	Date last read
	Date last written
LUNIT	Drive last used
CSTEP	Creating step name
CDDNAME	Creating ddname
TRERRC	--
OUTDATE	Movement tracking date
OUTCODE	Location
	Storage location
SLOT	Bin number
TWERRC	--
BTHDATE	Create date
VENDOR	--
COUNT	Volume use count
TRERRI	Temporary read
DATECLN	--
USECLN	--
CLNCNT	--
TWERRI	Temporary write
VOLSEQ	Volume sequence

<i>Table 6 (Page 2 of 2). CA-1 and DFSMSrmm Volume and Data Set Record Displays</i>	
CA-1	DFSMSrmm
EDMID	--
PRERRC	--
1STVOL	--
NEXTVOL	Next volume
PREVVOL	Previous volume
PWERRC	--
NUMDSNB	--
1STDSNB	--
LSTDSNB	--
PRERRI	Permanent read
LABEL	Label
DEN	Density
TRTCH	Recording format
PWERRI	Permanent write
RECFM	Record format
LRECL	Logical record length
BLKSIZE	Block size
BLKCNT	Block count
AUDATE	--
AUTIME	--
USERID	Last changed by
AUCODE	--
AUFLAG1	--
CPUID	System ID
FILE SEQ	Data set sequence
	Physical file sequence number
DSNB	--

4.2.5 Data Fields

During the extract process, the CA-1 records are read, and some fields are extracted to be translated in the DFSMSrmm database.

Some CA-1 fields are not used by the EXTRACT program, and some DFSMSrmm fields are not present in the CA-1 equivalent record.

Other DFSMSrmm fields have a different name in the CA-1 database but have a similar or equivalent meaning in the DFSMSrmm database.

4.2.5.1 Data Unavailable for Conversion

The following are new terms in DFSMSrmm that are not available in the CA-1 TMC (see Table 3 on page 66 for the field definitions):

- Data set size
- Date data set last read*
- Date data set last written*
- Date volume last read*
- Date volume last written*
- Original expiration date
- Home location
- Owner
- Rack
- Physical file sequence number
- Security level
- SMS data class name
- SMS storage class name
- SMS storage group name
- Tape media type
- Tape usage in kilobytes

*In CA-1 only the date last used is stored.

4.2.5.2 Data Not Converted

The following are the fields that the extract program does not extract from the CA-1 TMC (see Table 3 on page 66 for the field definitions):

- Job name that last used volume
- Data set security password
- Number of times tape cleaned
- Use count at last cleaning
- Date tape was last cleaned
- Number of times tape opened since birthdate
(In DFSMSrmm only the count of opens since last assigned is stored)
- EDMID (External data manager ID)
- Number of temporary read errors since cleaned
- Number of temporary write errors since cleaned
- Number of permanent read errors since cleaned
- Number of permanent write errors since cleaned
- Creating program name
- Last used program name

4.2.5.3 Data Converted, but Not Quite the Same

The following fields are present in the CA-1 TMC but not in DFSMSrmm (see Table 3 on page 66 for the field definitions):

- Creation date
- Creation time
- Data set sequence
- Date last used
- Date tape was first used
- Expiration date
- Number of times tape opened since birthdate

4.2.6 Cross-Reference

In this section we present tables that map data from the EDGCNVT input records to CA-1.

4.2.6.1 Volume Information

Table 7 shows the EDGCLREC volume information.

<i>Table 7 (Page 1 of 4). EDGCLREC Volume Information</i>		
	CA-1	Status
Volume serial number	√	Req
Volume owner ¹		Req
Volume owner user ID		
Volume owner node ID		
Volume security level		
Volume sequence number	√	Req
Volume create date	√	Req
Volume create time	√	
Volume create user ID		
Volume create job name	√	
UCB type (copy of UCBTYP)		
Tape media type information		Cond
Tape recording format	√	
Not 3480		
18 track		
36 track		
Tape media type		
Not 3480		
Standard capacity		
Enhanced capacity		
Tape compaction technique	√	
Unknown		
Not compact		
IDRC		
Tape special attributes		
None		
18 track read-only		
Total volume kilobytes used		
JCL volume expiration date	√	
Volume expiration date	√	Req
Volume retention date	√	
Count of data sets on volume	√	Req
Temporary read errors	√	
Temporary write errors	√	
Permanent read errors	√	
Permanent write errors	√	
Tape density (copy of JFCBDEN)	√	

Table 7 (Page 2 of 4). EDGCLREC Volume Information		
	CA-1	Status
BPI indicator 1600 bpi 6250 bpi 3480 3480 compact (IDRC) Undefined	√	Req
Volume status indicators Volume is master Volume pending release Vital record - do not release User tape (assigned by librarian) Tape is on loan Tape open - not yet closed Volume is scratch Volume recorded by open, close, or EOV	√ √ √	Req
Additional volume status indicators Scratch volume claimed by getvol Scratch volume has never been initialized Scratch volume with init action pending Scratch volume waiting to enter ATL Abend in process when a data set closed Abend probably in open, close, or EOV	√	
Volume label type Default retention period used Label type is NL Label type is AL Label type is SL Tape last written with BLP SL or AL with user labels	√ √ √ √ √	Req
Volume release actions Return to scratch pool - default Replace tape on release Reinitialize Degauss/security erase Return to owner Notify owner		
Volume access information Owner may read volume Owner may update volume Owner may alter volume Read-only protection Update protection Can be used on MVS systems Can be used on VM systems	√	Req
Media name		
Unit address used for creation	√	
Last used unit address	√	
Loan location	√	
Date volume last read		
Date volume last written		

<i>Table 7 (Page 3 of 4). EDGCLREC Volume Information</i>		
	CA-1	Status
Assigned from scratch date	√	
Assigned from scratch time	√	
Copy of JFCBLTYP	√	
From JFCTRTCH - IDRC support DSN used 3480 IDRC No compaction	√	Req
Current location of volume	√	Req
Destination name		
Home location name		
Storage group name		
Accounting information	√	
User description		
Authorized user IDs area		
Rack number ²		Req
Remote store bin number	√	Cond
Remote store bin date	√	Cond
Remote store bin media name		Cond
Label number of first file	√	Req
Name of first data set on volume	√	
First data set's owner's name		
Sequence number of first data set		Cond
First data set's RECFM	√	Cond
First data set's LRECL	√	Cond
First data set's BLKSIZE	√	Cond
First data set's block count	√	Cond
First data set's kilobytes		
First data set's security level		
DSN1 create date	√	Req
DSN1 create time	√	
DSN1 last read		Cond
DSN1 last written		Cond
DSN1 flag Data set cataloged Abend in progress when data set closed	√ √	
VRS management value ³		Cond
First file creating step name	√	
First file creating DD name	√	
First file retention date	√	
Name of last data set on volume		Cond
Creating system ID		

Table 7 (Page 4 of 4). EDGCLREC Volume Information		
	CA-1	Status
- Data that can be found in TMS/CA-1 is marked with a $\checkmark$. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹We recommend that you use the job name or the data set high-level qualifier (HLQ) as the owner. ²We recommend that you use the VOLSER as the rack number. ³We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

4.2.6.2 Data Set Name Information

Table 8 shows the EDGCDREC data set name information.

Table 8 (Page 1 of 2). EDGCDREC Data Set Name Information		
	CA-1	Status
Data set create date	$\checkmark$	Req
Data set create time	$\checkmark$	
Data set name	$\checkmark$	Req
Compaction status Data set used 3480 IDRC No compaction		
File sequence number on volume		Req
VOLSER holding data set	$\checkmark$	Req
Volume sequence within set of volumes		Req
Unit where data set was written		
Data set RECFM	$\checkmark$	Req
Data set LRECL	$\checkmark$	Req
Data set BLKSIZE	$\checkmark$	Req
Data set block count	$\checkmark$	Req
Data set size in kilobytes		
Data set owner name ¹		Req
Logical file sequence number	$\checkmark$	Req
Data set security level		
Data set date last read		Req
Data set date last written		Req
VRS management value ²		Cond
DSN flag Data set cataloged Abend in progress when data set closed	$\checkmark$ $\checkmark$	
Creating system ID		
Creating job name	$\checkmark$	
Creating step name	$\checkmark$	
Creating DD name	$\checkmark$	
Retention date	$\checkmark$	

Table 8 (Page 2 of 2). EDGCDREC Data Set Name Information		
	CA-1	Status
• Data that can be found in TMS/CA-1 is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹EDGCNVT ensures that all data sets on the same volume have the same owner as the VOLSER's owner. ²We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

4.2.6.3 Vital Record Information

Table 9 shows the EDGCKREC policy information.

Table 9. EDGCKREC Policy Information		
	CA-1	Status
Record create date		Req
Record create time		
VRS type		Req
Data set name	√	Cond
Format of data set name	√	Cond
Type of retention - Cycles - Cycles by days - Elapsed days - Extra days - Days unreferenced - While cataloged - Until expired	√	Cond
Volume serial number	√	Cond
Delay days		
Total count of number of cycles or days	√	Req
VRS name		Cond
Store keep number	√	Cond
VRS owner		Req
Automatic delete date		
VRS description		
Location name	√	Req
Name of NEXT VRS definition		
Name of AND VRS definition		
Release option		
Job name	√	
• Data that can be found in TMS/CA-1 is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>.		

4.2.6.4 Empty Rack and Bin Information

Table 10 shows the EDGCEREC empty rack and bin information.

Table 10. EDGCEREC Empty Rack and Bin Information		
	CA-1	Status
Shelf status Shelf is rack in library Shelf is bin in storage location		Req
Empty shelf location's name		Req
First of empty shelves start		Req
Count of contiguous shelves		
Media name		
• Data that DFSMSrmm requires is marked with <i>Req</i> .		

4.2.6.5 Owner Record Information

An owner record is required only for actual owners for whom you want to supply owner details and for those who are to be owners for VRS. Other owner records will be created by CONVERT postprocessing (EDGCVOVL). Table 11 shows the EDGCOREC owner information.

Table 11. EDGCOREC Owner Information		
	CA-1	Status
Owner Identifier		Req
Owner's surname		
Owner's first name		
Owner's department ¹		Req
Owner's first address line		
Owner's second address line		
Owner's third address line		
Owner's internal telephone number		
Owner's external telephone number		
Owner's user ID		
Owner's node ID		
• Data that DFSMSrmm requires is marked with <i>Req</i> . ¹ Not strictly required for conversion. Only the RMM ADDOWNER command enforces the owner's department.		

4.3 Reviewing the Current Environment

See Figure 8 on page 21 to review the flow of events that we follow in converting from CA-1 to DFSMSrmm. You must collect and then analyze all information about the current tape management environment.

Once you have reviewed the environment and understand the main differences between CA-1 and DFSMSrmm, you will better understand what you must change during the conversion process.

4.3.1 Collecting Information

The information that you collect about the current environment should include:

- CA-1 reports used daily or occasionally

Every day there should be a CA-1 report that the operator uses to move the cartridges between the desired locations and to return tapes to the scratch pools. You can use the DFSMSrmm supplied reports, but if you want to avoid changes to the layout of these reports, using REXX or DFSORT you can tailor the DFSMSrmm reports to make them similar to the CA-1 reports. Chapter 14, “Extending DFSMSrmm Reporting” on page 533 provides some useful examples of how to create the reports.

- Exits installed

Collect information about CA-1 exits installed in your environment so that you can ensure that you can achieve the same function in an DFSMSrmm environment. For example, you must check the DFSMSshm exit, ARCTVEXT, if you have DFSMSshm installed.

With DFSMS/MVS Version 1 Release 4, DFSMSrmm no longer uses the DFSMSshm tape volume exit ARCTVEXT to manage tapes that DFSMSshm uses. DFSMSrmm at the Version 1 Release 4 level provides the programming interface EDGTVEXT that is used from DFSMSshm.

If you are using a level of DFSMSrmm prior to Version 1 Release 4 refer to Appendix D, “Sample Exits and Reports” on page 603 to merge the ARCTVEXT code from CA-1 with DFSMSrmm.

- PARMLIB options used

You must review CA-1 PARMLIB options before you start DFSMSrmm because you may have to modify the DFSMSrmm options according to the CA-1 environment.

- Interfaces with other products

You want to collect information about the interfaces that CA-1 might have with other products (such as CICS, ROSCOE, and TSO).

- Housekeeping (batch inventory update) process

Collect information on the daily inventory management process so that you can synchronize it with the DFSMSrmm housekeeping function.

4.3.2 Analyzing the Record Layouts

Now you analyze the information you have collected to achieve the equivalent or acceptable alternative function in DFSMSrmm environment.

DFSMSrmm requires certain data to be available to build its CDS. You must analyze CA-1 databases and reports to know which types of data are available. There is not a one-to-one correspondence between DFSMSrmm and CA-1 data. DFSMSrmm may have data that CA-1 does not have and vice versa. If DFSMSrmm requires data that CA-1 does not provide, use the extract programs or the EDGCNVT SYSIN file statements to provide a value.

After analyzing the data, you must determine which types of data you want to have in the DFSMSrmm CDS in order to be consistent with the current operating environment. Knowing the types of data needed helps determine whether you can use CA-1 databases or reports as the sole input to the extract program (see Section 4.6, “TMS/CA-1 Data Extraction” on page 124 for a functional overview of the extract program).

In addition, you must review the current policies regarding retention management, storage location management, and tape pool management. DFSMSrmm manages these processes differently from CA-1

Note: Beginning with CA-1 Release 5.0, the sizes of the TMC volume record and the DSNB record differ from the sizes of previous releases. The record sizes do not differ between Release 5.0 and Release 5.2; only some reserved fields are activated.

4.3.2.1 Volume Ownership

DFSMSrmm requires that all private volumes have an owner. DFSMSrmm uses the RACF user ID of the batch job creating the first file. As CA-1 does not have this ownership concept, during conversion you must decide how to provide ownership information. We recommend that you use either the creating job name or the HLQ of the first file data set name as the volume owner. As tapes are recycled and reused, DFSMSrmm will record the new owner. Using job name or the HLQ of the data set provides you with an easy way of listing and reporting on volumes during the initial implementation of DFSMSrmm.

4.3.2.2 Rack Numbers

DFSMSrmm requires that all volumes have a rack number. DFSMSrmm uses the rack number to identify the volume to the operator and the library location where the volume is stored. As CA-1 does not have this concept, during conversion you must decide how to provide rack number information. We recommend that you use the same value as the VOLSER, unless you have any better information on which to base this value. For example, you may have an installation convention, or keep a file of information that relates an external label VOLSER to an internal label VOLSER.

4.3.2.3 Keyword Date Retention

Any data sets or volumes that are retained and managed using special keyword dates and that you want to continue to manage in this way must be converted to DFSMSrmm VRS management values. You can set the VRS management value in the records that the extract program produces.

4.3.3 CA-1 User Exits

Table 12 lists some CA-1 exits and their equivalent DFSMSrmm functions.

<i>Table 12 (Page 1 of 4). CA-1 User Exits and Equivalent DFSMSrmm Functions</i>		
	CA-1 Function	DFSMSrmm Equivalent
TMMUSER	Use to define user accounting fields in the TMC volume record.	No need to convert in DFSMSrmm. DFSMSrmm automatically stores the first 40 characters of the accounting information specified in your job statement or in a EXEC statement.
TMMVOLDF	Generate exit definitions for the TMSUX1U and TMSUX1E exits.	No need to convert in DFSMSrmm.

Table 12 (Page 2 of 4). CA-1 User Exits and Equivalent DFSMSrmm Functions

	CA-1 Function	DFSMSrmm Equivalent
TMSMSGEX	Use to display the first eight characters of the scratch pool name on tape drives.	In the EDGRMMxx PARMLIB member you can specify the MNTMSG operand to tailor mount and fetch messages so that they display the VOLSER and pool prefix, rack number, or pool name.
TMSUX1A	Automatically bypass CA-1 real-time tape tracking without using EXPDT=98000 in the JCL.	DFSMSrmm ignores all volumes not defined to DFSMSrmm other than when the EDGRMMxx PARMLIB option REJECT ANYUSE(*) is used. Define no REJECT parameter in the EDGRMMxx PARMLIB member. Note: For duplicate VOLSERs you must use EXPDT=98000 in the JCL, and you need read or update access to the profile: STGADMIN.EDG.IGNORE.TAPE.volser in the RACF FACILITY class. To bypass DFSMSrmm use the sample user exit EDGUX100.
TMSUX1B	Establish retention other than the assigned ABEND Default Retention specified in the CA-1 system options member, TMOOPTxx.	In DFSMSrmm you can specify ABEND retention using a VRS named ABEND. You can have one ABEND VRS without jobname, and an additional ABEND VRS for each jobname. You can define an ABEND VRS by using the TSO RMM subcommand, for example: RMM ADDVRS DSNAME('ABEND') - DAYS COUNT(3) This defines a default retention of three days for tape data sets OPEN at the time an application or system abend occurs.
TMSUX1C	Copy accounting information in the TMC volume record created by the TMSUX1J user exit.	You have no need for this function, because DFSMSrmm automatically stores all accounting information.
TMSUX1D	Use to more specifically define data sets that are to be controlled by an external data manager (EDM).	You can specify VRS for fully qualified data set names or generic data set names by using the Interactive Storage Management Facility (ISMF) naming convention. Use VRSs to permanently retain EDM data until the EDM releases them.
TMSUX1E	Convert internal numeric VOLSER to an external alphanumeric value.	You have no need for this function because you can specify a one- to six-character alphanumeric internal VOLSER and a six-character alphanumeric external VOLSER (rack number).

Table 12 (Page 3 of 4). CA-1 User Exits and Equivalent DFSMSrmm Functions

	CA-1 Function	DFSMSrmm Equivalent
TMSUX1F	Allow or disallow real-time double opens or re-create data sets.	In the EDGRMMxx PARMLIB member you can specify the MASTEROVERWRITE(<i>value</i>) operand to allow you to select whether existing files on tapes in master status can be used for output. <i>value</i> ADD New data can be created, but no existing data can be destroyed. LAST Only the last file on a MASTER volume can be overwritten. The data set name must match the the existing data set name. MATCH An existing file can be overwritten. The data set name must match the existing data set name. All files which are higher in sequence are destroyed. USER Any data set can be rewritten as many times as the user wishes until the volume expires.
TMSUX1G	Validate scratch tape mounted in response to a scratch subpool request.	You define pools to DFSMSrmm in the EDGRMMxx PARMLIB member that the EDGUX100 installation exit selects for new tape data sets which are to be created on nonspecific volumes (see the <i>DFSMSrmm Implementation and Customization Guide</i> , SC26-4932, "Using DFSMSrmm Installation Exits," for more details). DFSMSrmm always ensures that volumes are from the correct scratch pool.
TMSUX1J	Capture accounting data specified in the job or step statements in the JCL.	DFSMSrmm automatically stores the first 40 characters of the accounting information specified in your job statement. The accounting information of the step statement can optionally be stored based on a PARMLIB option.
TMSUX1L	Determine which tapes should have external gummed labels.	The EDGUX100 installation exit can be used to create labels and send to dedicated console.
TMSUX1S	Conjunction with the CA-1 external security interface.	DFSMSrmm provides support for RACF standard tape volume security protection, using any combination of RACF TAPEVOL and TAPEDSN options. DFSMSrmm requires no exit for this function.
TMSUX1U	Convert external alphanumeric VOLSER to an internal numeric value.	You have no need for this function, because you can specify a one- to six-character alphanumeric internal VOLSER and a six-character alphanumeric external VOLSER (rack number). The internal and external VOLSER can be different.
TMSUXCO	Modify fields during conversion of data to TMC format or CA-1 control statements.	Unnecessary to modify fields in DFSMSrmm.
TMSUXCV	Convert from the CA-1 Rel. 4.x TMC to the new CA-1 Rel. 5.1 TMC format.	Unnecessary to convert in DFSMSrmm.

Table 12 (Page 4 of 4). CA-1 User Exits and Equivalent DFSMSrmm Functions

	CA-1 Function	DFSMSrmm Equivalent
TMSUXID	Convert system catalog information to CA-1 TMC format.	You can use the conversion programs to create or expand your DFSMSrmm CDS. See Chapter 9, "Building the DFSMSrmm CDS" on page 403 for more information.
TMSXCLN1	Modify the expiration date and scratch indicator.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCLN2	Reset the expiration date of tape volumes using USER/nnn expiration dates during scratch processing.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCTLG	Allow the expiration date and expired by Catalog Control indicator to be modified.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCYCL	Modify the expiration date or bypass the TMC record update.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXEXP	Allow modification of the expiration date and eligible for RDS indicator or the TMC record update to be bypassed.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXOCAT	Use for MVS catalog and TMC synchronization.	--
TMSXTPNT	Reject volume for label processing.	You can define and delete volumes at any time without having to take DFSMSrmm out of service. In DFSMSrmm you need not define volume groups or volume ranges.
TMSXTPPR	Provide various options to control the processing of tapes.	--
TMSXVLT1	Allow TMC record selection for vault management system processing.	The retention and movement policies you define to DFSMSrmm are known as VRSS. You use them to say how long and where you want to keep data sets or volumes. You also use them to define how volumes are to be moved among the libraries DFSMSrmm supports, and the storage locations defined for vital records, and disaster recovery purposes. All information about storage location management processing is stored in the DFSMSrmm CDS.

4.3.4 CA-1 PARMLIB Options

Review carefully all of the CA-1 PARMLIB options before proceeding. Pay particular attention to the CA1.PPOPTION members, which include most of the system options.

You must modify the DFSMSrmm options to reflect the same defaults as specified in the CA-1 options. The most important DFSMSrmm options for the conversion are:

- **OPTION** statement
 - OPMODE parameter
 - BLP parameter
 - MASTEROVERWRITE parameter
 - RETPD parameter
 - TPRACF parameter
 - UNCATALOG parameter
 - VRSJOBNAME parameter
- **VLPOOL** statement

Only if you use multiple scratch pool support
- **LOCDEF** statement

Only if you use location names different from the installation defaults.

The TMS PARMLIB options should be input to the conversion tools so that the relevant values for policy creation can be used when the DFSMSrmm VRS records are built. Other than the VRS creation, the CA-1 PARMLIB options must be manually converted to DFSMSrmm options.

Table 13 lists the CA-1 PARMLIB options and their DFSMSrmm equivalences.

Table 13 (Page 1 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences	
Option	
CA-1	DFSMSrmm
Preferred date format	
DATEFMT Specifies the default format of the date	DATEFORM Specifies the default format of the date A mm/dd/yyyy - American E dd/mm/yyyy - European I yyyy/mm/dd - ISO J yyyy/ddd - Julian
SCRPOOL scratch pool management	
SCRPOOL Defines ranges or subranges of scratch pools	VLPOOL Use the VLPOOL command in PARMLIB to define pools. When you add a new volume to the library, DFSMSrmm assigns it a shelf location from the specified pool.

Table 13 (Page 2 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
Assignment control statement	
DSN / JOB / POOL Assigns a data set to a tape pool	EDGUX100 Installationwide exit, EDGUX100, could be used to assign data sets to a specified scratch pool. See the <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, or refer to section 4.6.5.1, "How the Conversion Process Works" on page 158 for an example of how the VRSTABLE is used by EDGUX100.
Specific mount message intercept	
XMSG Prefixes specific mounts with special message IDs	MNTMSG Use the MNTMSG command to tailor mount and fetch messages so they display the volume serial number and pool name or rack number.
Label change action	
LCHG Changes label if the label type of a tape mounted to satisfy a request is different from that specified in the JCL	BLP Specifies how DFSMSrmm controls BLP. If BLP(RMM) is specified, you can use BLP for input from and output to volumes in user status and for input from volumes in master status. When the BLP(NORMM) option is specified, you can use BLP for input from and output to volumes in user and master status and for output to scratch tapes. If the label is being changed and you are not using BLP, you also need access to the profile in the RACF FACILITY class: STGADMIN.EDG.LABEL.volser or STGADMIN.EDG.NOLABEL.volser in the RACF FACILITY class.
Density change action	
DCHG Enforces that the density be changed for a tape	-- The density of a tape can always be changed. You can also set the VOLNSNS option in PARMLIB to YES to allow a 36-trk formatted cartridge mounted on a 3480 device to be relabeled to 18-trk format. All IBM products, including DFSMSrmm, support this option.
Recording technique change action	
TCHG Changes the recording technique for tape	-- The recording technique of a tape can always be changed.

Table 13 (Page 3 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences	
Option	
CA-1	DFSMSrmm
Print gummed label action	
LAB Automatically generates a gummed label for a tape	Installationwide exit, EDGUX100, can be used to create label and send to dedicated console. See the <i>DFSMSrmm Implementation and Customization Guide</i> , SC26-4932, for more information.
Mixed EXPDT action	
MIXEXP Handles different expiration dates for a tape	-- You can always use mixed expiration dates when a tape is not AL.
Abend retention	
ABE Specifies the number of days to retain a volume closed during an abend	VRS In DFSMSrmm you can specify an ABEND retention using a VRS, ABEND. You can define multiple ABEND VRS data set vital record definitions by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('ABEND') - DAYS COUNT(3) JOBNAME(JOB12*)</pre> which defines a default retention of three days for tape data sets open at the time an application or system abend occurs and the JOBNAME begins with JOB12.
3480 message display action	
MG3480 Displays the NOT SCRATCH reason code on a 3480 and 3490 cartridge unit.	--
TMC DSNB - maximum threshold	
TMC DSNB Specifies a limit in percent of the total number of TMC DSNB records that can be written before a warning message is issued	The only limit for creating DFSMSrmm records in the DFSMSrmm CDS is the limit of VSAM.
TMC AUDIT - maximum threshold	
TMC AUDIT Specifies a limit in percent of the total number of AUDIT records that can be written before a warning message is issued	If the DFSMSrmm journal is full, a WTOR is displayed on the system console. For DFSMSrmm 1.3 and later releases there is a PARMLIB threshold value for the journal.

Table 13 (Page 4 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
Extended CYCLE retention	
RC Adds an additional number of days to provide extended protection	To add an additional number of days you must define a VRS chain by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('SCHLUM.EXTRA') - NEXTVRS(EXTRA) CYCLES COUNT(7) RMM ADDVRS NAME(EXTRA) - EXTRADAYS COUNT(2)</pre> For extended CYCLE retention the conversion routines build a VRS called RC and any keyword dates or RDS entries converted to VRS use these values as their NEXTVRS. For example <pre>RMM ADDVRS DSNAME('SCHLUM.EXTRA') - NEXTVRS(RC) CYCLES COUNT(7) RMM ADDVRS NAME(RC) EXTRADAYS COUNT(2)</pre> The RC VRS is equivalent to RC 2.
Extended CATALOG retention	
R9 Adds an additional number of days to provide extended protection Additional retention control is available using the RDS.	To add an additional number of days you must define a VRS chain by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('SCHLUM.BACKUP') - NEXTVRS(EXTRA) WHILECATALOG</pre> <pre>RMM ADDVRS NAME(EXTRA) - EXTRADAYS COUNT(2)</pre> For extended catalog retention the conversion routines build a VRS called R9 and any keyword dates or RDS entries converted to VRS uses these values as their NEXTVRS. For example <pre>RMM ADDVRS DSNAME('SCHLUM.CATLG') - NEXTVRS(R9) WHILECATALOG RMM ADDVRS NAME(R9) EXTRADAYS COUNT(3)</pre> The R9 VRS is equivalent to R9 3.
Extended CA-1 retention	
RR Adds an additional number of days to provide extended protection Additional retention control is available using the RDS. See vendor product documentation.	The equivalent in DFSMSrmm is to use NEXTVRS on any required VRS to chain to a NEXTVRS that specifies EXTRADAYS retention type.

Table 13 (Page 5 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
Extended EDM retention	
REDM Adds an additional number of days to provide extended protection Additional retention control is available using the RDS.	The equivalent in DFSMSrmm is to use NEXTVRS on any required VRS to chain to a NEXTVRS that specifies EXTRADAYS retention type.
Default retention	
RP Specifies the default retention period. Can be a retention period or a keyword date.	RETPD Specifies the default retention in days only. For example, RETPD(5)=retain for five days. To have a default such as retain while cataloged, you must use the DFSMSrmm VRSs; for example, <pre>RMM ADDVRS DSN('**') WHILECATALOG.</pre> If the RP value is a retention period, EDGCSRDS builds an entry in the EDGUX100 table for <pre>EDGCVRSRSG DSN=*,RETPD=ddd,R0=N0</pre> If the RP value is a keyword date EDGCSRDS builds an entry in the EDGUX100 table for <pre>EDGCVRSRSG DSN=*,VRSVAL=keydate</pre> The default retention period entry would be equivalent to an DFSMSrmm PARMLIB RETPD value, but the default keydate entry ensures that all data sets not specifying a value in the JCL get a VRS MV assigned to control retention. For more information, refer to 4.6.5.1, "How the Conversion Process Works" on page 158 for an example of how the VRSTABLE is used by EDGUX100.
RDS override action	
RO Allows JCL-supplied retention to be overridden by TMSEXPDT Additional retention control is available using the RDS.	MAXRETPD Specifies the maximum retention period that a user can request for data sets on volumes. Specify a value between 0 and 9999 days, or NOLIMIT. The specified value is added to the current date to determine the maximum allowed expiration date. If the calculated date is 31 December 1999, the expiration date 1 January 2000 is used. The RO value from the CA-1 PARMLIB is used during conversion to determine whether the VRS RELEASEOPTION(EXPIRYDATEIGNORE) option is used on VRSs built for retention, and whether the table entries built by EDGCSRDS specify RO=YES.

Table 13 (Page 6 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
TMSCLEAN uncatalog action	
UNCATA Specifies which data sets TMSCLEAN can uncatalog	UNCATALOG Specifies the type of catalog support to provide: N - Do not use DFSMSrmm catalog processing. DFSMSrmm does not uncatalog data sets under any circumstances. Y - Always uncatalog data sets. Use DFSMSrmm catalog processing when: • A volume is returned to scratch status • The RMM TSO DELETEVOLUME FORCE command is issued • The RMM TSO CHANGEVOLUME DSNAME command is issued • The RMM TSO DELETEDATASET command is issued • A tape data set is overwritten S - Only uncatalog data sets when the volume on which they reside is returned to scratch status.
CA-1 EXPDT keyword range override ddname	
KEYDD Specifies the ddname to be placed in the JCL indicating that a date specified in the JCL is not a CA-1 keyword	EDGUX100 Users often specify special dates in their JCL using the JCL EXPDT keyword. It is possible that they might want to code a real expiration date that matches one of the special dates your installation uses. You need to be able to distinguish what the user requires in these cases. See the <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, for more information. The default coded in the EDGUX100 exit is set to NOTKEYD8. Customize the exit to change name as needed.
Auto-checkin default retention	
KEYTAP Specifies the default retention if the automatic checkin facility is used for a data set	RETPD Specifies the default retention period for all new data sets on volumes. Specify a value between 0 and 9999 days. The specified value is added to the current date to determine the expiration date. Note that the MAXRETPD value you specify in the PARMLIB limits the calculated expiration date. DFSMSrmm uses the default in EDGRMMxx because neither RETPD nor EXPDT is specified.

Table 13 (Page 7 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
Cycle control cycle definitions (CDAY)	
CDAY DSN—Each explicit data set is counted as one cycle. DAY—All tapes of the same data set name and creation date are retained as one cycle.	VRS Specifying a VRS by using the BYDAYSCYCLE keyword. This means that all instances of a data set created on a single calendar day are considered to be a single cycle. DFSMSrmm retains data sets based on cycles or copies of a data set. For DFSMSrmm levels prior to APAR OW30969 (the VRS enhancements), installing APAR OW17377 plus a patch available through the IBM service team provides this capability. For GDG data set VRSs, the retention is based on GDG cycles. For non-GDG data sets, each occurrence of a data set is considered to be a cycle. CDAY is one of the PARMLIB options used by the conversion tools to build the correct VRSs for DFSMSrmm.
Cycle control cycle definitions (CJOB)	
CJOB DSN—The job name is ignored for cycles. JOB—A separate cycle is maintained for each different job name.	ADDVRS JOBNAME Specifies the job name for the VRS. You can specify a specific job name or a mask where '%' matches any one character and '*' matches any character string. Fully qualified names take precedence over generic names. JOBNAME(*) specified in the VRS is the same as CJOB=JOB. Specify VRSJOBNAME in PARMLIB to select whether DFSMSrmm uses the job name from VRSs as the primary or secondary match. Specify VRSJOBNAME(1) so that the job name takes precedence over the data set name during VRS match processing. Specify VRSJOBNAME(2) so that the data set name takes precedence over job name during VRS match processing. This is the equivalent to the CA-1 JOBNAME qualification. The CJOB parameter is used during conversion to build VRSs from the RDS and from any keyword dates, and the correct value for JOBNAME is determined and used.

Table 13 (Page 8 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences

Option	
CA-1	DFSMSrmm
Vault management cycle definition	
CYD Specifies the rotation of cycle control tapes to and from the data center to storage locations by creation date or cycle	VRS The equivalent to CYD is BYDAYSCYCLE. The conversion programs use this PARMLIB option to build the correct VRSS. Prior to OW30969 (VRS enhancements) OW17377 can be used to get cycle by days as described here. For DFSMSrmm levels prior to OW30969 (the VRS enhancements) APAR OW17377 provides cycle by days as described here.
TMC data set name specification	
TMCDN Specifies the fully qualified data set name of the TMC	DSNAME Specifies the name of the DFSMSrmm CDS
AUDIT data set name specification	
AUDDSN Specifies the fully qualified data set name of the AUDIT data set	JRNNAME Specifies the name of the DFSMSrmm journal
Data set security ddname	
DSSN Specifies the DD statement for the CA-1 internal security	RACF DFSMSrmm does not control authorization of access to tape data. DFP and RACF are used.
Auxiliary disposition ddname	
DSN Performs special real-time postprocessing functions, such as automatically marking a volume out of area	Installationwide exit, EDGUX100, can be used to create label and send to dedicated console.
TIQ console time out	
TW Specifies the elapsed time before a session will time out	--
Report customization	
TMSPRINT Customizes your reporting	EDGRPTD You can use the EXEC parameter to specify the security heading text, lines per page, and date format for each report. See the <i>DFSMSrmm Implementation and Customization Guide</i>, SC26-4932, for more information.

Table 13 (Page 9 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences	
Option	
CA-1	DFSMSrmm
Security interface specifications (OCEOV)	
OCEOV Specifies the external security processing	TPRACF Specifies the type of RACF tape support for DFSMSrmm to provide. Use this operand to request that DFSMSrmm maintain the security profiles that protect tape volumes.
Security interface specifications (DSNB)	
DSNB Specifies the external security processing for any secondary data sets	TPRACF Specifies the type of RACF tape support for DFSMSrmm to provide. Use this operand to request that DFSMSrmm maintain the security profiles that protect tape volumes.
YSVC call processing	
YSVC Controls the direct access to the TMC and AUDIT data sets for batch jobs and online users	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Batch call processing	
BATCH Controls access to the TMC volume and DSNB records by batch utilities	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Password call processing	
PSWD Limits the use of internal passwords	There is no support in DFSMSrmm specifically for use of passwords.
Command processing	
CMD Controls commands for online users	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Function call processing	
FUNC Controls special realtime functions (NL, NSL, and BLP)	RACF All access for a tape will be checked by RACF.
Scratch processing	
SCRATCH Passes control to the TMSUX1S security user exit	--

Table 13 (Page 10 of 10). CA-1 PARMLIB Options and DFSMSrmm Equivalences	
Option	
CA-1	DFSMSrmm
Undefined processing	
UNDEF Allows or rejects access to undefined resources	--
Create processing	
CREATE Defines the level of access required to create a data set on tape.	RACF You can protect your tape data sets by using normal RACF processing.

4.3.5 CA-1 Pattern Masking

In CA-1 releases 5.1 and later there is a pattern matching method which is similar to that used in DFSMSrmm and DFSMSdss. You need to translate the CA-1 pattern masks into DFSMS/MVS Removable Media Manager data set name masks.

The following special characters can be used in CA-1.

(?)	Question mark
(!)	Exclamation mark
(#)	Pound sign
(@)	At sign
(*)	Asterisk
(-)	Dash
(\)	Back slash

Table 14 lists the CA-1 pattern masks and their DFSMSrmm equivalences.

Note: The DFSMSrmm conversion tools only handle the conversion of the '-' partial masking character, converging it to '*.*.*'.

Table 14 (Page 1 of 2). CA-1 Pattern Masks and DFSMSrmm Equivalences	
Pattern Character	
CA-1	DFSMSrmm
Question mark (?)	
The absence of any single character matches	% for a single character * for absence of 1 or more characters
Exclamation mark (!)	
A place holder for a single character	(% percent sign) A place holder for a single character (¬ not sign) A place holder for a single character in a pseudo-gdg data set name mask
Pound sign (#)	
A place holder for a numeric value.	Closest equivalent is %

Table 14 (Page 2 of 2). CA-1 Pattern Masks and DFSMSrmm Equivalences	
Pattern Character	
CA-1	DFSMSrmm
At sign (@)	
A place holder for any alphabetic character	Closest equivalent is %
Asterisk (*)	
Represents zero or more characters	Represents zero or more characters
Dash (-)	
Represents zero or any number of characters before and after its position	*.*.* Represents zero or more characters or qualifiers
Back slash (\)	
Indicates that the following character is not a pattern character	--

4.3.6 Inventory Management and Reports

CA-1 provides batch facilities to perform the following functions:

- Batch update and reporting
- Tape retention and expiration
- Storage location management
- Backup, recovery, and verification of databases

Except for batch update and reporting, DFSMSrmm provides similar functions with the EDGHSKP, EDGBKUP, and EDGUTIL utilities (for more details, see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932).

EDGHSKP must be executed with the DFSMSrmm subsystem active. EDGBKUP and EDGUTIL can be executed independently of the subsystem.

The utilities have the following functions:

- EDGHSKP
 - Processes vital records
 - Performs storage location management
 - Performs expiration processing
 - Creates an extract data set
 - Backs up the DFSMSrmm CDS
 - Backs up and clears the DFSMSrmm journal
- EDGBKUP
 - Backs up the DFSMSrmm CDS
 - Backs up the DFSMSrmm journal
 - Recovers and restores the DFSMSrmm CDS
 - Reorganizes the DFSMSrmm CDS
- EDGUTIL
 - Verifies the DFSMSrmm CDS.²
 - Mending the DFSMSrmm CDS.

² The EDGUTIL function can also provide consistency checking between the CDS and the TCDB in a system-managed tape library.

DFSMSrmm does not provide any batch update utility. However, you can execute the DFSMSrmm TSO commands ADD, CHANGE, and DELETE in batch to achieve the same result. The ISPF dialog has the option to generate and group together TSO commands for later batch processing.

CA-1 provides batch reports. These may be supplemented with the use of the optional CA-EARL program, which allows you to customize your own reports from the TMC.

You can also use the CA-1 program, TMSGRW, to produce batch reports, but CA-1 Release 5.1 will be the last release that provides TMSGRW.

DFSMSrmm provides two reporting programs and one reporting REXX:

- EDGRPTD—processes the report extract file to produce:
 - Volume movement reports
 - Moved from storage locations
 - Ready-to-scratch sorted by RACK
 - Sorted by BIN number
 - Sorted by OWNER
 - Moved to storage locations
 - Sorted by RACK number
 - Sorted by OWNER
 - Volume inventory reports.
 - Moved from storage locations
 - Sorted by volume serial number
 - Sorted by RACK number or BIN number
 - Sorted by OWNER
- EDGAUD—processes DFSMSrmm SMF records to produce:
 - Security report (tracking information for classified data)
 - Audit report (CDS changes).
- EDGJRPT—processes the extended extract file
 - Pull list for scratch tapes sorted by volume serial number
 - Pull list for scratch tapes sorted by data set name
 - Inventory list by volume serial number
 - Inventory list by data set name
 - Inventory of data set including number of KB used
 - Inventory of volume serial number by location
 - Inventory of data set names by location
 - Inventory of BIN numbers by location
 - List of all data set names at loan location
 - List of all volume serial numbers at loan location
 - List of all multiple volume, multiple data sets
 - Movement report including the first data set name on the volume
 - Movement report sorted by storage location bin number
 - Movement report sorted by volume serial number
 - Inventory list sorted by volume serial number including volume count

You can execute the DFSMSrmm TSO commands, LIST and SEARCH, in batch to produce a wide variety of reports. You can also use DFSORT's ICETOOL to write your own customized reports.

EDGRPTD and EDGAUD sample jobs, REXX programs, and DFSORT's ICETOOL reports are provided in Chapter 14, "Extending DFSMSrmm Reporting" on page 533 and Appendix D, "Sample Exits and Reports" on page 603.

The CA-1 scratch list report is shifted one day in the DFSMSrmm environment because the DFSMSrmm report lists volumes that are in PENDING RELEASE status. Those volumes are returned to SCRATCH status during the next housekeeping process, which usually runs the next day.

We suggest that you move to the scratch pool the cartridges that will return to scratch only when they are no longer in PENDING RELEASE status. You can accomplish this in your batch housekeeping jobstream by using a step that lists the volumes in PENDING RELEASE status before running the EDGHSKP step. These are the volumes that are new scratch volumes according to the CA-1 report. Alternatively, you can execute EDGHSKP with the EXPROC parameter a second time to return pending release volumes to scratch. If you specify the RELEASE(SCRATCHIMMEDIATE) operand on the ADDVRS subcommand and the VRSEL(NEW) PARMLIB option, only a single run of inventory management EXPROC processing is required to return volumes to scratch. When the VRSEL(OLD) PARMLIB option is in use, DFSMSrmm ignores any VRS release options that may have been set for a volume.

4.3.7 Exits

DFSMSrmm provides source code for the installationwide exits installed as part of DFSMSrmm during the SMP/E APPLY process. The source code is supplied for:

ARCTVEXT	DFSMSShsm tape volume programming interface
IGXMSGEX	DFSMSrmm uses the DFSMSdfp MSGDISP exit to update tape drive displays
CBRUXENT	Object Access Method (OAM) library control system (LCS) cartridge entry exit
CBRUXEJC	OAM cartridge eject exit
CBRUXCUA	OAM cartridge change use attributes exit
CBRUXVNL	OAM volume not in library exit
EDGUX100	RMM installationwide exit; two samples are provided: EDGUX100, and EDGCVRSX in SAMPLIB

Before the SMP/E ACCEPT, the source code is in SMPSTS; after the ACCEPT, the exits are in AEDGSR1.

If your installation is already using one or more of these exits in the CA-1 environment, you must ensure that the correct action is performed during conversion. Chapter 10, "Parallel Running and Validation" on page 427 and Chapter 11, "Cutover to Production" on page 455 describe the procedure to follow.

CA-1 uses the CBRUXENT exit to interface the IBM 3494 and IBM 3495 Automated Tape Library Dataservers. You can test the DFSMSrmm version of this exit once you have completed the conversion.

The CBRUXVNL exit is optional. If you use it, however, as with the CBRUXENT exit, you must test it in the DFSMSrmm environment after conversion.

The installationwide exits that you installed with DFSMSrmm for either ARCTVEXT, IGXMSGEX, CBRUXENT, CBRUXEJC, and CBRUXCUA can replace any existing exits you had previously installed. Decide whether you still require the function your previous exits provided, even if for a short time while introducing DFSMSrmm. If you no longer need the function, proceed with the

next implementation step. If you still require the function, consider how to implement it, while still providing the function required by DFSMSrmm.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more details on the DFSMSrmm programming interface that you use with EDGTVEXT, IGXMSGEX, and the OAM exits. The supplied source code for these exits shows how to call the programming interface. You can either modify the supplied exit to call a modified version of your old exit, or modify your old exit to call the DFSMSrmm programming interface. Consider how you might provide an option to disable your code without another code modification. In either case, install the changed software by using an SMP/E USERMOD.

If your installation has DFSMSHsm installed and CA-1 is using the ARCTVEXT exit and you have a DFSMSrmm prior to Version 1 Release 4, you will need to merge the CA-1 code for the exit with the DFSMSrmm code in order to notify both tape management products when running in parallel. We recommend that you convert your code to run when called by the DFSMSrmm exit and modify the supplied exit to call your code. An example can be found in Appendix D, "Sample Exits and Reports" on page 603.

Based on the decisions you have taken about VRS management values and scratch pooling, you can update the supplied sample installation exit, EDGUX100, and the dynamic table produced by the conversion programs, to perform the function you require. We recommend you use the sample EDGUX100 exit that is shipped by DFSMSrmm as SAMPLIB member EDGCVRSX. This sample exit uses a dynamic table that is initially built by the sample job EDGJSRDS. The sample job creates a table called UXTABLE. It is a list of assembler macro statements that is easily customizable to add or remove entries. You can add your pooling decisions to the table by modifying the entries built by EDGCSRDS, or by adding new entries. Follow these steps:

1. Copy the sample installationwide exit, EDGCVRSX, and rename it to EDGUX100, and use this copy as a base for your exit.
2. Update the exit if needed. You should only need to update the exit to add entries to the EDMTAB to identify data that requires only the first file of each volume to be recorded, or if you want to exploit the 'PL100_ITS_CLOSE' call to implement any TMSDISP-like functions, like sticky labels.
3. Update the dynamic table for EDGUX100. You can do this only after running EDGCSRDS using the sample job EDGJSRDS. Copy the three source parts of the dynamic table:

EDGCVRSF The start of the table from SAMPLIB
VRSTABLE The main table, created by EDGCSRDS
EDGCVRSE The end of the table from SAMPLIB

You can update the table with data set names that should have VRS management values assigned, or special retention periods, or special scratch pools, or with additional keyword dates that the exit should support (refer to Figure 63 on page 158).

4. Build an SMP/E USERMOD to apply the updated source code for EDGUX100, and UXTABLE, including the necessary JCLIN statements to get the EDGUX100 load module added to the LINKLIB target library. Following is a sample usermod:

```

++USERMOD (VMRMM01) REWORK(1998066) .
++VER (Z038) FMID(JDZ11B4) PRE(UW19315) .
++JCLIN .
//EDGUX100 EXEC PGM=IEWL,PARM=' LET,NCAL,RENT,REUS,REFR,LIST,XREF'
//SYSLMOD DD DISP=SHR,DSN=SYS1.LINKLIB
//SRCLIB DD DISP=SHR,DSN=TERRI.SAMPLIB
//AEDGMOD1 DD DISP=SHR,DSN=SYS1.AEDGMOD1
//SYSPRINT DD SYSOUT=*
//SYSLIN DD *
INCLUDE AEDGMOD1(EDGUX100)
ENTRY EDGUX100
NAME EDGUX100(R)
//UXTABLE EXEC PGM=IEWL,PARM=' LET,NCAL,LIST,XREF'
//SYSLMOD DD DISP=SHR,DSN=SYS1.LINKLIB
//SRCLIB DD DISP=SHR,DSN=TERRI.SAMPLIB
//AEDGMOD1 DD DISP=SHR,DSN=SYS1.AEDGMOD1
//SYSPRINT DD SYSOUT=*
//SYSLIN DD *
INCLUDE AEDGMOD1(UXTABLE)
NAME UXTABLE(R)
++SRC(UXTABLE) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .
++SRC(EDGUX100) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .
++SAMP(EDGUX100) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .

```

5. Receive and apply the USERMOD.

6. Your new exit is now ready for use on your system.

You can use the same process to install the EDGUX200 code if necessary.

For more information about DFSMSrmm exits refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

4.3.7.1 Data Set Overwrite

CA-1 supports two volume statuses:

PRIVATE The volume contains valid data.

SCRATCH The volume can be used for scratch output requests.

In an DFSMSrmm environment you have three different volume statuses:

MASTER A private volume that contains data that is available for write processing based on the MASTEROVERWRITE PARMLIB option.

ADD New data can be created, but no existing data can be destroyed.

LAST Only the last file on a MASTER volume can be overwritten. The data set name must match the existing data set name.

MATCH An existing file can be overwritten. The data set name must match the existing data set name. All files which are higher sequence are destroyed.

USER The same as a volume in USER status.

SCRATCH The volume can be used for scratch output requests.

USER The volume is assigned to a user, and it can be overwritten, rewritten, and used repeatedly until the user returns it to the scratch pool.

DFSMSrmm can manage the overwrite of a MASTER volume with the PARMLIB option, MASTEROVERWRITE (see TMSUX1F in Table 12 on page 91). You can

tailor this option to meet your installation's needs. The option `MASTEROVERWRITE(LAST)` is equivalent to the function provided by CA-1. Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for a detailed explanation of the function.

4.3.7.2 BLP and NL Processing

In a CA-1 environment, because NL tapes do not have internal labels and BLP tapes bypass standard labels, you must supply the VOLSER in the JCL or from the operator console.

BLP and NL processing differs in an DFSMSrmm environment. All scratch tapes must be standard label. When DFSMSrmm successfully reads the VOL1 label for a volume mounted as a result of a nonspecific request, it provides the actual VOLSER to OPEN for use in related processing such as cataloging of the data. This replaces the logical VOLSER (Lnnnnn) generated by OPEN in these circumstances.

For BLP, further processing is determined by the BLP option set in PARMLIB. Processing is as follows:

RMM	This is the default and allows BLP processing of USER status tapes and BLP input from MASTER status tapes.
NORMM	BLP can be used for reading and writing of master and user status tapes and for output to scratch tapes. BLP reading of scratch tapes is not supported.

When BLP or NL is used with scratch tapes, DFSMSrmm still changes the volume to master status but sets the initialize release action to ensure that the volume has a valid standard label before it is returned to scratch status.

Some changes are required for how and when DFSMSrmm records information for tapes and data read and written using BLP:

- For BLP output to a nonspecific volume, the file sequence number is not checked. For example, `LABEL=(2,BLP)` could be used. For non-BLP requests, the restriction of using `LABEL=(1,label_type)` is still enforced.
- The data set information for files processed with BLP is only updated for output to the first file on a volume. All other types of BLP requests change only the volume information such as the date last read and date last written. For BLP output requests the data set information is only updated in the DFSMSrmm CDS when the data set is closed.

You can enable the use of NL output to scratch volumes by creating a profile in the RACF FACILITY class: `STGADMIN.EDG.NOLABEL.volser`. Once the profile is created, the function is enabled, and only those who have access to the profile can use this function. DFSMSrmm normally supports only standard label scratch tapes for NL tapes, and relabels them at time of use to NL and standard label at return to scratch. To use NL scratch tapes you can use the system tape exit `IFG019VM`. DFSMSrmm now ships a sample `IFG019VM` exit as member `EDG019VM` in `SAMPLIB`. The sample exit checks for a supported request and issues a WTOR to the operator to obtain the volume serial number. After installing the exit, testing should be done to ensure the exit supports the installation's need for NL tape.

4.3.7.3 DFSMSHsm Interface

CA-1 and DFSMSrmm process the release of DFSMSHsm tapes differently. When CA-1 is informed through the ARCTVEXT exit that a DFSMSHsm volume is no longer required, it uses the extended EDM retention value from PARMLIB to calculate the retention date for the volume.

When DFSMSrmm is informed, through the EDGTVEXT general-use programming interface, that a DFSMSHsm volume is no longer required, the volume is immediately marked *PENDING RELEASE* and will be returned to the scratch pool following the next run of the housekeeping expiration process "*EDGHSKP,PARM=(EXPROC)*". Thus DFSMSrmm can release DFSMSHsm tapes at a different time than CA-1. You must take this difference into account when comparing scratch reports for the two systems.

4.3.7.4 Tape Initialization

In an DFSMSrmm environment the initialization process is different. The EDGINERS utility should be used instead of IEHINITT for the tapes defined in the DFSMSrmm database. You must ensure that your operational procedures are updated in order to use the new DFSMSrmm utility after the conversion process.

DFSMSrmm does not require that you initialize tape volumes before they are available as scratch, but it prevents you from using as scratch a volume that requires initialization and will not include the volume in a scratch pull list. When adding scratch volumes to DFSMSrmm, you have the choice of deciding whether or not they need to be initialized.

Operators should use the DFSMSrmm EDGINERS utility (instead of IEHINITT) to initialize and erase tape volumes. EDGINERS automatically defines new volumes it initializes or erases to the DFSMSrmm CDS. For an existing volume, DFSMSrmm ensures that the proper volume is mounted and the requested action is actually required. You can request DFSMSrmm to automatically initialize scratch volumes when you add them to DFSMSrmm.

4.3.7.5 Tape Pool Management

CA-1 allows you to define ranges of tape volumes in a tape library to a specific pool. It restricts scratch requests to a specific pool based on the creating user and/or data set name. For example, you define ranges of tape volumes dedicated to a specific application use only.

Multiple Scratch Pool Support: CA-1 supports multiple scratch pool addressing by using a table, TMONSMxx, where you can specify a data set name and the range of volumes associated with it. If the volume opened for output does not fall within the specified range, CA-1 denies the request and dismounts the volume.

If your installation uses the TMONSMxx table to access different scratch pools, you must use the DFSMSrmm multiple scratch pools facility. With this support, you will be able to direct new tape data sets to specific scratch pools on the basis of job name and data set name. Thus you can discretely manage sets of volumes on a client, application, or other basis.

Only an installation using an DFSMSrmm installationwide exit can request multiple scratch pool support. The exit can be used to set a specific scratch pool for nonspecific tape volume output requests.

Implementation for DFSMSrmm: DFSMSrmm decides which pool to use on the basis of the VLPOOL definitions in the DFSMSrmm PARMLIB member. This function is dependent on the correct VLPOOL definitions, but the customer can decide which pool to select by using the EDGUX100 installation exit.

DFSMSrmm intercepts the mount messages issued on the system and, for those for nonspecific volume requests, calls the installation exit to enable it to make a decision. DFSMSrmm saves the scratch pool selected by the exit for use in updating tape drive displays, updating the mount message, and for validating the volume when one is mounted.

The exit can optionally request that DFSMSrmm prevent the cartridge loader from being indexed. Thus drives can be selectively preloaded with scratch volumes from specific pools, but scratch volumes from other pools can be manually mounted without the cartridge loader being emptied.

When the exit selects a pool, the only acceptable volumes are those in that pool. When DFSMSrmm selects the scratch pool without use of the exit, any scratch volume from any eligible scratch pool on that system is acceptable.

DFSMSrmm provides a working sample exit that implements a scratch pooling system based on job names and data set names; all you have to do is update a table containing job names, data set names, and pool prefixes in the EDGUX100 exit.

You can also specify different data set or job name pool tables for each system. If you specify the NAME operand in the VLPOOL command in PARMLIB member EDGRMMxx, the tape drive will display this name rather than the pool prefix.

Note: The specification of a NAME value allows DFSMSrmm to drive basic tape library support (BTLS) scratch pool selection. You can use data set names as well as job names to direct scratch pool selection. Use pool NAME values that match the BTLS pool names, such as SCRATCH1.

4.4 Retention and Movement

In the sections that follow we discuss some basic tape management retention and movement concepts and suggest some factors to consider when converting from CA-1 to DFSMSrmm.

4.4.1 Retention Management

You define and use a retention management policy to indicate how long to retain a tape data set in both a tape library and the storage locations.

4.4.1.1 CA-1 Policies

CA-1 uses specific or unique Julian dates (90000-90365 and 98000-99366) to control tape data set retention and expiration. It also provides the ability to differentiate the dates with true expiration dates using a unique JCL DD statement when creating the tape data set.

EXPDT=90ddd Specifies that a tape data set be retained for a minimum number of days (ddd), after which it will be retained for as long as it is cataloged

EXPDT=98000 Specifies that the tape volume being processed is a foreign tape, not under CA-1 management

EXPDT=98ddd Specifies that a tape data set be retained until the specified number of days (ddd) since the last reference has elapsed

EXPDT=99000 Specifies that a tape data set be retained for as long as it is cataloged

EXPDT=99ccc Specifies that a tape data set be retained for the total number of cycles (ccc) or generations

EXPDT=99365 Specifies that a tape data set be retained permanently

EXPDT=99366 Specifies that a tape data set be retained permanently.

In addition, CA-1 allows you to change the tape data set's retention and expiration value, after it has been created, using a retention data set. A batch program, which runs during inventory management, processes the retention data set and compares the entry to the TMC volume record. When the condition is met, the retention specified in the retention data set overrides the retention and expiration information in the TMC volume record.

With the retention data set, you can qualify by data set name and partial data set name and further qualify by job name.

CA-1 catalog retention is enforced through checking whether a data set is cataloged. CA-1 accepts output from the IDCAMS LISTCAT command rather than use a locate to determine whether a data set is still cataloged. Therefore, you do not need to share catalogs. With DFSMSrmm prior to DFSMS/MVS Version 1.5, you must share the ICF catalogs because DFSMSrmm always does a locate.

CA-1 treats DFSMSHsm as an EDM, so it does not maintain volume sequence information in its database for multi-volume DFSMSHsm dump tapes, whereas it does for non-DFSMSHsm-managed tapes. During the conversion process the extract program has no choice but to assume that each and every DFSMSHsm dump tape contains a single volume file instead of a multi-volume file. In contrast, DFSMSrmm maintains volume sequence information for multi-volume DFSMSHsm dump tapes so it can recognize a set of dump volumes as being one cycle. If you plan to use DFSMSrmm cycle control to manage the DFSMSHsm dumps, you can get details of the multi-volume set from DFSMSHsm by using the command

```
HSEND LIST PRIMARYVOLUME(vvvvvv) ALLDUMPS BCDS
```

to find the correct volume sequence order. You can use DFSMSrmm CHANGEVOLUME with the PREVVOL operand to build the correct volume chains once the conversion is completed and DFSMSrmm is started with the converted CDS. This should only be necessary if you want to use CYCLE control for off-site movement of DFSMSHsm tapes.

4.4.1.2 DFSMSrmm Policies

DFSMSrmm does not recognize that EXPDT dates of 90000-90365 and 98000-99364 have special meanings; but, by using the installationwide exit, EDGUX100, or a management class with VRS, you can get DFSMSrmm to provide the same function.³ DFSMSrmm uses VRSSs, which define the policy to control the tape data set retention and tape volume retention.

³ Expiration dates of 99365 and 99366 or 1999/365 and 1999/366 are considered "never-scratch" dates in DFSMSrmm.

DFSMSrmm allows you to set a system default retention period if the retention or expiration is not specified in the JCL during data set and volume creation. You can extend the retention period for the data set on the volume or for the volume, after it has been created using a VRS definition. The retention period specified in the VRS can override the expiration date in the DFSMSrmm CDS.

DFSMSrmm supports many different types of retention. For a data set, you can request the following types of retention using a specific or a generic data set name:

- *Retention by cycles or retention by cycles while cataloged*, which defines how many cycles or generations of a data set should be retained. Cycles can either be by data set or by day, as in CYCLES or BYDAYSCYCLE.
- *Retention by number of elapsed days since creation or retention by number of elapsed days since creation while cataloged*, which indicates that a data set be retained until the specified number of days since creation have elapsed
- *Retention by number of elapsed days since last reference or retention by number of elapsed days since last reference while cataloged*, which indicates that a data set be retained until the specified number of days since last reference have elapsed
- *Retention by catalog*, which indicates that a data set be retained as long as it is cataloged.
- *Retention until expired*, which means that a data set can now optionally be managed separately for retention. A vital record specification management value or a SMS management class can identify retention criteria, and the data set name can be used to identify movement criteria. Up to two VRSs can be used to provide policy information for a data set. The first VRS is called the *primary VRS*, and if there is a secondary VRS, it is called the *secondary VRS*.

When new data is created, and no vital record specification management value or management class is assigned because the jobs do not specify keyword dates and are not SMS managed, VRSEL processing finds only the data set VRS to match to, which specifies UNTILEXPIRED. DFSMSrmm checks the volume expiration date and the UNTILEXPIRED option makes DFSMSrmm drop the data set from retention.

The volume expiration date is also used if the following conditions are true and the PARMLIB option VRSEL(NEW) is specified:

- When a data set matches to a single VRS.
- When UNTILEXPIRED is used in the secondary VRS.

The VRS enhancements introduced with APAR OW30969 enable until expired retention to be based on any type of retention. When a primary and secondary VRS match to a data set, the use of UNTILEXPIRED in the primary uses the secondary VRS chain to determine overall retention. If there is no secondary VRS, or UNTILEXPIRED is used in a secondary VRS or a VRS management value VRS, and PARMLIB option VRSEL(NEW) is used, only the volume expiration date is used. If VRSEL(OLD) is specified in PARMLIB, UNTILEXPIRED retention uses the volume expiration date or catalog status to determine for how long to retain a data set.

Any policy used for retention or movement, or both retention and movement, can be a chain of multiple VRSs, where each VRS in a chain can use a different type of retention. Both the primary and secondary VRS can be VRS chains, offering enough flexibility to create quite complex retention and movement policies.

A secondary VRS can be only a VRS management class or a VRS management value. For more information how the new VRS processing works see Chapter 13, “Additional DFSMSrmm Function” on page 495.

For a volume, you can request the following types of retention using a specific or generic VOLSER:

- *Retention by number of elapsed days since creation*, which indicates that a volume be retained until the specified number of days since creation have elapsed
- *Retention by number of volumes*, which indicates how many volumes should be retained

For all data sets for which you do not specify a retention date in JCL, DFSMSrmm uses a default retention period. You use VRSs to extend this date. For data sets that are left open or that are closed by abend processing, DFSMSrmm provides special ABEND and OPEN VRSs to allow you to control the retention separate from the normal retention for those data sets.

Figure 38 and Figure 39 on page 116 show how to add a data set VRS to set a default retention management period that protects the data set as long as it is cataloged.

Panel Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . SCHLUM.**

Job name mask . . SAVE*

Volume serial . . (May be generic)

VRS name

Figure 38. Define Data Set Name Retention VRS

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : SCHLUM.**	GDG . . . NO
Job name mask : SAVE	
Count 99999	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location OFFSITE	
Number in location . 99999	
Priority 0	
Next VRS in chain . . :	Release options:
Chain using :	Expiry date ignore NO
	Scratch immediate NO
Owner SIEGEL	
Description . . .	
Delete date . . . 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 39. Define While Cataloged Retention VRS

DFSMSrmm provides support for the specific and unique Julian dates of 98000-99364 having special meanings. The support is provided through an installationwide exit, which you can use to interrogate and recognize the dates. When the installationwide exit detects a data set using the Julian dates, it assigns a VRS management value that identifies the retention management policy. A sample EDGUX100, called EDGCVRSX, is available in SAMPLIB that uses a dynamic table driven assignment method based on data set names and keyword dates. We recommend that those converting to DFSMSrmm use the EDGCVRSX sample and the table generated by the conversion tools.

When the VRSEL(NEW) PARMLIB option is in use, retention types can be combined to meet complex retention need. A data set can now be retained by a data set name VRS and a (SMS) VRS management class or (EDGXU100) VRS management value together. DFSMSrmm processes both primary and secondary VRSs to retain a data set.

DFSMSrmm uses the following criteria to retain the data set:

- If a data set name matches to a data set name VRS. DFSMSrmm will use this VRS as the primary VRS.
- The secondary VRS is based on a matching vital record specification management class or vital record specification management value
- If a match on data set name is not found, the vital record specification management class or vital record specification management value will be called the primary VRS; that means a vital record specification management class or a vital record specification management value can be either a primary VRS (if no matching data set name VRS is found) or a secondary VRS.

When the VRSEL(OLD) PARMLIB option is in use, during inventory management processing, if a match on data set name is not found, DFSMSrmm checks the data set's management value. If the value exists, DFSMSrmm matches the

assigned VRS management value with the defined VRS. If there is a match, DFSMSrmm uses the retention specified in the VRS overriding the expiration from the DFSMSrmm CDS. For example, if you want to support the expiration date of 99000 (retention while cataloged), you need to define a VRS to manage the retention while cataloged. In the installationwide exit, you intercept the tape data set using the 99000 expiration date and assign a VRS management value matching the defined VRS.

4.4.2 Storage Location Management

You define and use your installation's management policies for those tape volumes and data sets that contain installation-critical data for disaster recovery. These policies define:

- Critical data—data that is vital to your installation
- Storage location—where to store critical data
- Retention criteria—how long to keep the critical data in the storage location
- Movement criteria—how to move critical data to and from the data center and the storage locations.

Because DFSMSrmm and CA-1 differ in terms of how they define and process storage location management policies, it is important that you review the current storage location management policy to understand retention and movement criteria.

4.4.2.1 CA-1

CA-1 manages tape volume movement to and among vaults using your installation-defined vaulting policy. You store this policy in a VPD, which a batch program processes during the batch update, comparing the entry to the TMC volume record. When there is a match, the tape volume is marked for movement to and among the vaults. The policy is based on data set name, fully or partially qualified, and further qualified by job name. CA-1 supports many vaults and does not restrict movement to and among vaults.

The batch program automatically updates the TMC volume record for tape volumes meeting the criteria with the proper location ID (vault code) and generates the store bin (slot) numbers to use in the storage locations. In addition, it generates a report that indicates the current location, the destination location, and the store bin numbers for the tape volumes that require movement. If you delete a retention or vaulting policy, CA-1 continues to retain data based on the deleted entry.

CA-1 uses the predefined outcode location name of LIBR or LIB to delay volume movement until the specified number of days is reached. You can accomplish the same with DFSMSrmm by defining a VRS with LOCATION(HOME) and use NEXTVRS to chain VRS definitions together. Do not use the DELAY option of VRS unless the delay is only to apply to the latest copy or cycle. For example, you could have defined in the VPDD:

```
V=LIBR,CR=3  
V=VLT1,CR=10
```

The above definitions indicate that the volumes are to be kept in the tape library for three days after creation before they are moved to storage location VLT1.

The DFSMSrmm equivalent to the above VPDD definitions, which is produced by the conversion programs, is:

```
RMM ADDVRS DSN(...) DAYS COUNT(3) -  
    LOCATION(HOME) STORENUMBER(3) -  
    NEXTVRS(VRS00001)
```

```
RMM ADDVRS NAME(VRS00001) DAYS COUNT(10)  
    LOCATION(VLT1) STORENUMBER(10)
```

4.4.2.2 DFSMSrmm

Using your installation-defined VRSSs, DFSMSrmm automatically controls the movement of the data sets on a tape volume or the tape volume from the tape library to and among the storage locations. You can specify a data set name or a data set name mask and, optionally, a job name or job name mask to define a VRS. Additionally, you can specify a VOLSER or a generic VOLSER.

Figure 40 and Figure 41 on page 119 show how to add a data set VRS defining the storage location and the retention period to retain one copy for as long as the data set is cataloged.

Panel

Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . MY.CRITICAL.DATA

Job name mask . . D094746

Volume serial . . (May be generic)

VRS name

Figure 40. Define Data Set Name Movement VRS

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : MY.CRITICAL.DATA	GDG . . . NO
Job name mask : D094746	
Count 5	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location OFFSITE	
Number in location . 5	
Priority 0	
Next VRS in chain . . NXT1	Release options:
Chain using AND	Expiry date ignore YES
	Scratch immediate YES
Owner SIEGEL	
Description	
Delete date 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 41. Define Application Movement Policy VRS

Here is an explanation of two new VRS subparameters **ANDVRS** and **RELEASE**:

ANDVRS

Specifies ANDVRS to create a chain of VRS where all the retention conditions must be true for the data set to be retained. DFSMSrmm uses the STORENUMBER and LOCATION from the first VRS in the ANDVRS subchain.

RELEASE

Specifies special release options after a data set is released. RELEASE options take affect only for data sets retained by a VRS.

- EXPIRYDATEIGNORE is specified so DFSMSrmm ignores the volume expiration date when VRSEL(NEW) is used and no other vital VRS matches to the data set.
- SCRATCHIMMEDIATE is specified so DFSMSrmm returns the volume to scratch status in a single run of inventory management EXPROC processing when the only pending release action for the volume is to return to scratch.

DFSMSrmm does not restrict the movement of the data set on the tape volume or the tape volume to and among the storage locations. When a data set on the tape volume, or the tape volume, is moved from a tape library to and among the storage locations for disaster recovery or vital records, DFSMSrmm assigns a storage location bin number, storage location name, and store date.

DFSMSrmm allows you to specify one location name in each data set or volume VRS. If you require that a data set or volume be moved to more storage locations, you can chain together two or more VRSs by using the NEXTVRS, NAME, and ANDVRS keywords. DFSMSrmm does not restrict any sequence of data set or volume moves among storage locations and system-managed libraries. For example, if your installation's policy is to initially retain the data set in the *remote* storage location and then later retain it in the *distant* storage

location, you first define a data set VRS for the *remote* storage location and indicate that the data set requires multiple moves by specifying a name VRS. You then indicate the next storage location, the *distant* storage location, through that name VRS.

Figure 42 and Figure 43 on page 121 show how to add a data set VRS defining the first storage location (remote), the retention (one cycle), and the name of the next storage location.

Panel

Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . MY.CRITICAL.DATA

Job name mask . . D094746

Volume serial . . (May be generic)

VRS name

Figure 42. Define GDG Data Set Name VRS

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : MY.CRITICAL.DATA	GDG . . . NO
Job name mask : D094746	
Count 5	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location REMOTE	
Number in location . 1	
Priority 0	
Next VRS in chain . . NEXT002	Release options:
Chain using . . . NEXT	Expiry date ignore YES
	Scratch immediate YES
Owner SCHLUM	
Description	
Delete date . . . 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 43. Define Application First Store VRS

Figure 44 and Figure 45 on page 122 show how to add a name VRS defining the second storage location (*distant*) and the retention period (one cycle). This VRS is chained by the VRS definition in Figure 43.

Panel Help	

EDGPV200	DFSMSrmm Add Vital Record Specification
Command ==>	
Specify one of the following:	
Data set mask . .	
Job name mask . .	
Volume serial . .	(May be generic)
VRS name NEXT002	

Figure 44. Define Name VRS

Panel Help	

EDGPV240	DFSMSrmm Add Name VRS
Name	NEXT002
Count	1
Retention type	CYCLES
While cataloged	no
Until expired	no
Location	DISTANT
Number in location .	2
Next VRS in chain . .	
Chain using . . .	
Owner	SCHLUM
Description . . .	
Delete date . . .	1999/365 (YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	
Command ==>	

Figure 45. Define Application Second Store VRS

During inventory management processing, DFSMSrmm identifies the volumes that must be moved to and among the storage locations and assigns the exact store bin numbers where the volumes are to be stored. DFSMSrmm provides a batch utility (EDGRPTD), which you can use to generate a pull list for the volumes requiring movement. When the volumes have been moved, you must confirm that the moves have taken place. The confirmation process updates the DFSMSrmm CDS to reflect the most current information about where the volumes are stored.

DFSMSrmm uses the retention criteria from the VRS to manage tape volume retention in the storage locations. See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information about DFSMSrmm retention management.

When you convert to DFSMSrmm, you can continue to use the same vault names, but, because DFSMSrmm supports longer names, you may want to change the vault names to be more meaningful. For more detailed information to adding or changing VRS (see Chapter 10, “Parallel Running and Validation” on page 427 for a functional overview of the new VRS commands and panels).

Note: Each time you run inventory management VRSEL processing, DFSMSrmm recalculates retention and movement needs. If you delete a VRS, any data sets it retained can match to another VRS or are eligible for release processing.

4.5 Running the Extract Program

In this section we describe the sample programs and JCL for extracting data for conversion to DFSMSrmm from CA-1. The samples are supplied in the SAMPLIB member EDGCMM01. This book describes them as they were when the book was written. Before running the extract program, you should read and be familiar with the *Data Extraction Programs User's Guide* supplied with the current version of EDGCMM01, as it will be maintained to reflect changes and enhancements.

Table 15 lists the samples for the extraction of data from CA-1.

Table 15. IBM-Provided Data Extraction Samples, Programs, and JCL		
Name	Location	Function
EDGCMM01	SAMPLIB	Source and formatted User's Guide
EDGCSRDS	SAMPLIB	Source for TMS extract program
EDGCSVDS	SAMPLIB	Source for TMS extract program
EDGC4BIN	SAMPLIB	Source for TMS 4.9 extract program
EDGC4LDR	SAMPLIB	Source for TMS 4.9 extract program
EDGC5BIN	SAMPLIB	Source for TMS 5.0 and 5.1 extract program
EDGC5LDR	SAMPLIB	Source for TMS 5.0 and 5.1 extract program
EDGJSRDS	SAMPLIB	JCL to execute EDGCSRDS
EDGJSVDS	SAMPLIB	JCL to execute EDGCSVDS
EDGJ4BIN	SAMPLIB	JCL to execute EDGC4BIN
EDGJ4LDR	SAMPLIB	JCL to execute EDGC4LDR
EDGJ5BIN	SAMPLIB	JCL to execute EDGC5BIN
EDGJ5LDR	SAMPLIB	JCL to execute EDGC5LDR
EDGCVRSX	SAMPLIB	Sample EDGUX100 exit
EDGCVRSG	SAMPLIB	Macro used for EDGCVRSX/F
EDGCVRSF	SAMPLIB	Part of sample EDGUX100 table
EDGCVRSE	SAMPLIB	Part of sample EDGUX100 table
EDGCVRSL	SAMPLIB	Sample to load EDGUX100 table
EDGCVRSP	SAMPLIB	Procedure to run EDGCVRSL
EDGC1PRM	LINKLIB	Processes TMS PARMLIB
EDGC1PPL	MODGEN	Macro for EDGC1PRM input

The sample JCL is set up to work in a "typical" installation. There is sample JCL, some of which is commented out so that you choose to use it or not as is your preference. Procedures, which are normally part of other IBM products, are used to assemble and compile the sample code. Thus you do not have to assemble, link, and compile code before you can use it. If these procedures are not available to you, you can edit the JCL to uncomment the statements that execute the programs from a STEPLIB. If you choose to link-edit the programs yourself, you can use the standard link-edit options. None of the programs is reentrant.

Note: The new sample JCL uses the High Level Assembler (HLASM). If you are running on a system which does not yet have HLASM, you can modify the sample JCL to use Assembler H V2 procedures instead.

4.6 TMS/CA-1 Data Extraction

The data extraction programs extract the proper information from the TMC volume records and DSNB records and convert the information about retention and vaulting contained in the RDS and VPDD. Use the vendor-provided utility, TMSDATA, to unload volume records and DSNB records from the TMC database. (See Section 4.6.2.4, “Input Data Sets” on page 135 for detailed information about the processing requirements of the TMC volume records and the DSNB records.) If TMSDATA is not available, see sample EDGJ5BIN for the JCL to extract the records using DFSORT Release 11.1 or higher.

4.6.1 Preparing to Run the Extract

Before executing TMSDATA, you must ensure that there are no broken volume chains or data set chains. TMS/CA-1 provides the following utilities to accomplish this function:

- TMSPTRS (at the 9308 level or higher for 5.0)
- TMSFVSN

In addition, check that the expiration dates used under 5.0 or later levels of CA-1 are not set to STATS/xxx. CA-1 treats these dates as permanent retention, but you should correct them to be valid expiration dates before conversion. If you do not, the extract programs issue an information message and set a permanent retention date.

In addition, consider running other CA-1 utilities to ensure that the data being converted to RMM is current and will probably remain so long enough to complete the conversion process and validate the results. For example, consider running the following:

TMSEXPDT	To set the correct expiration dates for volumes on the basis of your RDS definitions and CA-1 parameters
TMSCTLG	To check data set catalog status
TMSCYCLE	To remove volumes from cycle control
TMSCLEAN	To return volumes to scratch status
TMSVMxx	To make retention decisions based on the VPDD and assign volumes to vaults and vault slots.

You should obtain a copy of the TMS PARMLIB, CA1.PPOPTION, for input to the conversion programs. There are some PARMLIB options used when creating the DFSMSrmm VRS policies which are obtained from the PARMLIB by EDGC1PRM.

You must unload the TMC database to create the input data sets for the data extraction programs. Use the TMSDATA utility, which creates two sequential data sets, one each for the TMC volume records and DSNB records. If TMSDATA is not available, you can also use DFSORT (Release 11.1 or higher) to unload the TMC database.

Figure 46 on page 125 shows sample DFSORT JCL.

```

//CVT50SRT JOB ,140.SCHLUMBERGER,MSGCLASS=H,REGION=6M,
//      MSGLEVEL=(1,1),NOTIFY=SCHLUM
//* ****
//* * SPLIT CA-1 DATABASE *
//* *      PRODUCES SEQUENTIAL DATA SETS FROM TMC VOLUME RECORDS AND *
//* *      DATA SET NAME BLOCKS (DSNB) RECORDS *
//* * *
//* * PREREQUISITE DFSORT REL. 11.1 OR REL. 11.0 INCLUDE THE *
//* *      APAR PL82191 *
//* * *
//* * CA-1 REL. 5.0 or higher *
//* *      TMC CONTROL RECORDS *
//* *      EXCLUDING THE FIRST 3 OR 4 CA-1 CONTROL RECORDS *
//* *      TMC VOLUME RECORD *
//* *      SORTED BY FIRST VOLSER, VOL SEQ AND VOLSER *
//* *      1. FIRST VOLSER OF DATASET      OFFSET 56   LENGTH 6 *
//* *      2. VOLUME SEQUENCE NUMBER      OFFSET 54   LENGTH 2 *
//* *      3. VOLUME SERIAL NUMBER        OFFSET 0    LENGTH 6 *
//* *      DSNB DATA SET RECORD *
//* *      1. FIRST VOLSER NUMBER OF THE 1ST. OFFSET 16   LENGTH 6 *
//* *      2. FILE SEQUENCE NUMBER        OFFSET 28   LENGTH 2 *
//* * *
//* ****
//DELETE EXEC PGM=IDCAMS
//SYSPRINT DD  SYSOUT=*
//SYSIN DD  *
DELETE DFRMM.CUSTOMER.CA1.VOLUME1 NONVSAM SCRATCH
DELETE DFRMM.CUSTOMER.CA1.DSNB1 NONVSAM SCRATCH
/*
//SORT01 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD  SYSOUT=*
//SORTIN DD  DISP=SHR,DSN=DFRMM.CUSTOMER.CA1
//SORTWK01 DD  UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD  DSN=DFRMM.CUSTOMER.CA1.VOLUME1,DISP=(,CATLG,DELETE),
//      DCB=(RECFM=FB,LRECL=340,BLKSIZE=0),UNIT=SYSDA,
//      SPACE=(27880,(160),RLSE,,ROUND)
//SYSOUT DD  SYSOUT=*
//* -----
//*      TMC VOLUME RECORDS
//*      SORTED BY FIRST VOLSER, VOLUME SEQUENCE AND VOLSER
//*      EXCLUDE TMC CONTROL RECORDS, DSNB RECORDS AND
//*      VOLUME IN DELETE (INACTIVE) STATUS
//* -----
//SYSIN DD  *
      SORT FIELDS=(57,6,CH,A,55,2,BI,A,1,6,CH,A)
      OMIT COND=(1,7,CH,EQ,C' TMSCTL#',
                  OR,1,1,CH,EQ,X' FF',
                  OR,85,1,BI,EQ,B'.....1.')
```

Figure 46 (Part 1 of 2). Sample DFSORT JCL

```

//SORT02 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=DFRMM.CUSTOMER.CA1
//SORTWK01 DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD DSN=&&TEMP,DISP=(,PASS,DELETE),
//          DCB=(RECFM=FB,LRECL=340,BLKSIZE=0),UNIT=SYSDA,
//          SPACE=(27880,(900),RLSE,,ROUND)
//SYSOUT DD SYSOUT=*
//* -----
//*      DSNB DATA SET RECORD
//*      SORTED BY COMPLETE RECORD
//*      INCLUDE ONLY DSNB RECORDS
//* -----
//SYSIN DD *
        SORT FIELDS=(1,340,CH,A)
        INCLUDE COND=(1,1,CH,EQ,X' FF')
/*
//SORT03 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=(OLD,DELETE),DSN=&&TEMP,
//          DCB=(RECFM=FB,LRECL=170,
//          BLKSIZE=0)
//SORTWK01 DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD DSN=DFRMM.CUSTOMER.CA1.DSNB1,DISP=(,CATLG,DELETE),
//          DCB=(RECFM=FB,LRECL=170,BLKSIZE=27880),UNIT=SYSDA,
//          SPACE=(27880,(900),RLSE,,ROUND)
//SYSOUT DD SYSOUT=*
//* -----
//*      DSNB DATA SET RECORD
//*      SORTED BY FIRST VOLSER AND FILE SEQUENCE
//*      INCLUDE ONLY DSNB RECORDS WITH DSNB USED INDICATOR
//* -----
//SYSIN DD *
        SORT FIELDS=(17,6,CH,A,29,2,BI,A)
        INCLUDE COND=(1,2,CH,EQ,X' FF80')
//

```

Figure 46 (Part 2 of 2). Sample DFSORT JCL

Note: In STEP03 each input record of 340 bytes will be split into two single 170 byte records. This means to split the left and right DSNB record into a single 170 byte length file.

The data extraction programs also convert installation-defined tape management policies (retention and outcode location movement) stored in the RDS and VPDD. In addition, the data extraction programs create empty bin records to represent unassigned bins (slots) in VMS-managed vaults (see Figure 60 on page 147).

The data extraction programs create a series of sequential output data sets that become the input to the EDGCNVT program. The output from EDGCNVT is used to build the DFSMSrmm CDS.

The data extraction jobs and programs provide the following functions:

- The EDGCxLDR program converts the TMC volume records and the DSNB records to L-Records and D-Records and creates K-Records for the special expiration dates supplied in the VRS management value (VRSMGMT DD)

data set. It can optionally create empty bin records. It also creates a file of keyword dates for input to EDGCSRDS.

- The EDGCxBIN program creates E-Records representing the empty bins in each of the vaults. When run with the CHECK execution parameter it can produce a list of OUTCODES you are using and classify them as STORES (VAULTS) and LOAN (all other outcodes). Use these EDGCxBIN created lists as input to EDGCxLDR and EDGCSVDS.
- The EDGCSRDS program converts the RDS retention policies to K-Records. EDGCSRDS also converts the keyword dates and RDS values into a table for use with EDGUX100.
- The EDGCSVDS program converts the VPDD movement policies to K-Records.

4.6.2 Extracting Data from the TMC Records Using EDGCxLDR

Two versions of the volume and data set data extraction programs are provided to accommodate the differences in the TMC volume records and the DSNB records between TMS/CA-1 Releases 4.9, 5.0, and 5.1:

- EDGC4LDR for TMS/CA-1 Release 4.9
- EDGC5LDR for TMS/CA-1 Release 5.0, 5.1, and 5.2

EDGCxLDR converts the TMC volume records to L-Records representing volumes and their first data sets and the DSNB records to D-Records representing the second and subsequent data sets. For each private volume, it validates the data set name on the volume. If it encounters an invalid data set name on a private volume (that is, a data set name beginning with x'00'), it writes an error message to the ERROR DD data set unless the volume is owned by an EDM identified through the EDMNAME DD data set. You must not proceed with the conversion process until you resolve the error conditions reported in the ERROR DD data set.

Optionally, EDGCxLDR creates K-Records that ultimately become DFSMSrmm VRSs. These K-Records are built from data supplied in the VRSMGMT DD data set and their names are placed in the VRS management value fields of the appropriate D-Records. You can also code VRSMV=YES in the VRSMGMT file instead of coding any keyword dates, and conversion processing will treat all 9998ddd and 9999ddd expiration dates as keyword dates. See Section 4.6.2.6, "VRSMGMT Management Value Data Set Record Format" on page 136 for detailed information.

EDGCxLDR calls EDGC1PRM to obtain R9=nnn, RC=nnn from the CA-1 PARMLIB (CA1.PPOPTION). K-records generated for keyword dates for catalog or cycles retention are built including a NEXTVRS(RC) or NEXTVRS(R9) and K-records generated for RC and R9 that specify EXTRADAYS COUNT(nnn). RC is extra days after cycles retention, and R9 is extra days after catalog retention.

EDGCxLDR can also verify the chaining integrity of data sets spanning multiple volumes, multiple data sets on a volume, and multiple data sets spanning multiple volumes. When EDGCxLDR detects a broken data set chain in the input data, it attempts to fix the error in working storage, and the attempt is reported in the ERROR DD data set. However, when EDGCxLDR detects a broken volume chain, it does not attempt to fix the error. TMSPTRS and TMSFVSN must be used. TMSDATA and EDGCxLDR must then be rerun until all chaining errors are removed.

If you currently use TMS 4.9 and have been specifying expiration dates into the next century, you need to undertake special activities to get the correct information to DFSMSrmm. CA-1 sets the expiration date to 99366 to prevent expiration, and it sets the correct 21st century date in the audit record it creates. During conversion, EDGC4LDR issues a message for each occurrence of 99366 that it encounters. You must obtain the expiration date from the audit records and use the RMM CHANGEVOLUME volser EXPDT subcommand to set the correct date once you have started DFSMSrmm with the CDS created from the conversion.

4.6.2.1 Sample EDGCxLRD JCL

Figure 47 shows sample execution JCL for TMS/CA-1 Release 4.9, and Figure 48 on page 130 shows sample execution JCL for TMS/CA-1 Release 5.0, Release 5.1, or Release 5.2. See SAMPLIB member EDGJ4LDR and EDGJ5LDR for the supplied sample JCL.

Note: To help identify some of the specifics about your current TMS/CA-1 environment, execute EDGCxBIN with PARM='CHECK'. This creates files of the storage and loan locations which can be input to other conversion programs.

EDGCxBIN externalizes the following:

- Outcode location names currently in use to help you define which outcode locations become DFSMSrmm storage locations and which become DFSMSrmm loan locations
- Volumes occupying outcode slot number zero, which is not compatible with DFSMSrmm
- Outcode slots with more than one volume assigned, which is also not compatible with DFSMSrmm.

```

/*JOBNAME  JOB
/*
/*PROPRIETARY V3 STATEMENT
/*LICENSED MATERIALS - PROPERTY OF IBM
/*"RESTRICTED MATERIALS OF IBM"
/*5695-DF1
/*(C) COPYRIGHT 1994  IBM CORP.
/*END PROPRIETARY V3 STATEMENT
/*
/* $02=spesloc,110,940519,MWW: LRECL for DEXTOUT increased to 512
/* $04=OW11151,110,941219,MWW: Implement macros
/* $03=OW10124,110,950109,MWW: match VRS MV names to EDGUX100
/* $S7=OW30969,140,971125,RS : VRS Enhancements @S7A
/* $K1=KVE0031,140,980223,OG : VRS Enh code review rework @K1A
/*
/* Sort by first VOLSER, volume sequence, and VOLSER
/*

```

Figure 47 (Part 1 of 3). Sample EDGC4LDR JCL: CA-1 Release 4.9


```

//SORTTMC EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(200,(50,10),RLSE),
//          DSN=&&SORTTMC,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(101,6,CH,A,55,2,BI,A,72,6,CH,A),DYNALLOC
/*
/* Sort by VOLSER which file 2 starts and file sequence number
/*
//SORTDSNB EXEC PGM=SORT,REGION=4M,COND=(0,NE)
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.DSNBRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(100,(50,10),RLSE),
//          DSN=&&SORTDSNB,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(2,6,CH,A,8,2,BI,A),DYNALLOC
/*
//DEL EXEC PGM=IEFBRI4
//D1 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDR.DEXTOUT.DATA
/*
/* The following step assembles and executes the supplied
/* EDGC4LDR assembler code from SAMPLIB. Alternatively
/* you can remove this step and use a load module you have
/* created into RMM.LOAD
/*
/* EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K1C
/* PARM.G='CONSOLIDATE/200', @K1A
// EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K1C@02C
// REGION.G=4M @S7A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC4LDR) @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB @K1C
// DD DISP=SHR,DSN=SYS1.MODGEN @K1C
//G.SYSUDUMP DD SYSOUT=* @K1M

```

Figure 47 (Part 2 of 3). Sample EDGC4LDR JCL: CA-1 Release 4.9

```

/**EDGC4LDR EXEC PGM=EDGC4LDR,REGION=4M,PARM='CONSOLIDATE/200'
/**EDGC4LDR EXEC PGM=EDGC4LDR,REGION=4M
/**STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/**SYSUDUMP DD SYSOUT=*
//REPORT DD SYSOUT=*
//ERROR DD SYSOUT=*
//STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
/**STORE DD * @K1C
/** cols 1-4 Current storage location name
/** 5-12 Owner Id (optional)
/** 13-20 New storage location name
//LOAN DD DISP=SHR,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA @K1C
/**LOAN DD * @K1C
/** cols 1-4 Loan location name
/** 5-12 Owner Id (optional)
//VRSMGMT DD *
99000D99000
99001D99001
99002D99002
99003D99003
98001D98001
98002D98002
98003D98003
/** cols 1-5 Special meaning expiration date
/** 6-13 VRS management value name
//EDMNAME DD *
DFHSM
/** cols 1-8 EDM job name
/** 9-16 EDM RACF ID (optional)
//TMC DD DISP=(OLD,DELETE),DSN=&&SORTTMC
//DSNB DD DISP=(OLD,DELETE),DSN=&&SORTDSNB
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700, @04C
// RECFM=VB,SPACE=(700,(200,100),RLSE),AVGREC=K, @04C
// DSN=STSGWD.EDGCXLDR.DEXTOUT.DATA

```

Figure 47 (Part 3 of 3). Sample EDGC4LDR JCL: CA-1 Release 4.9

```

/** Sort by first VOLSER, volume sequence, and VOLSER
/**
//SORTTMC EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS50.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(340,(50,10),RLSE),
// DSN=&&SORTTMC,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(57,6,CH,A,55,2,BI,A,1,6,CH,A),DYNALLO
/**
/** Sort by VOLSER which file 2 starts and file sequence number
/**

```

Figure 48 (Part 1 of 3). EDGC5LDR: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2

```

//SORTDSNB EXEC PGM=SORT,REGION=4M,COND=(0,NE)
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS50.DSNBRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(170,(50,10),RLSE),
//          DSN=&&SORTDSNB,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(17,6,CH,A,29,2,BI,A),DYNALLO
//*
//DEL EXEC PGM=IEFBRI4
//D1 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDL.DEXTOUT.DATA
//D2 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDL.VRSVALUE.DATA
//D3 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDL.DEXTOUTK.DATA
//*
//EDGC5LDR EXEC PGM=EDGC5LDR,REGION=4M
//STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//SYSUDUMP DD SYSOUT=*
//REPORT DD SYSOUT=*
//TMSPARM DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA
//VRSVALUE DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//            AVGREC=U,SPACE=(TRK,(1,1),RLSE),UNIT=SYSDA,
//            DSN=STSGWD.EDGCXLDL.VRSVALUE.DATA
//ERROR DD SYSOUT=*
//STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//*STORE DD *
//* cols 1-4 Current storage location name
//*      5-12 Owner Id (optional)
//*      13-20 New storage location name
//LOAN DD DISP=SHR,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*LOAN DD *
//* cols 1-4 Loan location name
//*      5-12 Owner Id (optional)
//VRSMGMT DD *
VRSMV=YES
//* cols 1-5 Special meaning expiration date
//*      6-13 VRS management value name
//EDMNAME DD *
DFHSM
//* cols 1-8 EDM job name
//*      9-16 EDM RACF ID (optional)
//TMC DD DISP=(OLD,DELETE),DSN=&&SORTTMC
//DSNB DD DISP=(OLD,DELETE),DSN=&&SORTDSNB
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700,
//            RECFM=VB,SPACE=(500,(200,100),RLSE),AVGREC=K,
//            DSN=STSGWD.EDGCXLDL.DEXTOUT.DATA
//DEXTOUTK DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700,
//            RECFM=VB,SPACE=(500,(200,100),RLSE),AVGREC=K,
//            DSN=STSGWD.EDGCXLDL.DEXTOUTK.DATA

```

Figure 48 (Part 2 of 3). EDGC5LDR: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2

```

/*
/* Step sorts the keyword dates in VRSVALUE file
/*
//SORTKEYS EXEC PGM=SORT,REGION=6M
//SYSPRINT DD  SYSOUT=*
//SORTIN  DD  DISP=OLD,DSN=STSGWD.EDGCXLD.R.VRSVALUE.DATA
//SORTOUT DD  DISP=OLD,DSN=STSGWD.EDGCXLD.R.VRSVALUE.DATA
//SYSOUT  DD  SYSOUT=*
//SYSIN   DD  *
SORT  FIELDS=(1,5,CH,A),DYNALLOC
END

```

Figure 48 (Part 3 of 3). EDGC5LDR: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2

4.6.2.2 Execution Parameters

Figure 49 shows the syntax of the execution parameters.

```

//EDGCxLDR EXEC PGM=EDGCxLDR,PARM=' x/y'

```

Figure 49. EDGCxLDR Execution Parameter Syntax

In the syntax, **x** is one of the following execution parameters:

- REASSIGN - requests that:
 - Volumes assigned to specific outcode location names have their names translated, name for name, to DFSMSrmm storage location names
 - Those volumes' assigned slot numbers be ignored and new bin numbers be generated.
- CONSOLIDATE - requests that:
 - Volumes residing in multiple outcode locations be combined into fewer RMM storage locations
 - Those volumes' slot numbers be ignored and new bin numbers be generated

and **y** is the number of empty bins requested. You must create empty bin records to provide space for RMM to move additional volumes to the storage locations after conversion. With this method you do not have to add the empty bin numbers using RMM ADDBIN commands.

Note: When CONSOLIDATE or REASSIGN is requested, EDGCxLDR builds the necessary empty bin records and thus eliminates the need to execute EDGCxBIN.

When creating the empty bin records EDGCxLDR sets the media name to CART3480. If you use subvaulting and plan to do the same under DFSMSrmm, you may have to modify EDGCxLDR to set the media names you want to use for each location. If CART3480 is not the value you want to use, update EDGCxLDR to set a suitable value for each location.

The execution options are provided to help in the following situations:

- Volumes assigned to slot 0:
 - If this is the only situation to be resolved, it is probably easier to force CA-1 to move the volumes to a nonzero slot in the outcode location.
- Multiple volumes per slot number:

If only a few slots are occupied by more than one volume, consider moving the necessary volumes to alleviate the situation.

- Total number of outcode locations to be converted to RMM storage locations is greater than the number you want to use under DFSMSrmm.

Attention: Do not use any of the preceding options if any outcode location depends on an accurate outcode slot number.

Figure 50 shows a sample consolidation of two outcode locations to LOCAL that also creates 100 empty bin records for each RMM storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='CONSOLIDATE/100'  
//STORE DD *  
VLT1 LOCAL  
VLT2 LOCAL  
VLT3 REMOTE  
VLT4 DISTANT
```

Figure 50. EDGCxLDR Execution Parameters: Sample 1

Figure 51 shows a sample that reassigns bin numbers, translates the outcode location names to RMM storage location names, and creates 200 empty bin records in each RMM storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='REASSIGN/200'  
//STORE DD *  
VLT1 LOCAL  
VLT2 REMOTE  
VLT3 DISTANT
```

Figure 51. EDGCxLDR Execution Parameters: Sample 2

Figure 52 shows a sample that reassigns bin numbers (without translation of outcode location names to storage location names). The translation of the outcode location names will have to be done later through EDGCNVT as shown in Figure 53 on page 134.

Note: EDGCNVT can neither consolidate nor ensure that not more than one volume will occupy any given bin.

The sample in Figure 52 also shows how to create 200 empty bin records in each storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='REASSIGN/00000200'  
//STORE DD *  
VLT1  
VLT2  
VLT3
```

Figure 52. EDGCxLDR Execution Parameters: Sample 3

```
//EDGCNVT EXEC PGM=EDGCNVT
//SYSIN DD *
IF STORLOC EQUALS VLT1 THEN RMMSTORE EQUALS LOCAL
IF STORLOC EQUALS VLT2 THEN RMMSTORE EQUALS REMOTE
IF STORLOC EQUALS VLT3 THEN RMMSTORE EQUALS DISTANT
```

Figure 53. EDGCNVT Storage Location Name Reassignment

Note: In Figure 53 the STORLOC keyword must start in column 4, and the outcode location name must start in column 19. The RMMSTORE keyword must start in column 33, and the DFSMSrmm storage location name must start in column 49.

4.6.2.3 Required Data

A number of fields have been set to default values within EDGCxLDR (Table 16). You have the option of modifying the defaults in the EDGCxLDR assembler source or waiting until the execution of the EDGCNVT program. It may be easier to wait until execution of the EDGCNVT program, when the changes can be driven through the SYSIN DD data set.

Table 16. EDGCxLDR Modifiable Fields

Field	Current Value	Description
HOMLOC	SHELF	Home location name
DFLTOWN	LIBRARY	Owner name
TAPE1600	TAPE1600	Media name
TAPE6250	TAPE6250	Media name
CART3480	CART3480	Media name
CART3490	CART3490	Media name

Note: The media names specified must match the RMM VLPOOL and LOCDEF definitions in the EDGRMMxx member of PARMLIB. If they do not match, errors will occur when DFSMSrmm is active.

Figure 54 shows an example of how to change the default media values to your installation definitions through the SYSIN DD data set for the EDGCNVT program. The sample shows how to change the pool names:

- From TAPE1600 to OLDREEL
- From TAPE6250 to REELS
- From CART3480 to SQUARE
- From CART3490 to BGSQUARE

```
IF VMEDIA EQUALS TAPE1600 THEN VLPOOLNAME EQUALS OLDREEL
IF VMEDIA EQUALS TAPE6250 THEN VLPOOLNAME EQUALS REELS
IF VMEDIA EQUALS CART3480 THEN VLPOOLNAME EQUALS SQUARE
IF VMEDIA EQUALS CART3490 THEN VLPOOLNAME EQUALS BGSQUARE
```

Figure 54. EDGCNVT SYSIN Control Statement Format for VMEDIA

Note: The VMEDIA keyword must start in column 4, and the value must start in column 18. The VLPOOLNAME must start in column 32, and the pool name must start in column 50.

Note: Any volumes defined in the TMC, but never used for data, are defined to TMS as tape reels. As a result, the conversion programs can convert a volume with the wrong media name value. You can correct this before conversion by updating the TMCDEN or TMCTRTCH fields, or during conversion by using the EDGCNVT SYSIN statement:
 IF VOLRANGETHEN MEDIANAME EQUALS

EDGC5LDR now uses the TMC last change user ID as the volume owner, and only uses DFLTOWN when there is no last change user ID.

Figure 55 shows an example of how to change the owner name from LIBRARY to OPERATOR and add more information relative to the default owner through the SYSIN DD data set for the EDGCNVT program.

```

IF OWNER EQUALS 'LIBRARY' THEN NEWOWNR EQUALS 'OPERATOR'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNS EQUALS 'DATA CENTER'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNF EQUALS 'COMPUTER ROOM'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWND EQUALS 'DEPARTMENT 999'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNT EQUALS '999-9999'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNU EQUALS 'OPERID '
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNN EQUALS 'SYSTEMA '

```

Figure 55. EDGCNVT SYSIN Control Statement Format for OWNER

Note: The OWNER keyword must start in column 4, and the value must start in column 18. The definition keywords must start in column 32, and their values must start in column 49.

4.6.2.4 Input Data Sets

EDGCxLDR has six input data sets (Table 17).

Table 17. EDGCxLDR Input Data Sets		
ddname	Description	Explanation
TMC	TMC volume record data set (sorted)	Required
DSNB	DSNB data set (sorted)	Required
EDMNAME	EDM job name	Optional
VRSMGMT	VRS management value	Optional
STORE	Outcode location and storage location name	Optional (unless outcodes are used and will become DFSMSrmm storage locations)
LOAN	Outcode location and loan location name	Optional (unless outcodes are used that are not already mapped to storage locations)
TMSPARM	The TMS PARMLIB options member	Optional

Note: If you do not use an optional input data set, you must specify DD DUMMY for it.

For TMC and DSNB records, first you must use the vendor-provided utility programs, TMSPTRS (9308 or newer level) and TMSFVSN, to verify and ensure the integrity of the TMS database (for example, no broken volume or data set chains). Do not rely on EDGCxLDR to fix the broken chains. The existence of broken chains *may not* be a problem during “dry runs” when you are only

testing the conversion process to gain experience and do not plan to use the converted database in production mode.

You then use TMSDATA or, if TMSDATA is not available, DFSORT (see Figure 46 on page 125) to offload the TMC database to two data sets:

- Sequential data set containing the TMC volume records
- Sequential data set containing the DSNB records

Sort the sequential TMC volume data set and DSNB data set as follows (see sample JCL and sort control statements in Figure 48 on page 130):

- Sort the TMC volume records by the first VOLSER of the volume set, the volume sequence number, and the volume VOLSER in ascending order.
- Sort the DSNB records by the VOLSER on which file 2 starts, and the file sequence number in ascending order.

4.6.2.5 EDMNAME Identification Data Set Record Format

Columns	Field Description
---------	-------------------

1-8	EDM job name
-----	--------------

9-16	Owner ID
------	----------

17-80	Comments
-------	----------

Figure 56 shows a sample of how to identify EDM job names.

```
//EDMNAME DD *  
DFHSM SYSOPER job name is DFHSM and owner id is SYSOPER  
/* cols 1-8 EDM job name  
/* 9-16 Default volume owner (optional)
```

Figure 56. EDGCxLDR: EDM Identification Sample

The maximum number of EDM job names that can be identified through the EDMNAME DD data set is 20, as EDGCxLDR is originally coded. If this number is too small, you must alter the EDGCxLDR program to increase it.

Note: Each EDM managed volume will be converted to USER status; this indicates that it is a private volume which can be overwritten by any data set. If you do not identify EDM job names, the EDM volumes are converted as normal volumes. If any of these are multi-file volumes, but with only the first file recorded in TMS, you should convert them as EDM volumes, or else DFSMSrmm will only allow file 1 to be used.

4.6.2.6 VRSMGMT Management Value Data Set Record Format

Columns	Field Description
---------	-------------------

1-5	Expiration date
-----	-----------------

6-13	One- to eight-character VRS management value name
------	---

14-80	Comments
-------	----------

Figure 57 on page 137 shows a sample of how to create VRS management value names. Any keyword dates specifically included in VRSMGMT can be defined with a VRS management value to use instead of building one using this default method. In our sample we show an expiration date of 99000 and a management value name WHILECAT.


```
//VRSMGMT DD *
VRSMV=YES
99000WHILECAT While catalog retention
99001D99001 1 cycle/copy retention
99002D99002 2 cycles/copies retention
99003D99003 3 cycles/copies retention
98001D98001 1 day since last referenced
98002D98002 2 days since last referenced
98003D98003 3 days since last referenced
//* cols 1-5 Special meaning expiration date
//* 6-13 VRS management value
```

Figure 57. EDGCxLDR: VRS Management Value Name Sample

The maximum number of VRS management value names supported by EDGCxLDR is 1000. If this number is too small, you must alter EDGCxLDR to increase it.

Note: The VRS management value names specified in the VRSMGMT DD data set are the names that your EDGUX100 exit should pass to DFSMSrmm when it executes once you have DFSMSrmm in record, warning, or protect mode.

If you plan to use any VRS management values, you must install the EDGUX100 exit.

If VRSMV=YES is not specified, only those keyword dates specified in VRSMGMT DD are converted to VRS management values. This enables you to move away from keyword dates.

If VRSMV=YES is specified, all keyword dates are converted, and by default the VRS management value name is determined by the conversion programs, unless you also specify the keyword date and a management value to use. The conversion tools use some special VRS naming formats based on the keyword date value. The names are similar to those used by TMS:

- Catalog control EXPDT=99000, uses 'CATALOG'
- Cycle control EXPDT=99ddd, uses 'CYCLEddd'
- Last reference control EXPDT=98ddd, uses 'LDATEddd'
- Catalog and days control EXPDT=90ddd, uses 'CATLGddd' with a NEXTVRS(CATALOG)

All management values used are written as K records, and written to a new file VRSVALUE, which is input to EDGCSRDS processing.

4.6.2.7 STORE Location Name Data Set Record Format

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (optional)
13-20	Storage location name (optional)
21-80	Comments

Figure 58 on page 138 shows a sample of how to define outcode and storage location names.

```
//STORE DD *
ARC LOCAL
DIS ownerid DISTANT
/* You must use DD DUMMY if no data is supplied
/* cols 1 - 4 Outcode location name
/* 5 - 12 Default volume owner (optional)
/* 13 - 20 Storage location name (optional)
```

Figure 58. EDGCxLDR: STORE DDNAME Sample

EDGCxLDR is coded to a maximum number of 1000 outcode location names to be translated to DFSMSrmm's storage locations. If this number is too small, you must alter EDGCxLDR to increase it. Use EDGCxBIN with the CHECK option to generate the list of vaults to convert as storage locations. You can customize the list as required.

4.6.2.8 LOAN Location Name Data Set Record Format

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (optional)
12-20	Loan location name (optional)
21-80	Comments

Figure 59 shows a sample of how to define loan location names.

```
//LOAN DD *
BANK Loan Location for Payroll Tapes
MFCH Loan Location for Tapes sent to Microfiche Vendor
/* You must use DD DUMMY if no data is supplied
/* cols 1 - 4 Outcode location name
/* 5 - 12 Default volume owner (optional)
/* 12 - 20 Loan location name (optional)
```

Figure 59. EDGCxLDR: LOAN DDNAME Sample

You can have many DFSMSrmm loan location names, and the names are not mandated by DFSMSrmm. Therefore, the outcode location names are used as DFSMSrmm loan location names when converting outcodes to loan locations unless a loan location name is specified.

The maximum number of outcode location names that can be specified to be set to RMM loan locations by EDGCxLDR is 200. If this number is too small, you must alter EDGCxLDR to increase it. Use EDGCxBIN with the CHECK option to generate the list of vaults to convert as loan locations. You can customize the list as required.

4.6.2.9 Output Data Sets

EDGCxLDR creates output data sets containing the converted volumes and data set records, an error report file, and a warning report file (Table 18 on page 139).

<i>Table 18. EDGCxLDR Output Data Sets</i>	
ddname	Description
DEXTOUT	L- and D-Records data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
DEXTOUTK	K-Records data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Error report data set
REPORT	Warning report data set
VRSVALUE	Data set containing all created management values K-records. This data set is input to EDGCSRDS processing.

4.6.2.10 Return Codes

EDGCxLDR sets a return code when it ends (Table 19).

<i>Table 19. EDGCxLDR Return Codes</i>	
Return Code	Description
0	Normal processing
4	- A broken DSNB chain has been encountered and an attempt has been made to fix it. - A VRS management value name in the VRSMGMT DD data set begins with a blank character and is therefore unusable. - A Special Meaning TMC EXPDT was encountered that does not have a corresponding statement in the VRSMGMT DD data set. Permanent retention is forced in the Expiration Date field of the L-Record - A Special Meaning TMC EXPDT was encountered that does not conform to the format 98xxx or 99xxx. Permanent retention is forced in the Expiration Date Field of the L-Record. - A TMC outcode of blanks (x'40404040') was encountered: - If the slot assigned was zero, the current location in the L-Record is forced to SHELF. - If the slot assigned was nonzero, an error message is written stating that the outcode name was blank. The current location in the L-Record is forced to SHELF. - A TMC EXPDT has a format error.
8	There are extra DSNB records for a volume; that is, there are actually more DSNB records than indicated in the TMC volume record's DSNB count field.
12	Too many keyword dates were in use in the TMC. A table for 1000 maximum entries is used by EDGC5LDR. If more are used, the table size must be increased and EDGC5LDR re-assembled.
16	- A broken TMC volume chain has been encountered. - There are missing DSNB records; that is, fewer DSNB records were found than it was indicated in the TMC volume record's DSNB count field. - A private (active) volume TMC record contains an invalid data set name. - The slot number assigned to one or more volumes that is being translated to storage location is zero. - A STORE DD control statement has no storage location name specified.

4.6.2.11 Messages

UNABLE TO PROCESS KEYWORD DATE yyddd - VRS TABLE IS FULL

Severity: ERROR

Explanation: Too many keyword dates were processed.

System Action: Sets return code 12

User Response: Update the EDGC5LDR source code to use a value larger than 1,000 for the number of keyword dates supported, reassemble the updated code and re-run the job.

EXTRA DAYS AFTER CATALOG RETENTION VRS CREATED

Severity: INFORMATION

Explanation: The R9 VRS has been created based on the TMS R9 PARMLIB option.

User Response: None

EXTRA DAYS AFTER CYCLES RETENTION VRS CREATED

Severity: INFORMATION

Explanation: The RC VRS has been created based on the TMS RC PARMLIB option.

User Response: None

EMPTY BIN COUNT GREATER THAN 99999999

Severity: ERROR

Explanation: You have requested to create more than 99999999 empty bins in this run. When CONSOLIDATE or REASSIGN is requested, you can specify how many empty bin records should be added to the storage locations.

System Action: None

User Response: Specify a value between 1 and 99999999 and rerun the job.

Destination: PARMERR

VRS MGMT VALUE DATE ERROR: *input card*- RC=4

Severity: ERROR

Explanation: EXPDT must be either 98xxx or 99xxx - check DDNAME(VRSMGMT)

System Action: None

User Response: You can only specify a VRS management value either 98xxx or 99xxx. For the 90000 date you have to specify 99000, because the program will change the date to 90000 internally.

The maximum number of VRS management value names supported by EDGC5LDR is 1,000. Rerun the job.

Destination: VRSERR

VRS MGMT VALUE DATE ERROR: *input card* - RC=4

Severity: Warning

Explanation: VRS management name must begin in column 6 - check DDNAME(VRSMGMT).

System Action: None

User Response: Specify your one- to eight-character VRS management value name in column 6 in the VRSMGMT DDNAME and rerun the job.

Destination: VRSERR2

NEW STORE NAME MISSING FOR CONSOLIDATE OPTION, CHECK DDNAME(STORE) - RC=16

Severity: ERROR

Explanation: Also check EXEC PARM if 'CONSOLIDATE' option is specified.

System Action: None

User Response: Specify your new one- to eight-character storage location name in column 13 in the STORE DDNAME and rerun the job.

Destination: STORERR

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff). RC = 8

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: ERROR

Explanation: An additional DSNB record was found for volume vvvvvv.

System Action: None

User Response: Check your CA-1 TMC and rerun your job.

Destination: XDSNBCK

vvvvvv ddd.ddd -> MULTI-VOLUMES BROKEN CHAIN ERROR - RC = 16

where:

vvvvvv is the volume serial number
dddddd is the data set name

Severity: ERROR

Explanation: Check if the volume being processed is the next volume in the multi-volumes chain.

System Action: None

User Response: Correct your CA-1 TMC and rerun the job.

Destination: CHAINCK

PREVIOUS VOLSER (pppppp) AND NEXT VOLSER (nnnnnn) DSNB COUNT IS ZERO BUT DSNB RECORD FOUND

where:

vvvvvv volume serial number
pppppp previous volume serial number of data set
nnnnnn next volume serial number of data set

Severity: ERROR

Explanation: Check if the volume being processed is the next volume in the multi-volumes chain and the number of data sets is greater than one.

System Action: None

User Response: Correct your CA-1 TMC and rerun the job.

Destination: CHAINCK

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff) AND EDGC5LDR FIXED IT.

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: Warning

Explanation: None
System Action: None
User Response: NONE
Destination: XTRAERR

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff).

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: ERROR
Explanation: The extra DSNB record was ignored.
System Action: None
User Response: Correct your CA-1 TMC and rerun the job.
Destination: DSNXTRA

VOLSER WITH DSNBVOL(vvvvvv) AND DSNBFVOL(dddddd) HAS A BROKEN DSNB CHAINED, AND EDGC5LDR FIXED IT.

where:

vvvvvv volume serial number
dddddd volume serial number of the extra DSNB record was found

Severity: Warning
Explanation: None
System Action: None
User Response: None
Destination: CONTFIX

SLOT NUMBER 0000 WAS ASSIGNED TO VOLSER vvvvvv STORED IN //// DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM - RC=16

where:

vvvvvv volume serial number
bbbbbb storage location bin number
11111111 storage location name

Severity: ERROR
Explanation: A volume assigned to a storage location was found, but the bin number (SLOT) is 0000.
System Action: None
User Response: Correct the error in the CA-1 TMC and rerun the job.
Destination: BIN0

0000 STORE WAS NOT DEFINED IN STORE DD OR LOAN DD. VOLSER vvvvvv IS MARKED AS BEING STORED IN THE LIBRARY (CURRENT LOCATION IS BEING FORCED TO SHELF). RC=4

where:

0000 name of the out code or vault location
vvvvvv volume serial number

Severity: Warning
Explanation: A volume assigned to an out-of-area location or a vault location was found, but the location name was either specified in the STOR DD-statement or the LOAN DD-statement.
System Action: None

User Response: Add additional information to the STORE DD-statement or to the LOAN DD-statement.

Destination: PUTLOCK

**0000 STORE WAS NOT DEFINED IN STORE DD OR LOAN DD
VOLSER vvvvvv WILL BE MARKED UNDER MANUAL MOVE CONTROL BY EDGCNVT -
CURRENT LOCATION SET TO OUTCODE. RC=4**

where:

0000 name of the out code or vault location
vvvvvv volume serial number

Severity: Warning

Explanation: A volume assigned to an out-of-area location or a vault location was found, but the location name was either specified in the STOR DD-statement or the LOAN DD-statement.

System Action: None

User Response: Add additional information to the STORE DD-statement or to the LOAN DD-statement. Or you can check the result after the conversion is finished and change the MOVEMODE to AUTO.

Destination: LIBMSG

**VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.
* CONVERTED AS NON-EDM VOLUME - RC=16**

where:

vvvvvv volume serial number
jjjjjjjj creating job name

Severity:

Explanation: The out-of-area information was ignored.

System Action: None

User Response: Correct your CA-1 TMC and rerun the job.

Destination: RESVOL

*** VOLSER vvvvvv CONTAINS INVALID DSNAME (IE. 1ST BYTE IS X'00'); dsname
* DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM
RC=16.**

where:

vvvvvv volume serial number
dsname data set name
jjjjjjjj creating job name

Severity: ERROR

Explanation: An invalid data set name dsname on volume vvvvvv was found and the volume status was not scratch.

System Action: None

User Response: Correct your CA1- TMC and rerun the job.

Destination: INVNAME1

**VOLSER: vvvvvv DSN: dsname
NO VRS MGMT VALUE SPECIFIED FOR DATE: ddddd CHECK DDNAME (VRSMGMT)
THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4**

where:

vvvvvv volume serial number
dsname data set name
dddddd VRS management value

Severity: Warning

Explanation: A volser with a special meaning expiration date was found, but the expiration date was not specified in the VRSMGMT DD-statement.

System Action: None

User Response: Add the missing VRS management value to the VRSMGMT DD-statement.

Destination: PUTEVRS

AN INVALID SPECIAL EXPIRATION DATE OF xxxxxxxx HAS BEEN ENCOUNTERED FOR:
dddddd

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

xxxxxxx	special meaning expiration date
vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: You have specified a special meaning expiration date outside the range.

System Action: None

User Response: You can only specify VRS management value names in the range 98ddd and 99ccc.

Destination: PUTCVMSG

NO VRS MGMT VALUE SPECIFIED FOR DATE: ddddd CHECK DDNAME (VRSMGMT)

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: A DSNB record with a special meaning expiration date was found, but the expiration date was not specified in the VRSMGMT DD-statement.

System Action: None

User Response: Add the missing VRS management value to the VRSMGMT DD-statement

Destination: PUTVRSE

AN INVALID SPECIAL EXPIRATION DATE OF xxxxxxxx HAS BEEN ENCOUNTERED FOR:
dddddd

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

xxxxxxx	special meaning expiration date
vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: You have specified a special meaning expiration date outside the range.

System Action: None

User Response: You can only specify VRS management value names in the range 98ddd and 99ccc.

Destination: PUTCDMSG

*** VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.**
*** DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM**
RC=16.

where:

vvvvvv	volume serial number
dsname	data set name
jjjjjjjj	creating job name

Severity:

Explanation: Read the previous message.

System Action: None

User Response: Correct the error in the CA-1 TMC and rerun the job.

Destination: PUTEND

*** VOLSER vvvvvv CONTAINS INVALID DSNAME (IE. 1ST BYTE IS X'00'); dsname ***
RESVOL+14 DSNBCJOB
*** VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.**
*** DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM**
RC=16.

where:

vvvvvv	volume serial number
dsname	data set name
jjjjjjjj	creating job name

Severity: ERROR

Explanation: An invalid data set name dsname on volume vvvvvv was found and the volume status was not scratch.

System Action: None

User Response: Correct your CA1- TMC and rerun the job.

Destination: INVNAME2

vvvvvv PREVIOUS VOLSER pppppp (N/A) AND NEXT VOLSER nnnnnn (N/A)
-> DSNB RECORD MISSING OR INCOMPLETE DSNB CHAINED RC=16

where:

vvvvvv	volume serial number
pppppp	volume serial number for the previous volume in the chain
nnnnnn	volume serial number for the next volume in the chain

Severity: ERROR

Explanation: The previous and next volume information not found.

System Action: None

User Response: Correct the error in the CA-1 TMC and rerun the job. (You can use the TMSPTRS CA-1 utility to fix the problem.)

Destination: DSNERR

BIN (SLOT) NUMBER WAS RESEQUENCED, STARTING WITH 000001
*** DUE TO 'REASSIGN' OR 'CONSOLIDATE' OPTION WAS SELECTED**
RC=4

Severity: Warning

Explanation: You have selected the REASSIGN or CONSOLIDATE execution parameter, and the conversion program has changed some of the slot numbers.

System Action: None

User Response: You must create empty bin records to provide space for RMM to move additional volumes to the storage locations after conversion.

Destination: CLOSE

4.6.3 Creating the Empty Bin Records Using EDGCxBIN

Attention: The empty bin program, EDGCxBIN, should not be run, other than with the CHECK parameter, if the reassign or consolidate option is used in the execution of EDGCxLDR. The necessary empty bin records (E-Records) are provided by EDGCxLDR when the reassign or consolidate parameter is used.

Two versions of the empty bin program are provided to accommodate the differences in the TMC volume records between TMS/CA-1 Releases 4.9, and 5.0 and 5.1:

- EDGC4BIN for TMS/CA-1 Release 4.9
- EDGC5BIN for TMS/CA-1 Release 5.0, Release 5.1, and Release 5.2.

DFSMSrmm keeps track of all of the bins (slots) that exist in each of its shelf-managed storage locations. Each bin has a status of either “in-use” or “empty”. TMS/CA-1 only keeps track of the in-use slots. Therefore, the purpose of EDGCxBIN is to build E-Records that represent the empty DFSMSrmm bins by recognizing gaps in the CA-1 in-use slot locations. It also verifies the slot numbers and the outcode location names. If it encounters slot number 0 or multiple volumes in a single slot, it writes an error message to the ERROR DD data set.

Note: DFSMSrmm does not support a bin number of 0. The lowest bin number is 1. DFSMSrmm does not allow more than one volume to be assigned to a given bin in a DFSMSrmm storage location.

If the outcode location name in the TMC volume record does not match one of the outcode location names specified in the STORE DD data set and the volume has a vault slot number assigned, EDGCxBIN writes an error message to the ERROR DD data set.

An execution parameter option is provided to allow the EDGCxBIN program to report all of the outcode locations and the in-use and empty slots in each outcode location. See Section 4.6.3.3, “Execution Parameter” on page 154 for more information.

4.6.3.1 Planning

Before executing EDGCxBIN, you must review the preprocessing considerations described in Section 4.6.1, “Preparing to Run the Extract” on page 124. If you decide to reassign bin numbers and/or consolidate outcode locations through EDGCxLDR, do not execute EDGCxBIN without the CHECK parameter.

EDGCxBIN requires that the records in the SLOTNO input file be sorted by outcode and slot number. This is achieved in the sort step. The sort step is also used to include only those records required for each type of media; the volumes are selected as either tape reels or tape cartridges. When running with the CHECK parameter you can pass all volume records for processing. When creating the empty bin records, without the CHECK parameter, as long as all vaults are named in the STORE DD data set, you can also pass all volume records. If you want to process only part of the vaults, use the SORT INCLUDE statement to select only the correct volume records by vault name.

If you have both tape reels and tape cartridges being sent to vault outcodes, CA-1 automatically subvaults the tape cartridges. DFSMSrmm can handle this using the MEDIANAME value, but you must run EDGCxBIN twice; once to process tape reels, and once to process cartridges. The sample EDGJxBIN does this. You can separate the reels and cartridges at sort time by using the TMC DEN field. If you do not have both types of media you need only execute EDGCxBIN once.

4.6.3.2 Sample JCL

Figure 60 shows sample execution JCL for TMS/CA-1 Release 4.9; and Figure 61 on page 151, shows sample execution JCL for TMS/CA-1 Release 5.0, Release 5.1 and Release 5.2. Sample JCL is supplied with DFSMSrmm as members EDGJ4BIN and EDGJ5BIN of SAMPLIB.

```

/*JOBNAME  JOB
/*
/*PROPRIETARY V3 STATEMENT
/*LICENSED MATERIALS - PROPERTY OF IBM
/*"RESTRICTED MATERIALS OF IBM"
/*5695-DF1
/*(C) COPYRIGHT 1994,1995  IBM CORP.
/*END PROPRIETARY V3 STATEMENT
/*
/* $01=OW06598,110,940712,MWW: Do bin checking based on type
/*      of media; split into 3420 and 3480 styles.
/* $02=OW11151,110,941219,MWW: Implement Macros
/* $S7=OW30969,140,971125,RS : VRS Enhancements           @S7A
/* $K1=KVE0031,140,980223,OG : VRS Enh code review rework  @K1A
/* $K2=KVE0066,140,980429,RS : Remove 2nd exec of HLASMCLG Proc @K2A
/*
/* This Job is setup to use CHECK option for EDGC4BIN
/* When running without the CHECK option remember to remove the
/* LOANOUT and STOREOUT DD statements and provide valid STORE DDs
/*
//DEL      EXEC PGM=IEFBRI4
//DEXTOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.LOANOUT.DATA

```

Figure 60 (Part 1 of 4). EDGC4BIN: Sample JCL, CA-1 Release 4.9

```

/**
/**  SORT BY OUTCODE AND VAULT SLOT NUMBER
/**  INCLUDING ONLY VALID STORAGE LOCATION NAME
/**  FIRST SORT SELECTS JUST 3480 STYLE VOLUMES
/**
//SORT      EXEC PGM=SORT,REGION=4M
//SYSOUT    DD  SYSOUT=*
//SORTIN    DD  DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT   DD  DISP=(,PASS),AVGREC=K,SPACE=(200,(3,3),RLSE),
//            DSN=&&SLOT34,UNIT=SYSDA
//SYSIN     DD  *
SORT FIELDS=(137,4,CH,A,53,2,BI,A),DYNALLO
* INCLUDE just the cartridge media
INCLUDE COND=((64,1,CH,EQ,X'E3',OR,64,1,CH,EQ,X'E7'))    3480/90
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
*       Only use the following if a subset of vaults are to be used
* INCLUDE COND=((137,4,CH,EQ,C'ARCS',OR,137,4,CH,EQ,C'VLT1',
*               OR,137,4,CH,EQ,C'OUTS'),AND,
*               (64,1,CH,EQ,X'E3',OR,64,1,CH,EQ,X'E7'))    3480/90
/**
/** OPTION: PARM=CHECK
/**  Use this option to quickly check what vaults are in use
/**  and whether there are any potential problems.
/**
/** The following step assembles and executes the supplied
/** EDGC4BIN assembler code from SAMPLIB. Alternatively
/** you can remove this step and use a load module you have
/** created into RMM.LOAD
/**
//G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',PARM.G='CHECK',    @K2C
//*G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',                @K2C
//      REGION.G=4M                                           @S7A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC4BIN)              @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB                        @K1C
//      DD DISP=SHR,DSN=SYS1.MODGEN                          @K1C
//G.SYSUDUMP DD SYSOUT=*                                       @K1M
/**EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE),PARM='CHECK'
/**EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE)
/**STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/**SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34

```

Figure 60 (Part 2 of 4). EDGC4BIN: Sample JCL, CA-1 Release 4.9

```

//DEXTOUT DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
//* COLS 1-4 STORAGE LOCATION NAME
//*      5-12 OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
//*
//* SORT BY OUTCODE AND VAULT SLOT NUMBER
//* INCLUDING ONLY VALID STORAGE LOCATION NAME
//* 2ND SORT SELECTS JUST 3420 STYLE VOLUMES
//*
//SORT EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(200,(3,3),RLSE),
//          DSN=&&SLOT34,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(137,4,CH,A,53,2,BI,A),DYNALLO
* INCLUDE just the tape reel media
INCLUDE COND=((64,1,CH,NE,X' E3',AND,64,1,CH,NE,X' E7')) -3480/90
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
* Only use the following if a subset of vaults are to be used
* INCLUDE COND=((137,4,CH,EQ,C' ARCS',OR,137,4,CH,EQ,C' VLT1',
* OR,137,4,CH,EQ,C' OUTS'),AND,
* (64,1,CH,NE,X' E3',AND,64,1,CH,NE,X' E7')) -3480/90
//*
//* OPTION: PARM=CHECK
//* Use this option to quickly check what vaults are in use
//* and whether there are any potential problems.
//*
//* The following step executes the previously assembled and @K2C
//* linked EDGC4BIN assembler code from SAMPLIB. @K2C
//* Alternatively you can remove this step and use a @K2C
//* load module you have created into RMM.LOAD @K2C
//*

```

Figure 60 (Part 3 of 4). EDGC4BIN: Sample JCL, CA-1 Release 4.9

```

//G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M,PARM=' CHECK' @K2C
//*G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M @K2C
//*EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE),PARM=' CHECK'
//*EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE)
//*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//*SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
//DEXTOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
/* COLS 1-4 STORAGE LOCATION NAME
/* 5-12 OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
/* Step removes duplicate names created by running two CHECK steps.
//SORTSTR EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/* Step removes duplicate names created by running two CHECK steps.
//SORTLOAN EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END

```

Figure 60 (Part 4 of 4). EDGC4BIN: Sample JCL, CA-1 Release 4.9

```

/**JOBNAME JOB
/**
/**PROPRIETARY V3 STATEMENT
/**LICENSED MATERIALS - PROPERTY OF IBM
/**"RESTRICTED MATERIALS OF IBM"
/**5695-DF1
/**(C) COPYRIGHT 1994,1995 IBM CORP.
/**END PROPRIETARY V3 STATEMENT
/**
/** $01=OW06598,110,940712,MWW: Do bin checking based on type
/** of media; split into 3420 and 3480 styles.
/** $02=OW11151,110,941219,MWW: Implement macros
/** $03=OW14495,120,950727,MWW: Add TMSDATA replacement
/** $04=OW27659,120,970612,MWW: Improve carts/reel selection @04A
/** $S7=OW30969,140,971125,RS : VRS Enhancements @S7A
/** $K1=KVE0031,140,980223,OG : VRS Enh code review rework @K1A
/** $K2=KVE0066,140,980429,RS : Remove 2nd exec of HLASMCLG Proc @K2A
/**
/**
/** This Job is setup to use CHECK option for EDGC5BIN
/** When running without the CHECK option remember to remove the
/** LOANOUT and STOREOUT DD statements and provide valid STORE DDs
/**
//DEL EXEC PGM=IEFBR14
//DEXTOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
/**
/** SORT BY OUTCODE AND VAULT SLOT NUMBER
/** INCLUDING ONLY VALID STORAGE LOCATION NAME
/** FIRST SORT SELECTS JUST 3480 STYLE VOLUMES
/**
//SORT EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS51.VOLRECS.DATA @04C
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(340,(3,3),RLSE),
// DSN=&&SLOT34,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(109,4,CH,A,113,4,BI,A),DYNALLOC
* INCLUDE just the cartridge media
INCLUDE COND=((90,1,CH,EQ,X'E3', 38K BPI @04C
OR,90,1,CH,EQ,X'E7', 38K BPI IDRC @04A
OR,89,1,CH,EQ,X'C0', 3480 18TRK @04A
OR,89,1,CH,EQ,X'E0', 3480 36TRK @04A
OR,89,1,CH,EQ,X'E1')) 3490 E 36TRK @04A

```

Figure 61 (Part 1 of 4). EDGC5BIN: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2.

```

* INCLUDE selected vaults by name ARCS VLT1 and OUTS
*       Only use the following if a subset of vaults are to be used
* INCLUDE COND=((109,4,CH,EQ,C' VLT1',OR,
*               109,4,CH,EQ,C' ARCS',OR,
*               109,4,CH,EQ,C' OUTS'),AND,
*               (90,1,CH,EQ,X'E3',
*               OR,90,1,CH,EQ,X'E7',
*               OR,89,1,CH,EQ,X'CO',
*               OR,89,1,CH,EQ,X'E0',
*               OR,89,1,CH,EQ,X'E1'))
*               38K BPI @04C
*               38K BPI IDRC @04A
*               3480 18TRK @04A
*               3480 36TRK @04A
*               3490 E 36TRK @04A
**
**/* OPTION: PARM=CHECK
**/* Use this option to quickly check what vaults are in use
**/* and whether there are any potential problems.
**/*
**/* The following step assembles and executes the supplied
**/* EDGC5BIN assembler code from SAMPLIB. Alternatively
**/* you can remove this step and use a load module you have
**/* created into RMM.LOAD
**/*
**G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',PARM.G='CHECK', @K2C
**/*G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K2C
**      REGION.G=4M @S7A
**C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC5BIN) @S7C
**C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB @K1C
**      DD DISP=SHR,DSN=SYS1.MODGEN @K1C
**G.SYSUDUMP DD SYSOUT=* @K1M
**/*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE),PARM='CHECK'
**/*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE)
**/*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
**/*SYSUDUMP DD SYSOUT=*
**ERROR DD SYSOUT=*
**SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
**DEXTOUT DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
**          AVGREC=K,SPACE=(512,(1,1),RLSE),UNIT=SYSDA,
**          DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
**STOREOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
**          AVGREC=K,SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
**          DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
**LOANOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
**          AVGREC=K,SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
**          DSN=STSGWD.EDGCXBIN.LOANOUT.DATA

```

Figure 61 (Part 2 of 4). EDGC5BIN: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2.


```

/**STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
/**STORE DD * @K1C
/** COLS 1-4 STORAGE LOCATION NAME
/** 5-12 OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
/**
/** SORT BY OUTCODE AND VAULT SLOT NUMBER
/** INCLUDING ONLY VALID STORAGE LOCATION NAME
/** 2ND SORT SELECTS JUST 3420 STYLE VOLUMES
/**
/**SORT EXEC PGM=SORT,REGION=4M
/**SYSOUT DD SYSOUT=*
/**SORTIN DD DISP=SHR,DSN=STSGWD.TMS51.VOLRECS.DATA @04C
/**SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(340,(3,3),RLSE),
// DSN=&&SLOT34,UNIT=SYSDA
/**SYSIN DD *
SORT FIELDS=(109,4,CH,A,113,4,BI,A),DYNALLO
* INCLUDE just the tape reel media
INCLUDE COND=((90,1,CH,NE,X'E3', -38K BPI @04C
AND,90,1,CH,NE,X'E7', -38K BPI IDRC @04A
AND,89,1,CH,NE,X'CO', -3480 18TRK @04A
AND,89,1,CH,NE,X'EO', -3480 36TRK @04A
AND,89,1,CH,NE,X'E1')) -3490 E 36TRK @04A
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
* Only use the following if a subset of vaults are to be used
* INCLUDE COND=((109,4,CH,EQ,C'VLT1',OR,
* 109,4,CH,EQ,C'ARCS',OR, X
* 109,4,CH,EQ,C'OUTS'),AND,
* (90,1,CH,NE,X'E3', -38K BPI @04C
* AND,90,1,CH,NE,X'E7', -38K BPI IDRC @04A
* AND,89,1,CH,NE,X'CO', -3480 18TRK @04A
* AND,89,1,CH,NE,X'EO', -3480 36TRK @04A
* AND,89,1,CH,NE,X'E1')) -3490 E 36TRK @04A
/**
/** OPTION: PARM=CHECK
/** Use this option to quickly check what vaults are in use
/** and whether there are any potential problems.
/**
/** The following step executes the previously assembled and @K2C
/** linked EDGC5BIN assembler code from SAMPLIB. @K2C
/** Alternatively you can remove this step and use a @K2C
/** load module you have created into RMM.LOAD @K2C
/**

```

Figure 61 (Part 3 of 4). EDGC5BIN: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2.

```

//G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M,PARM=' CHECK' @K2C
//*G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M @K2C
//*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE),PARM=' CHECK'
//*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE)
//*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//*SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
//DEXTOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
/* Step removes duplicate names created by running two CHECK steps.
//SORTSTR EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/* Step removes duplicate names created by running two CHECK steps.
//SORTLOAN EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/*

```

Figure 61 (Part 4 of 4). EDGC5BIN: Sample JCL, CA-1 Release 5.0, 5.1, and 5.2.

4.6.3.3 Execution Parameter

Table 20 describes the optional execution parameter of EDGCxBIN program.

Table 20. EDGCxBIN Execution Parameter	
Parameter	Description
CHECK	Use the CHECK parameter to identify all of your outcodes and produce input for the EDGCxLDR program. A pass is made through the TMC volume records to identify: • All of the outcode locations in use and classify them as either store or loan type. See 4.6.3.5, “Output Data Sets” on page 155 for the information that can be generated. • Each occurrence of slot zero’s use. • Each occurrence of more than one volume in a given slot within a given outcode location. • A summary of in-use and empty slot numbers for each outcode location.

Note: The STORE DD data set is not read when you use the CHECK parameter, which allows full analysis of all volume records, nor are E-Records produced.

4.6.3.4 Input Data Sets

EDGCxBIN has two input data sets (Table 21).

Table 21. EDGCxBIN Input Data Sets	
ddname	Description
SLOTNO	TMC volume records, representing all volumes, sorted by outcode location name and slot number in ascending order.
STORE	Outcode location names with corresponding (optional) storage location names. The EDGCxBIN program does not use the Default Volume Owner field. Thus this data set can be the same data set that is used with EDGCxLDR. It should have the same data in it, if it is not the same data set. A maximum of 49 vaults can be specified.

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (ignored)
13-20	Storage location name (optional)
21-80	Comments

To obtain the list of used vaults, for use as storage locations, you can run EDGCxBIN with the CHECK parameter.

Figure 62 shows a sample of how to change outcode location names to storage location names.

```

/* You must supply the names of the Outcode Location names
/* that will become Storage Locations when RMM is implemented.
//STORE DD *
SAFE      LOCAL  <---  SAFE Outcode to be translated to RMM's LOCAL
ARC        DISTANT  --- Storage Location, ARC to DISTANT, and VLT1
VLT1      REMOTE <---  to REMOTE.
/* cols 1-4 Outcode location name
/*      5-12 Default volume owner (ignored)
/*      13-20 Storage location name (optional)
/*      21-80 Comments

```

Figure 62. EDGCxBIN: STORE DDNAME Sample

4.6.3.5 Output Data Sets

EDGCxBIN creates an output data set containing E-Records that represent the empty bin in each of the outcode locations and an error report data set (Table 22 on page 156). When you use the CHECK parameter, a list of all outcodes by classification can be produced in the LOANOUT DD and STOREOUT DD data sets.

<i>Table 22. EDGCxBIN Output Data Sets</i>	
ddname	Description
DEXTOUT	E-Record data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Error messages
STOREOUT	A list of all outcodes using slot numbers. This should match the vaults defined to VMS. Use the file as input to EDGCxLDR STORE DD. This file is optional and is used only if CHECK is requested.
LOANOUT	A list of all outcodes that do not use slot numbers. This should match the codes you use outside VMS. Use the file as input to the EDGCxLDR LOAN DD. This file is optional and is used only if CHECK is requested.

4.6.3.6 Return Codes

EDGCxBIN sets a return code when it terminates (Table 23).

<i>Table 23. EDGCxBIN Return Codes</i>	
Return Code	Description
0	Normal processing
4	Outcode location name from the TMC volume record did not match any of the outcode location names specified in the outcode location name fields in the STORE DD data sets.
16	- Slot number 0 is in use. - More than one volume is assigned a single slot number.
20	Both return code 4 and 16 problems were found.

4.6.4 Overview of Policy Conversion

Conversion of retention and movement policies consists of multiple parts:

1. Conversion of TMC expiration dates, which may be real dates or keyword dates.
2. Determining the CA-1 PARMLIB options that affect policies, so that they can be built-in to the DFSMSrmm policies conversion creates.
3. Conversion of the RDS entries to both policies and values for use in the EDGUX100 installation exit.
4. Conversion of the VPD entries to DFSMSrmm VRS policies.

The successful running of DFSMSrmm policy management after conversion relies on the use of the new vital records selection enhancements which requires the use of the DFSMSrmm PARMLIB option VRSEL(NEW). This option supports up to two policies per data set:

1. For identifying retention
2. For identifying movement/vaulting

This is particularly important when keyword dates are used to identify retention and VPD entries also exist for the same data sets.

Conversion proceeds as follows:

1. TMC expiration dates, which are keyword dates, are used to generate a DFSMSrmm VRS management value. The VRS management value is assigned to the data set, and the expiration date is set to the assign/creation date. A K record is created for each generated VRS management value. EDGC5LDR handles this automatically when the VRSMV=YES option is coded in the VRSMGMT file.
2. Each of the programs, EDGC5LDR, EDGCSRDS, and EDGCSVDS obtain the CA-1 PARMLIB options that affect policies. They build them into the DFSMSrmm policies they create.
3. EDGCSRDS converts the RDS entries to DFSMSrmm VRS policies (K records at this stage) using the data set name in the VRS as a VRS management value. Only keyword dates and special CA-1 retention options are converted to management values. Any RETPD retention values are not converted to VRSs.

EDGCSRDS also converts each of the RDS entries, together with the VRSVALUE file input from EDGC5LDR, into a table for use with the EDGUX100 installation exit. The table, UXTABLE, is an assembler source code table which you can later customize and modify independently of the EDGUX100 installation exit.

4. EDGCSVDS converts the VPD entries to K records, which become DFSMSrmm VRS policies later in the conversion process.

The K records will be equivalent to the rules defined in the CA-1 VPD.

EDGCNVT converts the K records to DFSMSrmm VRS policies, and when you run EDGHSKP with the VRSEL execution parameter, DFSMSrmm determines which policies to use for each data set and updates the data set information in the DFSMSrmm CDS with the VRS details it determines, such as: retention date, matching primary and secondary VRS, and whether the data set is VRS retained.

For correct and complete policy conversion, you must:

- Run each of the three programs EDGC5LDR, EDGCSRDS, and EDGCSVDS, using the sample JCL.
- Convert the K records using EDGCNVT.
- Use the EDGCVRSX sample as your EDGUX100 installation exit.
- Ensure that UXTABLE is loaded at IPL time for use by EDGUX100. Use the sample EDGCVRSX procedure to do this.
- Specify the VRSEL(NEW) DFSMSrmm PARMLIB option.

4.6.5 Creating VRS Records Using EDGCSRDS

The EDGCSRDS program converts the CA-1 retention policy definitions from the Retention Data Set (RDS) to K-Records and source for a dynamic table for the EDGUX100 installation exit. The created K-Records are "data set name Name Vital Record Specifications" types, but use a single data set name qualifier that identifies the VRS management value. Also, the EDGCSRDS program will analyze the CA-1 PARMLIB member (*CA1.PPOPTION*) to convert the following information in DFSMSrmm CDS K-Records:

RC Extended CYCLE retention
R9 Extended CATALOG retention
RP Default retention
CYD cycle control option
RO retention override option

CJOB cycle control by jobname

No reference is made to the Vault Pattern Description Data Set (VPD), so no matching information is provided or required.

4.6.5.1 How the Conversion Process Works

The Retention Data Set (RDS) input is converted to values to be used by a modified EDGUX100 installation exit sample to assign retention at OPEN time. This includes an entry as a default for all data sets using the RP management value. If RO YES is specified, the converted values will be used to replace any JCL retention values, including keyword dates.

Figure 63 shows the table used by the EDGUX100 installation exit.

* start of RDS entries			
EDGCVRSG DSN=A.B*,			X
JOBNAME=J1,			X
RO=YES			X
VRSVAL=LDATE003			
EDGCVRSG DSN=BACKUP.DATA.G%%V%,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CYCLE002			
EDGCVRSG DSN=BACKUP.DATA,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CYCLE002			
EDGCVRSG DSN=OFFSITE*,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CATLG005			
* default RP value			
EDGCVRSG DSN=*,	default RP value		X
RO=NO,			X
VRSVAL=CATALOG			
* start of keyword dates from EDGCxLDR			
EDGCVRSG KEYDATE=99000,			X
VRSVAL=WHILECAT			
EDGCVRSG KEYDATE=99005,			X
VRSVAL=CYCLE005			

Figure 63. Example of Data Produced from the RDS

The table is assembled by the EDGJSRDS sample job into a separate load module called UXTABLE, that can be changed dynamically using the EDGCVRSL program from the EDGCVRSP procedure. The sample jobs, procedures, and programs are in SAMPLIB.

The following keywords are available for the EDGCVRSG macro in the EDGUX100 installation exit:

ACLOPT You can use the ACLOPT keyword to disable the cartridge loader for this request.

- 1** Cartridge loader is disabled for this request.
- 0** Cartridge loader is enabled. This is the normal processing.

DSECT	YES/NO Specifies whether to produce a map for a table entry.
DSN	One to forty four characters, following MVS data set naming conventions, including % and *. % can be used to ignore a positional character in the job name. * can be used to ignore all remaining characters in the data set name. A <i>DSN=*</i> means that the entry applies to all data sets. The use of * is not the same as in the generic data set names supported by DFSMSrmm for VRSs and search data set masks. Here the * works like the *.* might in a generic data set name mask. Either KEYDATE, or DSN and JOBNAME may be specified.
JOBNAME	One to eight alphanumeric or national characters including % and *. % can be used to ignore a positional character in the job name * can be used to ignore all remaining characters in the job name. A <i>JOBNAME=*</i> means that the entry applies to all jobs. Either KEYDATE, or DSN and JOBNAME may be specified.
KEYDATE	Specifies an EXPDT value used in JCL. A 5-character numeric value, such as 99000, 99003, 98012. Either KEYDATE, or DSN and JOBNAME may be specified.
POOL	Specifies the RMM VLPOOL prefix to be assigned to all matching data sets. When a POOL is specified, you can also specify ACLOPT.
RO	Specifies the retention override option. YES means to override any specified JCL RETPD or EXPDT value. NO means to accept the JCL value. When YES is specified, any matching data set has the expiration date and management value set using the current table entry values. If a data set matches to an entry with RO=NO, EDGUX100 continues to scan the table for any KEYDATE values that match the JCL specified EXPDT so that it can be converted to a VRS management value. RO value has no meaning for the KEYDATE keyword because all keyword expiration dates are cleared and are set using the RETPD keyword value from the table. If no value is specified for RETPD, the RMM PARMLIB default retention period value will be used by RMM.
VRSVAL	Specifies the name of a vital record specification management value. A vital record specification management value can be 1 to 8 alphanumeric or national characters. The value corresponds to the data set name used in a VRS DSNAME policy, such as DSN('D99000').

Note: The **POOL** and **ACLOPT** keywords will not be used by a CA-1 conversion. You can add them later if you wish to use EDGUX100 to control scratch sub-pooling using data set names.

If the RP keyword is specified in the CA-1 PARMLIB member, a table entry for DSN=* is created, and either the RETPD=ddd is coded, or the VRSVAL=keydate is coded, depending on whether the default is a retention period or a keyword date. The keyword date management value uses the same naming convention as described earlier.

The last K-records in a chain generated for catalog or cycles retention are built including a *NEXTVRS(RC)* or *NEXTVRS(R9)*. The K-records for RC and R9 that specify *EXTRADAYS COUNT(nnn)* are assumed to be generated by the EDGCxLDR conversion program. RC is extra days after cycles retention, and R9 is extra days after catalog retention.

If the RO=YES option is specified in the CA-1 PARMLIB member, each management value K-record generated includes the RELEASE(EXPIRYDATEIGNORE) option.

Any cycle control K-records use BYDAYSCYCLE as determined by the CA-1 PARMLIB option *CDAY=DAY*, and include JOBNAME(*) as determined by CJOB if *CJOB=JOB* is specified.

Figure 64 shows examples of VRSs built by the EDGCSRDS conversion program.

```

RMM ADDVRS DSN(LDATE003) -
    LASTREFERENCE          -
    COUNT(3)               -

RMM ADDVRS DSN(CYCLE002) -
    CYCLES                 -
    COUNT(2)               -
    NEXTVRS(RC)            -

RMM ADDVRS DSN(CATLG005) -
    DAYS                   -
    COUNT(5)               -
    NEXTVRS(CATALOG)      -

RMM ADDVRS NAME(CATALOG) -
    WHILECATALOG          -
    NEXTVRS(R9)           -

RMM ADDVRS DSN(CATALOG) -
    WHILECATALOG          -
    NEXTVRS(R9)           -

```

Figure 64. VRS Built by EDGCSRDS

4.6.5.2 Sample JCL

Figure 65 on page 161 shows the sample execution JCL. See SAMPLIB member EDGJSRDS for the supplied sample JCL.


```

/*
//DEL      EXEC PGM=IEFBRI4
//DEXTOUT  DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.DEXTOUT.DATA
//VRSTABLE DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA
/* REFORMAT THE RDS RECORDS FOR INPUT TO SORT
// EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL'
//C.SYSIN  DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCRFMT)
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//G.IFILE  DD DISP=SHR,DSN=STSGWD.TMS.RDS.DATA
//G.OFILE  DD DISP=(,PASS),DSN=&&RFMRDS,UNIT=SYSDA,
//          AVGREC=K,SPACE=(80,(1,1),RLSE)
/*
//EDGCSRDS EXEC PGM=EDGCSRDS,REGION=4M,PARM=' BOTH'
//STEPLIB  DD DISP=SHR,DSN=RMM.LOAD
//SYSUDUMP DD SYSOUT=*
//ERROR    DD SYSOUT=*
//PERMRPT  DD SYSOUT=*
//RDS      DD DISP=(MOD,PASS),DSN=&&SORTRDS
//VRSVALUE DD DISP=SHR,DSN=STSGWD.EDGCXLD.RVRSVALUE
//TMSPARM  DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA
//DEXTOUT  DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
//          AVGREC=K,SPACE=(512,(2,2),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.DEXTOUT.DATA
//VRSTABLE DD DISP=(,CATLG),AVGREC=K,SPACE=(80,(100,100),RLSE),
//          DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA,UNIT=SYSDA
/*
/* FOLLOWING STEPS ASSEMBLES AND LINKAGE EDITS THE UXTABLE.
/* THE UXTABLE CONSISTS OF 3 PARTS:
/* 1. EDGCVRSF (FROM SAMPLIB)
/* 2. THE ABOVE CREATED VRSTABLE
/* 3. EDGCVRSE (FROM SAMPLIB)
/*
// EXEC HLASMCLG
//C.SYSIN  DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCVRSF)
//          DD DISP=SHR,DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA
//          DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCVRSE)
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//          DD DISP=SHR,DSN=SYS1.SAMPLIB
//LINKEDIT EXEC PGM=IEWL,PARM=' LIST'
//SYSLMOD  DD DSN=SYS1.LINKLIB,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSLIN   DD DSN=&&OBJ,DISP=(OLD,DELETE)
//          DD *
//          NAME UXTABLE(R)
/*

```

Figure 65 (Part 1 of 2). Sample EDGCSRDS JCL

4.6.5.3 EDGCSRDS Execution Parameters

The data set names specified in the RDS and VPDD can be used to match to both standard data set names and generations of a GDG where the base name matches the data set mask. This processing by TMS is independent of whether a fully qualified name or an index is specified.

In DFSMSrmm, if a data set name mask is specified fully qualified, it can only match to a standard data set name unless the GDG operand is also specified for the VRS. If the GDG operand is specified, standard data set names can no longer match to the mask. In DFSMSrmm you can get the same results in these ways:

- Specify a generic data set name mask, so that both GDGs and standard data set names can match:

```
ADDVRS DSN('A.B.*') .....
```

- Create a VRS specifying the specific data set name without the GDG operand, and create a second mask using a pseudo-GDG mask to manage the GDG generations:

```
ADDVRS DSN('A.B') NOGDG .....  
ADDVRS DSN('A.B.G-----V---') NOGDG .....
```

The extract programs use the latter method to most specifically match the retention and movement policies of TMS. However, you can control what EDGCSRDS does by specifying one of the execution parameters—GDG, NONGDG, or BOTH. When BOTH is specified for EDGCSRDS, the UXTABLE table entries are created using DSN=A.B.G%%%%V%%.

EDGCSRDS optionally requires an execution parameter (Table 24).

Table 24. EDGCSRDS Execution Parameter Definition

Parameter	Description
GDG	Specify this parameter if the fully qualified data set names in the RDS are only used to match the GDG base names. If any of the data set names match the standard data sets, use the BOTH parameter.
NONGDG	Specify this parameter if the fully qualified data set names in the RDS never match a GDG base name. If any of the data set names match GDG base names, use the BOTH parameter (default).
BOTH	Specify this parameter if the fully qualified data set names in the RDS can potentially match either standard data set names or GDG base names. If you know more specifically what they match, consider either the GDG or NONGDG parameter.

4.6.5.4 Input Data Sets

EDGCSRDS has three input data sets (see Table 25).

Table 25 (Page 1 of 2). EDGCSRDS Input Data Sets

ddname	Description
VRSVALUE	Data set containing names of all created management value K-records. This data set is created by the EDGCxLDR processing.
RDS	The RDS sorted by data set and job name in descending order (required). The RDS data must be that created by EDGCRFMT processing.

Table 25 (Page 2 of 2). EDGCSRDS Input Data Sets	
ddname	Description
TMSPARM	This is the TMS PARMLIB options member.

Note: If you do not use an optional input data set, you must specify DD DUMMY for it.

4.6.5.5 Output Data Sets

EDGCSRDS creates output data sets and error reports (see Table 26).

Table 26. EDGCSRDS Output Data Sets	
ddname	Description
DEXTOUT	K-Records The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Information and error messages
PERMRPT	Permanent Retention Report
VRSTABLE	All the new EDGUX100 dynamic table entries are written to this file. They are later merged with samples EDGCVRSF and EDGCVRSE and assembled into the UXTABLE load module.

4.6.5.6 Return Codes

EDGCSRDS sets the return code accordingly when it ends (see Table 27).

Table 27. EDGCSRDS Return Codes	
Return Code	Description
0	Normal processing
4	Permanent retention policy found but K-Record is created.
16	Unable to create K-Records because of RDS or VPDD control statement error.

4.6.6 Creating VRS Records Using EDGCSVDS

EDGCSVDS converts the vaulting policies from the VPDD into K-Records representing the DFSMSrmm storage location movement policies. DFSMSrmm does not require that the retention type of all locations in a multiple location movement policy be the same.

The EDGCSVDS program converts the CA-1 vault pattern description (VPDD) options by the following rules:

- OR** The EDGCSVDS program detects multiple options on a single **V=** statement and converts them using ANDVRS. This matches the CA-1 'OR' condition.
- AND** Multiple options on separate **V=** statements are converted using NEXTVRS. This matches the CA-1 'AND' condition.

VR= VR= in other than the first vault is converted to EXTRADAYS in a NAME vital record specification (VRS). VR= in the first vault is still converted as DAYS (CR=) in a DSNNAME vital record specification (VRS).

CJOB= Any cycle control K-records use BYDAYSCYCLES as determined by CYD=YES and include JOBNAME(*) as determined by the VPD entry.

Note: All K-records created include retention types:

```

BYDAYSCYCLES
CYCLES
DAYS
LASTREFERENCEDAYS
EXTRADAYS
UNTILEXPIRED
WHILECATALOG

```

and count value *COUNT(days/cycles)*.

If you decide to reassign the storage location names when running EDGCxLDR, EDGCSVDS needs to know the new name of each of the storage locations; see Section 4.6.6.3, “Input Data Sets” on page 165 for more information.

The EXP value is converted to UNTILEXPIRED, and should work the same way as it did under TMS. If a data set matches to a data set name VRS, converted from the VPD, with UNTILEXPIRED, and also has a VRS management value assigned, the VRS management value determines how long the data set name VRS retains the data set. For data sets that have no VRS management value the volume expiration date is used by DFSMSrmm.

Note: You must use the DFSMSrmm PARMLIB option VRSEL(NEW) after converting from TMS. If you do not take this into consideration, DFSMSrmm may use an incorrect retention.

4.6.6.1 Sample JCL

Figure 66 shows the sample execution JCL. See SAMPLIB member EDGJSVDS for the supplied sample JCL.

```

/*
//DEL      EXEC PGM=IEFBRI4
//DEXTOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSVDS.DEXTOUT.DATA
/*
//EDGCSVDS EXEC PGM=EDGCSVDS,PARM=' BOTH'
//STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//SYSUDUMP DD SYSOUT=*
//ERROR    DD SYSOUT=*
//EXPRPT   DD SYSOUT=*
//STORE    DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//VDS      DD DISP=SHR,DSN=STSGWD.TMS.VPDD.DATA
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=512,
//          RECFM=VB,SPACE=(500,(200,100),RLSE),
//          DSN=STSGWD.EDGCSVDS.DEXTOUT.DATA
//TMSPARM DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA

```

Figure 66. Sample EDGCSVDS JCL

4.6.6.2 Execution Parameters

The data set names specified in the RDS and VPDD can be used to match to both standard data set names and generations of a GDG where the base name matches the data set mask. This processing by CA-1 is independent of whether a fully qualified name or an index is specified.

In DFSMSrmm, if a data set name mask is specified fully qualified, it can only match to a standard data set name unless the GDG operand is also specified for the VRS. If the GDG operand is specified, standard data set names can no longer match to the mask. In DFSMSrmm you can get the same results in these ways:

- Specify a generic data set name mask, so that both GDGs and standard data set names can match:

```
ADDVRS DSN('A.B.*') .....
```

- Create a VRS specifying the specific data set name without the GDG operand, and create a second mask using a pseudo-GDG mask to manage the GDG generations:

```
ADDVRS DSN('A.B') NOGDG .....  
ADDVRS DSN('A.B.G-----V---') NOGDG .....
```

The extract programs use the latter method to most specifically match the retention and movement policies of CA-1. However, you can control what EDGCSVDS does by specifying one of the execution parameters—GDG, NONGDG, or BOTH.

EDGCSVDS optionally requires an execution parameter (Table 28).

Table 28. EDGCSVDS Execution Parameters	
Parameter	Description
GDG	Specify this parameter if the fully qualified data set names in the VPDD are only used to match the GDG base names. If any of the data set names match the standard data sets, use the BOTH parameter.
NONGDG	Specify this parameter if the fully qualified data set names in the VPDD never match a GDG base name. If any of the data set names match GDG base names, use the BOTH parameter (default).
BOTH	Specify this parameter if the fully qualified data set names in the VPDD can potentially match either standard data set names or GDG base names. If you know more specifically what they match, consider either the GDG or NONGDG parameter.

4.6.6.3 Input Data Sets

EDGCSVDS has three input data sets (Table 29). If you do not use an optional input data set, you must specify DD DUMMY for it.

Table 29. EDGCSVDS Input Data Sets	
ddname	Description
VDS	VPDD unsorted (required)
STORE	Storage location name input data set (see Section 4.6.2.7, “STORE Location Name Data Set Record Format” on page 137 for information on data set format). A maximum of 1000 vaults can be specified.
TMSPARM	The TMS PARMLIB option member

4.6.6.4 Output Data Sets

EDGCSVDS creates an output data set containing the converted retention and storage location movement policies from the VPDD, the RDS, and the TMC volume records having the special meaning Julian dates and an error report file (Table 30).

Table 30. EDGCSVDS Output Data Sets	
ddname	Description
DEXTOUT	K-Records The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Information and error messages
EXPRPT	Until Expired Retention Report This file lists all occurrences of data sets in the VPDD that specify a retention of EXP.

4.6.6.5 Return Codes

EDGCSVDS sets the return code accordingly when it ends (see Table 31).

Table 31. EDGCSVDS Return Codes	
Return Code	Description
0	Normal processing
4	Unable to support a VPDD policy option but K-Record is written
12	Unable to create K-Record because the "V=" control statement is missing
16	Unable to create K-Record because of a VPDD control statement error

4.7 Interfaces

Before the conversion process you must consider some information about interfaces.

4.7.1 SMS Interface

If you have an IBM 3494 or an IBM 3495 Automated Tape Library Dataserver, you must ensure that it works correctly in the DFSMSrmm environment. You have to change the OAM exits, using the DFSMSrmm version and provide the correct input to the extract program for those volumes that belong to the tape library.

You also must define an DFSMSrmm location with the same name as the tape library name, using a location type of ATL. Thus, the CA-1 location name that you are using for the ATL tapes must be translated to this DFSMSrmm location.

4.7.2 BTLS Interface

If your installation uses BTLS to drive IBM 3494 and IBM 3495 Automated Tape Library Dataserver operations, you will need to notify BTLS when a volume has to return to SCRATCH status. In a CA-1 environment this is done with a batch job that gets information from the CA-1 database and then updates the BTLS catalog. In DFSMSrmm you can use a simple REXX CLIST that synchronizes the DFSMSrmm database with the BTLS catalog.

You can find a sample CLIST in Appendix D, “Sample Exits and Reports” on page 603.

4.7.3 Short-on-Scratch Processing Interface

DFSMSrmm offers an integrated SMS interface that updates the TCDB automatically.

In addition, you can free volumes that are in PENDING RELEASE status when a short-on-scratch condition is detected inside the automated tape library dataserver without manual intervention. Use the PARMLIB option, SCRATCHPROC(RMMSCR), where RMMSCR is the name of the procedure DFSMSrmm starts to replenish scratch volumes in the automated tape library dataserver.

4.7.4 TSO and ISPF Interfaces

DFSMSrmm has a simple TSO interface that uses a set of powerful commands to drive all tape activities.

There is also an ISPF interface that offers menus for end users, librarians, administrators, and support personnel. You can choose a full DFSMSrmm dialog or a local defined dialog. You can tailor the appropriate ISPF dialog for each user who will have access to DFSMSrmm.

4.7.5 Other Interfaces

CA-1 provides an online transaction that accesses CICS, TSO, ISPF, and ROSCOE environments. You can use inquiry and update processing through one of these interfaces, or you can use an operator console to issue CA-1 commands.

DFSMSrmm can use any DFSMSrmm command in batch mode using the IKJEFT01 or IKJEFT1A TSO batch interface. It can also process any DFSMSrmm command through the DFSMSrmm API. To use the DFSMSrmm API you need to code in the High Level Assembler language. A preferred method is to use the DFSMSrmm commands from a REXX environment where you can retrieve information back as variables.

If you have any of the above interfaces running in your CA-1 tape environment, you must change them to reflect the equivalent function in DFSMSrmm.

4.8 Am I Ready to Continue?

In this chapter we have covered all of the activities that you should have considered before converting from CA-1 to DFSMSrmm. After completing the activities in this chapter, you should have the following valid and tested input to load the DFSMSrmm CDS:

- Input to exits
- Correct PARMLIB options
- Housekeeping jobs to use in an DFSMSrmm environment
- Changed reports
- Extract and convert control statements
- Extracted data from the CA-1 TMC.

If the data extract program has error return codes, read the error messages to determine the actions required to correct the errors. Then rerun the extraction process (see Section 4.5, “Running the Extract Program” on page 123).

To have a valid DFSMSrmm CDS, you must stop all tape activity when you get the data from the CA-1 database until you start DFSMSrmm in record-only mode in parallel with CA-1.

Before producing a valid conversion, you can use a copy of the CA-1 database to test the extraction and conversion process, so you do not have to stop the tape activities every time you find that you have to go back and re-run the extract jobs.

Once you have tested the environment, you should have the correct tailored input to the Extract program.

You can now stop tape activities and produce a valid data extraction from the CA-1 TMC, RDS, and VPDD.

If your previous tests were satisfactory, you should obtain the correct input to the EDGCNVT program and proceed to the next step.

Chapter 5. Converting from TLMS to DFSMSrmm

In this chapter we describe how to convert from CA-DYNAM/TLMS (Tape Library Management System; hereafter referred to as TLMS), a tape management product of Computer Associates, to DFSMSrmm.

Objectives of Chapter

- Explain differences between TLMS and DFSMSrmm
- Describe how to customize DFSMSrmm for equivalent TLMS function
- List the functions where there is no equivalence
- Extract data and prepare input for EDGCNVT
- Provide hints and tips

Audience

- System programmers
- Storage administrators and librarians

At the end of this chapter you will have reviewed the implementation of your current TLMS environment and will understand the detailed changes required to customize DFSMSrmm to meet your tape management needs.

5.1 TLMS Overview

TLMS uses installation-defined retention management, storage location management, and tape pool management policies to manage tape volumes in tape libraries and local and remote storage locations.

TLMS has two databases containing volume inventory and management data:

- Volume master file (VMF) - volume and data set information
- Retention master file (RMF) - location and retention management criteria

The tape retention system (TRS) component, which processes information from the RMF and VMF databases, performs retention and location management.

To maintain the volume inventory, the online recorder component of TLMS accesses and updates the VMF. The online recorder processes transactions from open, close, EOVS; online inquiry and update, batch update and reporting, and TRS.

5.2 TLMS and DFSMSrmm

To ensure a successful conversion, it is important that you understand some of the differences between the products and how those differences affect your conversion project.

5.2.1 Terminology

Table 32 compares and defines some TLMS and DFSMSrmm terms.

<i>Table 32. TLMS and DFSMSrmm Terminology</i>		
TLMS	DFSMSrmm	Definition
Volume master file (VMF)	Control data set (CDS)	The tape management system database containing information about tape volumes and data sets
Retention master file (RMF)	– –	The database containing the management rules of tape data sets. In DFSMSrmm these rules are in the CDS.
Alternate log file	Journal	The sequential data set that keeps track of the changes in the database. It can be used for recovery purposes.
Online recorder	RMM subsystem	The address space that controls the real-time processing of the tape management system. It must be active to use the product functions.
Retention type number	Vital record specification (VRS)	The installation-defined policy that manages the life of a tape data set
Data center (DC)	SHELF location	The location nearest the tape drives
Cabinet and/or box	Bin number	The individual shelf location in any of the storage locations outside the data center
PRIVATE volume	MASTER volume	The status of a volume containing active data
Creation date	Assigned date	The date when a PRIVATE or MASTER volume was first added or when a volume changed status either to or from scratch
Keep date	Retention date	The expiration date or calculated by the tape management system
IBM expiration date	Expiration date	The JCL-specified or calculated default expiration date
TRS run	Housekeeping	The batch utility that updates the database and establishes the retention criteria for data sets
Hooks Intercepts	– –	The TLMS user modifications that you must install to interface TLMS with open, close, EOVS processing. In DFSMSrmm this interface is integrated within the DFSMS/MVS code.

5.2.2 Functions

Table 33 on page 171 summarizes the key tape management functions of TLMS and DFSMSrmm differences.

Table 33 (Page 1 of 3). TLMS and DFSMSrmm Functions

Function	
TLMS	DFSMSrmm
Tape security	
• Tape must be scratch for output scratch request • Data set name on tape must match inventory-specific request • 44-character data set name checked against VMF on input processing	• Tape must be scratch for output scratch request • Data set name on tape must match inventory-specific request • 44-character data set name checked against CDS on input processing • RACF interface for tape profile management • Access list for non-RACF system • Security erase on scratch • Prevent read of scratch tapes • Security classification by generic data set name • Master tape overwrite support controlled by PARMLIB option • If the DFSMSrmm address space is not started, you cannot use tapes. By using the S DFRMM,OPT=RESET command or the EDGRESET utility, you can permit tapes to be used without DFSMSrmm control.
Retention management	
• Based on RETPD or EXPDT • By cycle • By days since last reference • By days since creation • By cataloged • By a combination of cycles and days • By job name • Can specify default retention installationwide, and by specific or group data set name • Retention policies based on full or partial data set name on tape must match inventory-specific request	• Based on RETPD or EXPDT • By cycle • By days since last reference • By days since creation • By cataloged • By a combination of cycles and days • By job name • Can specify default and maximum retention installationwide. Default retention can also be specified using generic data set mask in VRS. • Retention policies based on full or generic data set name or VOLSER

Table 33 (Page 2 of 3). TLMS and DFSMSrmm Functions

Function	
TLMS	DFSMSrmm
Mixed retention management	
	DFSMSrmm allows you to combine any of its retentions together so that a data set must meet all criteria, or the types can be chained in sequence so that a complex and mixed retention can be defined. • By CYCLE WHILECATALOG DFSMSrmm retains data sets based on cycles or copies of a data set. For a GDG data set VRS, the retention is based on GDG cycles. For non-GDG data sets, each occurrence of a data set is considered to be a cycle. • By RETPD or EXPDT WHILECATALOG, retains the data set for as long as it is specified in the JCL RETPD parameter. After that period DFSMSrmm retains the data set as long as it is cataloged. • By days since last reference while cataloged • By days since creation while cataloged • By until expired • By BYDAYSCYCLE (cycles by day)
EXPDT with special meanings	
• 97000-99366	• Dates of 1999365 and 1999366 mean never scratch¹ • Installationwide exit can be used to support special dates • Full support of dates beyond twentieth century
Vault or storage location management	
• Supports up to five storage locations² • Define management by data set name – Full or partially qualified – Qualified by job name • Manages by store bin or slot number and box number • Unrestricted movement	• Supports an infinite number of storage locations and system-managed libraries • Define management by data set name, VOLSER, or job name – Full or generic (ISMF syntax), GDG or pseudo-GDG • Manages by store bin number • Unrestricted movement • Optional qualification by catalog status (while cataloged) • Supports loan location • Can manage movement based on media name
Security	
• Options table defines authorized users and passwords	• RACF control (through SAF interface) to authorize different functions
Exit for DFSMSHsm and ADSM	
• Supplied	• Supplied • Full support for DFHSM use of scratch pools

Table 33 (Page 3 of 3). TLMS and DFSMSrmm Functions	
Function	
TLMS	DFSMSrmm
Interfaces	
Supplied	- Integrated with DFSMS/MVS - Can test while TLMS in control
Initialization of new tapes	
Interface uses WTOR for integrity, updates inventory³	Interface ensures integrity, updates inventory⁴
Label support	
SL, AL, NL, BLP (BLP requires user exit), NSL	- SL, AL, NL, BLP, SUL, AUL - For BLP, further processing is determined by the BLP option set in PARMLIB.
Volume pools	
VOLSER ranges associated with data set name or user	VOLSER ranges associated with system or based on job name and data set name. Installationwide exit EDGUX100 can be used to support multiple scratch pools based on data set and job name.
Operator messages	
Scratch mount messages modified with pool name	- Scratch mount messages modified with pool name - Specific requests modified with rack (slot) number in library 3480, 3490, and 3590 message display modified with pool ID or rack (slot) number
Optional sticky labels	
All, none, nonscratch, out-of-service, out-of-range	Interface provided through EDGUX100 exit
Identify tapes out of library	
– –	Loan location support
¹DFSMSrmm provides permanent retention using 99365/99366 or 1999/365 and 1999/366. ²TLMS includes the data center in the count of five. ³TLMS provides a modification, TLMSIEHI, to IEHINITT that checks the volume in the TLMS VMF. Also, an operator reply to WTOR is needed. ⁴DFSMSrmm utility EDGINERS includes IEHINITT functions, but DFSMSrmm does not use IEHINITT; DFSMSrmm supports system-managed tape and requires no operator intervention, that is, no WTOR.	

5.2.3 Retention Methods

To help you understand the way in which TLMS and DFSMSrmm manage the retention of tape data sets and the movement of volumes between the storage locations, we compare the management rules of the two products.

Table 34 on page 174 presents the TLMS retention methods and the equivalent VRS definitions that you should use in the DFSMSrmm environment.

<i>Table 34. TLMS Retention Methods and DFSMSrmm VRS Equivalences</i>	
TLMS	DFSMSrmm VRS
Type 1 - Catalog control	Whilecatalog
Type 2 - Date control	Days or untilexpired
Type 3 - Catalog / date control	Whilecatalog and Days
Type 4 - Cycle control	Cycles
Type 5 - Elapsed days control	Days or extradays
Type 6 - Move immediate	None required
Type 7 - Manual control	Manual move function
Type 8 - Days since last used	Days last reference
Type 9 - Exp. date keyword control	VRS management value and UNTILEXPIRED

Below we explain the differences between each TLMS retention method and the DFSMSrmm VRS.

Type 1 - Catalog Control: The data set is kept until it is uncataloged. TLMS ignores the keep date, and the tape containing this data is scratched when the data set is uncataloged.

In the DFSMSrmm environment the data set is kept while it is cataloged and the expiration date is ignored using `RELEASE(EXPIRYDATEIGNORE)`.

Type 2 - Date Control: The data set is retained at the current location until the keep date has passed. The keep date can be determined from the `TLMSDTAB` table or by specifying a number of days in the count field of the RMF definition. The latter alternative prevents the keep date from being overridden. If the RMF specifies a value, a `DAYS` retention type is used, and 1 added to it to allow for TRS/RMM differences; otherwise, `UNTILEXPIRED` is used so that the volume expiration date is considered.

Optionally, DFSMSrmm ignores the expiration date of the data set, set by the JCL or by installation default, when the `RELEASE(EXPIRYDATEIGNORE)` operand on the `ADDVRS` subcommand is specified.

Type 3 - Catalog/Date Control: In TLMS the retention method is a combination of Types 1 and 2.

In DFSMSrmm two VRSs are used; the first with `DAYS` and the second with `WHILECATALOG`. The `DAYS` VRS is to be built as a type 2. The VRSs are chained using `NEXTVRS`.

Type 4 - Cycle Control: In TLMS, data sets are retained until the number of versions exceeds the quantity specified.

In DFSMSrmm there is an equivalent retention type specifying `CYCLES` in the `ADDVRS` command.

Differences can be found in GDG management processing. TLMS allows the specification of full or qualified data set names, whereas DFSMSrmm allows full generic specification qualified by GDG and non-GDG.

Type 5 - Elapsed Days Control: In TLMS, data sets are retained at the current location for the number of days specified.

When converted to DFSMSrmm, DAYS is used in the DSNAME VRS, and EXTRADAYS used in NAME VRSS. The number specified in the RMF is increased by 1 to allow for the difference between TLMS TRS and DFSMSrmm VRS.

Type 6 - Move Immediate: Tapes are moved from the data center the first time you run TRS after the data set is created.

In DFSMSrmm this function is available in every VRS definition by specifying the location name desired.

Type 7 - Manual Control: In TLMS, a data set is retained until it is either scratched or moved by specific manual instruction.

In DFSMSrmm you can place a volume under manual move control and it remains in a storage location until you manually return it to the data center. For an RMF entry, a VRS with LOCATION(*location*) and DAYS COUNT(99999) is built.

Type 8 - Days Since Last Used: In TLMS, a data set is retained for a specified number of days since it was last used for input or output.

In DFSMSrmm the equivalent is the LASTREFERENCEDAYS operand. The number specified in the RMF is increased by 1 to allow for the difference between TLMS TRS and DFSMSrmm VRS.

Type 9 - Exp. Date Keyword Control: In TLMS, data set retention is controlled by special values coded on the JCL parameter LABEL=EXPDT.

In DFSMSrmm you must use a VRS management value in conjunction with the EDGUX100 exit to manage special meaning expiration dates. At conversion time, EDGCDYNM builds a VRS using UNTILEXPIRED so that any vital record specification management value used for the data set determines retention.

5.2.4 Data Fields

When you display information for a volume or a data set, the data fields differ. In this section we show the differences.

5.2.4.1 Volume

Table 35 compares the TLMS and DFSMSrmm data fields.

Table 35 (Page 1 of 2). TLMS and DFSMSrmm Volume Data Fields	
TLMS	DFSMSrmm
Volume chain	Previous and next volume
Keep date	Expiration date
Scratch date	Assign date
Abend indicator	ABEND VRS retention
CPU number	--
Volume sequence number	Sequence number
--	Media name

<i>Table 35 (Page 2 of 2). TLMS and DFSMSrmm Volume Data Fields</i>	
TLMS	DFSMSrmm
--	Rack number
--	Original expiration date
--	Description
--	Media type
Recording format (available in TLMS 5.4 but not for 3590 cartridges)	Recording format
--	Availability
--	Release actions
--	Owner
--	Owner access
Creation date and time	Assigned date and time
--	Security name and classification
--	MVS or VM use
--	Loan location
--	Volume access list
--	Volume usage (KB)
--	Temporary and permanent errors (read and write)
--	Actions pending
--	Location type, destination and transit flag
--	Storage group
--	Home and required location
--	Move mode
--	Bin number and old bin number
--	Product information
--	UNTILEXPIRED
--	EXTRADAYS

5.2.4.2 Data Set

Table 36 compares the TLMS and DFSMSrmm data set data fields.

<i>Table 36 (Page 1 of 2). TLMS and DFSMSrmm Data Set Data Fields</i>	
TLMS	DFSMSrmm
Abend indicator	ABEND VRS Retention
Create job name and step name	Job name and step name
Last read drive	Included in volume display. Data set display contains creating drive.
Account number	Included in volume display
Data set keep date	Retention date
CPU number	Creating system ID
--	Owner

Table 36 (Page 2 of 2). TLMS and DFSMSrmm Data Set Data Fields	
TLMS	DFSMSrmm
--	Security name and classification

5.2.5 Cross-Reference

In this section we present tables that map data from the EDGCNVT input records to TLMS.

During the EXTRACT process, the TLMS records are read in input, and some fields are extracted to be converted in the DFSMSrmm database.

Some TLMS fields are not used by the EXTRACT program, and some DFSMSrmm fields are not present in the TLMS equivalent record. Also, some fields have a different name in the TLMS database but have a similar or equivalent meaning in the DFSMSrmm database.

5.2.5.1 Volume Information

Table 37 maps data from the EDGCNVT input records to TLMS.

Table 37 (Page 1 of 4). EDGCLREC Volume Information		
	TLMS	Status
Volume serial number	√	Req
Volume owner ¹		Req
Volume owner user ID		
Volume owner node ID		
Volume security level		
Volume sequence number	√	Req
Volume create date	√	Req
Volume create time	√	
Volume create user ID		
Volume create job name	√	
UCB type (copy of UCBTYP)		
Tape media type information		Cond
Tape recording format	√	
Not 3480		
18 track		
36 track		
Tape media type		
Not 3480		
Standard capacity		
Enhanced capacity		
Tape compaction technique	√	
Unknown		
Not compact		
IDRC		
Tape special attributes		
None		
18 track read-only		
Total volume kilobytes used		

Table 37 (Page 2 of 4). EDGCLREC Volume Information		
	TLMS	Status
JCL volume expiration date	√	
Volume expiration date	√	Req
Volume retention date	√	
Count of data sets on volume	√	Req
Temporary read errors	√	
Temporary write errors	√	
Permanent read errors	√	
Permanent write errors	√	
Tape density (copy of JFCBDEN)	√	
BPI indicator 1600 bpi 6250 bpi 3480 3480 compact (IDRC) Undefined	√	Req
Volume status indicators Volume is master Volume pending release Vital record - do not release User tape (assigned by librarian) Tape is on loan Tape open - not yet closed Volume is scratch Volume recorded by open, close, or EOV	√ √	Req
Additional volume status indicators Scratch volume claimed by getvol Scratch volume has never been initialized Scratch volume with init action pending Scratch volume waiting to enter ATL Abend in process when a data set closed Abend probably in open, close, or EOV	√	
Volume label type Default retention period used Label type is NL Label type is AL Label type is SL Tape last written with BLP SL or AL with user labels	√ √ √ √ √	Req
Volume release actions Return to scratch pool - default Replace tape on release Reinitialize Degauss/security erase Return to owner Notify owner		

Table 37 (Page 3 of 4). EDGCLREC Volume Information		
	TLMS	Status
Volume access information Owner may read volume Owner may update volume Owner may alter volume Read-only protection Update protection Can be used on MVS systems Can be used on VM systems	√	Req
Media name		
Unit address used for creation	√	
Last used unit address	√	
Loan location		
Date volume last read		
Date volume last written		
Assigned from scratch date		
Assigned from scratch time		
Copy of JFCBLTYP	√	
From JFCTRTCH - IDRC support DSN used 3480 IDRC No compaction	√	Req
Current location of volume	√	Req
Destination name		
Home location name		
Storage group name		
Accounting information	√	
User description		
Authorized user IDs area		
Rack number ²		Req
Remote store bin number	√	Cond
Remote store bin date	√	Cond
Remote store bin media name		Cond
Label number of first file		Req
Name of first data set on volume	√	
First data set's owner's name		
Sequence number of first data set		Cond
First data set's RECFM	√	Cond
First data set's LRECL	√	Cond
First data set's BLKSIZE	√	Cond
First data set's block count	√	Cond
First data set's kilobytes		
First data set's security level		
DSN1 create date	√	Req

Table 37 (Page 4 of 4). EDGCLREC Volume Information		
	TLMS	Status
DSN1 create time	√	
DSN1 last read		Cond
DSN1 last written		Cond
DSN1 flag Data set cataloged Abend in progress when data set closed	√	
VRS management value ³		Cond
First file creating step name	√	
First file creating DD name	√	
First file retention date	√	
Name of last data set on volume		Cond
Creating system ID	√	
• Data that can be found in TLMS is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹We recommend that you use the job name or the data set HLQ as the owner. ²We recommend that you use the VOLSER as the rack number. ³We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

5.2.5.2 Data Set Name Information

Table 38 maps data set information.

Table 38 (Page 1 of 2). EDGCDREC Data Set Name Information		
	TLMS	Status
Data set create date	√	Req
Data set create time	√	
Data set name	√	Req
Compaction status Data set used 3480 IDRC No compaction		
File sequence number on volume		Req
VOLSER holding data set	√	Req
Volume sequence within set of volumes		Req
Unit where data set was written		
Data set RECFM	√	Req
Data set LRECL	√	Req
Data set BLKSIZE	√	Req
Data set block count	√	Req
Data set size in kilobytes		
Data set owner name ¹		Req
Logical file sequence number	√	Req

Table 38 (Page 2 of 2). EDGCDREC Data Set Name Information		
	TLMS	Status
Data set security level		
Data set date last read		Req
Data set date last written		Req
VRS management value ²		Cond
DSN flag		
Data set cataloged	√	
Abend in progress when data set closed		
Creating system ID	√	
Creating job name	√	
Creating step name	√	
Creating DD name	√	
Retention date	√	
• Data that can be found in TLMS is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹CONVERT ensures that all data sets on the same volume have the same owner as the VOLSER's owner. ²We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

5.2.5.3 Vital Record Information

Table 39 maps the vital record Information.

Table 39 (Page 1 of 2). EDGCKREC Policy Information		
	TLMS	Status
Record create date		Req
Record create time		
VRS type	√	Req
Data set name	√	Cond
Format of data set name	√	Cond
Type of retention	√	Cond
Cycles		
Cycles by days		
Elapsed days		
Extra days		
Days unreferenced		
While cataloged		
Until expired		
Volume serial number	√	Cond
Delay days		
Total count of number of cycles or days	√	Req
VRS name		Cond
Store keep number	√	Cond
VRS owner		Req

Table 39 (Page 2 of 2). EDGCKREC Policy Information		
	TLMS	Status
Automatic delete date		
VRS description		
Location name	√	Req
Name of NEXT VRS definition		
Name of AND VRS definition		
Release options		
Job name		
• Data that can be found in TLMS reports is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>.		

5.2.5.4 Empty Rack and Bin Information

Table 40 shows empty rack and bin information.

Table 40. EDGCEREC Empty Rack and Bin Information		
	TLMS	Status
Shelf status Shelf is rack in library Shelf is bin in storage location		Req
Empty shelf location's name		Req
First of empty shelves start		Req
Count of contiguous shelves		
Media name		
• Data that DFSMSrmm requires is marked with <i>Req</i>.		

5.2.5.5 Owner Record Information

An owner record is required only for actual owners for whom you want to supply owner details and for those who are to be owners for VRS. Other owner records will be created by CONVERT postprocessing (EDGCVOL).

Table 41 lists the owner record information that DFSMSrmm requires and is not present in the TLMS database.

Table 41 (Page 1 of 2). EDGCOREC Owner Information		
Description	TLMS	Status
Owner identifier		Req
Owner's surname		
Owner's first name		
Owner's department ¹		Req
Owner's first address line		
Owner's second address line		
Owner's third address line		

Table 41 (Page 2 of 2). EDGCOREC Owner Information		
Description	TLMS	Status
Owner's internal telephone number		
Owner's external telephone number		
Owner's user ID		
Owner's node ID		
Data that DFSMSrmm requires is marked with <i>Req.</i> ¹Not strictly required for conversion. Only the RMM ADDOWNER command enforces the owner's department.		

5.2.5.6 Data Unavailable for Conversion

The following are new terms in DFSMSrmm that are not available in the TLMS VMF (see Table 3 on page 66 for the field definitions):

- Data set size
- Date data set last read*
- Date data set last written*
- Date volume last read*
- Date volume last written*
- Original expiration date
- Owner
- Rack
- Physical file sequence number
- Security level
- SMS data class name
- SMS storage class name
- SMS storage group name
- Tape media type
- Tape usage in kilobytes

*In TLMS only the date last used is stored.

5.2.6 Differences

To complete the comparison of TLMS and DFSMSrmm, we must discuss the differences in system structure.

5.2.6.1 Structure

TLMS works with two address spaces: CAIRIM and OMS/MONITOR. The CAIRIM address space is the Resource Initialization Manager, which TLMS uses to dynamically initialize and install the TLMS operating system management modules. The OMS/MONITOR address space receives information and controls all tape activity whenever a tape is used. It has a subtask called the Online Recorder, which controls the real-time processing of TLMS and updates the VMF catalog.

DFSMSrmm has only one address space that manages all tape activity and the updates to the CDS. DFSMSrmm has a subsystem interface that must be active in order to use tape management functions and ensure tape data set protection. If this interface is defined to MVS in the IEFSSNxx PARMLIB member but is not started, you cannot use tapes in your system. You can also control the inactivation of the subsystem interface using the DFSMSrmm EDGRESET utility

through RACF. Thus you can ensure that no tape activity is allowed unless DFSMSrmm is active.

5.2.6.2 Open, Close, and EOVS Interface

In a TLMS environment, if you stop the OMS/MONITOR address space, you can use tapes without any knowledge of the tape management system. TLMS requires changes to open, close, and EOVS IBM modules to interface tape activity. The changes are accomplished with zaps (called hooks) or dynamic updates to DFSMSdfp modules. You can apply these hooks permanently by using an SMP USERMOD, directly modifying code with the AMASPZAP program, or using the Dynamic Zap Method. The Dynamic Zap Method does not permanently alter the system libraries; the required hooks are dynamically applied to the storage copies of the IBM modules.

In a DFSMSrmm environment, the code that interfaces the open, close, and EOVS modules is integrated in DFSMSdfp, so you do not have to make any changes.

5.2.6.3 Shelf Management

TLMS uses cabinets and boxes to hold tapes offsite. Cabinets are defined with a unique two-character alphanumeric identifier and a slot position for the cartridge. Boxes are defined with a unique four-character alphanumeric identifier and a number indicating the total capacity of the box. You can move complete boxes only, not individual tapes, from one storage location to another.

In DFSMSrmm, bin numbers define individual shelf locations in offsite storage locations. Bin numbers are six characters and can only be assigned by DFSMSrmm that uses a sequential filling method of the bins in the selected location. Once a volume no longer uses a bin number, DFSMSrmm can reuse that number for another volume that needs to be placed in the same location. Thus you do not have to worry about the physical location of the tape in the offsite storage location, because DFSMSrmm tells you which free bin number you can use for the volumes that you are moving.

5.2.6.4 Retention

In most cases, TLMS uses a control file, which is the first file on the tape and drives the retention of all other files on the volume. The retention method is applied only to the first data set; your application must write the other data sets according to the retention method applied to the first.

DFSMSrmm applies a VRS definition to every data set on the volume, according to the data set or job name mask defined in the VRS, without considering the file position on the tape. This method should be carefully considered when planning the conversion process, because the VRS definition generated could be different from the management that you want. DFSMSrmm can return a single volume in a multivolume set of volumes to scratch—it does not always operate on the complete set of volumes. Consequently, you may see DFSMSrmm return a volume to scratch that is still retained by TLMS. Each data set in a multivolume set can be retained individually using VRS. If there are data sets in the volume set that have different retention requirements, DFSMSrmm can attempt to release them earlier. For a volume to be released, all data sets on that volume must no longer be retained by VRS. If you use a control file to cause the complete set to be retained, you may require additional VRS definitions under DFSMSrmm to cause the same volumes to be retained. Alternatively you can benefit from DFSMSrmm releasing volumes early when the data is not required to be retained.

5.2.6.5 Deletion Policies

When you return a volume to scratch (manually or by the inventory management process) in the TLMS database, all data set records, except file one, are deleted immediately, so you cannot recover the other data set.

In the DFSMSrmm CDS, the data set records are not deleted. They are deleted only when the volume is reused. Therefore you can reclaim volumes from scratch and still have all data set details.

5.3 Reviewing the Current Environment

See Figure 8 on page 21 to review the flow of events that we follow in converting from TLMS to DFSMSrmm. You must collect and then analyze all information about the current tape management environment. Once you have reviewed the environment and understand the main differences between TLMS and DFSMSrmm, you will better understand what you must change during the conversion process.

5.3.1 Collecting Information

The information that you collect about the current environment should include:

- TLMS reports used daily or occasionally

Every day there should be a TLMS report that the operator uses to move the cartridges between the desired locations and to return tapes to the scratch pools. You can use the DFSMSrmm supplied reports, but if you want to avoid changes to the layout of this report, using REXX or DFSORT you can tailor the DFSMSrmm reports to make them similar to the TLMS reports. Chapter 14, “Extending DFSMSrmm Reporting” on page 533 provides some useful examples how to create the reports.

- Exits installed

Collect information about TLMS exits installed in your environment so that you can ensure that you can achieve the same function in DFSMSrmm environment. For example, you must check the DFSMSshm exit, ARCTVEXT, if you have DFSMSshm installed. For an example of how to merge the ARCTVEXT code from TLMS with DFSMSrmm refer to D.1.1, “Sample DFSMSshm ARCTVEXT Router Exit” on page 603.

- PARMLIB options used

You must review TLMS PARMLIB options before you start DFSMSrmm because you may have to modify the DFSMSrmm options according to the TLMS environment.

- Interfaces with other products

You want to collect information about the interfaces that TLMS might have with other products (such as CICS, ROSCOE, and TSO).

- Housekeeping (batch inventory update) process

Collect information on the daily inventory management process so that you can synchronize it with the DFSMSrmm housekeeping function.

5.3.2 Analyzing the Record Layouts

Now you analyze the information you have collected to achieve the equivalent or acceptable alternative function in the DFSMSrmm environment.

DFSMSrmm requires certain data to be available to build its CDS. You must analyze TLMS databases and reports to know which types of data are available. There is not a one-to-one correspondence between DFSMSrmm and TLMS data. DFSMSrmm may have data that TLMS does not have and vice versa. If DFSMSrmm requires data that TLMS does not provide, use the extract programs or the EDGCNVT SYSIN fill statements to provide a value.

After analyzing the data, you must determine which types of data you want to have in the DFSMSrmm CDS in order to be consistent with the current operating environment. Knowing the types of data needed helps determine whether you can use TLMS databases or reports as the sole input to the extract program.

In addition, you must review the current policies regarding retention management, storage location management, and tape pool management. DFSMSrmm manages these processes differently from TLMS.

5.3.3 Retention Policies

Both TLMS and DFSMSrmm provide comprehensive retention criteria; however, the criteria are defined and processed in different ways.

Default retention periods can be specified in a user table, TLMSDTAB; this specification allows definition of:

- Minimum days for specific data sets
- Minimum days for groups of data sets
- Installationwide default for any other data sets

The RMF allows nine different retention policies, which are listed in Section 5.2.3, “Retention Methods” on page 173.

TLMS allows three types of data set names. Table 42 describes them and gives their equivalents in DFSMSrmm.

Table 42. TLMS Data Set Name Types		
	Description	DFSMSrmm Equivalent
F	Fully qualified data set name	DSN('dsname')
Q	Qualified data set name ending with a period	DSN('dsname.**')
P	Partially qualified data set name	DSN('dsname*.**')

If the data set type is P or Q and the retention type is 4 (cycle control), the data set name may be a GDG. DFSMSrmm provides more precise control by allowing a type of GDG to be specified along with a GDG base data set name. It checks for the standard version qualifier of GnnnnVnn during VRS processing. Also, a pseudo-GDG allows a customer to define a pattern that DFSMSrmm treats like a GDG.

You must use your knowledge of your installation to decide whether an equivalent VRS definition should use the .** suffix with the GDG or NOGDG operand, because TLMS does not specifically identify GDG data set names.

TLMS allows an additional qualifier, job name, to its selection criteria.

DFSMSrmm can tailor this selection criteria by using a mask for the job name and specifying in PARMLIB whether the job name should take precedence over the data set name during VRS match processing.

TLMS allows several combinations of retention types for a single definition, for example, elapsed days retention followed by cycle retention followed by days since created and cataloged. DFSMSrmm provides equivalent function, and many additional retention and movement options.

The EDGCDYNM extract program converts TLMS retention methods to the best equivalent rules in VRS definitions as explained in 5.2.3, "Retention Methods" on page 173.

To avoid VRS definitions with duplicate data set and job names being generated, EDGCDYNM processes only the first occurrence of a data set and job name combination and ignores all others.

A warning message is produced for each ignored data set. These should be reviewed after conversion to ensure that the desired retention criteria are being applied.

In addition, TLMS can use specific or unique Julian dates to control tape data set expiration and retention. These keyword dates can be overridden by the installation's retention policies, which are defined in the RMF:

EXPDT=98000	Specifies that the tape volume being processed is a foreign tape, not under TLMS control
EXPDT=98ddd	Specifies that a tape data set be retained until the specified number of days (ddd) since last reference have elapsed. This is equivalent to RMF type 8.
EXPDT=99000	Specifies that a tape data set be retained for as long as it is cataloged. This is equivalent to RMF type 1.
EXPDT=990cc	Specifies that the total number of cycles (cc) or generations of the data set be kept. This is equivalent to RMF type 4.
EXPDT=991dd	Specifies catalog and date control, where dd is the number of days since creation. This is equivalent to RMF type 3.
EXPDT=992dd	Specifies date control, where dd is the number of days since creation. This is equivalent to RMF type 2.
EXPDT=99365	Specifies that a tape data set be retained permanently. This is equivalent to RMF type 7.
EXPDT=99366	Specifies that a tape data set be retained permanently. This is equivalent to RMF type 7.

For the retention hierarchy see 12.1.1, "The Best Match Fit Sequence" on page 474.

If a job abends while creating a tape data set, TLMS flags the corresponding VMF record and sets a default retention period of one day. The volume may then be returned to the scratch pool after the next run of TRS. The DFSMSrmm equivalent is to use an ABEND VRS as follows:

```
RMMADDVRS DSN(' ABEND') -  
          DAYS COUNT(1) -  
          RELEASE(EXPIRYDATEIGNORE)
```

By using VRSs and the EDGUX100 exit, DFSMSrmm supports the JCL special expiration dates honored by other tape management systems. You define VRSs that can be used to provide management function equivalent to the special Julian dates of TLMS. You then assign that VRS to the data set using either a management class or the installationwide exit to set a VRS management value.

For instance:

```
//MYDSN DD DSN=MY.DATA,DISP=(,CATLG),LABEL=(,SL,EXPDT=99000),  
// additional parameters
```

You define the VRS:

```
RMM ADDVRS DSN('D99000') WHILECATALOG
```

At open time the DFSMSrmm EDGUX100 installationwide exit requests a VRS management value of D99000, having determined that you specified a “special date” in your JCL.

During inventory management VRS processing, if there is no VRS data set mask covering the data set, DFSMSrmm matches the data set VRS management value with the ADDVRS DSN('D99000') definition and retains the data set for as long as it is cataloged.

You can list the management classes and VRS management values assigned to data sets using either the RMM LISTDATASET command or the ISPF dialog or by reading the report extract file data set records.

However, you have three alternatives to gradually convert away from using special dates:

- Using system-managed tape, an ACS routine can assign a management class corresponding to a special date. DFSMSrmm uses this management class to relate to an associated VRS definition to provide the required retention criteria.
- A DFSMSrmm installationwide exit can provide a VRS management value that is used in a similar way to the management class.
- Use management class RETENTION LIMIT.

Note: The SMS management class takes precedence over the VRS management value.

The retention hierarchy is:

ABEND/OPEN ⇒ VRS DSN ⇒ MC / MV ⇒ defaults

While the EDGCDYNM program allows for processing of existing special expiration dates and the creation of VRS management values and definitions, it is the installation’s responsibility to ensure that:

- Any new retention criteria are defined as VRS management values and VRS definitions
- Either an appropriate DFSMS ACS routine or DFSMSrmm installation exit EDGUX100 is installed and active to set VRS management values from special expiration dates.

With APAR OW30969 a new sample of the EDGUX100 installation exit, called EDGCVRSX, is available in SAMPLIB that uses a dynamic table driven assignment method based on data set names and keyword dates. We

recommend that those converting to DFSMSrmm use the EDGCVRSX sample and the table generated by the conversion tools.

DFSMSrmm flags a data set if the job that writes to it leaves the data set open. This can occur only through failure of the MVS system, and not as a result of abends. Open data sets are retained by VRS processing, and the specified or calculated expiration date is kept. Thus manual intervention might be required to release the volume if you specified a long retention period. Otherwise, you can use the special OPEN VRS retention to establish how to manage this situation. The new VRS option RELEASE(EXPIRYDATEIGNORE) can be used on both ABEND and OPEN VRSS so that any default or specified expiration date is ignored when a job abends or leaves a data set open.

Pay attention to the GDG processing, which sometimes differs between the two products. For example, if you generate a duplicate generation for a data set, TLMS uses both generations with the same name in the count of the GDG versions. DFSMSrmm uses only the latest generation and releases the other one.

TLMS allows the specification of full or qualified data set names, whereas DFSMSrmm allows full generic specification qualified by GDG or non-GDG.

The conversion process produces VRS definitions for non-GDG data sets. Review these after conversion for any data sets that may be GDGs, particularly if you are using CYCLES control. A NOGDG VRS definition can retain GDG data sets, but only if the data set mask is generic and includes a generic character that covers the last qualifier of the GDG data set; for example:

Q A.B.

is translated to
DSN('A.B.**')

this matches to both 'A.B.C', a non-GDG; and 'A.B.G0010V00', a GDG.

You can find an example of a sample conversion of various RMF retention combinations in D.4, "Sample VRS Conversion from TLMS RMF" on page 668.

5.3.4 Movement Policies

TLMS supports up to five storage locations. Each location can be one of three types of storage:

- Undefined—usually standard racking as in the tape library
- Cabinets—one or more cabinets, each containing slots. These would normally correspond to fireproof safes.
- Boxes—these would normally be located in an offsite store.

TLMS manages the movement of tape volumes between locations through a batch program that checks on your installation-defined storage location policies. These policies are based on data set name, fully or partially qualified, and are further qualified by job name or owner. The storage location policies are defined as part of the retention policies in the RMF.

TLMS automatically checks tape volumes meeting the criteria and assigns the appropriate storage location and slot or box number. In addition, it generates a report indicating the current location, the destination location, and the slot or box numbers for tape volumes that require movement between locations.

DFSMSrmm uses your installation-defined VRSs (location and retention management policies) to control the movement of tape volumes between the storage locations that you have defined.

To define a VRS, you can use:

- A data set name or data set mask
- A job name or job mask
- A VOLSER number or VOLSER mask
- The management class or VRS management value

When a tape volume is to be moved from a tape library to a remote storage location for disaster recovery, DFSMSrmm assigns a store bin (slot) number, a store ID (location name), and a store date.

This target location does not become the current location of the volume until confirmed by the operator or tape librarian.

Further volume movement of a volume is prohibited until the existing movement is confirmed.

Movement confirmation can be for one of the following:

- Individual volume
- All volumes from one location to another
- All volumes from one location
- All volumes to a location
- All volumes ready to be scratched
- All volumes not ready to be scratched
- Any combination of the above

DFSMSrmm supports an almost unlimited number of storage and loan locations. You should be able to convert your TLMS storage locations to DFSMSrmm without hitting a limit caused by their number. In DFSMSrmm there are three built-in storage locations, all managed by bin number (slot). By using LOCDEF commands in the DFSMSrmm PARMLIB, however, you can use installation-defined names, including those you are familiar with under TLMS, and decide whether or not to have them shelf managed.

When you convert to DFSMSrmm you have the choice of using the storage location names you currently use with TLMS, or of using different names. You may want to use different names, because DFSMSrmm allows up to eight-character names rather than two-character names. If you decide to use the DFSMSrmm built-in storage locations, local, distant, and remote, you have to convert the location names and restrict them to three storage locations. Using a combination of built-in and installation-defined storage locations requires some conversion of names. When you decide to use different names under DFSMSrmm, you can translate the TLMS storage location names to DFSMSrmm storage location names during either the EDGCDYNM run or the EDGCNVT run. To convert the names with EDGCDYNM, use the TLMSLOCN RMMLOC keyword to specify a different name. To convert the names with EDGCNVT, use the LOCDEF SYSIN statement and the IF STORLOC statement to specify a different name.

DFSMSrmm also supports an almost unlimited number of loan locations. If you want to use these as a replacement for any of the TLMS storage locations, use the LOCDEF statement in EDGCNVT to identify the location as a type of loan.

When running EDGCDYNM, treat it as a storage location and use the TLMSLOCN and DFRMMLOC statements.

All volumes defined to DFSMSrmm have a rack number assigned to provide shelf management while in the data center and a unique external identifier for the volume. EDGCDYNM and EDGCNVT use the TLMS VOLSER as the DFSMSrmm rack number.

Shelf management in DFSMSrmm storage locations is optional, but if provided, it is done using bin numbers. Each bin number represents a single slot in which to store a volume. The EDGCDYNM OPTIONS statement lets you control what the conversion process does for bin numbers. You know, from the TLMS015 report, whether your storage locations are managed as cabinets or boxes or are undefined storage. You must decide how best to convert the information to DFSMSrmm. Both the VOLLOCN=PASS and VOLLOCN=PROCBIN options pass on bin information; PASS passes on the cabinet and box numbers, and PROCBIN generates bin numbers starting from 000001. Here are some options:

- Specify VOLLOCN=PROCLOC and use DFSMSrmm inventory management to assign bin numbers and locations.
- Use VOLLOCN=PROCBIN and have bin numbers generated for you. These are passed from EDGCDYNM to EDGCNVT, and, although they are simply numeric bin numbers starting from 000001 through 999999, you can use them either with the built-in or installation-defined storage locations. If you decide on non-shelf-managed storage locations, EDGCNVT simply ignores the numbers.
- Specify VOLLOCN=PASS and use the existing cabinet and box numbers under DFSMSrmm. Use of this option requires you to make some changes to the sample extract and conversion programs, as the cabinet and box numbers are only four digits, and DFSMSrmm requires six digits.

Cabinet numbers could be expanded to six digits and used as bin numbers with installation-defined storage locations. For example:

CCnn could be expanded to either
CC00nn or
CCnn00

You could then define bin numbers as CC0001 through CC0099, or CC0100 to CC9900, always leaving the last two digits as zero (00). A small change to EDGCDYNM just before label PVMFS120 could insert 00 into the cabinet number. DFSMSrmm shelf management is very similar to the cabinet management provided by TLMS. If the location is NL and the cabinet number is AA23, the DFSMSrmm bin number becomes NLAA23. This is a value that can readily be identified by your operations staff. If you would rather use a different approach for converting cabinet numbers or you use box numbers, you must modify the sample code.

Box slot numbers are not kept in the VMF, but you could process them in a similar way to the cabinet numbers and generate the last two digits in numerical sequence for each volume found in the same box. For example:

BBBB could be expanded to
BBBBnn

You could then define bin numbers BBBB01 through BBBB99 to DFSMSrmm. Be sure to define the location as an installation-defined storage location by using LOCDEF. Remember that DFSMSrmm does not provide box

management, and can only use the box and slot numbers as conventional open shelving.

5.3.5 Inventory Management and Reports

TLMS provides batch facilities to perform the following functions:

- Batch update and reporting
- Tape retention and expiration
- Storage location management
- Backup, recovery, and verification of databases

DFSMSrmm provides similar functions, except batch update and reporting, with the EDGHSKP, EDGBKUP, and EDGUTIL utilities (for more details, see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932).

You must execute EDGHSKP with the DFSMSrmm subsystem active. You can execute EDGBKUP and EDGUTIL independently of the subsystem.

The utilities have the following functions:

- EDGHSKP
 - Processes vital records
 - Performs storage location management
 - Performs expiration processing
 - Creates an extract data set
 - Backs up the DFSMSrmm CDS
 - Backs up and clears the DFSMSrmm journal
- EDGBKUP
 - Backs up the DFSMSrmm CDS
 - Backs up the DFSMSrmm journal
 - Recovers and restore DFSMSrmm CDS
 - Reorganizes DFSMSrmm CDS
- EDGUTIL
 - Verifies DFSMSrmm CDS⁴
 - Mending the DFSMSrmm CDS

DFSMSrmm does not provide any batch update utility. However, the DFSMSrmm TSO commands ADD, CHANGE, and DELETE can be executed in batch to achieve the same result. The ISPF dialog has the option to generate and group together TSO commands for later batch processing.

TLMS provides batch reports. These may be supplemented with the use of the optional CA-EARL product, which allows you to customize your own reports from the VMF.

DFSMSrmm provides three reporting programs:

- EDGRPTD—which processes the report extract file to produce:
 - Volume movement reports
 - Moved from storage locations
 - Ready-to-Scratch sorted by RACK
 - Sorted by BIN number
 - Sorted by OWNER

⁴ The EDGUTIL function can also provide consistency between the CDS and the TCDB in a system-managed tape library.

- Moved to storage locations
 - Sorted by RACK number
 - Sorted by OWNER
- Volume inventory reports
 - Moved from storage locations
 - Sorted by volume serial number
 - Sorted by RACK number or BIN number
 - Sorted by OWNER
- Volume inventory reports
- EDGAUD—which processes DFSMSrmm SMF records to produce:
 - Security report (tracking information for classified data)
 - Audit report (CDS changes)
- EDGJRPT—processes the extended extract file
 - Pull list for scratch tapes sorted by volume serial number
 - Pull list for scratch tapes sorted by data set name
 - Inventory list by volume serial number
 - Inventory list by data set name
 - Inventory of data set including number of KB used
 - Inventory of volume serial number by location
 - Inventory of data set names by location
 - Inventory of BIN numbers by location
 - List of all data set names at loan location
 - List of all volume serial numbers at loan location
 - List of all multiple volume, multiple data sets
 - Movement report including the first data set name on the volume
 - Movement report sorted by storage location bin number
 - Movement report sorted by volume serial number
 - Inventory list sorted by volume serial number including volume count

The DFSMSrmm TSO commands LIST and SEARCH may be executed in batch to produce a wide variety of reports. You may additionally use DFSORT and ICETOOL to write your own customized reports.

Examples of EDGRPTD and EDGAUD sample jobs, REXX programs, and DFSORT's ICETOOL reports are in Chapter 14, "Extending DFSMSrmm Reporting" on page 533 and Appendix D, "Sample Exits and Reports" on page 603.

The TLMS scratch list report is shifted one day in the DFSMSrmm environment because the DFSMSrmm report lists volumes that are in *PENDING RELEASE* status. Those volumes are then returned to *SCRATCH* status during the next housekeeping process, which usually runs next day.

We recommend that you move to the scratch pool the cartridges that will return to scratch only when they are no longer in *PENDING RELEASE* status. You can accomplish this in your batch housekeeping jobstream by using a step that lists the volumes in *PENDING RELEASE* status before you run the EDGHSKP step. These are the volumes that the TLMS scratch report lists as "new scratched volumes" because of the DFSMSrmm *PENDING RELEASE* status, which shifts by one day the matching of the TLMS and DFSMSrmm reports. Alternatively, you can execute EDGHSKP with the EXPROC parameter a second time to return pending release volumes to scratch. If you specify the *RELEASE(SCRATCHIMMEDIATE)* operand on the ADDVRS subcommand and the *VRSEL(NEW) PARMLIB* option, only a single run of inventory management EXPROC processing is required to return volumes to scratch. When the

VRSEL(OLD) PARMLIB option is in use, DFSMSrmm ignores any VRS release options that may have been set for a volume.

5.3.6 Exits

DFSMSrmm provides source code for the installationwide exits installed as part of DFSMSrmm during the SMP/E apply process. The source code is supplied for:

- IGXMSGEX (DFSMSrmm uses the DFSMSdfp MSGDISP exit to update tape drive displays)
- CBRUXENT (OAM cartridge entry exit)
- CBRUXEJC (OAM cartridge eject exit)
- CBRUXCUA (OAM cartridge change use attributes exit)
- CBRUXVNL (OAM volume not in library exit)

Before the SMP/E accept, the source code is in SMPSTS; after the accept, the exits are in AEDGSRC1.

If your installation already uses one or more of these exits in the TLMS environment, you must ensure that the correct action will be performed during the conversion. The steps to follow are described detailed in Chapter 10, “Parallel Running and Validation” on page 427 and Chapter 11, “Cutover to Production” on page 455.

TLMS uses the CBRUXENT exit to interface the IBM 3494 and 3495 Automated Tape Library Dataserver. The DFSMSrmm version of this exit can be tested once you have completed the conversion process.

The CBRUXVNL exit is optional, but if you use it you must follow the same process as the CBRUXENT exit to test it in the DFSMSrmm environment.

The installationwide exits that you installed with DFSMSrmm for IGXMSGEX, CBRUXENT, CBRUXEJC, and CBRUXCUA replace any existing exits you had previously installed. Decide whether you still require the function your previous exits provided, even if for a short time while introducing DFSMSrmm. If you no longer need the function, proceed with the next implementation step. If you still require the function, consider how to implement it, while still providing the function required by DFSMSrmm.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more details on the DFSMSrmm programming interface to use with the IGXMSGEX and the OAM exits. The supplied source code for these exits shows how to call the programming interface. You can either modify the supplied exit to call a modified version of your old exit or modify your old exit to call the DFSMSrmm programming interface. Consider how you might provide an option to disable your code without another code modification. In either case, install the changed software using an SMP/E USERMOD.

If your installation has DFSMSshm installed and TLMS is using the ARCTVEXT exit, you do not need to merge the TLMS coding for the exit with the DFSMSrmm coding to notify both tape management products when running in parallel. Both ARCTVEXT and EDGTVEXT can run in parallel. EDGTVEXT is the DFSMSshm interface to call DFSMSrmm to release tapes that are no longer needed by DFSMSshm. If you are using a level DFSMSrmm prior to Version 1 Release 4, refer to Appendix D, “Sample Exits and Reports” on page 603 to merge the ARCTVEXT code from CA-1 with DFSMSrmm. See Appendix D, “Sample Exits and Reports” on page 603.

On the basis of the decisions you have made about VRS management values and scratch pooling, you can update the supplied sample installation exit, EDGUX100, to perform the function you require. Follow these steps:

1. Copy the sample installation exit, EDGUX100, and use this copy as a base for your exit.
2. Update the exit if needed.
3. Build an SMP/E USERMOD to apply the updated source code for EDGUX100, including the necessary JCLIN statements to get the EDGUX100 load module added to the LINKLIB target library.
4. Receive and apply the USERMOD.
5. Your new exit is now ready for use on your system.

You can use the same process to install the EDGUX200 code, if needed. If you have a non-IBM robotic tape library, you can update the sample EDGUX200 to communicate with the library host software to ensure that the external inventory is updated dynamically as volumes return to scratch status.

For more information about DFSMSrmm exits refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

5.3.7 PARMLIB Options

During the analysis process you must check the TLMS PARMLIB options that can affect the DFSMSrmm environment. Carefully review all TLMS parameters before proceeding to the next step.

Use the TLMSOPTS program to automatically produce a listing of TLMS options. Figure 67 shows the sample output.

your company name CA-DYNAM/TLMS 5.4 9506TL54	INSTALLATION OPTIONS MAINTENANCE PROGRAM	TLMSOPTS PAGE 2 09 006 1995 11.59.03
---	--	---

OPTIONS SETTING LISTING

```

DATACENTER=DC
DATE=MM/DD/YY
DBLDRIVE=NO
DBLTIME=000000
LABELS=NO
LOGID=240
MODIFY=YES
OVERRIDE=NO
PAGESIZE=58
PROTECT=ALL
QSIZE=20
RECOVERY=ALTLOG
ROUTEAUX=2
ROUTEINQ=14
ROUTELBL=13
SMC=NO
SUBTASK=NOAUTO
VSNREQD=YES
WAIT=20

```

Figure 67. Sample TLMSOPTS Output

Here are the options to check:

DATACENTER=____ Is the name of the location nearest the tape drives. In DFSMSrmm the equivalent is the SHELF location.

DATE=____	Is the date format used. You can specify the same date format used by TLMS in the DFSMSrmm OPTIONS with the DATEFORM parameter.
DBLDRIVE=____	Indicates whether or not you want to ensure that double-opens of a data set are on the same physical drive. There is no equivalence in DFSMSrmm.
DBLTIME=____	Tells how much time you want between two opens for output for the same data set. There is no equivalence in DFSMSrmm.
LABELS=____	Indicates whether or not you want to print gummed labels. In DFSMSrmm you can use the EDGUX100 exit.
LOGID=____	Is the SMF record type used by TLMS if the RECOVERY=SMF options in effect. The default is 240. You can use the same record type for DFSMSrmm auditing and security SMF records, but be careful when selecting records for processing during times when both products are running.
MODIFY=____	Indicates whether or not you want to use the OMS/MONITOR subsystem command interface. An equivalent interface is already integrated in DFSMSrmm.
OVERRIDE=____	Indicates whether or not to allow the operator to override the data set protection feature when a nonscratch volume is detected for use as an output data set. In DFSMSrmm the equivalent is the MASTEROVERWRITE parameter of the EDGRMMxx PARMLIB member.
PAGESIZE=____	Sets the maximum number of lines on a printed page. In DFSMSrmm the equivalent is the LINECOUNT parameter of the OPTION control statement.
PROTECT=____	Restricts the "crash-protected" processing of TLMS to selected data sets. In DFSMSrmm you can control this process with the ABEND and OPEN VRS retention policies.
QSIZE=____	Is the size of the CSA storage area that will contain TLMS messages. There is no equivalence in DFSMSrmm.
RECOVERY=____	Indicates whether to create backup records and where to write them. In DFSMSrmm you use journal records and the EDGHSKP or EDGBKUP utilities for backup and restore purposes.
ROUTEAUX=____	Sets a route code to use for the auxiliary message console. There is no equivalence in DFSMSrmm. However, you can set a route code for every DFSMSrmm message by using the MVS MPF facility.
ROUTEINQ=____	Sets a route code to use for the inquiry console. There is no equivalence in DFSMSrmm. However, you can set a route code for every DFSMSrmm message by using the MVS MPF facility.
ROUTELBL=____	Sets a route code to use for the gummed-labels printer. There is no equivalence in DFSMSrmm.
SMC=____	Indicates whether or not you want to use the system message control option. There is no equivalence in DFSMSrmm.
SUBTASK=____	Indicates whether or not you want to start the inquiry module automatically with the online recorder. In DFSMSrmm all functions are integrated in the same started task.

VSNREQD=____ Indicate whether or not you want to require that a VOLSER be supplied before the volume is opened for output. There is no equivalence in DFSMSrmm.

WAIT=____ Selects a time interval for the wait time to avoid S522 abends. There is no equivalence in DFSMSrmm.

You must pay attention to the TLMSDTAB table that specifies the default retention days for the data sets not covered by an RMF definition. The default retention is three days. You must modify the DFSMSrmm option RETPD to reflect the same default retention period as the TLMS TLMSDTAB definition.

You must also review the DFSMSrmm PARMLIB definitions. The most important for the conversion are:

- OPTION statement
 - OPMODE parameter
 - RETPD parameter
 - BLP parameter
 - VRSJOBNAME parameter
 - UNCATALOG parameter.
 - VRSEL(NEW) Specify to again advantages of the new retention and policy options on a VRS
- VLPOOL statement
(only if you use multiple scratch pool support)
- LOCDEF statement
(only if you use location names different from the installation defaults).

5.3.7.1 Data Set Overwrite

TLMS supports two volume statuses:

- PRIVATE
The volume contains valid data.
- SCRATCH
The volume can be used for scratch output requests.

In a DFSMSrmm environment you have three different volume statuses:

- MASTER
The volume contains valid data.
- SCRATCH
The volume can be used for scratch output requests.
- USER
The volume is assigned to a user, and it can be overwritten, rewritten, and used repeatedly until the user returns it to the scratch pool.

DFSMSrmm can manage the overwrite of a MASTER volume with the PARMLIB option MASTEROVERWRITE. You can tailor this option to meet your installation's needs.

Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for a detailed explanation of the function.

5.3.7.2 BLP and NL Processing

In a TLMS environment, because NL tapes do not have internal labels and BLP tapes bypass standard labels, you must supply the VOLSER in the JCL or from the operator console.

BLP and NL processing differs in an DFSMSrmm environment. All scratch tapes must be standard label. When DFSMSrmm successfully reads the VOL1 label for a volume mounted as a result of a nonspecific request, it provides the actual VOLSER to OPEN for use in related processing such as cataloging of the data. This replaces the logical VOLSER (Lnnnnn) generated by OPEN in these circumstances.

You can enable the use of NL output to scratch volumes by creating a profile in the RACF FACILITY class: STGADMIN.EDG.NOLABEL.volser. Once the profile is created, the function is enabled, and only those who have access to the profile can use this function. DFSMSrmm normally supports only standard label scratch tapes for NL tapes, and relabels them at time of use to NL and standard label at return to scratch. To use NL scratch tapes use the system tape exit IFG019VM. DFSMSrmm now ships a sample IFG019VM exit as member EDG019VM in SAMPLIB. The sample exit checks for a supported request and issues a WTOR to the operator to obtain the volume serial number. After installing the exit, testing should be done to ensure that the exit supports the installation's needs for NL tape.

For BLP, further processing is determined by the BLP option set in PARMLIB. Processing is as follows:

RMM	This is the default and allows BLP processing of USER status tapes and BLP input from MASTER status tapes.
NORMM	BLP can be used for reading and writing of master and user status tapes and for output to scratch tapes. BLP reading of scratch tapes is not supported.

When BLP or NL is used with scratch tapes, DFSMSrmm still changes the volume to master status, but sets the initialize release action to ensure that the volume has a valid standard label before it is returned to scratch status.

Some changes are required for how and when DFSMSrmm records information for tapes and data read and written using BLP.

- For BLP output to a nonspecific volume, the file sequence number is not checked. For example, LABEL=(2,BLP) could be used. For non-BLP requests, the restriction of using LABEL=(1,label_type) is still enforced.
- The data set information for files processed with BLP is only updated for output to the first file on a volume. All other types of BLP requests change only the volume information such as the date last read and date last written. For BLP output requests, the data set information is only updated in the DFSMSrmm CDS when the data set is closed.

You can enable the use of NL output to standard label (SL) scratch volumes by creating a profile in the RACF FACILITY class: STGADMIN.EDG.NOLABEL.volser. Once the profile is created, the function is enabled, and only those with access to the profile can use this function.

5.3.7.3 DFSMSHsm Interface

TLMS and DFSMSrmm process the release of DFSMSHsm tapes differently. When TLMS is informed through the ARCTVEXT exit that a DFSMSHsm volume is no longer required, it changes the volumes creating job name to a different predefined name (by default, either HSMCOPY, HSMSCRDC, or HSMSCROF). Job name-qualified RMF definitions are then used to retain the data set for typically one day before being released to the scratch pool.

When DFSMSrmm is informed by DFSMSHsm, through the EDGTVEXT programming interface, the volume is immediately marked PENDING RELEASE and will be returned to the scratch pool following the next run of the housekeeping EXPROC process.

Thus DFSMSrmm can release DFSMSHsm tapes at a different time than TLMS. You must take this difference into account when comparing scratch reports for the two systems.

5.3.7.4 Tape Initialization

TLMS provides an interface to IEHINITT, the initialization utility. This interface consists of a hook that passes control to a TLMS routine when a tape initialization request is received.

If the VOLSER is not yet present in the VMF database, the initialization request is allowed.

If the VOLSER is known, two situations are possible:

- **SCRATCH**—The operator is prompted to allow or not allow the initialization of the volume.
- **PRIVATE**—TLMS sends a message to notify the operator to notify that the volume is not scratch, but still prompts the operator to allow the initialization to continue.

In a DFSMSrmm environment, the initialization process is different. A utility, EDGINERS, should be used instead of IEHINITT for the tapes defined in the DFSMSrmm database. You must ensure that your operational procedures are updated in order to use the new DFSMSrmm utility after the conversion process.

DFSMSrmm does not require that you initialize tape volumes before they are available as scratch, but it prevents you from using as scratch a volume that requires initialization and will not include the volume in a scratch pull list. When adding scratch volumes to DFSMSrmm, you have the choice of deciding whether they need to be initialized or not.

Operators should use the DFSMSrmm EDGINERS utility (instead of IEHINITT) to initialize and erase tape volumes. EDGINERS automatically defines new volumes it initializes or erases to the DFSMSrmm CDS. For an existing volume, DFSMSrmm ensures that the proper volume is mounted and the requested action is actually required. You can request DFSMSrmm to automatically initialize scratch volumes when you add them to DFSMSrmm.

5.3.7.5 Security

TLMS provides security options that control online inquiry and update. To change these options or add new users after installation, you must build an SMP USERMOD and then restart TLMS.

DFSMSrmm uses profiles in the FACILITY class of RACF to determine a user's authority to request actions in DFSMSrmm. DFSMSrmm uses profiles with the STGADMIN.EDG. prefix. You can control every command or DFSMSrmm function with this integrated RACF interface.

TLMS can use a RACF interface called TLMSRACF that allows you to automatically delete the tape volume from the RACF database when a tape is scratched.

The RACF tape profile must be already defined.

If in your actual environment you are using the RACF TAPEVOL or TAPEDSN classes to protect tape volumes, you no longer need to define the tape volumes to RACF because DFSMSrmm can do this task automatically.

You must specify either TPRACF(A) option in the DFSMSrmm PARMLIB member or use TPRACF(P), which allows you to use predefined RACF tape profiles. Otherwise you must use TPRACF(N).

Additional information can be found in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

5.3.7.6 Interfaces

Before the conversion there is additional information that you need to consider about interfaces.

SMS Interface: If you have an IBM 3494 or 3495 Automated Tape Library Dataserver, you must ensure that it works correctly in the DFSMSrmm environment. Change the OAM exits using the DFSMSrmm version and provide the correct input to the extract program for those volumes that belong to the tape library.

Define an DFSMSrmm location with the same name as the tape library name, using a location type of ATL. The TLMS location name that you are using for the ATL tapes must be translated to this DFSMSrmm location.

BTLS Interface: If your installation uses BTLS to drive IBM 3494 and/or 3495 Automated Tape Library Dataserver operations, you will need to notify BTLS when a volume must be returned to scratch status. In a TLMS environment this is done through a batch job that gets information from the TLMS database and then updates the BTLS catalog. In DFSMSrmm you can use a simple REXX CLIST that synchronizes the BTLS catalog with the DFSMSrmm CDS. For a sample CLIST see Appendix D, "Sample Exits and Reports" on page 603.

Short-on-Scratch Processing Interface: DFSMSrmm offers an integrated SMS interface that updates the TCDB automatically. In addition, without manual intervention you can free volumes that are in PENDING RELEASE status when a short-on-scratch condition is detected inside the automated tape library dataservers. Use the SCRATCHPROC(RMMSCR) PARMLIB option, where RMMSCR is the name of the procedure DFSMSrmm starts to replenish scratch volumes in the automated tape library dataservers.

TSO and ISPF Interfaces: DFSMSrmm has a simple TSO interface that uses a set of powerful commands to drive all tape activities.

There is also an ISPF interface that offers menus for end users, librarians, administrators, and support personnel. You can choose a full DFSMSrmm dialog or a local defined dialog.

You can tailor the correct ISPF dialog for each user who will have access to DFSMSrmm.

DFSMSrmm can use any DFSMSrmm command in batch mode using the IKJEFT01 or IKJEFT1A TSO batch interface. It can also process any DFSMSrmm command via the DFSMSrmm API. To use the DFSMSrmm API you need to code in the High Level Assembler language. A preferred method is to use the DFSMSrmm commands from a REXX environment where you can retrieve information back as variables.

Other Interfaces: TLMS provides an online transaction that accesses CICS, TSO, ISPF, and ROSCOE environments. You can use inquiry and update processing through one of these interfaces, or you can use an operator console to issue TLMS commands.

If you have any of these interfaces running in your TLMS tape environment, you must change them to reflect the equivalent function in DFSMSrmm.

5.3.7.7 Tape Pool Management

TLMS allows you to define ranges of tape volumes in a tape library to a specific pool. It restricts scratch requests to a specific pool based on the creating user and/or data set name. For example, you define ranges of tape volumes dedicated to a specific application use only.

Multiple Scratch Pool Support: TLMS supports multiple scratch pool addressing by using a table called TLMSRTAB where you can specify a data set name and the range of volumes associated with it. If the volume opened for output does not fall within the specified range, TLMS denies the request and dismounts the volume.

If your installation uses the TLMSRTAB table to access different scratch pools, you must use the DFSMSrmm multiple scratch pools facility. With this support, you will be able to direct new tape data sets to specific scratch pools on the basis of job name and data set name. Thus you can discretely manage sets of volumes on a client, application, or other basis.

DFSMSrmm can manage system managed tapes as well as any other pooling driven by an external product. You can use the DFSMSrmm installation exit EDGUX100 to support multiple scratch pools. The EDGUX100 exit can select a specific scratch pool for nonspecific tape volume output requests by using the data set name and jobname as the selection criteria.

Implementation for DFSMSrmm: DFSMSrmm decides which pool to use on the basis of the VLPOOL definitions in the DFSMSrmm PARMLIB member. This function is dependent on the correct VLPOOL definitions, but you can decide which pool to select either by system — by using the SYSID operand of the VLPOOL statement in the DFSMSrmm Parmlib member, or by using the EDGUX100 installation exit.

DFSMSrmm intercepts the mount messages issued on the system, and, for those for nonspecific volume requests, calls the installation exit to enable it to make a decision. DFSMSrmm saves the scratch pool selected by the exit for use in updating tape drive displays, updating the mount message, and validating the volume when one is mounted.

The exit can optionally request that DFSMSrmm prevent the cartridge loader from being indexed. Thus drives can be selectively preloaded with scratch volumes from specific pools, but scratch volumes from other pools can be manually mounted without the cartridge loader being emptied.

When the exit selects a pool, the only acceptable volumes are those in that pool. When DFSMSrmm selects the scratch pool without use of the exit, any scratch volume from any eligible scratch pool on that system is acceptable.

DFSMSrmm provides a working sample exit that implements a scratch pooling system based on job names and data set names; all you have to do is update a table containing job names, data set names, and pool prefixes in the EDGUX100 exit.

You can also specify different data set or job name pool tables for each system. If you specify the NAME operand in the VLPOOL command in PARMLIB member EDGRMMxx, the drive will display this name rather than the pool prefix.

Note: This specification allows DFSMSrmm to drive BTLS scratch pool selection. You can use data set names as well as job names to direct scratch pool selection.

5.4 Running the Extract Program

In this section we describe the sample programs and JCL for extracting data for conversion to DFSMSrmm from TLMS. The samples are as supplied in the DFSMSrmm product, EDGCMM01, at the time this book was written. Before running the extract program, you should read and be familiar with the *Data Extraction Programs User's Guide* supplied with the current version of EDGCMM01, as it will be maintained to reflect changes and enhancements.

Table 43 lists the samples for the extraction of data from TLMS.

Table 43. IBM-Provided Samples for TLMS		
Name	Location	Function
EDGCDYNM	SAMPLIB	Source for TLMS extract
EDGCMM01	SAMPLIB	Source and formatted user's guide
EDGJDYNM	SAMPLIB	JCL to execute EDGCDYNM

The sample JCL is set up to work in a "typical" installation. Some of the sample JCL is commented out so that you choose to use it or not, as is your preference. Procedures, which are normally part of other IBM products, are used to assemble and compile the sample code. Thus you do not have to assemble, link, and compile code before you can use it. If these procedures are not available to you, you can edit the JCL to uncomment the statements that execute the programs from a STEPLIB. If you choose to link-edit the programs yourself, you can use the standard link-edit options. None of the programs is reentrant.

5.4.1 Functional Description

The data extraction programs are designed to extract relevant information from the TLMS VMF and RMF TLMS016 report (Retention Management File report). The information is used to create four sequential output files containing a series of records that are input to the EDGCVNT program before they are loaded into the DFSMSrmm CDS (Table 44).

The LRECL of the output files is overridden during program execution. The value included in the JCL examples and samples is just an indication of the approximate record length.

<i>Table 44. EDGCDYNM Output Data Sets and Records</i>		
ddname	Output Description	Output Record
DEXTLIBD	• Data sets and volumes	• D (data set) record • L (volume) record
DEXTRBIN	• Empty bin numbers in the storage locations and empty rack numbers in the tape library	• E (empty rack/bin) record
DEXTVRS	Retention and storage location management policies	• K (VRS) record
DEXTOWN	• VRS and default owner information	• O (owner) record

Note that the data set and volume (D-type and L-type) records are sequence dependent — see Chapter 9, “Building the DFSMSrmm CDS” on page 403 for details.

5.4.2 Preparing to Run the Extract Program

Before you begin the extraction process, you should consider verifying the integrity of the TLMS VMF database to check for broken volume chains, expiration dates in error, or any other condition that can cause an error during data extraction.

You must run the appropriate TLMS utilities to verify the integrity of the database and correct the errors. The TLMS volume chaining verification system (TLMSV CVS) is a monitoring system that checks the validity of the links between volume base records. It explains the errors found in the VMF database and the actions taken for every error message.

Once you are sure that that you have a valid VMF database, you can proceed to the next step.

5.4.3 The EDGCDYNM Program

This section describes the setup and control of the TLMS EDGCDYNM data extraction program.

EDGCDYNM supports all current releases of TLMS. However, the correct level of TLMS must be specified at the time you assemble the sample source code. You use the *SYSPARM* assembler option to specify:

TLMS53 - for TLMS Release 5.3

or

TLMS54 - for TLMS Release 5.4.

See Figure 77 on page 211 for an example that specifies *SYSPARM(TLMS53)*. If you do not specify a SYSPARM value, the code is assembled for TLMS 5.4.

5.4.3.1 Input Files

You need the following DD statements for the input files:

SYSIN	Contains optional input parameters to control the execution of EDGCDYNM and the data extraction process
TLMSVMF	Defines the TLMS VMF, which contains the tape volume and data set information to be converted
TLMS016	Defines the TLMS Retention Selection Report file (TLMS report 016) which contains the volume retention and movement rules for converting to DFSMSrmm VRS definitions.

In addition, you need TLMS report 015, which lists the storage locations you use with TLMS. You will use the information in the report to define the TLMSLOCN and DFRMMLOC statements in the SYSIN data set.

5.4.3.2 Output Files

You need the following DD statements for the output files:

SYSPRINT	Contains informational and error messages associated with the data extraction process
DEXTLIBD	Library and data set records for input to the EDGCNVT program
DEXTOWN	Owner records for input to the EDGCNVT program
DEXTRBIN	Empty rack and bin records for input to the EDGCNVT program
DEXTVRS	VRS records for input to the EDGCNVT program

5.4.3.3 Input Parameters

The EDGCDYNM program accepts a number of optional control statements input in the SYSIN file. These statements are identified by a keyword starting in position 1, followed by one or more parameters starting in position 9. The statements are fixed format; keywords and operands must start in the indicated positions. Optional keywords and operands can be left blank.

Statements can be supplied in any order. Comment statements are identified by an asterisk (*) in position 1.

If you do not specify the control statements in the exact format, the EDGCDYNM program attempts to use the data you enter, and results could be unpredictable, including ABEND0C7, and ignoring of bin counts.

Processing Options: The OPTIONS statement is used to control the way in which EDGCDYNM processes the TLMS VMF and RMF data.

Figure 68 shows the format of the statement:

OPTIONS	OUTSERV=00000000	VOLCHAIN=cccccc	VOLLOCN=1111111
*	*	*	*
1	18	36	52

Figure 68. EDGCDYNM SYSIN: OPTIONS Control Statement

where:

- 00000000 specifies the processing for TLMS out-of-service volumes:

- Ignore - Specifies that out-of-service volumes are ignored and only empty rack records will be created.
- Process - Specifies that out-of-service volumes are processed as standard scratch or nonscratch volumes.
- *loanloc* - Specifies a one- to eight-character loan location name. Out-of-service volumes are processed as scratch or nonscratch and have their loan location set to the specified value.
- *cccccc* specifies the processing for volume chains:
 - Ignore - Specifies that EDGCDYNM process the VMF sequentially and ignore any volume chaining information. All volumes will have a volume sequence number and count of 1.

This option ensures that all volumes are converted, but it loses all multivolume count and chaining information.
 - Process - Specifies that EDGCDYNM attempts to process volume chaining information to produce correct volume sequencing and counts

Any errors in the VMF volume records may result in missing volumes in the converted data.
- *//////* specifies the processing for the volume's home and current location:
 - Pass - Specifies that EDGCDYNM does not convert the volume's current TLMS location. The DFSMSrmm current location and bin number are set to the TLMS location and cabinet or box number. Similarly, the TLMS RMF locations are forwarded into the corresponding DFSMSrmm VRS definitions.

The EDGCNVT program can be used to convert the TLMS information to DFSMSrmm data.

If TLMSLOCN statements are not supplied, the default processing option is PASS.
 - Procloc - Specifies that EDGCDYNM ignore the volume's current TLMS location and set an DFSMSrmm home and current location of SHELF. The TLMS RMF locations are converted according to the TLMSLOCN specifications.

The DFSMSrmm housekeeping processes, VRSEL and DSTORE, are used after conversion to assign current locations and bin numbers according to the converted VRS definitions.

If TLMSLOCN statements are supplied, the default processing option is procloc.
 - Procbn - Specifies that EDGCDYNM convert the TLMS locations according to the TLMSLOCN specifications. For those volumes in offsite locations, EDGCDYNM assigns a bin number. The bin numbers start at 1 for each location and are incremented by 1 for each volume assigned to that location.

Only one OPTIONS statement should be supplied; if not, the last statement encountered in the SYSIN file is used.

The default values for the OPTIONS statement are shown in the following example:

```
OPTIONS  OUTSERV=IGNORE  VOLCHAIN=PROCESS VOLLOCN=PROCLOC
```

VRS Name Prefix: TLMS retention management policies can define multiple volume movements between storage locations. The equivalent DFSMSrmm function requires the use of VRS name definitions to chain together the separate movement definitions. The EDGCDYNM extraction process automatically generates these VRS names according to an installation-defined prefix concatenated with an incrementing number to make it unique.

Figure 69 shows the format of the statement:

VRSNAMEP	PREFIX=ppppp
*	*
1	17

Figure 69. EDGCDYNM SYSIN: VRSNAMEP Control Statement

where:

- *ppppp* is a one- to five-character alphanumeric field beginning with an alphabetic.

The generated VRS NAME is extended to eight characters by appending a unique numeric portion.

If a VRSNAMEP statement is not supplied, a default prefix of VRSN is used.

Only one VRSNAMEP statement should be specified; if not, the last statement encountered is used.

VRS Management Value Prefix: TLMS allows the specification of retention management policies through the use of special expiration dates. The equivalent RMM function requires the use of VRS management values and user exit EDGUX100. The EDGCDYNM extraction process optionally can automatically generate these VRS management values according to an installation-defined prefix concatenated with the special Julian date. It also generates the corresponding VRS definition.

Figure 70 shows the format of the statement:

VRSMGMTP	PREFIX=ppp
*	*
1	17

Figure 70. EDGCDYNM SYSIN: VRSMGMTP Control Statement

where:

- *ppp* is a one- to three-character alphanumeric field beginning with an alphabetic.

If a VRSMGMTP statement is not supplied, special Julian dates are not processed and VRS management values or definitions are not produced.

Only one VRSMGMTP statement should be specified; if not, the last statement encountered is used. For example, a control statement of:

VRSMGMTP PREFIX=VDT

would produce, with a volume whose JCL expiry date was 98023, a VRS management value of:

VDT98023

and a VRS definition equivalent to the DFSMSrmm TSO subcommand:

```
ADDVRS DSN('VDT98023') LASTREFERENCEDAYS COUNT(23)
```

Note: The conversion process can only generate management values and VRS definitions for those special dates already in the VMF. For continued use of special dates the DFSMSrmm user exit EDGUX100 (or an appropriate ACS routine) must be used to set the management value for new tape data sets, and the associated VRS definitions must exist.

Storage Location Mapping: The TLMSLOCN control statement optionally specifies the mapping between TLMS location IDs and DFSMSrmm storage locations.

Figure 71 shows the format of the statement:

TLMSLOCN	TLMSID=tt	RMMLOC=llllllll
*	*	*
1	17	27

Figure 71. EDGCDYNM SYSIN: TLMSLOCN Control Statement

where:

- *tt* is the TLMS storage location ID.
- *llllllll* is the corresponding DFSMSrmm location name. Specify any one- to eight-character name that is the location name you plan to use under DFSMSrmm. You can specify the TLMS location ID and optionally use EDGCNVT to translate it to the value you want to use.

One TLMSLOCN statement should be supplied for each TLMS location ID to be translated.

The following example illustrates a typical mapping:

```
TLMSLOCN TLMSID=TL RMMLOC=SHELF
TLMSLOCN TLMSID=DC RMMLOC=SHELF
TLMSLOCN TLMSID=CL RMMLOC=SHELF
TLMSLOCN TLMSID=D2 RMMLOC=D2
TLMSLOCN TLMSID=FS RMMLOC=FS
TLMSLOCN TLMSID=OS RMMLOC=OS
```

Storage Location Values: The DFRMMLOC control statement optionally defines a DFSMSrmm storage location. Its principal use is to define system-managed tape libraries (automated and manual) and the maximum number of bins to be used for DFSMSrmm storage locations (local, distant, and remote).

Figure 72 shows the format of the statement:

DFRMMLOC	RMMLOC=lllllllll	RMMTYP=rrrrrrrrr	BINS=nnnnnn
*	*	*	*
1	17	33	47

Figure 72. EDGCDYNM SYSIN: DFRMMLOC Control Statement

where:

- *////////* is the DFSMSrmm location name. Specify any one- to eight-character name that is the location name you plan to use under DFSMSrmm. You can specify the TLMS location ID and optionally use EDGCNVT to convert it to the value you want to use.
- *rrrrrrrr* is the type of DFSMSrmm location, one of SHELF, STORE, MTL, or ATL.
- *nnnnnn* is the six-digit number, from 0 to 999999, of storage bin numbers (slots) in the DFSMSrmm storage location.

It should be specified only for shelf-managed storage locations. You do not need to specify a BINS value as EDGCDYNM generates its own numbers as required starting from 000001. Rack numbers are automatically generated for all volumes no matter where the volume is located.

If DFRMMLOC statements are not supplied, only the basic DFSMSrmm locations shelf, local, distant, and remote are recognized, and no empty bin numbers are generated for the DFSMSrmm storage locations.

The following example illustrates a typical mapping:

```
DFRMMLOC RMMLOC=FS      RMMTYP=STORE    BINS=001000
DFRMMLOC RMMLOC=D2      RMMTYP=STORE    BINS=000500
DFRMMLOC RMMLOC=OS      RMMTYP=STORE    BINS=
DFRMMLOC RMMLOC=ROBBY   RMMTYP=ATL
```

Volume Media Names: The VOLMEDIA control statement optionally specifies a volume prefix and the corresponding media name, type and home location. If VOLMEDIA statements are not supplied, a default media name of 3480 is assigned to all volumes.

Prefixes are scanned in the order of the control statements; to ensure correct processing, the most restrictive prefixes should be specified first.

Figure 73 shows the format of the statement:

VOLMEDIA VOLPFX=vvvvvv VMEDIA=mmmmmmmm VMTYPE=****t HOMELOC=11111111				
*	*	*	*	*
1	17	31	47	61

Figure 73. EDGCDYNM SYSIN: VOLMEDIA Control Statement

where:

- *vvvvvv* is the one- to six-character VOLSER prefix.
- *mmmmmmmm* is the corresponding DFSMSrmm media name. This should match the media names you specify on the VLPOOL and LOCDEF commands in the DFSMSrmm PARMLIB member.
- *****t* is the optional media type.

This is one of:

- REEL - non-cartridge-type tape reel
- 3480 - 3480 type cartridge (18 track)
- CST - 3490 Cartridge System Tape (36 track)
- ECCST - 3490 Enhanced Capacity Cartridge System Tape (36 track),

and, if specified, overrides any values obtained from the TLMS VMF records.

- *////////* is the optional home location name. If specified, this location name must match one of the DFSMSrmm location names, SHELF or any system-managed library name. Storage locations cannot be specified for a home location—EDGCNVT ignores them.

The default home location is SHELF.

Note: The VMEDIA names must match the DFSMSrmm VLPOOL definitions in the EDGRMMxx member of PARMLIB. Otherwise, errors will occur when DFSMSrmm is active.

Skip Processing VOLSERS: The SKIPVOL control statement optionally provides a means to skip processing one or more VOLSERS in the TLMS VMF. This skipping might be required if, for example, there was an error that could not be corrected in the TLMS data. The skipped volumes can then be manually added after conversion.

Figure 74 shows the format of the statement:

```

SKIPVOL  VOLSER=vvvvvv
*          *
1          17

      OR:

SKIPVOL  VOLRNG=vvvvvv-vvvvvv
*          *      *
1          17      24

      OR:

SKIPVOL  VOLPFX=pppppp
*          *
1          17

```

Figure 74. SKIPVOL Control Statement

where:

- *VOLSER* specifies a single VOLSER.
- *VOLRNG* specifies a range of VOLSERS.
- *VOLPFX* specifies a VOLSER prefix.
- *vvvvvv* is a six-character VOLSER.
- *pppppp* is a one- to six-character VOLSER prefix.

Default Volume Owner ID: DFSMSrmm requires that all nonscratch volumes be assigned an owner, represented by a one- to eight-character owner identifier. Because this function is not present in TLMS, a default owner ID must be provided. The DEFOWNER statement lets you provide an owner ID different from the default of LBRARIAN.

Figure 75 on page 210 shows the format of the statement:

DEFOWNER	00000000
*	*
1	10

Figure 75. EDGCDYNM SYSIN: DEFOWNER Control Statement

where:

- 00000000 is a one- to eight-character alphanumeric field beginning with an alphabetic.
- If a DEFOWNER statement is not supplied, a default owner of LBRARIAN is used.

Only one DEFOWNER statement should be specified; if not, the last statement encountered is used.

VRS Owner ID: DFSMSrmm requires that all VRS definitions be assigned an owner, represented by a one- to eight-character owner identifier. TLMS provides a 25-character authorization field in the RMF report. EDGCDYNM can attempt to use this field by extracting the first eight nonblank characters to form an owner ID. This ID may not always be suitable or valid, so the VRSOWNER statement gives the option of providing a fixed default owner ID.

Figure 76 shows the format of the statement:

VRSOWNER	00000000
*	*
1	10

Figure 76. EDGCDYNM SYSIN: VRSOWNER Control Statement

where:

- 00000000 is a one- to eight-character alphanumeric field beginning with an alphabetic. A value of AUTH indicates that the TLMS authorization field should be used.
- If a VRSOWNER statement is not supplied, the TLMS authorization field is used if valid; otherwise the default volume owner ID is used.

Only one VRSOWNER statement should be specified; if not, the last statement encountered is used.

Program Execution: Figure 77 on page 211 shows the sample JCL to execute the extract process for TLMS. See SAMPLIB member EDGJDYNM for the supplied sample JCL.

```

/*JOBNAME JOB .....
/*
/*PROPRIETARY V3 STATEMENT
/*LICENSED MATERIALS - PROPERTY OF IBM
/*"RESTRICTED MATERIALS OF IBM"
/*5695-DF1
/*(C) COPYRIGHT 1994,1995 IBM CORP.
/*END PROPRIETARY V3 STATEMENT
/*
/* $01= ,110,940509,MWW: Add region size
/* $02=OW07055,110,940802,MWW: Support new function
/* $03=OW08201,110,940916,MWW: Correct ,PASS to ,CATLG
/* $04=OW10356,110,941227,MWW: TLMS 5.4 support
/* $06=OW11151,110,941216,MWW: Implement macros
/* $05=OW10124,110,950109,MWW: Make VRSMGMTM match EDGUX100 prefix
/* $S7=OW30969,140,971125,RS : VRS Enhancements @S7A
/*
//DEL EXEC PGM=IEFBRI4
//LIBLIST DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCDYNM.DEXTLIB.DATA
//OWNLIST DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCDYNM.DEXTOWN.DATA
//BINLIST DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCDYNM.DEXTBIN.DATA
//VRSLIST DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCDYNM.DEXTVRS.DATA
/*
/* The following step assembles and executes the supplied
/* EDGCDYNM assembler code from SAMPLIB. Alternatively
/* you can remove this step and use a load module you have
/* created into RMM.LOAD
/*
/* If your system does not use High Level Assembler please @S7A
/* change the program name to the name of your assembler. @S7A
/*
// EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',REGION.G=5M, @S7C
// PARM.C=(SYSPARM(TLMS54)) @S7C@06A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCDYNM) @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB @S7A
// DD DISP=SHR,DSN=SYS1.MODGEN @S7A
//G.SYSUDUMP DD SYSOUT=* @S7C
/*DEXTRACT EXEC PGM=EDGCDYNM,REGION=5M
/*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/*SYSUDUMP DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
/*

```

Figure 77 (Part 1 of 2). Sample JCL to Execute EDGCDYNM

```

/*      TLMSVMF is the TLMS volume master file
//TLMSVMF DD DISP=SHR,DSN=STSGWD.TLMS.VMF.DATA
/*
/*      TLMS016 is the TLMS retention selection report (Report 016)
//TLMS016 DD DISP=SHR,DSN=STSGWD.TLMS.TLMS016.DATA
//DEXTLIBD DD DISP=(,CATLG),DSN=STSGWD.EDGCDYNM.DEXTLIB.DATA,
//      RECFM=VB,LRECL=700,AVGREC=K,UNIT=SYSDA,          @06C
//      SPACE=(700,(50,20),RLSE)                          @06C
//DEXTOWN DD DISP=(,CATLG),DSN=STSGWD.EDGCDYNM.DEXTOWN.DATA,
//      RECFM=VB,LRECL=512,AVGREC=K,UNIT=SYSDA,
//      SPACE=(512,(1,1),RLSE)
//DEXTRBIN DD DISP=(,CATLG),DSN=STSGWD.EDGCDYNM.DEXTRBIN.DATA,
//      RECFM=VB,LRECL=512,AVGREC=K,UNIT=SYSDA,
//      SPACE=(512,(10,5),RLSE)
//DEXTVRS DD DISP=(,CATLG),DSN=STSGWD.EDGCDYNM.DEXTVRS.DATA,
//      RECFM=VB,LRECL=512,AVGREC=K,UNIT=SYSDA,
//      SPACE=(512,(1,1),RLSE)
//SYSIN DD *
OPTIONS OUTSERV=IGNORE VOLCHAIN=PROCESS VOLLOCN=PROCLOC
VRSNAMEP PREFIX=VRSN
VRSMGMT PREFIX=D
TLMSLOCN TLMSID=TL RMMLOC=SHELF
TLMSLOCN TLMSID=DC RMMLOC=SHELF
TLMSLOCN TLMSID=CL RMMLOC=SHELF
TLMSLOCN TLMSID=D2 RMMLOC=D2
TLMSLOCN TLMSID=FS RMMLOC=FS
TLMSLOCN TLMSID=OS RMMLOC=OS
DFRMMLOC RMMLOC=D2 RMMTYP=STORE BINS=001000
DFRMMLOC RMMLOC=FS RMMTYP=STORE BINS=002000
DFRMMLOC RMMLOC=OS RMMTYP=STORE BINS=001000
/*

```

Figure 77 (Part 2 of 2). Sample JCL to Execute EDGCDYNM

5.4.4 EDGCDYNM Program Limitations

The following are the current program implementation restrictions for EDGCDYNM:

- A maximum of 100 TLMSLOCN statements are allowed.
- A maximum of 100 DFRMMLOC statements are allowed.
- A maximum of 100 SKIPVOL statements are allowed.
- A maximum of 100 VOLMEDIA statements are allowed.

5.4.5 EDGCDYNM Program Return Codes

A return code is set as follows:

- 0** Normal completion. No errors or warnings detected.
- 4** One or more warning messages issued. A valid, converted CDS is produced, but the warning messages should be reviewed to ensure that results are as expected. For example, default values may have been used for conversion when not intended.
- 8** One or more error messages issued. The CDS may contain invalid data. The errors should be corrected and the conversion process rerun.

- 16 An unrecoverable error has been encountered causing the conversion process to end early. The errors must be corrected and the conversion process rerun.

5.4.6 EDGCDYNM Program Messages

DXT002 END OF PROGRAM EDGCDYNM. HIGHEST RETURN CODE WAS xx.

where:

xx is the highest return code set by the program.

Severity: Information

Explanation: The EDGCDYNM program has completed execution.

System Action: None

User Response: If a return code greater than 0 is set, the previous messages should be reviewed and any errors corrected before rerunning the program.

DXT010 TLMS DATA EXTRACTION PARAMETERS USED:

Severity: Information

Explanation: The SYSIN control statements are listed following this message for information purposes.

System Action: None

User Response: None

DXT011 *ERROR* IN PREVIOUS CONTROL STATEMENT - PARAMETERS IGNORED.

Severity: Error

Explanation: An error was encountered in the previous control statement. This could include an unidentified keyword or keyword value, or a value out of range.

System Action: The indicated control statement is ignored, and default values are assumed where appropriate.

User Response: Correct the control statement and rerun the data extraction program.

DXT012 *ERROR* UNIDENTIFIED DFRMM STORAGE LOCATION - XXXXXXXX.

where:

xxxxxxx is an DFSMSrmm storage location defined in a TLMSLOCN statement.

Severity: Error

Explanation: The identified storage location name was specified on a TLMSLOCN statement, but there is no corresponding DFRMMLOC statement.

System Action: The TLMS location IDs corresponding to the identified DFSMSrmm location will be defaulted to a location of LIBRARY.

User Response: Correct the corresponding TLMSLOCN statements and/or supply appropriate DFRMMLOC statements and rerun the data extraction process.

DXT013 **CRITICAL ERROR UNIDENTIFIED RECORD TYPE=t IN THE VMF FILE**

where:

t is the record type identifier

Severity: Terminal

Explanation: An unidentified record type was encountered in the VMF data set.

System Action: The process is ended with a return code of 16.

User Response: Correct the TLMSVMF data set records and rerun the data extraction program. EDGCDYNM is designed to run against the actual or image copy of the VMF. Do not use a backup copy taken using the TLMS utility as input.

DXT014 **CRITICAL ERROR VOLUME xxxxxx IS NOT IN ANY DEFINED VOLUME RANGE**

where:

xxxxxx is the volume serial number

Severity: Terminal

Explanation: Specified volume is not in any defined volume range.

System Action: The process is ended with a return code of 16.

User Response: Correct the TLMSVMF control records and rerun the data extraction program. EDGCDYNM is designed to run against the actual or image copy of the VMF. Do not use a backup copy taken using the TLMS utility as input.

DXT020 *WARN* DUPLICATE RMF DSNAME IGNORED - t ddd.....ddd jjjjjjj

where:

t is the type of RMF entry, Full, Partial or Qualified

ddd.....ddd is the 44-character duplicate data set name

jjjjjjj is the optional 8-character job name qualifier

Severity: Warning

Explanation: An RMF entry with an identical data set and job name combination to a previous entry was encountered.

System Action: The indicated RMF entry is ignored.

User Response: Review the generated VRS definitions to ensure these produce the same results as the RMF specifications.

DXT022 *ERROR* UNIDENTIFIED TLMS LOCATION=ll RMF=rrr.....rrr

where:

ll is the unidentified TLMS location

rrr.....rrr is the data set name / job name identifying the RMF definition

Severity: Error

Explanation: An RMF entry with an unidentified location identifier was encountered. If mapping of TLMS locations to DFSMSrmm locations is required, the locations must be defined in TLMSLOCN statements.

System Action: The unidentified location ID is translated to an DFSMSrmm location of HOME.

User Response: Review the indicated RMF entry and supply an appropriate TLMSLOCN statement to map the TLMS ID to an DFSMSrmm location. Alternatively, set the location processing option (VOLLOCN on the OPTIONS statement) to PASS and use the EDGCNV program to map the location identifiers.

**DXT024 *WARN* CRASH PROTECTED DATA SET ddd.....ddd DETECTED ON
NON-SCRATCH VOLUME vvvvvv. PLEASE CHECK DATA SET NAME.**

where:

ddd.....ddd is the truncated 21 character data set name

vvvvvv is the 6 character volume serial number

Severity: Warning

Explanation: A TLMS crash protected data set was detected on a non—scratch volume. It is assumed that crash protection applies only to the first data set on a multi-file and/or multi-volume set.

System Action: The TLMS prefix 'TLMSII-CRASH-PROTECTED-' has been removed from the start of the data set name. The data set name may have been truncated by TLMS. If EDGCDYNM recognizes the data set name is not truncated, the volume is marked as recorded by O/C/EOV, else the text 'PLEASE CHECK DATA SET NAME' appears in the message and the volume is not marked as recorded by O/C/EOV.

User Response: If the data set name is not truncated, you can continue with the conversion. If it may be truncated, you must check for the correct data set name and if necessary, correct the data set name in TLMS and rerun EDGCDYNM to correct the data set name, or use the DFSMSrmm CHANGEVOLUME command to correct the data set name once DFSMSrmm is started with the converted CDS.

**DXT030 *WARN* INSUFFICIENT BIN NUMBERS SUPPLIED FOR LOCATION -
XXXXXXXX. USED=uuuuuu SUPPLIED=ssssss**

where:

xxxxxxxx is an DFSMSrmm storage location

uuuuuu is the number of bin numbers allocated to volumes

ssssss the count of bin numbers supplied

Severity: Warning

Explanation: The PROCBIN processing option was specified on the OPTIONS statement to force allocation of bin numbers to volumes in storage locations. Either no or insufficient bin numbers were specified on the DFRMMLOC statement defining the location.

System Action: Bin numbers will be correctly assigned, but no empty bins will be generated.

User Response: Either supply DFRMMLOC statements with appropriate bin numbers or manually add empty bins after conversion using the DFSMSrmm dialog or ADDBIN TSO subcommand. If no empty bins are generated or added, the RMM housekeeping process DSTORE may subsequently fail through lack of empty bins.

DXT031 *WARN* NO TLMSLOCN STATEMENTS SUPPLIED - LOCATION PROCESSING REQUESTED.

Severity: Warning

Explanation: You requested, or used by default, the PROCLOC option but did not specify any TLMSLOCN statements.

System Action: A processing option of PASS is forced; no mapping of TLMS storage locations will be performed.

User Response: Either supply TLMSLOCN statements with appropriate mappings or specifically request a processing option of PASS on the OPTIONS statement.

DXT032 *ERROR* UNIDENTIFIED TLMS LOCATION=II VOLSER=vvvvvv

where:

II is the unidentified TLMS location

vvvvvv is the VOLSER

Severity: Error

Explanation: A volume entry with an unidentified TLMS location identifier was encountered. If mapping of TLMS locations to DFSMSrmm locations is required, the locations must be defined in TLMSLOCN statements.

System Action: The unidentified location ID is translated to an DFSMSrmm location of SHELF.

User Response: Review the indicated volume entry and supply an appropriate TLMSLOCN statement to map the TLMS ID to a DFSMSrmm location. Alternatively, set the location processing option (VOLLOCN on the OPTIONS statement) to PASS and use the EDGCNVT program to map the location identifiers.

DXT040 *WARN* ZERO VOLUME SEQUENCE NUMBER SET TO ONE - VOLSER=vvvvvv

where:

vvvvvv is the VOLSER

Severity: Warning

Explanation: A nonscratch volume entry with a zero volume sequence number was encountered. This could cause errors in EDGCNVT and/or DFSMSrmm processing.

System Action: The zero volume sequence number is set to 1 to prevent possible later errors.

User Response: Review the indicated volume entry to ensure that it contains valid information.

DXT092 *ERROR* INSUFFICIENT NUMBER OF ENTRIES IN VRSMTAB.

Severity: Error.

Explanation: The message is issued when more than the specified VRS management values are in use. VRS management values are built during conversion from the special keyword dates detected in the VMF. A VRS management value is built for each unique date detected.

System Action: A VRS management value K record is built but duplicates are not tracked.

User Response: Correct the data extraction program EDGCDYNM: find the line marked as " = = = = > VRSMTAB SIZE DEFINITION" and increase the number of entries.

Then assemble it and re-run the program. If the number of entries will not be increased, the load of the DFSMSrmm CDS will fail because of duplicate K records.

**DXT093 *ERROR* INCORRECT CHAIN POINTER(S) ENCOUNTERED. RECORDS
SKIPPED = nnnnnnn**

where:

*nnnnnnn is the number of records skipped due to
errors in chain pointers.*

Severity: Error

Explanation: A VMF entry with an incorrect TLMS chain pointer has been encountered.

System Action: The volume records pointed to by the incorrect chain pointers are skipped.

User Response: Run the TLMSV CVS utility to correct chain pointers and re-run the data extraction program.

DXT094 RUN TLMSV CVS UTILITY TO CORRECT CHAIN POINTERS

Severity: Information

Explanation: This message will be follows the DXT093 message.

System Action: None

User Response: Run the TLMSV CVS utility to correct chain pointers and re-run the data extraction program.

DXT095 EDGCDYNM INPUT RECORD COUNTS:

Severity: Information

Explanation: This message will be followed by one or more DXT096 messages indicating the number of records read of a particular record type.

System Action: None

User Response: Information

DXT096 NUMBER OF rrrrrrrrrr RECORDS READ = nnnnnnnn

where:

rrrrrrrr is the type of record

nnnnnnnn is the number of records read

Severity: Information

Explanation: Information

System Action: None

User Response: Information

DXT097 EDGCDYNM OUTPUT RECORD COUNTS:

Severity: Information

Explanation: This message will be followed by one or more DXT098 messages indicating the number of records output for a particular record type.

System Action: None

User Response: Information

DXT098 NUMBER OF rrrrrrr RECORDS WRITTEN = nnnnnnnn

where:

rrrrrrr is the type of record

nnnnnnn is the number of records output

Severity: Information

Explanation: Information

System Action: None

User Response: Information

5.5 Am I Ready to Continue?

In this chapter we have covered all of the activities that you should have considered before converting from TLMS to DFSMSrmm. After completing the activities in this chapter, you should have the following valid and tested input to load the DFSMSrmm CDS:

- Input to exits
- Correct PARMLIB options
- Housekeeping jobs to use in a DFSMSrmm environment
- Changed reports
- Extract and convert control statements
- Extracted data from TLMS database

Before continuing, we suggest that you ensure that the extraction of data is error free. The output of the extract program may have error return codes. Read the corresponding error messages to determine the actions required to correct the errors. Then rerun the extraction process (see Section 5.4, “Running the Extract Program” on page 202).

To have a valid DFSMSrmm CDS, you must stop all tape activity from the time you get the data from the TLMS database until you start DFSMSrmm in record-only mode in parallel with TLMS.

Before producing a valid conversion, you can use a copy of the TLMS database to test the extraction and conversion process, so you do not have to stop the tape activities every time you find that you have to go back and rerun the extract jobs.

Once you have sufficiently tested the environment, you should have correct, tailored input to the extract program. You can now stop tape activities and produce a valid data extraction from the VMF.

If your previous tests were satisfactory, you should obtain the correct input to the EDGCNVT program and proceed to the next step.

Chapter 6. Converting from Control-T to DFSMSrmm

In this chapter we describe how to convert from Control-T, a tape management program of Computer Associates, to DFSMSrmm. We compare Control-T and DFSMSrmm terms and functions and explain the data extraction process.

Objectives of Chapter

- Explain differences between Control-T and DFSMSrmm
- Describe how to customize DFSMSrmm for equivalent Control-T functions
- List the functions where there is no equivalence
- Extract data and prepare input for EDGCNVT
- Provide hints and tips

Audience

- System programmers
- Storage administrators and librarians

At the end of this chapter, you will have reviewed the implementation of your current tape management system and will understand the detailed changes required to customize DFSMSrmm to meet your tape management needs.

6.1 Control-T Overview

Control-T uses your installation's retention management, vault management, and tape pool management policies to manage tape volumes in tape libraries and storage locations.

The Control-T tape management catalog (TMC) contains two record types with tape volume inventory information:

- TMC volume record
- Data set name block (DSNB)

Control-T assigns a default retention period defined by your installation to any tape data set that does not have a retention period or an expiration date specified in the JCL EXPDT or RETPD parameter. It also provides unique Julian dates, which have special meanings to manage tape data set retention. Additionally, it manages tape data set retention using a retention data set (RDS). When the condition is met, the retention specified in the retention data set overrides the expiration value in the TMC volume record. Thus, you can change the retention period after a tape data set has been created.

Control-T uses a vaulting policy to manage volume movement from the library to and among the storage locations. The vaulting policy specifies the data set name, the movement criteria, the storage location ID, and the retention criteria. This policy information is stored in the VPDD, which Control-T's batch program uses during batch update processing. When the batch program executes, it matches the data set name entry in the vault pattern description (VPD) with the TMC volume record. If it finds a match, it updates the TMC volume record and generates a pull list report, which you can use to move tape volumes to and among the storage locations.

Control-T allows you to assign a tape pool to a specific group of users or an application. On a scratch mount request, Control-T issues a mount message with the pool name indicating the tape pool from which the scratch volume should be selected. It matches the VOLSER of the mounted volume with the ranges defined in the tape pool. If the VOLSER is within the defined ranges, it honors the request. Otherwise, it rejects the volume and requests another scratch mount.

6.2 Control-T and DFSMSrmm

To ensure a successful conversion, it is important that you understand some of the differences between the products and how those differences affect your conversion project.

6.2.1 Terminology

Table 45 compares and defines some Control-T and DFSMSrmm terms.

<i>Table 45 (Page 1 of 9). Control-T and DFSMSrmm Terminology</i>		
Control-T	DFSMSrmm	Definition
Address of creation unit	Device number	The device number of the drive on which the volume was mounted when DFSMSrmm recorded information about the data set
Address of unit last used	Last use drive	The device number of the drive on which the volume was last mounted when DFSMSrmm recorded information about the data set
Audit file Audit log	Journal	A sequential data set that contains a chronological record of changes made to the DFSMSrmm control data set
Birth date	Creation date	The volume creation date
Control record	Control record	The first CDS record containing key information for DFSMSrmm
Creating DD name	Creating DD name	The DD statement that created the volume
Creating job name	Job name	The name of the job that created the volume
Creating step name	Creating step name	The name of the step that created the volume
Creation date	Assign date	Volume: The date when the volume changed status; either to or from scratch
	Create date	Data set: The date when the data set was first written to tape
Creation time	Assign time	Volume: The time when the volume changed status; either to or from scratch
	Create time	Data set: The time when the data set was first written to tape

Table 45 (Page 2 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
CATALOG	WHILECATALOG (VRS)	DFSMSrmm retains the data set only as long as it is cataloged. Issue the RMM TSO subcommand: RMM ADDVRS DSNAME('**') - WHILECATALOG This means to retain all versions of a data set while they are cataloged. The options <i>CYCLES</i> and <i>COUNT(99999)</i> are default.
CATLG/ddd	WHILECATALOG and DAYS in VRS combination (VRS)	DFSMSrmm retains the data set for a minimum of days since creation, and after that period, DFSMSrmm retains the data set for as long as it is cataloged. Issue the RMM TSO subcommand: RMM ADDVRS DSNAME('**') DAYS COUNT(5) - NEXT(WHILECATALOG) RMM ADDVRS NAME(WHILECATALOG) WHILECATALOG Note: DFSMSrmm PARMLIB option VRSEL(NEW) is required.
CYCLE/ccc Cycle control	CYCLE (VRS)	DFSMSrmm retains data sets based on cycles or copies of a data set. For GDG data set VRS, the retention is based on GDG cycles. For non-GDG data sets, each occurrence of a data set is considered to be a cycle. To retain 20 cycles for each data set issue the TSO RMM subcommand: RMM ADDVRS DSNAME('CYCLE20') - CYCLES COUNT(20)
Data set block count	Block count	The number of data blocks used by the data set
Data set security password	- -	Controls access to tape data sets. Password is stored only in the Control-T TMC.
Data set name	Data set name	The name of the data set
- -	Data set size	The size of the data set in kilobytes
Data set sequence	Physical file sequence number Data set sequence number	The physical file sequence number field specifies the relative position of a data set on a volume. The data set sequence number field corresponds to the IBM standard HDR1 Data set sequence number field.
Date last used	Read date Write date	The date when the data set was last read The date when the data set was last written to
Date tape was first used	Create date Create time	The date when a volume was created The time of day when the volume was created
Date tape was last cleaned	- -	The date when a volume was last cleaned

Table 45 (Page 3 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
Days since last used control LDATE/ddd	LASTREFERENCEDAYS (VRS)	Specifies that all copies of the data set be retained according to the number of elapsed days since the data set was last read or written to. For example, to retain a data set that has not been used for 14 days, issue the TSO RMM subcommand: RMM ADDVRS DSNAME('LDATE014') - LASTREFERENCE COUNT(14)
DSNB record	Data set record	Data set information. In Control-T, one for each file greater than the first file on volume ¹ . In DFSMSrmm, one for each file on the volume.
Expiration date	Expiration date Original expiration date	The Expiration date field displays the date when the volume will expire. The original Expiration date field displays a volume's original expiration date. This expiration date is set using the JCL EXPDT or RETPD keyword when the volume was used during open, close, EOVS processing.

Table 45 (Page 4 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
EDMID (External data managed)	Data Set Name / Jobname (VRS)	Specifies tapes controlled by an external data manager, such as DFSMSHsm. DFSMSrmm does not recognize EDM's, but: You can retain all DFSMSHsm and/or ADSM tapes that require no movement by specifying the TSO RMM subcommands: <pre> RMM ADDVRS DSNAME('mprefix.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('bprefix.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('authid.**') - COUNT(99999)- DELETEDATE(1999/365) RMM ADDVRS DSNAME('adsmprefix.**') - COUNT(99999)- DELETEDATE(1999/365) </pre> In addition, you can define VRS to control the retention of data sets that are not created normally by using the TSO RMM subcommand: <pre> RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(hsm_proc) RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(abars_proc) RMM ADDVRS DSNAME('ABEND') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(adsm_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(hsm_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(abars_proc) RMM ADDVRS DSNAME('OPEN') - COUNT(99999) CYCLES - DELETEDATE(1999/365) - JOBNAME(adsm_proc) </pre> See "Running DFSMSrmm with DFSMSHsm," in <i>DFSMSrmm Implementation and Customization Guide</i> for more information. Note: All ADSM tapes are managed in the same way as DFSMSHsm tapes.
Hooks	Subsystem interface	DFSMSrmm requires no changes to open, close, or EOVS instructions. DFSMSrmm uses the subsystem interface.
Job that last used volume	--	The name of the job that last used the volume

<i>Table 45 (Page 5 of 9). Control-T and DFSMSrmm Terminology</i>		
Control-T	DFSMSrmm	Definition
--	K-Record	The VRS information
Location ID	Loan location	A user-defined location of a volume that is on loan or not in location SHELF, a system-managed library, or any of the storage locations.
Location ID	Storage location	An onsite library or an offsite storage location, also known as vault or store.
Logical record length	Logical record length	The length, in bytes, of the longest record in the data set
Maximum block size	Block size	The block size of the data set
Number of permanent read errors since initialized	Permanent read	The number of permanent errors that occurred while reading the volume since it was created (Permanent errors are not recoverable.)
Number of permanent read errors since cleaned	--	The number of permanent errors that occurred while reading the volume since it was last cleaned
Number of permanent write errors since initialized	Permanent write	The number of permanent errors that occurred while writing to the volume since it was created (Permanent errors are not recoverable.)
Number of permanent write errors since cleaned	--	The number of permanent errors that occurred while writing to the volume since it was last cleaned
Number of temporary read errors since initialized	Temporary read	The number of temporary errors that occurred while writing to the volume since it was created (Temporary errors are recoverable.)
Number of temporary read errors since cleaned	--	The number of temporary errors that occurred while writing to the volume since it was last cleaned
Number of temporary write errors since initialized	Temporary write	The number of temporary errors that occurred while writing to the volume since it was created (Temporary errors are recoverable.)
Number of temporary write errors since cleaned	--	The number of temporary errors that occurred while writing to the volume since it was last cleaned.
Number of times tape was cleaned	--	The number of times the volume was cleaned
Number of times tape opened since birthday.	Volume use count	The number of times that the volume was opened for either read and write operations In DFSMSrmm the field will be reset to zero if the status changed from SCRATCH to MASTER or USER.

Table 45 (Page 6 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
--	OWNER	Users are defined to DFSMSrmm through owner IDs. Owners can be individuals, departments, or any logical grouping of people. General users can change information for their own owner ID once they have been defined to DFSMSrmm. DFSMSrmm automatically creates an owner record if a user who is not defined to DFSMSrmm creates a data set on a scratch volume managed by DFSMSrmm. DFSMSrmm uses the RACF user ID that created the data set as the DFSMSrmm owner ID.
Out-of-area	Loan location Storage location	A user-defined location of a volume that is on loan or not in location SHELF, a system-managed library, or any of the storage locations An onsite library or an offsite storage location, also known as vault or store
--	O-Record	Owner information
PERM/PERMANENT permanent hold	1999/365 or 1999/366	Expiration dates of 99365 and 99366 or 1999/365 and 1999/366 are considered "never-scratch" dates.
	A VRS with a COUNT(99999)	Data sets with these expiration dates must be released manually. Use a VRS with COUNT(99999) to automatically manage data sets with permanent retention.
--	Physical file sequence number	The Physical file sequence number field specifies the relative position of a data set on a volume.
PRIVATE	MASTER USER	A MASTER volume contains valid user data and cannot be overwritten when the same data set name is used. A USER volume is assigned to a user and is in use. It can hold any type of data and can be overwritten, rewritten, and used repeatedly until the volume has expired.
--	P-Record	The software product information
--	Rack number	In the removable media library, you store your volumes in shelves, where each volume occupies a single shelf location. This shelf location is called a rack number. A rack number matches the volume's external label. DFSMSrmm uses the external VOLSER to assign a rack number when adding a volume, unless you specify otherwise. The rack number is the external VOLSER. Thus, both must be six alphanumeric characters. The format of the VOLSER you define to DFSMSrmm must be one to six alphanumeric characters. The rack number must be six alphanumeric characters.

Table 45 (Page 7 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
Record format	Record format	The format and characteristics of the records in the data set
Recording density	Recording density	The volume's recording density. For a 3420 tape reel, you can use DENSITY as 1600 or 6250. For a 3480 tape cartridge, use a value of 3480.
Recording technique	Recording format	The basic recording format for tape volumes
Retention data set (RDS)	Data Set Name (VRS)	Use VRSs to set policies for retaining all removable media. You can also use VRSs to move data sets and volumes for disaster recovery and vital records within the removable media library and among your storage locations. You can use EDGUX100 to assign VRS management values or retention periods to data sets based on the converted RDS. See the <i>DFSMSrmm Implementation and Customization Guide</i> for more information.
--	R-Record	The library shelf location (rack) information
SCRATCH	SCRATCH	The volume is free and available for use. It contains no data, or the data on the volume is expired or invalid.
SCRATCH POOL	VLPOOL Volume pool	You can use pool IDs to add one or more rack numbers to a specific library in the removable media library. See "Using DFSMSrmm Installation Exits," in the <i>DFSMSrmm Implementation and Customization Guide</i> for more information.
--	Security level	You can define security classes for data sets and volumes. These security classes appear in reports and RMM TSO subcommand output. DFSMSrmm records the security classes in the DFSMSrmm CDS; it does not make RACF aware of them. There is no connection between these definitions and any similar definitions or function provided in RACF, but you can use similar values for overall consistency. Each time a data set is opened or closed DFSMSrmm automatically records its security level. DFSMSrmm determines the security classification of a tape volume with multiple data sets by the highest classification found for a single data set.
Slot	Bin number	Displays the shelf location in one of the storage locations assigned to the volume as part of the requirements of vital records processing. If the volume is moving, this is the bin number in the target location.
--	SMS data class name	A list of data set allocation attributes that SMS assigns to a data set when it is created

Table 45 (Page 8 of 9). Control-T and DFSMSrmm Terminology

Control-T	DFSMSrmm	Definition
--	SMS storage class name	A list of data set storage service attributes that identify performance and availability requirements. SMS uses these attributes to control data placement.
--	SMS storage group name	A list of volumes used for allocation of new system-managed data sets
--	S-Record	The storage location bin information
Tape label type	Label	The volume's label type: AL - ANSI label AUL - ASCII user label BLP - Bypass label processing NL - Non labeled SL - Standard label SUL - Standard user label
--	Tape media type	Specifies the volume's physical media type. Use one of the following: * - The volume is not a cartridge. CST - The volume is a 3480 or 3490 cartridge. ECCST - The volume is an enhanced capacity 3480 or 3490 cartridge. HPCT - High Performance Cartridge Tape (3590) EHPCT - Reserved for Extended High Performance Cartridge Tape
Time last used	--	The time when the data set was last used
TMC (Tape management catalog)	CDS (Control data set)	The DFSMSrmm CDS is a VSAM KSDS that contains the complete inventory of the removable media library. DFSMSrmm records all changes made to the inventory, such as adding or deleting volumes, in the CDS.
TMC record	--	Volume and first data set information
TMSCLEAN	EDGHSKP (Expiration processing)	DFSMSrmm checks the expiration date for volumes not managed by a vital record specification. If the expiration date has been reached, DFSMSrmm changes the volume status to "pending release." DFSMSrmm checks for any actions that should be taken before the volume can be released. DFSMSrmm returns volumes to scratch that are already in "pending release" status and require no action other than being returned to scratch.
TMSGRW (Generalized report writer)	EDGRPDT (Inventory and movement reports)	Create inventory reports for auditing the physical contents of the installation media library and storage locations
System option (CAI.PPOTION)	EDGRMMxx	The DFSMSrmm PARMLIB EDGRMMxx member includes many options for setting up DFSMSrmm

<i>Table 45 (Page 9 of 9). Control-T and DFSMSrmm Terminology</i>		
Control-T	DFSMSrmm	Definition
Use count at last cleaning	--	The use count since the volume was last cleaned
Vault	Storage location	A storage location is a place where you send a store volume. Optionally, it is comprised of shelf locations that you define to DFSMSrmm.
--	V-Record	The volume information
--	USER	A USER volume is assigned to a user and is in use. It can hold any type of data and can be overwritten, rewritten, and used repeatedly until the volume has expired.
Vault slot number	Bin number	Specifies the shelf location in one of the storage locations assigned to the volume as part of the requirements of vital records processing
Volume serial number	Volume VOLSER	The VOLSER of the volume
Vault pattern description data set (VPDS)	VRS	Use VRSs to set policies for retaining all removable media. You can also use VRSs to move data sets and volumes for disaster recovery and vital records within the removable media library and among your storage locations. <i>See the DFSMSrmm Implementation and Customization Guide for more information.</i>
VMS (Vault management system)	EDGHSKP Storage location management	Request storage location management processing when you are using vital record specifications to identify data sets and volumes to be moved between locations.
¹ The information about the first file is stored in the volume record. The information about the second file and up (file sequence numbers from 2 through 9999 of a volume) is stored in DSNB records.		

6.2.2 Functions

Table 46 on page 229 compares the key tape management functions of Control-T and DFSMSrmm.

Table 46 (Page 1 of 4). Control-T and DFSMSrmm Functions

Function	
Control-T	DFSMSrmm
Tape security	
• Tape must be scratch for output scratch request • Data set name on tape must match inventory specific request • Own password protection, as well as a security system interface that uses SAF to interface to the installed security system.	• Tape must be scratch for output scratch request • Data set name on tape must match inventory specific request • 44-character data set name checked against CDS on input processing • RACF interface for tape profile management • Access list for non-RACF system • Security erase on scratch • Prevent read of scratch tapes • Security classification by generic data set name • Master tape overwrite support controlled by PARMLIB option
Retention management	
• Based on RETPD or EXPDT • By cycle or cycles by day • By days since last reference • By nonmanaged • By catalog status • Permanent • Can default retention by data set name • By days and catalog	• Based on RETPD or EXPDT • By cycle or by BYDAYSCYCLE For DFSMSrmm releases 1.2 through 1.4 which do not have OW30969 (VRS enhancements) installed, APAR OW17377 is required, plus a patch available through the IBM service team. For DFSMSrmm releases 1.3 and 1.4 with OW30969 installed there is the BYDAYSCYCLE option for VRSS. • By days since last reference • By days since creation • By catalog status • Permanent • Can specify default and maximum retention installation-wide. Default retention can also be specified using generic data set mask in VRS¹ • Retention policies based on full or generic data set name or VOLSER • By cycle while cataloged • By job name • By days since last reference while cataloged • By days since creation while cataloged • By until expired

Table 46 (Page 2 of 4). Control-T and DFSMSrmm Functions

Function	
Control-T	DFSMSrmm
EXPDT with Special Meanings	
• 90ddd • 98000 • 98ddd • 99000 • 99ccc • 99365 and 99366 • Supports up to 2155/366	• Dates of 1999/365 and 1999/366 mean never scratch • Full support of dates beyond twentieth century • Installation-wide exit can be used to support special dates
ACCODE JCL parameter	
• xCAUSER • xCANORES • xCACATLG • xCAPERM	• The EDGUX100 installation-wide exit can be used to support the ACCODE keywords. APAR OW33428 is required.
Security	
• Password data set and SAF interface (see tape security function)	• RACF control (through SAF interface) to authorize different functions
Exit for DFSMSHsm and ADSM	
• Supplied	• Supplied With DFSMSrmm 1.4 and later releases this is a built-in interface with no need for exits. Prior to DFSMSrmm 1.4, ARCTVEXT is required. • Full support for DFSMSHsm use of scratch pools
Interfaces	
• Supplied	• Integrated with DFSMS/MVS • Can test while Control-T in control

Table 46 (Page 3 of 4). Control-T and DFSMSrmm Functions

Function	
Control-T	DFSMSrmm
Vault or storage location management	
• Supports up to 48 vaults (no limit in release 5.1 or higher) • Define management by data set name For Control-T 5.2 and later releases, there is now an option to specify volumes in the VPD. – Full or partially qualified For Control-T 5.1 and later there is an option to use generic data set masks. The masks are not compatible with those used in the IBM products DFSMSdss, DFSMSdtp, ISPF, and DFSMSrmm. – Qualified by job name • Manages by vault slot number • Unrestricted movement • Selection for management based on file 1 on the volume	• Supports up to three built-in storage location names LOCAL, DISTANT and REMOTE, unlimited installation-selected storage locations (any eight-character name can be defined), and any number of system-managed libraries • Define management by data set name, VOLSER, or job name – Full or generic (ISMF syntax), GDG or pseudo-GDG • Manages by store bin number • Unrestricted movement • Selection for management based on both a VRS for all data sets on the volume and a hierarchy • Movement hierarchy (PRIORITY) • Supports loan location
Initialization of new tapes	
• Uses “inactive” VOLSERS, updates inventory² • Uses WTOR for integrity	• Interface ensures integrity, updates inventory³ • Full support of system-managed libraries
Label support	
• SL, AL, NL, NSL, BLP (NSL, NL, BLP require user exit routine or SAF)	• SL, AL, NL, BLP, SUL, AUL
Volume pools (scratch request)	
• VOLSER ranges associated with data set name or job name or both	• VOLSER ranges associated with system; can also restrict input • Installationwide exit can be used to support scratch pools based on data set name and job name or any other installation-selected criteria.
Operator messages	
• Scratch mount messages issued with pool name • Uses 3480 display for “NOT SCRATCH” reason codes • Can modify with a usermod the 3480 and 3490 message display with pool name	• Scratch mount messages modified with pool name • Specific requests modified with rack (slot) number in library • 3480, 3490, and 3590 message display modified with pool ID, rack (slot) number, or pool name

Table 46 (Page 4 of 4). Control-T and DFSMSrmm Functions	
Function	
Control-T	DFSMSrmm
Optional sticky labels	
All, none, or selected (JCL or online request)	Installation-wide exit EDGUX100 can be used to create a label and send to dedicated console
Identify tapes out of library	
Out-of-area support	Loan location support
¹DFSMSrmm provides permanent retention using 99365/99366 or 1999/365 and 1999/366. ²In releases prior to Control-T 5.1, the Control-T utility TMSTPNIT, is a front-end to IEHINITT. An operator reply to WTOR is required. With Control-T 5.1 the TMSTPNIT utility is no longer a front-end to IEHINITT — IEHINITT is not required. ³DFSMSrmm utility EDGINERS includes IEHINITT functions, but DFSMSrmm does not use IEHINITT; DFSMSrmm supports system-managed tape, and requires no operator intervention, that is, no WTOR.	

6.2.3 Retention Methods

To help you understand the way in which Control-T and DFSMSrmm manage the retention of tape data sets, we compare the management rules of the two products.

Table 47 presents the Control-T retention methods and the equivalent VRS definitions that you should use in the DFSMSrmm environment.

Table 47 (Page 1 of 3). Control-T Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
Control-T	DFSMSrmm
Retained by days and catalog	
LABEL=EXPDT=90ddd Specifies that a tape data set will be retained for a minimum number of days (ddd), after which it will be retained for as long as it is cataloged	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSs. For example: <pre>RMM ADDVRS DSN('CATLG005') DAYS COUNT(5) - NEXTVRS(CATALOG) RMM ADDVRS NAME(CATALOG) - WHILECATALOG</pre> The catalog and days control EXPDT=90ddd will be translated to a MV of 'CATLGddd' and a NEXTVRS of 'CATALOG', as in the above example. If VRSMV=YES is specified in the EDGCxLDR data extraction programs, it will convert the 90ddd key word dates to a new special VRS naming format.

Table 47 (Page 2 of 3). Control-T Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
Control-T	DFSMSrmm
Ignoring duplicate or undefined VOLSERs	
LABEL=EXPDT=98000 LABEL=EXPDT=1998/000 ACCODE=xCANORES Specifies that a volume should be ignored Control-T also uses TMSUX0A to allow volumes to be ignored	When you use the EDGUX100 installation exit, DFSMSrmm calls the exit each time a volume is opened, allowing you to decide whether you want to ignore the volume. If you choose to use ignore the volume, the installation exit checks the JCL-specified EXPDT value for a special date of 98000, which indicates that the volume should be ignored. If this special date is found, the exit then uses the installation exit parameter list to request that DFSMSrmm ignore the volume. The sample EDGUX100 exit includes this function. You may have to customize the supplied EDGUX100 to get ignore support using a trigger other than EXPDT=98000. APAR OW33428 is required to support the ACCODE=xCANORES JCL parameter. Note: For this function you need read or update access to the profile: STGADMIN.EDG.IGNORE.TAPE.volser The RACF profile is stored in the FACILITY class.
Days since last used	
LABEL=EXPDT=98ddd Specifies that a tape data set be retained until the specified number of days (ddd) since last reference has elapsed	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSSs. DFSMSrmm uses the D98ddd VRS management value to retain all copies of the data set based on the number of elapsed days (ddd) since the data set was last read or For example, you can use the VRS management value LDATE020 for the special date 98020 and issue the following TSO RMM subcommand to add a new VRS: RMM ADDVRS DSNAME(' LDATE020') - LASTREFERENCEDAYS COUNT(20) If VRSMV=YES is specified in the EDGCxLDR data extraction programs, it will convert the 98ddd key word dates to a new special VRS naming format.

Table 47 (Page 3 of 3). Control-T Retention Methods and DFSMSrmm VRS Equivalences	
Retention methods	
Control-T	DFSMSrmm
Catalog control	
LABEL=EXPDT=99000 LABEL=EXPDT=1999/000 ACCODE=xCACATLG Specifies that a tape data set be retained for as long as it is cataloged	For 99000 the sample user exit EDGUX100 clears the expiration date field, so that DFSMSrmm uses the DFSMSrmm PARMLIB default retention period to calculate an expiration date. This prevents DFSMSrmm from treating a special expiration date as an actual expiration date. Also if the exit parameter list indicates that the exit can specify a VRS management value, EDGUX100 returns a VRS management value of D99000. APAR OW33428 is required, to support the ACCODE JCL parameter. The following example defines a VRS for managing the special date, 99000: <pre>RMM ADDVRS DSNAME(' CATALOG') - WHILECATALOG</pre>
Permanent Retention	
LABEL=EXPDT=99365 or 99366 LABEL=EXPDT=1999/365 or 1999/366 ACCODE=xCAUSER ACCODE=xCAPERM These parameters will retain a tape permanently.	Use the sample user exit EDGUX100 to detect the special JCL parameters and return a VRS management value of 'PERM' APAR OW33428 is required, to support the ACCODE JCL parameter. The following example defines a VRS for managing permanently protected tapes: <pre>RMM ADDVRS DSNAME(' PERM')</pre>
Cycle control	
LABEL=EXPDT=99ccc Specifies that a tape data set be retained for the total number of cycles (ccc) or generations.	EDGUX100 assigns a VRS management value for data sets with special dates. VRS management values must be added using VRSs. Using the VRS management value DFSMSrmm retains all copies, or cycles, of the data set based on the number of elapsed days (ddd) since the data set was last read or written to. For example, you can use the VRS management value CYCLE005 for the special date 99005 and issue the TSO RMM subcommand to add a new VRS: <pre>RMM ADDVRS DSNAME(' CYCLE005') - CYCLES COUNT(5)</pre>

The extract programs provided with DFSMSrmm for Control-T build K-Records for the keyword dates that you identify. The K-Records are converted to VRS definitions so you do not need to use the ADDVRS command. Also, the keyword dates and the RDS entries are converted into a table that can be changed dynamically and used with EDGUX100 (the EDGCVRSX sample) to assign VRS management values for new data sets. The table is a combination of JCL keyword values and RDS entries.

6.2.4 Volume and Data Set Record Display

When you display information for a volume or a data set, you will find that some Control-T fields differ from those of DFSMSrmm. Table 48 lists those differences.

<i>Table 48 (Page 1 of 2). Control-T and DFSMSrmm Volume and Data Set Record Displays</i>	
Control-T	DFSMSrmm
VOLSER	Volume
DSN	Data set name
DSN17	--
EXPDT	Expiration date
	Original expiration date
ACCT	Account number
FLAG1	--
HOOKID	--
FLAG2	--
BATCHID	--
FLAG3	--
FLAG4	--
CDATE	Assign date
CJOB	Job name
CTIME	Assign time
CUNIT	Unit address
LJOB	--
LTIME	Date last read
	Date last written
LUNIT	Drive last used
CSTEP	Creating step name
CDDNAME	Creating ddname
TRERRC	--
OUTDATE	Movement tracking date
OUTCODE	Location
	Storage location
SLOT	Bin number
TWERRC	--
BTHDATE	Create date
Vendor	--
COUNT	Volume use count
TRERRI	Temporary read
DATECLN	--
USECLN	--
CLNCNT	--
TWERRI	Temporary write
VOLSEQ	Volume sequence

<i>Table 48 (Page 2 of 2). Control-T and DFSMSrmm Volume and Data Set Record Displays</i>	
Control-T	DFSMSrmm
EDMID	--
PRERRC	--
1STVOL	--
NEXTVOL	Next volume
PREVVOL	Previous volume
PWERRC	--
NUMDSNB	--
1STDNSB	--
LSTDNSB	--
PRERRI	Permanent read
LABEL	Label
DEN	Density
TRTCH	Recording format
PWERRI	Permanent write
RECFM	Record format
LRECL	Logical record length
BLKSIZE	Block size
BLKCNT	Block count
AUDATE	--
AUTIME	--
USERID	Last changed by
AUCODE	--
AUFLAG1	--
CPUID	System ID
FILE SEQ	Data set sequence
	Physical file sequence number
DSNB	--

6.2.5 Data Fields

During the extract process, the Control-T records are read, and some fields are extracted to be translated in the DFSMSrmm database.

Some Control-T fields are not used by the EXTRACT program, and some DFSMSrmm fields are not present in the Control-T equivalent record.

Other DFSMSrmm fields have a different name in the Control-T database but have a similar or equivalent meaning in the DFSMSrmm database.

6.2.5.1 Data Unavailable for Conversion

The following are new terms in DFSMSrmm that are not available in the Control-T TMC (see Table 45 on page 220 for the field definitions):

- Data set size
- Date data set last read*
- Date data set last written*
- Date volume last read*
- Date volume last written*
- Original expiration date
- Home location
- Owner
- Rack
- Physical file sequence number
- Security level
- SMS data class name
- SMS storage class name
- SMS storage group name
- Tape media type
- Tape usage in kilobytes

*In Control-T only the date last used is stored.

6.2.5.2 Data Not Converted

The following are the fields that the extract program does not extract from the Control-T TMC (see Table 45 on page 220 for the field definitions):

- Job name that last used volume
- Data set security password
- Number of times tape cleaned
- Use count at last cleaning
- Date tape was last cleaned
- Number of times tape opened since birthdate
(In DFSMSrmm only the count of opens since last assigned is stored)
- EDMID (External data manager ID)
- Number of temporary read errors since cleaned
- Number of temporary write errors since cleaned
- Number of permanent read errors since cleaned
- Number of permanent write errors since cleaned
- Creating program name
- Last used program name

6.2.5.3 Data Converted, but Not Quite the Same

The following fields are present in the Control-T TMC but not in DFSMSrmm (see Table 45 on page 220 for the field definitions):

- Creation date
- Creation time
- Data set sequence
- Date last used
- Date tape was first used
- Expiration date
- Number of times tape opened since birthdate

6.2.6 Cross-Reference

In this section, we present tables that map data from the EDGCNVT input records to Control-T.

6.2.6.1 Volume Information

Table 49 shows the EDGCLREC volume information.

<i>Table 49 (Page 1 of 4). EDGCLREC Volume Information</i>		
	Control-T	Status
Volume serial number	√	Req
Volume owner ¹		Req
Volume owner user ID		
Volume owner node ID		
Volume security level		
Volume sequence number	√	Req
Volume create date	√	Req
Volume create time	√	
Volume create user ID		
Volume create job name	√	
UCB type (copy of UCBTYP)		
Tape media type information		Cond
Tape recording format	√	
Not 3480		
18 track		
36 track		
Tape media type		
Not 3480		
Standard capacity		
Enhanced capacity		
Tape compaction technique	√	
Unknown		
Not compact		
IDRC		
Tape special attributes		
None		
18 track read-only		
Total volume kilobytes used		
JCL volume expiration date	√	
Volume expiration date	√	Req
Volume retention date	√	
Count of data sets on volume	√	Req
Temporary read errors	√	
Temporary write errors	√	
Permanent read errors	√	
Permanent write errors	√	
Tape density (copy of JFCBDEN)	√	

Table 49 (Page 2 of 4). EDGCLREC Volume Information		
	Control-T	Status
BPI indicator 1600 bpi 6250 bpi 3480 3480 compact (IDRC) Undefined	√	Req
Volume status indicators Volume is master Volume pending release Vital record - do not release User tape (assigned by librarian) Tape is on loan Tape open - not yet closed Volume is scratch Volume recorded by open, close, or EOV	√ √ √	Req
Additional volume status indicators Scratch volume claimed by getvol Scratch volume has never been initialized Scratch volume with init action pending Scratch volume waiting to enter ATL Abend in process when a data set closed Abend probably in open, close, or EOV	√	
Volume label type Default retention period used Label type is NL Label type is AL Label type is SL Tape last written with BLP SL or AL with user labels	√ √ √ √ √	Req
Volume release actions Return to scratch pool - default Replace tape on release Reinitialize Degauss/security erase Return to owner Notify owner		
Volume access information Owner may read volume Owner may update volume Owner may alter volume Read-only protection Update protection Can be used on MVS systems Can be used on VM systems	√	Req
Media name		
Unit address used for creation	√	
Last used unit address	√	
Loan location	√	
Date volume last read		
Date volume last written		

<i>Table 49 (Page 3 of 4). EDGCLREC Volume Information</i>		
	Control-T	Status
Assigned from scratch date	√	
Assigned from scratch time	√	
Copy of JFCBLTYP	√	
From JFCTRTCH - IDRC support DSN used 3480 IDRC No compaction	√	Req
Current location of volume	√	Req
Destination name		
Home location name		
Storage group name		
Accounting information	√	
User description		
Authorized user IDs area		
Rack number ²		Req
Remote store bin number	√	Cond
Remote store bin date	√	Cond
Remote store bin media name		Cond
Label number of first file	√	Req
Name of first data set on volume	√	
First data set's owner's name		
Sequence number of first data set		Cond
First data set's RECFM	√	Cond
First data set's LRECL	√	Cond
First data set's BLKSIZE	√	Cond
First data set's block count	√	Cond
First data set's kilobytes		
First data set's security level		
DSN1 create date	√	Req
DSN1 create time	√	
DSN1 last read		Cond
DSN1 last written		Cond
DSN1 flag Data set cataloged Abend in progress when data set closed	√ √	
VRS management value ³		Cond
First file creating step name	√	
First file creating DD name	√	
First file retention date	√	
Name of last data set on volume		Cond
Creating system ID		

Table 49 (Page 4 of 4). EDGCLREC Volume Information		
	Control-T	Status
- Data that can be found in TMS/Control-T is marked with a $\checkmark$. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹We recommend that you use the job name or the data set high-level qualifier (HLQ) as the owner. ²We recommend that you use the VOLSER as the rack number. ³We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

6.2.6.2 Data Set Name Information

Table 50 shows the EDGCDREC data set name information.

Table 50 (Page 1 of 2). EDGCDREC Data Set Name Information		
	Control-T	Status
Data set create date	$\checkmark$	Req
Data set create time	$\checkmark$	
Data set name	$\checkmark$	Req
Compaction status Data set used 3480 IDRC No compaction		
File sequence number on volume		Req
VOLSER holding data set	$\checkmark$	Req
Volume sequence within set of volumes		Req
Unit where data set was written		
Data set RECFM	$\checkmark$	Req
Data set LRECL	$\checkmark$	Req
Data set BLKSIZE	$\checkmark$	Req
Data set block count	$\checkmark$	Req
Data set size in kilobytes		
Data set owner name ¹		Req
Logical file sequence number	$\checkmark$	Req
Data set security level		
Data set date last read		Req
Data set date last written		Req
VRS management value ²		Cond
DSN flag Data set cataloged Abend in progress when data set closed	$\checkmark$ $\checkmark$	
Creating system ID		
Creating job name	$\checkmark$	
Creating step name	$\checkmark$	
Creating DD name	$\checkmark$	
Retention date	$\checkmark$	

Table 50 (Page 2 of 2). EDGCDREC Data Set Name Information		
	Control-T	Status
- Data that can be found in TMS/Control-T is marked with a $\checkmark$. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹EDGCNVT ensures that all data sets on the same volume have the same owner as the VOLSER's owner. ²We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

6.2.6.3 Vital Record Information

Table 51 shows the EDGCKREC policy information.

Table 51. EDGCKREC Policy Information		
	Control-T	Status
Record create date		Req
Record create time		
VRS type		Req
Data set name	$\checkmark$	Cond
Format of data set name	$\checkmark$	Cond
Type of retention - Cycles - Cycles by days - Elapsed days - Extra days - Days unreferenced - While cataloged - Until expired	$\checkmark$	Cond
Volume serial number	$\checkmark$	Cond
Delay days		
Total count of number of cycles or days	$\checkmark$	Req
VRS name		Cond
Store keep number	$\checkmark$	Cond
VRS owner		Req
Automatic delete date		
VRS description		
Location name	$\checkmark$	Req
Name of NEXT VRS definition		
Name of AND VRS definition		
Release option		
Job name	$\checkmark$	
- Data that can be found in TMS/Control-T is marked with a $\checkmark$. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>.		

6.2.6.4 Empty Rack and Bin Information

Table 52 shows the EDGCEREC empty rack and bin information.

Table 52. EDGCEREC Empty Rack and Bin Information		
	Control-T	Status
Shelf status Shelf is rack in library Shelf is bin in storage location		Req
Empty shelf location's name		Req
First of empty shelves start		Req
Count of contiguous shelves		
Media name		
• Data that DFSMSrmm requires is marked with <i>Req</i> .		

6.2.6.5 Owner Record Information

An owner record is required only for actual owners for whom you want to supply owner details and for those who are to be owners for VRS. Other owner records will be created by CONVERT postprocessing (EDGCVOVL). Table 53 shows the EDGCOREC owner information.

Table 53. EDGCOREC Owner Information		
	Control-T	Status
Owner Identifier		Req
Owner's surname		
Owner's first name		
Owner's department ¹		Req
Owner's first address line		
Owner's second address line		
Owner's third address line		
Owner's internal telephone number		
Owner's external telephone number		
Owner's user ID		
Owner's node ID		
• Data that DFSMSrmm requires is marked with <i>Req</i> . ¹ Not strictly required for conversion. Only the RMM ADDOWNER command enforces the owner's department.		

6.3 Reviewing the Current Environment

See Figure 8 on page 21 to review the flow of events that we follow in converting from Control-T to DFSMSrmm. You must collect and then analyze all information about the current tape management environment.

Once you have reviewed the environment and understand the main differences between Control-T and DFSMSrmm, you will better understand what you must change during the conversion process.

6.3.1 Collecting Information

The information that you collect about the current environment should include:

- Control-T reports used daily or occasionally

Every day there should be a Control-T report that the operator uses to move the cartridges between the desired locations and to return tapes to the scratch pools. You can use the DFSMSrmm supplied reports, but if you want to avoid changes to the layout of these reports, using REXX or DFSORT, you can tailor the DFSMSrmm reports to make them similar to the Control-T reports. Chapter 14, “Extending DFSMSrmm Reporting” on page 533 provides some useful examples of how to create the reports.

- Exits installed

Collect information about Control-T exits installed in your environment so that you can ensure that you can achieve the same function in an DFSMSrmm environment. For example, you must check the DFSMSshm exit, ARCTVEXT, if you have DFSMSshm installed.

With DFSMS/MVS Version 1 Release 4, DFSMSrmm no longer uses the DFSMSshm tape volume exit ARCTVEXT to manage tapes that DFSMSshm uses. DFSMSrmm at the Version 1 Release 4 level provides the programming interface EDGTVEXT that is used from DFSMSshm.

If you are using a level of DFSMSrmm prior to Version 1 Release 4 refer to Appendix D, “Sample Exits and Reports” on page 603 to merge the ARCTVEXT code from Control-T with DFSMSrmm.

- PARMLIB options used

You must review Control-T PARMLIB options before you start DFSMSrmm because you may have to modify the DFSMSrmm options according to the Control-T environment.

- Interfaces with other products

You want to collect information about the interfaces that Control-T might have with other products (such as CICS, ROSCOE, and TSO).

- Housekeeping (batch inventory update) process

Collect information on the daily inventory management process so that you can synchronize it with the DFSMSrmm housekeeping function.

6.3.2 Analyzing the Record Layouts

Now you analyze the information you have collected to achieve the equivalent or acceptable alternative function in DFSMSrmm environment.

DFSMSrmm requires certain data to be available to build its CDS. You must analyze Control-T databases and reports to know which types of data are available. There is not a one-to-one correspondence between DFSMSrmm and Control-T data. DFSMSrmm may have data that Control-T does not have and vice versa. If DFSMSrmm requires data that Control-T does not provide, use the extract programs or the EDGCNVT SYSIN file statements to provide a value.

After analyzing the data, you must determine which types of data you want to have in the DFSMSrmm CDS in order to be consistent with the current operating environment. Knowing the types of data needed helps you determine whether you can use Control-T databases or reports as the sole input to the extract program (see Section 6.6, “TMS/CA-1 Data Extraction” on page 278 for a functional overview of the extract program).

In addition, you must review the current policies regarding retention management, storage location management, and tape pool management. DFSMSrmm manages these processes differently from Control-T.

Note: Beginning with Control-T Release 5.0, the sizes of the TMC volume record and the DSNB record differ from the sizes of previous releases. The record sizes do not differ between Release 5.0 and Release 5.2; only some reserved fields are activated.

6.3.2.1 Volume Ownership

DFSMSrmm requires that all private volumes have an owner. DFSMSrmm uses the RACF user ID of the batch job creating the first file. As Control-T does not have this ownership concept, during conversion you must decide how to provide ownership information. We recommend that you use either the creating job name or the HLQ of the first file data set name as the volume owner. As tapes are recycled and reused, DFSMSrmm will record the new owner. Using job name or the HLQ of the data set provides you with an easy way of listing and reporting on volumes during the initial implementation of DFSMSrmm.

6.3.2.2 Rack Numbers

DFSMSrmm requires that all volumes have a rack number. DFSMSrmm uses the rack number to identify the volume to the operator and the library location where the volume is stored. As CA-1 does not have this concept, during conversion you must decide how to provide rack number information. We recommend that you use the same value as the VOLSER, unless you have any better information on which to base this value. For example, you may have an installation convention, or keep a file of information that relates an external label VOLSER to an internal label VOLSER.

6.3.2.3 Keyword Date Retention

Any data sets or volumes that are retained and managed using special keyword dates, and that you want to continue to manage in this way, must be converted to DFSMSrmm VRS management values. You can set the VRS management value in the records that the extract program produces.

6.3.3 Control-T User Exits

Table 54 lists some Control-T exits and their equivalent DFSMSrmm functions.

<i>Table 54 (Page 1 of 4). Control-T User Exits and Equivalent DFSMSrmm Functions</i>		
	Control-T Function	DFSMSrmm Equivalent
TMMUSER	Use to define user accounting fields in the TMC volume record.	No need to convert in DFSMSrmm. DFSMSrmm automatically stores the first 40 characters of the accounting information specified in your job statement or in a EXEC statement.
TMMVOLDF	Generate exit definitions for the TMSUX1U and TMSUX1E exits.	No need to convert in DFSMSrmm.

Table 54 (Page 2 of 4). Control-T User Exits and Equivalent DFSMSrmm Functions

	Control-T Function	DFSMSrmm Equivalent
TMSMSGEX	Use to display the first eight characters of the scratch pool name on tape drives.	In the EDGRMMxx PARMLIB member you can specify the MNTMSG operand to tailor mount and fetch messages so that they display the VOLSER and pool prefix, rack number, or pool name.
TMSUX1A	Automatically bypass Control-T real-time tape tracking without using EXPDT=98000 in the JCL.	DFSMSrmm ignores all volumes not defined to DFSMSrmm other than when the EDGRMMxx PARMLIB option REJECT ANYUSE(*) is used. Define no REJECT parameter in the EDGRMMxx PARMLIB member. Note: For duplicate VOLSERs you must use EXPDT=98000 in the JCL, and you need read or update access to the profile: STGADMIN.EDG.IGNORE.TAPE.volser in the RACF FACILITY class. To bypass DFSMSrmm use the sample user exit EDGUX100.
TMSUX1B	Establish retention other than the assigned ABEND Default Retention specified in the Control-T system options member, TMOOPTxx.	In DFSMSrmm you can specify ABEND retention using a VRS named ABEND. You can have one ABEND VRS without jobname, and an additional ABEND VRS for each jobname. You can define an ABEND VRS by using the TSO RMM subcommand, for example: RMM ADDVRS DSNAME('ABEND') - DAYS COUNT(3) This defines a default retention of three days for tape data sets OPEN at the time an application or system abend occurs.
TMSUX1C	Copy accounting information in the TMC volume record created by the TMSUX1J user exit.	You have no need for this function because DFSMSrmm automatically stores all accounting information.
TMSUX1D	Use to more specifically define data sets that are to be controlled by an external data manager (EDM).	You can specify VRS for fully qualified data set names or generic data set names by using the Interactive Storage Management Facility (ISMF) naming convention. Use VRSs to permanently retain EDM data until the EDM releases them.
TMSUX1E	Convert internal numeric VOLSER to an external alphanumeric value.	You have no need for this function because you can specify a one- to six-character alphanumeric internal VOLSER and a six-character alphanumeric external VOLSER (rack number).

Table 54 (Page 3 of 4). Control-T User Exits and Equivalent DFSMSrmm Functions

	Control-T Function	DFSMSrmm Equivalent
TMSUX1F	Allow or disallow real-time double opens or re-create data sets.	In the EDGRMMxx PARMLIB member, you can specify the MASTEROVERWRITE(<i>value</i>) operand to allow you to select whether existing files on tapes in master status can be used for output. <i>value</i> ADD New data can be created, but no existing data can be destroyed. LAST Only the last file on a MASTER volume can be overwritten. The data set name must match the the existing data set name. MATCH An existing file can be overwritten. The data set name must match the existing data set name. All files which are higher in sequence are destroyed. USER Any data set can be rewritten as many times as the user wishes until the volume expires.
TMSUX1G	Validate scratch tape mounted in response to a scratch subpool request.	You define pools to DFSMSrmm in the EDGRMMxx PARMLIB member that the EDGUX100 installation exit selects for new tape data sets which are to be created on nonspecific volumes (see the <i>DFSMSrmm Implementation and Customization Guide</i> , "Using DFSMSrmm Installation Exits," for more details). DFSMSrmm always ensures that volumes are from the correct scratch pool.
TMSUX1J	Capture accounting data specified in the job or step statements in the JCL.	DFSMSrmm automatically stores the first 40 characters of the accounting information specified in your job statement. The accounting information of the step statement can be optionally stored based on a PARMLIB option.
TMSUX1L	Determine which tapes should have external gummed labels.	The EDGUX100 installation exit can be used to create labels and send to dedicated console.
TMSUX1S	Conjunction with the Control-T external security interface.	DFSMSrmm provides support for RACF standard tape volume security protection, using any combination of RACF TAPEVOL and TAPEDSN options. DFSMSrmm requires no exit for this function.
TMSUX1U	Convert external alphanumeric VOLSER to an internal numeric value.	You have no need for this function, because you can specify a one- to six-character alphanumeric internal VOLSER and a six-character alphanumeric external VOLSER (rack number). The internal and external VOLSER can be different.
TMSUXCO	Modify fields during conversion of data to TMC format or Control-T control statements.	Unnecessary to modify fields in DFSMSrmm.
TMSUXCV	Convert from the Control-T Rel. 4.x TMC to the new Control-T Rel. 5.1 TMC format.	Unnecessary to convert in DFSMSrmm.

Table 54 (Page 4 of 4). Control-T User Exits and Equivalent DFSMSrmm Functions

	Control-T Function	DFSMSrmm Equivalent
TMSUXID	Convert system catalog information to Control-T TMC format.	You can use the conversion programs to create or expand your DFSMSrmm CDS. See Chapter 9, "Building the DFSMSrmm CDS" on page 403 for more information.
TMSXCLN1	Modify the expiration date and scratch indicator.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCLN2	Reset the expiration date of tape volumes using USER/nnn expiration dates during scratch processing.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCTLG	Allow the expiration date and expired by Catalog Control indicator to be modified.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXCYCL	Modify the expiration date or bypass the TMC record update.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXEXP	Allow modification of the expiration date and eligible for RDS indicator or the TMC record update to be bypassed.	You can use either the TSO RMM subcommands or the ISPF dialog to change any information in the DFSMSrmm CDS. To change the volume status from MASTER or USER to SCRATCH you need the batch utility, EDGHSKP,PARM='EXPROC'.
TMSXOCAT	Use for MVS catalog and TMC synchronization.	--
TMSXTPNT	Reject volume for label processing.	You can define and delete volumes at any time without having to take DFSMSrmm out of service. In DFSMSrmm you do not need to define volume groups or volume ranges.
TMSXTPPR	Provide various options to control the processing of tapes.	--
TMSXVLT1	Allow TMC record selection for vault management system processing.	The retention and movement policies you define to DFSMSrmm are known as VRSS. You use them to say how long and where you want to keep data sets or volumes. You also use them to define how volumes are to be moved among the libraries DFSMSrmm supports, and the storage locations that are defined for vital records, and disaster recovery purposes. All information about storage location management processing is stored in the DFSMSrmm CDS.

6.3.4 Control-T PARMLIB Options

Review carefully all of the Control-T PARMLIB options before proceeding. Pay particular attention to the CA1.PPOPTION members, which include most of the system options.

You must modify the DFSMSrmm options to reflect the same defaults as specified in the Control-T options. The most important DFSMSrmm options for the conversion are:

- OPTION statement
 - OPMODE parameter
 - BLP parameter
 - MASTEROVERWRITE parameter
 - RETPD parameter
 - TPRACF parameter
 - UNCATALOG parameter
 - VRSJOBNAME parameter
- VLPOOL statement

Only if you use multiple scratch pool support
- LOCDEF statement

Only if you use location names different from the installation defaults.

The TMS PARMLIB options should be input to the conversion tools so that the relevant values for policy creation can be used when the DFSMSrmm VRS records are built. Other than the VRS creation, the Control-T PARMLIB options must be manually converted to DFSMSrmm options.

Table 55 lists the Control-T PARMLIB options and their DFSMSrmm equivalences.

Table 55 (Page 1 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences	
Option	
Control-T	DFSMSrmm
Preferred date format	
DATEFMT Specifies the default format of the date	DATEFORM Specifies the default format of the date A mm/dd/yyyy - American E dd/mm/yyyy - European I yyyy/mm/dd - ISO J yyyy/ddd - Julian
SCRPOOL scratch pool management	
SCRPOOL Defines ranges or subranges of scratch pools	VLPOOL Use the VLPOOL command in PARMLIB to define pools. When you add a new volume to the library, DFSMSrmm assigns it a shelf location from the specified pool.

Table 55 (Page 2 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
Assignment control statement	
DSN / JOB / POOL Assigns a data set to a tape pool	EDGUX100 Installationwide exit, EDGUX100, could be used to assign data sets to a specified scratch pool. See the <i>DFSMSrmm Implementation and Customization Guide</i> or refer to section 6.6.5.1, "How the Conversion Process Works" on page 312 for an example of how the VRSTABLE is used by EDGUX100.
Specific mount message intercept	
XMSG Prefixes specific mounts with special message IDs	MNTMSG Use the MNTMSG command to tailor mount and fetch messages so they display the volume serial number and pool name or rack number.
Label change action	
LCHG Changes label if the label type of a tape mounted to satisfy a request is different from that specified in the JCL	BLP Specifies how DFSMSrmm controls BLP. If BLP(RMM) is specified, you can use BLP for input from and output to volumes in user status and for input from volumes in master status. When the BLP(NORMM) option is specified, you can use BLP for input from and output to volumes in user and master status and for output to scratch tapes. If the label is being changed and you are not using BLP, you also need access to the profile in the RACF FACILITY class: STGADMIN.EDG.LABEL.volser or STGADMIN.EDG.NOLABEL.volser in the RACF FACILITY class.
Density change action	
DCHG Enforces that the density be changed for a tape	-- The density of a tape can always be changed. You can also set the VOLNSNS option in PARMLIB to YES to allow a 36-trk formatted cartridge mounted on a 3480 device to be relabeled to 18-trk format. All IBM products, including DFSMSrmm, support this option.
Recording technique change action	
TCHG Changes the recording technique for tape	-- The recording technique of a tape can always be changed.

Table 55 (Page 3 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences	
Option	
Control-T	DFSMSrmm
Print gummed label action	
LAB Automatically generates a gummed label for a tape	Installationwide exit, EDGUX100, can be used to create label and send to dedicated console. See the <i>DFSMSrmm Implementation and Customization Guide</i> for more information.
Mixed EXPDT action	
MIXEXP Handles different expiration dates for a tape	-- You can always use mixed expiration dates when a tape is not AL.
Abend retention	
ABE Specifies the number of days to retain a volume closed during an abend	VRS In DFSMSrmm you can specify an ABEND retention using a VRS, ABEND. You can define multiple ABEND VRS data set vital record definitions by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('ABEND') - DAYS COUNT(3) JOBNAME(JOB12*)</pre> which defines a default retention of three days for tape data sets open at the time an application or system abend occurs and the JOBNAME begins with JOB12.
3480 message display action	
MG3480 Displays the NOT SCRATCH reason code on a 3480 and 3490 cartridge unit.	--
TMC DSNB - maximum threshold	
TMC DSNB Specifies a limit in percent of the total number of TMC DSNB records that can be written before a warning message is issued	The only limit for creating DFSMSrmm records in the DFSMSrmm CDS is the limit of VSAM.
TMC AUDIT - maximum threshold	
TMC AUDIT Specifies a limit in percent of the total number of AUDIT records that can be written before a warning message is issued	If the DFSMSrmm journal is full, a WTOR is displayed on the system console. For DFSMSrmm 1.3 and later releases there is a PARMLIB threshold value for the journal.

Table 55 (Page 4 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
Extended CYCLE retention	
RC Adds an additional number of days to provide extended protection	To add an additional number of days you must define a VRS chain by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('SCHLUM.EXTRA') - NEXTVRS(EXTRA) CYCLES COUNT(7) RMM ADDVRS NAME(EXTRA) - EXTRADAYS COUNT(2)</pre> For extended CYCLE retention the conversion routines build a VRS called RC, and any keyword dates or RDS entries converted to VRS use these values as their NEXTVRS. For example, <pre>RMM ADDVRS DSNAME('SCHLUM.EXTRA') - NEXTVRS(RC) CYCLES COUNT(7) RMM ADDVRS NAME(RC) EXTRADAYS COUNT(2)</pre> The RC VRS is equivalent to RC 2.
Extended CATALOG retention	
R9 Adds an additional number of days to provide extended protection Additional retention control is available using the RDS.	To add an additional number of days you must define a VRS chain by using the RMM TSO subcommands: <pre>RMM ADDVRS DSNAME('SCHLUM.BACKUP') - NEXTVRS(EXTRA) WHILECATALOG</pre> <pre>RMM ADDVRS NAME(EXTRA) - EXTRADAYS COUNT(2)</pre> For extended catalog retention the conversion routines build a VRS called R9, and any keyword dates or RDS entries converted to VRS use these values as their NEXTVRS. For example, <pre>RMM ADDVRS DSNAME('SCHLUM.CATLG') - NEXTVRS(R9) WHILECATALOG RMM ADDVRS NAME(R9) EXTRADAYS COUNT(3)</pre> The R9 VRS is equivalent to R9 3.
Extended CA-1 retention	
RR Adds an additional number of days to provide extended protection Additional retention control is available using the RDS. See vendor product documentation.	The equivalent in DFSMSrmm is to use NEXTVRS on any required VRS to chain to a NEXTVRS that specifies EXTRADAYS retention type.

Table 55 (Page 5 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
Extended EDM retention	
REDM Adds an additional number of days to provide extended protection Additional retention control is available using the RDS.	The equivalent in DFSMSrmm is to use NEXTVRS on any required VRS to chain to a NEXTVRS that specifies EXTRADAYS retention type.
Default retention	
RP Specifies the default retention period. Can be a retention period or a keyword date.	RETPD Specifies the default retention in days only. For example, RETPD(5)=retain for five days. To have a default such as retain while cataloged, you must use the DFSMSrmm VRSs; for example, <pre>RMM ADDVRS DSN('**') WHILECATALOG.</pre> If the RP value is a retention period, EDGCSRDS builds an entry in the EDGUX100 table for <pre>EDGCVRSRSG DSN=*,RETPD=ddd,R0=N0</pre> If the RP value is a keyword date EDGCSRDS builds an entry in the EDGUX100 table for <pre>EDGCVRSRSG DSN=*,VRSVAL=keydate</pre> The default retention period entry would be equivalent to an DFSMSrmm PARMLIB RETPD value, but the default keydate entry ensures that all data sets not specifying a value in the JCL get a VRS MV assigned to control retention. For more information, refer to 6.6.5.1, "How the Conversion Process Works" on page 312 for an example of how the VRSTABLE is used by EDGUX100.
RDS override action	
RO Allows JCL-supplied retention to be overridden by TMSEXPDT Additional retention control is available using the RDS.	MAXRETPD Specifies the maximum retention period that a user can request for data sets on volumes. Specify a value between 0 and 9999 days, or NOLIMIT. The specified value is added to the current date to determine the maximum allowed expiration date. If the calculated date is 31 December 1999, the expiration date 1 January 2000 is used. The RO value from the Control-T PARMLIB is used during conversion to determine whether the VRS RELEASEOPTION(EXPIRYDATEIGNORE) option is used on VRSs built for retention, and whether the table entries built by EDGCSRDS specify RO=YES.

Table 55 (Page 6 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
TMSCLEAN uncatalog action	
UNCATA Specifies which data sets TMSCLEAN can uncatalog	UNCATALOG Specifies the type of catalog support to provide: N - Do not use DFSMSrmm catalog processing. DFSMSrmm does not uncatalog data sets under any circumstances. Y - Always uncatalog data sets. Use DFSMSrmm catalog processing when: • A volume is returned to scratch status • The RMM TSO DELETEVOLUME FORCE command is issued • The RMM TSO CHANGEVOLUME DSNAME command is issued • The RMM TSO DELETEDATASET command is issued • A tape data set is overwritten S - Only uncatalog data sets when the volume on which they reside is returned to scratch status.
Control-T EXPDT keyword range override ddname	
KEYDD Specifies the ddname to be placed in the JCL indicating that a date specified in the JCL is not a Control-T keyword	EDGUX100 Users often specify special dates in their JCL using the JCL EXPDT keyword. It is possible that they might want to code a real expiration date that matches one of the special dates your installation uses. You need to be able to distinguish what the user requires in these cases. See the <i>DFSMSrmm Implementation and Customization Guide</i> for more information. The default coded in the EDGUX100 exit is set to NOTKEYD8. Customize the exit to change name as needed.
Auto-checkin default retention	
KEYTAP Specifies the default retention if the automatic checkin facility is used for a data set	RETPD Specifies the default retention period for all new data sets on volumes. Specify a value between 0 and 9999 days. The specified value is added to the current date to determine the expiration date. Note that the MAXRETPD value you specify in the PARMLIB limits the calculated expiration date. DFSMSrmm uses the default in EDGRMMxx because neither RETPD nor EXPDT is specified.

Table 55 (Page 7 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
Cycle control cycle definitions (CDAY)	
CDAY DSN—Each explicit data set is counted as one cycle. DAY—All tapes of the same data set name and creation date are retained as one cycle.	VRS Specify a VRS using the BYDAYSCYCLE keyword. This means that all instances of a data set created on a single calendar day are considered to be a single cycle. DFSMSrmm retains data sets based on cycles or copies of a data set. For DFSMSrmm levels prior to APAR OW30969 (the VRS enhancements), installing APAR OW17377 plus a patch available through the IBM service team provides this capability. For GDG data set VRSs, the retention is based on GDG cycles. For non-GDG data sets, each occurrence of a data set is considered to be a cycle. CDAY is one of the PARMLIB options used by the conversion tools to build the correct VRSs for DFSMSrmm.
Cycle control cycle definitions (CJOB)	
CJOB DSN—The job name is ignored for cycles. JOB—A separate cycle is maintained for each different job name.	ADDVRS JOBNAME Specifies the job name for the VRS. You can specify a specific job name or a mask where '%' matches any one character and '*' matches any character string. Fully qualified names take precedence over generic names. JOBNAME(*) specified in the VRS is the same as CJOB=JOB. Specify VRSJOBNAME in PARMLIB to select whether DFSMSrmm uses the job name from VRSs as the primary or secondary match. Specify VRSJOBNAME(1) so that the job name takes precedence over the data set name during VRS match processing. Specify VRSJOBNAME(2) so that the data set name takes precedence over job name during VRS match processing. This is the equivalent to the Control-T JOBNAME qualification. The CJOB parameter is used during conversion to build VRSs from the RDS and from any keyword dates, and the correct value for JOBNAME is determined and used.

Table 55 (Page 8 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences

Option	
Control-T	DFSMSrmm
Vault management cycle definition	
CYD Specifies the rotation of cycle control tapes to and from the data center to storage locations by creation date or cycle	VRS The equivalent to CYD is BYDAYSCYCLE. The conversion programs use this PARMLIB option to build the correct VRSS. Prior to OW30969 (VRS enhancements) OW17377 can be used to get cycle by days as described here. For DFSMSrmm levels prior to OW30969 (the VRS enhancements) APAR OW17377 provides cycle by days as described here.
TMC data set name specification	
TMCDN Specifies the fully qualified data set name of the TMC	DSNAME Specifies the name of the DFSMSrmm CDS
AUDIT data set name specification	
AUDDSN Specifies the fully qualified data set name of the AUDIT data set	JRNNAME Specifies the name of the DFSMSrmm journal
Data set security ddname	
DSSN Specifies the DD statement for the Control-T internal security	RACF DFSMSrmm does not control authorization of access to tape data. DFP and RACF are used.
Auxiliary disposition ddname	
DSN Performs special real-time postprocessing functions, such as automatically marking a volume out of area	Installationwide exit, EDGUX100, can be used to create label and send to dedicated console.
TIQ console time out	
TW Specifies the elapsed time before a session will time out	--
Report customization	
TMSPRINT Customizes your reporting	EDGRPTD You can use the EXEC parameter to specify the security heading text, lines per page, and date format for each report. See the <i>DFSMSrmm Implementation and Customization Guide</i> for more information.

Table 55 (Page 9 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences	
Option	
Control-T	DFSMSrmm
Security interface specifications (OCEOV)	
OCEOV Specifies the external security processing	TPRACF Specifies the type of RACF tape support for DFSMSrmm to provide. Use this operand to request that DFSMSrmm maintain the security profiles that protect tape volumes.
Security interface specifications (DSNB)	
DSNB Specifies the external security processing for any secondary data sets	TPRACF Specifies the type of RACF tape support for DFSMSrmm to provide. Use this operand to request that DFSMSrmm maintain the security profiles that protect tape volumes.
YSVC call processing	
YSVC Controls the direct access to the TMC and AUDIT data sets for batch jobs and online users	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Batch call processing	
BATCH Controls access to the TMC volume and DSNB records by batch utilities	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Password call processing	
PSWD Limits the use of internal passwords	There is no support in DFSMSrmm specifically for use of passwords.
Command processing	
CMD Controls commands for online users	RACF All DFSMSrmm resources are protected with RACF profiles in the FACILITY class. Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG.
Function call processing	
FUNC Controls special realtime functions (NL, NSL, and BLP)	RACF All access for a tape will be checked by RACF.
Scratch processing	
SCRATCH Passes control to the TMSUX1S security user exit	--

Table 55 (Page 10 of 10). Control-T PARMLIB Options and DFSMSrmm Equivalences	
Option	
Control-T	DFSMSrmm
Undefined processing	
UNDEF Allows or rejects access to undefined resources	--
Create processing	
CREATE Defines the level of access required to create a data set on tape.	RACF You can protect your tape data sets by using normal RACF processing.

6.3.5 Control-T Pattern Masking

In Control-T releases 5.1 and later there is a pattern matching method which is similar to that used in DFSMSrmm and DFSMSdss. You need to translate the Control-T pattern masks into DFSMS/MVS Removable Media Manager data set name masks.

The following special characters can be used in Control-T.

(?)	Question mark
(!)	Exclamation mark
(#)	Pound sign
(@)	At sign
(*)	Asterisk
(-)	Dash
(\)	Back slash

Table 56 lists the Control-T pattern masks and their DFSMSrmm equivalences.

Note: The DFSMSrmm conversion tools only handle the conversion of the '-' partial masking character, converging it to '*.*.*'.

Table 56 (Page 1 of 2). Control-T Pattern Masks and DFSMSrmm Equivalences	
Pattern Character	
Control-T	DFSMSrmm
Question mark (?)	
The absence of any single character matches	% for a single character * for absence of 1 or more characters
Exclamation mark (!)	
A place holder for a single character	(% percent sign) A place holder for a single character (¬ not sign) A place holder for a single character in a pseudo-gdg data set name mask
Pound sign (#)	
A place holder for a numeric value.	Closest equivalent is %

Table 56 (Page 2 of 2). Control-T Pattern Masks and DFSMSrmm Equivalences	
Pattern Character	
Control-T	DFSMSrmm
At sign (@)	
A place holder for any alphabetic character	Closest equivalent is %
Asterisk (*)	
Represents zero or more characters	Represents zero or more characters
Dash (-)	
Represents zero or any number of characters before and after its position	*.* Represents zero or more characters or qualifiers
Back slash (\)	
Indicates that the following character is not a pattern character	--

6.3.6 Inventory Management and Reports

Control-T provides batch facilities to perform the following functions:

- Batch update and reporting
- Tape retention and expiration
- Storage location management
- Backup, recovery, and verification of databases

Except for batch update and reporting, DFSMSrmm provides similar functions with the EDGHSKP, EDGBKUP, and EDGUTIL utilities (for more details, see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932).

EDGHSKP must be executed with the DFSMSrmm subsystem active. EDGBKUP and EDGUTIL can be executed independently of the subsystem.

The utilities have the following functions:

- EDGHSKP
 - Processes vital records
 - Performs storage location management
 - Performs expiration processing
 - Creates an extract data set
 - Backs up the DFSMSrmm CDS
 - Backs up and clears the DFSMSrmm journal
- EDGBKUP
 - Backs up the DFSMSrmm CDS
 - Backs up the DFSMSrmm journal
 - Recovers and restores the DFSMSrmm CDS
 - Reorganizes the DFSMSrmm CDS
- EDGUTIL
 - Verifies the DFSMSrmm CDS.⁵
 - Mending the DFSMSrmm CDS.

⁵ The EDGUTIL function can also provide consistency checking between the CDS and the TCDB in a system-managed tape library.

DFSMSrmm does not provide any batch update utility. However, you can execute the DFSMSrmm TSO commands ADD, CHANGE, and DELETE in batch to achieve the same result. The ISPF dialog has the option to generate and group together TSO commands for later batch processing.

Control-T provides batch reports. These may be supplemented with the use of the optional CA-EARL program, which allows you to customize your own reports from the TMC.

You can also use the Control-T program, TMSGRW, to produce batch reports, but Control-T Release 5.1 will be the last release that provides TMSGRW.

DFSMSrmm provides two reporting programs and one reporting REXX:

- EDGRPTD—processes the report extract file to produce:
 - Volume movement reports
 - Moved from storage locations
 - Ready-to-scratch sorted by RACK
 - Sorted by BIN number
 - Sorted by OWNER
 - Moved to storage locations
 - Sorted by RACK number
 - Sorted by OWNER
 - Volume inventory reports.
 - Moved from storage locations
 - Sorted by volume serial number
 - Sorted by RACK number or BIN number
 - Sorted by OWNER
- EDGAUD—processes DFSMSrmm SMF records to produce:
 - Security report (tracking information for classified data)
 - Audit report (CDS changes).
- EDGJRPT—processes the extended extract file
 - Pull list for scratch tapes sorted by volume serial number
 - Pull list for scratch tapes sorted by data set name
 - Inventory list by volume serial number
 - Inventory list by data set name
 - Inventory of data set including number of KB used
 - Inventory of volume serial number by location
 - Inventory of data set names by location
 - Inventory of BIN numbers by location
 - List of all data set names at loan location
 - List of all volume serial numbers at loan location
 - List of all multiple volume, multiple data sets
 - Movement report including the first data set name on the volume
 - Movement report sorted by storage location bin number
 - Movement report sorted by volume serial number
 - Inventory list sorted by volume serial number including volume count

You can execute the DFSMSrmm TSO commands, LIST and SEARCH, in batch to produce a wide variety of reports. You can also use DFSORT's ICETOOL to write your own customized reports.

EDGRPTD and EDGAUD sample jobs, REXX programs, and DFSORT's ICETOOL reports are provided in Chapter 14, "Extending DFSMSrmm Reporting" on page 533 and Appendix D, "Sample Exits and Reports" on page 603.

The Control-T scratch list report is shifted one day in the DFSMSrmm environment because the DFSMSrmm report lists volumes that are in PENDING RELEASE status. Those volumes are returned to SCRATCH status during the next housekeeping process, which usually runs the next day.

We suggest that you move to the scratch pool the cartridges that will return to scratch only when they are no longer in PENDING RELEASE status. You can accomplish this in your batch housekeeping jobstream by using a step that lists the volumes in PENDING RELEASE status before running the EDGHSKP step. These are the volumes that are new scratch volumes according to the Control-T report. Alternatively, you can execute EDGHSKP with the EXPROC parameter a second time to return pending release volumes to scratch. If you specify the RELEASE(SCRATCHIMMEDIATE) operand on the ADDVRS subcommand and the VRSEL(NEW) PARMLIB option, only a single run of inventory management EXPROC processing is required to return volumes to scratch. When the VRSEL(OLD) PARMLIB option is in use, DFSMSrmm ignores any VRS release options that may have been set for a volume.

6.3.7 Exits

DFSMSrmm provides source code for the installation-wide exits installed as part of DFSMSrmm during the SMP/E APPLY process. The source code is supplied for:

IGXMSGEX	DFSMSrmm uses the DFSMSdfp MSGDISP exit to update tape drive displays
CBRUXENT	Object Access Method (OAM) library control system (LCS) cartridge entry exit
CBRUXEJC	OAM cartridge eject exit
CBRUXCUA	OAM cartridge change use attributes exit
CBRUXVNL	OAM volume not in library exit.
EDGUX100	RMM installation-wide exit; two samples are provided: EDGUX100, and EDGCVRSX in SAMPLIB

Before the SMP/E ACCEPT, the source code is in SMPSTS; after the ACCEPT, the exits are in AEDGSR1.

If your installation is already using one or more of these exits in the Control-T environment, you must ensure that the correct action is performed during conversion. Chapter 10, “Parallel Running and Validation” on page 427, and Chapter 11, “Cutover to Production” on page 455 describe the procedure to follow.

Control-T uses the CBRUXENT exit to interface the IBM 3494 and IBM 3495 Automated Tape Library Dataservers. You can test the DFSMSrmm version of this exit once you have completed the conversion.

The CBRUXVNL exit is optional. If you use it, however, as with the CBRUXENT exit, you must test it in the DFSMSrmm environment after conversion.

The installation-wide exits that you installed with DFSMSrmm for either ARCTVEXT, IGXMSGEX, CBRUXENT, CBRUXEJC, and CBRUXCUA can replace any existing exits you had previously installed. Decide whether you still require the function your previous exits provided, even if for a short time while introducing DFSMSrmm. If you no longer need the function, proceed with the next implementation step. If you still require the function, consider how to implement it, while still providing the function required by DFSMSrmm.

See the *DFSMSrmm Implementation and Customization Guide* for more details on the DFSMSrmm programming interface that you use with EDGTVEXT, IGXMSGEX, and the OAM exits. The supplied source code for these exits shows how to call the programming interface. You can either modify the supplied exit to call a modified version of your old exit, or modify your old exit to call the DFSMSrmm programming interface. Consider how you might provide an option to disable your code without another code modification. In either case, install the changed software by using an SMP/E USERMOD.

If your installation has DFSMSHsm installed and Control-T is using the ARCTVEXT exit and you have a DFSMSrmm prior to Version 1 Release 4, you will need to merge the Control-T code for the exit with the DFSMSrmm code in order to notify both tape management products when running in parallel. We recommend that you convert your code to run when called by the DFSMSrmm exit, and modify the supplied exit to call your code. An example can be found in Appendix D, "Sample Exits and Reports" on page 603.

Based on the decisions you have taken about VRS management values and scratch pooling, you can update the supplied sample installation exit, EDGUX100, and the dynamic table produced by the conversion programs, to perform the function you require. We recommend you use the sample EDGUX100 exit that is shipped by DFSMSrmm as SAMPLIB member EDGCVR SX. This sample exit uses a dynamic table that is initially built by the sample job EDGJSRDS. The sample job creates a table called UXTABLE. It is a list of assembler macro statements that is easily customizable to add or remove entries. You can add your pooling decisions to the table by modifying the entries built by EDGCSRDS, or by adding new entries. Follow these steps:

1. Copy the sample installation-wide exit, EDGCVR SX, and rename it to EDGUX100, and use this copy as a base for your exit.
2. Update the exit if needed. You should only need to update the exit to add entries to the EDMTAB to identify data that requires only the first file of each volume to be recorded, or if you want to exploit the 'PL100_ITS_CLOSE' call to implement any TMSDISP-like functions, like sticky labels.
3. Update the dynamic table for EDGUX100. You can do this only after running EDGCSRDS using the sample job EDGJSRDS. Copy the three source parts of the dynamic table:

EDGCVR SF The start of the table from SAMPLIB

VRSTABLE The main table, created by EDGCSRDS

EDGCVR SE The end of the table from SAMPLIB

You can update the table with data set names that should have VRS management values assigned, or special retention periods, or special scratch pools, or with additional keyword dates that the exit should support (refer to Figure 103 on page 312).

4. Build an SMP/E USERMOD to apply the updated source code for EDGUX100, and UXTABLE, including the necessary JCLIN statements to get the EDGUX100 load module added to the LINKLIB target library. Following is a sample usermod:

```

++USERMOD (VMRMM01) REWORK(1998066) .
++VER (Z038) FMID(JDZ11B4) PRE(UW19315) .
++JCLIN .
//EDGUX100 EXEC PGM=IEWL,PARM=' LET,NCAL,RENT,REUS,REFR,LIST,XREF'
//SYSLMOD DD DISP=SHR,DSN=SYS1.LINKLIB
//SRCLIB DD DISP=SHR,DSN=TERRI.SAMPLIB
//AEDGMOD1 DD DISP=SHR,DSN=SYS1.AEDGMOD1
//SYSPRINT DD SYSOUT=*
//SYSLIN DD *
INCLUDE AEDGMOD1(EDGUX100)
ENTRY EDGUX100
NAME EDGUX100(R)
//UXTABLE EXEC PGM=IEWL,PARM=' LET,NCAL,LIST,XREF'
//SYSLMOD DD DISP=SHR,DSN=SYS1.LINKLIB
//SRCLIB DD DISP=SHR,DSN=TERRI.SAMPLIB
//AEDGMOD1 DD DISP=SHR,DSN=SYS1.AEDGMOD1
//SYSPRINT DD SYSOUT=*
//SYSLIN DD *
INCLUDE AEDGMOD1(UXTABLE)
NAME UXTABLE(R)
++SRC(UXTABLE) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .
++SRC(EDGUX100) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .
++SAMP(EDGUX100) TXLIB(SRCLIB) DISTLIB(ASAMPLIB) .

```

5. Receive and apply the USERMOD.

6. Your new exit is now ready for use on your system.

You can use the same process to install the EDGUX200 code if necessary.

For more information about DFSMSrmm exits refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

6.3.7.1 Data Set Overwrite

Control-T supports two volume statuses:

PRIVATE The volume contains valid data.

SCRATCH The volume can be used for scratch output requests.

In an DFSMSrmm environment you have three different volume statuses:

MASTER A private volume that contains data that is available for write processing based on the MASTEROVERWRITE PARMLIB option.

ADD New data can be created, but no existing data can be destroyed.

LAST Only the last file on a MASTER volume can be overwritten. The data set name must match the existing data set name.

MATCH An existing file can be overwritten. The data set name must match the existing data set name. All files which are a higher sequence are destroyed.

USER The same as a volume in USER status.

SCRATCH The volume can be used for scratch output requests.

USER The volume is assigned to a user, and it can be overwritten, rewritten, and used repeatedly until the user returns it to the scratch pool.

DFSMSrmm can manage the overwrite of a MASTER volume with the PARMLIB option, MASTEROVERWRITE (see TMSUX1F in Table 54 on page 245). You can tailor this option to meet your installation's needs. The option MASTEROVERWRITE(LAST) is equivalent to the function provided by Control-T. Refer to the *DFSMSrmm Guide and Reference* for a detailed explanation of the function.

6.3.7.2 BLP and NL Processing

In a Control-T environment, because NL tapes do not have internal labels and BLP tapes bypass standard labels, you must supply the VOLSER in the JCL or from the operator console.

BLP and NL processing differs in an DFSMSrmm environment. All scratch tapes must be standard label. When DFSMSrmm successfully reads the VOL1 label for a volume mounted as a result of a nonspecific request, it provides the actual VOLSER to OPEN for use in related processing such as cataloging of the data. This replaces the logical VOLSER (Lnnnnn) generated by OPEN in these circumstances.

For BLP, further processing is determined by the BLP option set in PARMLIB. Processing is as follows:

- | | |
|--------------|---|
| RMM | This is the default and allows BLP processing of USER status tapes and BLP input from MASTER status tapes. |
| NORMM | BLP can be used for reading and writing of master and user status tapes and for output to scratch tapes. BLP reading of scratch tapes is not supported. |

When BLP or NL is used with scratch tapes, DFSMSrmm still changes the volume to master status, but it sets the initialize release action to ensure that the volume has a valid standard label before it is returned to scratch status.

Some changes are required for how and when DFSMSrmm records information for tapes and data read and written using BLP:

- For BLP output to a nonspecific volume, the file sequence number is not checked. For example, LABEL=(2,BLP) could be used. For non-BLP requests, the restriction of using LABEL=(1,label_type) is still enforced.
- The data set information for files processed with BLP is only updated for output to the first file on a volume. All other types of BLP requests change only the volume information, such as the date last read and date last written. For BLP output requests the data set information is only updated in the DFSMSrmm CDS when the data set is closed.

You can enable the use of NL output to scratch volumes by creating a profile in the RACF FACILITY class STGADMIN.EDG.NOLABEL.volser. Once the profile is created, the function is enabled, and only those who have access to the profile can use this function. DFSMSrmm normally supports only standard label scratch tapes for NL tapes, and it relabels them at time of use to NL and to standard label at return to scratch. To use NL scratch tapes you can use the system tape exit IFG019VM. DFSMSrmm now ships a sample IFG019VM exit as member EDG019VM in SAMPLIB. The sample exit checks for a supported request and issues a WTOR to the operator to obtain the volume serial number. After installing the exit, testing should be done to ensure the exit supports the installation's need for NL tape.

6.3.7.3 DFSMSHsm Interface

Control-T and DFSMSrmm process the release of DFSMSHsm tapes differently. When Control-T is informed through the ARCTVEXT exit that a DFSMSHsm volume is no longer required, it uses the extended EDM retention value from PARMLIB to calculate the retention date for the volume.

When DFSMSrmm is informed, through the EDGTVEXT general-use programming interface, that a DFSMSHsm volume is no longer required, the volume is immediately marked *PENDING RELEASE* and will be returned to the scratch pool following the next run of the housekeeping expiration process "*EDGHSKP,PARM=(EXPROC)*". Thus DFSMSrmm can release DFSMSHsm tapes at a different time than Control-T. You must take this difference into account when comparing scratch reports for the two systems.

6.3.7.4 Tape Initialization

In an DFSMSrmm environment the initialization process is different. The EDGINERS utility should be used instead of IEHINITT for the tapes defined in the DFSMSrmm database. You must ensure that your operational procedures are updated in order to use the new DFSMSrmm utility after the conversion process.

DFSMSrmm does not require that you initialize tape volumes before they are available as scratch, but it prevents you from using as scratch a volume that requires initialization and will not include the volume in a scratch pull list. When adding scratch volumes to DFSMSrmm, you have the choice of deciding whether or not they need to be initialized.

Operators should use the DFSMSrmm EDGINERS utility (instead of IEHINITT) to initialize and erase tape volumes. EDGINERS automatically defines new volumes it initializes or erases to the DFSMSrmm CDS. For an existing volume, DFSMSrmm ensures that the proper volume is mounted and the requested action is actually required. You can request DFSMSrmm to automatically initialize scratch volumes when you add them to DFSMSrmm.

6.3.7.5 Tape Pool Management

Control-T allows you to define ranges of tape volumes in a tape library to a specific pool. It restricts scratch requests to a specific pool based on the creating user and/or data set name. For example, you define ranges of tape volumes dedicated to a specific application use only.

Multiple Scratch Pool Support: Control-T supports multiple scratch pool addressing by using a table, TMONSMxx, where you can specify a data set name and the range of volumes associated with it. If the volume opened for output does not fall within the specified range, Control-T denies the request and dismounts the volume.

If your installation uses the TMONSMxx table to access different scratch pools, you must use the DFSMSrmm multiple scratch pools facility. With this support, you will be able to direct new tape data sets to specific scratch pools on the basis of job name and data set name. Thus you can discretely manage sets of volumes on a client, application, or other basis.

Only an installation using an DFSMSrmm installation-wide exit can request multiple scratch pool support. The exit can be used to set a specific scratch pool for nonspecific tape volume output requests.

Implementation for DFSMSrmm: DFSMSrmm decides which pool to use on the basis of the VLPOOL definitions in the DFSMSrmm PARMLIB member. This function is dependent on the correct VLPOOL definitions, but the customer can decide which pool to select by using the EDGUX100 installation exit.

DFSMSrmm intercepts the mount messages issued on the system and, for those for nonspecific volume requests, calls the installation exit to enable it to make a decision. DFSMSrmm saves the scratch pool selected by the exit for use in updating tape drive displays, updating the mount message, and for validating the volume when one is mounted.

The exit can optionally request that DFSMSrmm prevent the cartridge loader from being indexed. Thus drives can be selectively preloaded with scratch volumes from specific pools, but scratch volumes from other pools can be manually mounted without the cartridge loader being emptied.

When the exit selects a pool, the only acceptable volumes are those in that pool. When DFSMSrmm selects the scratch pool without use of the exit, any scratch volume from any eligible scratch pool on that system is acceptable.

DFSMSrmm provides a working sample exit that implements a scratch pooling system based on job names and data set names; all you have to do is update a table containing job names, data set names, and pool prefixes in the EDGUX100 exit.

You can also specify different data set or job name pool tables for each system. If you specify the NAME operand in the VLPOOL command in PARMLIB member EDGRMMxx, the tape drive will display this name rather than the pool prefix.

Note: The specification of a NAME value allows DFSMSrmm to drive basic tape library support (BTLS) scratch pool selection. You can use data set names as well as job names to direct scratch pool selection. Use pool NAME values that match the BTLS pool names, such as SCRATCH1.

6.4 Retention and Movement

In the sections that follow we discuss some basic tape management retention and movement concepts and suggest some factors to consider when converting from Control-T to DFSMSrmm.

6.4.1 Retention Management

You define and use a retention management policy to indicate how long to retain a tape data set in both a tape library and the storage locations.

6.4.1.1 Control-T Policies

Control-T uses specific or unique Julian dates (90000-90365 and 98000-99366) to control tape data set retention and expiration. It also provides the ability to differentiate the dates with true expiration dates using a unique JCL DD statement when creating the tape data set.

EXPDT=90ddd Specifies that a tape data set be retained for a minimum number of days (ddd), after which it will be retained for as long as it is cataloged

EXPDT=98000 Specifies that the tape volume being processed is a foreign tape, not under Control-T management

EXPDT=98ddd Specifies that a tape data set be retained until the specified number of days (ddd) since the last reference has elapsed

EXPDT=99000 Specifies that a tape data set be retained for as long as it is cataloged

EXPDT=99ccc Specifies that a tape data set be retained for the total number of cycles (ccc) or generations

EXPDT=99365 Specifies that a tape data set be retained permanently

EXPDT=99366 Specifies that a tape data set be retained permanently.

In addition, Control-T allows you to change the tape data set's retention and expiration value, after it has been created, using a retention data set. A batch program, which runs during inventory management, processes the retention data set and compares the entry to the TMC volume record. When the condition is met, the retention specified in the retention data set overrides the retention and expiration information in the TMC volume record.

With the retention data set, you can qualify by data set name and partial data set name and further qualify by job name.

Control-T catalog retention is enforced through checking whether a data set is cataloged. Control-T accepts output from the IDCAMS LISTCAT command rather than use a locate to determine whether a data set is still cataloged. Therefore, you do not need to share catalogs. With DFSMSrmm prior to DFSMS/MVS Version 1.5, you must share the ICF catalogs because DFSMSrmm always does a locate.

Control-T treats DFSMSHsm as an EDM, so it does not maintain volume sequence information in its database for multi-volume DFSMSHsm dump tapes, whereas it does for non-DFSMSHsm-managed tapes. During the conversion process the extract program has no choice but to assume that each and every DFSMSHsm dump tape contains a single volume file instead of a multi-volume file. In contrast, DFSMSrmm maintains volume sequence information for multi-volume DFSMSHsm dump tapes so it can recognize a set of dump volumes as being one cycle. If you plan to use DFSMSrmm cycle control to manage the DFSMSHsm dumps, you can get details of the multi-volume set from DFSMSHsm by using the command:

```
HSEND LIST PRIMARYVOLUME(vvvvvv) ALLDUMPS BCDS
```

to find the correct volume sequence order. You can use DFSMSrmm CHANGEVOLUME with the PREVVOL operand to build the correct volume chains once the conversion is completed and DFSMSrmm is started with the converted CDS. This should only be necessary if you want to use CYCLE control for off-site movement of DFSMSHsm tapes.

6.4.1.2 DFSMSrmm Policies

DFSMSrmm does not recognize that EXPDT dates of 90000-90365 and 98000-99364 have special meanings; but, by using the installationwide exit, EDGUX100, or a management class with VRS, you can get DFSMSrmm to provide the same function.⁶ DFSMSrmm uses VRSSs, which define the policy to control the tape data set retention and tape volume retention.

⁶ Expiration dates of 99365 and 99366 or 1999/365 and 1999/366 are considered "never-scratch" dates in DFSMSrmm.

DFSMSrmm allows you to set a system default retention period if the retention or expiration is not specified in the JCL during data set and volume creation. You can extend the retention period for the data set on the volume or for the volume, after it has been created using a VRS definition. The retention period specified in the VRS can override the expiration date in the DFSMSrmm CDS.

DFSMSrmm supports many different types of retention. For a data set, you can request the following types of retention using a specific or a generic data set name:

- *Retention by cycles or retention by cycles while cataloged*, which defines how many cycles or generations of a data set should be retained. Cycles can either be by data set or by day, as in CYCLES or BYDAYSCYCLE.
- *Retention by number of elapsed days since creation or retention by number of elapsed days since creation while cataloged*, which indicates that a data set be retained until the specified number of days since creation have elapsed.
- *Retention by number of elapsed days since last reference or retention by number of elapsed days since last reference while cataloged*, which indicates that a data set be retained until the specified number of days since last reference have elapsed.
- *Retention by catalog*, which indicates that a data set be retained as long as it is cataloged.
- *Retention until expired*, which means that a data set can now optionally be managed separately for retention. A vital record specification management value or a SMS management class can identify retention criteria, and the data set name can be used to identify movement criteria. Up to two VRSs can be used to provide policy information for a data set. The first VRS is called the *primary VRS*, and if there is a secondary VRS, it is called the *secondary VRS*.

When new data is created, and no vital record specification management value or management class is assigned because the jobs do not specify keyword dates and are not SMS managed, VRSEL processing finds only the data set VRS to match to, which specifies UNTILEXPIRED. DFSMSrmm checks the volume expiration date and the UNTILEXPIRED option makes DFSMSrmm drop the data set from retention.

The volume expiration date is also used if the following conditions are true and the PARMLIB option VRSEL(NEW) is specified:

- When a data set matches to a single VRS.
- When UNTILEXPIRED is used in the secondary VRS.

The VRS enhancements introduced with APAR OW30969 enable until expired retention to be based on any type of retention. When a primary and secondary VRS match to a data set, the use of UNTILEXPIRED in the primary uses the secondary VRS chain to determine overall retention. If there is no secondary VRS, or UNTILEXPIRED is used in a secondary VRS or a VRS management value VRS, and PARMLIB option VRSEL(NEW) is used, only the volume expiration date is used. If VRSEL(OLD) is specified in PARMLIB, UNTILEXPIRED retention uses the volume expiration date or catalog status to determine for how long to retain a data set.

Any policy used for retention or movement, or both retention and movement, can be a chain of multiple VRSs, where each VRS in a chain can use a different type of retention. Both the primary and secondary VRS can be VRS chains, offering enough flexibility to create quite complex retention and movement policies.

A secondary VRS can be only a VRS management class or a VRS management value. For more information about how the new VRS processing works, see *Chapter 13, "Additional DFSMSrmm Function" on page 495*.

For a volume, you can request the following types of retention using a specific or generic VOLSER:

- *Retention by number of elapsed days since creation*, which indicates that a volume be retained until the specified number of days since creation have elapsed.
- *Retention by number of volumes*, which indicates how many volumes should be retained.

For all data sets for which you do not specify a retention date in JCL, DFSMSrmm uses a default retention period. You use VRSs to extend this date. For data sets that are left open or that are closed by abend processing, DFSMSrmm provides special ABEND and OPEN VRSs to allow you to control the retention separate from the normal retention for those data sets.

Figure 78 and Figure 79 on page 270 show how to add a data set VRS to set a default retention management period that protects the data set as long as it is cataloged.

Panel Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . SCHLUM.**

Job name mask . . SAVE*

Volume serial . . (May be generic)

VRS name

Figure 78. Define Data Set Name Retention VRS

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : SCHLUM.**	GDG . . . NO
Job name mask : SAVE	
Count 99999	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location OFFSITE	
Number in location . 99999	
Priority 0	
Next VRS in chain . . :	Release options:
Chain using :	Expiry date ignore NO
	Scratch immediate NO
Owner SIEGEL	
Description . . .	
Delete date . . . 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 79. Define While Cataloged Retention VRS

DFSMSrmm provides support for the specific and unique Julian dates of 98000-99364 having special meanings. The support is provided through an installation-wide exit, which you can use to interrogate and recognize the dates. When the installation-wide exit detects a data set using the Julian dates, it assigns a VRS management value that identifies the retention management policy. A sample EDGUX100, called EDGCVRSX, is available in SAMPLIB that uses a dynamic table driven assignment method based on data set names and keyword dates. We recommend that those converting to DFSMSrmm use the EDGCVRSX sample and the table generated by the conversion tools.

When the VRSEL(NEW) PARMLIB option is in use, retention types can be combined to meet complex retention need. A data set can now be retained by a data set name VRS and a (SMS) VRS management class or (EDGXU100) VRS management value together. DFSMSrmm processes both primary and secondary VRSs to retain a data set.

DFSMSrmm uses the following criteria to retain the data set:

- If a data set name matches to a data set name VRS, then DFSMSrmm will use this VRS as the primary VRS.
- The secondary VRS is based on a matching vital record specification management class or vital record specification management value.
- If a match on data set name is not found, the vital record specification management class or vital record specification management value will be called the primary VRS; that means a vital record specification management class or a vital record specification management value can be either a primary VRS (if no matching data set name VRS is found) or a secondary VRS.

When the VRSEL(OLD) PARMLIB option is in use, during inventory management processing, if a match on data set name is not found, DFSMSrmm checks the data set's management value. If the value exists, DFSMSrmm matches the

assigned VRS management value with the defined VRS. If there is a match, DFSMSrmm uses the retention specified in the VRS overriding the expiration from the DFSMSrmm CDS. For example, if you want to support the expiration date of 99000 (retention while cataloged), you need to define a VRS to manage the retention while cataloged. In the installation-wide exit, you intercept the tape data set using the 99000 expiration date and assign a VRS management value matching the defined VRS.

6.4.2 Storage Location Management

You define and use your installation's management policies for those tape volumes and data sets that contain installation-critical data for disaster recovery. These policies define:

- Critical data—data that is vital to your installation
- Storage location—where to store critical data
- Retention criteria—how long to keep the critical data in the storage location
- Movement criteria—how to move critical data to and from the data center and the storage locations.

Because DFSMSrmm and Control-T differ in terms of how they define and process storage location management policies, it is important that you review the current storage location management policy to understand retention and movement criteria.

6.4.2.1 Control-T

Control-T manages tape volume movement to and among vaults using your installation-defined vaulting policy. You store this policy in a VPD, which a batch program processes during the batch update, comparing the entry to the TMC volume record. When there is a match, the tape volume is marked for movement to and among the vaults. The policy is based on data set name, fully or partially qualified, and further qualified by job name. Control-T supports many vaults and does not restrict movement to and among vaults.

The batch program automatically updates the TMC volume record for tape volumes meeting the criteria with the proper location ID (vault code), and generates the store bin (slot) numbers to use in the storage locations. In addition, it generates a report that indicates the current location, the destination location, and the store bin numbers for the tape volumes that require movement. If you delete a retention or vaulting policy, Control-T continues to retain data based on the deleted entry.

Control-T uses the predefined outcode location name of LIBR or LIB to delay volume movement until the specified number of days is reached. You can accomplish the same with DFSMSrmm by defining a VRS with LOCATION(HOME) and use NEXTVRS to chain VRS definitions together. Do not use the DELAY option of VRS unless the delay is only to apply to the latest copy or cycle. For example, you could have defined in the VPDD:

```
V=LIBR,CR=3  
V=VLT1,CR=10
```

The above definitions indicate that the volumes are to be kept in the tape library for three days after creation before they are moved to storage location VLT1.

The DFSMSrmm equivalent to the above VPDD definitions, which is produced by the conversion programs, is:

```
RMM ADDVRS DSN(...) DAYS COUNT(3) -  
    LOCATION(HOME) STORENUMBER(3) -  
    NEXTVRS(VRS00001)
```

```
RMM ADDVRS NAME(VRS00001) DAYS COUNT(10)  
    LOCATION(VLT1) STORENUMBER(10)
```

6.4.2.2 DFSMSrmm

Using your installation-defined VRSSs, DFSMSrmm automatically controls the movement of the data sets on a tape volume or the tape volume from the tape library to and among the storage locations. You can specify a data set name or a data set name mask and, optionally, a job name or job name mask to define a VRS. Additionally, you can specify a VOLSER or a generic VOLSER.

Figure 80 and Figure 81 on page 273 show how to add a data set VRS defining the storage location and the retention period to retain one copy for as long as the data set is cataloged.

Panel

Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . MY.CRITICAL.DATA

Job name mask . . D094746

Volume serial . . (May be generic)

VRS name

Figure 80. Define Data Set Name Movement VRS

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : MY.CRITICAL.DATA	GDG . . . NO
Job name mask : D094746	
Count 5	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location OFFSITE	
Number in location . 5	
Priority 0	
Next VRS in chain . . NXT1	Release options:
Chain using AND	Expiry date ignore YES
	Scratch immediate YES
Owner SIEGEL	
Description . . .	
Delete date . . . 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 81. Define Application Movement Policy VRS

Here is an explanation of two new VRS subparameters **ANDVRS** and **RELEASE**:

ANDVRS

Specifies ANDVRS to create a chain of VRS where all the retention conditions must be true for the data set to be retained. DFSMSrmm uses the STORENUMBER and LOCATION from the first VRS in the ANDVRS subchain.

RELEASE

Specifies special release options after a data set is released. RELEASE options take affect only for data sets that are retained by a VRS.

- EXPIRYDATEIGNORE is specified so DFSMSrmm ignores the volume expiration date when VRSEL(NEW) is used and no other vital VRS matches to the data set.
- SCRATCHIMMEDIATE is specified so DFSMSrmm returns the volume to scratch status in a single run of inventory management EXPROC processing when the only pending release action for the volume is to return to scratch.

DFSMSrmm does not restrict the movement of the data set on the tape volume or the tape volume to and among the storage locations. When a data set on the tape volume, or the tape volume, is moved from a tape library to and among the storage locations for disaster recovery or vital records, DFSMSrmm assigns a storage location bin number, storage location name, and store date.

DFSMSrmm allows you to specify one location name in each data set or volume VRS. If you require that a data set or volume be moved to more storage locations, you can chain together two or more VRSs by using the NEXTVRS, NAME, and ANDVRS keywords. DFSMSrmm does not restrict any sequence of data set or volume moves among storage locations and system-managed libraries. For example, if your installation's policy is to initially retain the data set in the *remote* storage location and then later retain it in the *distant* storage

location, you first define a data set VRS for the *remote* storage location and indicate that the data set requires multiple moves by specifying a name VRS. You then indicate the next storage location, the *distant* storage location, through that name VRS.

Figure 82 and Figure 83 on page 275 show how to add a data set VRS defining the first storage location (remote), the retention (one cycle), and the name of the next storage location.

Panel Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . MY.CRITICAL.DATA

Job name mask . . D094746

Volume serial . . (May be generic)

VRS name

Figure 82. Define GDG Data Set Name VRS

```

Panel  Help
-----
EDGPV210                      DFSMSrmm Add Data Set VRS
Command ==>

Data set mask : MY.CRITICAL.DATA  GDG . . . NO
Job name mask  : D094746

Count . . . . 5                Retention type . . . . . CYCLES
Delay . . . . 0 Days           While cataloged . . . . . YES
Until expired . . . . . NO

Location . . . . . REMOTE
Number in location . 1
Priority . . . . . 0

Next VRS in chain . . NEXT002   Release options:
Chain using . . . . . NEXT      Expiry date ignore . . . . . YES
Scratch immediate . . . . . YES

Owner . . . . . SCHLUM
Description . . .
Delete date . . . 1999/365      ( YYYY/DDD )

Press ENTER to ADD the VRS, or END command to CANCEL.

```

Figure 83. Define Application First Store VRS

Figure 84 and Figure 45 on page 122 show how to add a name VRS defining the second storage location (*distant*) and the retention period (one cycle). This VRS is chained by the VRS definition in Figure 83.

```

Panel  Help
-----
EDGPV200                      DFSMSrmm Add Vital Record Specification
Command ==>

Specify one of the following:
  Data set mask . .
  Job name mask . .

Volume serial . . . . . ( May be generic )

VRS name . . . . . NEXT002

```

Figure 84. Define Name VRS

Panel Help	

EDGPV240	DFSMSrmm Add Name VRS
Name	NEXT002
Count	1
Retention type	CYCLES
While cataloged	no
Until expired	no
Location	DISTANT
Number in location .	2
Next VRS in chain . .	
Chain using . . .	
Owner	SCHLUM
Description . . .	
Delete date . . .	1999/365 (YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	
Command ==>	

Figure 85. Define Application Second Store VRS

During inventory management processing, DFSMSrmm identifies the volumes that must be moved to and among the storage locations and assigns the exact store bin numbers where the volumes are to be stored. DFSMSrmm provides a batch utility (EDGRPTD), which you can use to generate a pull list for the volumes requiring movement. When the volumes have been moved, you must confirm that the moves have taken place. The confirmation process updates the DFSMSrmm CDS to reflect the most current information about where the volumes are stored.

DFSMSrmm uses the retention criteria from the VRS to manage tape volume retention in the storage locations. See the *DFSMSrmm Implementation and Customization Guide* for more information about DFSMSrmm retention management.

When you convert to DFSMSrmm, you can continue to use the same vault names, but, because DFSMSrmm supports longer names, you may want to change the vault names to be more meaningful. For more detailed information about adding or changing VRSS, see Chapter 10, "Parallel Running and Validation" on page 427 for a functional overview of the new VRS commands and panels.

Note: Each time you run inventory management VRSEL processing, DFSMSrmm recalculates retention and movement needs. If you delete a VRS, any data sets it retained can match to another VRS or are eligible for release processing.

6.5 Running the Extract Program

In this section we describe the sample programs and JCL for extracting data for conversion to DFSMSrmm from Control-T. The samples are supplied in the SAMPLIB member EDGCMM01. This book describes them as they were when the book was written. Before running the extract program, you should read and be familiar with the *Data Extraction Programs User's Guide* supplied with the current version of EDGCMM01. It will be maintained to reflect changes and enhancements.

Table 57 lists the samples for the extraction of data from Control-T.

Table 57. IBM-Provided Data Extraction Samples, Programs, and JCL		
Name	Location	Function
EDGCMM01	SAMPLIB	Source and formatted User's Guide
EDGCSRDS	SAMPLIB	Source for TMS extract program
EDGCSVDS	SAMPLIB	Source for TMS extract program
EDGC4BIN	SAMPLIB	Source for TMS 4.9 extract program
EDGC4LDR	SAMPLIB	Source for TMS 4.9 extract program
EDGC5BIN	SAMPLIB	Source for TMS 5.0 and 5.1 extract program
EDGC5LDR	SAMPLIB	Source for TMS 5.0 and 5.1 extract program
EDGJSRDS	SAMPLIB	JCL to execute EDGCSRDS
EDGJSVDS	SAMPLIB	JCL to execute EDGCSVDS
EDGJ4BIN	SAMPLIB	JCL to execute EDGC4BIN
EDGJ4LDR	SAMPLIB	JCL to execute EDGC4LDR
EDGJ5BIN	SAMPLIB	JCL to execute EDGC5BIN
EDGJ5LDR	SAMPLIB	JCL to execute EDGC5LDR
EDGCVRSX	SAMPLIB	Sample EDGUX100 exit
EDGCVRSG	SAMPLIB	Macro used for EDGCVRSX/F
EDGCVRSF	SAMPLIB	Part of sample EDGUX100 table
EDGCVRSE	SAMPLIB	Part of sample EDGUX100 table
EDGCVRSL	SAMPLIB	Sample to load EDGUX100 table
EDGCVRSP	SAMPLIB	Procedure to run EDGCVRSL
EDGC1PRM	LINKLIB	Processes TMS PARMLIB
EDGC1PPL	MODGEN	Macro for EDGC1PRM input

The sample JCL is set up to work in a typical installation. There is sample JCL, some of which is commented out so that you can choose to use it or not, as is your preference. Procedures, which are normally part of other IBM products, are used to assemble and compile the sample code. Thus, you do not have to assemble, link, and compile code before you can use it. If these procedures are not available to you, you can edit the JCL to uncomment the statements that execute the programs from a STEPLIB. If you choose to link-edit the programs yourself, you can use the standard link-edit options. None of the programs are reentrant.

Note: The new sample JCL uses the High Level Assembler (HLASM). If you are running on a system which does not yet have HLASM, you can modify the sample JCL to use Assembler H V2 procedures instead.

6.6 TMS/CA-1 Data Extraction

The data extraction programs extract the proper information from the TMC volume records and DSNB records and convert the information about retention and vaulting contained in the RDS and VPDD. Use the vendor-provided utility, TMSDATA, to unload volume records and DSNB records from the TMC database. See 6.6.2.4, “Input Data Sets” on page 289 for detailed information about the processing requirements of the TMC volume records and the DSNB records. If TMSDATA is not available, see sample EDGJ5BIN for the JCL to extract the records using DFSORT Release 11.1 or higher.

6.6.1 Preparing to Run the Extract

Before executing TMSDATA, you must ensure that there are no broken volume chains or data set chains. TMS/CA-1 provides the following utilities to accomplish this function:

- TMSPTRS (at the 9308 level or higher for 5.0)
- TMSFVSN

In addition, check that the expiration dates used under 5.0 or later levels of Control-T are not set to STATS/xxx. Control-T treats these dates as permanent retention, but you should correct them to be valid expiration dates before conversion. If you do not, the extract programs issue an information message and set a permanent retention date.

In addition, consider running other Control-T utilities to ensure that the data that is being converted to RMM is current, and will probably remain so long enough to complete the conversion process and validate the results. For example, consider running the following:

TMSEXPD	To set the correct expiration dates for volumes on the basis of your RDS definitions and Control-T parameters
TMSCTLG	To check data set catalog status
TMSCYCLE	To remove volumes from cycle control
TMSCLEAN	To return volumes to scratch status
TMSVMxx	To make retention decisions based on the VPDD and assign volumes to vaults and vault slots.

You should obtain a copy of the TMS PARMLIB, CA1.PPOPTION, for input to the conversion programs. There are some PARMLIB options that are used when creating the DFSMSrmm VRS policies, which are obtained from the PARMLIB by EDGC1PRM.

You must unload the TMC database to create the input data sets for the data extraction programs. Use the TMSDATA utility, which creates two sequential data sets, one each for the TMC volume records and DSNB records. If TMSDATA is not available, you can also use DFSORT (Release 11.1 or higher) to unload the TMC database.

Figure 86 on page 279 shows sample DFSORT JCL.

```

//CVT50SRT JOB ,140.SCHLUMBERGER,MSGCLASS=H,REGION=6M,
//      MSGLEVEL=(1,1),NOTIFY=SCHLUM
//* *****
//* * SPLIT Control-T DATABASE *
//* *      PRODUCES SEQUENTIAL DATA SETS FROM TMC VOLUME RECORDS AND *
//* *      DATA SET NAME BLOCKS (DSNB) RECORDS *
//* * *
//* * PREREQUISITE DFSORT REL. 11.1 OR REL. 11.0 INCLUDE THE *
//* *      APAR PL82191 *
//* * *
//* * Control-T REL. 5.0 or higher *
//* *      TMC CONTROL RECORDS *
//* *      EXCLUDING THE FIRST 3 OR 4 Control-T CONTROL RECORDS *
//* *      TMC VOLUME RECORD *
//* *      SORTED BY FIRST VOLSER, VOL SEQ AND VOLSER *
//* *      1. FIRST VOLSER OF DATASET          OFFSET 56  LENGTH 6 *
//* *      2. VOLUME SEQUENCE NUMBER          OFFSET 54  LENGTH 2 *
//* *      3. VOLUME SERIAL NUMBER            OFFSET 0   LENGTH 6 *
//* *      DSNB DATA SET RECORD *
//* *      1. FIRST VOLSER NUMBER OF THE 1ST. OFFSET 16  LENGTH 6 *
//* *      2. FILE SEQUENCE NUMBER            OFFSET 28  LENGTH 2 *
//* * *
//* *****
//DELETE EXEC PGM=IDCAMS
//SYSPRINT DD  SYSOUT=*
//SYSIN DD  *
DELETE DFRMM.CUSTOMER.CA1.VOLUME1 NONVSAM SCRATCH
DELETE DFRMM.CUSTOMER.CA1.DSNB1 NONVSAM SCRATCH
/*
//SORT01 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD  SYSOUT=*
//SORTIN DD  DISP=SHR,DSN=DFRMM.CUSTOMER.CA1
//SORTWK01 DD  UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD  DSN=DFRMM.CUSTOMER.CA1.VOLUME1,DISP=(,CATLG,DELETE),
//            DCB=(RECFM=FB,LRECL=340,BLKSIZE=0),UNIT=SYSDA,
//            SPACE=(27880,(160),RLSE,,ROUND)
//SYSOUT DD  SYSOUT=*
//* -----
//*      TMC VOLUME RECORDS
//*      SORTED BY FIRST VOLSER, VOLUME SEQUENCE AND VOLSER
//*      EXCLUDE TMC CONTROL RECORDS, DSNB RECORDS AND
//*      VOLUME IN DELETE (INACTIVE) STATUS
//* -----
//SYSIN DD  *
SORT FIELDS=(57,6,CH,A,55,2,BI,A,1,6,CH,A)
OMIT COND=(1,7,CH,EQ,C'TMSCTL#',
           OR,1,1,CH,EQ,X'FF',
           OR,85,1,BI,EQ,B'.....1.')
/*

```

Figure 86 (Part 1 of 2). Sample DFSORT JCL

```

//SORT02 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=DFRMM.CUSTOMER.CA1
//SORTWK01 DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD DSN=&&TEMP,DISP=(,PASS,DELETE),
//          DCB=(RECFM=FB,LRECL=340,BLKSIZE=0),UNIT=SYSDA,
//          SPACE=(27880,(900),RLSE,,ROUND)
//SYSOUT DD SYSOUT=*
//* -----
//*      DSNB DATA SET RECORD
//*      SORTED BY COMPLETE RECORD
//*      INCLUDE ONLY DSNB RECORDS
//* -----
//SYSIN DD *
        SORT FIELDS=(1,340,CH,A)
        INCLUDE COND=(1,1,CH,EQ,X' FF')
/*
//SORT03 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=(OLD,DELETE),DSN=&&TEMP,
//          DCB=(RECFM=FB,LRECL=170,
//          BLKSIZE=0)
//SORTWK01 DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD DSN=DFRMM.CUSTOMER.CA1.DSNB1,DISP=(,CATLG,DELETE),
//          DCB=(RECFM=FB,LRECL=170,BLKSIZE=27880),UNIT=SYSDA,
//          SPACE=(27880,(900),RLSE,,ROUND)
//SYSOUT DD SYSOUT=*
//* -----
//*      DSNB DATA SET RECORD
//*      SORTED BY FIRST VOLSER AND FILE SEQUENCE
//*      INCLUDE ONLY DSNB RECORDS WITH DSNB USED INDICATOR
//* -----
//SYSIN DD *
        SORT FIELDS=(17,6,CH,A,29,2,BI,A)
        INCLUDE COND=(1,2,CH,EQ,X' FF80')
//

```

Figure 86 (Part 2 of 2). Sample DFSORT JCL

Note: In step SORT03 each input record of 340 bytes will be split into two single 170 byte records. This means to split the left and right DSNB record into a single 170 byte length file.

The data extraction programs also convert installation-defined tape management policies (retention and outcode location movement) stored in the RDS and VPDD. In addition, the data extraction programs create empty bin records to represent unassigned bins (slots) in VMS-managed vaults (see Figure 100 on page 301).

The data extraction programs create a series of sequential output data sets that become the input to the EDGCNVT program. The output from EDGCNVT is used to build the DFSMSrmm CDS.

The data extraction jobs and programs provide the following functions:

- The EDGCxLDR program converts the TMC volume records and the DSNB records to L-Records and D-Records and creates K-Records for the special expiration dates supplied in the VRS management value (VRSMGMT DD)

data set. It can optionally create empty bin records. It also creates a file of keyword dates for input to EDGCSRDS.

- The EDGCxBIN program creates E-Records representing the empty bins in each of the vaults. When run with the CHECK execution parameter, it can produce a list of OUTCODES you are using and classify them as STORES (VAULTS) and LOAN (all other outcodes). Use these EDGCxBIN created lists as input to EDGCxLDR and EDGCSVDS.
- The EDGCSRDS program converts the RDS retention policies to K-Records. EDGCSRDS also converts the keyword dates and RDS values into a table for use with EDGUX100.
- The EDGCSVDS program converts the VPDD movement policies to K-Records.

6.6.2 Extracting Data from the TMC Records Using EDGCxLDR

Two versions of the volume and data set data extraction programs are provided to accommodate the differences in the TMC volume records and the DSNB records between TMS/CA-1 Releases 4.9, 5.0, and 5.1:

- EDGC4LDR for TMS/CA-1 Release 4.9
- EDGC5LDR for TMS/CA-1 Release 5.0, 5.1, and 5.2

EDGCxLDR converts the TMC volume records to L-Records representing volumes and their first data sets, and the DSNB records to D-Records representing the second and subsequent data sets. For each private volume, it validates the data set name on the volume. If it encounters an invalid data set name on a private volume (that is, a data set name beginning with x'00'), it writes an error message to the ERROR DD data set unless the volume is owned by an EDM identified through the EDMNAME DD data set. You must not proceed with the conversion process until you resolve the error conditions reported in the ERROR DD data set.

Optionally, EDGCxLDR creates K-Records that ultimately become DFSMSrmm VRSs. These K-Records are built from data supplied in the VRSMGMT DD data set, and their names are placed in the VRS management value fields of the appropriate D-Records. You can also code VRSMV=YES in the VRSMGMT file instead of coding any keyword dates, and conversion processing will treat all 9998ddd and 9999ddd expiration dates as keyword dates. See 6.6.2.6, "VRSMGMT Management Value Data Set Record Format" on page 290 for detailed information.

EDGCxLDR calls EDGC1PRM to obtain R9=nnn, RC=nnn from the Control-T PARMLIB (CAI.PPOPTION). K-records generated for keyword dates for catalog or cycles retention are built, including a NEXTVRS(RC) or NEXTVRS(R9) and K-records generated for RC and R9 that specify EXTRADAYS COUNT(nnn). RC is extra days after cycles retention, and R9 is extra days after catalog retention.

EDGCxLDR can also verify the chaining integrity of data sets spanning multiple volumes, multiple data sets on a volume, and multiple data sets spanning multiple volumes. When EDGCxLDR detects a broken data set chain in the input data, it attempts to fix the error in working storage, and the attempt is reported in the ERROR DD data set. However, when EDGCxLDR detects a broken volume chain, it does not attempt to fix the error. TMSPTRS and TMSFVSN must be used. TMSDATA and EDGCxLDR must then be rerun until all chaining errors are removed.

If you currently use TMS 4.9 and have been specifying expiration dates into the next century, you need to undertake special activities to get the correct information to DFSMSrmm. Control-T sets the expiration date to 99366 to prevent expiration, and it sets the correct 21st century date in the audit record it creates. During conversion, EDGC4LDR issues a message for each occurrence of 99366 that it encounters. You must obtain the expiration date from the audit records and use the RMM CHANGEVOLUME volser EXPDT subcommand to set the correct date once you have started DFSMSrmm with the CDS created from the conversion.

6.6.2.1 Sample EDGCxLRD JCL

Figure 87 shows sample execution JCL for TMS/CA-1 Release 4.9, and Figure 88 on page 284 shows sample execution JCL for TMS/CA-1 Release 5.0, Release 5.1, or Release 5.2. See SAMPLIB member EDGJ4LDR and EDGJ5LDR for the supplied sample JCL.

Note: To help identify some of the specifics about your current TMS/CA-1 environment, execute EDGCxBIN with PARM='CHECK'. This creates files of the storage and loan locations which can be input to other conversion programs.

EDGCxBIN externalizes the following:

- Outcode location names currently in use to help you define which outcode locations become DFSMSrmm storage locations and which become DFSMSrmm loan locations
- Volumes occupying outcode slot number zero, which is not compatible with DFSMSrmm
- Outcode slots with more than one volume assigned, which is also not compatible with DFSMSrmm.

```

/*JOBNAME  JOB
/*
/*PROPRIETARY V3 STATEMENT
/*LICENSED MATERIALS - PROPERTY OF IBM
/*"RESTRICTED MATERIALS OF IBM"
/*5695-DF1
/*(C) COPYRIGHT 1994  IBM CORP.
/*END PROPRIETARY V3 STATEMENT
/*
/* $02=spesloc,110,940519,MWW: LRECL for DEXTOUT increased to 512
/* $04=OW11151,110,941219,MWW: Implement macros
/* $03=OW10124,110,950109,MWW: match VRS MV names to EDGUX100
/* $S7=OW30969,140,971125,RS : VRS Enhancements @S7A
/* $K1=KVE0031,140,980223,OG : VRS Enh code review rework @K1A
/*
/* Sort by first VOLSER, volume sequence, and VOLSER
/*

```

Figure 87 (Part 1 of 3). Sample EDGC4LDR JCL: Control-T Release 4.9

```

//SORTTMC EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(200,(50,10),RLSE),
//          DSN=&&SORTTMC,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(101,6,CH,A,55,2,BI,A,72,6,CH,A),DYNALLOC
/*
/* Sort by VOLSER which file 2 starts and file sequence number
/*
//SORTDSNB EXEC PGM=SORT,REGION=4M,COND=(0,NE)
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.DSNBRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(100,(50,10),RLSE),
//          DSN=&&SORTDSNB,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(2,6,CH,A,8,2,BI,A),DYNALLOC
/*
//DEL EXEC PGM=IEFBRI4
//D1 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDR.DEXTOUT.DATA
/*
/* The following step assembles and executes the supplied
/* EDGC4LDR assembler code from SAMPLIB. Alternatively
/* you can remove this step and use a load module you have
/* created into RMM.LOAD
/*
/* EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K1C
/* PARM.G='CONSOLIDATE/200', @K1A
// EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K1C@02C
// REGION.G=4M @S7A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC4LDR) @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB @K1C
// DD DISP=SHR,DSN=SYS1.MODGEN @K1C
//G.SYSUDUMP DD SYSOUT=* @K1M

```

Figure 87 (Part 2 of 3). Sample EDGC4LDR JCL: Control-T Release 4.9

```

/**EDGC4LDR EXEC PGM=EDGC4LDR,REGION=4M,PARM='CONSOLIDATE/200'
/**EDGC4LDR EXEC PGM=EDGC4LDR,REGION=4M
/**STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/**SYSUDUMP DD SYSOUT=*
//REPORT DD SYSOUT=*
//ERROR DD SYSOUT=*
//STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
/**STORE DD * @K1C
/** cols 1-4 Current storage location name
/** 5-12 Owner Id (optional)
/** 13-20 New storage location name
//LOAN DD DISP=SHR,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA @K1C
/**LOAN DD * @K1C
/** cols 1-4 Loan location name
/** 5-12 Owner Id (optional)
//VRSMGMT DD *
99000D99000
99001D99001
99002D99002
99003D99003
98001D98001
98002D98002
98003D98003
/** cols 1-5 Special meaning expiration date
/** 6-13 VRS management value name
//EDMNAME DD *
DFHSM
/** cols 1-8 EDM job name
/** 9-16 EDM RACF ID (optional)
//TMC DD DISP=(OLD,DELETE),DSN=&&SORTTMC
//DSNB DD DISP=(OLD,DELETE),DSN=&&SORTDSNB
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700, @04C
// RECFM=VB,SPACE=(700,(200,100),RLSE),AVGREC=K, @04C
// DSN=STSGWD.EDGCXLDREXTOUT.DATA

```

Figure 87 (Part 3 of 3). Sample EDGC4LDR JCL: Control-T Release 4.9

```

/** Sort by first VOLSER, volume sequence, and VOLSER
/**
//SORTTMC EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS50.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(340,(50,10),RLSE),
// DSN=&&SORTTMC,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(57,6,CH,A,55,2,BI,A,1,6,CH,A),DYNALLOC
/**
/** Sort by VOLSER which file 2 starts and file sequence number
/**

```

Figure 88 (Part 1 of 3). EDGC5LDR: Sample JCL, Control-T Release 5.0, 5.1, and 5.2

```

//SORTDSNB EXEC PGM=SORT,REGION=4M,COND=(0,NE)
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS50.DSNBRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(170,(50,10),RLSE),
//          DSN=&&SORTDSNB,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(17,6,CH,A,29,2,BI,A),DYNALLO
//*
//DEL EXEC PGM=IEFBRI4
//D1 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDR.DEXTOUT.DATA
//D2 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDR.VRSVALUE.DATA
//D3 DD DISP=(MOD,DELETE),UNIT=SYSDA,SPACE=(TRK,1),
//    DSN=STSGWD.EDGCXLDR.DEXTOUTK.DATA
//*
//EDGC5LDR EXEC PGM=EDGC5LDR,REGION=4M
//STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//SYSUDUMP DD SYSOUT=*
//REPORT DD SYSOUT=*
//TMSPARM DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA
//VRSVALUE DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//            AVGREC=U,SPACE=(TRK,(1,1),RLSE),UNIT=SYSDA,
//            DSN=STSGWD.EDGCXLDR.VRSVALUE.DATA
//ERROR DD SYSOUT=*
//STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//*STORE DD *
//* cols 1-4 Current storage location name
//*      5-12 Owner Id (optional)
//*      13-20 New storage location name
//LOAN DD DISP=SHR,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*LOAN DD *
//* cols 1-4 Loan location name
//*      5-12 Owner Id (optional)
//VRSMGMT DD *
VRSMV=YES
//* cols 1-5 Special meaning expiration date
//*      6-13 VRS management value name
//EDMNAME DD *
DFHSM
//* cols 1-8 EDM job name
//*      9-16 EDM RACF ID (optional)
//TMC DD DISP=(OLD,DELETE),DSN=&&SORTTMC
//DSNB DD DISP=(OLD,DELETE),DSN=&&SORTDSNB
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700,
//            RECFM=VB,SPACE=(500,(200,100),RLSE),AVGREC=K,
//            DSN=STSGWD.EDGCXLDR.DEXTOUT.DATA
//DEXTOUTK DD DISP=(,CATLG),UNIT=SYSDA,LRECL=700,
//            RECFM=VB,SPACE=(500,(200,100),RLSE),AVGREC=K,
//            DSN=STSGWD.EDGCXLDR.DEXTOUTK.DATA

```

Figure 88 (Part 2 of 3). EDGC5LDR: Sample JCL, Control-T Release 5.0, 5.1, and 5.2

```

/*
/* Step sorts the keyword dates in VRSVALUE file
/*
//SORTKEYS EXEC PGM=SORT,REGION=6M
//SYSPRINT DD  SYSOUT=*
//SORTIN  DD  DISP=OLD,DSN=STSGWD.EDGCXLDR.VRSVALUE.DATA
//SORTOUT DD  DISP=OLD,DSN=STSGWD.EDGCXLDR.VRSVALUE.DATA
//SYSOUT  DD  SYSOUT=*
//SYSIN   DD  *
SORT  FIELDS=(1,5,CH,A),DYNALLOC
END

```

Figure 88 (Part 3 of 3). EDGC5LDR: Sample JCL, Control-T Release 5.0, 5.1, and 5.2

6.6.2.2 Execution Parameters

Figure 89 shows the syntax of the execution parameters.

```

//EDGCxLDR EXEC PGM=EDGCxLDR,PARM=' x/y'

```

Figure 89. EDGCxLDR Execution Parameter Syntax

In the syntax, **x** is one of the following execution parameters:

- REASSIGN - requests that:
 - Volumes assigned to specific outcode location names have their names translated, name for name, to DFSMSrmm storage location names
 - Those volumes assigned slot numbers be ignored and that new bin numbers be generated.
- CONSOLIDATE - requests that:
 - Volumes residing in multiple outcode locations be combined into fewer RMM storage locations
 - Those volumes slot numbers be ignored and that new bin numbers be generated

and **y** is the number of empty bins requested. You must create empty bin records to provide space for RMM to move additional volumes to the storage locations after conversion. With this method, you do not have to add the empty bin numbers using RMM ADDBIN commands.

Note: When CONSOLIDATE or REASSIGN is requested, EDGCxLDR builds the necessary empty bin records, and eliminates the need to execute EDGCxBIN.

When creating the empty bin records, EDGCxLDR sets the media name to CART3480. If you use subvaulting and plan to do the same under DFSMSrmm, you may have to modify EDGCxLDR to set the media names you want to use for each location. If CART3480 is not the value you want to use, update EDGCxLDR to set a suitable value for each location.

The execution options are provided to help in the following situations:

- Volumes assigned to slot 0:
 - If this is the only situation to be resolved, it is probably easier to force Control-T to move the volumes to a nonzero slot in the outcode location.
- Multiple volumes per slot number:

If only a few slots are occupied by more than one volume, consider moving the necessary volumes to alleviate the situation.

- Total number of outcode locations to be converted to RMM storage locations is greater than the number you want to use under DFSMSrmm.

Attention: Do not use any of the preceding options if any outcode location depends on an accurate outcode slot number.

Figure 90 shows a sample consolidation of two outcode locations to LOCAL that also creates 100 empty bin records for each RMM storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='CONSOLIDATE/100'  
//STORE DD *  
VLT1 LOCAL  
VLT2 LOCAL  
VLT3 REMOTE  
VLT4 DISTANT
```

Figure 90. EDGCxLDR Execution Parameters: Sample 1

Figure 91 shows a sample that reassigns bin numbers, translates the outcode location names to RMM storage location names, and creates 200 empty bin records in each RMM storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='REASSIGN/200'  
//STORE DD *  
VLT1 LOCAL  
VLT2 REMOTE  
VLT3 DISTANT
```

Figure 91. EDGCxLDR Execution Parameters: Sample 2

Figure 92 shows a sample that reassigns bin numbers (without translation of outcode location names to storage location names). The translation of the outcode location names will have to be done later through EDGCNVT, as shown in Figure 93 on page 288.

Note: EDGCNVT can neither consolidate nor ensure that not more than one volume will occupy any given bin.

The sample in Figure 92 also shows how to create 200 empty bin records in each storage location.

```
//EDGCxLDR EXEC PGM=EDGCxLDR,PARM='REASSIGN/00000200'  
//STORE DD *  
VLT1  
VLT2  
VLT3
```

Figure 92. EDGCxLDR Execution Parameters: Sample 3

```
//EDGCNVT EXEC PGM=EDGCNVT
//SYSIN DD *
IF STORLOC EQUALS VLT1 THEN RMMSTORE EQUALS LOCAL
IF STORLOC EQUALS VLT2 THEN RMMSTORE EQUALS REMOTE
IF STORLOC EQUALS VLT3 THEN RMMSTORE EQUALS DISTANT
```

Figure 93. EDGCNVT Storage Location Name Reassignment

Note: In Figure 93, the STORLOC keyword must start in column 4, and the outcode location name must start in column 19. The RMMSTORE keyword must start in column 33, and the DFSMSrmm storage location name must start in column 49.

6.6.2.3 Required Data

A number of fields have been set to default values within EDGCxLDR (Table 58). You have the option of modifying the defaults in the EDGCxLDR assembler source or waiting until the execution of the EDGCNVT program. It may be easier to wait until execution of the EDGCNVT program, when the changes can be driven through the SYSIN DD data set.

Table 58. EDGCxLDR Modifiable Fields

Field	Current Value	Description
HOMLOC	SHELF	Home location name
DFLTOWN	LIBRARY	Owner name
TAPE1600	TAPE1600	Media name
TAPE6250	TAPE6250	Media name
CART3480	CART3480	Media name
CART3490	CART3490	Media name

Note: The media names specified must match the RMM VLPOOL and LOCDEF definitions in the EDGRMMxx member of PARMLIB. If they do not match, errors will occur when DFSMSrmm is active.

Figure 94 shows an example of how to change the default media values to your installation definitions through the SYSIN DD data set for the EDGCNVT program. The sample shows how to change the pool names:

- From TAPE1600 to OLDREEL
- From TAPE6250 to REELS
- From CART3480 to SQUARE
- From CART3490 to BGSQUARE

```
IF VMEDIA EQUALS TAPE1600 THEN VLPOOLNAME EQUALS OLDREEL
IF VMEDIA EQUALS TAPE6250 THEN VLPOOLNAME EQUALS REELS
IF VMEDIA EQUALS CART3480 THEN VLPOOLNAME EQUALS SQUARE
IF VMEDIA EQUALS CART3490 THEN VLPOOLNAME EQUALS BGSQUARE
```

Figure 94. EDGCNVT SYSIN Control Statement Format for VMEDIA

Note: The VMEDIA keyword must start in column 4, and the value must start in column 18. The VLPOOLNAME must start in column 32, and the pool name must start in column 50.

Note: Any volumes defined in the TMC, but never used for data, are defined to TMS as tape reels. As a result, the conversion programs can convert a volume with the wrong media name value. You can correct this before conversion by updating the TMCDEN or TMCTRTCH fields, or during conversion by using the EDGCNVT SYSIN statement:
 IF VOLRANGETHEN MEDIANAME EQUALS

EDGC5LDR now uses the TMC last change user ID as the volume owner, and it only uses DFLTOWN when there is no last change user ID.

Figure 95 shows an example of how to change the owner name from LIBRARY to OPERATOR and add more information relative to the default owner through the SYSIN DD data set for the EDGCNVT program.

```

IF OWNER EQUALS 'LIBRARY' THEN NEWOWNR EQUALS 'OPERATOR'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNS EQUALS 'DATA CENTER'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNF EQUALS 'COMPUTER ROOM'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWND EQUALS 'DEPARTMENT 999'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNT EQUALS '999-9999'
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNU EQUALS 'OPERID '
IF OWNER EQUALS 'LIBRARY' THEN NEWOWNN EQUALS 'SYSTEMA '

```

Figure 95. EDGCNVT SYSIN Control Statement Format for OWNER

Note: The OWNER keyword must start in column 4, and the value must start in column 18. The definition keywords must start in column 32, and their values must start in column 49.

6.6.2.4 Input Data Sets

EDGCxLDR has six input data sets (Table 59).

Table 59. EDGCxLDR Input Data Sets		
ddname	Description	Explanation
TMC	TMC volume record data set (sorted)	Required
DSNB	DSNB data set (sorted)	Required
EDMNAME	EDM job name	Optional
VRSMGMT	VRS management value	Optional
STORE	Outcode location and storage location name	Optional (unless outcodes are used and will become DFSMSrmm storage locations)
LOAN	Outcode location and loan location name	Optional (unless outcodes are used that are not already mapped to storage locations)
TMSPARM	The TMS PARMLIB options member	Optional

Note: If you do not use an optional input data set, you must specify DD DUMMY for it.

For TMC and DSNB records, first you must use the vendor-provided utility programs, TMSPTRS (9308 or newer level) and TMSFVSN, to verify and ensure the integrity of the TMS database (for example, no broken volume or data set chains). Do not rely on EDGCxLDR to fix the broken chains. The existence of broken chains may not be a problem during dry runs when you are only testing

the conversion process to gain experience and do not plan to use the converted database in production mode.

You then use TMSDATA or, if TMSDATA is not available, DFSORT (see Figure 86 on page 279) to offload the TMC database to two data sets:

- Sequential data set containing the TMC volume records
- Sequential data set containing the DSNB records

Sort the sequential TMC volume data set and DSNB data set as follows (see sample JCL and sort control statements in Figure 88 on page 284):

- Sort the TMC volume records by the first VOLSER of the volume set, the volume sequence number, and the volume VOLSER in ascending order.
- Sort the DSNB records by the VOLSER on which file 2 starts, and the file sequence number in ascending order.

6.6.2.5 EDMNAME Identification Data Set Record Format

Columns	Field Description
---------	-------------------

1-8	EDM job name
-----	--------------

9-16	Owner ID
------	----------

17-80	Comments
-------	----------

Figure 96 shows a sample of how to identify EDM job names.

```
//EDMNAME DD *  
DFHSM SYSOPER job name is DFHSM and owner id is SYSOPER  
//* cols 1-8 EDM job name  
//* 9-16 Default volume owner (optional)
```

Figure 96. EDGCxLDR: EDM Identification Sample

The maximum number of EDM job names that can be identified through the EDMNAME DD data set is 20, as EDGCxLDR is originally coded. If this number is too small, you must alter the EDGCxLDR program to increase it.

Note: Each EDM managed volume will be converted to USER status; this indicates that it is a private volume which can be overwritten by any data set. If you do not identify EDM job names, the EDM volumes are converted as normal volumes. If any of these are multi-file volumes, but with only the first file recorded in TMS, you should convert them as EDM volumes, or else DFSMSrmm will only allow file 1 to be used.

6.6.2.6 VRSMGMT Management Value Data Set Record Format

Columns	Field Description
---------	-------------------

1-5	Expiration date
-----	-----------------

6-13	One- to eight-character VRS management value name
------	---

14-80	Comments
-------	----------

Figure 97 on page 291 shows a sample of how to create VRS management value names. Any keyword dates specifically included in VRSMGMT can be defined with a VRS management value to use instead of building one using this default method. In our sample, we show an expiration date of 99000 and a management value name WHILECAT.

```
//VRSMGMT DD *
VRSMV=YES
99000WHILECAT While catalog retention
99001D99001 1 cycle/copy retention
99002D99002 2 cycles/copies retention
99003D99003 3 cycles/copies retention
98001D98001 1 day since last referenced
98002D98002 2 days since last referenced
98003D98003 3 days since last referenced
//* cols 1-5 Special meaning expiration date
//*      6-13 VRS management value
```

Figure 97. EDGCxLDR: VRS Management Value Name Sample

The maximum number of VRS management value names supported by EDGCxLDR is 1000. If this number is too small, you must alter EDGCxLDR to increase it.

Note: The VRS management value names specified in the VRSMGMT DD data set are the names that your EDGUX100 exit should pass to DFSMSrmm when it executes, once you have DFSMSrmm in record, warning, or protect mode.

If you plan to use any VRS management values, you must install the EDGUX100 exit.

If VRSMV=YES is not specified, only those keyword dates specified in VRSMGMT DD are converted to VRS management values. This enables you to move away from keyword dates.

If VRSMV=YES is specified, all keyword dates are converted, and by default the VRS management value name is determined by the conversion programs, unless you also specify the keyword date and a management value to use. The conversion tools use some special VRS naming formats based on the keyword date value. The names are similar to those used by TMS:

- Catalog control EXPDT=99000, uses 'CATALOG'
- Cycle control EXPDT=99ddd, uses 'CYCLEddd'
- Last reference control EXPDT=98ddd, uses 'LDATEddd'
- Catalog and days control EXPDT=90ddd, uses 'CATLGddd' with a NEXTVRS(CATALOG)

All management values used are written as K records and are written to a new file VRSVALUE, which is input to EDGCSRDS processing.

6.6.2.7 STORE Location Name Data Set Record Format

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (optional)
13-20	Storage location name (optional)
21-80	Comments

Figure 98 on page 292 shows a sample of how to define outcode and storage location names.

```
//STORE DD *
ARC LOCAL
DIS ownerid DISTANT
/* You must use DD DUMMY if no data is supplied
/* cols 1 - 4 Outcode location name
/* 5 - 12 Default volume owner (optional)
/* 13 - 20 Storage location name (optional)
```

Figure 98. EDGCxLDR: STORE DDNAME Sample

EDGCxLDR is coded to a maximum number of 1000 outcode location names to be translated to DFSMSrmm's storage locations. If this number is too small, you must alter EDGCxLDR to increase it. Use EDGCxBIN with the CHECK option to generate the list of vaults to convert as storage locations. You can customize the list as required.

6.6.2.8 LOAN Location Name Data Set Record Format

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (optional)
12-20	Loan location name (optional)
21-80	Comments

Figure 99 shows a sample of how to define loan location names.

```
//LOAN DD *
BANK Loan Location for Payroll Tapes
MFCH Loan Location for Tapes sent to Microfiche Vendor
/* You must use DD DUMMY if no data is supplied
/* cols 1 - 4 Outcode location name
/* 5 - 12 Default volume owner (optional)
/* 12 - 20 Loan location name (optional)
```

Figure 99. EDGCxLDR: LOAN DDNAME Sample

You can have many DFSMSrmm loan location names, and the names are not mandated by DFSMSrmm. Therefore, the outcode location names are used as DFSMSrmm loan location names when converting outcodes to loan locations unless a loan location name is specified.

The maximum number of outcode location names that can be specified to be set to RMM loan locations by EDGCxLDR is 200. If this number is too small, you must alter EDGCxLDR to increase it. Use EDGCxBIN with the CHECK option to generate the list of vaults to convert as loan locations. You can customize the list as required.

6.6.2.9 Output Data Sets

EDGCxLDR creates output data sets containing the converted volumes and data set records, an error report file, and a warning report file (Table 60 on page 293).

<i>Table 60. EDGCxLDR Output Data Sets</i>	
ddname	Description
DEXTOUT	L- and D-Records data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
DEXTOUTK	K-Records data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Error report data set
REPORT	Warning report data set
VRSVALUE	Data set containing all created management values K-records. This data set is input to EDGCSRDS processing.

6.6.2.10 Return Codes

EDGCxLDR sets a return code when it ends (Table 61).

<i>Table 61. EDGCxLDR Return Codes</i>	
Return Code	Description
0	Normal processing
4	- A broken DSNB chain has been encountered and an attempt has been made to fix it. - A VRS management value name in the VRSMGMT DD data set begins with a blank character and is therefore unusable. - A Special Meaning TMC EXPDT was encountered that does not have a corresponding statement in the VRSMGMT DD data set. Permanent retention is forced in the Expiration Date field of the L-Record - A Special Meaning TMC EXPDT was encountered that does not conform to the format 98xxx or 99xxx. Permanent retention is forced in the Expiration Date Field of the L-Record. - A TMC outcode of blanks (x'40404040') was encountered: - If the slot assigned was zero, the current location in the L-Record is forced to SHELF. - If the slot assigned was nonzero, an error message is written stating that the outcode name was blank. The current location in the L-Record is forced to SHELF. - A TMC EXPDT has a format error.
8	There are extra DSNB records for a volume; that is, there are actually more DSNB records than indicated in the TMC volume record's DSNB count field.
12	Too many keyword dates were in use in the TMC. A table for 1000 maximum entries is used by EDGC5LDR. If more are used, the table size must be increased and EDGC5LDR re-assembled.
16	- A broken TMC volume chain has been encountered. - There are missing DSNB records; that is, fewer DSNB records were found than it was indicated in the TMC volume record's DSNB count field. - A private (active) volume TMC record contains an invalid data set name. - The slot number assigned to one or more volumes that is being translated to storage location is zero. - A STORE DD control statement has no storage location name specified.

6.6.2.11 Messages

UNABLE TO PROCESS KEYWORD DATE yyddd - VRS TABLE IS FULL

Severity: ERROR

Explanation: Too many keyword dates were processed.

System Action: Sets return code 12

User Response: Update the EDGC5LDR source code to use a value larger than 1,000 for the number of keyword dates supported, reassemble the updated code and re-run the job.

EXTRA DAYS AFTER CATALOG RETENTION VRS CREATED

Severity: INFORMATION

Explanation: The R9 VRS has been created based on the TMS R9 PARMLIB option.

User Response: None

EXTRA DAYS AFTER CYCLES RETENTION VRS CREATED

Severity: INFORMATION

Explanation: The RC VRS has been created based on the TMS RC PARMLIB option.

User Response: None

EMPTY BIN COUNT GREATER THAN 99999999

Severity: ERROR

Explanation: You have requested to create more than 99999999 empty bins in this run. When CONSOLIDATE or REASSIGN is requested, you can specify how many empty bin records should be added to the storage locations.

System Action: None

User Response: Specify a value between 1 and 99999999 and rerun the job.

Destination: PARMERR

VRS MGMT VALUE DATE ERROR: *input card*- RC=4

Severity: ERROR

Explanation: EXPDT must be either 98xxx or 99xxx - check DDNAME(VRSMGMT)

System Action: None

User Response: You can only specify a VRS management value either 98xxx or 99xxx. For the 90000 date you have to specify 99000, because the program will change the date to 90000 internally.

The maximum number of VRS management value names supported by EDGC5LDR is 1,000. Rerun the job.

Destination: VRSERR

VRS MGMT VALUE DATE ERROR: *input card* - RC=4

Severity: Warning

Explanation: VRS management name must begin in column 6 - check DDNAME(VRSMGMT).

System Action: None

User Response: Specify your one- to eight-character VRS management value name in column 6 in the VRSMGMT DDNAME and rerun the job.

Destination: VRSERR2

NEW STORE NAME MISSING FOR CONSOLIDATE OPTION, CHECK DDNAME(STORE) - RC=16

Severity: ERROR

Explanation: Also check EXEC PARM if 'CONSOLIDATE' option is specified.

System Action: None

User Response: Specify your new one- to eight-character storage location name in column 13 in the STORE DDNAME and rerun the job.

Destination: STORERR

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff). RC = 8

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: ERROR

Explanation: An additional DSNB record was found for volume vvvvvv.

System Action: None

User Response: Check your Control-T TMC and rerun your job.

Destination: XDSNBCK

vvvvvv ddd.ddd -> MULTI-VOLUMES BROKEN CHAIN ERROR - RC = 16

where:

vvvvvv is the volume serial number
dddddd is the data set name

Severity: ERROR

Explanation: Check if the volume being processed is the next volume in the multi-volumes chain.

System Action: None

User Response: Correct your Control-T TMC and rerun the job.

Destination: CHAINCK

PREVIOUS VOLSER (pppppp) AND NEXT VOLSER (nnnnnn) DSNB COUNT IS ZERO BUT DSNB RECORD FOUND

where:

vvvvvv volume serial number
pppppp previous volume serial number of data set
nnnnnn next volume serial number of data set

Severity: ERROR

Explanation: Check if the volume being processed is the next volume in the multi-volumes chain and the number of data sets is greater than one.

System Action: None

User Response: Correct your Control-T TMC and rerun the job.

Destination: CHAINCK

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff) AND EDGC5LDR FIXED IT.

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: Warning

Explanation: None
System Action: None
User Response: NONE
Destination: XTRAERR

EXTRA DSNB RECORD FOUND, DSNBVOL(vvvvvv) AND DSNBFVOL(ffffff).

where:

vvvvvv volume serial number of the extra DSNB record was found
ffffff volume serial number on which this extra DSNB record starts

Severity: ERROR

Explanation: The extra DSNB record was ignored.

System Action: None

User Response: Correct your Control-T TMC and rerun the job.

Destination: DSNXTRA

VOLSER WITH DSNBVOL(vvvvvv) AND DSNBFVOL(dddddd) HAS A BROKEN DSNB CHAINED, AND EDGC5LDR FIXED IT.

where:

vvvvvv volume serial number
dddddd volume serial number of the extra DSNB record was found

Severity: Warning

Explanation: None

System Action: None

User Response: None

Destination: CONTFIX

SLOT NUMBER 0000 WAS ASSIGNED TO VOLSER vvvvvv STORED IN //// DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM - RC=16

where:

vvvvvv volume serial number
bbbbbb storage location bin number
11111111 storage location name

Severity: ERROR

Explanation: A volume assigned to a storage location was found, but the bin number (SLOT) is 0000.

System Action: None

User Response: Correct the error in the Control-T TMC and rerun the job.

Destination: BIN0

**0000 STORE WAS NOT DEFINED IN STORE DD OR LOAN DD.
VOLSER vvvvvv IS MARKED AS BEING STORED IN THE LIBRARY (CURRENT LOCATION IS BEING FORCED TO SHELF). RC=4**

where:

0000 name of the out code or vault location
vvvvvv volume serial number

Severity: Warning

Explanation: A volume assigned to an out-of-area location or a vault location was found, but the location name was either specified in the STOR DD-statement or the LOAN DD-statement.

System Action: None

User Response: Add additional information to the STORE DD-statement or to the LOAN DD-statement.

Destination: PUTLOCK

0000 STORE WAS NOT DEFINED IN STORE DD OR LOAN DD
VOLSER vvvvvv WILL BE MARKED UNDER MANUAL MOVE CONTROL BY EDGCNVT -
CURRENT LOCATION SET TO OUTCODE. RC=4

where:

0000 name of the out code or vault location
vvvvvv volume serial number

Severity: Warning

Explanation: A volume assigned to an out-of-area location or a vault location was found, but the location name was either specified in the STOR DD-statement or the LOAN DD-statement.

System Action: None

User Response: Add additional information to the STORE DD-statement or to the LOAN DD-statement. Or you can check the result after the conversion is finished and change the MOVEMODE to AUTO.

Destination: LIBMSG

VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.
*** CONVERTED AS NON-EDM VOLUME - RC=16**

where:

vvvvvv volume serial number
jjjjjjjj creating job name

Severity:

Explanation: The out-of-area information was ignored.

System Action: None

User Response: Correct your Control-T TMC and rerun the job.

Destination: RESVOL

*** VOLSER vvvvvv CONTAINS INVALID DSNAME (IE. 1ST BYTE IS X'00'); dsname**
*** DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM**
RC=16.

where:

vvvvvv volume serial number
dsname data set name
jjjjjjjj creating job name

Severity: ERROR

Explanation: An invalid data set name dsname on volume vvvvvv was found and the volume status was not scratch.

System Action: None

User Response: Correct your CA1- TMC and rerun the job.

Destination: INVNAME1

VOLSER: vvvvvv DSN: dsname
NO VRS MGMT VALUE SPECIFIED FOR DATE: ddddd CHECK DDNAME (VRSMGMT)
THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

vvvvvv volume serial number
dsname data set name
dddddd VRS management value

Severity: Warning

Explanation: A volser with a special meaning expiration date was found, but the expiration date was not specified in the VRSMGMT DD-statement.

System Action: None

User Response: Add the missing VRS management value to the VRSMGMT DD-statement.

Destination: PUTEVRS

AN INVALID SPECIAL EXPIRATION DATE OF xxxxxxxx HAS BEEN ENCOUNTERED FOR:
dddddd

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

xxxxxxx	special meaning expiration date
vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: You have specified a special meaning expiration date outside the range.

System Action: None

User Response: You can only specify VRS management value names in the range 98ddd and 99ccc.

Destination: PUTCVMSG

NO VRS MGMT VALUE SPECIFIED FOR DATE: ddddd CHECK DDNAME (VRSMGMT)

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: A DSNB record with a special meaning expiration date was found, but the expiration date was not specified in the VRSMGMT DD-statement.

System Action: None

User Response: Add the missing VRS management value to the VRSMGMT DD-statement

Destination: PUTVRSE

AN INVALID SPECIAL EXPIRATION DATE OF xxxxxxxx HAS BEEN ENCOUNTERED FOR:
dddddd

VOLSER: vvvvvv **DSN:** dsname

THE EXPIRATION DATE HAS BEEN FORCED TO PERM RETENTION RC=4

where:

xxxxxxx	special meaning expiration date
vvvvvv	volume serial number
dsname	data set name
dddddd	VRS management value

Severity: Warning

Explanation: You have specified a special meaning expiration date outside the range.

System Action: None

User Response: You can only specify VRS management value names in the range 98ddd and 99ccc.

Destination: PUTCDMSG

*** VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.**
*** DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM**
RC=16.

where:

vvvvvv	volume serial number
dsname	data set name
jjjjjjjj	creating job name

Severity:

Explanation: Read the previous message.

System Action: None

User Response: Correct the error in the Control-T TMC and rerun the job.

Destination: PUTEND

*** VOLSER vvvvvv CONTAINS INVALID DSNAME (IE. 1ST BYTE IS X'00'); dsname ***
RESVOL+14 DSNBCJOB
*** VOLSER vvvvvv MAY BE AN EDM OWNED VOLUME, ITS CREATION JOB NAME IS jjjjjjjj.**
*** DO NOT PROCEED WITH CONVERSION, PLEASE VERIFY AND CORRECT PROBLEM**
RC=16.

where:

vvvvvv	volume serial number
dsname	data set name
jjjjjjjj	creating job name

Severity: ERROR

Explanation: An invalid data set name dsname on volume vvvvvv was found and the volume status was not scratch.

System Action: None

User Response: Correct your CA1- TMC and rerun the job.

Destination: INVNAME2

vvvvvv PREVIOUS VOLSER pppppp (N/A) AND NEXT VOLSER nnnnnn (N/A)
-> DSNB RECORD MISSING OR INCOMPLETE DSNB CHAINED RC=16

where:

vvvvvv	volume serial number
pppppp	volume serial number for the previous volume in the chain
nnnnnn	volume serial number for the next volume in the chain

Severity: ERROR

Explanation: The previous and next volume information not found.

System Action: None

User Response: Correct the error in the Control-T TMC and rerun the job. (You can use the TMSPTRS Control-T utility to fix the problem.)

Destination: DSNERR

BIN (SLOT) NUMBER WAS RESEQUENCED, STARTING WITH 000001
*** DUE TO 'REASSIGN' OR 'CONSOLIDATE' OPTION WAS SELECTED**
RC=4

Severity: Warning

Explanation: You have selected the REASSIGN or CONSOLIDATE execution parameter, and the conversion program has changed some of the slot numbers.

System Action: None

User Response: You must create empty bin records to provide space for RMM to move additional volumes to the storage locations after conversion.

Destination: CLOSE

6.6.3 Creating the Empty Bin Records Using EDGCxBIN

Attention: The empty bin program, EDGCxBIN, should not be run, other than with the CHECK parameter, if the reassign or consolidate option is used in the execution of EDGCxLDR. The necessary empty bin records (E-Records) are provided by EDGCxLDR when the reassign or consolidate parameter is used.

Two versions of the empty bin program are provided to accommodate the differences in the TMC volume records between TMS/CA-1 Releases 4.9, and 5.0 and 5.1:

- EDGC4BIN for TMS/CA-1 Release 4.9
- EDGC5BIN for TMS/CA-1 Release 5.0, Release 5.1, and Release 5.2.

DFSMSrmm keeps track of all of the bins (slots) that exist in each of its shelf-managed storage locations. Each bin has a status of either “in-use” or “empty”. TMS/CA-1 only keeps track of the in-use slots. Therefore, the purpose of EDGCxBIN is to build E-Records that represent the empty DFSMSrmm bins by recognizing gaps in the Control-T in-use slot locations. It also verifies the slot numbers and the outcode location names. If it encounters slot number 0 or multiple volumes in a single slot, it writes an error message to the ERROR DD data set.

Note: DFSMSrmm does not support a bin number of 0. The lowest bin number is 1. DFSMSrmm does not allow more than one volume to be assigned to a given bin in a DFSMSrmm storage location.

If the outcode location name in the TMC volume record does not match one of the outcode location names specified in the STORE DD data set and the volume has a vault slot number assigned, EDGCxBIN writes an error message to the ERROR DD data set.

An execution parameter option is provided to allow the EDGCxBIN program to report all of the outcode locations and the in-use and empty slots in each outcode location. See Section 6.6.3.3, “Execution Parameter” on page 308 for more information.

6.6.3.1 Planning

Before executing EDGCxBIN, you must review the preprocessing considerations described in Section 6.6.1, “Preparing to Run the Extract” on page 278. If you decide to reassign bin numbers and/or consolidate outcode locations through EDGCxLDR, do not execute EDGCxBIN without the CHECK parameter.

EDGCxBIN requires that the records in the SLOTNO input file be sorted by outcode and slot number. This is achieved in the sort step. The sort step is also used to include only those records required for each type of media; the volumes are selected as either tape reels or tape cartridges. When running with the CHECK parameter you can pass all volume records for processing. When creating the empty bin records, without the CHECK parameter, as long as all vaults are named in the STORE DD data set, you can also pass all volume records. If you want to process only part of the vaults, use the SORT INCLUDE statement to select only the correct volume records by vault name.

If you have both tape reels and tape cartridges being sent to vault outcodes, Control-T automatically subvaults the tape cartridges. DFSMSrmm can handle this using the MEDIANAME value, but you must run EDGCxBIN twice; once to process tape reels, and once to process cartridges. The sample EDGJxBIN does this. You can separate the reels and cartridges at sort time by using the TMC DEN field. If you do not have both types of media you need only execute EDGCxBIN once.

6.6.3.2 Sample JCL

Figure 100 shows sample execution JCL for TMS/CA-1 Release 4.9; and Figure 101 on page 305, shows sample execution JCL for TMS/CA-1 Release 5.0, Release 5.1 and Release 5.2. Sample JCL is supplied with DFSMSrmm as members EDGJ4BIN and EDGJ5BIN of SAMPLIB.

```

/*JOBNAME  JOB
/*
/*PROPRIETARY V3 STATEMENT
/*LICENSED MATERIALS - PROPERTY OF IBM
/*"RESTRICTED MATERIALS OF IBM"
/*5695-DF1
/*(C) COPYRIGHT 1994,1995  IBM CORP.
/*END PROPRIETARY V3 STATEMENT
/*
/* $01=OW06598,110,940712,MWW: Do bin checking based on type
/*      of media; split into 3420 and 3480 styles.
/* $02=OW11151,110,941219,MWW: Implement Macros
/* $S7=OW30969,140,971125,RS : VRS Enhancements           @S7A
/* $K1=KVE0031,140,980223,OG : VRS Enh code review rework  @K1A
/* $K2=KVE0066,140,980429,RS : Remove 2nd exec of HLASMCLG Proc  @K2A
/*
/* This Job is setup to use CHECK option for EDGC4BIN
/* When running without the CHECK option remember to remove the
/* LOANOUT and STOREOUT DD statements and provide valid STORE DDs
/*
//DEL      EXEC PGM=IEFBRI4
//DXTOUT DD  DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT  DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCXBIN.LOANOUT.DATA

```

Figure 100 (Part 1 of 4). EDGC4BIN: Sample JCL, Control-T Release 4.9

```

/**
/**  SORT BY OUTCODE AND VAULT SLOT NUMBER
/**  INCLUDING ONLY VALID STORAGE LOCATION NAME
/**  FIRST SORT SELECTS JUST 3480 STYLE VOLUMES
/**
//SORT      EXEC PGM=SORT,REGION=4M
//SYSOUT    DD  SYSOUT=*
//SORTIN    DD  DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT    DD  DISP=(,PASS),AVGREC=K,SPACE=(200,(3,3),RLSE),
//              DSN=&&SLOT34,UNIT=SYSDA
//SYSIN     DD  *
SORT FIELDS=(137,4,CH,A,53,2,BI,A),DYNALLO
* INCLUDE just the cartridge media
INCLUDE COND=((64,1,CH,EQ,X'E3',OR,64,1,CH,EQ,X'E7'))      3480/90
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
*       Only use the following if a subset of vaults are to be used
* INCLUDE COND=((137,4,CH,EQ,C'ARCS',OR,137,4,CH,EQ,C'VLT1',
*              OR,137,4,CH,EQ,C'OUTS'),AND,
*              (64,1,CH,EQ,X'E3',OR,64,1,CH,EQ,X'E7'))      3480/90
/**
/** OPTION: PARM=CHECK
/**  Use this option to quickly check what vaults are in use
/**  and whether there are any potential problems.
/**
/** The following step assembles and executes the supplied
/** EDGC4BIN assembler code from SAMPLIB. Alternatively
/** you can remove this step and use a load module you have
/** created into RMM.LOAD
/**
//G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',PARM.G='CHECK',      @K2C
//*G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',                  @K2C
//      REGION.G=4M                                              @S7A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC4BIN)                @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB                          @K1C
//      DD DISP=SHR,DSN=SYS1.MODGEN                             @K1C
//G.SYSUDUMP DD SYSOUT=*                                         @K1M
/**EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE),PARM='CHECK'
/**EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE)
/**STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/**SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34

```

Figure 100 (Part 2 of 4). EDGC4BIN: Sample JCL, Control-T Release 4.9

```

//DEXTOUT DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//
//          DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
//* COLS 1-4 STORAGE LOCATION NAME
//*      5-12 OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
//*
//* SORT BY OUTCODE AND VAULT SLOT NUMBER
//* INCLUDING ONLY VALID STORAGE LOCATION NAME
//* 2ND SORT SELECTS JUST 3420 STYLE VOLUMES
//*
//SORT EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS49.VOLRECS.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(200,(3,3),RLSE),
//          DSN=&&SLOT34,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(137,4,CH,A,53,2,BI,A),DYNALLO
* INCLUDE just the tape reel media
INCLUDE COND=((64,1,CH,NE,X' E3',AND,64,1,CH,NE,X' E7')) -3480/90
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
* Only use the following if a subset of vaults are to be used
* INCLUDE COND=((137,4,CH,EQ,C' ARCS',OR,137,4,CH,EQ,C' VLT1',
* OR,137,4,CH,EQ,C' OUTS'),AND,
* (64,1,CH,NE,X' E3',AND,64,1,CH,NE,X' E7')) -3480/90
//*
//* OPTION: PARM=CHECK
//* Use this option to quickly check what vaults are in use
//* and whether there are any potential problems.
//*
//* The following step executes the previously assembled and @K2C
//* linked EDGC4BIN assembler code from SAMPLIB. @K2C
//* Alternatively you can remove this step and use a @K2C
//* load module you have created into RMM.LOAD @K2C
//

```

Figure 100 (Part 3 of 4). EDGC4BIN: Sample JCL, Control-T Release 4.9

```

//G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M,PARM=' CHECK' @K2C
//*G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M @K2C
//*EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE),PARM=' CHECK'
//*EMPTYBIN EXEC PGM=EDGC4BIN,REGION=4M,COND=(0,NE)
//*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//*SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
//DEXTOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
/* COLS 1-4 STORAGE LOCATION NAME
/* 5-12 OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
/* Step removes duplicate names created by running two CHECK steps.
//SORTSTR EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/* Step removes duplicate names created by running two CHECK steps.
//SORTLOAN EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END

```

Figure 100 (Part 4 of 4). EDGC4BIN: Sample JCL, Control-T Release 4.9


```

/**JOBNAME JOB
/**
/**PROPRIETARY V3 STATEMENT
/**LICENSED MATERIALS - PROPERTY OF IBM
/**"RESTRICTED MATERIALS OF IBM"
/**5695-DF1
/**(C) COPYRIGHT 1994,1995 IBM CORP.
/**END PROPRIETARY V3 STATEMENT
/**
/** $01=OW06598,110,940712,MWW: Do bin checking based on type
/** of media; split into 3420 and 3480 styles.
/** $02=OW11151,110,941219,MWW: Implement macros
/** $03=OW14495,120,950727,MWW: Add TMSDATA replacement
/** $04=OW27659,120,970612,MWW: Improve carts/reel selection @04A
/** $S7=OW30969,140,971125,RS : VRS Enhancements @S7A
/** $K1=KVE0031,140,980223,OG : VRS Enh code review rework @K1A
/** $K2=KVE0066,140,980429,RS : Remove 2nd exec of HLASMCLG Proc @K2A
/**
/**
/** This Job is setup to use CHECK option for EDGC5BIN
/** When running without the CHECK option remember to remove the
/** LOANOUT and STOREOUT DD statements and provide valid STORE DDs
/**
//DEL EXEC PGM=IEFBR14
//DEXTOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
/**
/** SORT BY OUTCODE AND VAULT SLOT NUMBER
/** INCLUDING ONLY VALID STORAGE LOCATION NAME
/** FIRST SORT SELECTS JUST 3480 STYLE VOLUMES
/**
//SORT EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.TMS51.VOLRECS.DATA @04C
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(340,(3,3),RLSE),
// DSN=&&SLOT34,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(109,4,CH,A,113,4,BI,A),DYNALLOC
* INCLUDE just the cartridge media
INCLUDE COND=((90,1,CH,EQ,X'E3', 38K BPI @04C
OR,90,1,CH,EQ,X'E7', 38K BPI IDRC @04A
OR,89,1,CH,EQ,X'C0', 3480 18TRK @04A
OR,89,1,CH,EQ,X'E0', 3480 36TRK @04A
OR,89,1,CH,EQ,X'E1')) 3490 E 36TRK @04A

```

Figure 101 (Part 1 of 4). EDGC5BIN: Sample JCL, Control-T Release 5.0, 5.1, and 5.2.

```

* INCLUDE selected vaults by name ARCS VLT1 and OUTS
*       Only use the following if a subset of vaults are to be used
* INCLUDE COND=((109,4,CH,EQ,C' VLT1',OR,
*               109,4,CH,EQ,C' ARCS',OR,
*               109,4,CH,EQ,C' OUTS'),AND,
*               (90,1,CH,EQ,X'E3',
*               OR,90,1,CH,EQ,X'E7',
*               OR,89,1,CH,EQ,X'CO',
*               OR,89,1,CH,EQ,X'EO',
*               OR,89,1,CH,EQ,X'E1'))
*               38K BPI      @04C
*               38K BPI IDRC @04A
*               3480 18TRK   @04A
*               3480 36TRK   @04A
*               3490 E 36TRK @04A
/*
/** OPTION: PARM=CHECK
/** Use this option to quickly check what vaults are in use
/** and whether there are any potential problems.
/**
/** The following step assembles and executes the supplied
/** EDGC5BIN assembler code from SAMPLIB. Alternatively
/** you can remove this step and use a load module you have
/** created into RMM.LOAD
/**
//G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL',PARM.G='CHECK', @K2C
//*G1 EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL', @K2C
//      REGION.G=4M @S7A
//C.SYSIN DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGC5BIN) @S7C
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB @K1C
//      DD DISP=SHR,DSN=SYS1.MODGEN @K1C
//G.SYSUDUMP DD SYSOUT=* @K1M
/*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE),PARM='CHECK'
/*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE)
/*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
/*SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
//DEXTOUT DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
//      AVGREC=K,SPACE=(512,(1,1),RLSE),UNIT=SYSDA,
//      DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//      AVGREC=K,SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
//      DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,DSORG=PS,
//      AVGREC=K,SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
//      DSN=STSGWD.EDGCXBIN.LOANOUT.DATA

```

Figure 101 (Part 2 of 4). EDGC5BIN: Sample JCL, Control-T Release 5.0, 5.1, and 5.2.

```

/**STORE    DD  DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA          @K1C
/**STORE    DD  *                                                  @K1C
/** COLS 1-4  STORAGE LOCATION NAME
/**      5-12  OWNER NAME AND IT MUST HAVE A VALID RACF PROFILE
/**
/** SORT BY OUTCODE AND VAULT SLOT NUMBER
/** INCLUDING ONLY VALID STORAGE LOCATION NAME
/** 2ND SORT SELECTS JUST 3420 STYLE VOLUMES
/**
/**SORT      EXEC PGM=SORT,REGION=4M
/**SYSOUT    DD  SYSOUT=*
/**SORTIN    DD  DISP=SHR,DSN=STSGWD.TMS51.VOLRECS.DATA            @04C
/**SORTOUT   DD  DISP=(,PASS),AVGREC=K,SPACE=(340,(3,3),RLSE),
/**          DSN=&&SLOT34,UNIT=SYSDA
/**SYSIN     DD  *
SORT FIELDS=(109,4,CH,A,113,4,BI,A),DYNALLO
* INCLUDE just the tape reel media
INCLUDE COND=((90,1,CH,NE,X'E3',          -38K BPI          @04C
              AND,90,1,CH,NE,X'E7',        -38K BPI IDRC    @04A
              AND,89,1,CH,NE,X'CO',        -3480 18TRK     @04A
              AND,89,1,CH,NE,X'EO',        -3480 36TRK     @04A
              AND,89,1,CH,NE,X'E1'))        -3490 E 36TRK   @04A
* INCLUDE selected vaults by name ARCS VLT1 and OUTS
*      Only use the following if a subset of vaults are to be used
* INCLUDE COND=((109,4,CH,EQ,C'VLT1',OR,
*      109,4,CH,EQ,C'ARCS',OR,              X
*      109,4,CH,EQ,C'OUTS'),AND,
*      (90,1,CH,NE,X'E3',          -38K BPI          @04C
*      AND,90,1,CH,NE,X'E7',        -38K BPI IDRC    @04A
*      AND,89,1,CH,NE,X'CO',        -3480 18TRK     @04A
*      AND,89,1,CH,NE,X'EO',        -3480 36TRK     @04A
*      AND,89,1,CH,NE,X'E1'))        -3490 E 36TRK   @04A
/**
/*** OPTION: PARM=CHECK
/** Use this option to quickly check what vaults are in use
/** and whether there are any potential problems.
/**
/*** The following step executes the previously assembled and      @K2C
/** linked EDGC5BIN assembler code from SAMPLIB.                  @K2C
/** Alternatively you can remove this step and use a               @K2C
/** load module you have created into RMM.LOAD                     @K2C
/**

```

Figure 101 (Part 3 of 4). EDGC5BIN: Sample JCL, Control-T Release 5.0, 5.1, and 5.2.

```

//G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M,PARM='CHECK' @K2C
//*G2 EXEC PGM=*.G1.L.SYSLMOD,REGION=4M @K2C
//*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE),PARM='CHECK'
//*EMPTYBIN EXEC PGM=EDGC5BIN,REGION=4M,COND=(0,NE)
//*STEPLIB DD DISP=SHR,DSN=RMM.LOAD
//*SYSUDUMP DD SYSOUT=*
//ERROR DD SYSOUT=*
//SLOTNO DD DISP=(OLD,DELETE),DSN=&&SLOT34
//DEXTOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//STOREOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//LOANOUT DD DISP=MOD,
// DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//*STORE DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA @K1C
//*STORE DD * @K1C
/* Step removes duplicate names created by running two CHECK steps.
//SORTSTR EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/* Step removes duplicate names created by running two CHECK steps.
//SORTLOAN EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SORTOUT DD DISP=OLD,DSN=STSGWD.EDGCXBIN.LOANOUT.DATA
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(1,8,CH,A),DYNALLOC
SUM FIELDS=NONE removes duplicates
END
/*

```

Figure 101 (Part 4 of 4). EDGC5BIN: Sample JCL, Control-T Release 5.0, 5.1, and 5.2.

6.6.3.3 Execution Parameter

Table 62 describes the optional execution parameter of EDGCxBIN program.

Table 62. EDGCxBIN Execution Parameter	
Parameter	Description
CHECK	Use the CHECK parameter to identify all of your outcodes and produce input for the EDGCxLDR program. A pass is made through the TMC volume records to identify: • All of the outcode locations in use and classify them as either store or loan type. See 6.6.3.5, “Output Data Sets” on page 309 for the information that can be generated. • Each occurrence of slot zero’s use. • Each occurrence of more than one volume in a given slot within a given outcode location. • A summary of in-use and empty slot numbers for each outcode location.

Note: The STORE DD data set is not read when you use the CHECK parameter, which allows full analysis of all volume records, nor are E-Records produced.

6.6.3.4 Input Data Sets

EDGCxBIN has two input data sets (Table 63).

Table 63. EDGCxBIN Input Data Sets	
ddname	Description
SLOTNO	TMC volume records, representing all volumes, sorted by outcode location name and slot number in ascending order.
STORE	Outcode location names with corresponding (optional) storage location names. The EDGCxBIN program does not use the Default Volume Owner field. Thus, this data set can be the same data set that is used with EDGCxLDR. It should have the same data in it, if it is not the same data set. A maximum of 49 vaults can be specified.

Columns	Field Description
1-4	Outcode location name
5-12	Default volume owner (ignored)
13-20	Storage location name (optional)
21-80	Comments

To obtain the list of used vaults, for use as storage locations, you can run EDGCxBIN with the CHECK parameter.

Figure 102 shows a sample of how to change outcode location names to storage location names.

```

/* You must supply the names of the Outcode Location names
/* that will become Storage Locations when RMM is implemented.
//STORE DD *
SAFE      LOCAL  <---  SAFE Outcode to be translated to RMM's LOCAL
ARC        DISTANT  --- Storage Location, ARC to DISTANT, and VLT1
VLT1      REMOTE <---  to REMOTE.
/* cols 1-4  Outcode location name
/*      5-12 Default volume owner (ignored)
/*      13-20 Storage location name (optional)
/*      21-80 Comments

```

Figure 102. EDGCxBIN: STORE DDNAME Sample

6.6.3.5 Output Data Sets

EDGCxBIN creates an output data set containing E-Records that represent the empty bin in each of the outcode locations and an error report data set (Table 64 on page 310). When you use the CHECK parameter, a list of all outcodes by classification can be produced in the LOANOUT DD and STOREOUT DD data sets.

<i>Table 64. EDGCxBIN Output Data Sets</i>	
ddname	Description
DEXTOUT	E-Record data set The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Error messages
STOREOUT	A list of all outcodes using slot numbers. This should match the vaults defined to VMS. Use the file as input to EDGCxLDR STORE DD. This file is optional and is used only if CHECK is requested.
LOANOUT	A list of all outcodes that do not use slot numbers. This should match the codes you use outside VMS. Use the file as input to the EDGCxLDR LOAN DD. This file is optional and is used only if CHECK is requested.

6.6.3.6 Return Codes

EDGCxBIN sets a return code when it terminates (Table 65).

<i>Table 65. EDGCxBIN Return Codes</i>	
Return Code	Description
0	Normal processing
4	Outcode location name from the TMC volume record did not match any of the outcode location names specified in the outcode location name fields in the STORE DD data sets.
16	- Slot number 0 is in use. - More than one volume is assigned a single slot number.
20	Both return code 4 and 16 problems were found.

6.6.4 Overview of Policy Conversion

Conversion of retention and movement policies consists of multiple parts:

1. Conversion of TMC expiration dates, which may be real dates or keyword dates.
2. Determining the Control-T PARMLIB options that affect policies, so that they can be built-in to the DFSMSrmm policies conversion creates.
3. Conversion of the RDS entries to both policies and values for use in the EDGUX100 installation exit.
4. Conversion of the VPD entries to DFSMSrmm VRS policies.

The successful running of DFSMSrmm policy management after conversion relies on the use of the new vital records selection enhancements which requires the use of the DFSMSrmm PARMLIB option VRSEL(NEW). This option supports up to two policies per data set:

1. For identifying retention
2. For identifying movement/vaulting

This is particularly important when keyword dates are used to identify retention and VPD entries also exist for the same data sets.

Conversion proceeds as follows:

1. TMC expiration dates, which are keyword dates, are used to generate a DFSMSrmm VRS management value. The VRS management value is assigned to the data set, and the expiration date is set to the assign/creation date. A K record is created for each generated VRS management value. EDGC5LDR handles this automatically when the VRSMV=YES option is coded in the VRSMGMT file.
2. Each of the programs, EDGC5LDR, EDGCSRDS, and EDGCSVDS obtain the Control-T PARMLIB options that affect policies. They build them into the DFSMSrmm policies they create.
3. EDGCSRDS converts the RDS entries to DFSMSrmm VRS policies (K records at this stage) using the data set name in the VRS as a VRS management value. Only keyword dates and special Control-T retention options are converted to management values. Any RETPD retention values are not converted to VRSs.

EDGCSRDS also converts each of the RDS entries, together with the VRSVALUE file input from EDGC5LDR, into a table for use with the EDGUX100 installation exit. The table, UXTABLE, is an assembler source code table which you can later customize and modify independently of the EDGUX100 installation exit.

4. EDGCSVDS converts the VPD entries to K records, which become DFSMSrmm VRS policies later in the conversion process.

The K records will be equivalent to the rules defined in the Control-T VPD.

EDGCNVT converts the K records to DFSMSrmm VRS policies, and when you run EDGHSKP with the VRSEL execution parameter, DFSMSrmm determines which policies to use for each data set and updates the data set information in the DFSMSrmm CDS with the VRS details it determines, such as: retention date, matching primary and secondary VRS, and whether the data set is VRS retained.

For correct and complete policy conversion, you must:

- Run each of the three programs EDGC5LDR, EDGCSRDS, and EDGCSVDS, using the sample JCL.
- Convert the K records using EDGCNVT.
- Use the EDGCVRSX sample as your EDGUX100 installation exit.
- Ensure that UXTABLE is loaded at IPL time for use by EDGUX100. Use the sample EDGCVRSX procedure to do this.
- Specify the VRSEL(NEW) DFSMSrmm PARMLIB option.

6.6.5 Creating VRS Records Using EDGCSRDS

The EDGCSRDS program converts the Control-T retention policy definitions from the Retention Data Set (RDS) to K-Records and source for a dynamic table for the EDGUX100 installation exit. The created K-Records are "data set name Name Vital Record Specifications" types, but they use a single data set name qualifier that identifies the VRS management value. Also, the EDGCSRDS program will analyze the Control-T PARMLIB member (*CAI.PPOPTION*) to convert the following information in DFSMSrmm CDS K-Records:

RC Extended CYCLE retention
R9 Extended CATALOG retention
RP Default retention
CYD cycle control option
RO retention override option

CJOB cycle control by jobname

No reference is made to the Vault Pattern Description Data Set (VPD), so no matching information is provided or required.

6.6.5.1 How the Conversion Process Works

The Retention Data Set (RDS) input is converted to values to be used by a modified EDGUX100 installation exit sample to assign retention at OPEN time. This includes an entry as a default for all data sets using the RP management value. If RO YES is specified, the converted values will be used to replace any JCL retention values, including keyword dates.

Figure 103 shows the table used by the EDGUX100 installation exit.

* start of RDS entries			
EDGCVRSG DSN=A.B*,			X
JOBNAME=J1,			X
RO=YES			X
VRSVAL=LDATE003			
EDGCVRSG DSN=BACKUP.DATA.G%%V%,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CYCLE002			
EDGCVRSG DSN=BACKUP.DATA,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CYCLE002			
EDGCVRSG DSN=OFFSITE*,			X
JOBNAME=*,			X
RO=YES			X
VRSVAL=CATLG005			
* default RP value			
EDGCVRSG DSN=*,	default RP value		X
RO=NO,			X
VRSVAL=CATALOG			
* start of keyword dates from EDGCxLDR			
EDGCVRSG KEYDATE=99000,			X
VRSVAL=WHILECAT			
EDGCVRSG KEYDATE=99005,			X
VRSVAL=CYCLE005			

Figure 103. Example of Data Produced from the RDS

The table is assembled by the EDGJSRDS sample job into a separate load module called UXTABLE, that can be changed dynamically using the EDGCVRSL program from the EDGCVRSP procedure. The sample jobs, procedures, and programs are in SAMPLIB.

The following keywords are available for the EDGCVRSG macro in the EDGUX100 installation exit:

ACLOPT You can use the ACLOPT keyword to disable the cartridge loader for this request.

1 Cartridge loader is disabled for this request.

0 Cartridge loader is enabled. This is the normal processing.

DSECT	YES/NO Specifies whether to produce a map for a table entry.
DSN	One to forty four characters, following MVS data set naming conventions, including % and *. % can be used to ignore a positional character in the job name. * can be used to ignore all remaining characters in the data set name. A <i>DSN=*</i> means that the entry applies to all data sets. The use of * is not the same as in the generic data set names supported by DFSMSrmm for VRSs and search data set masks. Here the * works like the *.* might in a generic data set name mask. Either KEYDATE, or DSN and JOBNAME may be specified.
JOBNAME	One to eight alphanumeric or national characters including % and *. % can be used to ignore a positional character in the job name * can be used to ignore all remaining characters in the job name. A <i>JOBNAME=*</i> means that the entry applies to all jobs. Either KEYDATE, or DSN and JOBNAME may be specified.
KEYDATE	Specifies an EXPDT value used in JCL. A 5-character numeric value, such as 99000, 99003, 98012. Either KEYDATE, or DSN and JOBNAME may be specified.
POOL	Specifies the RMM VLPOOL prefix to be assigned to all matching data sets. When a POOL is specified, you can also specify ACLOPT.
RO	Specifies the retention override option. YES means to override any specified JCL RETPD or EXPDT value. NO means to accept the JCL value. When YES is specified, any matching data set has the expiration date and management value set using the current table entry values. If a data set matches to an entry with RO=NO, EDGUX100 continues to scan the table for any KEYDATE values that match the JCL specified EXPDT so that it can be converted to a VRS management value. RO value has no meaning for the KEYDATE keyword, because all keyword expiration dates are cleared and are set using the RETPD keyword value from the table. If no value is specified for RETPD, the RMM PARMLIB default retention period value will be used by RMM.
VRSVAL	Specifies the name of a vital record specification management value. A VRS management value can be 1 to 8 alphanumeric or national characters. The value corresponds to the data set name used in a VRS DSNNAME policy, such as DSN('D99000').

Note: The **POOL** and **ACLOPT** keywords will not be used by a Control-T conversion. You can add them later if you wish to use EDGUX100 to control scratch sub-pooling using data set names.

If the RP keyword is specified in the Control-T PARMLIB member, a table entry for DSN=* is created, and either the RETPD=ddd is coded, or the VRSVAL=keydate is coded, depending on whether the default is a retention period or a keyword date. The keyword date management value uses the same naming convention as described earlier.

The last K-records in a chain generated for catalog or cycles retention are built including a *NEXTVRS(RC)* or *NEXTVRS(R9)*. The K-records for RC and R9 that specify *EXTRADAYS COUNT(nnn)* are assumed to be generated by the EDGCxLDR conversion program. RC is extra days after cycles retention, and R9 is extra days after catalog retention.

If the RO=YES option is specified in the Control-T PARMLIB member, each management value K-record generated includes the RELEASE(EXPIRYDATEIGNORE) option.

Any cycle control K-records use BYDAYSCYCLE, as determined by the Control-T PARMLIB option *CDAY=DAY*, and include JOBNAME(*), as determined by CJOB, if *CJOB=JOB* is specified.

Figure 104 shows examples of VRSs built by the EDGCSRDS conversion program.

```

RMM ADDVRS DSN(LDATE003) -
    LASTREFERENCE          -
    COUNT(3)               -

RMM ADDVRS DSN(CYCLE002) -
    CYCLES                 -
    COUNT(2)               -
    NEXTVRS(RC)            -

RMM ADDVRS DSN(CATLG005) -
    DAYS                   -
    COUNT(5)               -
    NEXTVRS(CATALOG)      -

RMM ADDVRS NAME(CATALOG) -
    WHILECATALOG          -
    NEXTVRS(R9)           -

RMM ADDVRS DSN(CATALOG) -
    WHILECATALOG          -
    NEXTVRS(R9)           -

```

Figure 104. VRS Built by EDGCSRDS

6.6.5.2 Sample JCL

Figure 105 on page 315 shows the sample execution JCL. See SAMPLIB member EDGJSRDS for the supplied sample JCL.

```

/*
//DEL      EXEC PGM=IEFBRI4
//DEXTOUT  DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.DEXTOUT.DATA
//VRSTABLE DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA
/* REFORMAT THE RDS RECORDS FOR INPUT TO SORT
// EXEC HLASMCLG,PARM.L='MAP,LET,LIST,CALL'
//C.SYSIN  DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCRFMT)
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//G.IFILE  DD DISP=SHR,DSN=STSGWD.TMS.RDS.DATA
//G.OFILE  DD DISP=(,PASS),DSN=&&RFMTDRS,UNIT=SYSDA,
//          AVGREC=K,SPACE=(80,(1,1),RLSE)
/*
//EDGCSRDS EXEC PGM=EDGCSRDS,REGION=4M,PARM=' BOTH'
//STEPLIB  DD DISP=SHR,DSN=RMM.LOAD
//SYSUDUMP DD SYSOUT=*
//ERROR    DD SYSOUT=*
//PERMRPT  DD SYSOUT=*
//RDS       DD DISP=(MOD,PASS),DSN=&&SORTRDS
//VRSVALUE  DD DISP=SHR,DSN=STSGWD.EDGCXLD.RVRSVALUE
//TMSPARM   DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA
//DEXTOUT   DD DISP=(,CATLG),LRECL=512,RECFM=VB,DSORG=PS,
//          AVGREC=K,SPACE=(512,(2,2),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCSRDS.DEXTOUT.DATA
//VRSTABLE  DD DISP=(,CATLG),AVGREC=K,SPACE=(80,(100,100),RLSE),
//          DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA,UNIT=SYSDA
/*
/* FOLLOWING STEPS ASSEMBLES AND LINKAGE EDITS THE UXTABLE.
/* THE UXTABLE CONSISTS OF 3 PARTS:
/* 1. EDGCVRSF (FROM SAMPLIB)
/* 2. THE ABOVE CREATED VRSTABLE
/* 3. EDGCVRSE (FROM SAMPLIB)
/*
// EXEC HLASMCLG
//C.SYSIN  DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCVRSF)
//          DD DISP=SHR,DSN=STSGWD.EDGCSRDS.VRSTABLE.DATA
//          DD DISP=SHR,DSN=SYS1.SAMPLIB(EDGCVRSE)
//C.SYSLIB DD DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//          DD DISP=SHR,DSN=SYS1.SAMPLIB
//LINKEDIT EXEC PGM=IEWL,PARM=' LIST'
//SYSLMOD  DD DSN=SYS1.LINKLIB,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SYSLIN   DD DSN=&&OBJ,DISP=(OLD,DELETE)
//          DD *
//          NAME UXTABLE(R)
/*

```

Figure 105 (Part 1 of 2). Sample EDGCSRDS JCL

6.6.5.3 EDGCSRDS Execution Parameters

The data set names specified in the RDS and VPDD can be used to match to both standard data set names and generations of a GDG where the base name matches the data set mask. This processing by TMS is independent of whether a fully qualified name or an index is specified.

In DFSMSrmm, if a data set name mask is specified fully qualified, it can only match to a standard data set name unless the GDG operand is also specified for the VRS. If the GDG operand is specified, standard data set names can no longer match to the mask. In DFSMSrmm you can get the same results in these ways:

- Specify a generic data set name mask, so that both GDGs and standard data set names can match:

```
ADDVRS DSN('A.B.*') .....
```

- Create a VRS specifying the specific data set name without the GDG operand, and create a second mask using a pseudo-GDG mask to manage the GDG generations:

```
ADDVRS DSN('A.B') NOGDG .....  
ADDVRS DSN('A.B.G-----V---') NOGDG .....
```

The extract programs use the latter method to most specifically match the retention and movement policies of TMS. However, you can control what EDGCSRDS does by specifying one of the execution parameters—GDG, NONGDG, or BOTH. When BOTH is specified for EDGCSRDS, the UXTABLE table entries are created using DSN=A.B.G%%%%V%%.

EDGCSRDS optionally requires an execution parameter (Table 66).

Table 66. EDGCSRDS Execution Parameter Definition

Parameter	Description
GDG	Specify this parameter if the fully qualified data set names in the RDS are only used to match the GDG base names. If any of the data set names match the standard data sets, use the BOTH parameter.
NONGDG	Specify this parameter if the fully qualified data set names in the RDS never match a GDG base name. If any of the data set names match GDG base names, use the BOTH parameter (default).
BOTH	Specify this parameter if the fully qualified data set names in the RDS can potentially match either standard data set names or GDG base names. If you know more specifically what they match, consider either the GDG or NONGDG parameter.

6.6.5.4 Input Data Sets

EDGCSRDS has three input data sets (see Table 67).

Table 67 (Page 1 of 2). EDGCSRDS Input Data Sets

ddname	Description
VRSVALUE	Data set containing names of all created management value K-records. This data set is created by the EDGCxLDR processing.
RDS	The RDS sorted by data set and job name in descending order (required). The RDS data must be that created by EDGCRFMT processing.

Table 67 (Page 2 of 2). EDGCSRDS Input Data Sets	
ddname	Description
TMSPARM	This is the TMS PARMLIB options member.

Note: If you do not use an optional input data set, you must specify DD DUMMY for it.

6.6.5.5 Output Data Sets

EDGCSRDS creates output data sets and error reports (see Table 68).

Table 68. EDGCSRDS Output Data Sets	
ddname	Description
DEXTOUT	K-Records The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Information and error messages
PERMRPT	Permanent Retention Report
VRSTABLE	All the new EDGUX100 dynamic table entries are written to this file. They are later merged with samples EDGCVRSF and EDGCVRSE and assembled into the UXTABLE load module.

6.6.5.6 Return Codes

EDGCSRDS sets the return code accordingly when it ends (see Table 69).

Table 69. EDGCSRDS Return Codes	
Return Code	Description
0	Normal processing
4	Permanent retention policy found but K-Record is created.
16	Unable to create K-Records because of RDS or VPDD control statement error.

6.6.6 Creating VRS Records Using EDGCSVDS

EDGCSVDS converts the vaulting policies from the VPDD into K-Records representing the DFSMSrmm storage location movement policies. DFSMSrmm does not require that the retention type of all locations in a multiple location movement policy be the same.

The EDGCSVDS program converts the Control-T vault pattern description (VPDD) options by the following rules:

- OR** The EDGCSVDS program detects multiple options on a single **V=** statement and converts them using ANDVRS. This matches the Control-T 'OR' condition.
- AND** Multiple options on separate **V=** statements are converted using NEXTVRS. This matches the Control-T 'AND' condition.
- VR=** VR= in other than the first vault is converted to EXTRADAYS in a NAME VRS. VR= in the first vault is still converted as DAYS (CR=) in a DSNAME VRS.

CJOB= Any cycle control K-records use BYDAYSCYCLES as determined by CYD=YES and include JOBNAME(*) as determined by the VPD entry.

Note: All K-records created include retention types:

BYDAYSCYCLES
CYCLES
DAYS
LASTREFERENCEDAYS
EXTRADAYS
UNTILEXPIRED
WHILECATALOG

and count value *COUNT(days/cycles)*.

If you decide to reassign the storage location names when running EDGCxLDR, EDGCSVDS needs to know the new name of each of the storage locations; see Section 6.6.6.3, “Input Data Sets” on page 319 for more information.

The EXP value is converted to UNTILEXPIRED, and should work the same way as it did under TMS. If a data set matches to a data set name VRS, converted from the VPD, with UNTILEXPIRED, and also has a VRS management value assigned, the VRS management value determines how long the data set name VRS retains the data set. For data sets that have no VRS management value the volume expiration date is used by DFSMSrmm.

Note: You must use the DFSMSrmm PARMLIB option VRSEL(NEW) after converting from TMS. If you do not take this into consideration, DFSMSrmm may use an incorrect retention.

6.6.6.1 Sample JCL

Figure 106 shows the sample execution JCL. See SAMPLIB member EDGJSVDS for the supplied sample JCL.

```
/*  
//DEL      EXEC PGM=IEFBR14  
//DEXTOUT DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,  
//          DSN=STSGWD.EDGCSVDS.DEXTOUT.DATA  
/*  
//EDGCSVDS EXEC PGM=EDGCSVDS,PARM=' BOTH'  
//STEPLIB DD DISP=SHR,DSN=RMM.LOAD  
//SYSUDUMP DD SYSOUT=*  
//ERROR    DD SYSOUT=*  
//EXPRPT   DD SYSOUT=*  
//STORE    DD DISP=SHR,DSN=STSGWD.EDGCXBIN.STOREOUT.DATA  
//VDS      DD DISP=SHR,DSN=STSGWD.TMS.VPDD.DATA  
//DEXTOUT DD DISP=(,CATLG),UNIT=SYSDA,LRECL=512,  
//          RECFM=VB,SPACE=(500,(200,100),RLSE),  
//          DSN=STSGWD.EDGCSVDS.DEXTOUT.DATA  
//TMSPARM DD DISP=SHR,DSN=STSGWD.TMS50.PARMLIB.DATA
```

Figure 106. Sample EDGCSVDS JCL

6.6.6.2 Execution Parameters

The data set names specified in the RDS and VPDD can be used to match to both standard data set names and generations of a GDG where the base name matches the data set mask. This processing by Control-T is independent of whether a fully qualified name or an index is specified.

In DFSMSrmm, if a data set name mask is specified fully qualified, it can only match to a standard data set name unless the GDG operand is also specified for the VRS. If the GDG operand is specified, standard data set names can no longer match to the mask. In DFSMSrmm you can get the same results in these ways:

- Specify a generic data set name mask, so that both GDGs and standard data set names can match:

```
ADDVRS DSN('A.B.*') .....
```

- Create a VRS specifying the specific data set name without the GDG operand, and create a second mask using a pseudo-GDG mask to manage the GDG generations:

```
ADDVRS DSN('A.B') NOGDG .....
```

```
ADDVRS DSN('A.B.G-----V---') NOGDG .....
```

The extract programs use the latter method to most specifically match the retention and movement policies of Control-T. However, you can control what EDGCSVDS does by specifying one of the execution parameters—GDG, NONGDG, or BOTH.

EDGCSVDS optionally requires an execution parameter (Table 70).

Table 70. EDGCSVDS Execution Parameters	
Parameter	Description
GDG	Specify this parameter if the fully qualified data set names in the VPDD are only used to match the GDG base names. If any of the data set names match the standard data sets, use the BOTH parameter.
NONGDG	Specify this parameter if the fully qualified data set names in the VPDD never match a GDG base name. If any of the data set names match GDG base names, use the BOTH parameter (default).
BOTH	Specify this parameter if the fully qualified data set names in the VPDD can potentially match either standard data set names or GDG base names. If you know more specifically what they match, consider either the GDG or NONGDG parameter.

6.6.6.3 Input Data Sets

EDGCSVDS has three input data sets (Table 71). If you do not use an optional input data set, you must specify DD DUMMY for it.

Table 71. EDGCSVDS Input Data Sets	
ddname	Description
VDS	VPDD unsorted (required)
STORE	Storage location name input data set (see Section 6.6.2.7, “STORE Location Name Data Set Record Format” on page 291 for information on data set format). A maximum of 1000 vaults can be specified.
TMSPARM	The TMS PARMLIB option member

6.6.6.4 Output Data Sets

EDGCSVDS creates an output data set containing the converted retention and storage location movement policies from the VPDD, the RDS, and the TMC volume records having the special meaning Julian dates and an error report file (Table 72).

Table 72. EDGCSVDS Output Data Sets	
ddname	Description
DEXTOUT	K-Records The LRECL of this file is overridden during program execution. The value in the JCL examples and samples is just an indication of the approximate record length.
ERROR	Information and error messages
EXRPT	Until Expired Retention Report This file lists all occurrences of data sets in the VPDD that specify a retention of EXP.

6.6.6.5 Return Codes

EDGCSVDS sets the return code accordingly when it ends (see Table 73).

Table 73. EDGCSVDS Return Codes	
Return Code	Description
0	Normal processing
4	Unable to support a VPDD policy option but K-Record is written
12	Unable to create K-Record because the "V=" control statement is missing
16	Unable to create K-Record because of a VPDD control statement error

6.7 Interfaces

Before the conversion process you must consider some information about interfaces.

6.7.1 SMS Interface

If you have an IBM 3494 or an IBM 3495 Automated Tape Library Dataserver, you must ensure that it works correctly in the DFSMSrmm environment. You have to change the OAM exits, using the DFSMSrmm version and provide the correct input to the extract program for those volumes that belong to the tape library.

You also must define a DFSMSrmm location with the same name as the tape library name, using a location type of ATL. Thus, the Control-T location name that you are using for the ATL tapes must be translated to this DFSMSrmm location.

6.7.2 BTLS Interface

If your installation uses BTLS to drive IBM 3494 and IBM 3495 Automated Tape Library Dataserver operations, you will need to notify BTLS when a volume has to return to SCRATCH status. In a Control-T environment this is done with a batch job that gets information from the Control-T database and then updates the BTLS catalog. In DFSMSrmm you can use a simple REXX CLIST that synchronizes the DFSMSrmm database with the BTLS catalog.

You can find a sample CLIST in Appendix D, “Sample Exits and Reports” on page 603.

6.7.3 Short-on-Scratch Processing Interface

DFSMSrmm offers an integrated SMS interface that updates the TCDB automatically.

In addition, you can free volumes that are in PENDING RELEASE status when a short-on-scratch condition is detected inside the automated tape library dataserver without manual intervention. Use the PARMLIB option, SCRATCHPROC(RMMSCR), where RMMSCR is the name of the procedure DFSMSrmm starts to replenish scratch volumes in the automated tape library dataserver.

6.7.4 TSO and ISPF Interfaces

DFSMSrmm has a simple TSO interface that uses a set of powerful commands to drive all tape activities.

There is also an ISPF interface that offers menus for end users, librarians, administrators, and support personnel. You can choose a full DFSMSrmm dialog or a local defined dialog. You can tailor the appropriate ISPF dialog for each user who will have access to DFSMSrmm.

6.7.5 Other Interfaces

Control-T provides an online transaction that accesses CICS, TSO, ISPF, and ROSCOE environments. You can use inquiry and update processing through one of these interfaces, or you can use an operator console to issue Control-T commands.

DFSMSrmm can use any DFSMSrmm command in batch mode using the IKJEFT01 or IKJEFT1A TSO batch interface. It can also process any DFSMSrmm command through the DFSMSrmm API. To use the DFSMSrmm API you need to code in the High Level Assembler language. A preferred method is to use the DFSMSrmm commands from a REXX environment where you can retrieve information back as variables.

If you have any of the above interfaces running in your Control-T tape environment, you must change them to reflect the equivalent function in DFSMSrmm.

6.8 Am I Ready to Continue?

In this chapter we have covered all of the activities that you should have considered before converting from Control-T to DFSMSrmm. After completing the activities in this chapter, you should have the following valid and tested input to load the DFSMSrmm CDS:

- Input to exits
- Correct PARMLIB options
- Housekeeping jobs to use in an DFSMSrmm environment
- Changed reports
- Extract and convert control statements
- Extracted data from the Control-T TMC.

If the data extract program has error return codes, read the error messages to determine the actions required to correct the errors. Then re-run the extraction process (see Section 6.5, “Running the Extract Program” on page 277).

To have a valid DFSMSrmm CDS, you must stop all tape activity when you get the data from the Control-T database until you start DFSMSrmm in record-only mode in parallel with Control-T.

Before producing a valid conversion, you can use a copy of the Control-T database to test the extraction and conversion process, so you do not have to stop the tape activities every time you find that you have to go back and re-run the extract jobs.

Once you have tested the environment, you should have the correct tailored input to the Extract program.

You can now stop tape activities and produce a valid data extraction from the Control-T TMC, RDS, and VPDD.

If your previous tests were satisfactory, you should obtain the correct input to the EDGCNVT program and proceed to the next step.

Chapter 7. Converting from EPIC to DFSMSrmm

In this chapter we describe how to convert from EPIC, a tape management product of Computer Associates, to DFSMSrmm. We review some product differences that you should consider before beginning the conversion process and explain how to prepare the input for the conversion program.

7.1.1.1 Objectives of Chapter

- Explain differences between EPIC and DFSMSrmm
- Describe how to customize DFSMSrmm for equivalent EPIC functions
- List the functions where there is no equivalence
- Extract data and prepare input for EDGCNVT
- Provide hints and tips.

7.1.1.2 Audience

- System programmers
- Storage administrators
- Librarians

At the end of this chapter you will have reviewed the implementation of your current EPIC environment and will understand the detailed changes required to customize DFSMSrmm to meet your tape management needs.

7.2 EPIC Overview

EPIC uses your installation-defined retention management, vault management, and tape pool management policies to manage tape data sets.

EPIC has two databases containing data set, volume, and management data:

- DSN catalog - contains data set retention information and data set details
- VMS catalog - contains the vaulting rules

EPIC is a data-set-name-based product, and its DSN catalog is like an extension to the MVS catalog. Tape volumes used for data sets defined to EPIC are automatically defined and managed.

Note: EPIC and Prevail/XP Media can be run in parallel with DFSMSrmm, but in order to use tape volumes successfully, you have to install a modification. The modification is necessary because of the way that EPIC modifies DFSMSdfp OPEN/CLOSE/EOV processing. The modification is to DFSMSrmm, and the details of this modification, including how to install and validate it, are in Appendix E, "Modification Required for EPIC and DFSMSrmm Parallel Run" on page 671. Use the information provided to temporarily update DFSMSrmm while running in parallel and using tape volumes.

7.3 EPIC and DFSMSrmm

To ensure a successful conversion, it is important that you understand some of the differences between the products and how those differences affect your conversion project.

7.3.1 Terminology

Table 74 compares and defines some EPIC and DFSMSrmm terms.

<i>Table 74 (Page 1 of 2). EPIC and DFSMSrmm Terminology</i>		
EPIC	DFSMSrmm	Definition
DSN catalog	Control data set (CDS)	The tape management system database containing information about tape volumes and data sets
VMS catalog	Control data set (CDS)	The database containing the management rules of tape data sets. In DFSMSrmm these rules are in the CDS.
Recorder file	Journal	The sequential data set that keeps track of changes in the database. It can be used for recovery purposes.
Retention rules	Vital record specification (VRS)	The rule that manages the life of a tape data set
Vault	Location	Represents where the volume physically resides; for example, in the data center or an offsite location
Box, case, and pallet	Bin number	The individual shelf location in any of the storage locations
PRIVATE volume	MASTER volume	The status of a volume containing active data
Creation date	Assigned date	The date when a PRIVATE/MASTER volume was first added or when a SCRATCH volume became a PRIVATE/MASTER
Expiration date	Expiration date	When the data set is created, the expiration date used in JCL or calculated by the tape management system
Version	Generation or copy	A unique instance of a tape data set. A generation is used when the data set is a GDS, and copy is used when the data set is not a GDS.
--	Rack number	A unique identifier for the shelf location where a volume is stored while in the library. Also known as the external label. It need not match the VOLSER.
Catalog maintenance facility	ISPF dialog	The online full-screen interface to the data in the DFSMSrmm database. Allows online inquiry, viewing, reporting, and updates. It need not match the VOLSER.
Groupings	Owners	DFSMSrmm allows data to be retrieved and grouped using the userid that created the data.

<i>Table 74 (Page 2 of 2). EPIC and DFSMSrmm Terminology</i>		
EPIC	DFSMSrmm	Definition
Main library	Home location	DFSMSrmm tracks the library location where the volume should be returned after vaulting; for example, a system-managed library.
Manual vault	Loan location or storage location	You can assign a volume to a location by command rather than use the automatic location management function.
Catalog entry	Data set record	A database record describing a file on a volume
Reel number	VOLSER, or volume serial number	The value recorded in the VOL1 label of a tape volume, or the unique name of the volume
Expiration date keywords	VRS management value	DFSMSrmm allows you to use the expiration date to assign a VRS management value to a data set. This identifies retention criteria.
Date last accessed	Last reference date, last read date, or last write date	A date when the data was last updated or read
AutoCat, AutoLog	Record	DFSMSrmm automatically records all data set information for managed volumes.

7.3.2 Functions

Table 75 compares the key tape management functions of EPIC and DFSMSrmm.

Table 75 (Page 1 of 3). EPIC and DFSMSrmm Functions	
Function	
EPIC	DFSMSrmm
Tape security	
• Tape must be scratch for output scratch request • Data set name on tape must match inventory-specific request • 44-character data set name checked against DSN catalog on input processing	• Tape must be scratch for output scratch request • Data set name on tape must match inventory-specific request • 44-character data set name checked against CDS on input processing • RACF interface for tape profile management • Access list for non-RACF system • Security erase on scratch • Prevent read of scratch tapes • Security classification by generic dsname • Overwrite of MASTER volumes controlled by PARMLIB option • If the DFSMSrmm address space is not started, you must disable the DFSMSrmm subsystem interface to allow the use of the tapes with the EDGRESET utility. Ability to disable the interface is RACF protected. • REJECT option is available at the volume level through parmlib options.
• All output tapes have an expiration date of 99365 to force operator involvement in tape usage when EPIC is not active. • Data sets can be LOCKed to prevent read or write access.	

Table 75 (Page 2 of 3). EPIC and DFSMSrmm Functions

Function	
EPIC	DFSMSrmm
Retention management	
• Based on RETPD or EXPDT • By cycle • By days since last reference • By days since creation • By cataloged • By combination of any retention criteria • Combined retention policies are OR values • No jobname on retention • Can specify default retention installationwide, and by specific or group dsname • Retention policies based on full or partial dsname. Data set name on tape must match inventory-specific request.	• Based on RETPD or EXPDT • By cycle or daily cycle • By days since last reference • By days since creation • By cataloged • By combination of any retention criteria • Combined retention policies can be AND or OR values¹ • By jobname • Can specify default retention installationwide. Default retention can also be specified using generic data set mask in VRS² • Can specify maximum retention installationwide. • Retention policies based on full or generic dsname or VOLSER • By cycle while cataloged • By days since last reference while cataloged • By days since creation while cataloged • By until expired
EXPDT with special meanings	
• 98000-99364	• Dates of 1999365 and 1999366 mean never scratch Installationwide exit required to support special dates
Vault or storage location management	
• Supports up to 65,535 vaults • Define management by dsname – Generic data set name – Qualified by job name • Manages by vault slot number and box number • Unrestricted movement • Can manage movement based on media type, that is, reel or cartridge	• Supports an almost unlimited number of storage locations and system-managed libraries • Define management by dsname, volser or jobname: – Full or generic (ISMF syntax), GDG or pseudo-GDG – Qualified by generic jobname • Manages by store bin number • Unrestricted movement • Supports loan location • Can manage movement based on media name
Security	
• Supports the use of RACF, ACF2, and TopSecret. Includes support for the use of passwords	• Uses the MVS SAF interface to interface with security products including RACF and any functionally equivalent product. The use of passwords is not supported.

Table 75 (Page 3 of 3). EPIC and DFSMSrmm Functions	
Function	
EPIC	DFSMSrmm
Initialization of new tapes	
Provides a utility that updates inventory³	Interface ensures integrity, updates inventory⁴
Volume pools	
Pools are assigned to new data sets by name using DSN catalog definitions or PARMLIB defaults. PARMLIB includes TAPEPOOL definitions for up to 36 volume ranges.	- Pools can be assigned to new data sets based on the system name or job and data set names. Pools are defined based on rack numbers using the VLPOOL options in PARMLIB. - Interface to BTLS to allow DFSMSrmm to select scratch pool. Allows the tape management system to control pool selection and use selection options not available under BTLS. The NAME operand on the VLPOOL option in PARMLIB identifies the BTLS scratch pool name.
Optional sticky labels	
Labeling capability is provided	Interface provided
Identify tapes out of library	
--	Loan location support
¹ Applies to DFSMS/MVS Version 1 Release 4.0 with the VRS SPE installed. ² DFSMSrmm provides permanent retention using 99365/99366 or 1999/365 and 1999/366. ³ The EPIC tape initialize utility is a front-end to IEHINITT. IEHINITT does not support system-managed tape in an automated tape library. Also, an operator reply to WTOR is needed. ⁴ DFSMSrmm utility EDGINERS includes IEHINITT functions, but DFSMSrmm does not use IEHINITT; DFSMSrmm supports system-managed tape, and requires no operator intervention, that is, no WTOR.	

7.3.3 Data Fields

When you list the information for a volume or a data set, there are differences in the data fields. Below we briefly describe the major data field differences.

During the conversion process, the EPIC records are read, and some fields are translated into the format required for the DFSMSrmm CDS.

Some EPIC fields are not used by the extract program, and some DFSMSrmm fields are not present in the EPIC equivalent record.

Other DFSMSrmm fields have a different name in the EPIC DSN catalog but have a similar or equivalent meaning in the DFSMSrmm CDS.

7.3.4 Data Not Converted

Table 76 on page 327 lists all fields that the extract program does not extract from the EPIC DSN catalog.

Table 76 (Page 1 of 2). EPIC Data Not Converted	
	Description
Locked data set flag	Controls access to the tape file on the volume

<i>Table 76 (Page 2 of 2). EPIC Data Not Converted</i>	
	Description
Small tape flag	Indicates mini tape reel volume
Revolving data set flag	Identifies a set of data sets that are part of a closed cycle of volumes.
Conditional catalog flag	Data set disposition
Tape pool ID	Identifies the TAPEPOOL where the data set resides
EDM controlled flag	Identifies externally managed data sets
Last accessed jobname	Jobname of last job to read and/or write the file
Current use count	Although DFSMSrmm has the same field, the extract and EDGCNVT programs do not yet convert this data.
Total use count	DFSMSrmm maintains use count each time a volume is assigned to a new user.
Total error count	DFSMSrmm maintains use count each time a volume is replaced.
Last cleaned date	DFSMSrmm does not have a cleaning function.
VMS return date	Volume retention date In DFSMSrmm the retention date is not always the date on which the volume returns from a storage location.

7.3.5 Data Unavailable for Conversion

The following are new terms in DFSMSrmm that are not available in the EPIC catalog (see Table 74 on page 324 for the field definitions):

- Data set size
- Date data set last read*
- Date data set last written*
- Date volume last read*
- Date volume last written*
- Original expiration date
- Home location
- Rack
- Security level
- SMS data class name
- SMS storage class name
- SMS storage group name
- Tape media type
- Tape usage in kilobytes
- Volume owner UID and node name
- Creation date and time
- Creation UID
- Temporary and permanent read and/or write errors
- Abend in process during close of data set
- Volume release actions
- Last used drive address
- Creating stepname and ddname

*In EPIC, only the date last used is stored.

7.3.6 Data Converted, but Not Quite the Same

Table 77 on page 329 is a list of fields that are present in EPIC but not in DFSMSrmm (see Table 74 on page 324 for the field definitions).

Table 77. Fields Not Quite the Same		
EPIC	DFSMSrmm	Explanation
Creation date and time	Assign date and time	The date on which the volume was assigned to a user
Data set sequence	Physical file sequence number Data set sequence number	The Physical file sequence number field specifies the relative position of a data set on the volume where it resides. The Data set sequence number field corresponds to the IBM Standard HDR1 Data set sequence number field. DFSMSrmm maintains both values. During conversion the physical file sequence number is calculated by EDGCNVT.
Date last used	Read date Write date	The date when the data set was last read. The date when the data set was last written to.
VMS out of area date	Movement tracking date	In DFSMSrmm this is the date when the volume last started to move or when the movement was confirmed. The exact reason can be determined by the volume's movement status.

7.4 Cross-Reference

In this section we present tables that map data from the EDGCNVT input records to EPIC.

7.4.1 Volume Information

Table 78 on page 329 shows the EDGCLREC volume information.

Table 78 (Page 1 of 4). EDGCLREC Volume Information		
	EPIC	Status
Volume serial number	√	Req
Volume owner ¹	√	Req
Volume owner user ID		
Volume owner node ID		
Volume security level		
Volume sequence number	√	Req
Volume create date	√	Req
Volume create time	√	
Volume create user ID		
Volume create job name	√	
UCB type (copy of UCBTYP)		

Table 78 (Page 2 of 4). EDGCLREC Volume Information		
	EPIC	Status
Tape media type information Tape recording format Not 3480 18 track 36 track Tape media type Not 3480 Standard capacity Enhanced capacity Tape compaction technique Unknown Not compact IDRC Tape special attributes None 18 track read-only	 √ √ √	Cond
Total volume kilobytes used		
JCL volume expiration date		
Volume expiration date		Req
Volume retention date		
Count of data sets on volume	√	Req
Temporary read errors		
Temporary write errors		
Permanent read errors		
Permanent write errors		
Tape density (copy of JFCBDEN)		
BPI indicator 1600 bpi 6250 bpi 3480 3480 compact (IDRC) Undefined		Req
Volume status indicators Volume is master Volume pending release Vital record - do not release User tape (assigned by librarian) Tape is on loan Tape open - not yet closed Volume is scratch Volume recorded by open, close, or EOV	 √	Req
Additional volume status indicators Scratch volume claimed by getvol Scratch volume has never been initialized Scratch volume with init action pending Scratch volume waiting to enter ATL Abend in process when a data set closed Abend probably in open, close, or EOV		

Table 78 (Page 3 of 4). EDGCLREC Volume Information		
	EPIC	Status
Volume label type Default retention period used Label type is NL Label type is AL Label type is SL Tape last written with BLP SL or AL with user labels		Req
Volume release actions Return to scratch pool - default Replace tape on release Reinitialize Degauss/security erase Return to owner Notify owner		
Volume access information Owner may read volume Owner may update volume Owner may alter volume Read-only protection Update protection Can be used on MVS systems Can be used on VM systems		Req
Media name		
Unit address used for creation	√	
Last used unit address		
Loan location		
Date volume last read	√	
Date volume last written		
Assigned from scratch date	√	
Assigned from scratch time	√	
Copy of JFCBLTYP		
From JFCTRTCH - IDRC support DSN used 3480 IDRC No compaction		Req
Current location of volume	√	Req
Destination name		
Home location name		
Storage group name		
Accounting information		
User description		
Authorized user IDs area		
Rack number ²		Req
Remote store bin number		Cond
Remote store bin date	√	Cond
Remote store bin media name		Cond

Table 78 (Page 4 of 4). EDGCLREC Volume Information		
	EPIC	Status
Label number of first file	√	Req
Name of first data set on volume	√	
First data set's owner's name		
Sequence number of first data set		Cond
First data set's RECFM	√	Cond
First data set's LRECL	√	Cond
First data set's BLKSIZE	√	Cond
First data set's block count	√	Cond
First data set's kilobytes		
First data set's security level		
DSN1 create date	√	Req
DSN1 create time	√	
DSN1 last read	√	Cond
DSN1 last written		Cond
DSN1 flag Data set cataloged Abend in progress when data set closed	√	
VRS management value ³		Cond
First file creating step name		
First file creating DD name		
First file retention date		
Name of last data set on volume		Cond
Creating system ID		
• Data that can be found in EPIC is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹We recommend that you use the job name or the data set HLQ as the owner. ²We recommend that you use the VOLSER as the rack number. ³We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

7.4.2 Data Set Name Information

Table 79 shows the EDGCDREC data set name information.

Table 79 (Page 1 of 2). EDGCDREC Data Set Name Information		
	EPIC	Status
Data set create date	√	Req
Data set create time	√	
Data set name	√	Req

Table 79 (Page 2 of 2). EDGCDREC Data Set Name Information		
	EPIC	Status
Compaction status Data set used 3480 IDRC No compaction		
File sequence number on volume	√	Req
VOLSER holding data set	√	Req
Volume sequence within set of volumes	√	Req
Unit where data set was written	√	
Data set RECFM	√	Req
Data set LRECL	√	Req
Data set BLKSIZE	√	Req
Data set block count	√	Req
Data set size in kilobytes		
Data set owner name ¹	√	Req
Logical file sequence number	√	Req
Data set security level		
Data set date last read	√	Req
Data set date last written		Req
VRS management value ²		Cond
DSN flag Data set cataloged Abend in progress when data set closed	√	
Creating system ID		
Creating jobname	√	
Creating step name		
Creating DD name		
Retention date		
- Data that can be found in EPIC is marked with a √. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹CONVERT ensures that all data sets on the same volume have the same owner as the VOLSER's owner. ²We recommend that you use the other vendor program's "keep date," if the keep date is a keyword date, and put a single character in front to create a VRS management value.		

7.4.3 Vital Record Information

Table 80 shows EDGCKREC policy information.

Table 80 (Page 1 of 2). EDGCKREC Policy Information		
	EPIC	Status
Record create date		Req
Record create time		
VRS type		Req

Table 80 (Page 2 of 2). EDGCKREC Policy Information		
	EPIC	Status
Data set name	√	Cond
Format of dsname	√	Cond
Type of retention Cycles Elapsed days Days unreferenced While cataloged Until expired	√	Cond
Volume serial number		Cond
Delay days		
Total count of number of cycles or days	√	Req
VRS name		Cond
Store keep number	√	Cond
VRS owner		Req
Automatic delete date		
VRS description		
Location name	√	Req
Name of next VRS definition		
Jobname		
• Data that can be found in EPIC reports is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>.		

7.4.4 Empty Rack and Bin Information

Table 81 shows EDGCEREC empty rack and bin information.

Table 81. EDGCEREC Empty Rack and Bin Information		
	EPIC	Status
Shelf status Shelf is rack in library Shelf is bin in storage location		Req
Empty shelf location's name		Req
First of empty shelves start		Req
Count of contiguous shelves		
Media name		
• Data that DFSMSrmm requires is marked with <i>Req</i>.		

7.4.5 Owner Record Information

An owner record is required only for actual owners for whom you want to supply owner details and for those who are to be owners for VRS. Other owner records will be created by CONVERT postprocessing (EDGCVOL). Table 82 shows EDGCOREC owner information.

Table 82. EDGCOREC Owner Information		
	EPIC	Status
Owner identifier	√	Req
Owner's surname		
Owner's first name		
Owner's department ¹		Req
Owner's first address line		
Owner's second address line		
Owner's third address line		
Owner's internal telephone number		
Owner's external telephone number		
Owner's user ID		
Owner's node ID		
Data that DFSMSrmm requires is marked with <i>Req</i>. ¹Not strictly required for conversion. Only the RMM ADDOWNER command enforces the owner's department.		

7.5 Differences

To complete the comparison of EPIC and DFSMSrmm, we must discuss the differences in system structure, open, close, EOVS interface, and shelf management.

7.5.1 Structure

EPIC runs as an MVS subsystem like DFSMSrmm. If EPIC is not active, the operator can allow use of tape volumes by replying to the IEC507D messages that open processing generates. The EPIC DSN catalog is like an extension to the ICF catalog. Data sets that are not cataloged in an ICF catalog can still be referenced, as if they were cataloged, because EPIC retrieves the volume information from its DSN catalog.

DFSMSrmm has only one address space that manages all tape activities and the updates to the CDS. DFSMSrmm has a subsystem interface that must be active in order to use tape management functions and ensure tape data set protection. If this interface is defined to MVS in the IEFSSNxx PARMLIB member but not started, you cannot use tapes in your system. You can also control the inactivation of the subsystem interface using the DFSMSrmm EDGRESET utility through RACF. Thus you will have no tape activity that is either not registered or disallowed by the tape management system.

7.5.2 Open, Close, and EOVS Interface

EPIC documentation claims that EPIC does not modify IBM code. However, its interface to tape processing is established by making dynamic changes to the supplied IBM code when EPIC is started.

To accommodate these modifications and allow EPIC and DFSMSrmm to run in parallel, some changes are required and available from IBM.

The modification is to DFSMSrmm, and details of this modification, including how to install and validate it, are in Appendix E, "Modification Required for EPIC and DFSMSrmm Parallel Run" on page 671. Use the information provided to temporarily update DFSMSrmm while running in parallel and using tape volumes.

Note: Without these changes or a fix from Computer Associates, you cannot use tape volumes while running DFSMSrmm and EPIC in parallel.

If this modification can not be installed for some reason, DFSMSrmm can be started in parallel, but ONLY in MANUAL mode. When running in parallel in manual mode, the DFSMSrmm control data set can be loaded from the extract file, DFSMSrmm housekeeping jobs can be tested, and, for example, scratch lists and movement lists can be compared to reports from EPIC, to verify functions.

In a DFSMSrmm environment, the code that interfaces to open, close, and EOVS is integrated in DFSMSdfp, so you do not have to make any changes.

7.5.3 Shelf Management

For onsite shelf management, EPIC does not provide functions to allocate to empty shelves' volumes with serial numbers outside the installation ranges. Although nonnumeric VOLSERS are supported, it is up to you to decide how they are shelf managed.

For onsite shelf management, DFSMSrmm provides pools of shelf space (rack numbers). DFSMSrmm can manage the assignment of these to volumes that are brought into your installation.

For offsite shelf management, EPIC uses boxes, pallets, and case. Slot numbers can be defined and selected based on type of media.

In DFSMSrmm, bin numbers define individual shelf locations in storage locations offsite. Bin numbers are six characters and can only be assigned by DFSMSrmm using a sequential method of filling the bins in the selected location. Once a volume no longer uses a bin number, DFSMSrmm can reuse that number for another volume that needs to be placed in the same location. Thus you do not have to worry about the physical location of the tape in the offsite storage location. DFSMSrmm tells you which free bin number you can use for the volumes that you are moving.

7.6 Reviewing the Current Environment

See Chapter 2, “Planning for Conversion” on page 19 to review the flow of events that we follow in converting from EPIC to DFSMSrmm. You must collect and then analyze all information about the current tape management environment. Once you have reviewed the environment and understand the main differences between EPIC and DFSMSrmm, you will better understand what you must change during the conversion process.

7.6.1 Collecting Information

The information that you collect about the current environment should include:

- EPIC reports used daily or occasionally

Every day there should be an EPIC report that the operator uses to move the cartridges between the desired locations and to return tapes to the scratch pools. You can use the reports provided by DFSMSrmm, but if you want to avoid changes to the layout of these reports, using REXX or DFSORT you can tailor the DFSMSrmm reports to make them similar to the EPIC reports. Chapter 14, “Extending DFSMSrmm Reporting” on page 533 provides some useful examples of how to create the reports.

- Exits installed

Collect information about EPIC exits installed in your environment so you can ensure that you can achieve the same function in the DFSMSrmm environment.

- PARMLIB options used

You must review the EPIC PARMLIB options before you start DFSMSrmm, because you may have to modify the DFSMSrmm options according to the EPIC environment.

- Interfaces with other products

You collect information about the interfaces that EPIC might have with other products (such as CICS, ROSCOE, and TSO).

- Housekeeping (batch inventory update) process

Collect information on the daily inventory management process so that you can synchronize it with the DFSMSrmm housekeeping function.

7.6.2 Analyzing the Record Layouts

Now you analyze the information you have collected to achieve the equivalent or acceptable alternative function in the DFSMSrmm environment.

DFSMSrmm requires certain data to be available to build its CDS. You must analyze EPIC databases and reports to know which types of data are available. There is not a one-to-one correspondence between DFSMSrmm and EPIC data. DFSMSrmm may have data that EPIC does not have and vice versa. If DFSMSrmm requires data that EPIC does not provide, you must use the extract programs or the EDGCNVT SYSIN statements to provide a value.

After analyzing the data, you must determine which types of data you want to have in the DFSMSrmm CDS in order to be consistent with the current operating environment. Knowing the types of data needed helps determine whether you can use EPIC databases or reports as the sole input to the extract programs.

In addition, you must review the current policies regarding retention management, storage location management, and tape pool management. DFSMSrmm manages these processes differently from EPIC.

7.6.3 Retention Policies

When DFSMSrmm VRS options are set in a single VRS, they are logically ANDed, so that if the options are A and B, then A AND B must be true for the volume to be retained. For example, for DAYS COUNT(5) WHILECATALOG, the data set is retained until either the five days have passed or the data set is uncataloged. However, when EPIC policy options are combined, they are logically ORed, so that if the options are A and B, then the volume will be retained if either A or B is true. For example, for RET=5,MVSCAT=YES, the data set is retained until both the five days have passed and the data set is uncataloged. In addition, all EPIC options can be combined, whereas only the DFSMSrmm WHILECATALOG and UNTILEXPIRED options can be combined with a basic retention type. DFSMSrmm VRS options can also be chained together using NEXTVRS or ANDVRS operands so that they can be logically ORed or ANDed together.

For a detailed description of new VRS-enhancements in DFSMSrmm, see 12.1, “Vital Record Selection” on page 473. The following examples apply to DFSMSrmm with the VRS SPE (APAR OW30969) installed.

In the following example:

```
RMM ADDVRS DSN('dsname') DAYS COUNT(5) NEXTVRS(WC)
RMM ADDVRS NAME(WC) WHILECATALOG
```

The data set is retained until the five days have passed, and then until the data set is uncataloged. All the DFSMSrmm VRS options can be combined by chaining them together using NEXTVRS or ANDing them together with ANDVRS.

When EPIC policy options are combined, they are logically ORed, so that if the options are A and B, then the volume will be retained if either A or B is true. For example, for RET=5,MVSCAT=YES, the data set is retained until both the five days have passed and the data set is uncataloged. In addition, all EPIC options can be combined, but only with OR combinations.

In EPIC you might specify CYC=10, MVSCAT=YES, indicating that all cataloged occurrences of the data set or GDG and the latest 10 cycles should be kept, whether or not they are cataloged. In DFSMSrmm, specifying equivalent options on a single VRS, CYCLES COUNT(10) WHILECATALOG, would result in keeping the last 10 cycles that are cataloged. DFSMSrmm allows you to chain VRSs together so you can specify

```
RMM ADDVRS DSN('dsname') CYCLES COUNT(10) NEXTVRS(WC)
RMM ADDVRS NAME(WC) WHILECATALOG
```

which achieves the same retention as EPIC: at least 10 cycles are kept.

If the DFSMSrmm VRSs are specified in the reverse order, you get different results:

```
RMM ADDVRS DSN('dsname') WHILECATALOG NEXTVRS(C10)
RMM ADDVRS NAME(C10) CYCLES COUNT(10)
```

This keeps all data sets that are cataloged, and then another 10 cycles.

Here is a further example: in EPIC, you can specify RET=28,CYC=10, indicating that all occurrences of the data set should be kept for 28 days, but the last 10 cycles must be kept. Such a specification would be of little relevance if the data set were created on a precisely regular basis; if the data set were created daily, the cycles specification would be redundant, and if the data set were created monthly, the retention specification would be redundant. However, if the data set is created intermittently, for example, with 15 generations created in one week, and the next generation is created several months later, such a policy specification could be vital to your installation's processing.

During conversion we can match exactly to the EPIC retention. In the above case, the following VRSs are equivalent:

```
RMM ADDVRS DSN(' dsname') CYCLES COUNT(10) NEXTVRS(D28)
RMM ADDVRS NAME(D28) DAYS COUNT(28)
```

The sample conversion program, EDGCPIC2, identifies all policies that contain multiple specifications and lists them in the file with ddname PLERRPRT. The one exception is for data sets kept permanently, by a specification of either RET=PERM or RET=nnnn, where nnnn is a large number such as 1800, being approximately 5 years. The large number is specified as variable PSEUDOPM in sample EDGCPIC2. If a data set is to be kept permanently, or pseudo-permanently, all other specifications are ignored by the conversion process.

During the conversion of the data you should check all multiple specification policies and make any necessary changes to the VRSs created so that the policies that DFSMSrmm uses will match your requirements.

EDGCPIC2 handles multiple specifications in the following order:

1. If one of the specifications is permanent or pseudo-permanent, a VRS specifying DAYS COUNT(99999) is created.
2. If one of the specifications is MVSCAT=YES, a VRS specifying CYCLES COUNT(99999) WHILECAT is created, unless the data set is a GDG and the policy also specifies CYC=nnn, in which case a VRS specifying CYCLES COUNT(nnn) is created.
3. If one of the specifications is MVSCAT=NO, a VRS specifying CYCLES COUNT(99999) WHILECAT is created, unless the data set is a GDG and the policy also specifies CYC=nnn, in which case a VRS specifying CYCLES COUNT(nnn) is created.
4. If one of the specifications is CYC=nnn, a VRS specifying CYCLES COUNT(nnn) is created.
5. If one of the specifications is RET=nnn, a VRS specifying DAYS COUNT(nnn) is created.
6. If one of the specifications is DLA=nnn, a VRS specifying LASTREF COUNT(nnn) is created.

Retention information in EPIC can be specified in either global policies in the EPIC startup options, or can be defined for specific, fully-qualified data sets in the DSN catalog. During conversion, EDGCPIC2 analyzes the information in the DSN catalog and the startup options to determine whether data sets are retained by global or specific policies, and builds K-records to describe the policies. The K-records are converted to VRS definitions by a EDGCNVT later in conversion. If you have many of specific policies defined to EPIC, there will be many of K-records generated. However, through analysis of the retention information, it is possible to determine where generic data set name masks could be used to

decrease the number of policies that are required. The POLPRINT file created by EDGCPIC2 lists the groups of like named data sets with common retention policies. You can use this information to create global policy definitions in the EPICPARM file input to EDGCPIC2. By specifying generic masks in global policies used just for conversion, you can dramatically reduce the number of K-records created. EDGCPIC2 checks to see whether a data set matches the global policy data set mask and retention information, and if it matches, EDGCPIC2 avoids creating a K-record.

During the conversion of policy information, EDGCPIC2 checks what locations a data set is stored in and uses this information to build the K-records. In EPIC, information about which vault or location to store a volume in is retained in the VMS catalog, which is not input to the conversion programs, because the format is not known. If a data set is retained in multiple locations, EDGCPIC2 cannot correctly determine which location name to use, so it lists this information in the PLERRPRT file. You must analyze this information and use it, along with your VMS catalog information, to update the policies that the conversion process generates.

Although DFSMSrmm can provide the same retention as EPIC, because the exact VMS locations and policies cannot be identified, it is important to validate that DFSMSrmm provides the retention you require. The easiest way to measure the effectiveness of the conversion is to compare the volumes known to both EPIC and DFSMSrmm as scratch, and the volume movement lists, after each run of housekeeping for both products. If the number of volumes known as scratch to one system but not the other is large, or increases with time, the conversion must be considered suspect. However, if the number is small and fairly constant over time, the conversion could be satisfactory, especially if the reasons for the volumes having different statuses can be determined.

To aid this checking process a sample program, EDGCXTRC, is provided, which extracts the volumes from EPIC and DFSMSrmm scratch volume lists and creates files that you can compare by using ISRSUPC, the program behind ISPF/PDF 3.12. See Section 7.7.6, "The EDGCPIC2 Program" on page 352 for details of this program and its use.

The comparisons of scratch volume lists could well lead to the requirement to change the conversion process, or modify VRSSs after conversion, so that conversion is likely to be an iterative process. The scenario could be that an initial conversion is run and subsequent analysis of the differences in the scratch volume lists points to the requirement for different VRSSs. A second conversion, incorporating the changes, is done, and analysis of the scratch volume differences indicates the need for more fine tuning of the VRSSs. Thus the process may be repeated many times until the count of differences is small and remains small as time passes.

EPIC can use specific or unique Julian dates to control tape data set expiration and retention. These keyword dates override the installation's retention policies, which are defined in the DSN catalog and PARMLIB options:

- EXPDT=98000** Specifies that the tape volume being processed is a foreign tape, not under EPIC control.
- EXPDT=98ddd** Specifies that a tape data set be retained until the specified number of days (ddd) has elapsed since the last reference.
- EXPDT=99000** Specifies that a tape data set be retained as long as it is cataloged.

EXPDT=990cc Specifies that the total number of cycles (cc) or generations of the data set be kept.

EXPDT=99365 Specifies that a tape data set be retained permanently

EXPDT=99366 Specifies that a tape data set be retained permanently.

By using VRSs and the EDGUX100 exit, DFSMSrmm supports the JCL special expiration dates honored by other tape management systems. You define VRSs that can be used to provide management function equivalent to the special Julian dates of EPIC. You then assign a VRS to the data set using either a management class or the installationwide exit, EDGUX100, to set a VRS management value.

For instance:

```
//MYDSN DD DSN=MY.DATA,DISP=(,CATLG),LABEL=(,SL,EXPDT=99000),
// additional parameters
```

You define the VRS:

```
RMM ADDVRS DSN('D99000') WHILECATALOG
```

At OPEN time the DFSMSrmm installationwide exit requests a VRS management value of D99000, having determined that you specified a “special date” in your JCL.

During inventory management VRS processing, if there is no VRS data set mask covering the data set, DFSMSrmm matches the data set VRS management value with the ADDVRS DSN('D99000') definition and retains the data set for as long as it is cataloged.

You can list the management classes and VRS management values assigned to data sets by using the RMM LISTDATASET command, reading the report extract file data set records, or using the ISPF dialog.

However, you have three alternatives to gradually migrate away from using special dates:

- Using system-managed tape, an ACS routine can assign a management class corresponding to a special date. DFSMSrmm uses this management class to relate to an associated VRS definition to provide the required retention criteria.
- A DFSMrmm installationwide exit can provide a VRS management value that is used in a similar way to the management class.
- Use management class RETENTION LIMIT.

Note: The SMS management class takes precedence over the VRS management value.

The retention hierarchy is:

```
ABEND/OPEN ⇒ VRS DSN ⇒ MC/MV ⇒ defaults
```

Note: For a detailed description of the best match fit sequence see 12.1.1, “The Best Match Fit Sequence” on page 474.

Although the EDGCPIC2 program allows for processing of existing special expiration dates and the creation of VRS management values and definitions, it is the installation’s responsibility to ensure that:

- Any new retention criteria are defined as VRS management values and VRS definitions
- Either an appropriate DFSMS ACS routine or the DFSMSrmm installation exit EDGUX100 is installed and active to set VRS management values from special expiration dates.

DFSMSrmm flags a data set if the job that writes to it leaves the data set OPEN. This can occur only through failure of the MVS system, and not as a result of abends. Open data sets are retained by VRS processing, and the specified or calculated expiration date is kept. Thus manual intervention might be required to release the volume if you specify a long retention period. Otherwise you can use the special OPEN VRS retention to establish how to manage this situation.

7.6.4 Inventory Management and Reports

EPIC provides batch facilities to perform the following functions:

- Batch update and reporting
- Tape retention and expiration
- Storage location management
- Backup, recovery, and verification of databases

Except for batch update and reporting, DFSMSrmm provides similar functions as the EDGHSKP, EDGBKUP, and EDGUTIL utilities (for more details, see 10.1.7, “Prepare Inventory Management” on page 438 and the *DFSMSrmm Implementation and Customization Guide*, SC26-4932).

EDGHSKP must be executed with the DFSMSrmm subsystem active. EDGBKUP and EDGUTIL can be executed independently of the subsystem.

The utilities have the following functions:

- EDGHSKP
 - Processes vital record
 - Performs storage location management
 - Performs expiration processing
 - Creates an extract data set
 - Backs up the DFSMSrmm CDS
 - Backs up and clears the DFSMSrmm journal
- EDGBKUP
 - Backs up the DFSMSrmm CDS
 - Backs up the DFSMSrmm journal
 - Recovers and restores the DFSMSrmm CDS
 - Reorganizes the DFSMSrmm CDS
- EDGUTIL
 - Verifies DFSMSrmm CDS.⁷
 - Mending the DFSMSrmm CDS.

DFSMSrmm does not provide batch update utility. However, you can execute the DFSMSrmm TSO commands ADD, CHANGE, and DELETE in batch to achieve the same result. Through the ISPF dialog you can generate and group together TSO commands for later batch processing.

⁷ The EDGUTIL function can also provide consistency between the CDS and the TCDB in a system-managed tape library.

EPIC provides batch reports. These may be supplemented with the use of the customized reports.

DFSMSrmm provides two reporting programs and one reporting REXX:

- EDGRPTD—processes the report extract file to produce:
 - Volume movement reports
 - Moved from storage locations
 - Ready-to-scratch sorted by RACK
 - Sorted by BIN number
 - Sorted by OWNER
 - Moved to storage locations
 - Sorted by RACK number
 - Sorted by OWNER
 - Volume inventory reports
 - Moved from storage locations
 - Sorted by volume serial number
 - Sorted by RACK number or BIN number
 - Sorted by OWNER
- EDGAUD—processes DFSMSrmm SMF records to produce:
 - Security report (tracking information for classified data)
 - Audit report (CDS changes)
- EDGJRPT—processes the extended extract file
 - Pull list for scratch tapes sorted by volume serial number
 - Pull list for scratch tapes sorted by data set name
 - Inventory list by volume serial number
 - Inventory list by data set name
 - Inventory of data set including number of KB used
 - Inventory of volume serial number by location
 - Inventory of data set names by location
 - Inventory of BIN numbers by location
 - List of all data set names at loan location
 - List of all volume serial numbers at loan location
 - List of all multiple volume, multiple data set
 - Movement report including the first data set name on the volume
 - Movement report sorted by storage location bin number
 - Movement report sorted by volume serial number
 - Inventory list sorted by volume serial number including volume count

You can execute the DFSMSrmm TSO commands, LIST and SEARCH, in batch to produce a wide variety of reports. You can also use DFSORT's ICETOOL to write your own customized reports.

EDGRPTD and EDGAUD sample jobs, REXX programs, and DFSORT's ICETOOL reports are provided in Chapter 14, "Extending DFSMSrmm Reporting" on page 533 and Appendix D, "Sample Exits and Reports" on page 603 and in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

The EPIC scratch list report is shifted one day in the DFSMSrmm environment because the DFSMSrmm report lists volumes that are in **PENDING RELEASE** status. Those volumes are returned to **SCRATCH** status during the next housekeeping process, which usually runs the next day.

With DFSMS/MVS 1.4.0 the first run of EDGHSKP EXPROC processing assigns the pending release status, and the second run moves volumes to scratch status as long as there are no release actions required on the volume.

Note: If the DFSMS/MVS 1.4.0 with SPE VRS is installed, the expiry of tapes can be set to immediate in VRS definitions, so some categories of tapes can be released immediately in the first EDGHSKP EXPROC run. However, there can still be differences in scratchlists between the tape management systems.

We suggest that you move to the scratch pool the cartridges that will return to scratch only when they are no longer in *PENDING RELEASE* status. You can accomplish this in your batch housekeeping jobstream by using a step that lists the volumes in *PENDING RELEASE* status before running the EDGHSKP step. These are the volumes that are new scratch volumes today, according to the EPIC report. Alternatively, you can execute inventory management EXPROC processing a second time to return pending release volumes to scratch.

7.6.5 Exits

DFSMSrmm provides programming interfaces and installationwide exits installed as part of DFSMSrmm during the SMP/E apply process.

DFSMSrmm provides following programming interfaces:

EDGDFHSM DFSMSHsm tape volume interface
EDGTVEXT DFSMSrmm tape volume interface
IGXMSGEX DFSMSrmm uses the DFSMSdfp MSGDISP interface to update tape drive displays
EDGLCSUX DFSMSrmm tape library volume interface

Source code is supplied for the following exits:

CBRUXENT OAM cartridge entry exit
CBRUXEJC OAM cartridge eject exit
CBRUXCUA OAM cartridge change use attributes exit
CBRUXVNL OAM volume not in library exit

Before the SMP/E accept, the source code is in SMPSTS; after the accept, the exits are in SYS1.AEDGSRC1.

If your installation is already using one or more of these exits in the EPIC environment, you must ensure that the correct action is performed during conversion.

EPIC uses the CBRUXENT exit to interface the IBM 3494 and 3495 Automated Tape Library Dataservers. You can test the DFSMSrmm version of this exit once you have completed the conversion.

The CBRUXVNL exit is optional. If you use it, however, as with the CBRUXENT exit, you must test it in the DFSMSrmm environment after conversion.

The installationwide exits that you installed with DFSMSrmm for:

- ARCTVEXT
- IGXMSGEX
- CBRUXENT
- CBRUXEJC
- CBRUXCUA
- CBRUXVNL

can replace any existing exits you had previously installed. Decide whether you still require the function your previous exits provided, even if for a short time

while introducing DFSMSrmm. If you no longer need the function, proceed with the next implementation step. If you still require the function, consider how to implement it, while still providing the function required by DFSMSrmm.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more details on the DFSMSrmm programming interfaces and the exits. The supplied source code for these exits shows how to call the programming interface. You can modify either the supplied exit to call a modified version of your old exit, or modify your old exit to call the DFSMSrmm programming interface. Consider how you might provide an option to disable your code without another code modification. In either case, install the changed software by using an SMP/E USERMOD.

The action you need to take depends upon what level of DFSMSrmm you install. With DFSMSshm and DFSMSrmm you can communicate directly without calling the ARCTVEXT exit.

If you have a release of DFSMSrmm prior to 1.4 and your installation has DFSMSshm installed and EPIC is using the ARCTVEXT exit, you may need to merge the EPIC code for the exit with the DFSMSrmm code for related exit, in order to notify both tape management products when running in parallel. Also see 3.1.1, “Update Installationwide Exits” on page 28, for more information about programming interfaces and exits. We recommend that you convert your code to run when called by the DFSMSrmm exit, and modify the supplied exit to call your code.

Based on the decisions you have taken about VRS management values and scratch pooling, you can update the supplied sample installation exit, EDGUX100, to perform the function you require. Follow these steps:

1. Copy the sample installation exit, EDGUX100, and use this copy as a base for your exit.
2. Update the exit if needed.
3. Build an SMP/E USERMOD to apply the updated source code for EDGUX100, including the necessary JCLIN statements to get the EDGUX100 load module added to the LINKLIB target library.
4. Receive and apply the USERMOD.
5. Your new exit is now ready for use on your system.

You can use the same process to install the EDGUX200 code if necessary. If you have a non-IBM robot tape library, you can update the sample EDGUX200 to communicate with the library host software to ensure that the external inventory is updated dynamically as volumes return to scratch status.

For more information about DFSMSrmm exits refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

7.6.6 PARMLIB Options

During the analysis process you must check the EPIC PARMLIB options that can affect the DFSMSrmm environment. Review carefully all EPIC parameters before you run the extract programs.

Figure 107 on page 346 shows sample PARMLIB options.

```

LICENSE 5782W86862ABC1234568
OPTION  ACL=NO
OPTION  AUTOCAT=SELECT
OPTION  AUTOLABL=NO
OPTION  AUTOLOG=NO
OPTION  AUTOUPD=NO
OPTION  CNTRL=NO
OPTION  DSNWARN=200,DSNMSG=10
OPTION  ESR=40
OPTION  EXPKEYWD=YES
OPTION  EXP98000=NOPROMPT
OPTION  EXTCAT=YES
OPTION  JCLOVRDE=NO
OPTION  LINECNT=050
OPTION  MULTCPU=NO
OPTION  MSG05=NO,MSG28=YES,MSG33=NO,MSG47=NO,MSG51=NO
OPTION  OEMTAPE=NO
OPTION  RECAT=YES
OPTION  RECWARN=200,RECMSG=10
OPTION  SECEXIT=NO,SECSIZE=1024
OPTION  UNCTL=NO
OPTION  VERIFY=YES
OPTION  PANELLIB=EPIC.V03R0100.GOALPDS
OPTION  CARTDEV=3480
OPTION  EDMF1=HSM.BACKTAPE
OPTION  EDMF2=HSM.DMP
OPTION  EDMF3=HSM.HMIG
OPTION  EDMPGM=ARCCTL
OPTION  PASSWD=EPICMVS1
OPTION  REELDEV=3400-6
OPTION  CYC=3,MVSCAT=NO,RET=7,UID=AC
OPTION SELECT DSN=AP*,SCOPE=INPUT,CYC=3,RET=31
OPTION SELECT DSN=AP*,SCOPE=OUTPUT,CYC=5,RET=90
OPTION SELECT DSN=DSNCDB2M.ARCHLOG*,SCOPE=BOTH,CYC=0,RET=1,SID=DB
OPTION SELECT DSN=DSNC130T.ARCHLOG*,SCOPE=BOTH,CYC=0,RET=7,SID=DB
OPTION SELECT DSN=DSNC130Q.ARCHLOG*,SCOPE=BOTH,CYC=0,RET=14,SID=DB
OPTION SELECT DSN=DSNCDB2P.ARCHLOG*,SCOPE=BOTH,CYC=0,RET=14,SID=DB
OPTION SELECT DSN=EPIC.BKUP.*,SCOPE=BOTH,CYC=5,RET=0,SID=EP
OPTION TAPEPOOL POOLID=0,LOW=002471,HIGH=002478,TPLSCR=YES
OPTION TAPEPOOL POOLID=0,LOW=0P0001,HIGH=0P0100,TPLSCR=YES
OPTION TAPEPOOL POOLID=0,LOW=000159,HIGH=000161,TPLSCR=YES
OPTION TAPEPOOL POOLID=B,LOW=CS0001,HIGH=CS1500,TPLSCR=NO

```

Figure 107. Sample EPIC PARMLIB Options

Table 83 lists the options to check.

Table 83 (Page 1 of 2). EPIC PARMLIB Options		
	Explanation	DFSMSrmm Equivalent
EXPKEYWD	Enables JCL-specified keyword dates	In DFSMSrmm you support use of keyword dates using the EDGUX100 installation exit, or you can force a maximum retention period using OPTION MAXRETPD(max).
LINECNT	Report lines per page	OPTION LINECOUNT

Table 83 (Page 2 of 2). EPIC PARMLIB Options		
	Explanation	DFSMSrmm Equivalent
ACL	Identifies whether robot library in use	You cannot identify robot libraries other than IBM system-managed libraries to DFSMSrmm. Use this parameter to help decide whether you need to obtain DFSMSrmm scratch and movement lists to pass to your robot software.
AUTOLABL	Identifies whether external labels are required	DFSMSrmm label support is through EDGUX100.
EDMPFX, EDMPGM	Identifies external data-manager-managed data sets	DFSMSrmm allows you to use VRS policies to retain data permanently until the external data manager releases the data.
TAPEPOOL	Identifies a pool of tape volumes	The DFSMSrmm equivalent is to use the VLPOOL command in PARMLIB. Each pool in DFSMSrmm must have a unique prefix.
SELECT	Defines data set retention policies	Retention information in DFSMSrmm is defined in the CDS and can also include the movement policies. Use DFSMSrmm VRS policies.
OEMTAPE	Identifies the existence of another tape management system on the same system	There is no DFSMSrmm equivalent. You can run DFSMSrmm in parallel with EPIC.
AUTOCAT, AUTOLOG	Controls whether data set details are recorded in the database	DFSMSrmm records data set details for all data created on the volumes managed by DFSMSrmm.

7.6.7 Data Set Overwrite

EPIC supports two volume statuses:

- PRIVATE** The volume contains valid data.
- SCRATCH** The volume can be used for scratch output requests.

In a DFSMSrmm environment you have three different volume statuses:

- MASTER** A private volume containing data that is available for write processing based on the MASTEROVERWRITE parmlib option.
 - ADD** New data can be created, but no existing data can be destroyed.
 - LAST** Only the last file on a MASTER volume can be overwritten. The data set name must match the existing data set name.
 - MATCH** An existing file can be overwritten. The data set name must match the existing data set name. All files which are higher sequence are destroyed.
 - USER** The same as a volume in USER status.
- SCRATCH** The volume can be used for scratch output requests.
- USER** The volume is assigned to a user, and it can be overwritten, rewritten, and used repeatedly until the user returns it to the scratch pool.

DFSMSrmm can manage the overwrite of a MASTER volume with the PARMLIB option, MASTEROVERWRITE. You can tailor this option to meet your installation's needs. Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for a detailed explanation of the function.

7.7 Running the Extract Programs

In this section we describe the sample programs and JCL for extracting data for conversion to DFSMSrmm from EPIC.

Table 84 on page 348 lists the samples for the extraction of data from EPIC.

<i>Table 84. IBM-Provided Samples for EPIC</i>		
Name	Location	Function
EDGCPIC1	SAMPLIB	Source for EPIC extract
EDGCPIC2	SAMPLIB	Source for EPIC reformatting program
EDGCVRST	SAMPLIB	Source for table used by EDGCVRSS
EDGCVRSS	SAMPLIB	Source for VRS value-setting routine called by EPIC EDGCPIC2
EDGCPICO	SAMPLIB	Source for owner-setting routine called by EPIC EDGCPIC2
EDGCXTRC	SAMPLIB	Source for EPIC validating scratch lists
EDGJPIC1	SAMPLIB	JCL to execute EPIC EDGCPIC1
EDGJPIC2	SAMPLIB	JCL to execute EPIC EDGCPIC2
EDGJPICA	SAMPLIB	JCL to assemble EPIC samples
EDGJPICV	SAMPLIB	JCL to execute EPIC scratch validation

The sample JCL is set up to work in a “typical” installation. Some of the sample JCL is commented out so that you choose to use it or not as is your preference. Procedures, which are normally part of other IBM products, are used to assemble and compile the sample code. Thus you do not have to assemble, link, and compile code before you can use it. If these procedures are not available to you, you can edit the JCL to uncomment the statements that execute the programs from a STEPLIB. If you choose to link-edit the programs yourself, you can use the standard link-edit options. None of the programs is reentrant.

7.7.1 Functional Description

The process uses the EPIC TSIDACC interface, through the EDGCPIC1 program, to extract policy and data set records from the EPIC DSN catalog. The extracted records are processed, through the EDGCPIC2 program, into formatted records ready for use with the DFSMSrmm convert program, EDGCNVT. See Chapter 9, “Building the DFSMSrmm CDS” on page 403 for details of the EDGCNVT program and completion of the data conversion.

Most information on volumes and data sets can be transferred from the DSN catalog to the DFSMSrmm CDS by using the EDGCPIC2 and EDGCNVT programs. Manual changes made to data in the DSN catalog can impact the extraction of accurate data. For example, EDGCPIC2 may be unable to build a correct multivolume or multifile set of data set and volume records. When EDGCPIC2 detects such problems, it writes an error message to the error log file, VLERRPRT, and writes out records that are its best guess as to what is actually recorded on the volumes. Conversion of VMS policy records is not quite so straightforward, because the vaulting information is not always able to be determined by EDGCPIC2. Use the PLERRPRT file to determine which policies need manual changes (see 7.6.3, “Retention Policies” on page 338).

7.7.2 Preparing to Run the Extract Program

The sample programs to assist with the extraction of data and the JCL to execute them are provided as samples in SAMPLIB. You must assemble and link-edit the sample programs using the correct levels of macros and interface programs (Figure 108).

```
//STSGMAAC JOB CLASS=M,MSGCLASS=X,NOTIFY=STSGMA,COND=(0,NE)
//*
//ASMPROC PROC P=*
//ASM      EXEC PGM=IEV90,REGION=5M,
//          PARM=' DECK,NOOBJECT,NORLD,XREF(SHORT)'
//SYSPRINT DD SYSOUT=&P
//SYSUT1   DD DISP=(OLD,PASS),DSN=*.ALLOC.WKF
//SYSUPUNCH DD DISP=(OLD,PASS),DSN=&&OBJ(&M)
//SYSLIB   DD DISP=SHR,DSN=SYS1.SAMPLIB
//          DD DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//          DD DISP=SHR,DSN=EPIC.MACLIB
//SYSIN    DD DISP=SHR,DSN=SYS1.SAMPLIB(&M)
//          PEND
//*
//ALLOC    EXEC PGM=IEFBR14
//OBJ       DD DISP=(,PASS),DSN=&&OBJ,UNIT=VIO,SPACE=(TRK,(5,1,5)),
//          RECFM=FB,LRECL=80,BLKSIZE=800
//WKF       DD DISP=(,PASS),UNIT=SYSDA,SPACE=(TRK,(5,1))
//*
//ASM1      EXEC ASMPROC,P='(*,,APC1)',M=EDGCPIC1
//ASM2      EXEC ASMPROC,P='(*,,APC2)',M=EDGCPIC2
//ASMVS     EXEC ASMPROC,P='(*,,APVS)',M=EDGCVRSS
//ASMT      EXEC ASMPROC,P='(*,,APVT)',M=EDGCVRST
//ASMO      EXEC ASMPROC,P='(*,,APOC)',M=EDGCPICO
//ASMVX     EXEC ASMPROC,P='(*,,APVX)',M=EDGCXTRC
//*
//LKED1     EXEC PGM=IEWL,
//          PARM='MAP,LIST,LET,NCAL'
//SYSPRINT  DD SYSOUT=(*,,LNK1)
//SYSUT1    DD DISP=(OLD,PASS),DSN=*.ALLOC.WKF
//OBJ        DD DISP=(OLD,DELETE),DSN=&&OBJ
//SYSLIB     DD DISP=SHR,DSN=EPIC.SYSLIB
//SYSMOD     DD DISP=SHR,DSN=STSGMA.RMM.EPICNVT.LOAD
//SYSLIN     DD *
//          INCLUDE OBJ(EDGCPIC1)
//          INCLUDE SYSLIB(TSIDACC)          Obtain EPIC interface program
//          ORDER  EDGCPIC1(P)              PAGE ALIGN TO ASSIST TESTING
//          SETCODE AC(1)
//          NAME   EDGCPIC1(R)
//          INCLUDE OBJ(EDGCPIC2,EDGCVRSS,EDGCVRST,EDGCPICO)
//          ORDER  EDGCPIC2(P)              PAGE ALIGN TO ASSIST TESTING
//          NAME   EDGCPIC2(R)
//          INCLUDE OBJ(EDGCXTRC)
//          NAME   EDGCXTRC(R)
```

Figure 108 (Part 1 of 2). Sample Preparation JCL

```

/* Compress load library
//CMPR EXEC PGM=IEBCOPY
//SYSPRINT DD SYSOUT=(*,,COPY)
//SYSUT3 DD DISP=(OLD,DELETE),DSN=*.ALLOC.WKF
//SYSUT4 DD UNIT=SYSDA,SPACE=(TRK,(5,1))
//SYSIN DD *
      C I=LOAD,0=LOAD
//LOAD DD DISP=SHR,DSN=STSGMA.RMM.EPICNVT.LOAD

```

Figure 108 (Part 2 of 2). Sample Preparation JCL

In Figure 108 on page 349 the following input is required:

- SAMPLIB** Contains both the sample macros and the sample source code for the extract programs.
- MODGEN** Contains the EDGCxREC macros that map the records that the extract generates. You must use the levels of these macros that match the maintenance level of the DFSMSrmm product you plan to run.
- EPIC.MACLIB** Should be changed to point to the EPIC library that contains the EPICTREC macro to ensure that the records produced by the EPIC TSIDACC interface program are correctly mapped to the extract programs.
- LKED1.SYSLIB** The EPIC library that contains the TSIDACC interface program. The link-edit includes this program in the EDGCPIC1 load module. If TSIDACC is available through the linklist libraries at run time, you can avoid including TSIDACC at link-edit time, and EDGCPIC1 will load the module when required.

The EDGCPIC1 and EDGCPIC2 sample programs, EDGCPICO, EDGCVRSS, and EDGCVRST, their subroutines, and the EDGCXTRC program are assembled and link-edited.

The source module and CSECT EDGCPICO will need some customization, although possibly not initially. The sample supplied sets the owner of all volumes and VRSs generated from EPIC discrete policies to DEFLTOWN and the owner of VRSs generated from EPIC global policies to GLOBLOWN. Comments in the source of EDGCPICO explain how the module can be modified to determine the desired owner definitions.

The source module and CSECT EDGCPIC2 may need some customization, although possibly not initially. The sample supplied sets the volume media type, density, recording format, and media name information as best it can. The records extracted from the DSN catalog contain no density or recording format information. The information they have is whether a volume is a REEL or a CARTRIDGE. Based on this information, EDGCPIC2, sets the media name for tape reels as TAPE6250 and the density as 'undefined', and the media name for cartridges as CART3480 and the TDSI information to indicate 36track, CST of unknown compaction. If you can better determine the media information, perhaps based on volume serial number, you can customize the source code to set the correct values. You can later, when running EDGCNVT, use SYSIN statements to set by volume range whether cartridges are CST or ECCST.

The source module and CSECT EDGCVRST may also need some customization, although, again, possibly not initially. The sample supplied only defines a minimum retention period of 3 days. Possible uses of EDGCVRST are

documented in comments in the source. Some of the facilities that you can select in EDGCVRST through the GENVV TAB macro are not relevant to the conversion process, but to an extension of the sample EDGUX100 exit.

Once you have prepared the programs and JCL and run the EPIC batch programs to perform retention, vaulting, and scratch processing you are ready to extract the data.

7.7.3 The EDGCPIC1 Program

This section describes the setup and control of the EDGCPIC1 data extraction program.

The EDGCPIC1 program can complete successfully only when EPIC is active. Figure 109 shows a sample EDGCPIC1 JCL.

```
//WOODMW1 JOB CLASS=5,MSGCLASS=X,NOTIFY=WOODMW,
//          COND=(0,NE)
//*
//AMS      EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN  DD *
//          DEL 'WOODMW.EDGCPIC1.*'
//          LISTDS 'WOODMW.RMM.EPICNVT.LOAD'
/*
//*
//CNVT1    EXEC PGM=EDGCPIC1
//STEPLIB  DD DISP=SHR,DSN=WOODMW.RMM.EPICNVT.LOAD
//SYSUDUMP DD SYSOUT=D
//CATOUT   DD DISP=(,PASS),DSN=&&CATOUT,
//          UNIT=SYSDA,SPACE=(CYL,(5,1),RLSE),BLKSIZE=23432
//*
//SORT1    EXEC PGM=SORT,REGION=4M
//SYSOUT   DD SYSOUT=*
//SORTIN   DD DISP=(OLD,DELETE),DSN=&&CATOUT
//SORTOUT  DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC1.CATOUT,
//          UNIT=SYSDA,SPACE=(CYL,(5,1),RLSE),BLKSIZE=23432
//SYSIN    DD *
//          SORT FIELDS=(3,44,CH,A,2,1,CH,A),DYNALLO
//          *
```

Figure 109. EDGCPIC1 Sample JCL

7.7.4 Input-Output Files

- | | |
|----------------|---|
| STEPLIB | This execution library contains the EDGCPIC1 load module. It must be in an authorized program facility (APF) library. |
| CATOUT | The extracted records are written to this output file. Before the EDGCPIC2 program is run, the records are sorted by data set name and record type. |
| SORTOUT | This file, created by the SORT1 step, is required as input to the EDGCPIC2 program. |

7.7.4.1 Input Parameters

There are no input parameters.

7.7.4.2 Limitations

The interface to the EPIC TSIDACC utility is limited.

7.7.5 Return Codes

A return code is set as follows:

0	Normal completion.
ABEND	There was an error calling the EPIC interface utility TSIDACC.

7.7.6 The EDGCPIC2 Program

This section describes the setup and control of the EDGCPIC2 data extraction program.

7.7.6.1 Policy Conversion

The POLPRINT, PLIGNPRT, and PLERRPRT output files provide information about conversion of policy information. No information is input to EDGCPIC2 on your vaulting rules, so you must use the EPIC VMS vaulting rules, the EDGCPIC2 output files, and the EPICPARM input to tailor the policy records created by EDGCPIC2.

The converted policies are referred to as VRSES; both discrete and global VRSES are converted. The global VRSES are created from your EPICPARM OPTION SELECT statements, and the discrete VRs are created from the retention detail records from the DSN catalog.

During the repeated runs of EDGCPIC2 you should edit the EPICPARM file to add or modify the OPTION SELECT statements as necessary to reduce the number of discrete VRSES generated. Start by creating an OPTION SELECT statement for your default retention; for example:

```
OPTION SELECT DSN=*,RET=5,CYC=0,DLA=0
```

This retains data not covered by a more specific policy, for 5 days.

Using the output from the most recent run of EDGCPIC2, look at the contents of the POLPRINT file. It lists a summary at the end of the file. This contains all the global policies and how many data sets use them, and for data sets that do not use them it lists a summary of policy groups and the size of the largest group. The remainder of the file lists the groups, the retention information and the data sets that are in those groups. Look at the largest group and perhaps by adding an OPTION SELECT policy to the EPICPARM file using the listed retention information, you can get the next run to use this instead of creating the discrete VRSES.

Only retention information is used to determine a group for the data sets which do not match to a global policy. The file also lists the required location (LOC=name). One reason why a global policy may not be used is that the required location is different than that specified on the global policy. The default location for global policies is the HOME location.

EDGCPIC2 uses all the versions of a data set to determine if it has any off-site storage requirement. If it does, it builds this information into the VRSES built.

Because the VMS vaulting information is not available, EDGCPIC2 cannot accurately determine this information. To help overcome this, EDGCPIC2 supports an addition keyword on the OPTION SELECT policy; this is the LOC= keyword. For example:

```
OPTION SELECT DSN=SYS1.BACKUP*,
              CYC=3,RET=0,DLA=0,LOC=RS
```

You use the information about locations in the POLPRINT file to determine the best cases to use this keyword. The listing of the global VRSES at the end of the file includes counts of how many data sets matched to the global VRS and a second count of those which could not use it because the location name is different. For those that have an 'IGNORED FOR' count, consider adding the LOC= keyword to the EPICPARM input.

The PLERRPRT file lists data sets retained by multiple options; which RMM does not support. It also lists those data sets which appear to have a mixed on-site and off-site retention requirement. You should use the information available to determine if the correct off-site storage count has been generated by EDGCPIC2. The simplest way to check this may be to run the next stage of conversion, the EDGNCVT program, and check the VRSCMDS output file. It lists the policy records in understandable format.

Once an acceptable level of policy conversion is reached, you can proceed to the next step. You may also be able to use the information gained by running EDGCPIC2 to customize the VRSCMDS file, and to use that to define the DFSMSrmm VRS policies, rather than the records that EDGNCVT produces.

The sample JCL in Figure 110 shows how to run the EDGCPIC2 program. The first step deletes any output files created by a previous run. As you use the extract and conversion programs, you are likely to rerun this job several times, so this step simplifies the rerunning of the job.

```
//WOODMW2 JOB CLASS=5,MSGCLASS=X,NOTIFY=WOODMW,
//          COND=(0,NE)
//*
//DEL      EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
           DEL 'WOODMW.EDGCPIC2.*'
           LISTDS 'WOODMW.RMM.EPICNVT.LOAD'
/*
```

Figure 110 (Part 1 of 2). EDGCPIC2 Sample JCL

```

/*
//CNVT2   EXEC PGM=EDGCPIC2
//STEPLIB DD DISP=SHR,DSN=WOODMW.EDGCPIC2.LOAD
//BADRECS DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.BADRECS,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//SCRPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.SCRLIST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//MSTPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.MSTLIST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//POLPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.POLLIST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//VLERRPRT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.ERRVLST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//PLERRPRT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.ERRPLST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//PLIGNPRT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.IGNPLST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//LOCPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.LOCLIST,
//          UNIT=SYSDA,SPACE=(TRK,(50,50),RLSE),BLKSIZE=23476
//SYSUDUMP DD DISP=(,DELETE,CATLG),DSN=WOODMW.EDGCPIC2.SYSUDUMP,
//          UNIT=SYSDA,SPACE=(TRK,(99,99),RLSE)
//CATIN    DD DISP=SHR,DSN=WOODMW.EDGCPIC1.CATOUT
//EPICPARM DD DISP=SHR,DSN=EPIC.PARMLIB(EPICPRMS)
//CONVOUT  DD DISP=(,CATLG),DSN=WOODMW.EDGCPIC2.CONVOUT,
//          UNIT=SYSDA,SPACE=(CYL,(5,1),RLSE),BLKSIZE=23476

```

Figure 110 (Part 2 of 2). EDGCPIC2 Sample JCL

7.7.7 Input Files

- STEPLIB** This execution library contains the EDGCPIC2 load module. It need not be an APF-authorized library.
- CATIN** This contains the sorted, extracted records from the EPIC DSN catalog. It was created by the EDGCPIC1 program as described in 7.7.3, "The EDGCPIC1 Program" on page 351.
- EPICPARM** This file must contain the EPIC global policy definitions. Initially, it should point to the EPIC startup parameters, but, as conversion continues, a file of desired global retention policies can be prepared that provides the best conversion results.

7.7.8 Output Files

- CONVOUT** The reformatted records are written to this output file. The record formats now match those that are defined for input to the EDGCNVT program.
- SCRPRINT** This file contains a list of all scratch volumes extracted from the DSN catalog records.
- MSTPRINT** This file contains a list of all private volumes extracted from the DSN catalog records.

Details of the data sets and volumes which make up each multi-volume, multi-data set volume set are listed, including records to show the records inserted by EDGCPIC2. EDGCPIC2 inserts data set records so that it can correctly reflect file and volume sequences. The data set records were deleted from the DSN catalog when the data set retention requirements had been met.

EDGCPIC2 uses the data set name 'EPIC.RELEASED' for data sets it inserts during processing.

VLERRPRT This file contains a list of errors found in private volumes. This information identifies where volume and sequence information cannot be correctly determined.

The records which the EDGCPIC2 program is analyzing to determine the correct sequence are written to the BADRECS output file, to allow further, manual analysis.

BADRECS This file contains a copy of the records that EDGCPIC2 could not correctly use because of conflicting information. Pairs of records are written to the BADRECS file.

The layout of the records in this file matches the layout of records in EDGCPICR macro. In addition, the first character is set to an error code. The following error code can be set:

- 1** Identical volume and file sequences found for two data sets with the same data set name. The data sets are converted as discrete data sets.
- 3** Two data sets recorded on the same volume, with same volume and file sequence number. The most recently created data set is converted.

POLPRINT This file contains a list of policies found.

EDGCPIC2 processes the retention policies in the type 1 DSN catalog records in sorted data set name sequence. It groups together data sets which are in sequence and have the same retention policy. You can use this information to manually build generic VRS definitions within DFSMSrmm instead of using the discrete policies created by EDGCPIC2. Use of generic data set names can significantly reduce the number of VRSS required.

PLERRPRT This file contains a list of problems found when building policy information.

The errors are basically those policies found by EDGCPIC2 which contain multiple retention types or move requirements which cannot be converted to a DFSMSrmm supported retention definition. For example, retention by days and whilecatalog.

PLIGNPRT This file contains a list of retention policies ignored by EDGCPIC2 and gives a reason for them being ignored.

LOCPRINT This file contains a list of all storage locations and loan locations found during processing.

7.7.8.1 Input Parameters

There are no input parameters.

7.7.8.2 Limitations

Processing of pool information has not been implemented.

7.7.9 Return Codes

A return code is set as follows:

0 Normal completion.

ABEND U0100 The record type of an input record from the EDGCPIC1 program was not supported. Only types 1 and 2 are supported.

ABEND U0200 An error was detected in the volume sequence for a multivolume data set.

ABEND U0300 An error was detected in the sequence of records read from the CATIN file. A detail data set record did not have the same data set name as the data set name in the header record. Check that the CATOUT file produced by EDGCPIC1 is correctly sorted by data set name and record type.

ABEND U0400 An incorrect chain of records was detected during processing.

ABEND U0500 A logic error detected in the program

ABEND U0601 The design limit of 9,999,999 input DSN catalog records is exceeded.

ABEND U0602 The design limit of 9,999 logical files on a volume is exceeded.

ABEND U0603 The design limit of 255 volumes in a multi-volume sequence is exceeded.

7.7.10 Messages

EDGCPIC2 STARTED, VERSION = *date time level*

Explanation: Execution is starting. The date, time and maintenance level of the EDGCPIC2 program is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 COUNTED ALL SUB-CHAINS

Explanation: Execution continues and all data sets are being processed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* RECORDS IN CATIN FILE

Explanation: Execution is continuing and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* POLICY RECORDS

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* data set RECORDS

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* AS SQUARE ROOT OF DATASET RECORD COUNT.

Explanation: This is information that helps analysis of the processing time of the program.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* VOLUMES IN SUB-CHAIN

Explanation: This is information that helps analysis of the processing time of the program.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* DIFFERENT VOLUMES

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* RECORDS FOR SCRATCH VOLUMES

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* RECORDS FOR UNCONNECTED VOLS

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* MISSING MULTI-VOL DSN RECS.

Explanation: Execution has been successful and the number of records processed is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* DUPLICATE VOLUME DSN RECS.

Explanation: Execution has been successful and the number of duplicate records is listed.

System Action: Each pair of records is written to the BADRECS output file.

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* DUPLICATE RECORD PAIRS

Explanation: Execution has been successful and the number of duplicate records is listed.

System Action: Each pair of records is written to the BADRECS output file.

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* VOLUMES NEEDING INSERTED FILES

Explanation: Execution is successful and the number of volumes for which data set records had to be inserted by EDGCPIC2 for missing CATIN data set records is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 FOUND *number* DATA SETS WITH LOW FILE SEQUENCE

Explanation: Execution is successful and the number of data set records with low file sequence number is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 BUILT *number* INSERTED FILES

Explanation: Execution is successful and the number of data set records that had to be inserted by EDGCPIC2 for missing CATIN data set records is listed.

System Action: None

User Response: None

Destination: Routcde 2

EDGCPIC2 WROTE *number* RECORDS TO CONVOUT FILE

Explanation: Execution is successful and the number of records created is listed.

System Action: None

User Response: None

Destination: Routcde 2

EPICNVT2 FOUND *number* MIXED LOCATION VRSES

Explanation: Execution is completed and the number of policy records that included multiple locations has been determined.

System Action: None

User Response: Use the information in the PLERRPRT file to analyze these policies and ensure that the correct VRSS are built manually.

Destination: Routcde 2

VOLUME *volser* SCRATCH

Explanation: A scratch volume is listed in the SCRPRINT file.

******* END OF CONNECTED VOLUMES LIST. *******

Explanation: This delimits the list of private volumes in the MSTPRINT file.

******* MISSING MULTI-VOLUME SEGMENT OF data set *data_set_name* FOR VOLUME *volser***

Explanation: EDGCPIC2 is processing a multivolume data set. It has detected missing information.

******* RECORD FOR data set *data_set_name* ON VOLUME *volser* HAS DUPLICATE ENTRIES**

Explanation: EDGCPIC2 is processing a data set and has found multiple records for the same data set.

******* DUPLICATE data set *data_set_name* FOR VOLUME *volser* IGNORED**

Explanation: EDGCPIC2 is processing a data set and has found multiple records for the same data set.

System Action: The record is ignored.

****INCOMPLETE* VRS *data_set_name* ***** MULTIPLE LOCATIONS FOR DSNAME/JOBNAME**

Explanation: Identifies a policy which routes volumes to two different locations for the same jobname. The VRS generated only specifies one location, and must be corrected manually, after EDGCNVT processing, to match the EPIC processing.

System Action: Processing continues. A best guess VRS is built.

*****CHECK**** VRS *data_set_name*
STORECOUNT CALCULATED, ERRORS POSSIBLE**

Explanation: Identifies a policy which routes volumes to both a store and the home location by dsname. The VRS STORENUMBER has been set less than the COUNT, but it should be checked manually against the EPIC vaulting policy to ensure that it will have the desired effect.

The STORENUMBER is set to the number of data sets determined to be in the storage location; this may not accurately reflect the true number that should be moved off-site.

System Action: Processing continues. A best guess VRS is built.

*****CHECK**** VRS *data_set_name***

JOB *job_name* STORECOUNT CALCULATED, ERRORS POSSIBLE

Explanation: Identifies a policy which routes volumes to both a store and the home location by *dsname* and *jobname* combination. The VRS STORENUMBER has been set less than the COUNT, but it should be checked manually against the EPIC vaulting policy to ensure that it will have the desired effect.

The STORENUMBER is set to the number of data sets determined to be in the storage location; this may not accurately reflect the true number that should be moved off-site.

System Action: Processing continues. A best guess VRS is built.

**GROUP START = *data_set_name* POOL= *p* *d* =DLA *r* =RET *c* =CYC
MVSCAT=*m* LOC=*l***

Explanation: Identifies the start of a group of data sets with common retention requirements. EDGCPIC2 processes data sets in data set name sequence and lists the groups of data sets that are not covered by the global policies defined in the EPICPARM input file. You can use the information in this and the following messages to modify the EPICPARM input OPTION SELECT statements to reduce the number of discrete VRSs built during conversion.

In the message:

p Is the pool identifier. It is blank if no specific pool is identified.

d Is the DLA count.

r Is the RET count.

c Is the CYC count.

m Is the MVSCAT value.

l Is the retention location.

System Action: Processing continues.

data_set_name* LOC=*l

Explanation: Identifies a member of a group of data sets with common retention requirements. EDGCPIC2 processes data sets in data set name sequence and lists the groups of data sets that are not covered by the global policies defined in the EPICPARM input file. You can use the information in this and the following messages to modify the EPICPARM input OPTION SELECT statements to reduce the number of discrete VRSs built during conversion.

In the message:

l Is the retention location.

System Action: Processing continues.

GROUP END = *data_set_name* DS NAMES IN GROUP= *group_count* LOC=*l*

Explanation: Identifies the end of a group of data sets with common retention requirements. EDGCPIC2 processes data sets in data set name sequence and lists the groups of data sets that are not covered by the global policies defined in the EPICPARM input file. You can use the information in this and the previous messages to modify the EPICPARM input OPTION SELECT statements to reduce the number of discrete VRSs built during conversion.

In the message:

group_count

Is the number of data sets in the group.

l Is the retention location.

System Action: Processing continues.

GROUPS DETECTED = *groups* DATASETS COVERED BY GLOBAL POLICIES = *dsn_count*

Explanation: This message is listed in the POLPRINT file and lists the total number of groups found for data sets that did not match to global policies, and also the number of data sets that matched to the global policies. No specific policy is converted for those matching to a global policy, as the global policy is converted and will be used by DFSMSrmm. You can use the information in this and the previous messages to modify the EPICPARM input OPTION SELECT statements to reduce the number of discrete VRSS built during conversion.

In the message:

groups

Is the number of groups for which discrete policies have been converted.

dsn_count

Is the number of data sets for which no specific policy was converted.

System Action: Processing continues.

LARGEST GROUP CONTAINED *group_count* DATASETS, INCLUDING *data_set*

Explanation: This message is listed in the POLPRINT file and lists the largest group of data sets found. Use the data set name to find the group, and consider adding a global policy to the EPICPARM input file using OPTION SELECT statements.

In the message:

group_count

Is the number of data sets in the group.

data_set

One of the data sets in the largest group.

System Action: Processing continues.

**GLOBAL VRS = *data_set_name* POOL= *p* d =DLA r =RET c =CYC
MVSCAT=*m* LOC=*l* USED FOR *used_count* IGNORED FOR *other_location_count*
DATASETS**

Explanation: Identifies a global policy set by an OPTION SELECT statement in the EPICPARM input. The retention policy, location name and matching information is listed. You can use the information in this and other messages in the POLPRINT file to modify the EPICPARM input OPTION SELECT statements to reduce the number of discrete VRSS built during conversion.

In the message:

data_set_name

The policy data set name mask.

p Is the pool identifier. It is blank if no specific pool is identified.

d Is the DLA count.

r Is the RET count.

c Is the CYC count.

m Is the MVSCAT value.

l Is the retention location.

used_count

Is the number of data sets which matched to the retention and location information and for which no discrete policy was converted.

other_location_count

Is the number of data sets which matched to the retention information but which could not use this global policy because their required location does not match. A discrete policy was converted for each of these data sets.

System Action: Processing continues.

7.7.11 The EDGCXTRC Program

This section describes the setup and control of the EDGCXTRC scratch list validation program. Use this program to help validate the successful conversion of data and policies.

The first sample is for use with EPIC/MVS, and the second is for use with Prevail/XP Media which has a new utility to list scratch volumes. You must customize the EDGJPCV sample JCL, as described within the sample, for use with Prevail/XP Media.

Figure 111 shows the sample JCL for execution of the EDGCXTRC program (EPIC/MVS).

```
//WOODMWV JOB CLASS=5,MSGCLASS=X,MSGLEVEL=(1,1),
//          NOTIFY=WOODMW
//*
//DEL      EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN  DD *
//          DEL 'WOODMW.EDGCXTRC.*'
//          LISTDS 'WOODMW.RMM.EPICNVT.LOAD'
//
//
//STEP1    EXEC PGM=TSIDUTL,PARM=EDATE,REGION=48M
//SYSPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.EPICLIST,
//          UNIT=SYSDA,SPACE=(TRK,(9,3),RLSE),
//          RECFM=FBA,LRECL=133,BLKSIZE=23408
//SYSOUT   DD SYSOUT=*
//WORK     DD DSN=&&WORK,DISP=(,DELETE),UNIT=SYSDA,
//          SPACE=(TRK,(9,5),RLSE),
//          DCB=(RECFM=FB,LRECL=50,BLKSIZE=500)
//SYSIN    DD *
//          DSNLY SCRATCH,SEQ=SERIAL
//
//
//STEP2    EXEC PGM=IKJEFT01
//SYSTSPRT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.RMMLIST,
//          UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
//          RECFM=FBA,LRECL=133,BLKSIZE=23408
//SYSTSIN  DD *
//          PROFILE MSGID
//          RMM SEARCHVOLUME VOLUME(*) OWNER(*) STATUS(SCRATCH,REL) LIMIT(*)
//
//
```

Figure 111 (Part 1 of 2). EDGJPCV Sample JCL

```

//STEP3 EXEC PGM=EDGCXTRC
//STEPLIB DD DISP=SHR,DSN=WOODMW.RMM.EPICNVT.LOAD
//EPICIN DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICLIST
//RMMIN DD DISP=SHR,DSN=WOODMW.EDGCXTRC.RMMLIST
//EPICOUT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.EPICVOLS,
// UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
// RECFM=FB,LRECL=6,BLKSIZE=23472
//RMMOUT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.RMMVOLS,
// UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
// RECFM=FB,LRECL=6,BLKSIZE=23472
//*
//STEP4 EXEC PGM=ISRSUPC,PARM=(DELTA,LINECMP)
//NEWDD DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICVOLS
//OLDD DD DISP=SHR,DSN=WOODMW.EDGCXTRC.RMMVOLS
//OUTDD DD SYSOUT=*

```

Figure 111 (Part 2 of 2). EDGJPICV Sample JCL

Figure 112 shows the sample JCL for execution of the EDGCXTRC program (Prevail/XP Media).

```

//WOODMWV JOB CLASS=5,MSGCLASS=X,MSGLEVEL=(1,1),
// NOTIFY=WOODMW
//DEL EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
// DEL 'WOODMW.EDGCXTRC.*'
// LISTDS 'WOODMW.RMM.EPICNVT.LOAD'
//
//
//STEP1 EXEC PGM=TSIDMNT,REGION=48M
//SYSPRINT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.EPICLIST,
// UNIT=SYSDA,SPACE=(TRK,(9,3),RLSE),
// RECFM=FBA,LRECL=133,BLKSIZE=23408
//SYSOUT DD SYSOUT=*
//WORK DD DSN=&&WORK,DISP=(,DELETE),UNIT=SYSDA,
// SPACE=(TRK,(9,5),RLSE),
// DCB=(RECFM=FB,LRECL=50,BLKSIZE=500)
//SYSIN DD *
// DISPLAY ALL,SCRATCH
//

```

Figure 112 (Part 1 of 2). EDGJPICV Sample JCL for Prevail/XP Media

```

/*
//STEP2 EXEC PGM=IKJEFT01
//SYSTSPRT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.RMMLIST,
//          UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
//          RECFM=FBA,LRECL=133,BLKSIZE=23408
//SYSTSIN DD *
PROFILE MSGID
RMM SEARCHVOLUME VOLUME(*) OWNER(*) STATUS(SCRATCH,REL) LIMIT(*)
/*
/*
//STEP3 EXEC PGM=EDGCXTRC
//STEPLIB DD DISP=SHR,DSN=WOODMW.RMM.EPICNVT.LOAD
//EPICIN DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICLIST
//RMMIN DD DISP=SHR,DSN=WOODMW.EDGCXTRC.RMMLIST
//EPICOUT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.EPICVOLS,
//          UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
//          RECFM=FB,LRECL=6,BLKSIZE=23472
//RMMOUT DD DISP=(,CATLG),DSN=WOODMW.EDGCXTRC.RMMVOLS,
//          UNIT=SYSDA,SPACE=(TRK,(5,1),RLSE),
//          RECFM=FB,LRECL=6,BLKSIZE=23472
/*
/* SORT EPICOUT BY VOLSER
/*
//SORT EXEC PGM=SORT,REGION=4M
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICVOLS
//SORTOUT DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICVOLS
//SYSIN DD *
SORT FIELDS=(1,6,CH,A),DYNALLOC
/*
//STEP4 EXEC PGM=ISRSUPC,PARM=(DELTA,LINECMP)
//NEWDD DD DISP=SHR,DSN=WOODMW.EDGCXTRC.EPICVOLS
//OLDDDD DD DISP=SHR,DSN=WOODMW.EDGCXTRC.RMMVOLS
//OUTDD DD SYSOUT=*

```

Figure 112 (Part 2 of 2). EDGJPICV Sample JCL for Prevail/XP Media

7.7.11.1 Input Files

- STEPLIB** This execution library contains the EDGCXTRC load module. It need not be an APF-authorized library.
- EPICIN** This is the list of volumes created by the EPIC TSIDUTL program in STEP1.
- RMMIN** This is the list of volumes created by the DFSMSrmm SEARCHVOLUME subcommand in STEP2.

7.7.11.2 Output Files

- EPICOUT** This is a list of volumes taken from the EPICIN file records. It is created in STEP3.
- RMMOUT** This is a list of volumes taken from the RMMIN file records. It is created in STEP3.
- OUTDD** This is a list of mismatches as found by the ISRSUPC utility in STEP4. Review this list to determine whether there are differences.

7.7.11.3 Input Parameters

There are no input parameters.

7.7.11.4 Limitations

EDGCXTRC assumes that the input files are in the correct format.

7.7.11.5 Return Codes

A return code is set as follows:

0 Normal completion.

7.8 Am I Ready to Continue?

In this chapter we have covered all of the activities that you should have considered before converting from EPIC to DFSMSrmm. After completing the activities in this chapter, you should have the following valid and tested input to load the DFSMSrmm CDS:

- Input to exits
- Correct PARMLIB options
- Housekeeping jobs to use in an DFSMSrmm environment
- Changed reports
- Extract and convert control statements
- Extracted data from the EPIC TMC

If the data extract program has error return codes, read the error messages to determine the actions required to correct the errors. Then rerun the extraction process (see Section 7.7, “Running the Extract Programs” on page 348).

To have a valid DFSMSrmm CDS, you must stop all tape activity when you get the data from the EPIC database until you start DFSMSrmm in record-only mode in parallel with EPIC.

Before producing a valid conversion, you can use a copy of the EPIC database to test the extraction and conversion process, so you do not have to stop the tape activities every time you find that you have to go back and re-run the extract jobs.

Once you have tested the environment you should have the correct tailored input to the Extract program.

You can now stop tape activities and produce a valid data extraction from the DSN catalog.

If your previous tests were satisfactory, you should obtain the correct input to the EDGCNVT program and proceed to the next step.

Chapter 8. Converting from Manual Management to DFSMSrmm

In this chapter we describe how to convert from a manual tape management system to DFSMSrmm. In this case, the manual tape management system consists of one or more integrated catalog facility (ICF) catalogs.

Objectives of Chapter

- Compare ICF catalog information and DFSMSrmm terms
- Extract data and prepare input for RMM TSO subcommands
- Provide hints and tips

Audience

- System programmers
- Storage administrators and librarians

At the end of this chapter you will have reviewed the implementation of your manual tape management system and will understand the detailed changes required to customize DFSMSrmm to meet your tape management needs.

8.1 ICF Catalog Overview

The following information is stored in an ICF catalog:

- Volume
 - Serial number
 - Sequence number
 - Device type
- Data set
 - Name
 - Sequence number
 - Creation date
- SMS data
 - Storage class
 - Management class
 - Data class

This information is enough to create a complete DFSMSrmm CDS.

8.2 ICF Catalog and DFSMSrmm

To ensure a successful conversion it is important that you understand some of the differences between the ICF catalog and DFSMSrmm and how those differences affect your conversion project.

8.2.1 Terminology

Table 85 on page 368 compares and defines some ICF catalog and DFSMSrmm terms.

Table 85. ICF Catalog and DFSMSrmm Terminology		
ICF Catalog	DFSMSrmm	Definition
Creation	Assign date	Volume: The date when the volume was assigned to a user
	Create date	Data set: The date when the data set was first written to tape
Data set name	Data set name	The name of the data set
Data set sequence	Physical file sequence number Data set sequence number	The physical file sequence number field specifies the relative position of a data set on a volume. The data set sequence number field corresponds to the IBM standard HDR1 data set sequence number field.
Device type	MEDIANAME	Describes the shape or type of a volume. Media names are defined by your installation and must be one to eight characters. (See "Using the PARMLIB Member EDGRMMxx," in the <i>DFSMSrmm Implementation and Customization Guide</i> , SC26-4932.
SMSDATA DATACLASS	SMS data class name	A list of data set allocation attributes that SMS assigns to a data set when it is created
SMSDATA STORAGECLASS	SMS storage class name	A list of data set storage service attributes that identify performance and availability requirements. SMS uses these attributes to control data placement.
SMSDATA MANAGEMENTCLASS	SMS management class name	A collection of management attributes defined by the storage administrator that are used to control the retention, migration, backup, and release of allocated but unused space for data sets, or to control the retention, backup, and class transition of objects.
VOLSER	Volume VOLSER	The volume's VOLSER
--	Volume sequence number	The Volume Sequence field displays the volume sequence number of a volume in a multi volume data set.

8.2.2 Cross-Reference

In this section we present tables that map data from the RMM TSO subcommands to an ICF catalog.

8.2.2.1 Volume Information

Table 86 shows the EDGCLREC volume information.

Table 86 (Page 1 of 4). Volume Information		
	ICF Catalog	Status
Volume serial number	√	Req
Volume owner ¹		Req

Table 86 (Page 2 of 4). Volume Information		
	ICF Catalog	Status
Volume owner user ID		
Volume owner node ID		
Volume security level		
Volume sequence number		Req
Volume create date	√	Req
Volume create time		
Volume create user ID		
Volume create job name		
UCB type (copy of UCBTYP)		
Tape media type information Tape recording format Not 3480 18 track 36 track 128 track Tape media type Not 3480 Standard capacity Enhanced capacity High performance cartridge tape Extended high performance cartridge tape Tape compaction technique Unknown Not compact IDRC Tape special attributes None 18 track read-only		Cond
Total volume kilobytes used		
JCL volume expiration date		
Volume expiration date	√	Req
Volume retention date		
Count of data sets on volume	√	Req
Temporary read errors		
Temporary write errors		
Permanent read errors		
Permanent write errors		
Tape density (copy of JFCBDEN)		
BPI indicator 1600 bpi 6250 bpi 3480 3480 compact (IDRC) Undefined		Req

Table 86 (Page 3 of 4). Volume Information		
	ICF Catalog	Status
Volume status indicators Volume is master Volume pending release Vital record - do not release User tape (assigned by librarian) Tape is on loan Tape open - not yet closed Volume is scratch Volume recorded by open, close, or EOV	√	Req
Additional volume status indicators Scratch volume claimed by GETVOLUME Scratch volume has never been initialized Scratch volume with initialization action pending Scratch volume waiting to enter ATL Abend in process when a data set closed Abend probably in open, close, or EOV		
Volume label type Default retention period used Label type is NL Label type is AL Label type is SL Tape last written with BLP SL or AL with user labels		Req
Volume release actions Return to scratch pool - default Replace tape on release Reinitialize Degauss/security erase Return to owner Notify owner		
Volume access information Owner may read volume Owner may update volume Owner may alter volume Read-only protection Update protection Can be used on MVS systems Can be used on VM systems		Req
Media name		
Unit address used for creation		
Last used unit address		
Loan location		
Date volume last read		
Date volume last written		
Assigned from scratch date		
Assigned from scratch time		
Copy of JFCBLTYP		

Table 86 (Page 4 of 4). Volume Information		
	ICF Catalog	Status
From JFCTRTCH - IDRC support DSN used 3480 IDRC No compaction		Req
Current location of volume		Req
Destination name		
Home location name		
Storage group name		
Accounting information		
User description		
Authorized user IDs area		
Rack number ²		Req
Remote store bin number		Cond
Remote store bin date		Cond
Remote store bin media name		Cond
Label number of first file		Req
Name of first data set on volume	√	
First data set's owner's name		
Sequence number of first data set		Cond
First data set's RECFM		Cond
First data set's LRECL		Cond
First data set's BLKSIZE		Cond
First data set's block count		Cond
First data set's kilobytes		
First data set's security level		
DSN1 create date	√	Req
DSN1 create time		
DSN1 last read		Cond
DSN1 last written		Cond
DSN1 flag Data set cataloged Abend in progress when data set closed		
VRS management value		Cond
First file creating step name		
First file creating DD name		
First file retention date		
Name of last data set on volume		Cond
Creating system ID		
• Data that can be found in an ICF catalog is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹We recommend that you use the job name or the data set high-level qualifier (HLQ) as the owner. ²We recommend that you use the VOLSER as the rack number.		

8.2.2.2 Data Set Name Information

Table 87 shows the EDGCDREC data set name information.

Table 87. EDGCDREC Data Set Name Information		
	ICF Catalog	Status
Data set create date	√	Req
Data set create time		
Data set name	√	Req
Compaction status Data set used 3480 IDRC No compaction		
File sequence number on volume		Req
VOLSER holding data set	√	Req
Volume sequence within set of volumes		Req
Unit where data set was written		
Data set RECFM		Req
Data set LRECL		Req
Data set BLKSIZE		Req
Data set block count		Req
Data set size in kilobytes		
Data set owner name ¹		Req
Logical file sequence number	√	Req
Data set security level		
Data set date last read		Req
Data set date last written		Req
VRS management value		Cond
DSN flag Data set cataloged Abend in progress when data set closed		
Creating system ID		
Creating job name		
Creating step name		
Creating DD name		
Retention date		
• Data that can be found in an ICF catalog is marked with a √. • Data that DFSMSrmm requires is marked with <i>Req</i>. • Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>. ¹CONVERT ensures that all data sets on the same volume have the same owner as the VOLSER's owner.		

8.2.2.3 Vital Record Information

Table 88 shows EDGCKREC policy information.

Table 88. EDGCKREC Policy Information		
	ICF Catalog	Status
Record create date		Req
Record create time		
VRS type		Req
Data set name		Cond
Format of data set name		Cond
Type of retention BYDAYSCYCLE Cycles Elapsed days Elapsed days since last referenced While cataloged Until expired		Cond
Volume serial number		Cond
Delay days		
Total count of number of cycles or days		Req
VRS name		Cond
Store keep number		Cond
VRS owner		Req
Automatic delete date		
VRS description		
Location name		Req
Name of next VRS definition		
Job name		
- No information about VRSs is stored in an ICF catalog. - Data that DFSMSrmm requires is marked with <i>Req</i>. - Data that DFSMSrmm requires only when the environment exists is marked with <i>Cond</i>.		

8.2.2.4 Empty Rack and Bin Information

Table 89 shows the EDGCEREC empty rack and bin information.

Table 89. EDGCEREC Empty Rack and Bin Information		
	ICF Catalog	Status
Shelf status Shelf is rack in library Shelf is bin in storage location		Req
Empty shelf location's name		Req
First of empty shelves start		Req
Count of contiguous shelves		
Media name		
Data that DFSMSrmm requires is marked with <i>Req</i>.		

8.2.2.5 Owner Record Information

An owner record is required only for actual owners for whom you want to supply owner details and for those who are to be owners for VRS. Other owner records will be created by CONVERT post processing (EDGCVOVL). Table 90 shows the EDGCOREC owner information.

Table 90. EDGCOREC Owner Information		
	ICF Catalog	Status
Owner identifier		Req
Owner's surname		
Owner's first name		
Owner's department ¹		Req
Owner's first address line		
Owner's second address line		
Owner's third address line		
Owner's internal telephone number		
Owner's external telephone number		
Owner's user ID		
Owner's node ID		
Data that DFSMSrmm requires is marked with <i>Req</i>. ¹Not strictly required for conversion. Only the RMM ADDOWNER command enforces the owner's department.		

8.3 Reviewing the Current Environment

See Figure 8 on page 21 to review the flow of events that we followed in converting from ICF catalog to DFSMSrmm. You must collect and analyze all information about the current tape management environment.

8.3.1 Analyzing the Record Layouts

You must analyze the information you have collected to achieve the equivalent or acceptable alternative function in DFSMSrmm environment.

DFSMSrmm requires certain data to be available to build its CDS. You can create input records for the EDGCNVT program or RMM TSO subcommands and add the information directly through commands (see Chapter 9, "Building the DFSMSrmm CDS" on page 403).

8.3.2 Volume Ownership

DFSMSrmm requires that all private volumes have an owner. DFSMSrmm uses the RACF user ID of the batch job creating the first file. We recommend that you use either the creating job name or the HLQ of the first file data set name as the volume owner. As tapes are recycled and reused, DFSMSrmm will record the new owner. Using job name or the HLQ of the data set provides you with an easy way of listing and reporting on volumes during the initial implementation of DFSMSrmm.

8.3.3 Rack Numbers

DFSMSrmm requires that all volumes have a rack number. DFSMSrmm uses the rack number to identify the volume to the operator and the library location where the volume is stored. We recommend that you use the same value as the VOLSER, unless you have any better information on which to base this value. For example, you may have an installation convention or keep a file of information that relates an external label VOLSER to an internal label VOLSER.

8.4 Retention and Movement

In the sections that follow we discuss some basic tape management retention and movement concepts and suggest some factors to consider when converting to DFSMSrmm.

8.4.1 Retention Management

You define and use a retention management policy to indicate how long to retain a tape data set in both a tape library and the storage locations.

In DFSMSrmm the expiration dates of 99365 and 99366 or 1999/365 and 1999/366 are considered “never-scratch” dates. Data sets with these expiration dates are neither deleted nor overwritten.

DFSMSrmm allows you to set a system default retention period if the retention or expiration is not specified in the JCL during data set and volume creation. You can extend the retention period for the data set on the volume or for the volume, after it has been created using a VRS definition. The retention period specified in the VRS can override the expiration date in the DFSMSrmm CDS.

DFSMSrmm supports many different types of retention. For a data set, you can request the following types of retention using a specific or a generic data set name:

- *Retention by cycles* or *retention by cycles while cataloged*, which defines how many cycles or generations of a data set should be retained.
- *Retention by bydayscycle*, which defines how many cycles or generations of a data set should be retained. The difference from *cycles* is that all instances of a data set created on a single calendar day are considered to be a single cycle.
- *Retention by number of elapsed days since creation* or *retention by number of elapsed days since creation while cataloged*, which indicates that a data set be retained until the specified number of days since creation have elapsed.
- *Retention by number of elapsed days since last reference* or *retention by number of elapsed days since last reference while cataloged*, which indicates that a data set be retained until the specified number of days since last reference have elapsed.
- *Retention by catalog*, which indicates that a data set be retained as long as it is cataloged.

For a volume, you can request the following types of retention using a specific or generic VOLSER:

- *Retention by number of elapsed days since creation*, which indicates that a volume be retained until the specified number of days since creation has elapsed.

- *Retention by number of volumes*, which indicates how many volumes should be retained.

DFSMSrmm provides support for the specific and unique Julian dates of 98000-99366 having special meanings. The support is provided through an installationwide EDGUX100 exit, which you can use to interrogate and recognize the dates. When the installationwide exit detects a data set using the Julian dates, it assigns a VRS management value that identifies the retention management policy.

During inventory management processing, if a match on data set name is not found, DFSMSrmm checks the data set's management value. If the value exists, DFSMSrmm matches the assigned VRS management value with the defined VRS. If there is a match, DFSMSrmm uses the retention specified in the VRS overriding the expiration from the DFSMSrmm CDS. For example, if you want to support the expiration date of 99000 (retention while cataloged), you need to define a VRS to manage the retention while cataloged. In the installationwide exit, you intercept the tape data set using the 99000 expiration date and assign a VRS management value matching the defined VRS.

8.4.2 Storage Location Management

You define and use your installation's management policies for those tape volumes and data sets that contain installation-critical data for disaster recovery. These policies define:

- Critical data—data that is vital to your installation
- Storage location—where to store critical data
- Retention criteria—how long to keep the critical data in the storage location
- Movement criteria—how to move critical data to and from the data center and the storage locations.

Using your installation-defined VRSs, DFSMSrmm automatically controls the movement of the data set on the tape volume or the tape volume from the tape library to and among the storage locations. You can specify a data set name or a data set name mask to define a VRS. Additionally, you can specify a VOLSER or a generic VOLSER.

DFSMSrmm does not restrict the movement of the data set on the tape volume or the tape volume to and among the storage locations. When a data set on the tape volume, or the tape volume, is moved from a tape library to and among the storage locations for disaster recovery or vital records, DFSMSrmm assigns a storage location bin number, storage location name, and store date.

DFSMSrmm allows you to specify one location name in each data set or volume VRS. If you require that a data set or volume be moved to and among the storage locations, you can chain together two or more VRSs by using the NEXTVRS and NAME keywords. DFSMSrmm does not restrict any sequence of data set or volume moves among storage locations and system-managed libraries. For example, if your installation's policy is to initially retain the data set in the *remote* storage location and then later retain it in the *distant* storage location, you first define a data set VRS for the *remote* storage location and indicate that the data set requires multiple moves by specifying a name VRS. You then indicate the next storage location, the *distant* storage location, through that name VRS.

During inventory management processing, DFSMSrmm identifies the volumes that must be moved to and among the storage locations and assigns the exact store bin numbers where the volumes are to be stored. DFSMSrmm provides a batch utility, which you can use to generate a pull list for the volumes requiring movement. You then use the pull list report to confirm that the moves have taken place. The confirmation process updates the DFSMSrmm CDS to reflect the most current information about where the volumes are stored.

DFSMSrmm uses the retention criteria from the VRS to manage tape volume retention in the storage locations. The volume can be retained by cycles, days since creation or last reference, or while a data set is cataloged (see *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information about DFSMSrmm retention management).

8.4.3 Inventory Management and Reports

DFSMSrmm provides inventory management and reporting functions with the EDGHSKP, EDGBKUP, and EDGUTIL utilities.

EDGHSKP must be executed with the DFSMSrmm subsystem active. EDGBKUP and EDGUTIL can be executed independently of the subsystem.

The utilities have the following functions:

- EDGHSKP
 - Processes vital records
 - Performs storage location management
 - Performs expiration processing
 - Creates an extract data set
 - Backs up the DFSMSrmm CDS
 - Backs up and clears the DFSMSrmm journal
- EDGBKUP
 - Backs up the DFSMSrmm CDS
 - Backs up the DFSMSrmm journal
 - Recovers and restores the DFSMSrmm CDS
 - Reorganizes the DFSMSrmm CDS
- EDGUTIL
 - Verifies DFSMSrmm CDS

DFSMSrmm does not provide any batch update utility. However, you can execute the RMM TSO commands ADD, CHANGE, and DELETE in batch to achieve the same result. The ISPF dialog has the option to generate and group together TSO commands for later batch processing.

DFSMSrmm provides the following reporting programs:

- EDGRPTD—processes the report extract file to produce:
 - Volume movement reports
 - Volume inventory reports.
- EDGAUD—processes DFSMSrmm SMF records to produce:
 - Security report (tracking information for classified data)
 - Audit report (CDS changes).
- EDGJRPT—processes the extended extract file
 - Pull list for scratch tapes sorted by volume serial number
 - Pull list for scratch tapes sorted by data set name
 - Inventory list by volume serial number
 - Inventory list by data set name

- Inventory of data set including number of KB used
- Inventory of volume serial number by location
- Inventory of data set names by location
- Inventory of BIN numbers by location
- List of all data set names at loan location
- List of all volume serial numbers at loan location
- List of all multiple volume, multiple data set
- Movement report including the first data set name on the volume
- Movement report sorted by storage location bin number
- Movement report sorted by volume serial number
- Inventory list sorted by volume serial number including volume count

You can execute the DFSMSrmm TSO commands, LIST and SEARCH, in batch to produce a wide variety of reports. You can also use DFSORT's ICETOOL to write your own customized reports.

EDGRPTD, EDGJRPT, and EDGAUD sample jobs, REXX programs, and DFSORT's ICETOOL reports are provided in Chapter 14, "Extending DFSMSrmm Reporting" on page 533 and Appendix D, "Sample Exits and Reports" on page 603.

We suggest that you move to the scratch pool the cartridges that will return to scratch only when they are no longer in *PENDING RELEASE* status. You can accomplish this in your batch housekeeping jobstream by using a step that lists the volumes in *PENDING RELEASE* status before running the EDGHSKP step.

8.4.4 Exits

DFSMSrmm provides source code for the installationwide exits installed as part of DFSMSrmm during the SMP/E apply process. The source code is supplied for:

- EDGTVEXT—DFSMSshm tape volume programming interface ⁸
- IGXMSGEX—DFSMSrmm uses the DFSMSdfp MSGDISP exit to update tape drive displays
- CBRUXENT—OAM cartridge entry exit
- CBRUXEJC—OAM cartridge eject exit
- CBRUXCUA—OAM cartridge change use attributes exit
- CBRUXVNL—OAM volume not in library exit

Before the SMP/E accept, the source code is in SMPSTS; after the accept, the exits are in AEDGSR1.

If your installation is already using one or more of these exits in the current environment, you must ensure that the correct action is performed during conversion. Chapter 10, "Parallel Running and Validation" on page 427 and Chapter 11, "Cutover to Production" on page 455 describe the procedure to follow. You can test the DFSMSrmm version of this exit once you have completed the conversion.

The CBRUXVNL exit is optional. If you use it, however, as with the CBRUXENT exit, you must test it in the DFSMSrmm environment after conversion.

The installationwide exits that you installed with DFSMSrmm for either IGXMSGEX, CBRUXENT, CBRUXEJC, and CBRUXCUA can replace any existing

⁸ In DFSMSrmm releases prior to Version 1 Release 4 you must use the ARCTVEXT exit supplied by DFSMSrmm. The EDGTVEXT will be shipped in object code only.

exits you had previously installed. Decide whether you still require the function your previous exits provided, even if for a short time while introducing DFSMSrmm. If you no longer need the function, proceed with the next implementation step. If you still require the function, consider how to implement it, while still providing the function required by DFSMSrmm.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more details on the DFSMSrmm programming interface that you use with EDGTVEXT, IGXMSGEX, and the OAM exits. The supplied source code for these exits shows how to call the programming interface. You can modify either the supplied exit to call a modified version of your old exit or your old exit to call the DFSMSrmm programming interface. Consider how you might provide an option to disable your code without another code modification. In either case, install the changed software by using an SMP/E USERMOD.

Based on the decisions you have taken about VRS management values and scratch pooling, you can update the supplied sample installation exit, EDGUX100, to perform the function you require. Follow these steps:

1. Copy the sample installation exit, EDGUX100, and use this copy as a base for your exit.
2. Update the exit if needed.
3. Build an SMP/E USERMOD to apply the updated source code for EDGUX100, including the necessary JCLIN statements to get the EDGUX100 load module added to the LINKLIB target library.
4. Receive and apply the USERMOD.
5. Your new exit is now ready for use on your system.

You can use the same process to install the EDGUX200 code. If you have a non-IBM robot tape library, you can update the sample EDGUX200 to communicate with the library host software to ensure that the external inventory is updated dynamically as volumes return to scratch status.

For more information about DFSMSrmm exits refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

8.4.4.1 Data Set Overwrite

In a DFSMSrmm environment you have three different volume statuses:

- MASTER
The volume contains valid data.
- SCRATCH
The volume can be used for scratch output requests.
- USER
The volume is assigned to a user and it can be overwritten, rewritten, and used repeatedly until the user returns it to the scratch pool.

DFSMSrmm can manage the overwrite of a MASTER volume with the PARMLIB option, MASTEROVERWRITE (see Table 12 on page 91). You can tailor this option to meet your installation's needs.

8.4.4.2 BLP and NL Processing

BLP and NL processing differs in an DFSMSrmm environment. When DFSMSrmm successfully reads the VOL1 label for a volume mounted as a result of a nonspecific request, it provides the actual VOLSER to OPEN for use in related processing such as cataloging of the data. This replaces the logical VOLSER (Lnnnnn) generated by OPEN in these circumstances.

For BLP, further processing is determined by the BLP option set in PARMLIB. Processing is as follows:

RMM	This is the default and works as for current processing.
NORMM	BLP can be used for reading and writing of master and user status tapes and for output to scratch tapes. BLP reading of scratch tapes is not supported.

When BLP or NL is used with scratch tapes, DFSMSrmm still changes the volume to master status but sets the initialize release action to ensure that the volume has a valid standard label before it is returned to scratch status.

Some changes are required for how and when DFSMSrmm records information for tapes and data read and written using BLP:

- For BLP output to a nonspecific volume, file sequence number is not checked. For example, LABEL=(2,BLP) could be used. For non-BLP requests, the restriction of using LABEL=(1,label_type) is still enforced.
- The data set information for files processed with BLP is only updated for output to the first file on a volume. All other types of BLP requests change only the volume information, such as the date last read and date last written. For BLP output requests, the data set information is only updated in the DFSMSrmm CDS when the data set is closed.

You can enable the use of NL output to scratch volumes by creating a profile in the RACF FACILITY class: STGADMIN.EDG.NOLABEL.volser. Once the profile is created, the function is enabled, and only those with access to the profile can use this function.

8.4.4.3 DFSMShsm Interface

When DFSMSrmm is informed, through the EDGTVEXT general-use programming interface, that a DFSMShsm volume is no longer required, the volume is immediately marked *PENDING RELEASE*, and will be returned to the scratch pool following the next run of the housekeeping expiration process *"EDGHSKP,PARM=(EXPROC)"*.

8.4.4.4 Tape Initialization

In a DFSMSrmm environment the initialization process is different. A utility, EDGINERS, should be used instead of IEHINITT for the tapes defined in the DFSMSrmm database. EDGINERS automatically defines new volumes it initializes to or erases from the DFSMSrmm CDS. For an existing volume, DFSMSrmm ensures that the proper volume is mounted and the requested action is actually required. You can request DFSMSrmm to automatically initialize scratch volumes when you add them to DFSMSrmm. You must ensure that your operational procedures are updated in order to use the new DFSMSrmm utility after the conversion process.

DFSMSrmm does not require that you initialize tape volumes before they are available as scratch, but it prevents you from using as scratch a volume that

requires initialization, and it will not include the volume in a scratch pull list. When adding scratch volumes to DFSMSrmm, you have the choice of deciding whether they need to be initialized or not.

8.4.4.5 Tape Pool Management

DFSMSrmm decides which pool to use on the basis of the VLPOOL definitions in the DFSMSrmm PARMLIB member. This function is dependent on the correct VLPOOL definitions, but you can decide which pool to select by using the EDGUX100 installation exit.

DFSMSrmm intercepts the mount messages issued on the system and, for those for nonspecific volume requests, calls the installation exit to enable it to make a decision. DFSMSrmm saves the scratch pool selected by the exit for use in updating tape drive displays, updating the mount message, and validating the volume when one is mounted.

The exit can optionally request that DFSMSrmm prevent the cartridge loader from being indexed. Thus drives can be selectively preloaded with scratch volumes from specific pools, but scratch volumes from other pools can be manually mounted without the cartridge loader being emptied.

When the exit selects a pool, the only acceptable volumes are those in that pool. When DFSMSrmm selects the scratch pool without use of the exit, any scratch volume from any eligible scratch pool on that system is acceptable.

DFSMSrmm provides a working sample exit that implements a scratch pooling system based on job names and data set names; all you have to do is update a table containing job names, data set names, and pool prefixes in the EDGUX100 exit.

You can also specify different data set or job name pool tables for each system. If you specify the NAME operand in the VLPOOL command in PARMLIB member EDGRMMxx, the tape drive will display this name rather than the pool prefix. The specification allows DFSMSrmm to drive BTLS scratch pool selection. You can use data set names as well as job names to direct scratch pool selection.

8.4.4.6 Interfaces

Before the conversion process you must consider some information about interfaces.

SMS Interface: If you have an IBM 3494 or 3495 Automated Tape Library Dataserver, you must ensure that it works correctly in the DFSMSrmm environment. You must install the DFSMSrmm supplied OAM exits and provide the correct input to the extract program for those volumes that belong to the tape library.

You also must define an DFSMSrmm location with the same name as the tape library name, using a location type of ATL.

Basic Tape Library Support Interface: If your installation uses BTLS to drive IBM 3494 and/or 3495 Automated Tape Library Dataserver operations, you will need to notify BTLS when a volume has to return to SCRATCH status. In DFSMSrmm you can use a simple REXX CLIST that synchronizes the DFSMSrmm database with the BTLS catalog.

You can find a sample CLIST in Appendix D, “Sample Exits and Reports” on page 603.

Short-on-Scratch Processing Interface: DFSMSrmm offers an integrated SMS interface that updates the TCDB automatically.

In addition, without manual intervention, you can free volumes that are in *PENDING RELEASE* status when a short-on-scratch condition is detected inside the automated tape library dataserer.

Use the PARMLIB option, SCRATCHPROC(RMMSCR), where RMMSCR is the name of the procedure DFSMSrmm starts to replenish scratch volumes in the automated tape library dataserer.

TSO and ISPF Interfaces: DFSMSrmm has a simple TSO interface that uses a set of powerful commands to drive all tape activities.

There is also an ISPF interface that offers menus for end users, librarians, administrators, and support personnel. You can choose a full DFSMSrmm dialog or a local defined dialog.

You can tailor the appropriate ISPF dialog for each user who will have access to DFSMSrmm.

DFSMSrmm Application Programming Interface (API): Your application program can communicate with a DFSMSrmm Application Programming Interface (API) through a general-use programming interface. Use the DFSMSrmm API to read, extract, and update data in the DFSMSrmm control data set. You can use the data to create reports or implement automation.

The main benefits of the DFSMSrmm Application Programming Interface are:

- You use EDGXCI to define a parameter list for communicating between your user application program and DFSMSrmm.
- You call the DFSMSrmm Application Programming Interface module using EDGXCI.
- You receive data in either line format or field format as structured fields in your output buffer.
- All outputs are identified using a structured field format: Structured Field Introducer (SFI) and the output data.

EDGXCI Call DFSMSrmm Interface: You use the EDGXCI macro in your application program to set parameters in the list, to modify parameters in the list, and to call the DFSMSrmm API module, EDGXAPI. The module EDGXAPI and the macros EDGXCI and EDGXSF, which define structured fields, are general-use programming interfaces between your application program and DFSMSrmm.

Other Interfaces: DFSMSrmm can use any DFSMSrmm command in batch mode using the IKJEFT01 or IKJEFT1A TSO batch interface.

8.5 Running the Conversion Procedures

In this section we describe how to transfer ICF catalog information to DFSMSrmm. You can do this using REXX procedures and creating RMM TSO subcommands, you can extract ICF catalog information and create input records for the EDGCNVT program, or you can use an assembler program and call the DFSMSrmm API.

The conversion that we describe is only a sample. You can use the same process when converting from DFSMShsm, or any other manual or owner-written tape management system.

If you are not using the EDGCNVT program for your conversion, it is easier to define all your racks and volumes up front so DFSMSrmm can record all open and close information until the conversion process is running.

For the conversion that we describe, you must define resources to DFSMSrmm in the following sequence using RMM TSO subcommands or by calling the DFSMSrmm API:

1. ADDOWNER
2. ADDRACK
3. ADDVOLUME
4. CHANGEVOLUME (STATUS)
5. ADDDATASET
6. CHANGEVOLUME (PREVVOL).

8.5.1 Add Owner Information

Before you can use the RMM TSO subcommands ADDVOLUME, CHANGEVOLUME, and ADDDATASET you must add an owner. The ADDOWNER subcommand defines an owner to DFSMSrmm. An owner can be an individual or a group defined by a RACF group name, or any other value you choose. Figure 113 on page 384 shows an example to add an owner using the RMM TSO ADDOWNER subcommand in a batch job.

```

//LCLCVTCO JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP1    EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN  DD  *
           RMM ADDOWNER SCHLUM
           DEPT(' INTERNATIONAL PROJECTS (0460)')
           SNAME(' SCHLUMBERGER')
           FNAME(' NORBERT')
           ADDR1(' IBM DEUTSCHLAND INFORMATIONSSYSTEME GMBH')
           ADDR2(' AM KELTENWALD 1')
           ADDR3('71139 EHNINGEN (GERMANY)')
           INTEL('919-3579')
           EXTEL('++49-(0)7034-15-3579')
           USER(DEIBMPVS)
           NODE(IBMMAIL)
/*
//

```

Figure 113. JCL to Execute the RMM TSO ADDOWNER Subcommand

Figure 114 and Figure 115 on page 385 are examples of adding an owner using the ISPF menus.

EDGP0000	DFSMSrmm Owner Menu
OPTION ==> 2	
0	OPTIONS - Specify dialog options and defaults
1	DISPLAY - Display owner information
2	ADD - Add a new owner
3	CHANGE - Change owner information
4	DELETE - Delete an owner
Enter owner identification to be processed:	
Owner id ==>	SCHLUM
Enter selected option or END command. For more information, enter HELP or PF1.	
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Figure 114. ISPF DFSMSrmm Owner Menu Panel


```

EDGP0200          DFSMSrmm Change Owner Details - SCHLUM
COMMAND ==>

Surname    ==> Schlumberger
Forenames  ==> Norbert

Department ==> International Projects (0460)

Address:
Line 1     ==> IBM Deutschland Informationssysteme GmbH
Line 2     ==> Am Keltenwald 1
Line 3     ==> D-71139 Ehningen (Germany)

Telephone:
Internal   ==> 919-3579      External ==> ++49-(0)7034-15-3579

Electronic Mail:
Userid     ==> DEIBMPVS      Node      ==> IBMAIL

Press ENTER to CHANGE Owner Details, or END command to CANCEL.

```

Figure 115. ISPF DFSMSrmm Change Owner Details Menu

8.5.2 Add RACK Information

The next step is to use the RMM TSO ADDRACK subcommand shown in Figure 116 to add racks to DFSMSrmm. The racks must be added before you can add volume information.

```

//LCLCVTC1 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//* *****
//STEP1    EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN  DD  *
//          RMM ADDRACK rack_number COUNT(nnnnnn)
//*
//

```

Figure 116. JCL to Execute the RMM TSO ADDRACK Subcommand

You can also using the ISPF dialog to add the RACK and VOLUME information. Figure 117 on page 386 and Figure 118 on page 386 show how you can use the ISPF dialog to add the RACK information.

Panel Help	
EDGPR000	DFSMSrmm Rack and Bin Menu
Option ==> 2	
0 Options	- Specify dialog options and defaults
1 Display	- Display Racks and Bins
2 Add	- Add new Racks and Bins
3 Delete	- Delete Empty Racks and Bins
4 Search	- Search for Racks and Bins
Enter selected option or END command. For more info., enter HELP or PF1. 5695-DF1 (C) COPYRIGHT IBM CORPORATION 1993	

Figure 117. ISPF DFSMSrmm Rack and Bin Menu Panel

Panel Help	
EDGPR200	DFSMSrmm Add Racks and Bins
Command ==>	
Specify one of:	
Rack number rack	Enter first or only rack number to be added
or	
Bin number	Enter first or only bin number to be added. Leave blank when adding built-in storage location bin numbers.
Location	Specify storage location name. Leave blank to add rack numbers.
Media Name	Required only for adding installation-defined storage location bin numbers.
Count of racks or bins . . . nnnnnn	Specify number of rack or bin numbers to add. Leave blank when adding only one rack or bin number.

Figure 118. ISPF DFSMSrmm Add Racks and Bins Panel

COUNT Specifies how many rack numbers to add to the removable library. The value is 1 to 99999. The default value is 1.

LOCATION Specifies the location where you want to add shelf space. Use *SHELF* to add shelf locations to your removable media library. You can use *SHELF* or any installation defined storage location name defined with the LOCDEF command

definitions in the DFSMSrmm PARMLIB member. Use the LISTCONTROL VLPOOL RMM TSO subcommand to view the pool IDs defined to your installation. The default value is SHELF.

MEDIANAME Defines the media that can reside in a shelf location. When adding bin numbers, the media name must appear in the MEDIANAME value in the PARMLIB LOCDEF parameters for the LOCATION specified or DFSMSrmm rejects the request. MEDIANAME is not specified for adding rack numbers because the media name is obtained from the current VLPOOL definitions.

Use the LISTCONTROL VLPOOL RMM TSO subcommand to view the MEDIANAME defined for the requested LOCATION. The default value is 3480.

8.5.3 Add Volume Information

Now you are ready to define the volume information to the DFSMSrmm CDS. Figure 119 shows sample JCL using the RMM TSO ADDVOLUME subcommand.

```
//LCLCVTC2 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP1    EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN  DD *
//          RMM ADDVOLUME volser          -
//          COUNT(nnnnnn)                 -
//          STATUS(SCRATCH)                -
//          INITIALIZE(N)                  -
//          LOCATION(locname)              -
//          MEDIANAME(media_name)
//*
//
```

Figure 119. JCL to Execute the RMM TSO ADDVOLUME Subcommand

Figure 120 on page 388 and Figure 121 on page 388 show how you can use the ISPF dialog to add the volumes.

Panel Help	
EDGPT000	DFSMSrmm Volume Menu
Option ==>	7
0	OPTIONS - Specify dialog options and defaults
1	DISPLAY - Display volume information
2	ADD - Add a new volume
3	CHANGE - Change volume information
4	RELEASE - Delete or Release a volume
5	SEARCH - Search for volumes
6	REQUEST - Request a volume
7	ADDSCR - Add one or more SCRATCH volumes
8	CONFIRM - Confirm librarian or operator actions
Enter selected option or END command. For more info., enter HELP or PF1.	
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Figure 120. ISPF DFSMSrmm Volume Menu Panel

Panel Help	
EDGPT230	DFSMSrmm Add Scratch Volumes
Command ==>	
Volume	volser Pool
	or
Media name	Rack
	Location name . . .
Count	nnnnnn (Default is 1)
Description	
Assigned date . . .	YYYY/DDD MVS use
Assigned time . . .	VM use
Media type	
Label	(AL, NL or SL)
Density	(1600, 3480, 6250 or *)
Initialize	N (Default is YES)

Figure 121. ISPF DFSMSrmm Add Scratch Volumes Panel

COUNT Specifies how many volumes to be added. The value is 1 to 99999. The default value is 1.

DENSITY	Specifies the volume recording density. For a 3420 tape reel, you can use DENSITY as 1600 or 6250. For a 3480 tape cartridge, use a value of 3480. Use an asterisk if you do not know the density. The default value is an asterisk.
DESCRIPTION	Specifies descriptive text about the volume. Descriptive text is 1 to 30 characters. The default is blanks.
INITIALIZE	Specifies whether the volume must be initialized before it can be used. Specify Y to request initialization. Specify N to indicate that the volume does not need to be initialized. If you use STATUS(SCRATCH), the default is Y.
LABEL	Specifies the volume's label type, which can be SL IBM standard labels. This is the default. NL No labels AL ISP/ANSI/FIPS labels
LOCATION	Specifies the location where you want to add volumes. Use SHELF to add shelf locations to your removable media library. You can name a system-managed library where the volumes reside. The default value is SHELF.
MEDIANAME	Defines the media that can reside in a shelf location. If you add the volume to a pool, the media name you use must match the type of media defined for the pool in the current VLPOOL definitions. The default value is 3480.
MEDIATYPE	Specifies the volume's physical media type. Use one of the following: * The volume is not a cartridge CST Cartridge System Tape ECCST Enhanced Capacity Cartridge System Tape EHPT Extended High Performance Cartridge Tape HPCT High Performance Cartridge Tape The default value is *.
POOL/RACK	If you do not supply a pool prefix or a rack number, DFSMSrmm assigns the volume a rack number matching the volume serial number.
USE	Specifies the operating system where the volume can be used. You can select MVS, VM or both. The default is MVS.

8.5.4 Defining a VRS

You can define retention policies, known as VRSs, to retain data sets or volumes you own based on your needs. These policies override any expiration date associated with the data set or volume.

To specify that DFSMSrmm retain a data set as long as it is cataloged, you must define a VRS with the WHILECATALOG parameter. You can specify a global VRS to retain all your data as long as cataloged. This is the simplest way for a conversion from ICF user catalogs.

Figure 122 on page 390 shows an example of using the RMM TSO ADDVRS subcommand specifying the WHILECATALOG parameter to retain all data sets only while they are cataloged.

```
//LCLCVTC3 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP1    EXEC PGM=IKJEFT01
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN  DD  *
//          RMM ADDVRS DSNAME('**') -
//          DESCRIPTION('Retain all DSN while cataloged') -
//          WHILECATALOG
//
//
```

Figure 122. JCL to Execute the RMM TSO ADDVRS Subcommand

Figure 123 and Figure 124 on page 391 show how you can use the ISPF dialog to add the volumes.

Panel Help

EDGPV200 DFSMSrmm Add Vital Record Specification

Command ==>

Specify one of the following:

Data set mask . . *******

Job name mask . .

Volume serial . . (May be generic)

VRS name

Figure 123. ISPF DFSMSrmm Add Vital Record Specification Panel

Panel Help	

EDGPV210	DFSMSrmm Add Data Set VRS
Command ==>	
Data set mask : ***	GDG . . . NO
Job name mask :	
Count 99999	Retention type CYCLES
	While cataloged YES
Delay 0 Days	Until expired NO
Location HOME	
Number in location . 99999	
Priority 0	
Next VRS in chain . .	Release options:
Chain using . . .	Expiry date ignore NO
	Scratch immediate NO
Owner SCHLUM	
Description . . .	
Delete date . . . 1999/365	(YYYY/DDD)
Press ENTER to ADD the VRS, or END command to CANCEL.	

Figure 124. ISPF DFSMSrmm Add Data Set VRS Panel

COUNT	Specifies a retention amount, based on the retention type chosen: number of days or cycles for data sets, number of volumes for volumes. The COUNT default is 99999.
CYCLES	Specifies that DFSMSrmm retains data sets based on cycles or copies of a data set. CYCLES is the default for data set VRSS.
DELETEDATE	Specifies the date when DFSMSrmm deletes the VRS. The default value is 1999/365, which indicates that DFSMSrmm never deletes the vital record specification.
DSNAME	Identifies the type of VRS and gives a data set name for the VRS. There is no default.
LOCATION	Specifies a location where the volume should be retained. The default value is HOME.
GDG/NOGDG	Specifies that a data set name is a GDG base name. DFSMSrmm checks for the standard version qualifier GnnnnVnn. The default value is NOGDG.
STORENUMBER	Specifies how many data set cycles to retain. The range is 1-99999, where 99999 indicates that all remaining data sets are to be retained. The default value is 99999.
WHILECATALOG	Specifies that DFSMSrmm retains the data set only as long as it is cataloged. There is no default.

8.5.5 IDCAMS LISTCAT

Before you start your conversion from ICF user catalog to DFSMSrmm, find out how many ICF user catalogs are available on your system. Run the IDCAMS command LISTCAT UCAT on all of your available systems to get a complete list of your ICF user catalogs. All the ICF user catalogs will be input for the conversion procedure. Figure 125 on page 392 is a sample IDCAMS LISTCAT job.

```

//LCLCVTCO JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP1    EXEC PGM=IDCAMS
//SYSPRINT DD  SYSOUT=*
//SYSIN     DD  *
DELETE RMM.UK.CUSTOMER.UCAT NONVSAM SCRATCH
SET MACC=0
/*
//STEP2    EXEC PGM=IDCAMS
//SYSPRINT DD  DISP=(,CATLG),DSN=RMM.UK.CUSTOMER.UCAT,
//            SPACE=(TRK,(500,50),RLSE),
//            DCB=(RECFM=VBA,LRECL=125,BLKSIZE=6129)
//SYSIN     DD  *
LISTCAT UCAT
/*

```

Figure 125. Sample IDCAMS JCL to LIST User Catalogs

The sample JCL in Figure 126 on page 393 lists all the fields for each catalog entry in the specified ICF user catalogs. The sequential output file specified on the SYSPRINT DD statement is input for your conversion procedures.


```

//LCLCVTC3 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP1    EXEC PGM=IDCAMS
//SYSPRINT DD  SYSOUT=*
//SYSIN     DD  *
DELETE RMM.UK.CUSTOMER.LISTCAT NONVSAM SCRATCH
/*
//STEP2    EXEC PGM=IDCAMS
//SYSPRINT DD  DISP=(,CATLG),DSN=RMM.UK.CUSTOMER.LISTCAT,
//          SPACE=(TRK,(500,50),RLSE),
//          DCB=(RECFM=VBA,LRECL=125,BLKSIZE=6129)
//SYSIN     DD  *
LISTCAT CAT('ICF.TS000A') ALL
LISTCAT CAT('ICF.TS000B') ALL
LISTCAT CAT('ICF.USERCAT') ALL
LISTCAT CAT('ICF.MCAT') ALL
...
...
/*

```

Figure 126. Sample IDCAMS LISTCAT CATALOG JCL

8.5.6 Pre-Allocating Data Sets for Inventory Management

To check if a volume is predefined and O/C/EOV recorded, you must create a DFSMSrmm extract data set by using the EDGHSKP utility. You can also check this information in your conversion procedures using the RMM TSO subcommand or the API *LISTVOLUME volser* command for each volume.

Before running EDGHSKP, you need to define several data sets. At this point you need the MESSAGE and REPTTEXT data sets. These data sets must be predefined and cataloged. Figure 127 on page 394 contains sample JCL to pre-allocate the MESSAGE and REPTTEXT data sets.

```

//LCLCVTC4 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP01 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE RMM.UK.CUSTOMER.EXTRACT NONVSAM SCRATCH
DELETE RMM.UK.CUSTOMER.MESSAGE NONVSAM SCRATCH
/*
//STEP02 EXEC PGM=IEFBR14
//EXTRACT DD DISP=(,CATLG),DSN=RMM.UK.CUSTOMER.EXTRACT,
//          UNIT=SYSDA,
//          DCB=(LRECL=719,BLKSIZE=23476,RECFM=VB),
//          SPACE=(TRK,(500,50),RLSE)
//MESSAGE DD DISP=(,CATLG),DSN=RMM.UK.CUSTOMER.MESSAGE,
//          UNIT=SYSDA,
//          DCB=(LRECL=133,BLKSIZE=6118,RECFM=FBA),
//          SPACE=(TRK,(1,1))
//

```

Figure 127. Sample JCL to Pre-Allocate the Housekeeping Data Sets

8.5.7 Creating an Extract Data Set

The report extract data sets contain the extract copy of the DFSMSrmm CDS. DFSMSrmm provides the format of the records in the extract data set in mapping macros. See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information.

Figure 128 on page 395 shows the JCL to create an extract data set. The second step copies the MESSAGE data set into the joblog.

```

//LCLCVTC5 JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//*
//* IBM Deutschland Informationssysteme GmbH
//* SN Internationale Projekte (Dept. 0460)
//* Am Keltenwald 1
//* 71139 Ehningen (Germany)
//*
//* *****
//STEP01 EXEC PGM=EDGHSKP,PARM='RPTXT'
//SYSPRINT DD SYSOUT=*
//MESSAGE DD DSN=RMM.UK.CUSTOMER.MESSAGE,DISP=SHR
//REPTXT DD DSN=RMM.UK.CUSTOMER.EXTRACT,DISP=SHR
//* *****
//STEP02 EXEC PGM=IEBGENER
//SYSPRINT DD DUMMY
//SYSUT1 DD DSN=RMM.UK.CUSTOMER.MESSAGE,DISP=SHR
//SYSUT2 DD SYSOUT=*
//SYSIN DD DUMMY

```

Figure 128. Sample JCL to Copy MESSAGE Data Set

8.5.8 Temporary Sequential Data Set

Before you can create RMM TSO subcommands you have to create a sequential file containing the ICF catalog information, and then sort the input in the correct sequence. From the ICF catalog you get the following information:

- Data set name
- Volume serial number
- Create date
- Expiration date
- Management class

Once you have the temporary file created, you must sort the data in the following sequence:

1. Ascending by volume serial number
2. Ascending by data set sequence number
3. Descending by data set creation date

8.5.9 Using RMM TSO Subcommands for Conversion

Continue your conversion by reading your sorted temporary file and use the RMM TSO subcommands or the API to convert your information to DFSMSrmm. Until the ICF catalog to DFSMSrmm conversion is complete, you can check if a volume is O/C/EOV recorded or not and suppress the conversion for this information. You should record the information for all skipped records.

In this conversion, we will change the volume status from *SCRATCH* to *USER*, add the available data set information, and if available, build chains for multi-volume data sets.

To do this, you can use RMM TSO subcommands or the DFSMSrmm API to issue the following commands:

```
CHANGEVOLUME  
ADDDATASET
```

```
/* Schlumberger at IBM Germany
```

Figure 129 on page 397 shows a sample REXX procedure to create TSO RMM subcommands and write them into a sequential file. You can execute the commands in that sequential file at a later time.

```

/* REXX to convert a manual TMC to DFSMSrmm                                */
/*                                                                           */
/* add here your own REXX code                                           */
/*                                                                           */

...
"EXECIO" i_cnt "DISKR EXTRACT (STEM IN." /* read extract file */
lclcvtcg = "SHELF"
o_cnt = 0
...

if vol_rec = 'not found' then do
o_cnt = o_cnt + 1
out.o_cnt = "RMM AR "||vol||" LOCATION(SHELF)"
o_cnt = o_cnt + 1
out.o_cnt = "RMM AV "||vol||" STATUS(SCRATCH) INIT(N)",
"LOCATION("||lclcvtcg||")"
end
if file_seq eq '1' then do
if xvol eq ' ' then do
o_cnt = o_cnt + 1
out.o_cnt = "RMM CV "||vol||" STATUS("||lclcvtca||")",
"ASDATE("||crd||") RELEASEACTION("||lclcvtcf||")"
end
else do
o_cnt = o_cnt + 1
out.o_cnt = "RMM CV "||vol||" STATUS("||lclcvtca||")",
"PREVVOL("||xvol||") ASDATE("||crd||")",
"RELEASEACTION("||lclcvtcf||")"
end
end
else do
if file_seq ne save_file_seq then do
o_cnt = o_cnt + 1
out.o_cnt = "RMM AD 'DUMMY.FILE' ||pseq||".MISSING'",
"VOLUME("||vol||") SEQ("||pseq||")",
"LABELNUMBER("||pseq||")"
end
o_cnt = o_cnt + 1
out.o_cnt = "RMM AD '"||dsn||"' CRDATE("||crd||")",
"VOLUME("||vol||") SEQ("||pseq||")",
"LABELNUMBER("||pseq||")"
end

...

"EXECIO" o_cnt "DISKW RMMCMDS (STEM OUT. FINIS"
exit max_return_code

```

Figure 129. Sample REXX EXEC to Write RMM TSO subcommands to a File

Figure 130 on page 398 shows a sample assembler program that uses the DFSMSrmm API to read and store information in the DFSMSrmm CDS.

```

YOURPGM  CSECT
R0      EQU  0
R1      EQU  1
R3      EQU  3
R4      EQU  4
R9      EQU  9
R11     EQU  11
R12     EQU  12
R13     EQU  13
R15     EQU  15
*
    ..
    USING *,R11
    USING WORKDS,R12
    LA    R13,REGSAVE      Point to register save area
*
    ..
*
    LA    R0,OUTBUFVK      Save the
    ST    R0,APIOUTB@      address of output buffer
*****
*      Load the API module      **
*****
    LOAD  EP=EDGXAPI
    ST    R0,APIMOD@      Save API module address
*
    ..
    XC    MYTOKEN,MYTOKEN  Ensure no token yet
    LA    R4,LISTV@      List volume subcmd address
    BAL   R9,BEGINCMD     Begin the command
*
    ..
*****
*      Going to reuse the resources, instead of releasing**
*      resources obtained by the API for the 1st BEGIN  **
*****
    LA    R4,SEARCHD@      Search subcmd address
    BAL   R9,BEGINCMD     Begin the command
*
    ..
    BAL   R9,MOREDATA      Get more data for search
*
    ..
    BAL   R9,RELEASE      All done, release resources
*
    ..
*****
*      Delete the API module      **
*****
    DELETE EP=EDGXAPI
*
    ..
*****
*      Call API to begin a new subcommand      **
*****
BEGINCMD DS    OH
CALL1    EDGXCI  MF=(E,MYPL),PLISTVER=MAX,          X
          APIADR=APIMOD@,OPERATION=BEGIN,          X
          TOKEN=MYTOKEN,                          X
          SUBCMDADDR=(R4),OUTBUFADDR=(R3)
          BR     R9      RETURN

```

Figure 130 (Part 1 of 2). Sample Assembler Program Using the DFSMSrmm API

```

*****
*      Call API to get more data for current subcommand  **
*****
MOREDATA DS    OH
CALL2     EDGXCI MF=(E,MYPL,NOCHECK),PLISTVER=MAX,          X
              OPERATION=CONTINUE,TOKEN=MYTOKEN
              BR    R9          RETURN
*****
*      Call API to release resources such as storage and **
*      loaded modules                                     **
*****
RELEASE DS    OH
CALL2     EDGXCI MF=(E,MYPL,NOCHECK),PLISTVER=MAX,          X
              OPERATION=RELEASE,TOKEN=MYTOKEN
              BR    R9          RETURN
*****
*      SEARCHDATASET subcommand                          **
*****
SEARCHD  DS    OC
          DC    AL2(SEARCHDL)
          DC    C' SEARCHDATASET ..... '
SEARCHDL EQU    *-SEARCHD
SEARCHD@ DC    A(SEARCHD)
*****
*      LISTVOLUME subcommand                             **
*****
LISTV    DS    OC
          DC    AL2(SEARCHDL)
          DC    C' LISTVOLUME ..... '
LISTVL   EQU    *-LISTV
LISTV@   DC    A(LISTV)
*        ..
*****
*      Program work area                                **
*****
WORKDS   DSECT
APIOUTB@ DS    A          Pointer to output buffer
APIMOD@  DS    A          Address of the API module
REGSAVE  DS    18F        Save area
MYTOKEN  DS    CL4        Token from the API
*****
*      Parameter list definition                         **
*****
          EDGXCI MF=(L,MYPL,OD),PLISTVER=MAX      PLIST area
          DS    OD
OUTBUF@  DS    CL4096      Output buffer area
*****
*      Structured field definitions                     **
*****
SFDEFDS  DSECT
          EDGXSf
          END

```

Figure 130 (Part 2 of 2). Sample Assembler Program Using the DFSMSrmm API

Note: Macro continuation characters need to be in column 72. The continuation lines in this example are shortened for formatting purposes.

If the data set sequence is one, you can use the `CHANGEVOLUME` command to store all data set information. If the data set sequence is greater than one, you must use the `ADDDATASET` command.

When you add a data set, the preceding data set on the volume must already be defined. If the preceding data set is missing, add a dummy file named

```
DUMMY.FILEsequence_number.MISSING
```

or

```
DUMMY.LBLsequence_number.MISSING
```

so you can add your information.

Data sets preceding the data set you want to add must be already defined to DFSMSrmm, or your request fails.

If you have a duplicate-file-on-volume condition, check and correct your input file and rerun the job.

8.5.10 Executing the RMM TSO Subcommands

The last step is to execute the commands, if you have not done it directly in your REXX exec by using the RMM TSO subcommands or by your program by using the API.

8.6 Am I Ready to Continue?

In this chapter we have covered all of the activities that you should have considered before converting from ICF catalogs to DFSMSrmm. After completing the activities in this chapter, you should have the following valid and tested input to load into the DFSMSrmm CDS:

- Input to exits
- Correct PARMLIB options
- Housekeeping jobs to use in a DFSMSrmm environment
- Changed reports
- Data from the ICF catalogs
- DFSMSrmm commands to create the CDS
- Extracted data from the ICF catalogs and prepared input records to EDGCNVT

If you have any error return codes, read the error messages to determine the actions required to correct the errors. Then rerun the conversion process (see 8.5.1, “Add Owner Information” on page 383).

To have a valid DFSMSrmm CDS, you must stop all tape activity from the time you get the data from the ICF catalogs until you start DFSMSrmm in record-only mode.

Before producing a valid conversion, you can test the extraction and conversion process, so you do not have to stop the tape activities every time you find that you have to go back and rerun the extract jobs.

Once you have tested the environment you should have the correct tailored input to the REXX procedures.

You can now stop tape activities and produce a valid data extraction from the ICF catalogs.

If your previous tests were satisfactory, you should obtain the input to the EDGCNVT program and proceed to the next step. If you use RMM TSO subcommands, continue to Chapter 10, “Parallel Running and Validation” on page 427.

Chapter 9. Building the DFSMSrmm CDS

In this chapter we describe the DFSMSrmm EDGCNVT program and how to use it to build a DFSMSrmm CDS. The input to the EDGCNVT program consists of records that describe the volumes, data sets, and policies you want to convert to DFSMSrmm. In addition, you can control EDGCNVT program processing by using SYSIN control statements that can edit the input records.

The output of the EDGCNVT program includes:

- Records ready for the DFSMSrmm CDS
- DFSMSrmm ADDVRS subcommands that interpret the processed policies
- Control statements for use with the EDGUTIL utility
- Messages that describe processing problems and list statistics

Objectives of Chapter

- Explain how to use the EDGCNVT conversion program
- Convert extracted data
- Build a ready-to-use DFSMSrmm CDS containing converted data

Audience

- System programmers
- Storage administrators

At the end of this chapter you will have built a DFSMSrmm CDS ready for use in any of the following environments:

- Testing
- Parallel running
- Cutover to production

9.1 Preparing to Run EDGCNVT

Once you have the extracted data from the vendor database or any other data source, you must use the EDGCNVT program to convert the data to a format usable in the DFSMSrmm CDS.

Before running EDGCNVT, check your environment for:

- Migration of system-managed tape volumes
- Different pool prefixes
- The best conversion method for VRS policies

9.1.1 System-Managed Volumes

The EDGCNVT input records do not contain information to indicate whether a location is a system-managed library or a volume is system-managed. At the time you extracted data from your existing tape management system you may have supplied a location name for every volume belonging to an automated tape library. However, you probably do not have a specific location name for system-managed libraries in your previous tape management system.

You must supply the specific location information by using EDGCNVT control statements. There are three options:

- LOCDEF

Use this statement to define a location as being system-managed and whether it is an IBM 3494 or IBM 3495 Automated Tape Library (ATL) or a IBM 3495-M10 Manual Tape Library (MTL)

- IF VOLRANGE

Use this statement to identify the volumes that are system-managed. If you know the range of volumes that are inside the system-managed library or have the system-managed library as a home location, you can assign a specific location name to this range. You only need to do this if the input records do not already have the correct location and home location names.

- IF STORLOC

If you have a specific location name for the volumes belonging to the library and it is different from the name to be used under DFSMSrmm, you can translate it to the location name that you plan to use in the DFSMSrmm environment.

For an explanation of these statements, see Section 9.2, “EDGCNVT Processing” on page 405.

Remember that in a system-managed DFSMSrmm environment, the location name you specify for the tape library must match the name of the library defined to SMS.

9.1.2 Pool Prefixes

If you plan to move your volumes using media name criteria, you must ensure that you have correctly supplied media names in the extraction process by using the VOLMEDIA statement for TLMS migration and the hard coded values for CA-1.

You can optionally use the IF VMEDIA control statement of the EDGCNVT program to translate the media names in the input records to the values you plan to use with DFSMSrmm. For an explanation of these statements see Section 9.2, “EDGCNVT Processing” on page 405. Ensure that the media names you use match the values specified on the DFSMSrmm PARMLIB LOCDEF and VLPOOL options.

9.1.3 VRS Conversion

When you run EDGCNVT specifying the VRSCMDS DD statement, all ADDVRS commands are built from the input K-Records to match the retention and movement criteria. If you are not satisfied with the VRS records built by the combination of extract programs and EDGCNVT program, you can edit the commands. This could be an ideal opportunity to simplify your retention and movement policies to exploit the DFSMSrmm generic data set name mask capabilities.

If you edit the commands, you can execute the command stream once you have loaded the remaining data created by EDGCNVT into an DFSMSrmm CDS. Remember to remove the VRSLIST file from the EDGJLOAD job so that the VRS records created by EDGCNVT do not end up in the DFSMSrmm CDS.

See Sections 9.5.2, “Adding VRSs” on page 424 and 13.2.1, “Reduce VRS Policies” on page 505 for additional information on the VRS conversion process.

9.1.4 Movement Tracking Date

The data set primary VRS subchain start date for all data sets on a volume is set from the volume movement tracking date. The movement tracking date is currently set from the CA-1 outcode date (date assigned to a vault). EDGHSKP VRSEL processing uses this date and the volume current location to determine whether and how to process EXTRADAYS VRSS on the first VRSEL run after conversion.

9.2 EDGCNVT Processing

EDGCNVT is a batch program that takes the records from the input data set, reformats them, and makes them ready for the DFSMSrmm CDS. The input records are usually the records you created by running one of the sample extract processes, but you can construct the records using any technique or tool that you prefer.

As each record is read, it is passed to the EDGCNVT user exit, if you have supplied one, where it can optionally be updated or skipped. Upon return from the user exit, EDGCNVT compares each record encountered for applicability to the control statements in the SYSIN data set and modifies the records accordingly. It then generates the output records and data sets, which are ready for postprocessing and ultimately entered into the DFSMSrmm CDS.

The format of the EDGCNVT input records can be found in Appendix C, "Conversion Record Layouts" on page 589. The SYSIN data set record format can be found in Section 9.2.2.2, "SYSIN Control Statements" on page 408.

9.2.1 Running EDGCNVT

Figure 131 on page 406 shows the sample JCL for execution of EDGCNVT.

We describe the input and output files in the sections that follow.

```

//CONVERT JOB ....
//*
//STEP1 EXEC PGM=EDGCNVT,REGION=6M
//STEPLIB DD DISP=SHR,DSN=STSGWD.RMM.LINKLIB
//SYSPRINT DD SYSOUT=*
//SYSOUT DD DISP=(,CATLG),LRECL=80,RECFM=FB,AVGREC=K,
//          SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.SYSOUT.DATA
//VRSCMDS DD DISP=(,CATLG),LRECL=80,RECFM=FB,AVGREC=K,
//          SPACE=(80,(1,1),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.VRSCMDS.DATA
//DEXTRCT DD DISP=SHR,DSN=STSGWD.EDGCXLDL.DEXTOUT.DATA
//          DD DISP=SHR,DSN=STSGWD.EDGCSRDS.DEXTOUT.DATA
//          DD DISP=SHR,DSN=STSGWD.EDGCSVDS.DEXTOUT.DATA
//          DD DISP=SHR,DSN=STSGWD.EDGCXBIN.DEXTOUT.DATA
//LIBLIST DD DISP=(,CATLG),LRECL=1000,RECFM=VB,AVGREC=K,
//          SPACE=(1000,(70,10),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.LIBLIST.DATA
//OWNLIST DD DISP=(,CATLG),LRECL=400,RECFM=VB,AVGREC=K,
//          SPACE=(400,(1,1),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.OWNLIST.DATA
//BINLIST DD DISP=(,CATLG),LRECL=256,RECFM=VB,AVGREC=K,
//          SPACE=(256,(20,10),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.BINLIST.DATA
//VRSLIST DD DISP=(,CATLG),LRECL=256,RECFM=VB,AVGREC=K,
//          SPACE=(256,(20,10),RLSE),UNIT=SYSDA,
//          DSN=STSGWD.EDGCNVT.VRSLIST.DATA
//SYSIN DD *
IF OWNER EQUALS ' ' THEN DEFOWNER EQUALS 'USER001 '
LOCDEF MTLNAME MTL
LOCDEF ATLNAME ATL
IF VOLRANGE IS 200000 TO 205000 THEN SET LOCATION ATLNAME
IF VMEDIA EQUALS CART3480 THEN VLPOOLNAME EQUALS CARTS
IF VMEDIA EQUALS CART3490 THEN VLPOOLNAME EQUALS CARTS
IF STORLOC EQUALS LIB THEN RMMSTORE EQUALS SHELF
IF STORLOC EQUALS LIBR THEN RMMSTORE EQUALS SHELF
/*

```

Figure 131. Sample EDGCNVT Execution JCL

The sample JCL to run the EDGCNVT program is provided in SAMPLIB member EDGJCNT.

You should modify the data set names and data set attributes and specify the SYSIN control statements you need before using the sample jobstream.

9.2.1.1 Execution Parameter

EDGCNVT has a single, optional execution parameter (Table 91).

Table 91. EDGCNVT Execution Parameters

Parameter	Description
nnnn	A number of one or more digits that specifies the maximum number of LOCDEF, VOLRANGE or VOLPRE statements to support. The default value is 500 of each type. Specify a value if you want to increase the limit.

9.2.2 EDGCNVT Input Files

Table 92 lists the input files used by EDGCNVT.

<i>Table 92. EDGCNVT Input Files</i>	
ddname	Description
DEXTRCT	This file contains the records that are to be converted to DFSMSrmm format. See Section 9.2.2.1, “DEXTRCT” for information about the input records.
SYSIN	The SYSIN control statements describe the processing that you would like EDGCNVT to perform on the records to be converted. See Section 9.2.2.2, “SYSIN Control Statements” on page 408 for information about the control statements.

9.2.2.1 DEXTRCT

The DEXTRCT file contains the records that are input to EDGCNVT for conversion to DFSMSrmm record format. You can either use the DFSMSrmm-supplied sample extract programs to create the input records or generate them using any tool or method you prefer. This record-based interface provides you with a clearly defined way to convert volume tape resource data into a format supported by DFSMSrmm.

There are five different records that describe input to EDGCNVT:

Record Type	Description
--------------------	--------------------

- | | |
|----------|--|
| L | Library. An L-Record contains details of a single volume and the first data set on that volume. |
| D | Data set name. A D-Record contains details of a single data set. One D-Record is required for each additional data set on a volume. If you supply a D-Record for a file, all previous files on the same volume must be specified in D-Records that precede the record. |

The L-Records and D-Records have the following sequence requirements:

- Library (L) records for a multivolume set of volumes must appear in ascending volume sequence order, interrupted only by:
 - Data set name (D) records, in ascending file sequence order, after each L-Record that is created for the volumes from which the data set originates.
- The count of D-Records that follow after a nonscratch L-record must be one less than the count of data sets in the LVDSNCNT field of that L-Record, because each L-Record contains first file data set information.

- | | |
|----------|---|
| E | Empty rack and bin. The E-Record contains details of a single, empty shelf location. It can describe either a rack number or a bin number. |
| O | Owner. The O-Record contains details of a single owner ID. The owner may own resources that are being converted, but it need not own any resources. |
| K | VRS. The K-Record contains details of a single policy definition. |

The EDGCNVT program input records have the following content requirements:

- The intention is that you will use the DFSMSrmm installationwide exit, EDGUX100, to check the JCL-specified EXPDT value. You should also use that exit to check for a special date of 98000 (other vendor product's "ignore volume"). When the date is found, the exit requests, through the parameter list, that DFSMSrmm ignore the volume. The implication for migration, and therefore for extract processing, is that no volumes with identical VOLSERs can be input to EDGCNVT.
- Data sets that are retained by special keyword dates in other tape management systems and are to be associated with a special expiration date VRS management value must either:
 - Contain the VRS management value in either the D-Record DSVRSVAL field or the L-Record LDVRSVAL field in the extract program output data set
 - or
 - Have the VRS management value satisfied through the user exit (see Section 9.2.5, "EDGCNVT User Exit" on page 418) by updating the LDVRSVAL field in an L-Record or the DSVRSVAL in a D-Record.
- The security level of data sets and volumes should be specified through either the appropriate fields in the L-Records and D-Records or a series of policy and rules statements in SYSIN. The values for security level should be the same as those established through PARMLIB.

9.2.2.2 SYSIN Control Statements

EDGCNVT accepts a series of statements from SYSIN in the format shown in Figure 132.

```
IF fieldname1 EQUALS fieldname2 THEN fieldnamex EQUALS fieldnamey
      or
      valuexxxxx                      or
      valueyyyyy
```

Figure 132. Convert Program VRS Input Record Format

In all SYSIN control statements the position of identification fields and values is critical. You must specify the field names and values in the exact column that is identified for each type of control statement.

EDGCNVT Record Create Date, Time, and System Identification: EDGCNVT accepts three control statements as shown in Figure 133 on page 409. If these statements are not included, EDGCNVT acquires the values from the operating system at the time of each execution. It uses these values to date, time stamp, and system identify the output records.


```

RMMCRD EQUALS yyyyddd
RMMCRT EQUALS hhmsst
RMMCRS EQUALS systemid
*              *
1              15

```

Figure 133. EDGCNVT Record Create Date, Time, and System Identification

The control statement must start in column 1, and the values must start in column 15.

yyyyddd

Is the creation date used on all records converted by EDGCNVT.

hhmsst

Is the creation time used on all records converted by EDGCNVT. The time is specified as hh - hours, mm - minutes, ss - seconds, and t - tenths of seconds.

systemid

This is the system identifier used to identify the system on which a record is created. The value you specify should match the value you specify on the DFSMSrmm PARMLIB OPTION SYSID operand.

Assigning a Default Release Action: Use the statement in Figure 134 to assign a release action to any volume that does not have a release action specified.

This is the only IF RETACTN statement that is valid.

```

IF RETACTN EQUALS '00' THEN RETACTN EQUALS SCRPOOL
*              *              *
1              20              44

```

Figure 134. RETACTN Release Action

The keywords must be in the columns indicated.

Keyword	Description
00	A hexadecimal value that indicates there are no release actions.
SCRPOOL	Requests that return to scratch release action be assigned to all volumes that have no release action.

Translating Location Names: Use this statement to translate the name of locations in L-, E-, and K-Records in the DEXTRCT file to values that you plan to use with DFSMSrmm. Only do this if the location names are to be different.

If, for example, your current tape management system refers to the tape library by the name TLIBRARY, the statement shown in Figure 135 on page 410 will change that name to SHELF, as DFSMSrmm requires. If your current tape management system uses OFFSITE1, OFFSITE2, or OFFSITE3, the statement shown in Figure 135 on page 410 will change the names to LOCAL, DISTANT, and REMOTE, respectively.


```
IF STORLOC EQUALS IRONMTN THEN RMMSTORE EQUALS REMOTE
```

Figure 139. IF STORLOC Remote Example

Specifying Security Levels: If you would like to establish a security level for data sets that meet a specific naming convention, use the statement shown in Figure 140 as a sample.

```
IF DSN EQUALS SYS1.IC.IC.**          THEN SECLEVEL EQUALS 011
*          *                          *
4          15                        60
```

Figure 140. IF DSN SYSIN Statement Syntax

Note: Only asterisks (*) are supported as filters, and they must be last in the data set name or VOLSER.

The security level number assigned must be a three-digit number and have a value between 0 and 255 as defined by the DFSMSrmm SECLEV option in PARMLIB.

If you would like to establish a security level for volumes that meet a specific naming convention, use the statement shown in Figure 141 as a sample.

```
IF VOL EQUALS SY** THEN SECLEVEL EQUALS 009
*          *          *
4          15          41
```

Figure 141. IF VOL SYSIN Statement Syntax

Specifying and Translating Owner Information: If, as the EDGCNVT program is executing, you would like to have owner information added to those volumes and data sets that currently do not have owner information use the example shown in Figure 142 , with your default values beginning in column 49.

```
IF OWNER EQUALS '          ' THEN DEFOWNR EQUALS 'library '
IF OWNER EQUALS '          ' THEN DEFOWNS EQUALS 'sur name '
IF OWNER EQUALS '          ' THEN DEFOWNF EQUALS '1st name '
IF OWNER EQUALS '          ' THEN DEFOWND EQUALS 'dept num '
IF OWNER EQUALS '          ' THEN DEFOWNT EQUALS 'int tele'
IF OWNER EQUALS '          ' THEN DEFOWNU EQUALS 'userid '
IF OWNER EQUALS '          ' THEN DEFOWNN EQUALS 'nodeid '
*          *          *
1          18          49
```

Figure 142. IF OWNER SYSIN Statement Syntax for Default Owner

The keywords must be in the columns indicated.

Keyword Description

library 1- to 8-character owner ID to be used as a default value

sur name 1- to 20-character surname or family name for the default owner ID

1st name 1- to 20-character first name for the default owner ID

- dept num** 1- to 30-character department name for the default owner ID
 Note that in Figure 142, the space between the beginning and ending apostrophe is less than 30 characters.
- int tele** 1- to 8-character internal telephone number for the default owner ID
- userid** 1- to 8-character userid for the default owner ID
- nodeid** 1- to 8-character node name for the default owner ID

If, when the EDGCNVT program is executing, you would like to translate one existing owner to a new value, use the example shown in Figure 143 with your new values values beginning in column 49.

```

IF OWNER EQUALS 'ownerid ' THEN NEWOWNR EQUALS 'newown '
IF OWNER EQUALS '      ' THEN NEWOWNS EQUALS 'sur name      '
IF OWNER EQUALS '      ' THEN NEWOWNF EQUALS '1st name      '
IF OWNER EQUALS '      ' THEN NEWOWND EQUALS 'dept num      '
IF OWNER EQUALS '      ' THEN NEWOWNT EQUALS 'int tele'
IF OWNER EQUALS '      ' THEN NEWOWNU EQUALS 'userid '
IF OWNER EQUALS '      ' THEN NEWOWNN EQUALS 'nodeid '
*                *                *
1                18                49

```

Figure 143. IF OWNER SYSIN Statement Syntax for Translating Owner

The keywords must be in the columns indicated.

- | Keyword | Description |
|-----------------|---|
| ownerid | 1- to 8- character owner ID in the input records to be translated |
| newown | 1- to 8-character owner ID to be used instead of ownerid |
| sur name | 1- to 20-character surname or family name for the owner ID |
| 1st name | 1- to 20-character first name for the owner ID |
| dept num | 1- to 30-character department name for the owner ID
Note that in Figure 143, the space between the beginning and ending apostrophe is less than 30 characters. |
| int tele | 1- to 8-character internal telephone number for the owner ID |
| userid | 1- to 8-character user ID for the owner ID. |
| nodeid | 1- to 8-character node name for the owner ID |

Identifying Location Names and Types: Use the LOCDEF statement in Figure 144 to define locations other than the DFSMSrmm built-in storage locations (LOCAL, DISTANT, REMOTE, or SHELF) to the conversion program.

```

LOCDEF loc_name type
*      *      *
1      8      17

```

Figure 144. LOCDEF Statement

The keywords must be in the columns indicated.

Keyword	Description
---------	-------------

loc_name	1- to 8-character location name that will be known to DFSMSrmm after conversion. If the location names in the extract records that are input to EDGCNVT do not match these values, use the IF STORLOC statement to make EDGCNVT perform the translation.										
type	One of LOAN, BINS, NOBINS, MTL, or ATL: <table border="1"> <tbody> <tr> <td>LOAN</td><td>Defines the location as a loan location. If the extract records have the location as a storage location, EDGCNVT clears the assigned bin number field and sets the volume location to its home location and marks the volume on loan.</td></tr> <tr> <td>BINS</td><td>Identifies the location as an installation-defined shelf-managed storage location. The extract records must have set a bin number for the volume.</td></tr> <tr> <td>NOBINS</td><td>Identifies the location as an installation-defined non-shelf-managed storage location. If the extract records have set a bin number for the volume, it is ignored.</td></tr> <tr> <td>ATL</td><td>The location is identified as an automated system-managed library.</td></tr> <tr> <td>MTL</td><td>The location is identified as a manual system-managed library.</td></tr> </tbody> </table>	LOAN	Defines the location as a loan location. If the extract records have the location as a storage location, EDGCNVT clears the assigned bin number field and sets the volume location to its home location and marks the volume on loan.	BINS	Identifies the location as an installation-defined shelf-managed storage location. The extract records must have set a bin number for the volume.	NOBINS	Identifies the location as an installation-defined non-shelf-managed storage location. If the extract records have set a bin number for the volume, it is ignored.	ATL	The location is identified as an automated system-managed library.	MTL	The location is identified as a manual system-managed library.
LOAN	Defines the location as a loan location. If the extract records have the location as a storage location, EDGCNVT clears the assigned bin number field and sets the volume location to its home location and marks the volume on loan.										
BINS	Identifies the location as an installation-defined shelf-managed storage location. The extract records must have set a bin number for the volume.										
NOBINS	Identifies the location as an installation-defined non-shelf-managed storage location. If the extract records have set a bin number for the volume, it is ignored.										
ATL	The location is identified as an automated system-managed library.										
MTL	The location is identified as a manual system-managed library.										

Up to 500 LOCDEF statements can be defined in SYSIN.

Assigning Media Type Using IF VOLPRE: Use the statement shown in Figure 145 to assign a media type to a range of volumes based on the VOLSER prefix and suffix. The media type must be either CST or ECCST.

IF VOLPRE IS ppppp AND VOLSUF IS ssssss TO eeeee THEN SET CST
IF VOLPRE IS ppppp AND VOLSUF IS ssssss TO eeeee THEN SET ECCST
* * * *
1 14 34 44 60

Figure 145. VOLPRE SYSIN Statement

The keywords must be in the columns indicated.

Keyword	Description
---------	-------------

ppppp	VOLSER prefix. Specify from 1 to 5 characters as a prefix.
ssssss	The starting range of a VOLSER suffix. Specify up to 6 characters; the characters must be right aligned, and the total number of prefix and suffix characters must be 6.
eeeeee	The ending range of a VOLSER suffix. Specify up to 6 characters; the characters must be right aligned, and the total number of prefix and suffix characters must be 6.

Up to 500 VOLPRE statements can be defined in SYSIN.

Figure 146 on page 414 shows a sample to set the media type to CST for VOLSERS C00001 to C01000 and ECCST for VOLSERS E00001 to E01000.

```

IF VOLPRE IS C      AND VOLSUF IS 00001 TO 01000 THEN SET CST
IF VOLPRE IS E      AND VOLSUF IS 00001 TO 01000 THEN SET ECCST

```

Figure 146. VOLPRE Example

IF VOLRANGE with LOCATION: Use the VOLRANGE with LOCATION statement in Figure 147 to assign a location name to a set of volumes based on the VOLSER prefix. The location must be defined by using the LOCDEF statement, or it must be a built-in location name.

```

IF VOLRANGE IS fvolser TO lvolser THEN SET LOCATION loc_name
*           *           *           *
1           16          26          51

```

Figure 147. VOLRANGE with LOCATION Statement

The keywords must be in the columns indicated.

Keyword Description

fvolser The first VOLSER in the range. Specify a 1- to 6-character VOLSER.

lvolser The last VOLSER in the range. Specify a 1- to 6-character VOLSER. You can specify the same value as for fvolser to have a single volume range.

loc_name The 1- to 8-character location name that will be known to DFSMSrmm after conversion.

Up to 500 VOLRANGE with LOCATION statements can be defined in SYSIN.

IF VOLRANGE with PREFIX: Use the IF VOLRANGE with PREFIX statement (see Figure 148) to change the rack number that is used for a set of volumes based on the VOLSER. You can change as many characters in the rack number as you want, based on the volume VOLSER.

For example, you have volumes in the 013500—013999 range in a scratch pool, but you also have a scratch pool with a range of 013000-013499. The RMM VLPOOL prefix of 013* does not provide for two separate pools. However, using VOLRANGE with PREFIX, you can change the rack numbers so that the VLPOOL prefixes are different and get RMM to manage two pools:

```

.IF VOLRANGE IS 013000 TO 013496 THEN SET PREFIX P .IF VOLRANGE IS
013497 TO 013999 THEN SET PREFIX S

```

You can still use the significant parts of the VOLSERs to identify volumes, and the rack numbers allow the scratch pool to be identified as well as the VOLSER.

```

IF VOLRANGE IS fvolser TO lvolser THEN SET PREFIX r_pref
*           *           *           *
1           16          26          49

```

Figure 148. VOLRANGE with PREFIX Statement

The keywords must be in the columns indicated.

Keyword Description

fvolser The first VOLSER in the range. Specify a 1- to 6-character VOLSER.

- lvolsr** The last VOLSER in the range. Specify a 1- to 6-character VOLSER. You can specify the same value as for fvolser to have a single volume range.
- r_pref** 1- to 5-character prefix value that is to be substituted in the rack number passed in the L-Record to EDGCNVT.

Up to 500 VOLRANGE with PREFIX statements can be defined in SYSIN.

IF VOLRANGE with MEDIANAME: Use the VOLRANGE with MEDIANAME statement in Figure 149 to assign a media name to a set of volumes based on the VOLSER prefix. The media name should be one that matches those you will define on the VLPOOL and LOCDEF DFSMSrmm PARMLIB options.

```
IF VOLRANGE IS fvolser TO lvolsr THEN SET MEDIANAME med_name
*           *           *           *
1           16          26          52
```

Figure 149. VOLRANGE with MEDIANAME statement

The keywords must be in the columns indicated.

Keyword Description

- fvolser** The first VOLSER in the range. Specify a 1- to 6-character VOLSER.
- lvolsr** The last VOLSER in the range. Specify a 1- to 6-character VOLSER. You can specify the same value as for fvolser to have a single volume range.
- med_name** The 1- to 8-character media name that will be known to DFSMSrmm after conversion.

Up to 500 VOLRANGE with MEDIANAME statements can be defined in SYSIN if no execution parameter is specified. If more are required, you can increase the limit using the optional execution parameter.

Editing the Volume Media Name: Use this statement to change the media name, which is defined for a volume when the EDGCVNT input records contain a media name that you do not want to use.

For example, you have volumes to which the extract programs have given a media name of CART3480 and TAPE6250, but you want to use CARTS and REEL as the media names instead. You would code the statements:

```
IF VMEDIA EQUALS CART3480 THEN VLPOOLNAME EQUALS CARTS
IF VMEDIA EQUALS TAPE6250 THEN VLPOOLNAME EQUALS REEL
```

Figure 150 shows the statements that you can code.

```
IF VMEDIA EQUALS TAPE1600 THEN VLPOOLNAME EQUALS media
IF VMEDIA EQUALS TAPE6250 THEN VLPOOLNAME EQUALS media
IF VMEDIA EQUALS CART3480 THEN VLPOOLNAME EQUALS media
IF VMEDIA EQUALS CART3490 THEN VLPOOLNAME EQUALS media
IF VMEDIA EQUALS CART349E THEN VLPOOLNAME EQUALS media
*           *           *
1           18          50
```

Figure 150. VMEDIA Statement with VLPOOLNAME

Keyword	Description
---------	-------------

Values TAPE1600, TAPE6250, CART3480, CART3490, and CART349E are hard coded values. You can specify only one IF VMEDIA statement for each of these hard coded values. If you specify more than one statement, the last value is used.

IF VMEDIA EQUALS ' '	THEN MEDIANAME EQUALS media
* *	*
1 19	51

The keywords must be in the columns indicated.

Keyword	Description
---------	-------------

You can only specify one IF VMEDIA with MEDIANAME statement to set a default value. If you specify more than one statement, the last value is used.

Table 93 lists the output files created by EDGCNVT.

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9.2.3.1 SYSPRINT

The SYSPRINT file contains the messages generated during EDGCNVT processing. The messages contain statistics, information, and error details. Use the messages to determine whether processing has been successful and to follow up on any nonzero return code.

9.2.3.2 SYSOUT

At the end of processing EDGCNVT writes the EDGUTIL control statement to the SYSOUT file. The control statement is used during EDGUTIL CREATE processing to initialize the DFSMSrmm CDS control record and set the correct numbers for rack records. Figure 152 shows what the file might contain after processing.

```
CONTROL -  
  LBINNO(200) LBINFREE(63) -  
  RACKNO(100213) RACKFREE(10115)
```

Figure 152. Example of SYSOUT Contents

9.2.3.3 VRSCMDS

Each time that EDGCNVT successfully processes a K-Record, in addition to creating an DFSMSrmm VRS record, it can optionally create an ADDVRS subcommand that represents the converted policy (Figure 153).

```
RMM ADDVRS DSN('P$SECU.SG003*.**') NOGDG      +  
  CYCLES                                         +  
  COUNT(    3)                                  +  
  DELAY( 0)                                     +  
  OWNER(AAAAAAAA)                              +  
  DELETE(1999/365)                             +  
  LOCATION(SY      )                           +  
  STORENUM(    2)                               +  
  PRIORITY( 0)                                  +  
  NEXT(VRSN0001)                                +  
  
RMM ADDVRS NAME(VRSN0001)                       +  
  OWNER(AAAAAAAA)                              +  
  DELETE(1999/365)                             +  
  LOCATION(CO      )                           +  
  STORENUM(    1)
```

Figure 153. Example of VRSCMDS Contents

9.2.3.4 LIBLIST

The EDGCNVT input records together with modifications made by either the EDGCNVT policy and rules statements or the user exit, result in output data sets containing the following data:

From L-Records EDGCNVT produces:

- Volume information for all volumes in the library and all store locations
- Data set name information for the first data set on volumes.

From D-Records, EDGCNVT produces data set name records for each data set, after the first data set, that originates on a volume.

The output records are ready for loading into the DFSMSrmm CDS.

9.2.3.5 OWNLIST

From O-Records, EDGCNVT produces owner records for each owner that currently owns or could potentially own one or more volumes in the library or storage locations or VRS records.

9.2.3.6 BINLIST

From E-Records, EDGCNVT produces:

- Rack information for empty racks that exist in the library
- Bin information for empty bins that exist in any store.

From L-Records, EDGCNVT produces:

- Rack information for scratch and in-use volumes in the library
- Storage location bin information for those volumes that reside in any storage location.

9.2.3.7 VRSLIST

From the input K-Records, EDGCNVT produces a VRS record containing details of a single retention and movement policy.

9.2.4 EDGCNVT Return Codes

EDGCNVT sets a return code when it ends. Table 94 describes the EDGCNVT return codes.

Table 94. EDGCNVT Return Codes	
Return Code	Description
0	• Normal processing
4	• Loan location cannot be converted to a shelf-managed storage location. • Bin number 0 is set for a shelf-managed storage location. • The number of D-Records supplied for a volume was greater than the number expected.
8	• A location name was used that was not defined to the EDGCNVT SYSIN file.
1000	• An error occurred that must be corrected before the EDGCNVT program is rerun. See the SYSPRINT file for an explanatory message.

9.2.5 EDGCNVT User Exit

EDGCNXIT, a user exit called from EDGCNVT, can be used at your discretion. There are three calls to the user exit in the EDGCNVT program. The first call is when the exit is loaded and allows the exit to initialize for processing. For example, it can GETMAIN storage, open a file, and prepare for processing. The second call is made each time a record is read from the DEXTRCT file, allowing you to modify the record if necessary. Each record is passed to the exit before EDGCNVT has processed it. Upon return from the exit, processing within EDGCNVT continues, using the updated DEXTRCT record. The third call is made at end of processing to allow the exit to issue any messages or free any getmained storage.

The exit is not part of EDGCNVT, but you can write it to modify the input records before EDGCNVT begins to operate on them. A sample exit is provided by

DFSMSrmm as member EDGCNXIT of SAMPLIB. The macros that map the records passed to EDGCNXIT are described in Appendix C, “Conversion Record Layouts” on page 589.

You would use this exit, for example, to add information that the current tape management system does not provide. A specific case would be the addition of a VRS name (in the LDVRSVAL field of a library record or the DSVRSVAL field of a data set record) to be associated with a tape data set.

Other uses of this exit might be to alter volume ownership by updating the LVOWNER field of L-Records or to alter and/or specify the security level to be associated with a data set by updating the DSNSECLV field of a D-Record and the LVSECLEV field of an L-Record.

9.2.5.1 Input

EDGCNXIT is passed a parameter list. The address of the parameter list is in register 1 on entry. The format of the parameter list is:

Offset	Value
+ 0	Function code. A word containing the function code.
0	Initial call. No record is being passed. Use this exit call to perform any initialization that might be required. The sample exit obtains a work area for use on all exit calls.
4	A record call. The address of a record is passed.
8	Final call. No record is passed. Use this exit call to perform any cleanup necessary. The sample exit frees the work area it obtained.
+ 4	The return code. This is an output field to be set by the exit.
+ 8	The address of the record passed to the exit.
+12	A word you can use in the exit for any purpose. The word is set to zero on the initial call and is never changed by EDGCNVT. You can use it to save a work area address or remember any values you want to have from one exit call to the next.

9.2.5.2 Output

You can use the EDGCNXIT exit to update the contents of the record passed to you by means of the parameter list.

Set the return code in the parameter list to inform EDGCNVT of the processing you want performed:

Return code	Meaning
0	Process the record. The exit may or may not have updated the record
8	Skip this record. If you skip a volume record, remember to skip the data set records that follow it for data sets on the same volume.
16	Invalid parameter list was passed to the exit. This return code causes EDGCNVT processing to end; no more records are processed.

9.3 DFSMSrmm CDS Creation

After the successful execution of the EDGCNVT program, you must run the EDGJLOAD job and create the DFSMSrmm CDS control record to prepare the CDS for parallel running and validation of the conversion process.

9.3.1 Run EDGJLOAD

The SAMPLIB member EDGJLOAD sorts the output records from the EDGCNVT run, sorts the owners generated and removes duplicates, processes the records with the DFSMSrmm EDGCVOVL utility, and finally sorts them using the DFSORT program.

The job allocates the DFSMSrmm CDS and loads the records with an IDCAMS REPRO operation.

Figure 154 on page 421 shows the sample JCL.

```

/*JOBNAME JOB .....
/*
/* $01=          ,110,940422,MWW: make the JCL work
/*
//SORT1 EXEC PGM=SORT,REGION=4M
//NEWOWN DD DISP=(MOD,DELETE),SPACE=(TRK,1),UNIT=SYSDA,
//          DSN=STSGWD.EDGCVOL.INTERM.DATA
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.EDGCNVT.LIBLIST.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(1000,(70,10),RLSE),
//          DSN=&&SORTLIB,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(5,1,CH,D,213,8,CH,A,7,6,CH,A),DYNALLOC
/*
//SORT2 EXEC PGM=SORT,REGION=6M,COND=(0,NE)
//SYSPRINT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.EDGCNVT.OWNLIST.DATA
//SORTOUT DD DSN=&&SORTOWN,
//          DCB=*.SORTIN,
//          UNIT=SYSDA,DISP=(,PASS),
//          SPACE=(400,(1,1),RLSE),AVGREC=K
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(6,8,CH,A),DYNALLOC
SUM FIELDS=NONE          removes duplicates
END
//CVOVL EXEC PGM=EDGCVOL,REGION=4M,COND=(0,NE)
//STEPLIB DD DISP=SHR,DSN=SYS1.LINKLIB
//SYSPRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=*
//CTLOUT DD DISP=(OLD,PASS),DSN=&&SORTLIB
//OWNOUT DD DISP=(OLD,PASS),DSN=&&SORTOWN
//NEWOWN DD DISP=(,PASS),DSN=STSGWD.EDGCVOL.INTERM.DATA,
//          AVGREC=K,SPACE=(1000,(100,30),RLSE),UNIT=SYSDA
//MERGE EXEC PGM=SORT,REGION=4M,COND=(0,NE)
//SYSOUT DD SYSOUT=*
//SORTIN DD DISP=SHR,DSN=STSGWD.EDGCNVT.LIBLIST.DATA,LRECL=2044
//          DD DISP=SHR,DSN=STSGWD.EDGCNVT.BINLIST.DATA
//          DD DISP=SHR,DSN=STSGWD.EDGCVOL.INTERM.DATA
//          DD DISP=SHR,DSN=STSGWD.EDGCNVT.VRSLIST.DATA
//SORTOUT DD DISP=(,PASS),AVGREC=K,SPACE=(1000,(100,30),RLSE),
//          DSN=&&MERGE,UNIT=SYSDA
//SYSIN DD *
SORT FIELDS=(5,56,CH,A),DYNALLOC
/*

```

Figure 154 (Part 1 of 2). Sample EDGJLOAD JCL

```

//DELDEF EXEC PGM=IDCAMS,COND=(0,NE)
//SYSPRINT DD SYSOUT=*
//MASTER DD DISP=SHR,VOL=SER=DFRMMMA,UNIT=3390
//SYSIN DD *
DELETE STSGWD.RMM.CDS
DEFINE CLUSTER (NAME(STSGWD.RMM.CDS)
               FILE(MASTER) -
               FREESPACE(35 20) -
               KEY(56 0) -
               REUSE -
               RECSZ(512 9216) -
               SHR(3 3) -
               VOLUMES(DFRMMMA)) -
DATA (NAME(STSGWD.RMM.CDS.DATA) -
      KILOBYTES(13000 1300) -
      CISZ(10240) -
      FILE(MASTER)) -
INDEX (NAME(STSGWD.RMM.CDS.INDEX) -
       CISZ(2048) -
       KILOBYTES(1300 130) -
       FILE(MASTER))

SET MAXCC = 0
/*
//LOAD EXEC PGM=IDCAMS,COND=(0,NE)
//SYSPRINT DD SYSOUT=*
//IN DD DISP=(OLD,PASS),DSN=&&MERGE
//OUT DD DISP=SHR,DSN=STSGWD.RMM.CDS,
// AMP=(' BUFND=50,BUFNI=50')
//SYSIN DD *
REPRO INFILE(IN) OUTFILE(OUT)
/*

```

Figure 154 (Part 2 of 2). Sample EDGJLOAD JCL

Before running this EDGJLOAD job, you can tailor some parameters, such as the space allocation values for the DFSMSrmm CDS. To choose the space allocation values you can get information from the reports you generated during the environment analysis phase. You need to know how many volumes and data sets you have, how many locations you must define, and the number of VRSS. The output of the extract program can also be useful in getting the information.

After you have the information, you must calculate the amount of space required by using the space calculation rules in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

While you are loading the DFSMSrmm CDS using the IDCAMS utility, you may see the message **IDC3314I** - RECORDS OUT OF SEQUENCE.

This message may be issued when duplicate records are loaded into the DFSMSrmm CDS. The loading of duplicate records may occur if errors were detected during the extract process. Refer back to the output produced during the extract to see whether any return codes higher than 4 were produced and ensure that all corrective actions are taken before rerunning the programs.

Note also that IDCAMS REPRO continues on for only four errors and then quits. So you may have more errors to correct than those that IDCAMS identifies.

9.3.2 Create DFSMSrmm CDS Control Record

The DFSMSrmm CDS control record contains information about the contents of the DFSMSrmm CDS, such as the number of shelf locations in the library and built-in storage locations.

After you have run EDGJLOAD, you must create the DFSMSrmm CDS control record by using the output from the SYSOUT DD statement of the EDGCNVT program. The SYSOUT DD statement contains the SYSIN control statements that the EDGUTIL utility requires to create the DFSMSrmm CDS control record.

A sample job for creating the DFSMSrmm CDS control record can be found in the SAMPLIB member EDGJVERR (see Figure 155).

You run the EDGJVERR job to create the DFSMSrmm CDS control record and verify the DFSMSrmm CDS that was built.

Note: See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information about EDGUTIL.

9.4 Verifying the DFSMSrmm CDS

You must check the VSAM data set to ensure that all information stored in it is in the correct format, and without inconsistencies. Use the EDGUTIL utility with a job like that shown in Figure 155.

```
/*JOBNAME JOB .....
```

```
//CREATE EXEC PGM=EDGUTIL,PARM='CREATE'
```

```
//SYSPRINT DD SYSOUT=*
```

```
//SYSUDUMP DD SYSOUT=*
```

```
//MASTER DD DISP=OLD,DSN=STSGWD.RMM.CDS
```

```
//SYSIN DD DISP=SHR,DSN=STSGWD.EDGCNVT.SYSOUT.DATA
```

```
/* *****
```

```
/* SYSIN DD *
```

```
/* CONTROL MASTER(Y) RACKNO(41275) RACKFREE(0) -
```

```
/* RBINNO(3539) RBINFREE(435)
```

```
/* /*
```

```
/* *****
```

```
//VERIFY EXEC PGM=EDGUTIL,PARM='VERIFY(ALL)',
```

```
// COND=(0,NE),REGION=64M
```

```
//SYSPRINT DD SYSOUT=*
```

```
//SYSUDUMP DD SYSOUT=*
```

```
//MASTER DD DISP=OLD,DSN=STSGWD.RMM.CDS,
```

```
// AMP=('BUFND=50,BUFNI=50')
```

```
//SYSIN DD DUMMY
```

Figure 155. Sample EDGUTIL Execution JCL for VERIFY Processing

You can find sample JCL in the SYS1.SAMPLIB EDGJVERR.

Refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information on the parameters that you can use with the EDGUTIL program.

Review the output of the verification job carefully to see whether there are error messages that will affect the integrity of the DFSMSrmm CDS. If there are errors that do not allow you to proceed to the parallel running phase, you can try to

correct the errors and go back to the beginning of the conversion phase, after the extract process, change the necessary EDGCNVT control statements, and restart the conversion steps.

If you cannot correct the errors, it may be necessary to contact an IBM service representative who can help you in the use of the EDGUTIL MEND utility, which may correct the errors.

Note: It should not be necessary to use the MEND utility regularly. Limit its use to fixing DFSMSrmm CDS problems introduced by errors in the conversion process that initially created the DFSMSrmm CDS. Please ask for IBM assistance before you use the MEND utility.

9.5 Running DFSMSrmm Commands

Once you have successfully verified the integrity of the DFSMSrmm CDS, it may be necessary to run some DFSMSrmm commands to:

- Verify and eventually add bins to storage locations
- Add VRSs if the conversion process has not added them.

Note: These commands cannot be run before starting DFSMSrmm.

You must consider preparing the command stream that adds VRSs during this phase of the migration in order to be ready for the parallel running stage.

9.5.1 Adding Empty Bins

If you use shelf-managed storage locations, you might have been able to add empty bins during the extract process for every storage location defined. If you were not able to add bins, or if you are migrating from a manual tape management and do not have empty bins defined, you can add them now by using the DFSMSrmm ADDBIN command.

First you must determine how many bins you need for every storage location, and then execute the command:

```
RMM ADDBIN bin_number LOCATION(myloc) COUNT(n_of_bins)
```

where:

bin_number

is the bin number that you want to add

myloc

is the location name where you add bins

n_of_bins

is the number of bins that you want to add.

9.5.2 Adding VRSs

The VRS conversion can be done either manually or through the conversion process. It may be easier to create the VRS definitions manually than to use the conversion programs when:

- There is a small set of VRS definitions to convert

- The VRS was not built properly by the conversion program and does not meet your needs
- You are using the VRSCMDS file from the EDGCNVT program.

If your installation has only a few VRS rules that can be easily defined, you can use the DFSMSrmm ADDVRS command rather than the VRS conversion process.

In addition, if the VRS rules in your previous tape management environment cannot be easily converted, such as “cycle by days” retention or some particular GDG retention methods, you must consider using the DFSMSrmm ADDVRS command to add them.

The EDGCNVT program shipped with DFSMSrmm gives you an additional opportunity to build VRS definitions by using the output generated in the VRSCMDS DD file. This method can be useful if you want to check VRS definitions before adding them. You can review the definitions in this file and change the VRS commands as necessary before submitting the command stream.

Note: Remember to delete any previous VRS definition from the DFSMSrmm CDS before adding the VRSs from the VRSCMDS file to avoid duplication or error messages.

For more information about using the DFSMSrmm ADDVRS command, refer to the *DFSMSrmm Guide and Reference*, SC26-4931, and 13.2.1, “Reduce VRS Policies” on page 505.

Chapter 10. Parallel Running and Validation

Before you change the DFSMSrmm operation mode to PROTECT you should successfully test all of your production scenarios when DFSMSrmm is running in RECORD-ONLY or WARNING mode. When DFSMSrmm is running in WARNING mode, you can scan your job or system logs for DFSMSrmm messages. If DFSMSrmm is running in parallel, you can compare the vendor product and DFSMSrmm reports and check for differences.

This chapter describes the activities that occur when DFSMSrmm is running in parallel and not switched to PROTECT mode.

Objectives of Chapter

- List and explain all tasks required to achieve the conversion with minimum risk and no loss of data.
- Verify correct conversion.
- Execute DFSMSrmm and the vendor product together to manage the same volumes.
- Identify and explain tasks required to validate that DFSMSrmm and the vendor product are providing the same function.
- Identify when it is time to start cutover to production.

Audience

- System programmers
- Storage administrators and librarians
- Production controllers

At the end of this chapter you will be ready to deactivate the old tape management system and run DFSMSrmm in production. You will have validated that DFSMSrmm meets your needs and provides equivalent function.

10.1 Preparing to Run in Parallel

To prepare for parallel running, we recommend that you test and validate the conversion in a test environment to ensure that you are ready for parallel running in production.

Note: When running DFSMSrmm with CA-1 5.1 and later releases, be sure to start DFSMSrmm before CA-1 or to include the EDGSSSI program name in IEFSSNxx. Both products update LPA to get control during OPEN. The updates must be done in the correct order for processing to work correctly.

Please check that you have applied APAR OW20070.

10.1.1 Validate PARMLIB Members

Check that the following SYS1.PARMLIB members are correct updated before a start of DFSMSrmm.

- IEFSSNxx
- IKJTSOxx
- IFAPRDxx
- EDGRMMxx

- SMFPRMxx
- GRSRNLxx
- SCHEDxx

Perform this step once for each MVS image.

The method for updating the PARMLIB members is described in 3.1.2, “Update and Validate PARMLIB Members” on page 28.

10.1.2 IEFSSNxx

Enabling the DFSMSrmm subsystem interface ensures that the interface will start every time you IPL the system. This step ensures that users cannot use tapes before the DFSMSrmm subsystem starts. To enable DFSMSrmm, change the IEFSSNxx member of SYS1.PARMLIB. Add EDGSSSI, as the DFSMSrmm subsystem initialization program, as shown in Figure 156.

SUBSYS SUBNAME(JES2)	/* JES2 PRIMARY SUBSYSTEM START	*/
PRIMARY(YES) START(YES)		
SUBSYS SUBNAME(DFRM)	/* Name of the DFSMSrmm subsystem	*/
INITRTN(EDGSSSI)	/* DFSMSrmm initialization routine	*/
SUBSYS SUBNAME(AOPA)	/* Netview	*/

Figure 156. Defining DFSMSrmm to MVS through IEFSSNxx

Figure 156 shows the correct relative position of the DFSMSrmm subsystem, updated to include the initialization program.

10.1.3 EDGRMMxx

In this section we highlight which updates are important in EDGRMMxx when DFSMSrmm is running in parallel with another tape management system.

Update the EDGRMMxx PARMLIB member definition for DFSMSrmm as shown in Figure 157 on page 429. Set the following options:

- OPMODE(W)
- TPRACF(N)
- UNCATALOG(N)
- VRCHANGE(INFO)
- VRSEL(NEW)

OPTION	OPMODE(W)	/* Warning Mode */ -
	ACCOUNTING(J)	/* Accounting from JOB */ -
	BACKUPPROC(EDGBKUP)	/* Name of BACKUP-proc */ -
	BLP(RMM)	/* DFSMSrmm controls BLP */ -
	CATRETPD(0012)	/* catalog retention */ -
	CDSID(PROD)	/* control data set id */ -
	DATEFORM(J)	/* Date format */ -
	DSNAME(RMM.PROD.CDS)	/* CDS data set name */ -
	IPLDATE(N)	/* IPL date checking */ -
	JRNLLNAME(RMM.PROD.JRNL)	/* JRNL data set name */ -
	JOURNALFULL(75)	/* Percentage JRNL full*/ -
	LINECOUNT(054)	/* Lines per page */ -
	MASTEROVERWRITE(LAST)	/* Overwriting of a vol*/ -
	MAXHOLD(100)	/* Number of I/O oper. */ -
	MAXRETPD(9999)	/* Maximum retention */ -
	MSG(M)	/* case for message txt*/ -
	NOTIFY(Y)	/* Notify volume owners*/ -
	PDA(OFF)	/* PDA is disabled */ -
	PDABLKCT(255)	/* number of blocks */ -
	PDABLKCT(31)	/* blocksize in K */ -
	PDALOG(ON)	/* PDA output disabled */ -
	RETPD(0005)	/* Default retention */ -
	SCRATCHPROC(EDGXPROC)	/* ATL/MTL procedure */ -
	SMFAUD(248)	/* SMF audit records */ -
	SMFSEC(249)	/* SMF security records*/ -
	SYSID(EGZB)	/* Name of the system */ -
	TPRACF(N)	/* RACF tape support */ -
	UNCATALOG(N)	/* Catalog support */ -
	VRSCCHANGE(INFO)	/* No additional action*/ -
	VRSEL(NEW)	/* New VRS processing */ -
	VRJOBNAME(2)	/* DATASETNAME/JOBNAME */ -
	VRSMIN(0000000100,WARN)	/* Warn if < 100 VRSs */ -
	TLCSV1(N)	/* Call TLCS O/C/EOV */ -
/* ***** */		
MNTMSG	MSGID(IEF233A)	ID(001) VOLUME(016) RACK(016)
MNTMSG	MSGID(IEF233D)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E K')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E R')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E D')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(IEF455D)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(IEC501A)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E K')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E D')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E R')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E RD')	ID(001) VOLUME(017) RACK(999)
MNTMSG	MSGID(' IEC502E RK')	ID(001) VOLUME(017) RACK(999)
MNTMSG	MSGID(IAT5110)	ID(001) VOLUME(044) RACK(999)
MNTMSG	MSGID(IAT5210)	ID(001) VOLUME(050) RACK(999)
MNTMSG	MSGID(IAT5410)	ID(001) VOLUME(020) RACK(999)

Figure 157 (Part 1 of 3). Sample EDGRMMxx PARMLIB Member

```

/* ***** */
REJECT    ANYUSE(*)
/* ***** */
VLPPOOL   PREFIX(SC*)
          TYPE(S)
          MEDIANAME(3480)
          RACF(Y)
          EXPDTCHECK(N)
          DESCRIPTION('3480 DEFAULT SCRATCH POOL')
VLPPOOL   PREFIX(N*)
          TYPE(R)
          RACF(N)
          EXPDTCHECK(O)
          DESCRIPTION(' TEST POOL FOR NORBERT')
VLPPOOL   PREFIX(F*)
          TYPE(R)
          RACF(Y)
          EXPDTCHECK(Y)
          DESCRIPTION(' FOREIGN CARTRIDGES')
VLPPOOL   PREFIX(PP*)
          TYPE(R)
          RACF(Y)
          EXPDTCHECK(Y)
          DESCRIPTION(' SOFTWARE CARTRIDGES')
/* ***** */
LOCDEF    LOCATION(SAFE)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(SANJOSE)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(NEWYORK)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(NO BINS)
LOCDEF    LOCATION(IBMATL1)
          TYPE(LIBRARY)
          PRIORITY(4800)

```

Figure 157 (Part 2 of 3). Sample EDGRMMxx PARMLIB Member

```

/* ***** */
SECCLS  NUMBER(010) -
        NAME(UCL) -
        SMF(N) -
        MESSAGE(N) -
        ERASE(N) -
        DESCRIPTION(' UNCLASSIFIED') -
        MASK(**)
SECCLS  NUMBER(050) -
        NAME(IUO) -
        SMF(Y) -
        MESSAGE(N) -
        ERASE(N) -
        DESCRIPTION(' INTERNAL USE ONLY') -
        MASK(**.IUO.**,**.INTERNAL.**,**.INT.***)
SECCLS  NUMBER(100) -
        NAME(IC) -
        SMF(Y) -
        MESSAGE(N) -
        ERASE(N) -
        DESCRIPTION(' CONFIDENTIAL') -
        MASK(**.IC.***)
SECCLS  NUMBER(125) -
        NAME(ICR) -
        SMF(Y) -
        MESSAGE(N) -
        ERASE(Y) -
        DESCRIPTION(' CONFIDENTIAL RESTRICTED') -
        MASK(**.ICR.***)
SECCLS  NUMBER(150) -
        NAME(RIC) -
        SMF(Y) -
        MESSAGE(Y) -
        ERASE(Y) -
        DESCRIPTION(' REGISTERED CONFIDENTIAL') -
        MASK(**.RIC.***)

```

Figure 157 (Part 3 of 3). Sample EDGRMMxx PARMLIB Member

Here are the most important options of the EDGRMMxx PARMLIB member if DFSMSrmm is running in parallel to another tape management system:

OPMODE

Specifies the mode of DFSMSrmm

- R** Record-only mode. You can use DFSMSrmm TSO commands, ISPF dialog, and inventory management functions. In addition, DFSMSrmm records information about magnetic tape volumes used on the system, including details about volume owners and data set names. For volumes in an automated tape library dataserwer, DFSMSrmm records information about entry and eject activities. During entry processing, DFSMSrmm provides volume status information to OAM. If a volume is not already defined to DFSMSrmm, DFSMSrmm automatically defines it as long as there are no conflicts with either the VOLSER or corresponding rack number.

DFSMSrmm does not validate or reject volumes when in record-only mode. When you issue DFSMSrmm commands, the TCDB is updated if appropriate, and the volume is system-managed.

- W** Warning mode. In addition to performing the actions described for record-only mode, DFSMSrmm validates magnetic tape volumes as you use them. If DFSMSrmm discovers errors, it issues error messages instead of rejecting tapes.

BACKUPPROC	Specifies the name of the procedure you want to be started when the journal percentage full threshold is reached. If no name is specified, then no automatic command is issued.
DATEFORM	Specifies the date format for messages and reports. Specify the date format preferred in your computing center.
DSNAME	Specifies the name of the DFSMSrmm CDS. Specify a name up to 44 characters long. Default: If you do not specify DSNAME, you must specify the data set name in the MASTER DD statement in the DFSMSrmm started procedure. If you specify a name for both DSNAME and MASTER DD, DFSMSrmm ignores the MASTER DD statement.
JRNLMNAME	Specifies the name of the journal. Specify a name up to 44 characters long. Default: If you do not specify JRNLMNAME in EDGRMMxx, you can specify a name in the JOURNAL DD statement in the DFSMSrmm started procedure. If you do not specify a journal name in either EDGRMMxx or the started procedure, DFSMSrmm does not provide journaling. If you specify a name in both, DFSMSrmm uses the JRNLMNAME value and ignores the JOURNAL DD statement.
JOURNALFULL	Specify JOURNALFULL to define a percentage full threshold of the journal dataset. When DFSMSrmm detects that the journal has reached this threshold, DFSMSrmm issues message EDG2107E. DFSMSrmm also issues this message at startup if the journal has already reached the threshold specified. If you specify a backup procedure name on the BACKUPPROC operand, the procedure is started automatically. If you specify a value of 0, DFSMSrmm issues no warning on that system. You can specify different threshold values for sharing systems.
MAXHOLD	Specifies the maximum number of I/O operations to the CDS before the reserve is released and reacquired. Specify MAXHOLD to minimize the impact that long operations, such as DFSMSrmm TSO ADD subcommands with large count values, or large searches, will have on other users.
RETPD	Specifies the default retention period for all new data sets on volumes. Specify a value between 0 and 9999 days. The specified value is added to the current date to determine the expiration date.

	The value you specify for the RETPD operand should be the same as the value you use in your current tape management system.
MAXRETPD	Specifies the maximum retention period for all new data sets on tape volumes. Specify a value between 0 and 9999 days, or NOLIMIT. The specified value is added to the current date to determine the maximum allowable expiration date. If you wish to allow any user to specify a permanent expiration date, such as 99365, you must be sure to specify the NOLIMIT value. Any numeric value prevents the use of permanent retention dates for new tape data sets.
TPRACF	Do not create RACF tape profiles.
UNCATALOG	Do not use DFSMSrmm uncatalog processing. If UNCATALOG(N) is set, DFSMSrmm does not uncatalog data sets under any circumstances. Attention: DFSMSrmm will uncatalog data sets in each operating mode if the UNCATALOG function is set to Y or S.
VRSCCHANGE	Use VRSCCHANGE to specify the action DFSMSrmm should take during inventory management if you have made any changes to vital record specifications. If INFO is set, no action is taken during inventory management. With VERIFY, any change in vital record specifications must be verified by running EDGHSKP with the VERIFY parameter.
VRSJOBNAME	Specify this operand to select whether DFSMSrmm uses the job name from VRS as the primary or secondary match. Specify VRSJOBNAME(1) so that the job name takes precedence over the data set name during VRS match processing. Specify VRSJOBNAME(2) so that the data set name takes precedence over the job name during VRS match processing. This is the option that most closely matches the use of job names in vendor tape management systems.
VRSEL	Specify this operand to select the VRS processing method used by DFSMSrmm. Specify VRSEL(NEW) to exploit the new VRS features such as mixed retention type VRS chains, and VRS release options. We recommend the use of this option because the conversion programs build new types of VRSs. Specify VRSEL(OLD) for compatibility with any existing use of VRSs.
REJECT	DFSMSrmm can reject a range of tapes only if you first define them to DFSMSrmm. It can reject all tapes not defined to it, which you specify with a single asterisk for the prefix, but it cannot reject a range of tapes not defined to it. Note: If you specify REJECT(*), all tapes not defined to the DFSMSrmm CDS cause an error message to be issued by DFSMSrmm when the tapes are used.

SCRATCHPROC Specifies the name of the procedure DFSMSrmm starts to replenish scratch volumes in an automated tape library dataserer. Specify a procedure name one to eight characters long. When an automated tape library dataserer detects a low-on-scratch condition, OAM write a write-to-operator message (CBR3660A). DFSMSrmm intercepts the message and starts the SCRATCHPROC procedure. This process runs DFSMSrmm expiration processing to replenish the automated tape library dataserer's scratch volumes.

Default: SCRATCHPROC(EDGXPROC)

SECCLS Use the SECCLS command to define security classes for data sets and volumes. These security classes appear in reports and DFSMSrmm TSO subcommand output. DFSMSrmm records these security classes in the DFSMSrmm CDS only; it does not make RACF aware of them. There is no connection between these definitions and any similar definitions or function provided in RACF, but you can use similar values for overall consistency.

The examples of the SECCLS commands shown in Figure 157 on page 429 show the use of the following functions:

- Volumes must be erased on release.
- Issue operator confirmation messages.
- Write SMF security records each time a tape data set in this class is opened.

MNTMSG Use the MNTMSG command to tailor mount and fetch messages so they display the VOLSER and pool name or rack number. This information makes it easier for the operator to pull and mount volumes.

VLPOOL Use the VLPOOL command to define pools. When you add a new volume to the library, DFSMSrmm assigns it a shelf location from the specified pool.

In DFSMSrmm you can specify two categories of pools. The TYPE parameter specifies the type of pool:

- R** A rack pool is shelf space assigned to hold volumes that are generally read-only and that enter and leave your installation on an individual basis. These volumes are typically software product volumes and customer volumes and do not adhere to your installation's naming conventions.
- S** A scratch pool is shelf space assigned to hold all other volumes in the removable media library. The volumes assigned to this shelf space can be used to satisfy scratch requests as long as the volumes are in scratch status. Once the volume has been written to, it becomes a master volume until the data is no longer required by the installation. The volume remains in the same DFSMSrmm scratch pool in that it occupies the same shelf space regardless of status.

Note: We recommend that you specify all of your pools that are defined in your tape management system running in parallel.

10.1.4 Check the DFSMSrmm Procedure

Check the procedure in SYS1.PROCLIB to start the DFSMSrmm subsystem address space. Figure 158 contains sample JCL.

```
//DFRMM PROC M=00,OPT=MAIN
//IEFPROC EXEC PGM=EDG&OPT.,PARM='&M',TIME=1440,REGION=4096K
//PARMLIB DD DDNAME=IEFRDER
//IEFRDER DD DISP=SHR,DSN=SYS1.PARMLIB
//*MASTER DD DISP=SHR,DSN=RMM.CONTROL.DSET
//*JOURNAL DD DISP=SHR,DSN=RMM.JOURNAL.DSET
//EDGPDOX DD DISP=SHR,DSN=RMM.PDOX.DSET
//EDGPDOY DD DISP=SHR,DSN=RMM.PDOY.DSET
```

Figure 158. DFSMSrmm Procedure in SYS1.PROCLIB

In this sample JCL:

- DFRMM** Specifies the procedure name. You can use any procedure name from 1 to 8 characters long, subject to the restrictions documented under the IEFSSNxx PARMLIB member in the *DFSMS/MVS: Implementation and Customization Guide*, SC26-4932.
- M** Allows you to specify a PARMLIB member suffix. M=00 tells DFSMSrmm to use EDGRMM00.
- OPT** Allows you to specify:
- OPT=RESET to disable the subsystem interface
 - OPT=MAIN to execute the DFSMSrmm subsystem.
- Note:** Before DFSMSrmm disables the subsystem interface, it ensures that the user who made the request is authorized.
- See the *DFSMS/MVS: Implementation and Customization Guide*, SC26-4932, for more information.
- PARMLIB** Optional statement; if PARMLIB is not coded, then IEFRDER is redundant, and EDGRMMxx is defaulted to SYS1.PARMLIB, or your system PARMLIB concatenation.
- IEFRDER** Optional statement
- MASTER** Optional statement; if not specified, the control data set name must be specified in the EDGRMMxx PARMLIB member.
- JOURNAL** Optional statement; if not specified, the journal data set name must be specified in the EDGRMMxx PARMLIB member.
- EDGPDOx** Optional statement; required only for external trace output recording.
- EDGPDOy** Optional statement; required only for external trace output recording.

In the procedure example in Figure 158, the MASTER and JOURNAL DD statements are commented by intention. This is recommended, so you can disable the journal data set during special runs of inventory management. In

10.3.1, “First Run of EDGHSKP with PARM(VRSEL)” on page 444, a scenario of disabling the journal data set is described.

10.1.5 Exits

DFSMSrmm provides programming interfaces and source code for installationwide exits installed as part of DFSMSrmm during the SMP/E apply process.

DFSMSrmm provides the following programming interfaces.

EDGDFHSM DFSMShsm tape volume interface

EDGTVEXT Tape volume interface for other applications

EDGMSGEX DFSMSrmm uses the DFSMSdfp MSGDISP interface to update tape drive displays

Source code is supplied for:

CBRUXENT OAM cartridge entry exit

CBRUXEJC OAM cartridge eject exit

CBRUXCUA OAM cartridge change use attributes exit

CBRUXVNL OAM volume not in library exit

Before the SMP/E ACCEPT, the source code is in the SMPSTS. After the ACCEPT, the exits are in SYS1.AEDGSRCL.

If your installation is already using one or more of these exits in the tape management system environment, you must ensure that the correct action is performed during the parallel running.

If you will be sharing one or more exits between your tape management system and DFSMSrmm, you must check if some updates are needed in any of the exits to achieve coexistence when running in parallel.

The action you take depends on what level of DFSMSrmm you install. With DFSMS/MVS Version 1 Release 4.0, DFSMShsm and DFSMSrmm communicate directly without calling the ARCTVEXT exit. Appendix D, “Sample Exits and Reports” on page 603 contains an example ARCTVEXT exit; a front-end ARCTVEXT to call the CA1TVEXT and the RMMTVEXT.

Note: Create one front-end exit for each pair of exits if you want both DFSMSrmm and the other tape management system to be called.

You can also merge the two exits into one. See 3.1.1, “Update Installationwide Exits” on page 28 for more information.

10.1.5.1 SAMS:DISK Interface

You can modify your SAMS:Disk to support more than one tape management system. To do this, you have to update the SAMS:Disk SYSPARMS in the following way:

TAPCTLEXADSTH016 This will activate the SAMS:Disk DFSMSrmm support. Please be sure that the ARCTVEXT is available for this support.

TMSCTLEXADSTH014 Normally this parameter is already defined to support your running CA-1 tape management system.

Note: After the cutover you will change the **TMSCTLEX** from **ADSTH014** to **ADSTH016** value and delete the **TAPCTLEX** entry from your SAMS:Disk PARMLIB.

For more information read the SAMS:Disk books.

10.1.6 Verify Authorization and Security

To protect DFSMSrmm functions, you need to use an external security product, such as RACF. If RACF is not installed, you must provide equivalent function through the SAF interface.

When running on a system with RACF installed, check that DFSMSrmm is defined in the started procedures name table ICHRIN03 or in the RACF STARTED class. We recommend that the RACF user ID assigned should match the name of the DFSMSrmm procedure created, but any installation-selected RACF user ID is acceptable. As data sets are created for use by the DFSMSrmm procedure, add the DFSMSrmm RACF user ID to the access list for the data sets.

If you plan to use the EDGLABEL, EDGXPROC or the procedure you define as the BACKUPPROC procedure, you must define the procedure in the ICHRIN03 or RACF STARTED class.

- EDGLABEL requires UPDATE access to STGADMIN.EDG.OPERATOR.
- EDGXPROC requires READ access to STGADMIN.EDG.HOUSEKEEP.
- The defined BACKUPPROC requires ALTER access to STGADMIN.EDG.HOUSEKEEP.

For more information on updating ICHRIN03, the RACF STARTED class, and protecting DFSMSrmm resources, see Appendix A, "Security Considerations and Implementation" on page 561.

Each DFSMSrmm resource has an entity name prefixed with STGADMIN.EDG. Add each of the following RACF profiles in the FACILITY class and grant permission to all users who have a need for the functions:

- STGADMIN.EDG.FORCE
- STGADMIN.EDG.HOUSEKEEP
- STGADMIN.EDG.IGNORE.TAPE.volser
- STGADMIN.EDG.INERS.WRONGLABEL
- STGADMIN.EDG.LABEL.volser
- STGADMIN.EDG.LISTCONTROL
- STGADMIN.EDG.MASTER
- STGADMIN.EDG.NOLABEL.volser
- STGADMIN.EDG.OPERATOR
- STGADMIN.EDG.OWNER.userid
- STGADMIN.EDG.RELEASE
- STGADMIN.EDG.RESET.SSI
- STGADMIN.EDG.VRS
- STGADMIN.ADR.DUMP.CNCURRNT
- STGADMIN.EDG.VRS

Note: Grant the started task procedure (DFRMM) RACF user ID ALTER access authority to STGADMIN.EDG.RESET.SSI so you can disable the DFSMSrmm subsystem by using the command shown in Figure 159 on page 438.

```
S DFRMM,OPT=RESET  
P DFRMM
```

Figure 159. Disabling the DFSMSrmm Subsystem Interface

For more information on authorization and security, see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, 3.1.2.5, “EDGRMMxx” on page 31, and Appendix A, “Security Considerations and Implementation” on page 561.

10.1.7 Prepare Inventory Management

DFSMSrmm provides the EDGHSKP utility to help you perform inventory management.

You can use an existing job scheduling product to run the utilities, such as OPC/ESA. You can schedule each activity to run on its own, or you can schedule many activities to run together in sequence.

While DFSMSrmm is running in parallel, schedule the following DFSMSrmm functions:

- Vital record processing
- Expiration processing
- Backing up the DFSMSrmm CDS
- Backing up and clearing the DFSMSrmm journal
- Creating an extract data set
- Storage location management

All of the above functions should run with the same frequency and time as the equivalent functions in the tape management system running in parallel.

Note: You must run the EDGHSKP BACKUP function as it clears the DFSMSrmm journal data set when the backup of the DFSMSrmm CDS is successful.

In order to keep a record of the EDGHSKP utility processing you can copy the output files such as the MESSAGE file to SYSOUT. Copying the output files to SYSOUT enables you to view the output along with the job and prevents data loss due to multiple runs of EDGHSKP.

10.1.7.1 EDGHSKP Procedure

Figure 160 contains sample JCL for the EDGHSKP procedure.

```
//HSKP    EXEC PGM=EDGHSKP,  
//        PARM='VRSEL,EXPROC,DSTORE,BACKUP(DSS),RPTEXT,DATEFORM(E)'  
//ACTIVITY DD DISP=SHR,DSN=HSKP.ACTIVITY  
//MESSAGE DD DISP=SHR,DSN=HSKP.MESSAGES  
//REPORT  DD DISP=SHR,DSN=HSKP.VRS.REPORT  
//REPTXT  DD DISP=SHR,DSN=HSKP.REPTXT  
//SYSPRINT DD SYSOUT=A  
//BACKUP  DD DISP=(,CATLG),DSN=MASTER.BACKUP,UNIT=SYSDA  
//JRNLBKUP DD DISP=(,CATLG),DSN=JOURNAL.BACKUP,UNIT=SYSDA
```

Figure 160. Sample EDGHSKP Procedure

Before running EDGHSKP, several data sets need to be predefined. All data sets are not required for all functions. The following list shows the data sets you should define.

ACTIVITY	Contains detailed information about data set related changes that DFSMSrmm makes to the control data set during inventory management. This data set is required when you specify the VERIFY parameter and must be preallocated.
MESSAGE	Lists the messages the DFSMSrmm subsystem issues during inventory management. This data set is required and must be preallocated.
REPORT	Contains a detailed report of DFSMSrmm vital record specification processing. Specify if you want a report when you have specified the VRSEL parameter. This data set must be preallocated.
REPTXT	Contains the extract copy of the DFSMSrmm control data set. The extract copy is called the extract data set. The extract data set is required when you specify the RPTXT parameter and must be preallocated.
BACKUP	Contains the backup copy of the DFSMSrmm control data set. You can backup directly to tape when you specify the BACKUP(DSS) parameter. This data set is required when you specify the BACKUP parameter. You do not need to preallocate this data set.
JRNLBKUP	Contains the backup copy of the DFSMSrmm journal data set. You can backup directly to tape when you specify the BACKUP(DSS) parameter. This data set is optional when you specify the BACKUP parameter. You do not need to preallocate this data set.
SYSPRINT	Contains the utility program messages that IDCAMS and ADRDSSU issue when backing up the DFSMSrmm control data set. The SYSPRINT data set is required when you specify the BACKUP parameter. This data set can be a SYSOUT file.
DSSOPT	Contains DUMP or RESTORE command options used by DFSMSdss during backup processing.

10.1.7.2 EXEC Parameters in EDGHSKP

The following list shows the possible EXEC parameters for the EDGHSKP program.

Note: All of them should not be specified during the parallel-run phase of the conversion. Some of the parameters also depend on the setting of the related OPTION parameter in the PARMLIB member.

BACKUP(AMS DSS)	Specify BACKUP(DSS) when you want DFSMSdss to perform the backup. Using DFSMSdss concurrent copy permits the update of the control data set during backup processing so all tape activity can continue during CDS backup processing. To use DFSMSdss concurrent copy, you must have a concurrent copy environment set up. SnapShot capable devices are also supported by DFSMSdss.
------------------------	--

The DFSMSDss operands are specified in the DSSOPT DD statement, in the EDGHSKP and EDGBKUP utilities. When using BACKUP(DSS) without concurrent copy, you can still direct the output to tape, but all updates to the control data set will wait until the CDS backup is complete. BACKUP(AMS) is the default. DFSMSrmm uses access method services REPRO command to perform backup processing. Specifying BACKUP(AMS) prevents updating the control data set during CDS backup processing.

DATE(date,+nnnn)	Use this parameter to set the date for VERIFY processing. The current date is the default.
DATEFORM(A E I J D)	Use this parameter to override the date format in reports produced in inventory management. The default date format for all date fields is the value specified in the PARMLIB member EDGRMMxx. The value is initially set to J for Julian.
DSTORE	Specify this parameter for storage location management.
EXPROC	Specify this parameter for expiration processing.
RPTEXT	Specify this parameter to create records in extract dataset
VERIFY	Specify this parameter to request a trial run of vital record processing. DFSMSrmm performs vital record processing, but does not update the control dataset on this run of inventory management. You can use this operand to test new vital record specifications that you define to DFSMSrmm. The use of VERIFY in the EDGHSKP utility depends on the setting of the VRCHANGE parameter in the PARMLIB member EDGRMMxx. The default value of VRCHANGE is VERIFY. Note: The VRCHANGE parameter in PARMLIB member EDGRMMxx should be set to INFO during parallel processing.
VRSEL	Specify this parameter for vital record processing.

10.1.7.3 EDGRPTD Report Utility

Figure 161 on page 441 shows a sample of JCL for inventory and movement reporting using the EDGRPTD utility.


```

//RPTD      EXEC PGM=EDGRPTD,
//          PARM=' SEC' '( INTERNAL USE ONLY)'' , DATEFORM(I),LINECOUNT(54)'
//REPTXT DD DISP=SHR,DSN=HSCP.REPTXT
//SYSPRINT DD SYSOUT=*
//INSTVOL DD DISP=SHR,DSN=HSCP.INSTVOL
//INSTBIN DD DISP=SHR,DSN=HSCP.INSTBIN
//INSTOWN DD DISP=SHR,DSN=HSCP.INSTOWN
//TOSTRCK DD DISP=SHR,DSN=HSCP.TOSTRCK
//TOSTOWN DD DISP=SHR,DSN=HSCP.TOSTOWN
//FMSTBIN DD DISP=SHR,DSN=HSCP.FMSTBIN
//FMSTOWN DD DISP=SHR,DSN=HSCP.FMSTOWN
//RDYTOSCR DD DISP=SHR,DSN=HSCP.RDYTOSCR
//SYSOUT DD SYSOUT=*

```

Figure 161. DFSMSrmm Sample of JCL for Inventory and Movement Reporting

The input file in the EDGRPTD run is the REPTXT DD file created in the EDGHSKP Inventory Management run with PARM(RPTXT) specified.

The DD statements you can code for Inventory and Movement Report are the following:

INSTVOL	Output file for the report containing the inventory of volumes by location sorted by volume serial number.
INSTBIN	Output file for the report containing the inventory of volumes by location sorted by rack number or bin number. The storage location report is sorted by bin number. All other reports are sorted by rack number.
INSTOWN	Output file for the report containing the inventory of volumes by location sorted by owner.
TOSTRCK	Output file for movement reports sorted by rack number.
TOSTOWN	Output file for movement reports sorted by owner.
FMSTBIN	Output file for movement reports sorted by bin number.
FMSTOWN	Output file for movement reports sorted by owner.
RDYTOSCR	Output file for movement reports sorted in ascending order by rack report column which can be either a bin or rack number. Volumes to be moved from locations to home locations.
Note: Volumes listed in Ready-to-Scratch list are excluded from the TOSTRCK or FMSTBIN reports.	

You only need to code DD statements for the reports you intend to create.

Appendix D, “Sample Exits and Reports” on page 603 contains a JCL example of how to use EDGHSKP and EDGRPTD in inventory management reports.

10.1.7.4 DFSMSrmm Scratch List

Produce a list of scratch volumes from DFSMSrmm ordered by VOLSER, to be compared with the other products scratch list.

To obtain this list from DFSMSrmm you can use the following command:

```
RMM SV * STATUS(SCRATCH) OWNER(*) LIMIT(*)
```

Using the IKJEFT01 or IKJEFT1A TSO batch interface program, you can store the results of this command in a data set referenced by the SYSTSPRT DD statement.

10.1.8 Check Interfaces

Before DFSMSrmm can run in parallel to your old tape management system, there is some additional information that you must consider about interfaces.

10.1.8.1 SMS Interface

If you have an IBM 3494 or 3495 Automated Tape Library Dataserver, you must ensure that it works correctly in the DFSMSrmm environment. You must install the DFSMSrmm-supplied OAM exits using the DFSMSrmm version and provide the correct input to the extract program for those volumes that belong to the tape library.

You must define an DFSMSrmm location with the same name as the tape library name, using a location type of ATL. The name of the ATL is defined in your SMS environment.

10.1.8.2 BTLS Interface

If your installation uses BTLS to drive IBM 3494 and/or 3495 Automated Tape Library Dataserver operations, you will need to notify BTLS when a volume must be returned to scratch status. In a TLMS environment this is done through a batch job that gets information from the TLMS database and then updates the BTLS catalog. In DFSMSrmm you can use a simple REXX CLIST that synchronizes the DFSMSrmm database with the BTLS catalog.

For a sample CLIST see Appendix D, "Sample Exits and Reports" on page 603.

10.1.8.3 Short-on-Scratch Processing

DFSMSrmm offers an integrated SMS interface that updates the TCDB automatically. In addition, you can free volumes that are in PENDING RELEASE status when a short-on-scratch condition is detected inside the automated tape libraries without manual intervention. Use the SCRATCHPROC(EDGXPROC) PARMLIB option, where EDGXPROC is the name of the procedure DFSMSrmm starts to replenish scratch volumes in the automated tape libraries.

10.1.8.4 TSO and ISPF Interface

DFSMSrmm has a simple TSO interface that uses a set of powerful commands to drive all tape activities.

There is also an ISPF interface that offers menus for end users, librarians, administrators, and support personnel. You can choose a full DFSMSrmm dialog or a local defined dialog.

You can tailor the correct ISPF dialog for each user category and assign each the needed access to DFSMSrmm. See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information.

10.1.8.5 Other Interfaces

DFSMSrmm can use any DFSMSrmm command in batch mode, using the IKJEFT01 or IKJEFT1A batch TSO interface. It can also process any DFSMSrmm command through the DFSMSrmm API. To use the DFSMSrmm API, you need to code in the High Level Assembler language. A preferred method is to use the DFSMSrmm commands from a REXX environment where you can retrieve information back as variables.

If you have any of those interfaces running in your current tape environment, you must change them to reflect the equivalent function in DFSMSrmm.

10.2 Parallel Processing

The timing of the DFSMSrmm update to its CDS during OPEN processing is such that any volumes rejected by the vendor system should not result in updates to the DFSMSrmm CDS.

Also, the timing of the DFSMSrmm reject of a tape is such that recording by the other tape management system may not occur if DFSMSrmm rejects a volume. Therefore, parallel running problems are unlikely to be the result of volume rejection and control by one system or the other. More likely, the differences in inventory management functions such as expiration and retention control cause the problems. DFSMSrmm's uncataloging of tape data sets could cause the other vendor system to return a tape to scratch and vice versa.

Timing of inventory management and batch runs may vary, causing volumes to return to scratch at different times, or to move between storage locations at different times. This can cause problems if both systems are in protect mode at the same time. If DFSMSrmm is in record-only or warning mode, the differences should not be a significant problem because DFSMSrmm continues to allow tapes to be used and keeps track of the latest tape information.

Note: When running in parallel, you should prevent DFSMSrmm from uncataloging tape data sets, but continue to run both sets of inventory management. This can be accomplished by setting the PARMLIB option to **UNCATALOG(N)** in member EDGRMMxx.

One other method to consider for conversion is to run both DFSMSrmm and the other vendor product in protect mode in parallel, but use different ranges of volumes managed by each system. You can use the exits available in each product (EDGUX100 for DFSMSrmm) to control which volumes they manage and which volumes they are to ignore or treat as foreign. In time, as volumes return to scratch, or as new applications are added, you can move more and more volumes to DFSMSrmm control.

10.3 Starting Tape Management and Inventory Management

When you have checked the EDGRMMxx PARMLIB member you are ready to start DFSMSrmm. To start the DFSMSrmm subsystem, issue the command shown in Figure 162.

```
S DFRMM,M=xx
```

Figure 162. Starting DFSMSrmm

If you have not enabled the DFSMSrmm subsystem at IPL time, you get the message shown in Figure 163 on page 444.

```
EDG0103D DFSMSrmm SUBSYSTEM INTERFACE IS INACTIVE -  
ENTER "IGNORE","CANCEL" or "RETRY"
```

Figure 163. Starting DFSMSrmm Interface Inactive Message

Reply RETRY to this message. In response, DFSMSrmm initializes its subsystem interface. If you reply IGNORE, the DFSMSrmm tape recording function is not activated.

Before you can start your tape activities, run inventory management as follows:

1. Run vital record processing first before expiration and storage location management processing to identify which volumes to retain and where volumes should be moved, based on vital record specifications.
2. Run expiration processing to identify those volumes not required for vital records that are ready to expire. During expiration processing, release actions for volumes are noted.
3. Run storage location management to assign shelf locations in storage locations to volumes that are being stored as vital records.
4. You must run storage location management processing after vital record processing has been successfully run, but not necessarily in the same run of EDGHSKP. Storage location management assigns the exact shelf location to be used for the volume while it is in the storage location.

10.3.1 First Run of EDGHSKP with PARM(VRSEL)

The first run of EDGHSKP vital record selection processing will run longer than all subsequent runs of EDGHSKP. The reason for this discrepancy in runs is that DFSMSrmm must update all volume and data set records to set the correct vital record status and retention information.

Note: The first run will also generate many records in the journal data set. Prepare for this by allocating a large journal data set, or process the first EDGHSKP PARM(VRSEL) run with the journal data set disabled.

In the EDGRMMxx PARMLIB member you can define the JOURNALFULL parameter. If it is not specified the value defaults to 75, which means that when the journal data set is 75 % filled, the warning message in Figure 164 is written to the system log.

```
EDG2107E JOURNAL THRESHOLD REACHED -  
JOURNAL IS percentage_value% FULL.  
tracks TRACKS (kilobytesK) AVAILABLE
```

Figure 164. Journal Threshold Warning

If you set the value JOURNALFULL(0), DFSMSrmm will not issue any warning messages. See the EDGRMMxx PARMLIB member description on page 428 for an explanation of the JOURNALFULL and the relation to the BACKUPPPROC parameter.

We recommend that you set JOURNALFULL(0) in this stage of the conversion and exploit this function when DFSMSrmm is set in production, after removal of your old tape management product.

When a journal full situation occurs, the messages in Figure 165 are written to the system console:

```
EDG2104E JOURNAL FILE IS FULL - SCHEDULE CONTROL DATA SET
        BACKUP TO CLEAR IT.

EDG2103D PERMANENT JOURNAL ERROR - REPLY 'R' TO RETRY,
        'I' TO IGNORE, 'D' TO DISABLE, OR 'L' TO LOCK.
```

Figure 165. Journal Full Messages

When these messages appear, reply to EDG2103D with 'D', to disable the journal data set. The rest of the EDGHSKP run will then continue without journaling.

10.3.2 Disabling the Journal Data Set

Another possibility is to do the first run of EDGHSKP with the journal disabled from the beginning. The possibility of disabling the journaling depends on how the journal data set is defined in the DFSMSrmm procedure and in the PARMLIB member EDGRMMxx.

If the journal data set is not defined in the DFSMSrmm procedure, and defined in the PARMLIB member EDGRMMxx, it is possible to do the following steps to perform a EDGHSKP PARM(VRSEL) run with the journaling disabled.

1. Create a new member EDGRMMyy by copying the old member EDGRMMxx in PARMLIB.
2. In the new member EDGRMMyy remove the JRNLIBNAME statement in the OPTION parameter.
3. Then restart DFSMSrmm with the new EDGRMMyy member, by issuing the command
F DFRMM,M=yy
4. When DFSMSrmm is restarted without the journal data set the messages in Figure 166 are written in the system log.

```
EDG0204I DFSMSrmm BEING INITIALIZED FROM MEMBER EDGRMMyy IN SYS1.PARMLIB
EDG0122I NO JOURNAL FILE ALLOCATED - JOURNALING DISABLED
EDG0105I DFSMSrmm SUBSYSTEM INITIALIZATION COMPLETE
```

Figure 166. DFSMSrmm Restart Messages

You can also check the status of the journal data set by issuing the following command:

```
RMM LISTCONTROL OPTIONS
```

Figure 167 on page 446 shows the result of the LISTCONTROL command.

```

READY
RMM LISTCONTROL OPTION
System options:
Operating mode = W      Retention period Default = 5 Maximum = 99
                          Catalog = 24      hours
Control data set name   = RMM.PROD.CDS
Journal file data set name =
Journal threshold       = 75%
Scratch procedure name  = EDGXPROC
Backup procedure name   = EDGBKUP
IPL date check = N      Date format = J      RACF support = A
SMF audit = 248         SMF security = 249    CDS id =
MAXHOLD value = 100     Lines per page = 66   System ID = E7
BLP = RMM              TLCS V1 = N          Notify = Y
Uncatalog = Y          VRS job name = 2       Message case = M
MASTER overwrite= ADD   Accounting = J       PARMLIB Suffix = OJ
VRS change = INFO      VRSMIN action = FAIL   VRSMIN count = 10

READY
END

```

Figure 167. DFSMSrmm LISTCONTROL Command Output

The Journal file data set name field should be blank when the journal data set is disabled.

5. DFSMSrmm is now running with the journal data set disabled and the EDGHSKP PARM(VRSEL) job can start. Running EDGHSKP without journaling also reduces the execution time.

When the EDGHSKP run is complete, the journal data set should be activated again. To activate the journal data set do the following:

1. Restart DFSMSrmm with the EDGRMMxx PARMLIB member, with the following command:
F DFRMM,M=xx
2. Backup the DFSMSrmm CDS and the journal data set, with the EDGHSKP PARM(BACKUP) utility.

DFSMSrmm journal data set is now active, nullified, and synchronized with the DFSMSrmm CDS after the backup.

10.4 Validating DFSMSrmm Operation

In this section we show how to validate the DFSMSrmm operation. During the validation phase of the conversion process, you must create reports to compare and verify the correct conversion of the database and policies. We will explain how to obtain reports that are useful for product comparisons.

10.4.1 Data Set Name List of Nonscratch Volumes

In both tape management systems, you should produce a list of the data sets for every nonscratch volume. This list should include:

- Data set name
- VOLSER
- Retention date or equivalent in the vendor product
- Assigned or creation date or equivalent
- Job name
- Volume sequence number
- Data set sequence number

You can get this information with the DFSMSrmm extract file as input, and then use DFSMSrmm delivered REXX execs and JCL to create the report you need. How to create reports is described in chapter Chapter 14, “Extending DFSMSrmm Reporting” on page 533 and in the DFSMSrmm Implementation and Customization Guide.

For CA-1 use the TMSRPT2 utility report 10. If this utility is not available, use the TMSGRW utility to produce equivalent reports.

For TLMS use the TLMSRPTS utility with the TLMS004 report, or the TLMS018 report, which is a customizable report where you can pass the fields you want to include in the output.

When you have a report from DFSMSrmm and a report in identical format from your existing product, you can compare the two report files using a compare utility such as ISPF SuperC (ISPF/PDF option 3.13).

10.4.2 Scratch Volume Lists

A list of scratch volumes should be produced for both products, ordered by VOLSER, and then compared using the SuperC utility.

To obtain this list from DFSMSrmm you can use the following command:

```
RMM SV * STATUS(SCRATCH) OWNER(*) LIMIT(*)
```

Using the IKJEFT01 or IKJEFT1A TSO batch interface program you can store the results of this command in a data set referenced by the SYSTSPRT DD statement. Then you can compare the result with the list obtained with a similar utility or command of the other tape management system. For example, for CA-1 use the TMSCLEAN utility report 5, and for TLMS use the TLMSRPTS utility report TLMS003,ALL.

10.4.2.1 Inventory by Location

You can produce an inventory of the volumes present in every storage location defined and then compare it with an equivalent inventory of the vendor product volumes. Use the DFSMSrmm command:

```
RMM SV * LOCATION(loc_name) STATUS(MASTER) OWNER(*) LIMIT(*)
```

where **loc_name** is the storage location name you want to use for the comparison.

Alternatively, you can create REPORT06 described in 14.2.6, “REPORT06” on page 545 to list the volumes in every storage location sorted by VOLSER.

For CA-1 you can use the TMSVMLST utility report 25, and for TLMS you can use the TLMSRPTS utility with the TLMS006 report or the TLMS018 report appropriately customized.

10.4.3 Multifile Multivolume Reports

A multifile multivolume report is created using REPORT11 described in 14.2.11, "REPORT11" on page 550. The Multifile Multivolume report includes the following information:

- VOLSER
- Volume sequence
- Data set sequence
- Data set name
- Expiration date
- First volume
- Previous volume
- Next volume
- Creating user ID
- Creating job name
- Creating date
- Creating time

The report is sorted by the first file on the first volume, volume sequence, and file sequence.

10.4.4 Comparing Results

You can have differences between your old tape management system and DFSMSrmm in:

- Storage location bins
- Volume expiration date
- Logical and physical file sequence number
- Multivolume chain
- Job name creation
- Scratch lists

You must compare the results of DFSMSrmm inventory management processing with the equivalent processing in your existing product. You can compare the results from the reports produced using a product like ISPF/PDF SuperC.

When you have compared the results of processing of DFSMSrmm and your existing system and have verified that there are no differences, or that the differences are minimal, understood, and acceptable, then you are ready to cut over to production.

10.4.4.1 Storage Location Bins

Storage location management processing assigns a shelf location, known as the bin number, to volumes moving to and among storage locations and sets a destination location. Storage location management processing must be run to manage volume movement, whether or not the moves involve storage locations.

The algorithm used by DFSMSrmm for bin assignment can result in more bin numbers being required than with your existing system. DFSMSrmm processes volumes in ascending VOLSER sequence, and for each volume can confirm a move, freeing a bin number, and initiate a move, assigning a bin number. A

volume later in sequence can use the freed bin number. Free bin numbers are assigned in ascending bin number sequence.

DFSMSrmm does not assign a bin number to more than one volume at a time. Bin numbers are freed only when the move from the location has been confirmed and the volume updated in the DFSMSrmm CDS.

10.4.4.2 Volume Expiration Date

Any nonscratch volume defined to DFSMSrmm has an expiration date indicating when the volume is to be considered for release. This date can be set for the volume by:

- An expiration date or retention period, given in the JCL by the user when writing a data set to the volume
- The DFSMSrmm GETVOLUME command, when a user manually requests a scratch volume, or when information about the volume was manually added or changed
- A default retention period for data sets set by your installation with the RETPD option in EDGRMMxx in SYS1.PARMLIB.

Your installation can also set a maximum retention period in PARMLIB member EDGRMMxx. An expiration date or retention period set for a volume cannot exceed this value.

You can retain a volume beyond its expiration date through a VRS. You can also override an expiration date and release a volume manually before the expiration date is reached.

The volume expiration date is the highest expiration date of any one data set on this volume. The first data set on a volume has no priority to keep or move a volume.

Note: DFSMSrmm can handle mixed expiration dates without any problems. This means you can keep a volume by a VRS and a retention period. For more information see the *DFSMSrmm Guide and Reference*, SC26-4932.

In addition to the volume expiration date, DFSMSrmm calculates and saves a retention date for each data set and volume. The retention date probably is a more accurate match to the expiration or keep date known to your existing system.

10.4.4.3 Logical and Physical File Sequence Number

Logical File Sequence: Specifies the data set sequence number you have to enter on the LABEL JCL parameter for allocating the specific data set without using the catalog entry. The minimum allowable decimal value is 1; the maximum allowable decimal value is 9999.

Physical File Sequence: Specifies the relative position of the data set on the volume. The minimum allowable decimal value is 1; the maximum allowable decimal value is 9999. When you add a data set that is not the first data set on a volume, the preceding data sets on the volume must already be defined to DFSMSrmm.

Your existing system records only the logical file sequence number.

10.4.4.4 Multivolume Chain

DFSMSrmm manages individual data sets using VRS definitions. Each data set on a volume or multivolume set can have different retention criteria. DFSMSrmm does not attempt to manage the multivolume set based on a controlling file, but allows each volume to be retained separately based on the data sets it contains.

As a result, DFSMSrmm can expire and return to scratch one or more volumes from a multivolume set while still retaining the other volumes that have data sets managed by VRS, or that have later expiration dates.

You may have to add extra VRS definitions so that all data sets in a multivolume set are managed using common retention criteria if the data sets on the released volumes should also be retained.

10.4.4.5 Job Name Creation

You can have differences in your job name creation because, in the DFSMSrmm volume record, the JOBNAME is the name of the job that created the first data set on the volume and, in the data set record, the JOBNAME specifies the name of the job that created the data set.

10.4.4.6 Comparing Scratch Lists

You can have differences in your scratch lists because of the PENDING RELEASE status used by DFSMSrmm for volumes that have reached their expiration date and are not retained by VRS.

Prior to the VRS SPE, the first run of EDGHSKP EXPROC processing assigns the pending release status, and the second run moves volumes into scratch status as long as there are no release actions required on the volume.

With DFSMS/MVS 1.4.0 with the VRS SPE installed, the expiration of tapes can be set to immediate in VRS definitions, so some categories of tapes can be released immediately in the first EDGHSKP EXPROC run. However there can still be differences in scratchlists between the tape management systems.

You also can have differences in your scratch list because of volumes kept by an External Data Manager. The volume status of a volume released by DFSMSHsm is set to pending release by the ARCTVEXT/EDGDFHSM exit call and changed to scratch status by the first expiration processing. There is no option for extending the retention date when a volume is released in this way.

10.4.5 Checking Messages

You can check DFSMSrmm messages in your JOB LOG and SYSTEM LOG only if DFSMSrmm is running in WARNING mode. Check all warning messages and correct the error, if necessary. The most important messages are:

EDG4004I ****WARNING**** USE OF volser BY jobname, stepname, ddname ONLY PERMITTED BECAUSE DFSMSrmm RUNNING IN WARNING MODE

Explanation: DFSMSrmm determined that the specified volume should not be used, but allows the volume to be used because it is running in warning mode only. It is preceded by a message listing the error detected.

EDG4025I VOLUME volser REJECTED. READING OF SCRATCH VOLUMES IS NOT PERMITTED

Explanation: The specified volume is either a scratch volume or a scratch volume obtained using the DFSMSrmm TSO GETVOLUME subcommand. The volume cannot be used for input processing unless you write the first file.

Possible cause: The status of the volume in DFSMSrmm does not match the status known to the product from which you are converting. Perhaps the timing of return to scratch differs between the products.

EDG4026I VOLUME volser REJECTED. DATA SET NAME ON FIRST FILE DOES NOT MATCH RECORDED NAME

Explanation: The information recorded by DFSMSrmm for the specified volume does not match the current information recorded on the volume.

Possible cause: The data set name extracted and converted to DFSMSrmm does not match the data set name on the tape. DFSMSrmm validates the first file data set name when the OCEFLG is set to on in the L-Record input to EDGCNVT. You can avoid having DFSMSrmm check the data set name on the first file by either setting the OCEFLG to off in the L-Record at the time of conversion or using the following DFSMSrmm subcommand after conversion:

RMM CV volser INIT(Y) CRLSE(INIT)

EDG4027I VOLUME volser REJECTED. IT IS NOT A SCRATCH VOLUME AND MOUNT REQUEST WAS NON-SPECIFIC

Explanation: A nonspecific tape mount, either PRIVAT or SCRTCH, was issued. The operator mounted a nonscratch volume. A nonscratch volume is one that is not defined to DFSMSrmm or that is defined to DFSMSrmm in either MASTER or USER status. Only a scratch volume that is defined to DFSMSrmm is acceptable.

Possible cause: The status of the volume in DFSMSrmm does not match the status known to the product from which you are converting. Perhaps the timing of return to scratch differs between the products.

EDG4029I VOLUME volser REJECTED. DATA SET NAME DOES NOT MATCH FOR A RECORDED VOLUME

Explanation: The data set name does not match information in the recorded DFSMSrmm CDS during previous I/O operations. The data set names must match for input from a recorded volume. Because they do not match, the file cannot be read, and the specified volume is rejected.

Possible cause: The data set name extracted and converted to DFSMSrmm does not match the data set name on the tape. DFSMSrmm validates all existing data set names when the OCEFLG is set to on in the L-Record input to EDGCNVT. You can avoid having DFSMSrmm check the data set name by either setting the OCEFLG to off in the L-Record at the time of conversion or using the following DFSMSrmm subcommand after conversion:

```
RMM CV volser INIT(Y) CRLSE(INIT)
```

You must now provide the correct data set name information so that the correct retention can be determined using VRS, or you should set the retention period you would like for the volume. The following example shows how to set the first file data set name and add a second file to the volume:

```
RMM CV volser DSN('data set name')  
RMM AD 'data set name2' VOLUME(volser) SEQ(2)
```

EDG4032I VOLUME volser REJECTED. IT HAS EXPIRED AND IS PENDING RELEASE

Explanation: The volume VOLSER has expired and is pending release.

Possible cause: The status of the volume in DFSMSrmm does not match the status known to the product from which you are converting. Perhaps the timing of return to scratch differs between the products.

EDG4033I VOLUME volser REJECTED. THE VOLUME IS WAITING TO BE REINITIALIZED

Explanation: The specified volume must be initialized before it can be used.

Possible cause: The volume has been initialized with the utility provided with the product from which you are converting and you have not informed DFSMSrmm that the initialize has completed. You can use the DFSMSrmm CV volser CRLSE(INIT) subcommand to do this.

EDG4035I VOLUME volser REJECTED. VOLUME IS SCRATCH AND OUTPUT NOT TO FIRST FILE

Explanation: For scratch volumes, files on the volume must be written in sequence. The first file must be written to first. The request was rejected because the file that was being written was not the first file on the volume.

Possible cause: The status of the volume in DFSMSrmm does not match the status known to the product from which you are converting. Perhaps the timing of return to scratch differs between the products.

EDG4036I VOLUME volser REJECTED. REQUEST WAS FOR A SPECIFIC SCRATCH VOLUME

Explanation: The requested volume is a scratch volume and cannot be specifically requested by name. However, a personal volume can be obtained using the DFSMSrmm GETVOLUME subcommand, or by removing the VOL = parameter in the JCL.

Possible cause: The status of the volume in DFSMSrmm does not match the status known to the product from which you are converting. Perhaps the timing of return to scratch differs between the products.

EDG4037I VOLUME volser REJECTED. FILE BEING CREATED DOES NOT FOLLOW LAST FILE RECORDED

Explanation: The file created on the specified output volume does

not follow the last file recorded in the DFSMSrmm CDS. The files must be recorded in sequence. The information recorded by DFSMSrmm might be incorrect, or the request might not be valid because intervening files have not yet been written.

Possible cause: The number of data set names extracted and converted to DFSMSrmm does not match the number of files on the tape. DFSMSrmm validates the number of files when the OCEFLG is set to on in the L-Record input to EDGCNVT. You can avoid having DFSMSrmm check the data set name by either setting the OCEFLG to off in the L-Record at the time of conversion or using the following DFSMSrmm subcommand after conversion:

```
RMM CV volser INIT(Y) CRLSE(INIT)
```

You must now provide the correct data set name information so that the new file can be recorded by DFSMSrmm. The following example shows how to set the first file data set name and add a second file to the volume:

```
RMM CV volser DSN('data set name')  
RMM AD 'data set name2' VOLUM(volser) SEQ(2)
```

EDG4038I VOLUME volser REJECTED. ATTEMPT TO READ FILE THAT HAS NOT YET BEEN RECORDED

Explanation: While opening the specified input volume, DFSMSrmm found that it never recorded the file being read from. The information recorded by DFSMSrmm might be incorrect because the tape was created on another system, or the request might not be valid because the file was never created.

Possible cause: The number of data set names extracted and converted to DFSMSrmm does not match the number of files on the tape. DFSMSrmm validates the number of files when the OCEFLG is set to on in the L-Record input to EDGCNVT. You can avoid having DFSMSrmm check the data set name by either setting the OCEFLG to off in the L-Record at the time of conversion or using the following DFSMSrmm subcommand after conversion:

```
RMM CV volser INIT(Y) CRLSE(INIT)
```

You can now provide the correct data set name information so that DFSMSrmm has the details of all files. The following example shows how to set the first file data set name and add a second file to the volume:

```
RMM CV volser DSN('data set name')  
RMM AD 'data set name2' VOLUM(volser) SEQ(2)
```

EDG4040I VOLUME volser, PREVIOUS IN SEQUENCE TO prev_volser, NOT MANAGED BY DFSMSrmm, jobname, stepname, ddname

Explanation: While recording information about the volume being processed, DFSMSrmm found that the volume was part of a multiple volume sequence, and that the previous volume in the sequence was not managed by DFSMSrmm.

Possible cause: During the extract and conversion of volumes to DFSMSrmm a volume record was skipped, or the volume was never defined to the product from which you are converting.

EDG4041I VOLUME volser REJECTED, DATA SET NAME DOES NOT MATCH FOR A MASTER VOLUME

Explanation: The data set name specified does not match information recorded in the CDS. For output to a master volume, the data set names must match. Because they do not match, the file is not overwritten and the specified VOLSER is rejected.

Possible cause: The data set names extracted and converted to DFSMSrmm do not match the data set names on the tape. DFSMSrmm validates the data set names when the OCEFLG is set to on in the L-Record input to EDGCNVT. You can avoid having DFSMSrmm check the data set name by either setting the OCEFLG to off in the L-Record at the time of conversion or using the following DFSMSrmm subcommand after conversion:

```
RMM CV volser INIT(Y) CRLSE(INIT)
```

You must now provide the correct data set name information so that the correct retention can be determined using VRS, or you should set the retention period you would like for the volume. The following example shows how to set the first file data set name and add a second file to the volume:

```
RMM CV volser DSN('data set name')  
RMM AD 'data set name2' VOLUM(volser) SEQ(2)
```

EDG4047I VOLUME volser IGNORED. IGNORE REQUESTED BY INSTALLATION EXIT

Explanation: The DFSMSrmm EDGUX100 installation exit has requested that DFSMSrmm ignore this volume. DFSMSrmm does not record any information about the specified volume and permits the volume's use because either the user is authorized to request that the volume be ignored, or DFSMSrmm is running in record or warning mode.

Possible cause: The EDGUX100 installation exit has requested that the volume be ignored.

For more information, see *OS/390 MVS System Messages, Vol 2*, GC28-1785.

10.5 Keeping Databases Synchronized

If you run DFSMSrmm in parallel to your old tape management system, try to avoid manual changes. If you must change expiration dates, add or delete volumes, or release volumes in your old environment, you must make the same changes in the DFSMSrmm CDS. Only if you update both tape management systems can you compare them successfully.

If you change DFSMSrmm policies, PARMLIBs, or exits, keep track of them so that they are automatically included if a reconversion is performed. Make all changes through batch and save the TSO DFSMSrmm subcommands.

Chapter 11. Cutover to Production

This chapter explains how to cut over to production.

Objectives of Chapter

- List and explain all tasks required to run DFSMSrmm in production with minimum risk
- Get DFSMSrmm into production and deactivate vendor product
- Ensure that correct tasks are executed at the correct time
- Ensure that cutover will be performed only when ready

Audience

- System programmers
- Storage administrators
- Production controllers
- Librarians

At the end of this chapter you will have DFSMSrmm installed in production and will have removed the old tape management system. You will be ready to exploit additional DFSMSrmm functions or continue business as usual.

11.1 EDGRMMxx PARMLIB Options for Production Cutover

In this section we describe the steps you must follow to set up your system for production cutover and remove the vendor product.

Your DFSMSrmm system is probably running in *RECORD-ONLY* or *WARNING* mode, because during the parallel running and validation phase you had to check for differences and verify the correct conversion of the vendor database.

The next time you start DFSMSrmm you must change several PARMLIB options to use DFSMSrmm with the full set of functions active. Figure 168 on page 456 shows a sample EDGRMMxx PARMLIB member for running DFSMSrmm in protect mode. Parameters that affect protect mode running are highlighted.

OPTION	OPMODE(P)	/* Protect Mode */ -
	ACCOUNTING(J)	/* Accounting from JOB */ -
	BACKUPPROC(EDGBKUP)	/* Name of BACKUP-proc */ -
	BLP(RMM)	/* RMM controls BLP */ -
	CATRETPD(0012)	/* catalog retention */ -
	CDSID(PROD)	/* control data set id */ -
	DATEFORM(J)	/* Date format */ -
	DSNAME(RMM.PROD.CDS)	/* CDS data set name */ -
	IPLDATE(N)	/* IPL date checking */ -
	JRNLLNAME(RMM.PROD.JRNL)	/* JRNL data set name */ -
	JOURNALFULL(75)	/* Percentage JRNL full*/ -
	LINECOUNT(054)	/* Lines per page */ -
	MASTEROVERWRITE(LAST)	/* Overwriting of a vol*/ -
	MAXHOLD(100)	/* Number of I/O oper. */ -
	MAXRETPD(9999)	/* Maximum retention */ -
	MSG(M)	/* case for message txt*/ -
	NOTIFY(Y)	/* Notify volume owners*/ -
	PDA(OFF)	/* PDA is disabled */ -
	PDABLKCT(255)	/* number of blocks */ -
	PDABLKSZ(31)	/* blocksize in K */ -
	PDALOG(ON)	/* PDA output disabled */ -
	RETPD(0005)	/* Default retention */ -
	SCRATCHPROC(EDGXPROC)	/* ATL/MTL procedure */ -
	SMFAUD(248)	/* SMF audit records */ -
	SMFSEC(249)	/* SMF security records*/ -
	SYSID(EGZB)	/* Name of the system */ -
	TLCSV1(N)	/* Call TLCS 0/C/EOV */ -
	TPRACF(A)	/* RACF tape support */ -
	UNCATALOG(Y)	/* Catalog support */ -
	VRSCCHANGE(INFO)	/* No additional action*/ -
	VRSEL(NEW)	/* New VRS process */ -
	VRSJOBNAME(2)	/* DATASETNAME/JOBNAME */ -
	VRSMIN(0000000100,WARN)	/* Warn if < 100 VRSs */ -
	/* ***** */	
MNTMSG	MSGID(IEF233A)	ID(001) VOLUME(016) RACK(016)
MNTMSG	MSGID(IEF233D)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E K')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E R')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEF234E D')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(IEF455D)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(IEC501A)	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E K')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E D')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E R')	ID(001) VOLUME(016) RACK(999)
MNTMSG	MSGID(' IEC502E RD')	ID(001) VOLUME(017) RACK(999)
MNTMSG	MSGID(' IEC502E RK')	ID(001) VOLUME(017) RACK(999)
MNTMSG	MSGID(IAT5110)	ID(001) VOLUME(044) RACK(999)
MNTMSG	MSGID(IAT5210)	ID(001) VOLUME(050) RACK(999)
MNTMSG	MSGID(IAT5410)	ID(001) VOLUME(020) RACK(999)

Figure 168 (Part 1 of 3). Sample EDGRMMxx PARMLIB Member Running Protect Mode


```

/* ***** */
REJECT    ANYUSE(*)
/* ***** */
VLPPOOL   PREFIX(SC*)
          TYPE(S)
          MEDIANAME(3590)
          RACF(Y)
          EXPDTCHECK(N)
          DESCRIPTION('3350 DEFAULT SCRATCH POOL')
VLPPOOL   PREFIX(N*)
          TYPE(R)
          RACF(N)
          EXPDTCHECK(O)
          DESCRIPTION(' TEST POOL FOR NORBERT')
VLPPOOL   PREFIX(F*)
          TYPE(R)
          RACF(Y)
          EXPDTCHECK(Y)
          DESCRIPTION(' FOREIGN CARTRIDGES')
VLPPOOL   PREFIX(PP*)
          TYPE(R)
          RACF(Y)
          EXPDTCHECK(Y)
          DESCRIPTION(' SOFTWARE CARTRIDGES')
/* ***** */
LOCDEF    LOCATION(SAFE)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(SANJOSE)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(BINS)
LOCDEF    LOCATION(NEWYORK)
          TYPE(STORAGE)
          PRIORITY(2000)
          MEDIANAME(*)
          MANAGEMENTTYPE(NO BINS)
LOCDEF    LOCATION(IBMATL1)
          TYPE(LIBRARY)
          PRIORITY(4800)

```

Figure 168 (Part 2 of 3). Sample EDGRMMxx PARMLIB Member Running Protect Mode

```

/* ***** */
SECCLS    NUMBER(010)                -
          NAME(UCL)                  -
          SMF(N)                     -
          MESSAGE(N)                 -
          ERASE(N)                   -
          DESCRIPTION(' UNCLASSIFIED') -
          MASK(**)                   -
SECCLS    NUMBER(050)                -
          NAME(IUO)                  -
          SMF(Y)                     -
          MESSAGE(N)                 -
          ERASE(N)                   -
          DESCRIPTION(' INTERNAL USE ONLY') -
          MASK(**.IUO.**,**.INTERNAL.**,**.INT.**)-
SECCLS    NUMBER(100)                -
          NAME(IC)                   -
          SMF(Y)                     -
          MESSAGE(N)                 -
          ERASE(N)                   -
          DESCRIPTION(' CONFIDENTIAL') -
          MASK(**.IC.**)-
SECCLS    NUMBER(125)                -
          NAME(ICR)                  -
          SMF(Y)                     -
          MESSAGE(N)                 -
          ERASE(Y)                   -
          DESCRIPTION(' CONFIDENTIAL RESTRICTED') -
          MASK(**.ICR.**)-
SECCLS    NUMBER(150)                -
          NAME(RIC)                  -
          SMF(Y)                     -
          MESSAGE(Y)                 -
          ERASE(Y)                   -
          DESCRIPTION(' REGISTERED CONFIDENTIAL') -
          MASK(**.RIC.**)-

```

Figure 168 (Part 3 of 3). Sample EDGRMMxx PARMLIB Member Running Protect Mode

11.1.1 OPMODE PARMLIB Option

Once you have performed all testing, switch the DFSMSrmm operational mode to *PROTECT*.

In *PROTECT* mode, DFSMSrmm is fully operational, that is, it validates all magnetic tape requests and rejects tape volumes when it finds an error, instead of giving a message as in *WARNING* mode.

You must have completely tested your DFSMSrmm environment before switching to production mode. Review carefully Chapter 10, “Parallel Running and Validation” on page 427 and be sure to complete all tasks described in that chapter before proceeding.

Set the OPMODE PARMLIB parameter to:

OPMODE(P)

to allow full operation of DFSMSrmm.

11.1.2 UNCATALOG PARMLIB Option

The default of this parameter, if you have not specified it in the OPTION control statement, is:

UNCATALOG(N), when running in manual, record-only, and warning mode

UNCATALOG(Y), when running in protect mode

You need to change this parameter only if you have previously specified it in the OPTION control statement, because you now use

UNCATALOG(Y) or **UNCATALOG(S)**

DFSMSrmm needs to enforce its policies only when running in protect mode; when OPMODE(P) is set, the UNCATALOG operand default is set to Y, which specifies that data sets should be uncataloged under these conditions:

- When a volume is returned to scratch status, DFSMSrmm uncatalogs all the data sets on the volume.
- When the RMM TSO DELETEVOLUME FORCE command is issued for a volume, DFSMSrmm uncatalogs all data sets on the volume.
- When the RMM TSO CHANGEVOLUME DSNAME command is issued for a volume, DFSMSrmm uncatalogs all data sets on the volume. If the data set name specified on the CHANGEVOLUME command matches the data set name on the volume, DFSMSrmm only uncatalogs subsequent data sets.
- When the RMM TSO DELETEDATASET command is issued for a data set, DFSMSrmm uncatalogs the data set. Also, DFSMSrmm uncatalogs all data sets recorded on the same volume with higher data set sequence numbers.
- When a tape data set is overwritten, DFSMSrmm uncatalogs the data set. Also, all data sets recorded on the same volume with higher data set sequence numbers are uncataloged.

You can also set the UNCATALOG operand to S, which specifies that you will uncataloging data sets only under these conditions:

- When a volume is returned to scratch status, DFSMSrmm uncatalogs all the data sets on the volume.

11.1.3 TPRACF PARMLIB Option

The TPRACF option specifies the type of RACF tape support for DFSMSrmm to provide. If you use RACF, you must change this operand to *TPRACF(P)* or *TPRACF(A)* before going into production to request that DFSMSrmm maintain the security profiles that protect tape volumes.

When you use *TPRACF(P)*, DFSMSrmm ensures that nonscratch volumes have a TAPEVOL profile to protect them if the TAPEVOL class is active. If TAPEDSN is also in use, DFSMSrmm predefines scratch tape profiles for volumes in scratch status.

When you use *TPRACF(A)*, DFSMSrmm expects that tape security profiles are created during processing by options or functions that you use with RACF on your system.

In both cases, DFSMSrmm ensures that when a tape data set is closed, a TAPEVOL profile exists to protect that volume. If one does not exist, DFSMSrmm automatically creates one.

11.1.4 EXPDTCHECK PARMLIB Option

Enabling expiration date checking means that all the volumes in a specified pool will never be overwritten unless the expiration date is passed.

If you did not use expiration date checking before the conversion process, now you must consider changing this parameter to meet your production system requirements.

To enable expiration date checking, use:

EXPDTCHECK(Y) or EXPDTCHECK(O)

in the VLPOOL control statement of the DFSMSrmm PARMLIB member. If you want DFSMSrmm to automatically allow overwriting of unexpired files, specify:

EXPDTCHECK(N).

11.2 Control Access to DFSMSrmm Functions

You must ensure that the correct RACF protection will be in place in a production environment to prevent unauthorized access to DFSMSrmm resources (see Appendix A, "Security Considerations and Implementation" on page 561).

If you have not already done so, review the RACF and DFSMSrmm definitions in effect, and change them as necessary to meet your production needs. Be sure to grant RACF access to the correct users for the

STGADMIN.EDG.RESET.SSI

RACF FACILITY resource that prevents users from disabling the DFSMSrmm subsystem interface. Once in production, you do not want the OPT=RESET option to be used.

11.3 Activate Exits

During the conversion phase, you merged some exits to allow DFSMSrmm to run in parallel with the vendor product and test that the exits work correctly in an DFSMSrmm environment without affecting the production operations of the vendor product.

Now you can remove these merged exits using the DFSMSrmm-supplied version, because in a production environment, you will no longer be using the other tape management product.

If the code of one or all of the following exits has been merged to allow the use of both tape management products, you must consider removing the merged version of the exit and using the DFSMSrmm version when running in production:

ARCTVEXT	DFSMSShsm tape volume exit (only in DFSMSrmm prior Version 1 Release 4)
IGXMSGEX	DFSMSrmm uses the DFSMSdfp MSGDISP exit to update tape drive displays
CBRUXENT	OAM cartridge entry exit
CBRUXEJC	OAM cartridge eject exit
CBRUXCUA	OAM cartridge change use attributes exit

11.4 Remove Other Vendor's Product

Now you are ready to perform the following steps to remove the other vendor's product from the system:

- Deactivate program interfaces from the operating system.
- Stop the other vendor's product.
- Remove the code from the system libraries.

If your previous tape management product installation changed some the IBM modules, you must remove those changes and restore the IBM modules to the latest PTF level before the changes were made.

Use either SMP/E RESTORE processing or the AMASPZAP utility to remove those changes. You may need to IPL to activate the updated code. These modifications can be removed without an IPL if your vendor's product uses dynamic updates to IBM modules. Refer to the installation manual of the vendor's product if you are not sure which method to use.

Once you have removed the interfaces, you can stop tape activity for the vendor's product, and delete the libraries from the system.

Important note:

You can stop the other vendor's product only when you are ready to start DFSMSrmm in production. Stopping the other vendor's product must be done after stopping all tape activity, in a single step. Refer to 11.7, "Starting DFSMSrmm" on page 467 for the correct timing of this activity.

11.4.1 Remove CA-1

To deactivate CA-1 you should:

1. Determine your system password by referring to your local modification to the USERMOD CL05023. By default as shipped, the password is CA1(TMS).
2. Run TMSINIT with the PARM=SOFTIPL option.
3. Respond to the IEFTMS0 WTOR with the system password.

You will receive a message that CA-1 has been disabled.

We recommend that you run as described above for a certain period of time. Then, at some point, schedule removal of the CA-1 versions of the LPA aliases, or simply do not install CA-1 during the next system upgrade.

11.4.2 Remove TLMS

First you must stop TLMS after your normal flow of operation. Stopping TLMS should end TLMS address spaces, allowing tape activities to continue without the knowledge of TLMS. Every time there is tape activity, however, you should see a TLMS message indicating that CA-DYNAM/TLMS is not active. The TLMS hooks that are still installed on the DFSMSdfp modules cause a message to be issued.

The load modules you should check for modifications are:

IGC0001I	csects IFG0196Q and IFG0195H
IGC0005E	csects IFG0552B and IFG0551H
IFG0194A	csect IFG0194K
IGC00020	csect IFG0202F

You might have modified these modules with one of these methods:

- SMP/E USERMOD
- Superzap
- Dynamic zap

Refer to the CA-DYNAM/TLMS installation guide to determine how your TLMS environment was installed.

If you used the SMP/E method, you should now use SMP/E RESTORE to remove USERMODs from the system. Be sure to restore modules to the PTF level they had before they were modified by the hook.

If you used the superzap method to modify the modules, rerun the same zap operation with the VER and REP statements reversed to restore modules to the previous PTF levels.

Using the dynamic zap method should not affect system libraries because changes are made directly to the storage copies of the modules when you start the OMS/MONITOR address space. If you stop TLMS, the next time you IPL MVS, the changes no longer exist.

11.4.3 Modifying SAMS:Disk

You should modify your SAMS:Disk to support only the DFSMSrmm tape management system. To do this, you have to update the SAMS:Disk SYSPARMS.

Change the **TMSCTLEX** from **ADSTH014** to **ADSTH016** value and delete the **TAPCTLEX** entry from your SAMS:Disk PARMLIB.

For more information refer to the SAMS:Disk books.

11.5 Inventory Management

During the parallel running and validation phase, you should have properly customized your inventory management procedures to allow both DFSMSrmm and the other vendor product to update databases on a daily basis and perform retention and movement processing.

In the previous phase, you produced retention and movement reports from both tape management products. You no longer have a need for the vendor product reports, and now you use only the equivalent DFSMSrmm reports.

Because you no longer need to run housekeeping jobs on the vendor's tape management product, if you have a scheduling management system like the IBM Sysplex Operations Manager for MVS/ESA (OPC/ESA) or a similar product, you can remove the unwanted jobs or steps from the schedule plan.

In addition to inventory management jobs, there are other reports that you no longer need, such as retention and movement reports that you previously produced on a regular or periodic basis. Substitute these reports with the

equivalent DFSMSrmm reports, which now will be the only reports you need to produce.

Be sure that you have removed all of the vendor's product jobs from the schedule plan, because they can no longer run once the code of the old tape management product has been removed from the system.

If you have a scheduling management product, you probably changed the daily job scheduling plan to allow both DFSMSrmm and the vendor product to update their databases simultaneously when running in parallel. Now you must delete all references to the jobs of the vendor product, because once in production, these jobs will no longer run.

Production controllers should be aware of these changes to ensure that they will not impact the production scheduling of other jobs.

If you are migrating from CA-1, consider this hint: CA-11, the restart and rerun package, has a CA-1 interface to expire tape data sets that will be recreated on restart. Tapes expired by the CA-11 /CA-1 interface use the CA-1 abend retention period in calculating the expiration dates of the tapes created in the first run.

DFSMSrmm uses the special ABEND VRS retention policy to control tape data sets generated by mistake as result of an abend. You can achieve the same function in the DFSMSrmm environment.

11.6 Interface to System-Managed Tape Libraries

If you have an automated tape library installed in your production system, the only way you can test it before going into production with DFSMSrmm is to run DFSMSrmm in record or warning mode and prevent DFSMSrmm from updating the TCDB catalog. You can do this by activating only the interface provided with the other tape management product, and not running EDGHSKP EXPROC processing, or by running DFSMSrmm in record mode. In record mode DFSMSrmm makes no changes to the TCDB. When you switch DFSMSrmm to production, you now use DFSMSrmm data as input instead of the data from your existing product. A usermod is available to enable TCDB updates to be avoided in other OPMODES.

11.6.1 IBM Tape Libraries

You must use the DFSMSrmm version of the following exits:

CBRUXENT	OAM cartridge entry
CBRUXCUA	OAM cartridge change use attribute
CBRUXEJC	OAM cartridge eject
CBRUXVNL	OAM cartridge volume not in library

If you previously changed one or more of these exits with a merged version to allow DFSMSrmm and the other product to work together, you must now substitute them with the DFSMSrmm version.

If you have not yet tested your tape library with DFSMSrmm, you must test it in production to ensure that everything works correctly. Refer to 11.8, "Final Testing" on page 468 for more information.

11.6.2 Other Tape Libraries

Whatever tape library is installed in your system, DFSMSrmm can support it through a user interface that tells the library which volumes to return to scratch status and which volumes must be ejected and moved to another storage location.

With DFSMSrmm's reporting and command capabilities it is simple to build this interface, using REXX, DFSORT, DFSMSrmm utilities, and TSO RMM subcommands.

You must analyze what the vendor tape library requires as input for retention and movement processing and then customize DFSMSrmm to provide the library with the correct input.

Be sure to keep the vendor tape library catalog and the DFSMSrmm database synchronized to avoid any mismatch that would cause rejection of tape mounts.

11.6.2.1 STK Tape Libraries

Below we provide you with a simple interface example between an STK Silo tape library and DFSMSrmm to eject tapes that need to be moved to another location.

After running DFSMSrmm inventory management (storage location management processing DSTORE), you can create a list of all volumes that are moving by using DFSMSrmm commands. Follow these steps:

1. Use DFSMSrmm commands:

```
RMM SV VOLUME(*) DESTINATION(LOCAL) CLIST(' ','-')
RMM SV VOLUME(*) DESTINATION(REMOTE) CLIST(' ','-')
RMM SV VOLUME(*) DESTINATION(DISTANT) CLIST(' ','-')
RMM SV VOLUME(*) DESTINATION(my_loc) CLIST(' ','-')
```

Note: Unless you exploit the capability of APAR OW30790 and use a pre-allocated RMMCLIST DD name, you can use only one TSO RMM subcommand at a time, because each call to the CLIST option will overwrite the data created before.

The resulting list, which is stored in the RMMCLIST data set, or the data set

prefix.EXEC.RMM.CLIST

consists of VOLSERs only. The data set uses blocked, variable-length records, with a logical record length of 255. Where *prefix* is the TSO user PROFILE PREFIX that the user has currently defined. If you have specified *NOPREFIX*, DFSMSrmm allocates a data set using the RACF userid or the jobname, to avoid using EXEC.RMM.CLIST.

If you are using the TSO RMM subcommands in a batch job, you can set the prefix in front of the TSO RMM subcommand. Figure 169 on page 465 shows sample JCL:


```

//RMMMOVE JOB ....
/*
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
PROF PREFIX(prefix)
RMM SV VOLUME(*) DESTINATION(my_loc) CLIST(' ','-')
/*

```

Figure 169. Using TSO RMM Subcommands in a Batch JOB

2. Convert the list from variable-length records to fixed-length records.

HSC/MVS control statement syntax conventions: The control statement for each utility program consists of a command (indicating the utility function) followed by parameters, as applicable, in 80-character card-image records (see the *HSC/MVS Programmer's Guide, Volume 2* provided with the product).

3. Edit this list and create an EJECT statement by following this example:

```

EJECT VOLSER( -
volser1 -
volser2 -
volser3 -
..... -
) CAP(cap-list)

```

Please be aware that each single EJECT command causes an operator interaction.

4. Run the HSC update utility, **SLUADMIN** (see the *HSC/MVS System Programmer's Guide, Volume 1* provided with the product).

11.6.2.2 The HSC Initialize Cartridge Utility (SLUADMIN)

Figure 170 on page 466 shows sample JCL that you can use to initialize cartridges with the HSC SLUADMIN utility, by using DFSMSrmm EDGINERS processing.

```

//JOBINIT JOB (ACCOUNT),PROGRAMER
//*
//TINIT   EXEC PGM=SLUADMIN,TIME=1440,PARM=MIXED
//TAPE    DD   UNIT=(580,1,DEFER),DISP=NEW
//*                               EDGINERS TAPE UNIT DD
//*
//SLSTAPE DD   UNIT=AFF=TAPE           INIT CARTRIDGES INIT TAPE
//*                               UNIT DD.
//SYSIN   DD   UNIT=VIO,SPACE=(TRK,1) EDGINERS CONTROL DATASET
//SLSPRINT DD  SYSOUT=*                UTILITY MESSAGES
//SLSPRINT DD  SYSOUT=*                EDGINERS MESSAGES
//SLSINIT DD   *
INIT LABEL(SL) VOL(*****))
//*
//SLSIN   DD   *
//*
* WHEN USING THE DFSMSrmm TAPE INITIALIZATION PROGRAM EDGINERS, YOU
* MUST SPECIFY THE 'PROGRAM' PARAMETER ON THE INIT CARTRIDGES
* CONTROL STATEMENT.
INITIALIZE CAP(000) PROGRAM(EDGINERS) OPTION(SCRATCH)
/*

```

Figure 170. JCL for Initializing Cartridges with EDGINERS

11.6.2.3 The HSC Scratch Conversion SLUCONDB Utility

The HSC scratch conversion SLUCONDB program is provided to generate scratch transactions from DFSMSrmm tape management system. SLUCONDB calls the routine SLUDRRMM to read the DFSMSrmm report extract file if RMM is specified in the EXEC JCL statement. This program utilizes the DFSMSrmm EDGRVEXT macro to map the layout of the extract file. The following list shows the fields that are used:

- RVTYPE
- RVVOLSER
- RVLCDATE
- RVNAME
- RVSTATUS
- RVEXPDT
- RVEXPDTO
- RVLABEL

SLUDRRMM processes the extract and passes the volume record information to SLUCONDB, which builds scratch card images to input to SLUADMIN. Volume status scratch (scratch or nonscratch) is then updated in the HSC CDS for each volume record in the extract file.

Dates on the DFSMSrmm report must be in Julian date format (EDGHSKP run with PARM 'DATEFORM(J)'). Tapes listed on the DFSMSrmm report without expiration dates will be skipped by SLUDRRMM. Figure 171 on page 467 is an example of the DFSMSrmm and HSC synchronization JCL.

```

//JOBSCUP JOB (ACCOUNT),PROGRAMMER
//S1 EXEC PGM=EDGHSKP,PARM='RPTTEXT,DATEFORM(J)'
//MESSAGE DD DSN=RMM.MESSAGE.DATASET,DISP=SHR
//REPTTEXT DD DSN=RMM.REPORT.EXTRACT,DISP=SHR
//S2 EXEC PGM=SLUCONDB,PARM=('RMM,SCRPOOL(SL)',
'87100-87110,MIXED')
//SLSTMS DD DSN=RMM.REPORT.EXTRACT,DISP=SHR
//SLSSOUT DD DSN=&&SCUPINPT,DISP=(NEW,PASS),UNIT=,SPACE=DCB=
//SLSPRINT DD SYSOUT=A
//S3 EXEC PGM=SLUADMIN
//SLSIN DD DSN=&&SCUPINPT,DISP=(OLD,DELETE)
//SLSPRINT DD SYSOUT=A
//

```

Figure 171. DFSMSrmm and HSC Synchronization JCL

Note: The SLUCONDB utility, when PARM=RMM is specified, checks in the DFSMSrmm extract file to see if the MEDIANAME of the tapes being returned to scratch is either 3480 or 3490. For example, if you changed the default media name 3480, and you are using CARTS, you will never get your tapes selected as scratch by the utility.

11.6.2.4 Memorex Tape Libraries

Memorex has announced the availability of LMSrmm, which provides integration of the DFSMSrmm tape database with LMS automation tape library control functions.

LMSrmm provides DFSMSrmm support for all models of the Memorex/Telex 5100, 5200, and 5400 ATL product family.

Refer to the Memorex documentation on how to install this product. You will have to install some LMSrmm code and modify the EDGUX100 DFSMSrmm exit.

11.7 Starting DFSMSrmm

You may need to build a final version of the converted DFSMSrmm CDS to ensure that the production environment is in sync with the vendor product at this point.

To obtain a final updated DFSMSrmm database, you may have to rerun the entire conversion process.

By now, you should have a tailored job stream that can be run very quickly to reconvert the latest data. By using a prepared and tested job stream you can reduce to a minimum the elapsed time during which tape activity must be interrupted. The job stream uses as input the vendor database from the point of time where you stopped tape activity, and gives you a correctly converted DFSMSrmm CDS ready to be used in production.

Once you have obtained the final version of the DFSMSrmm CDS, be sure that you have performed all steps previously described in this chapter and detailed in Table 95 on page 468.

<i>Table 95. Steps Required to Go into Production with DFSMSrmm</i>	
1	Review and change DFSMSrmm PARMLIB options
2	Review and change DFSMSshm and other exits
3	Remove vendor product's IBM module modifications
4	Modify or remove inventory management jobs
5	Modify or remove tape management reports
6	Install or modify IBM tape library exits
7	Set up other vendor robots to work with DFSMSrmm
8	Review and modify automatic job scheduling plan
9	Stop tape activities
10	Final reconvert to DFSMSrmm from vendor product
11	Do an IPL if necessary
12	Start DFSMSrmm in production in PROTECT mode

You must ensure that the people who will be using DFSMSrmm in production have sufficient skills to correctly operate the tape management system in the new environment. The education sessions that you scheduled during the conversion phases should have included these people.

11.8 Final Testing

You are now in a production environment with DFSMSrmm started in protect mode. The most important thing to do now is to retest the vital functions of DFSMSrmm and test other functions and interfaces that you could not test before going into production, such as:

- DB2 logging and backup
- RACF tape protection
- Robot interface

The first simple test that you can do to verify DFSMSrmm functions in protect mode is to mount a wrong volume and see whether DFSMSrmm rejects it. Try to mount a specific master volume as input to a scratch request, or a volume with a different label when asking for a specific mount request. The volume mounted for the first test must be rejected by DFSMSrmm with the message:

```
EDG4027I VOLUME volser REJECTED. IT IS NOT A SCRATCH VOLUME
          AND MOUNT REQUEST WAS NON-SPECIFIC
```

and for the second test, it must be rejected with the message:

```
EDG4050I VOLUME volser REJECTED. IT IS NOT EQUAL
          TO THE VOLUME REQUESTED
```

With these simple tests, you can see whether DFSMSrmm will reject any wrong tape mount instead of issuing a warning message as it did while running in warning mode.

11.8.1 DB2

DB2 uses its own retention process for archive logs and image copy of tables and table spaces.

If you want to control DB2 tape activities with VRS policies, you must change the default retention period in the DB2 installation options to a date closer than 99365, which in DFSMSrmm means “never expire.” With this change, you can protect DB2 archive and image copy tape data sets with VRS, masking the data set name as desired.

You must synchronize the DB2 and DFSMSrmm expiration dates to avoid a mismatch between the DB2 system tables and DFSMSrmm CDS information. You can use a VRS definition with UNTILEXPIRED for DB2 archive log files to synchronize the retention period specified in the DSNZPARM DB2 initialization module with DFSMSrmm retention.

You can test the above VRS rules by running DB2 jobs that produce archive logs and image copy data sets on tape. Then, after running DFSMSrmm housekeeping, use the following RMM command:

```
RMM LV volser ALL
```

and check whether the AVAILABILITY field is “VITAL RECORD” and the RETENTION DATE is as desired.

11.8.2 RACF

The DFSMSrmm-RACF interface automatically handles the definition of RACF profiles using either the TPRACF(A) or TPRACF(P) OPTION PARMLIB control statement.

When you activate either of these options, you must test using RACF and DFSMSrmm to verify the correct working of this interface. You can submit test jobs that use a new tape data set name and then check whether the RACF profile has been created, using commands or RACF ISPF panels.

When you delete these data sets using DFSMSrmm commands, any discrete RACF data set profiles should be deleted automatically. Discrete TAPEVOL profiles are managed by DFSMSrmm as data is created and at return to scratch time.

Be sure to check RACF tape profiles for TAPEVOL class, and also DATASET class if TAPEDSN is used.

11.8.3 Robot Interface

If you have an IBM tape library, you can test the DFSMSrmm interface simply by releasing a volume from DFSMSrmm and then verifying that the TCDB record reflects the DFSMSrmm CDS information after running the housekeeping process.

Another test you can do is to use the following command:

```
RMM CV volser LOCATION(loc_name)
```

This will cause the specified volume to be ejected from the tape library with the following OAM message:

CBR3010I Volume *volser* ejected from library *library-name*.
Place in shelf location *shelfloc*.

The TCDB record will be deleted. It will be rebuilt once the volume is inserted into the library using the information from the DFSMSrmm CDS.

If you are using another vendor's tape library you must test the interface used with the library for retention and movement processing. Try using jobs or commands that cause ejection or release of volumes. Use the DFSMSrmm status information to drive input to the software that controls the robot tape library.

11.9 Commitment to DFSMSrmm

If the final testing of DFSMSrmm functions is successful, you can be sure that your conversion is complete. If, however, something goes wrong during the final testing phase, you have two options:

- Go back to the parallel running phase
- Fall back to another vendor product and plan reconversion when ready.

The option you select will depend on how serious the problem is that prevents DFSMSrmm from working correctly in a production environment.

Remember that in both situations you must use an old version of the vendor's product database, which might not be updated to reflect the tape activities since you started DFSMSrmm in production to remove the other tape management product.

You can avoid this situation by testing your production environment on a test system with DFSMSrmm in protect mode and the vendor product removed. You can do this early in the conversion process to anticipate and plan around any problems detected.

If some VRS definitions are wrong and you did not correct them during the parallel running and validation phase, you might have tapes that are assigned to SCRATCH even though they contain valid data for your production system.

Remember that when running DFSMSrmm in PROTECT mode, you cannot read a SCRATCH or PENDING RELEASE tape, so you need a method to read the contents of those tapes for recovery. To recover without fallback, you can use the following DFSMSrmm command if the volume is in SCRATCH status:

RMM CV *volser* STATUS(USER)

Use the following command if the volume is in PENDING RELEASE status:

RMM CV *volser* RETPD(1)

After you have issued these commands you can read either kind of tape.

Using DFSMSrmm commands, you can solve other mismatches in VRSs without the need to return to the other vendor's product. For example, if you have a wrong location name for a set of tapes, you can change the location for those tapes and confirm the movement to DFSMSrmm by issuing the following command:

RMM CV *volser* LOC(*new_loc*) CMOVE

This command will have the volume recorded as being in the *new_loc* location. Without the CMOVE operand, only the destination is set.

If you have many volumes, you can use the command:

```
RMM SV VOL(*) OWN(*) LIMIT(*) LOCATION(wrong_loc)
CLIST('RMM CV ',' DEST(new_loc) CMOVE')
```

Then execute the CLIST generated. Next, you can use either the TSO RMM subcommand:

```
RMM SV VOL(*) LOC(new_loc)
```

or use **EDGRPTD** to generate an inventory report.

If you cannot recover the error situation in your production environment and you have to fall back to the previous environment, you must follow the steps detailed in Table 96.

Table 96. Fallback Plan	
1	Stop DFSMSrmm and stop tape activities.
2	Reinstall vendor product code.
3	Restore the last version of vendor product's database.
4	Add the last tape activities not recorded, if possible.
5	Restore changes made after parallel running (DB2, RACF, robots.)
6	Start the vendor product tape management system.
7	Start DFSMSrmm in record or warning mode.
8	Solve the problem that caused the fallback.
9	Plan another cutover to production.

Remember that timing is the critical factor in deciding whether or not to fall back. For example, if you have had DFSMSrmm in production for five working days, it is difficult to fall back. If you have had DFSMSrmm in production for one hour, the fallback is quite easy.

You need to be aware of the tape activity in your installation. We strongly recommend that you consider looking at a fallback procedure in your planning process, regardless of whether you end up executing the fallback. You need to identify the point where you can back out and develop the fallback plan based on that point. Unfortunately, we are not able to recommend a specific point, because it varies from installation to installation.

If the other vendor tape management program can coexist with DFSMSrmm running in protect mode, we recommend that you continue to use it. We suggest that you test the process first. If you conclude that running DFSMSrmm and the other vendor tape management program in parallel is not going to affect your operating environment, you should proceed with such an approach.

Because DFSMSrmm can operate in several modes, we strongly recommend that you run the conversion process as many times as you like until you are comfortable with the result, before taking DFSMSrmm into production. This approach should help eliminate any concern for fallback from DFSMSrmm, but it does not necessarily mean that you do not have to plan the fallback.

11.10 Final Activities

After your successful DFSMSrmm cutover to production, you can uninstall and delete your old tape management system completely:.

1. Remove the patches.
2. Delete the procedure(s).
3. Delete the JCL.
4. Delete the old data sets.
5. Delete the RACF profiles.
6. Delete the old programs.

Chapter 12. VRS Processing

Objectives of Chapter

- Explain how DFSMSrmm processes data retention and movement policies
- Describe how DFSMSrmm processes vital records based on the EDGRMMxx VRSEL keyword.
- Show how to create a VRS chain to define retention and movement policies.

Audience

- System programmers
- Storage administrators

At the end of this chapter you will understand how DFSMSrmm uses VRSs to determine which data sets and volumes DFSMSrmm should retain, and whether a volume needs to move.

12.1 Vital Record Selection

To customize and use all the possible DFSMSrmm functions and tape data set management variations for your specific needs, it is important to understand how DFSMSrmm works with VRSs, the way DFSMSrmm uses VRSs, and how DFSMSrmm connects retention and tape movement policies to tape data sets and tape volumes.

DFSMSrmm compares the data set, job name, and volume information recorded in the control data set with information in the VRSs to determine which data sets and volumes to retain and what processing is required.

As part of vital record processing, DFSMSrmm checks that data set name masks and job name masks used in vital record specifications are specified according to DFSMSrmm naming conventions.

DFSMSrmm compares the data set, job name, and volume information recorded in the control data set with information in the VRSs to determine which data sets and volumes to retain and the processing required. This includes any volumes with special JCL specified expiration dates used by your installation.

Figure 172 on page 474 is an overview of how VRSEL works.

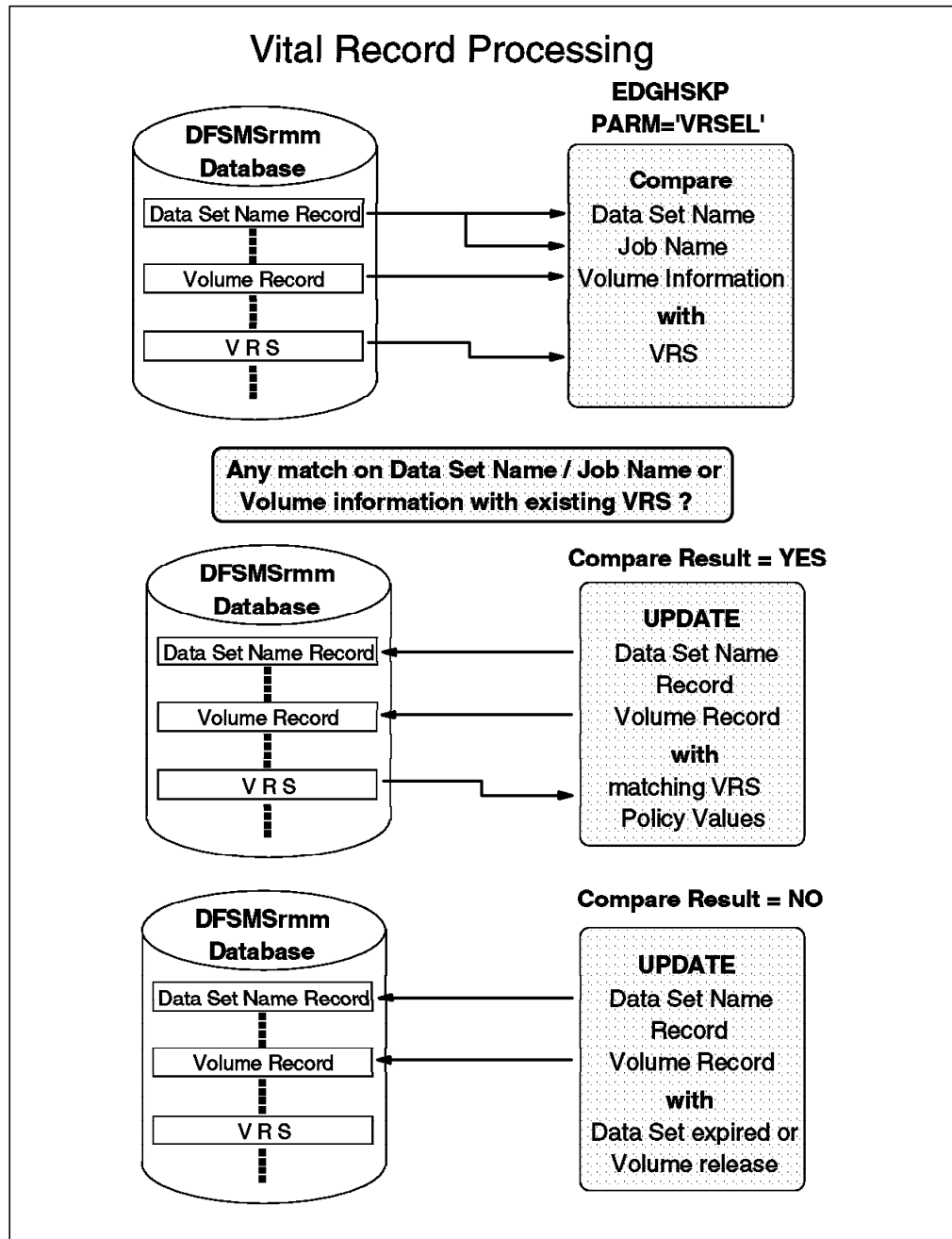


Figure 172. Vital Record Processing

12.1.1 The Best Match Fit Sequence

The next subject we will cover is how DFSMSrmm is designed to obtain a match between tape data sets and customer defined VRSs.

You select the type of vital record processing you want DFSMSrmm to perform by specifying in the EDGRMMxx PARMLIB member the OPTION VRSEL(OLD) or VRSEL(NEW) operand. If you use VRS retention and movement functions that are not supported by VRSEL(OLD), DFSMSrmm ignores the unsupported values, issues the message EDG2227I, and continues to process the remaining policy. If you use VRSEL(NEW), DFSMSrmm processes any VRSs that are available.

Which matching order DFSMSrmm uses depends on how the OPTION command in EDGRMMxx is specified. The default is VRSEL(OLD). Figure 173 on page 475 shows how DFSMSrmm matches data set names to the data set name masks defined in the VRS with VRSEL(OLD). We recommend that you use VRSEL(NEW).

Use the following information together with the *DFSMSrmm Guide and Reference*, SC26-4931, which describes in detail the matching rules.

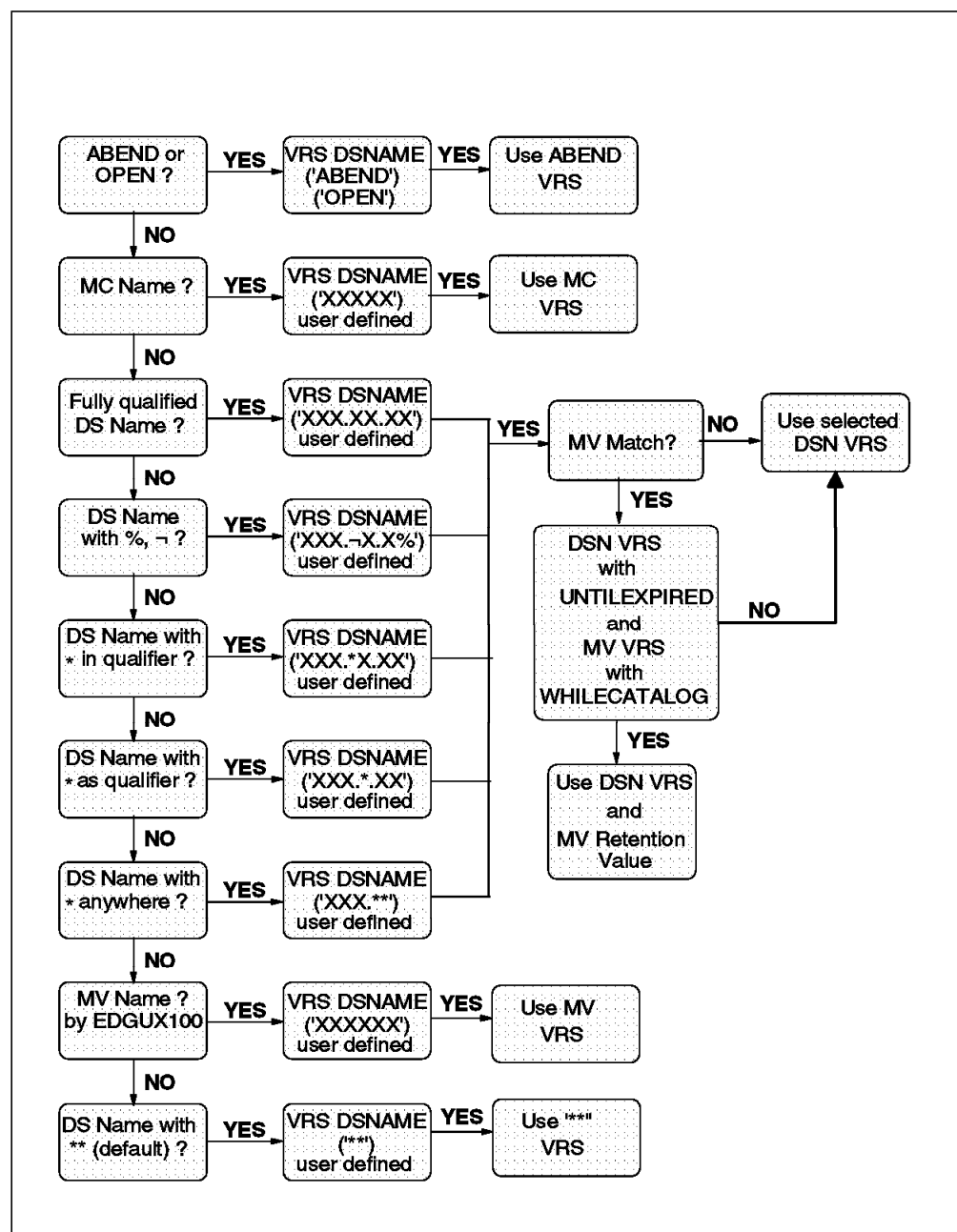


Figure 173. VRSEL(OLD) Best Match Fit Sequence

With VRSEL(OLD), if a data set name matches a data set name mask other than DSN('**'), in a data set VRS which specifies UNTILEXPIRED, and its VRS management value also matches to a VRS, which specifies WHILECATALOG and the data set is still cataloged, the UNTILEXPIRED is effectively replaced by the WHILECATALOG specification. This is the only situation when the retention

information in a management value VRS is used, if the data set names match to a data set VRS other than the default.

If a data set has a management class name assigned to it, and a VRS is defined for that management class name, only the management class VRS is used to determine the retention policy.

Figure 174 on page 477 shows the best match fit sequence with VRSEL(NEW) specified that DFSMSrmm uses to match data sets names to the data set name masks defined in the VRS. This processing is extended to handle any of the retention types that can be specified in a VRS management value. In addition, both the data set name VRS and the management value can be used either separately or together to control the movement and retention of a data set, even if UNTILEXPIRED is not specified on the data set name VRS.

The support for management class is enhanced to exactly match the new support available for VRS management value.

Use the information in Figure 174 on page 477 together with the *DFSMSrmm Guide and Reference*, SC26-4931, which describes in detail the matching rules.

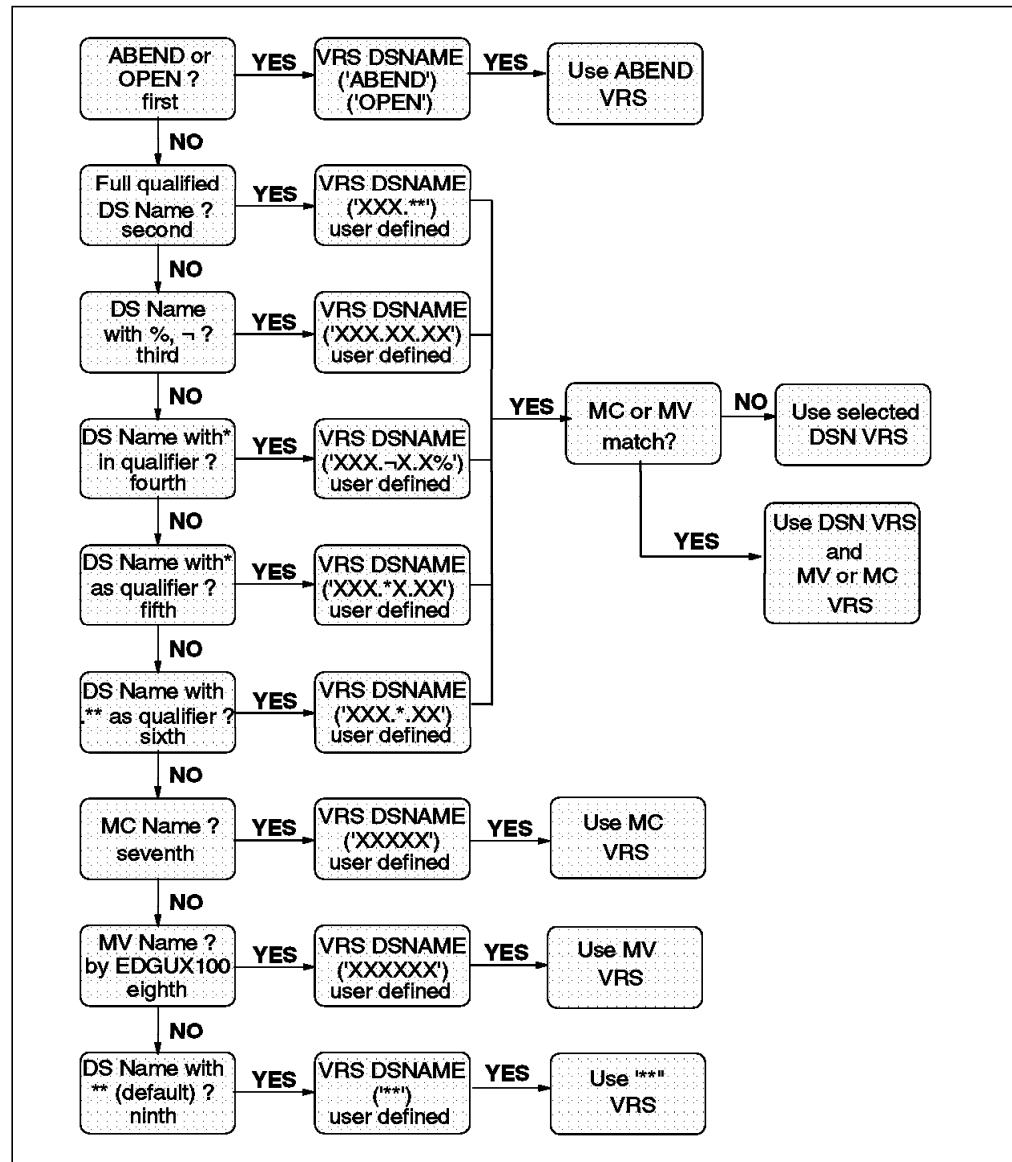


Figure 174. VRSEL(NEW) Best Match Fit Sequence

12.1.2 VRSEL Processing Example

The following example demonstrates the different match fit sequence when a VRS DSNAME and a VRS MV are assigned to a data set.

We first define the following data set name and management value VRSs:

```

RMM ADDVRS DSNAME('A.**') DAYS COUNT(100) +
    LOCATION(REMOTE)
RMM ADDVRS DSNAME(D99002) CYCLES COUNT(2) +
    LOCATION(DISTANT) STORENUMBER(1)
  
```

The following data sets are created and have a management value of D99002.

1. Data set name A.JIM creation date: 120 days ago
2. Data set name A.JIM creation date: 119 days ago
3. Data set name A.JIM creation date: 118 days ago
4. Data set name A.JIM creation date: 90 days ago

5. Data set name A.JIM creation date: 89 days ago
6. Data set name A.JIM creation date: 88 days ago

When VRSEL(NEW) is in use, numbers 4, 5, and 6 are retained in location REMOTE.

When VRSEL(OLD) is in use, numbers 4, 5, and 6 are retained in location REMOTE.

With VRS(OLD) only the data set name VRS is used. The MV VRS is ignored.

Now consider what happens when the VRS management value retains the data longer than the data set name vital record specification. Define the following vital record specifications:

```
RMM ADDVRS DSNAME('A.**') DAYS COUNT(100) +
                                LOCATION(REMOTE)
RMM ADDVRS DSNAME(D99004) CYCLES COUNT(4) +
                                LOCATION(DISTANT) STORENUMBER(1)
```

The following data sets are created and have a management value of D99004.

1. Data set name A.JIM creation date: 120 days ago
2. Data set name A.JIM creation date: 119 days ago
3. Data set name A.JIM creation date: 118 days ago
4. Data set name A.JIM creation date: 90 days ago
5. Data set name A.JIM creation date: 89 days ago
6. Data set name A.JIM creation date: 88 days ago

When VRSEL(NEW) is in use, numbers 4, 5, and 6 are retained in location REMOTE, and number 3 in HOME location.

When VRSEL(OLD) is in use, numbers 4, 5, and 6 are retained in location REMOTE.

With VRS(NEW) the MV VRS is honored when the data set name VRS no longer retains the data set. Both VRS are processed in parallel.

When VRSEL(NEW) is in use, whether UNTILEXPIRED is specified or not, DFSMSrmm always processes the retention and movement policies specified in the primary and secondary VRSs in parallel. DFSMSrmm evaluates both VRSs and then determines which policy to apply. If a policy applies to a data set and retains it, the retention is according to the criteria which is described in Figure 174 on page 477.

If a VRS in the data set name VRS chain specifies UNTILEXPIRED, retention by the secondary VRS determines whether or not the UNTILEXPIRED retention is true when that point in the retention chain is reached. All other retention types specified in the data set VRS subchain must also be true for the data set to be retained.

When VRSEL(NEW) is in use, the volume expiration date is not used during data set name VRS UNTILEXPIRED processing when a data set matches both a primary VRS and a secondary VRS. For example, if a retention type of UNTILEXPIRED is defined in the primary data set name VRS, and in the secondary management value (name vital record) VRS there is a retention type of DAYS with COUNT(30) defined, then the volume will be retained by the

secondary management value VRS. This definition retains the data set for 30 more days before being expired.

If UNTILEXPIRED is used in a secondary VRS, the volume expiration date is used to determine if the retention condition is true or not.

There is no restriction on the definition of either the primary or secondary VRS; both can be vital record specification chains containing any of the possible ADDVRS operands.

Matching VRS information is maintained in data set records while volumes are not released or scratched.

12.1.3 VRS Chain and Subchain

The VRS is key to how DFSMSrmm manages your tape data sets and volumes; retention and movement is driven by VRSs. There are many ways to build VRS chains and subchains. Even if you are able to meet all your retention and movement needs, your approach should be to keep the VRS chain and subchain as simple as possible.

There are some terms you should be familiar with before we discuss building VRS chains and subchains.

Data Set Group All the data sets with the same name that matches a VRS are treated as a single vital retention group.

You can use a GDG base name when defining a VRS to retain volumes. You must not supply the generation data set group suffix. You must specify CYCLES if you want DFSMSrmm to manage the data sets as a data set group. If you are using GDG version numbering, DFSMSrmm only keeps the latest version of each generation.

VRS chain A volume or data set VRS and all of the name VRSs chained from it.

VRS subchain A data set name VRS, NAME VRS with retention information, or AND VRS group and all the VRSs chained from it, up to but not including the next VRS in the chain that contains retention information. Both a VRS chain and subchain may comprise one or more VRSs. When a data set name VRS is chained only to NAME VRSs that have no retention information, there is no difference between a VRS chain and a VRS subchain.

Note: A data set VRS can only be used as the first definition in a chain. A data set VRS cannot be used in the NEXTVRS operand.

Following are examples of a VRS chain and subchain that we use to explain the difference between these concepts.

In the examples we use the default values for the following operands of the ADDVRS subcommand:

COUNT	The default value is COUNT(99999).
DELAY	The default value is DELAY(0).
DELETEDATE	The default value is DELETEDATE(1999/365).

LOCATION The default value is LOCATION(HOME).
PRIORITY The default value is PRIORITY(0).
STORENUMBER The default value is STORENUMBER(99999).

12.1.3.1 Adding a Single VRS

In the first example (see Figure 175) we are using a single VRS to retain and move a data set. The data sets shown in Figure 176 are retained only by this *PRIMARY* VRS.

The tasks for this DSN VRS are as follows::

- Retain 10 cycles of each matching data set group.
- All instances of a data set created on a single calendar day are considered to be a single cycle.
- The newest five cycles must be stored in the outside storage location *SAVE*, and all others in their *HOME* location.

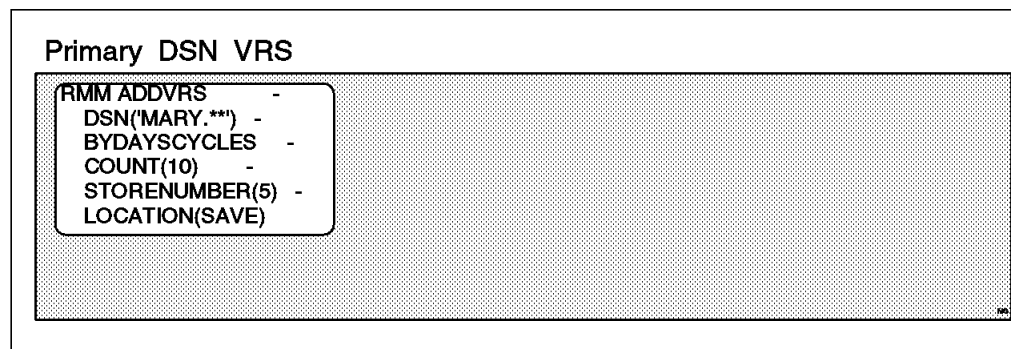


Figure 175. Single VRS Example

Figure 176 shows the data sets and the location retained by the single VRS definition:

data set name	creation date	location	creation occurrence
MARY.TSO.DATA.BACKUP	98123	HOME	-9
MARY.TSO.DATA.BACKUP	98124	HOME	-8
MARY.TSO.DATA.BACKUP	98125	HOME	-7
MARY.TSO.DATA.BACKUP	98126	HOME	-6
MARY.TSO.DATA.BACKUP	98127	HOME	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98127	HOME	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98130	SAVE	-4
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98132	SAVE	-2
MARY.TSO.DATA.BACKUP	98133	SAVE	-1
MARY.TSO.DATA.BACKUP	98134	SAVE	0

Figure 176. Single VRS Retention and Location Results

12.1.3.2 VRS Chain with One NEXTVRS Operand

In the second example (Figure 177) we are using the *NEXTVRS* operand to have an additional *LOCATION* definition. The request for this VRS is almost the same as the one in Figure 175 on page 480; the difference is that the newest five cycles must be always in the HOME location.

The tasks for this VRS specification are as follows:

- Retain 10 cycles of each matching data set group.
- All instances of a data set created on a single calendar day are considered to be a single cycle.
- The newest cycle (0) must be stored in the HOME location. You do not need to specify the *LOCATION(HOME)* operand, because HOME is the default.
- The next five cycles (-1 to -5) must be stored in the outside storage location SAVE.
- All other cycles (-6 to -9) should be retained in the HOME location.

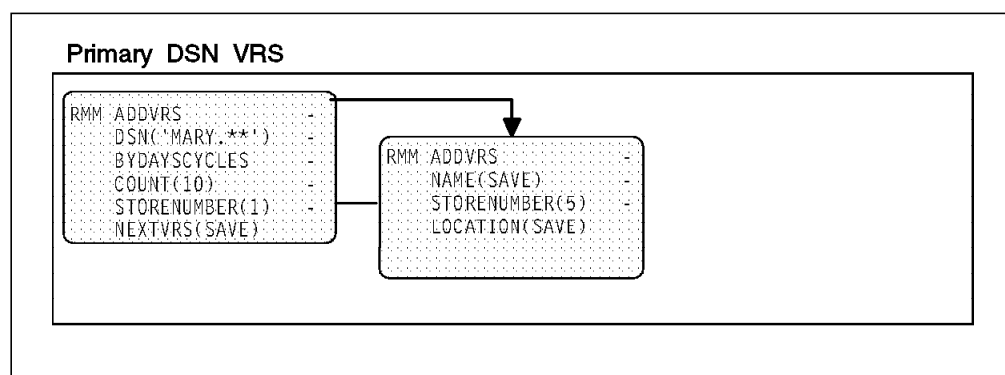


Figure 177. VRS Chain with One NEXTVRS Operand

Figure 178 shows the data sets and their location retained by this VRS definition:

data set name	creation date	location	creation occurrence
MARY.TSO.DATA.BACKUP	98123	HOME	-9
MARY.TSO.DATA.BACKUP	98124	HOME	-8
MARY.TSO.DATA.BACKUP	98125	HOME	-7
MARY.TSO.DATA.BACKUP	98126	HOME	-6
MARY.TSO.DATA.BACKUP	98127	SAVE	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98127	SAVE	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98130	SAVE	-4
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98132	SAVE	-2
MARY.TSO.DATA.BACKUP	98133	SAVE	-1
MARY.TSO.DATA.BACKUP	98134	HOME	0

Figure 178. Single VRS Using the NEXTVRS Operand Results

12.1.3.3 VRS Chain with Two NEXTVRS Definitions

The third example (see Figure 179) shows a VRS with two *NEXTVRS* definitions to have more than one outside *LOCATION* definitions. The only difference between the second example is, that the data sets must be moved to a secondary storage location.

The tasks for this VRS specification are as follows:

- Retain 10 cycles of each matching data set group
- All instances of a data set created on a single calendar day are considered to be a single cycle.
- The newest cycles (0) must be stored in the HOME location. You do not need to specify the *LOCATION*(HOME) operand, because HOME is the default.
- The next three cycles (-1 to -3) must be stored in the outside storage location SAVE.
- The cycles (-4 to -6) must be stored in the outside storage location MUNICH.
- All other cycles (-7 to -9) should be retained in the HOME location.

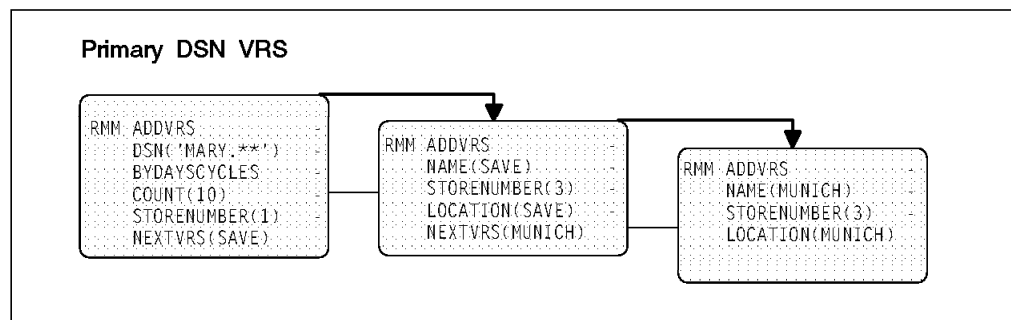


Figure 179. VRS Chain with Two NEXTVRS Operands

Figure 180 shows the data sets and their location retained by this VRS definition:

data set name	creation date	location	creation occurrence
MARY.TSO.DATA.BACKUP	98123	HOME	-9
MARY.TSO.DATA.BACKUP	98124	HOME	-8
MARY.TSO.DATA.BACKUP	98125	HOME	-7
MARY.TSO.DATA.BACKUP	98126	MUNICH	-6
MARY.TSO.DATA.BACKUP	98127	MUNICH	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98127	MUNICH	-5 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98130	MUNICH	-4
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98131	SAVE	-3 BYDAYSCYCLE
MARY.TSO.DATA.BACKUP	98132	SAVE	-2
MARY.TSO.DATA.BACKUP	98133	SAVE	-1
MARY.TSO.DATA.BACKUP	98134	HOME	0

Figure 180. Single VRS Using Two NEXTVRS Operands Results

12.1.3.4 VRS Chain with Two NEXTVRS Operands with Retention Information

Figure 181 shows a management class (MC) or a management value (MV) VRS using two NEXTVRS operands to have more than one retention criterias. All data sets retained by this VRS are retained in the HOME location. This example is a VRS chain with two additional subchains.

The tasks for this VRS specification are as follows:

- Retain a data set for 5 calendar days by specifying the operand DAYS.
- After 5 calendar days, retain the data set as long as it is cataloged by specifying the WHILECATALOGED operand.
- If the data set is no longer cataloged, retain the data set for one additional day by specifying the EXTRADAYS operand.

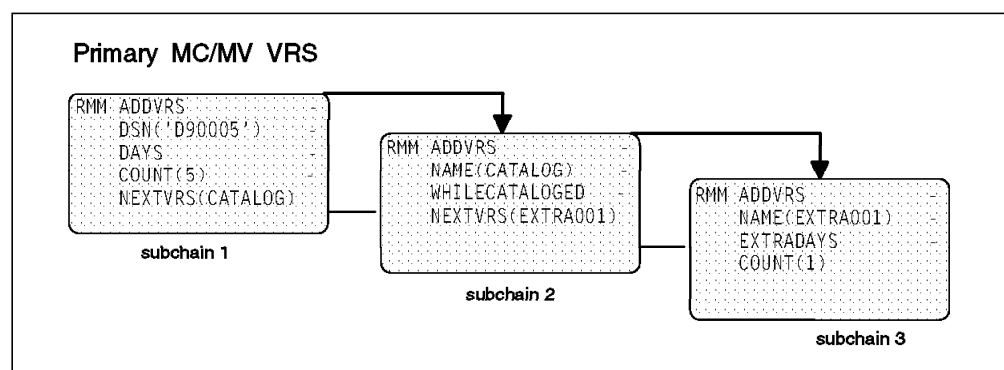


Figure 181. VRS Chain with Retention Information Example

In Figure 182 the generation data group (GDG) is defined with a limit of 10. Generation data set MARY.TSO.DATA.G0011V00 was uncataloged manually yesterday, and the the generation data set MARY.TSO.DATA.G0010V00 was uncataloged manually last week. The first of the two generation data sets MARY.TSO.DATA.G0013V00 was created incorrectly and not cataloged.

data set name	creation date	location	data set cataloged	retention criteria
MARY.TSO.DATA.G0009V00	98123	HOME	Y	WHILECATALOGED
MARY.TSO.DATA.G0010V00	98124	HOME	N	not retained
MARY.TSO.DATA.G0011V00	98125	HOME	N	EXTRADAYS
MARY.TSO.DATA.G0012V00	98126	HOME	Y	WHILECATALOGED
MARY.TSO.DATA.G0013V00	98127	HOME	N	not retained
MARY.TSO.DATA.G0013V00	98127	HOME	Y	WHILECATALOGED
MARY.TSO.DATA.G0014V00	98130	HOME	Y	DAYS
MARY.TSO.DATA.G0015V00	98131	HOME	Y	DAYS
MARY.TSO.DATA.G0015V00	98131	HOME	N	DAYS
MARY.TSO.DATA.G0016V00	98132	HOME	Y	DAYS
MARY.TSO.DATA.G0017V00	98133	HOME	Y	DAYS
MARY.TSO.DATA.G0018V00	98134	HOME	Y	DAYS

Figure 182. Single VRS Using Two Retention NEXTVRS Operands Results

12.1.3.5 VRS Chain with Two ANDVRS Operands

The fifth example (see Figure 183) shows a complex VRS with two ANDVRS operands, in order to have more than one retention criteria for retaining a data set. If you specify ANDVRS, all definitions must be true to retain the data set. If any definition is not true, the data set is no longer retained by this VRS chain.

The tasks for this VRS are as follows:

- Retain a data set for 3 cycles.
- Only do so if the data set has been used in the last 6 days.
- Also, the data set should be moved to the outside storage location STOREX until the volume expiration date is reached.

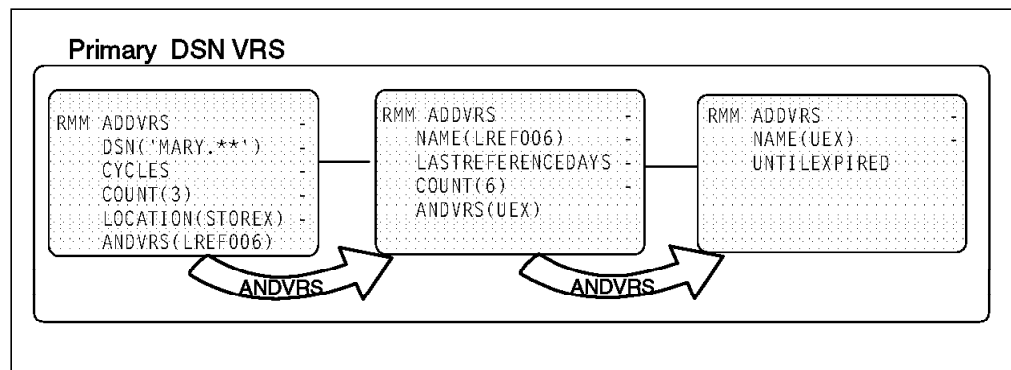


Figure 183. VRS Chain with Two ANDVRS Operands Example

Figure 184 shows the data sets and their location retained by the VRS definition in Figure 183. Only the generation data sets G0016V00, G0017V00 and G0018V00 are retained by the VRS because we specified CYCLES COUNT(3).

data set name	creation date	exp. date	last used date	retention criteria		
				WC	LR	UEx
MARY.TSO.DATA.G0009V00	1996/123	1998/123	1996/123	N	N	N
MARY.TSO.DATA.G0010V00	1996/124	1998/124	1996/124	N	N	N
MARY.TSO.DATA.G0011V00	1996/125	1998/125	1997/231	N	N	N
MARY.TSO.DATA.G0012V00	1997/126	1999/126	1997/126	N	N	Y
MARY.TSO.DATA.G0013V00	1997/127	1999/127	1997/127	N	N	Y
MARY.TSO.DATA.G0013V00	1997/127	2000/127	1998/132	N	Y	Y
MARY.TSO.DATA.G0014V00	1997/130	1999/130	1997/130	N	N	Y
MARY.TSO.DATA.G0015V00	1997/131	1999/131	1998/100	N	Y	Y
MARY.TSO.DATA.G0015V00	1997/131	1999/131	1997/131	N	N	Y
MARY.TSO.DATA.G0016V00	1998/132	2000/132	1998/132	Y	Y	Y <===
MARY.TSO.DATA.G0017V00	1998/133	2000/133	1998/133	Y	Y	Y <===
MARY.TSO.DATA.G0018V00	1998/134	2000/134	1998/134	Y	Y	Y <===

Figure 184. Single VRS Using Two ANDVRS Operands Results

12.1.3.6 VRS Chain with Two Subchains

Figure 185 on page 485 shows a complex VRS with an ANDVRS definition and a NEXTVRS VRS with retention information.

The tasks for this VRS are as follows:

- Retain a data set for 3 cycles.

- Only do so if the data set has been used in the last 6 calendar days.
- If the data set is no longer retained by the first subchain, the data set must be retained for 5 additional days.

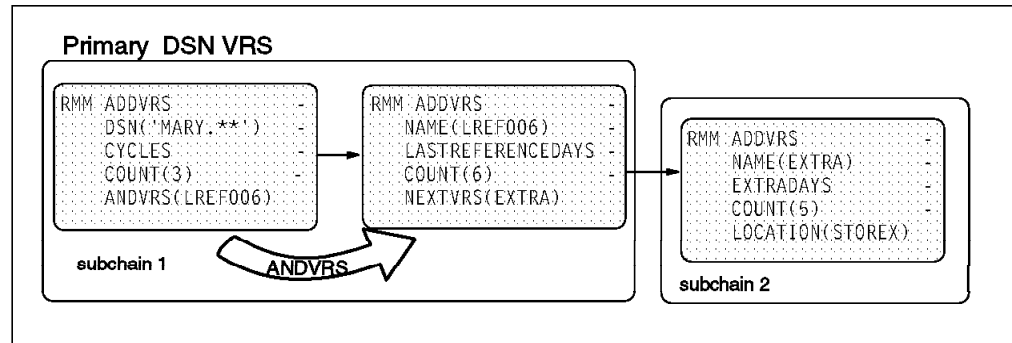


Figure 185. VRS Chain with Two Subchains Example

Figure 186 shows the data sets and their location retained by the VRS definition in Figure 185. The generation data sets G0016V00, G0017V00 and G0018V00 are retained by the first subchain and the generation data set G0015V00 is retained and moved to the storage location STOREX by the secondary subchain.

data set name	creation date	exp. date	last used date	retention criteria		
				CYC	LR	*
MARY.TSO.DATA.G0009V00	1996/123	1998/123	1996/123	N	N	
MARY.TSO.DATA.G0010V00	1996/124	1998/124	1996/124	N	N	
MARY.TSO.DATA.G0011V00	1996/125	1998/125	1997/231	N	N	
MARY.TSO.DATA.G0012V00	1997/126	1999/126	1997/126	N	N	
MARY.TSO.DATA.G0013V00	1997/127	1999/127	1997/127	N	N	
MARY.TSO.DATA.G0013V00	1997/127	2000/127	1998/132	N	Y	
MARY.TSO.DATA.G0014V00	1997/130	1999/130	1997/130	N	N	
MARY.TSO.DATA.G0015V00	1997/131	1999/131	1998/100	N	N	
MARY.TSO.DATA.G0015V00	1997/131	1999/131	1998/131	N	Y	<=== 2
MARY.TSO.DATA.G0016V00	1998/132	2000/132	1998/132	Y	Y	<=== 1
MARY.TSO.DATA.G0017V00	1998/133	2000/133	1998/133	Y	Y	<=== 1
MARY.TSO.DATA.G0018V00	1998/134	2000/134	1998/134	Y	Y	<=== 1

* = retained by VRS subchain

Figure 186. ANDVRS and NEXTVRS with Retention Operand Results

12.1.3.7 Complex VRS Chain with Three Subchains

Figure 187 on page 486 shows a complex VRS with a primary DSN VRS that has two NEXTVRS definitions followed by two subchains with retention information. The second subchain has a NEXTVRS definition and the third subchain is using the ANDVRS operand.

The tasks for this VRS are as follows::

1. The newest 20 cycles are retained by the first subchain.
 - The newest 5 cycles are always in the HOME location.
 - The next 5 cycles are moved to the storage location SAVE.
 - The cycles -10 to -14 are retained in the storage location MUNICH
 - The last 5 cycles are retained in the HOME location.

2. After the data sets are no longer retained by the first subchain, we will retain all data sets for 365 calendar days since creation.
 - The first 60 days, retain the data sets in the HOME location.
 - Then retain the data sets for 14 days in the storage location MAINZ.
 - The last 291 days, retain the data sets in the HOME location.
3. The third subchain retains the data set only under the following conditions:
 - If we have up to 30 cycles.
 - And if the data set has been read or written to in the last 400 calendar days.

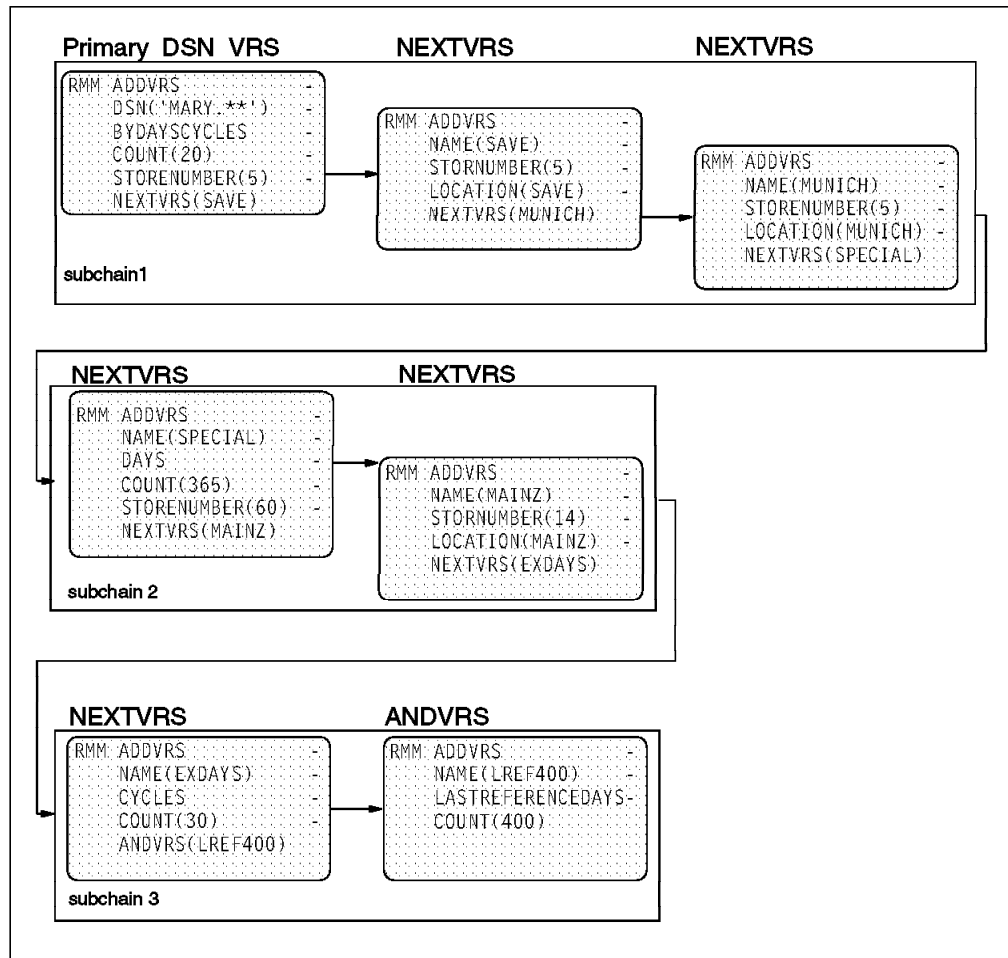


Figure 187. VRS Chain with Three Subchains Example

12.1.4 VRS Examples for Tape Data Set Retention

The following examples explain several VRS constructs and what happens with the tape data set they retain.

The following terminology is used in the table headers of the VRS examples in this section.

Terminology	Meaning
Data Set Age	Data Set Age is the data set sequence in the time period associated with the example. There is no fixed data associated with it.

Sequence Number	The Sequence Number is the relative position of the data set in the example time period.
Retention Day	The Retention Day is the number of the day in the retention time period that the data set was created in the example time period.
Retention Type	The retention type represents the RETENTION TYPE defined by VRS.
Location	The location is the cartridge location at a given point in time. The location is defined by the LOCATION parameter in the VRS.
Held by VRS	This is the VRS holding the data set at the given point in time.

Example 1

```
RMM ADDVRS DSN('WOODY.**') WHILECATALOG LOCATION(HOME) NEXTVRS(DAYS5)
RMM ADDVRS NAME(DAYS5) DAYS COUNT(5) LOCATION(HOME)
```

The data set will be retained for a minimum of 5 days even if it will never be cataloged.

Example 2

```
RMM ADDVRS DSN('GW.**') CYCLES COUNT(3) STORENUMBER(3) NEXTVRS(DAYS3)
RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) LOCATION(HOME)
```

DFSMSrmm retains the most recent 3 cycles and all additional cycles which are not older than 3 days.

Example 3

Example 3 is divided into parts A and B. Part 3A has a clearer structure, as compared to part 3B. Part 3B will have the same approach as part 3A, but it is more complex and is not as readable. Therefore, we recommend that you build VRS chains as described in Part 3A.

Example 3A:

```
RMM ADDVRS DSN('WK.**') CYCLES COUNT(1) LOCATION(HOME) NEXTVRS(REMC1)
RMM ADDVRS NAME(REMC1) CYCLES COUNT(1) LOCATION(REMOTE) NEXTVRS(HOMC1)
RMM ADDVRS NAME(HOMC1) CYCLES COUNT(1) LOCATION(HOME) NEXTVRS(DAYS3)
RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) LOCATION(HOME)
```

Retain 3 cycles and each additional cycle for at least 3 days. Retain the most recent cycle onsite, the next cycle in storage location REMOTE, and all remaining cycles onsite. Table 97 shows how the VRS in Example 3A works.

Table 97 (Page 1 of 2). VRS Example 3A

Data Set Age	Elapsed Days	Retention Type	Location	Held by VRS
Oldest	1	-	Home	The data set is not held by a VRS, the data set expired.
-	2	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + LOCATION(HOME)

Table 97 (Page 2 of 2). VRS Example 3A

Data Set Age	Elapsed Days	Retention Type	Location	Held by VRS
-	3	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + LOCATION(HOME)
-	4	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + LOCATION(HOME)
-	5	CYCLES	Home	RMM ADDVRS NAME(HOMC1) CYCLES COUNT(1) + LOCATION(HOME) NEXTVRS(DAYS3)
-	6	CYCLES	Remote	RMM ADDVRS NAME(REMC1) CYCLES COUNT(1) + LOCATION(REMOTE) NEXTVRS(HOMC1)
Youngest	7	CYCLES	Home	RMM ADDVRS DSN('WK.**') CYCLES COUNT(1) + LOCATION(HOME) NEXTVRS(REMC1)

Example 3B:

```
RMM ADDVRS DSN('WK.**') CYCLES COUNT(3) STORENUMBER(1) NEXTVRS(REMC1)
RMM ADDVRS NAME(REMC1) STORENUMBER(1) LOCATION(REMOTE) NEXTVRS(DAYS3)
RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) LOCATION(HOME)
```

Example 3B contains a NON-RETENTION type name VRS, instead of the RETENTION type VRS shown in Example 3A. Table 98 shows how the VRS chain in example 3B works.

Table 98. VRS Example 3B

Data Set Age	Sequence Number	Retention Type	Location	Held by VRS
Oldest	1	-	Home	The data set is not held by a VRS, the data set expired.
-	2	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + LOCATION(HOME)
-	3	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + LOCATION(HOME)
-	4	DAYS	Home	RMM ADDVRS NAME(DAYS3) DAYS COUNT(3) + 1 LOCATION(HOME)
-	5	CYCLES	Home ²	RMM ADDVRS NAME(REMC1) STORENUMBER(1) + 3 LOCATION(REMOTE) NEXTVRS(DAYS3)
-	6	CYCLES	Remote	RMM ADDVRS NAME(REMC1) STORENUMBER(1) + 4 LOCATION(REMOTE) NEXTVRS(DAYS3)
Youngest	7	CYCLES	Home	RMM ADDVRS DSN('WK.**') CYCLES COUNT(3) + STORENUMBER(1) NEXTVRS(REMC1)

¹ The new retention type DAYS took place after CYCLES(3) has reached three cycles.

² The move has occurred. The storage location is home again.

³ There is still the same retention type in place (CYCLES) from the previous VRS.

⁴ Although this name VRS has no retention type defined, the retention from the previous VRS is still in effect. In this case it is CYCLES. The VRS has a STORENUMBER(1) defined, which means that the data set will stay one day in that storage location. On the next move occurrence, the data set will move by default to HOME.

Example 4

The following example shows a structured VRS chain where each data set is retained in the HOME location for 3 days, then moved to VAULT1 for one CYCLES, then to STOREX for 30 days. Data sets are retained in the STOREX location as long as they are cataloged. Finally, data sets are retained for 2 days in the HOME location after the data set is uncataloged:


```

RMM ADDVRS DSN('PROD.OFF.**') DAYS COUNT(3) +
    STORENUMBER(3) NEXTVRS(V1C)
RMM ADDVRS NAME(V1C) CYCLES COUNT(1) +
    LOCATION(VAULT1) NEXTVRS(STEX)
RMM ADDVRS NAME(STEX) EXTRADAYS COUNT(30) +
    LOCATION(STOREX) NEXTVRS(STEXWC)
RMM ADDVRS NAME(STEXWC) WHILECATALOG +
    LOCATION(STOREX) NEXTVRS(HOLD2)
RMM ADDVRS NAME(HOLD2) EXTRADAYS COUNT(2)

```

Table 99 shows you how DFSMSrmm retains the data set. The example assumes that each day a data set matching the data set name VRS ('PROD.OFF.**') is created, and the data set cataloged for 39 days.

Table 99. VRS Example 4

Data Set Age	Sequence Number	Retention Type	Location	Held by VRS
Oldest	1	-	Home	The data set is not held by a VRS, the data set expired.
-	3	EXTRA-DAYS	HOLD2	RMM ADDVRS NAME(HOLD2) EXTRADAYS COUNT(2) 1
-	8	WHILE-CATAL.	STOREX	RMM ADDVRS NAME(STEXWC) WHILECATALOG + LOCATION(STOREX) NEXTVRS(HOLD2) 2
-	38	EXTRA-DAYS	Home	RMM ADDVRS NAME(STEX) EXTRADAYS COUNT(30) + LOCATION(STOREX) NEXTVRS(STEXWC) 3
-	39	CYCLES	VAULT1	RMM ADDVRS NAME(V1C) CYCLES COUNT(1) + LOCATION(VAULT1) NEXTVRS(STEX) 4
Youngest	42	DAYS	HOME	RMM ADDVRS DSN('PROD.OFF.**') DAYS + COUNT(3) STORENUMBER(3) NEXTVRS(V1C) 5

¹ After the data set is uncataloged, it is retained for 2 more days by DFSMSrmm. After EXTRADAYS is in effect DFSMSrmm no longer has the retention type information set by previous VRSs.

² The data set will be in storage location STOREX as long as it is cataloged, because the retention type WHILECATALOGED is used in the VRS.

³ The data set is retained by EXTRADAYS assigned by the VRS. DFSMSrmm starts counting the days based on the date that the EXTRADAYS VRS retained the data set. DFSMSrmm no longer has any information about the retention type set previously. If you chain a NEXTVRS to a VRS with retention type EXTRADAYS take care that the NEXTVRS has a retention type defined.

⁴ The data set is retained for 1 cycle in the storage location VAULT1. The retention type changed from days to cycles. This change keeps one data set in the vault storage location, even if no more data sets are created.

⁵ The data set is retained for 3 days in the storage location HOME, then moved to VAULT1. The retention type is DAYS with a count of 3.

Example 5

That example shows you how an ANDVRS VRS group compares with the Boolean AND function.

Assume that a data set will be created in ascending sequence and randomly referenced. Three criteria must be true in order for DFSMSrmm to retain the data set. The criteria are the number of cycles that exist, when the data set was last referenced, and if the data set is cataloged.

```

RMM ADDVRS DSN('ZERB.PRV.**') COUNT(2) CYCLES ANDVRS(ZER)
RMM ADDVRS NAME(ZER) COUNT(6)) LASTREFERENCEDAYS ANDVRS(WHCTL)
RMM ADDVRS NAME(WHCTL) WHILECATALOGED

```

Table 100 on page 490 shows you how the VRS will retain data sets.

Table 100. VRS Example 6

Data Set Age	Sequence Number	Retention Type	Location	Held by VRS
Oldest	1	-	HOME	The data set is not held by a VRS, the data set expired.
-	2	-	HOME	The data set is not held by a VRS, the data set expired.
-	3	-	HOME	The data set is not held by a VRS, the data set expired.
-	4	CYCLES	HOME	RMM ADDVRS DSN(' ZERB.PRV.**') + 1 COUNT(2) CYCLES ANDVRS(ZER) RMM ADDVRS NAME(ZER) COUNT(6) + LASTREFERENCEDAYS ANDVRS(WHCTL) RMM ADDVRS NAME(WHCTL) WHILECATALOGED
Youngest	5	CYCLES	HOME	RMM ADDVRS DSN(' ZERB.PRV.**') + 1 COUNT(2) CYCLES ANDVRS(ZER) RMM ADDVRS NAME(ZER) COUNT(6) + LASTREFERENCEDAYS ANDVRS(WHCTL) RMM ADDVRS NAME(WHCTL) WHILECATALOGED

¹ Combining the three results with the Boolean AND function, the result would be to retain data sets 4 and 5. DFSMSrmm checks each data set against all combined conditions before considering the data set for retention, so if you define multiple VRSs we recommended you use the ANDVRS function.

Table 101 can be used to determine if a ANDVRS retains a data set.

Table 101. ANDVRS Boolean AND Function Chart			
Data Set Name	CYCLES	LAST-REFERENCED	WHILE-CATALOGED
1. DSN referenced 1 day ago	N	Y	N
2. DSN referenced 12 day ago	N	N	Y
3. DSN referenced 5 day ago	N	Y	Y
4. DSN referenced 3 day ago	Y	Y	Y
5. DSN referenced 1 day ago	Y	Y	Y

Example 7: This example shows how you use the RELEASE option to return a cartridge early. When data sets are created by jobs thatabend, the tape data sets are to be retained for two days and then returned to scratch as quickly as possible. The default retention period in PARMLIB member EDGRMMxx is 5 days. Normally the data set would be retained as long as it is cataloged. The VRSs for this example are defined as follows:

```
RMM ADDVRS DSNAME(' ABEND') DAYS COUNT(2) +
      LOCATION(HOME) +
      RELEASE(EXPIRYDATEIGNORE SCRATCHIMMEDIATE)
RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOG
```

When created successfully, the data sets have an expiration date of 5 days from the creation date as a result of the default retention, and are cataloged. If the application creating the data set abends, the data set must not be retained. The ABEND VRS retains the data set for 2 days to allow the application to validate the data, and then the RELEASE options override the volume expiration date to set the volume pending release and return the volume immediately to scratch during the first inventory management run.

Figure 188 on page 491, Table 102 on page 491, and Table 103 on page 491 show how the release parameter and the *best match fit sequence* work if you define the VRSs for example 6.

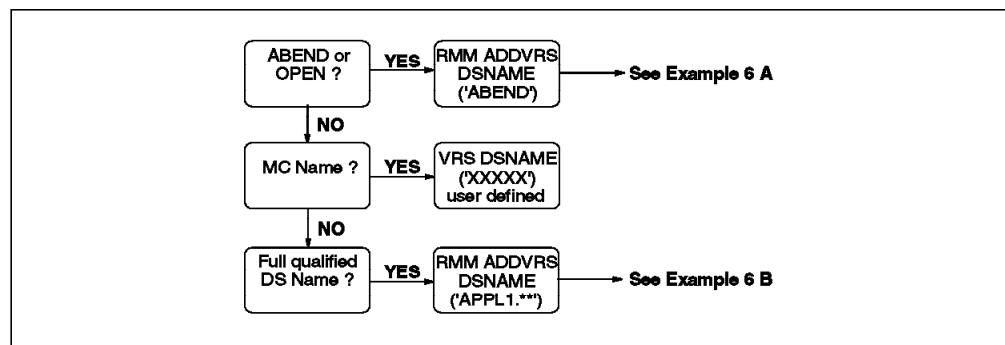


Figure 188. Best Match Fit Sequence

Table 102. VRS Example 6A

Data Set Age	Retention Day	Retention Type	Location	Held by VRS
Oldest	1	-	HOME	The data set is not held by an VRS, the data set expired
-	2	CYCLES	HOME	RMM ADDVRS DSNAME(' ABEND') DAYS COUNT(2) + 1 LOCATION(HOME) + RELEASE(EXPIRYDATEIGNORE SCRATCHIMMEDIATE)
Youngest	3	CYCLES	HOME	RMM ADDVRS DSNAME(' ABEND') DAYS COUNT(2) + 1 LOCATION(HOME) + RELEASE(EXPIRYDATEIGNORE SCRATCHIMMEDIATE)

¹ The ABEND DSNAME VRS is the first match in the sequence. The ABEND VRS assigns the retention type CYCLES and ignores the default expiration date, as well as returning the cartridge immediately to scratch.

Table 103. VRS Example 6B

Data Set Age	Retention Day	Retention Type	Location	Held by VRS
Oldest	1	-	HOME	The data set is not held by a VRS,the data set expired.
-	2	WHILE-CATALG.	HOME	RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOGED 1
-	3	WHILE-CATALG.	HOME	RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOGED 1
-	4	WHILE-CATALG.	HOME	RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOGED 1
-	5	WHILE-CATALG.	HOME	RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOGED 1
Youngest	6	WHILE-CATALG.	HOME	RMM ADDVRS DSNAME(' APPL1.**') WHILECATALOGED 1

¹ The DSNAME VRS is the first match in the sequence. The DSNAME VRS assigns the retention type WHILECATALOGED. The default cataloged period is 5 days; after that time, the cartridge returns to scratch.

Example 7: Example 7 describes a tape data set retention scenario where a VRS is assigned by the data set name, and a VRS by the EDGUX100 (management value) exit. You should be aware of which EDGRMMxx PARMLIB

option, VRSEL(NEW) or VRSEL(OLD), is in effect. This example requires VRSEL(NEW).

Using the following VRS definitions, data sets with a high-level qualifier of A are created and cataloged. The data sets are retained for a maximum of 10 days in the HOME location until the data set expires. The data set expiration date is specified in the JCL using EXPDT=nnnnn. Use the EDGUX100 exit to define a vital record management value and to assign the value to the data set. When data sets have reached their expiration date or after 10 days, the data sets are moved to the storage location MAINZ for 5 more days. The data sets finally move to the storage location WARWICK, where they remain until they are no longer cataloged. The data sets return to the HOME location when they are no longer cataloged. Table 104 provides more detail.

```
RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS COUNT(10) NEXTVRS(D2)
RMM ADDVRS NAME('D2') EXTRADAYS COUNT(5) LOCATION(MAINZ)
RMM ADDVRS DSNAME('MV**') WHILECATALOGED LOCATION(WARWICK)
```

Table 104. VRS Example 7A

Data Set Age	Retention Day	Retention Type	Location	Held by VRS
Oldest	3	-	HOME	The data set is not held by a VRS, the data set expired
-	5	WHILE-CATALOG.	HOME	RMM ADDVRS DSNAME('MV**') WHILECATALOGED + 1 LOCATION(WARWICK)
-	-	WHILE-CATALOG. EXTRA-DAYS	MAINZ	RMM ADDVRS DSNAME('A.**') EXTRADAYS COUNT(5) + 1 LOCATION(MAINZ) RMM ADDVRS DSNAME('MV**') WHILECATALOGED + LOCATION(WARWICK)
-	10	WHILE-CATALOG. EXTRA-DAYS	MAINZ	RMM ADDVRS DSNAME('A.**') EXTRADAYS COUNT(5) + 1 LOCATION(MAINZ) RMM ADDVRS DSNAME('MV**') WHILECATALOGED + LOCATION(WARWICK)
-	-	WHILE-CATALOG. UNTIL-EXPIRED	HOME	RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS + 1 COUNT(10) NEXTVRS(D2) RMM ADDVRS DSNAME('MV**') WHILECATALOGED + LOCATION(WARWICK)
-	-	WHILE-CATALOG. UNTIL-EXPIRED	HOME	RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS + 1 COUNT(10) NEXTVRS(D2) RMM ADDVRS DSNAME('MV**') WHILECATALOGED + LOCATION(WARWICK)
-	20	WHILE-CATALOG. UNTIL-EXPIRED	HOME	RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS + 1 COUNT(10) NEXTVRS(D2) RMM ADDVRS DSNAME('MV**') WHILECATALOGED + LOCATION(WARWICK)

¹ With the DFSMSrmm option VRSEL(NEW) DFSMSrmm, both VRss are processed in parallel. When the VRS assigned by data set name no longer has the data set under retention, the management value VRS assigned by the EDGUX100 exit retains the data set as long as defined. Both VRs have movement direction included.

Table 105 on page 493 is an example where a data set is uncataloged and expires before the date specified in the VRS.

In example 7B, the data set is uncataloged and expires after 7 days in the HOME storage location. The data set then moves to the MAINZ storage location, where the data set is retained for 5 days. After 5 days, the data set returns to the HOME location.

Table 105. VRS Example 7B

Data Set Age	Retention Day	Retention Type	Location	Held by VRS
Oldest	1	-	HOME	The data set is not held by a VRS, the data set expired.
-	6	EXTRA-DAYS	MAINZ	RMM ADDVRS DSNAME('A.**') EXTRADAYS COUNT(5) 1 LOCATION(MAINZ)
-	-	WHILE-CATALOG. UNTIL-EXPIRED	HOME	RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS + 2 COUNT(10) NEXTVRS(D2) RMM ADDVRS DSNAME('MV*') WHILECATALOGED + LOCATION(WARWICK)
Youngest	13	WHILE-CATALOG. UNTIL-EXPIRED	HOME	RMM ADDVRS DSNAME('A.**') UNTILEXPIRED DAYS + 2 COUNT(10) NEXTVRS(D2) RMM ADDVRS DSNAME('MV*') WHILECATALOGED + LOCATION(WARWICK)

¹ The data set is still retained by the VRS assigned by the data set name. The VRS assigned by EDGUX100 exit does not retain the data set since the data set is uncataloged.

² The data set is uncataloged and expired.

³ The data set is retained by both VRSs, the VRS assigned by data set name and the VRS assigned by EDGUX100 (management value) exit.

As you can see from the examples in this chapter, DFSMSrmm gives you the ability to setup your tape data set retention and movement policies to satisfy your installation requirements. DFSMSrmm allows you to change your tape data set retention and movement policies at any time.

12.2 Inventory Management Trial Run

DFSMSrmm provides the EDGHSKP utility to help you perform inventory management. The VERIFY parameter performs a trial run of vital record processing, so you can see how vital record processing changes affect data set and volume information before the CDS is updated. After adding or changing VRSs, run VERIFY and use the output to confirm your new retention and movement policies achieve the expected results. Figure 189 contains sample JCL for inventory management trial run processing.

```
//HSCP    EXEC PGM=EDGHSKP,
//        PARM='VRSEL,VERIFY'
//MESSAGE DD DISP=SHR,DSN=HSCP.MESSAGES
//REPORT  DD DISP=SHR,DSN=HSCP.VRS.REPORT
//ACTIVITY DD DISP=SHR,DSN=HSCP.ACTIVITY
```

Figure 189. Sample JCL for Inventory Management Trial Run Processing

The ACTIVITY file and REPORT file must be pre-allocated and cataloged. To use the VERIFY function, VRSCCHANGE(VERIFY) must be specified in the PARMLIB OPTION statements. The VRSCCHANGE operand default value is VERIFY.

With VRSCCHANGE(VERIFY) specified, each change in the VRS definitions must be followed by a EDGHSKP run, with VERIFY specified in the EXEC PARM of EDGHSKP. DATE can also be specified in the EXEC PARM to further exploit the features of the VERIFY command.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information about the VERIFY function.

Chapter 13. Additional DFSMSrmm Function

In this chapter we describe what you can do to gain additional benefit from functions that DFSMSrmm provides. We also explain how to merge and split DFSMSrmm CDSs, clean up your retention policies, and use DFSMSrmm with system-managed tape and BTLS.

Objectives of Chapter

- Explain how to enhance the use of DFSMSrmm functions
- Clean up the DFSMSrmm implementation
- Customize DFSMSrmm for use with system-managed tape and BTLS
- Merge and split the DFSMSrmm CDS

Audience

- System programmers
- Storage administrators

At the end of this chapter you will be able to gain additional benefit from DFSMSrmm functions. You will be ready to merge and split the DFSMSrmm CDSs, clean up your retention policies, and use system-managed tape or BTLS.

13.1 Using DFSMSrmm Functions

Not only does DFSMSrmm have a different database format from other media management products, it also records different information and provides different functions.

DFSMSrmm exploits its VSAM database format by enabling you to define resources to DFSMSrmm at any time without having to reformat the database. As long as there is free space in the database, or the database can extend in size, you can define new resources to DFSMSrmm. You can define and delete volumes at any time without having to take DFSMSrmm out of service.

You can also use the additional information that DFSMSrmm records and the different functions it provides to improve your media management services. In this chapter we explain ways in which you can use DFSMSrmm functions.

13.1.1 Movement Control

DFSMSrmm tracks the current location and destination of each volume and whether the volume is in transit to the destination. This function enables you to better track where volumes are and to use different movement cycles for each of your locations. For example, you might send volumes offsite on a daily basis to location DRVAULT, but only weekly to location SITEX. DFSMSrmm provides you with the capability to control and track these movements independently.

You can use the RMM CHANGEVOLUME subcommand to confirm tape movement. Figure 190 on page 496 lists the commands you would execute each day to confirm the daily movements once they have completed.

```
RMM CHANGEVOLUME * CMOVE(SHELF,DRVAULT)
RMM CHANGEVOLUME * CMOVE(DRVAULT,SHELF)
```

Figure 190. CHANGEVOLUME Commands for Daily Moves to DRVAULT

Figure 191 lists the commands you would execute each week to confirm the weekly movements once they have completed.

```
RMM CHANGEVOLUME * CMOVE(SHELF,SITEX)
RMM CHANGEVOLUME * CMOVE(SITEX,SHELF)
RMM CHANGEVOLUME * CMOVE(SHELF,DRVAULT)
RMM CHANGEVOLUME * CMOVE(DRVAULT,SHELF)
```

Figure 191. CHANGEVOLUME Commands for Weekly Moves to SITEX

During DFSMSrmm inventory management, the movement status of volumes is updated according to the move confirmations you have performed using DFSMSrmm commands. Those moves that you do not confirm are left outstanding and will be aggregated with any future moves that are identified by DFSMSrmm inventory management each time you run it.

Use the inventory management RPTEXT option to produce a report extract file and DFSMSrmm EDGRPTD utility to produce movement picking list reports.

You can also use the RMM SEARCHVOLUME subcommand or the DFSMSrmm dialog search volume function to list selected moving volumes at any time. For example, this command produces a list of all volumes currently moving to location SITEX:

```
RMM SEARCHVOLUME VOLUME(*) OWNER(*) LIMIT(*) DESTINATION(SITEX) INTRANSIT(Y)
```

The volume destination is set during DFSMSrmm inventory management DSTORE processing according to the policies defined in VRSs. In this way DFSMSrmm can automatically keep track of where volumes are moving and match volume locations with your defined policies. If you want to override the automatic movement, whether to cancel a move that is in progress or to assign a volume to a different location, you can use the RMM CHANGEVOLUME subcommand with the LOCATION operand and optionally the MANUALMOVE operand. To cancel a volume move, issue one of the following commands:

```
RMM CHANGEVOLUME volser MANUALMOVE
or
RMM CHANGEVOLUME volser LOCATION(current_loc)
```

where:

volser

is the VOLSER.

current_loc

is the volume's current location, for example, SHELF.

Once a volume is under manual move control, DFSMSrmm no longer automatically moves the volume according to the movement policies you have defined. You must initiate all moves with the RMM CHANGEVOLUME command or place the volume back under automatic move control by using the following command:

```
RMM CHANGEVOLUME volser AUTOMOVE
```


See the *DFSMSrmm Guide and Reference*, SC26-4931, for more information on managing volume movement.

As part of the conversion effort, you will already have set in place the necessary actions to confirm that movement of volumes has been completed. If not, you are likely to rapidly exhaust the scratch pool.

If you want to implement a movement tracking system that more closely matches the actual shipment of volumes, you can now do that. How you implement such a system depends on the facilities you have at your disposal.

If you simply want to change the confirmation of movement to be a different set of moves on different days, you could proceed as follows:

1. Place the DFSMSrmm commands you plan to use for confirming moves into a data set. Place each required set into a different member of a partitioned data set (PDS). For example, the commands in Figure 190 on page 496 and Figure 191 on page 496 are placed into members CMDAY and CMWEEK, respectively.
2. Create a procedure to execute the commands and the DFSMSrmm inventory management DSTORE process (see Figure 192).

```
//CMOVE   PROC TYPE=DAY
//TMP     EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD DISP=SHR,DSN=RMM.COMMANDS(CM&TYPE.)
//*
//HSCP    EXEC PGM=EDGHSKP,PARM=DSTORE
//MESSAGE DD DISP=SHR,DSN=RMM.MESSAGE
//        PEND
```

Figure 192. CMOVE Procedure

3. Use your scheduling system to submit a batch job that executes the CMOVE procedure with the correct keyword specified for TYPE (see Figure 193).

```
//STSGWD1 JOB (111,E1C,M4031MB),'CMOVE',CLASS=M,
//        NOTIFY=STSGWD,MSGCLASS=X,MSGLEVEL=(1,1)
//CM      EXEC CMOVE,TYPE=WEEK
```

Figure 193. CMOVE Job

4. Your scheduling system can be triggered to submit the job when the movement of the volumes has been completed.

If you have facilities such as a barcode scanner and all volumes are labeled with a barcode that matches the VOLSER, you could perhaps confirm moves at the volume level. For example, you could:

1. Use the EDGRPTD movement reports as a picking list to pick the volumes.
2. As each volume is picked, scan the barcode label.
3. Use the list of scanned VOLSERs as input to a process on your system to issue a TSO RMM command for each volume:

```
RMM CHANGEVOLUME volser CMOVE
```

When the command is issued, the volume information is updated in the DFSMSrmm CDS. There is no need to run DSTORE processing, which is required only when a global confirm move command is issued.

13.1.2 Foreign Tape Management

DFSMSrmm can manage all the volumes in your installation, including those that do not follow your naming convention. The only volumes that cannot be defined to DFSMSrmm are those that are duplicates of volumes already defined to DFSMSrmm.

DFSMSrmm records both a VOLSER and a rack number for each volume you have. These values do not have to be the same. For example, you can have volume DZ100B stored in rack number MW0005. The volume's external label would record the rack number MW0005. Also, at any time, you can define additional rack numbers and volumes without reformatting the DFSMSrmm CDS.

During the conversion it is likely that all volumes were defined with rack number identical to the VOLSERS. Now you can extend the span of control you have for tape volumes by adding to DFSMSrmm the volumes you have not previously managed. Typically these volumes are those that are sent to you either from another site or by a software supplier or customer.

Your objective should be to have DFSMSrmm manage all tape volumes that it can. As you define these foreign volumes to DFSMSrmm, you can have DFSMSrmm determine which empty shelf space is available to store the volume. For DFSMSrmm to provide this function, empty rack numbers must be defined to DFSMSrmm, and you may optionally have to direct DFSMSrmm as to which range of rack numbers to use.

When you add a volume to DFSMSrmm, by default, DFSMSrmm attempts to use the rack number that matches the VOLSER. If there is no match, you can direct DFSMSrmm to a particular pool of shelf space. You define pools of shelf space by using the VLPOOL definitions in the DFSMSrmm PARMLIB. Once the pool is defined and DFSMSrmm is using the definition, you can define empty rack numbers using the range of numbers covered by that pool. You can add a single rack number or any other quantity you want at any time.

DFSMSrmm manages foreign tapes no differently from any other tapes. It records data set names, provides VRS retention and movement controls, and has expiration release options to help you identify how to remove the foreign tapes from your library. When a volume mount is requested, DFSMSrmm updates the mount message and tape drive display with the rack number of the volume for easier retrieval by the operator.

To take advantage of DFSMSrmm's foreign tape management, you need to set up some DFSMSrmm PARMLIB options, define rack numbers, and change your tape library procedures to use a new check-in process. For example:

1. Decide on a range of rack numbers to be used for foreign tapes that enter your installation. Consider whether you want to have a separate range for each type of media. If you have both tape reels and tape cartridges to manage as foreign tapes and want each to have its own range of rack numbers, you need multiple pool definitions.

2. Define the pools. In this example we have selected PR as the prefix for reels and PC as the prefix for cartridges (you add the VLPOOL definitions to the DFSMSrmm PARMLIB member):

```
VLPOOL DESC('foreign tape reels') EXPDTCHECK(Y) -  
  MEDIANAME(REELS) PREFIX(PR*) RACF(N) TYPE(R)  
VLPOOL DESC('foreign tape cartridges') EXPDTCHECK(Y) -  
  MEDIANAME(CARTS) PREFIX(PC*) RACF(N) TYPE(R)
```

3. Restart DFSMSrmm with the new PARMLIB member, using the following operator command, assuming the member name is EDGRMM01:

```
F DFRMM,M=01
```

4. Define some empty rack numbers in each pool (we have chosen to start with space for 50 reels and 500 cartridges):

```
RMM ADDRACK PR0001 COUNT(50)  
RMM ADDRACK PC0001 COUNT(500)
```

5. Now, when a tape volume is brought into your installation, you can define it to DFSMSrmm and add it to either POOL(PR*) or POOL(PC*), depending on the type of media. For example:

```
RMM ADDVOLUME DZ100B POOL(PC*) MEDIANAME(CARTS) STATUS(MASTER) -  
  RETPD(90) RELEASEACTION(RETURN)
```

In this example volume DZ100B will be retained for 90 days and will then be released. The release action, RETURN, prevents the volume from going to the scratch pool. You can use it to identify volumes that should be removed from the library and either returned to their owner or destroyed.

6. To identify volumes waiting for RETURN release action you can use the SEARCHVOLUME subcommand:

```
RMM SEARCHVOLUME VOL(*) OWNER(*) LIMIT(*) STATUS(RELEASE) ACTION(RETURN)
```

Use the list generated to pull the volumes from the library and return them to their owner. Once the volume is removed from the library you can confirm to DFSMSrmm that it has been removed. Use the following command to confirm the release action for volume DZ100B:

```
RMM CHANGEVOLUME DZ100B CRLSE(RETURN)
```

When the RETURN action is confirmed, the volume is deleted from the CDS, and the rack it occupied is now empty.

13.1.3 Product Volume Management

Many of the foreign tapes will contain software products sent by the software company. Using the foreign tape support described in 13.1.2, "Foreign Tape Management" on page 498 you can now define and manage these volumes. However, DFSMSrmm goes a stage further for these software product tapes. You can define the product to DFSMSrmm, and DFSMSrmm then tracks the volume as a product volume and provides search capabilities to retrieve product volume information.

You define each product or product level to DFSMSrmm using the RMM ADDPRODUCT TSO subcommand. You can then use either the RMM ADDVOLUME subcommand or the DFSMSrmm Add Product Volume dialog to define product volumes to DFSMSrmm.

You can use the DFSMSrmm search product command and dialog options to quickly retrieve information about products and associated volumes. Your

system support people can use this information to determine which volume to use during software installation. You can use the DFSMSrmm retention management functions to ensure that the volume can be tracked and removed from the library when they are no longer required.

See the *DFSMSrmm Guide and Reference*, SC26-4931, for more information on defining software products.

To take advantage of product volume management, you need to set up some DFSMSrmm PARMLIB options, define rack numbers, define products to DFSMSrmm, and change your tape library procedures to use a new check-in process. For example:

1. Decide on a range of rack numbers to be used for product volumes that enter your installation. Consider whether you want to have a separate range for foreign and product volumes. If you plan to use the same range, just follow the steps defined in 13.1.2, "Foreign Tape Management" on page 498. Otherwise you need to define a new VLPOOL for your product volumes.

2. Define the pool. In this example we have selected PP as the prefix for product volumes. You add the VLPOOL definitions to the DFSMSrmm PARMLIB member.

```
VLPOOL DESC('software products') EXPDTCHECK(Y) -  
    MEDIANAME(CARTS) PREFIX(PP*) RACF(N) TYPE(R)
```

3. Restart DFSMSrmm with the new PARMLIB member, using the following operator command, assuming the member name is EDGRMM01:

```
F DFRMM,M=01
```

4. Define some empty rack numbers in the pool (we have chosen to start with space for 100 volumes):

```
RMM ADDRACK PP0001 COUNT(100)
```

5. Define the products you have or plan to have in your installation. You can use the ADDPRODUCT subcommand or the Add Product dialog to add products. If you use the Add Product dialog, when you add volumes, it checks to see whether the product is already defined. If the product is not defined, the dialog takes you through the definition steps as well as adding the volume. An example of adding a product by command is:

```
RMM ADDPRODUCT 5695-DF1 NAME('DFSMS/MVS') OWNER(WOODY) -  
    LEVEL(V01R03M00) DESCRIPTION('DFSMS/MVS Release 3')
```

6. Now, when a product volume is brought into your installation you can define it to DFSMSrmm and add it to POOL(PP*). For example:

```
RMM ADDVOLUME DZ11C0 POOL(PP*) MEDIANAME(CARTS) STATUS(MASTER) -  
    RETPD(90) RELEASEACTION(RETURN) -  
    NUMBER(5695-DF1) LEVEL(V01R03M00) FEATCD(1234)
```

In this example volume DZ11C0 will be retained for 90 days and will then be released. It is associated with product 5695-DF1. The release action RETURN prevents the volume from going to the scratch pool. You can use it to identify volumes that should be removed from the library and either returned to their owner or destroyed.

7. To identify volumes associated with a product, you can use the SEARCHPRODUCT subcommand:

```
RMM SEARCHPRODUCT NUMBER(5695-DF1) LIMIT(*)
```

Use the list generated to identify the volume for the level and feature code for which you are searching.

13.1.4 Notification

Sometimes it is useful to have the tape management system tell you of an event that has occurred or will occur. For example, you might want to know when a particular volume has been checked into the library or that one of your volumes has expired.

DFSMSrmm provides a notification facility that can send electronic messages to an identified user when such events occur. DFSMSrmm uses the TSO TRANSMIT command to send these messages, so they can be sent to any system that supports the receipt of messages or mail using that format, such as another TSO system or an OfficeVision/VM system.

Two specific events can result in electronic notification:

- Adding a product volume
- Expiration of a volume that has the NOTIFY release action.

DFSMSrmm has predefined messages for use in each situation, but you can customize these messages as described in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932,

You might want to exploit the notification function for foreign tapes and product volumes that are checked into the library.

The DFSMSrmm PARMLIB option, NOTIFY, controls whether DFSMSrmm performs the notification. If NOTIFY(N) is set, DFSMSrmm does not send any messages. The NOTIFY release action can still be used, but the notification must be done by you and the notify release action must be confirmed to DFSMSrmm to allow release processing to be completed.

Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for more information on the use of notify.

To use the notification function you must activate the NOTIFY option in PARMLIB and define an electronic address for users. Follow these steps:

1. Update the DFSMSrmm PARMLIB member OPTION command to include the NOTIFY operand:
OPTION NOTIFY(Y)
2. Restart DFSMSrmm with the new PARMLIB member, using the following operator command, assuming the member name is EDGRMM01:
F DFRMM,M=01
3. You must define an electronic address for those users to whom you want to send notify messages. The electronic address is defined to DFSMSrmm in the OWNER information. When a product volume is added, DFSMSrmm uses the product owner's OWNER information, and when a volume is released, DFSMSrmm uses the volume owner's OWNER information.

To define an electronic address for OWNER WOODY whose electronic address is WOODY at WARWICK issue the DFSMSrmm command:

```
RMM ADDOWNER WOODY DEPT('department name') USER(WOODY) NODE(WARWICK)
```

or, if the OWNER is already defined,
RMM CHANGEOWNER WOODY USER(WOODY) NODE(WARWICK)

Repeat this for each owner who will use the notify function.

4. For each product defined to DFSMSrmm, ensure that the product owner is the person who is to receive the notification. For example, this person might be the system programmer who will install the product.
5. Review the text of the messages that DFSMSrmm issues and follow the instructions in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, to customize them if necessary.

You are now ready to exploit the notify function. When you add new product volumes, the product owner is notified.

6. When you check in a tape volume to the library, consider specifying the NOTIFY release action as well as the RETURN release action so that the owner is notified when the expiration date is reached.

13.1.5 Tape Replacement Management

During conversion, any permanent and temporary I/O error counts were converted to the equivalent field in DFSMSrmm. When DFSMSrmm is running it tracks permanent and temporary I/O errors that occur on tape volumes and records the counts in the volume information. When one or more permanent errors are recorded, DFSMSrmm identifies the volume for replacement. DFSMSrmm sets the volume REPLACE release action. When the volume is no longer retained and has reached its expiration date, the volume release action must be activated.

DFSMSrmm does not let the volume return to scratch status for reuse until the replace action has been confirmed to DFSMSrmm and the new volume labeled.

Use of the replace action for permanent I/O errors is forced by DFSMSrmm, although you can say you replaced the volume but not actually do so. To further exploit the DFSMSrmm tape replacement management function, you may want to replace a volume after a certain number of temporary I/O errors have been recorded, or perhaps once the volume is older than an installation-selected date.

Extending the use of the REPLACE action requires you to decide when you want volumes replaced and then develop the process by which the replacement rules are implemented.

Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for more information on the use of replace.

You can use the information in the DFSMSrmm report extract file or any other information such as EREP reports or the IBM Service Director reports. For example, select volumes from the extract file that have high instances of temporary I/O errors.

When you have selected a volume to be replaced, you can use the RMM CHANGEVOLUME to identify the volume for replacement on release:

```
RMM CHANGEVOLUME volser RELEASEACTION(REPLACE)
```

To identify the volumes waiting to be replaced, use the search capabilities of the RMM TSO command or the DFSMSrmm ACTION dialog:

```
RMM SEARCHVOLUME VOL(*) OWNER(*) LIMIT(*) STATUS(RELEASE) ACTION(REPLACE)
```

Physically replace the volume with a new volume that has the same external label. Confirm to DFSMSrmm that the volume has been replaced:

```
RMM CHANGEVOLUME volser CRLSE(REPLACE)
```

DFSMSrmm marks the volume with the INIT action. You now should use the DFSMSrmm EDGINERS tape labeling utility to label the replacement volume.

13.1.6 Automatic CDS-Driven Tape Erasing and Labeling

The DFSMSrmm tape erase and initialization utility, EDGINERS, can use action information from the DFSMSrmm CDS, and it requires little or no operator intervention. You can benefit from scheduling the utility to run on a regular basis. Some events require that initialize or erase actions be performed on tape:

- Adding new scratch volumes
- Replacing volumes as a result of I/O errors or aging
- Erasing confidential data
- Relabeling a tape to create a standard label after it has been used for BLP output or as an NL tape

Other than the adding of new scratch volumes, these actions are identified for execution when a volume has expired and is pending release for return to scratch or to its owner.

These actions must be performed and confirmed to DFSMSrmm. The EDGINERS utility can both perform these actions and confirm to DFSMSrmm that they are completed. You will have to regularly check to see whether any of these actions is outstanding and run EDGINERS to perform them. Alternatively, you could schedule EDGINERS to run on a regular basis. If there are pending actions, it will perform them, but if there is none, it will do nothing.

You can schedule EDGINERS to run at a time when you have operators available to mount volumes. The operator responds to the mount messages by mounting the requested volume. In most cases operators do not have reply to a WTOR; when a volume contains data but has no VOL1 label, however, they have to confirm that the correct volume is mounted.

You do not have to use EDGINERS to perform the initialize and erase actions. You can use any equivalent utility or facility, but, if you do, you must confirm to DFSMSrmm that the action has been performed. If you initialize or erase a volume without telling DFSMSrmm what you have done, DFSMSrmm will reject the volume the next time it is used.

Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for more information on the use of the INIT and ERASE actions. Refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for more information on the use of EDGINERS.

Implement a regularly scheduled run of EDGINERS to take care of the INIT and ERASE release actions:

1. Consider the type of tape processing activities and events that occur on your systems that could result in INIT or ERASE actions.

2. Decide on the best time to schedule the EDGINERS utility to be run. Pick a time when operators are available to mount tapes and to respond to a WTOR if one is issued. If you have a facility to automatically mount tapes, you still need to consider the need for an operator to reply to the occasional WTOR, unless you have an IBM Automated Tape Library Dataserver.
3. Schedule a job like that shown in Figure 194 to be run under the control of your job scheduling system.

```
//STSGWDI JOB (111,E1C,M4031MB),'INERS',CLASS=M,  
//      NOTIFY=STSGWD,MSGCLASS=X,MSGLEVEL=(1,1)  
//INERS EXEC PGM=EDGINERS,  
//      PARM='COUNT(99),MEDIANAME(CARTS),INIT,ERASE,NOVERIFY'  
//SYSPRINT DD SYSOUT=*  
//TAPE DD UNIT=(3490,,DEFER)
```

Figure 194. Running EDGINERS Automatic Processing

Because the TAPE DD statement identifies a specific device type on which the volumes are to be mounted, you should specify processing options, such as MEDIANAME or POOL, to identify only the volumes that can be processed on that device type. If you have multiple types of media, you need to schedule a job to be run for each type.

With this job in place, volumes that expire can be returned to scratch status with minimal involvement on your part.

13.1.7 Owner Information

DFSMSrmm records the owner of all nonscratch volumes. The owner is the RACF userid of the task creating the first file on the volume, or the job name if the task is not RACF defined. You can use the owner ID to search for volumes and data sets belonging to particular users.

You do not have to define owner IDs to DFSMSrmm, as DFSMSrmm automatically defines the owner IDs as volumes are used if you do not predefine the IDs.

If you need to contact the owner of a volume or data set or other DFSMSrmm-managed resource, you must associate the owner ID with the person. You may already have a process or tool for doing this association, but you can use DFSMSrmm to record details of the owner ID along with contact information. You can define name, address, telephone, and electronic mail information. See Section 13.1.4, "Notification" on page 501 for how DFSMSrmm can use the electronic mail information.

You could use the owner information to contact the user or for input to reports or labels that you generate. Another way in which you could use owner IDs is to define loan location names as owner IDs. In this way you can keep details of a contact at the location and even use the owner IDs for input to printing mailing information. The printing function is not provided entirely by DFSMSrmm. DFSMSrmm provides the information, and you can use it when you create procedures and processes to help you manage volumes.

To use owner ID information you must define the information to DFSMSrmm. You do this by using the ADDOWNER or CHANGEOWNER subcommands.

Exploitation of owner information is dependent on your producing procedures that retrieve and use it.

For those users about whom you want to keep detailed information, you must add or modify the owner information. For example:

```
RMM ADDOWNER WOODY DEPT('RMM Development') USER(WOODY) NODE(WARWICK) -  
  ADDR1('Room 37B') ADDR2('Building 70B') -  
  FNAME(Mike) SNAME(Wood) -  
  INTEL('276-6822') EXTEL('1-408-256-6822')
```

or, if the OWNER is already defined,

```
RMM CHANGEOWNER WOODY USER(WOODY) NODE(WARWICK) -  
  ADDR1('Room 37B') ADDR2('Building 70B') -  
  FNAME(Mike) SNAME(Wood) -  
  INTEL('276-6822') EXTEL('1-408-256-6822')
```

If you have an existing database of contact information, you should be able use it to build the RMM subcommands and execute them in a batch job rather than add them manually through the DFSMSrmm dialog.

13.2 Cleaning up the DFSMSrmm Implementation

Cleaning up the DFSMSrmm implementation consists of:

1. Reducing VRS policies through exploitation of data set name masks
2. Centralizing control of retention and movement through VRS
3. Deleting volumes that no longer exist but were included in the conversion

13.2.1 Reduce VRS Policies

Vendor tape management systems have only recently started to provide generic or filter capability for defining data set names for retention and movement. DFSMSrmm has always included this capability. During conversion, most of the retention and movement policies being converted use one of the following:

- A fully qualified data set name
- A partially qualified data set name
- A data set name index

By using data set name masks as filters you can significantly reduce the number of VRS policies required as long as there is some commonality to the data set names used. For example, Figure 195 shows a single mask that is equivalent to all of the data set names defined in Figure 196 on page 506.

DAILY.BACKUP.V*

Figure 195. A Data Set Name Mask

```
DAILY.BACKUP.VVOL001  
DAILY.BACKUP.VVOL002  
DAILY.BACKUP.VVOL003  
DAILY.BACKUP.VVOL004  
DAILY.BACKUP.VVOL005  
DAILY.BACKUP.VVOL006  
DAILY.BACKUP.VVOL007
```

Figure 196. Fully Qualified Data Set Names

This is a very simple case, but through the same principles and use of the %, *, -, and ** generic characters, you can build masks in many ways.

You can only manage data sets through a common data set name filter if the data sets have identical retention and movement needs.

When you define a new VRS to DFSMSrmm, and the data set name mask is less specific than the mask that is already defined to DFSMSrmm, it is not used. So, as part of the consolidation you can define VRSSs with data set names without deleting the existing VRSSs. You can then run DFSMSrmm inventory management VRSEL processing and check the retention report to see whether any of the new VRSSs are used. If any of the new VRSSs are used, you are bringing additional data sets under VRS control. This may not be desirable, so you need to review the retention report carefully to see whether the correct results are obtained. It might be that these data sets should have been retained, but through an oversight in the past, they were not.

Once you are satisfied with the report, you can clean up the old VRSSs by deleting them from DFSMSrmm. Rerun the DFSMSrmm inventory management VRSEL processing to generate a report that shows which data sets are being retained by which VRS. The report should now show that the data sets covered by the VRSSs you deleted are covered by the new, generic VRSSs you defined.

Having generic VRSSs that cover many different data sets with the same retention requirements makes it simpler to modify the retention requirements. There are fewer changes to make.

13.2.1.1 VRS Cleanup after Conversion

When you implement the VRS cleanup after conversion you define new VRSSs and then delete the old VRSSs:

1. Analyze your current VRSSs. Consider the data set names and look for those with common naming conventions and identical retention and movement needs.
2. Build a list of new data set names and the retention criteria they need.
3. Use the ADDVRS subcommand to define the new VRSSs.
4. Run EDGHSKP with PARM=VRSEL and the REPORT DD statement included. Use the report output to ensure that the new VRSSs are not yet being used for any existing data sets.
5. Use the DELETEVRS subcommand to remove the policies that are no longer required.

6. Run EDGHSKP with PARM=VRSEL and the REPORT DD statement included. Use the report output to ensure that the new VRSSs are now being used for the data sets previously retained by the deleted VRSSs.

13.2.1.2 VRS Cleanup as Part of Conversion

When you do the VRS cleanup as part of the conversion process, you do not have VRSSs to delete. You have either the K-Records that are input to EDGCNVT or the ADDVRS commands generated by EDGCNVT. We recommend that you do the cleanup on the ADDVRS commands, by editing them to add data set names and removing the redundant names. When you load the DFSMSrmm CDS, you do not use the records that EDGCNVT writes to the VRSLIST output file; you execute the modified ADDVRS commands once DFSMSrmm is started with the new CDS. The VRS cleanup consists of the following steps:

1. Run EDGCNVT and validate that the converted retention policies are meeting your needs.
2. Review the ADDVRS commands in the VRSCMDS file created by EDGCNVT. Consider the data set names and look for those with common naming conventions and identical retention and movement needs.
3. Build a list of new data set names and the retention criteria they need.
4. Modify the ADDVRS subcommands to include the new VRSSs and remove the VRSSs that are not required.
5. Modify the LOADRMM job to remove the VRSLIST file created by EDGCNVT. After you start DFSMSrmm with the new CDS, you execute the ADDVRS commands you have prepared.
6. Run EDGHSKP with PARM=VRSEL and the REPORT DD statement included. Use the report output to ensure that the new VRSSs are now being used to provide the retention you need.

Each time that you reconvert, use the prepared ADDVRS commands, not the VRSLIST that EDGCNVT generates.

13.2.2 Centralize Control of Retention and Movement

Under your previous tape management system retention criteria could come from several places:

- JCL RETPD or EXPDT keywords
- JCL-specified keyword date, such as 99000
- SMS data class
- A default retention
- A retention policy

DFSMSrmm also provides support for the above ways of specifying retention. The first three ways of specifying retention allow the user to control retention, and the last two ways provide a way for the installation to control retention. Without a centrally controlled policy, and without enforcement of that policy, you have no way of controlling the retention of data on tape.

With DFSMSrmm you can enforce central control over all retention if you want. Use the VRSSs to define all retention needs. Use the RETPD and MAXRETPD values in the DFSMSrmm PARMLIB to set a default retention period and prevent the volume expiration date from overriding the retention information in the VRS.

Specifying MAXRETPD(0) forces the expiration date of all volumes to be set to the date they are assigned from scratch status. MAXRETPD cannot be less than the RETPD value, so RETPD(0) also must be specified. You may decide to specify just a small value so that new data is kept for a minimum time period, for example, MAXRETPD(2) RETPD(2).

Note: Applications such as the DFSMSHsm that use a permanent retention date like 99365 to retain data are affected by reducing the MAXRETPD value. DFSMSrmm no longer retains them forever. You must ensure that VRSs are defined that cause the data sets they create to be retained until the application releases them.

You can define a default retention period for all data sets using VRS as shown in Figure 197.

```
RMM ADDVRS DSN('**') WHILECATALOG
```

Figure 197. A Default VRS

The VRS in Figure 197 covers all data sets not covered by any other VRS. To prevent the VRS management value from being used, you can define a default VRS as in Figure 198.

```
RMM ADDVRS DSN('*.**') WHILECATALOG
```

Figure 198. A Default VRS to Prevent Use of VRS Management Value

See the *DFSMSrmm Guide and Reference*, SC26-4931, “Defining Vital Record Specifications,” for details on the VRS matching process.

To implement centralized control of retention follow these steps:

1. Define a VRS as shown in Figure 198.
2. Change the DFSMSrmm PARMLIB members MAXRETP and RETPD to be either 0 or a small value.
3. Restart DFSMSrmm with the new PARMLIB member, using the operator command, assuming the member name is EDGRMM01:
F DFRMM,M=01
4. For existing volumes that have an expiration date that has not yet been reached, change the volume expiration date to today’s date. You can identify these volumes using the DFSMSrmm search volume dialog and sort the list of volumes by the expiration date data column. However, it may be more practical to use the DFSMSrmm report extract file and process it with a tool like DFSORT to select just those volume records with an expiration date later than today’s date. Check that the volumes concerned are covered by a VRS or that it is alright for them to be expired. If any should be retained by VRS you should define a VRS to provide the retention that they require. If today is 1 June 1995, for each volume you can change the expiration date using the subcommand:
RMM CHANGEVOLUME volser EXPDT(1995/152)
5. Run EDGHSKP with PARM=VRSEL and the REPORT DD statement included. Use the report output to ensure that the VRSs are now being used to provide the retention you need.

6. When the retention report is acceptable, run EDGHSKP with PARM=EXPROC. Any volume not retained by VRS will be set to pending release status. Use the DFSMSrmm SEARCHVOLUME subcommand to list all volumes in pending release status. If any volumes should not have been released, you can reclaim them by using the CHANGEVOLUME subcommand to assign a new retention period. Ensure that they are retained by a VRS before rerunning VRSEL and EXPROC processing.
7. As new retention and movement requirements are identified, use VRS exclusively to specify the retention needs.

13.2.3 Delete Nonexistent Volumes

When you converted to DFSMSrmm you may have converted information about volumes that you no longer have. Some vendor tape management systems use databases that must be formatted for each range of volumes to be managed. You may have stopped using some of these ranges but never reformatted the database to remove the volumes.

Running under DFSMSrmm you have the flexibility to add and delete volumes or other resources at any time. Use the knowledge you have about your existing tape volume ranges to check that you have defined to DFSMSrmm only the volumes and rack numbers that you want to manage.

You can use the DFSMSrmm commands to build lists of commands to delete resources you do not want. When you delete volumes, consider deleting the rack numbers that are left in empty status.

For example, you know that volumes 100000 through 199999 no longer exist but that they were converted to DFSMSrmm. Use the commands in Figure 199 to delete the volumes and rack numbers.

```
RMM SEARCHVOLUME VOLUME(1*) OWNER(*) LIMIT(*) -  
    NOLIST CLIST('RMM DV ',' FORCE')  
EXEC EXEC.RMM LIST  
RMM DELETERACK 100000 COUNT(99999)  
RMM DELETERACK 199999 COUNT(1)
```

Figure 199. Deleting Nonexistent Volumes

Refer to the *DFSMSrmm Guide and Reference*, SC26-4931, for information about the commands used. You can repeat these commands for each range of volumes and rack numbers you want to delete.

You can avoid the need to do cleanup by deleting the volume information during conversion. Use the EDGCNXIT user exit from EDGCNVT to check for ranges of volumes you want deleted and tell EDGCNVT to skip the volumes and data set records within those ranges.

13.3 System-Managed Tape and Basic Tape Library Support

You can use either system-managed tape or BTLS to support the use of IBM 3494 and 3495 Automated Tape Library Dataservers. Both components provide support for selecting and mounting volumes in a library and for tracking control of volume status.

DFSMSrmm is integrated with system-managed tape but has no direct interface with BTLS. You can use DFSMSrmm with either component. Refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for information about using DFSMSrmm with system-managed tape.

In this section we briefly describe how you can customize your use of DFSMSrmm when using either of these components.

13.3.1 System-Managed Tape

You do not have to perform any customization unless you want to, because DFSMSrmm provides a complete set of functions to support the processing of system-managed volumes. However, you may want to change some of the processing that DFSMSrmm performs.

13.3.1.1 Customize the CBRUXVNL Exit

The volume not in library (VNL) exit provides you with a way of deciding whether a volume should be entered into a library to allow a job to process. DFSMSrmm provides a sample exit that attempts to get a volume into a library only if it is system managed. Refer to the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for a full description of the processing the sample exit provides. You can easily change the decisions that the sample code makes and use the WTOR interface to request that the operator enter the volume into a library.

13.3.1.2 Customize the CBRUXEJC Exit

DFSMSrmm provides a sample exit that always requests that a volume being ejected from a library be removed from the TCDB. If you would rather leave the volume entry in the TCDB, you can modify the exit to change the DFSMSrmm decision. Figure 200 shows a code fragment that ensures that all volumes are left in the TCDB.

```

...
LINKCSUX LR      R1,R11          PARAMETER LIST FOR EDGLCSUX
          LINK    EP=EDGLCSUX,ERRET=ERROR
          SPACE 1
          CH      R15,=H'12'      ENVIRONMENT ERROR?
          LA      R15,UXJFAIL      ASSUME YES - SET FAIL REQUEST @004
          BE      FREEMAIN         IF ERROR, EXIT WITH FAILURE
          L       R15,LCSUP_LCSRC  OTHERWISE, TAKE LCSUX RETURNED CODE
          LA      R1,UXJCHG        CHECK RETCODE 4 @RBA
          CR      R15,R1           AGAINST ACTUAL CODE @RBA
          BH      FREEMAIN         > 4 CONTINUE @RBA
          LTR     R15,R15          RETURN CODE 0 ?? @RBA
          BZ      FREEMAIN         YES.. ACCEPT USERS OPTION @RBA
          MVI     UXJVDISP,UXJKEEP FORCE TCDB ENTRY TO BE KEPT @RBA
          LA      R15,UXJCHG      SET RETCODE 4 - PLIST CHANGED @RBA
          B       FREEMAIN         RETURN TO CALLER
...

```

Figure 200. Customized Sample Section of CBRUXEJC

The changes flagged with @RBA have been added to the code to set the KEEP option and change the return code to tell OAM that the parameter list has been changed.

13.3.2 Customizing DFSMSrmm for Use with BTLS

You can use DFSMSrmm to work with BTLS in two areas:

- Selection of scratch pools
- Returning volumes to scratch status

13.3.2.1 Selection of Scratch Pools

BTLS provides a way of selecting from any one of its eight scratch pools using job names. If you already use the data set name to select a scratch pool, you can use DFSMSrmm to select the scratch pool, and DFSMSrmm tells BTLS which pool is to be used.

Use the DFSMSrmm VLPOOL command in PARMLIB to associate an DFSMSrmm scratch pool with the name of a BTLS scratch pool.

```
VLPOOL PREFIX(10*) TYPE(S) MEDIANAME(CARTS) NAME(SCRTCH1) -  
    DESCRIPTION('BTLS pool 1') EXPDTCHECK(N)  
VLPOOL PREFIX(2*) TYPE(S) MEDIANAME(3590) NAME(SCRTCH7) -  
    DESCRIPTION('BTLS pool 7') EXPDTCHECK(N)  
VLPOOL PREFIX(*) TYPE(S) MEDIANAME(TAPE) -  
    DESCRIPTION('default pool') EXPDTCHECK(N)
```

To use the data set name to select the scratch pool you must implement the EDGUX100 installation exit. The *DFSMSrmm Implementation and Customization Guide*, SC26-4932, provides details on how to customize, install, and use this exit.

When BTLS issues a mount to the library for a nonspecific volume, DFSMSrmm uses the IGXMSGEX installation exit to return the correct BTLS scratch pool name to BTLS, and BTLS uses that name instead of any pool it may have selected.

13.3.2.2 Returning Volumes to Scratch Status

Although BTLS tracks the status of volumes, changing them to private status when they are used for a nonspecific mount, it provides no method of deciding when to return the volume to scratch. You use your tape management system to do this. DFSMSrmm provides simple TSO commands that you can use to drive the BTLS LIBRARY command to change a volume back to scratch status.

When you have run DFSMSrmm inventory management, you list the volumes in scratch status and pass the list to the BTLS LIBRARY command as shown in Figure 201 on page 512.

```

//TMP      EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//SYSTSIN  DD *
RMM SEARCHVOLUME VOLUME(*) OWNER(*) LIMIT(*) STATUS(SCRATCH) -
  NOLIST CLIST
/*
//AMS      EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//LIBIN    DD DISP=SHR,DSN=userid.EXEC.RMM.CLIST
//SYSIN    DD *
  LIBRARY SETCAT UNIT(123) CATEGORY(SCRATCH)
/*

```

Figure 201. Returning Volumes to Scratch Using DFSMSrmm and BTLS

By customizing the DFSMSrmm command, you can select volumes from a single scratch pool if necessary so that BTLS only returns a small number of volumes to scratch status.

13.4 Merging and Splitting the DFSMSrmm CDS

The DFSMSrmm CDS is basically a VSAM KSDS. It is made up of multiple records, which have complex interrelationships that are managed by DFSMSrmm processing. Never attempt to update the records in the CDS, as it could adversely affect the processing that DFSMSrmm performs.

When DFSMSrmm is not active, the records in the CDS can be processed using IDCAMS REPRO. This is the tool that DFSMSrmm uses for backup, restore, and reorganization of the CDS. However, the DFSMSrmm utility that calls IDCAMS does special processing to ensure that it does not impact DFSMSrmm processing.

The CDS contains information about the resources managed by DFSMSrmm. Normally the resources are all those used on a single system or in a set of systems with a shared DASD environment. If you ever split these systems or merge them together because of changes in workload, you may also want to do the same to the DFSMSrmm CDS.

13.4.1 When to Merge or Split

You should not normally have to merge or split the contents of the DFSMSrmm CDS. However, either during conversion or when you are changing the number of MVS systems you run, you may want to merge or split the contents of the DFSMSrmm CDS.

13.4.1.1 During Conversion

When you are converting from an existing database to an DFSMSrmm CDS you may want to use an existing DFSMSrmm CDS. You may have one system already converted to DFSMSrmm, and now a second is to be converted, but you want to share the CDS with the first system.

During conversion you will be adding data produced by EDGCNVT to an existing CDS. You would not be splitting the conversion data stream into multiple CDSs. See 13.4.6, “Converting to an Existing CDS” on page 531 for details of how you merge the data you are converting into an existing DFSMSrmm CDS.

13.4.1.2 Changing the Number of Systems

If you are changing the number of systems or CDSs that you have and are not deleting the information, you will want to merge or split the contents of the DFSMSrmm CDS.

If you are increasing the number of CDSs, you are splitting the CDS. See 13.4.5, “Splitting the CDS” on page 524 for details on how you split an existing CDS into multiple DFSMSrmm CDSs.

If you are decreasing the number of CDSs, you are merging CDSs. See 13.4.4, “Merging the CDS” on page 520 for details on how you merge multiple DFSMSrmm CDSs.

13.4.2 Considerations

When you merge or split a CDS, you must consider the timing of events as well as the contents of the CDS.

DFSMSrmm must be inactive, and you must not be performing tape processing when you are processing the CDS. When you have stopped DFSMSrmm, you should back up existing CDSs and ensure that you have recovery plans in place in case you are unable to complete the process successfully. When you stop DFSMSrmm, do not use the OPT=RESET function, as it allows tapes to be processed without DFSMSrmm’s knowledge.

When merging or splitting a CDS, it is important to understand the contents of the CDS. For example, if you are merging two CDSs, you cannot merge if the same resource is defined in each CDS. You must analyze the contents of the CDS to ensure that there are no duplicate resources. Below we describe each resource type in the CDS and explain what you can do to handle duplicate resources.

The record types are:

key type	record type
C	Control
CA	Action
D	Data set
E	Empty rack
F	Scratch rack
K	VRS
O	Owner
P	Product
R	Empty bin
S	In-use bin
U	In-use rack
V	Volume

13.4.2.1 Control

Each DFSMSrmm CDS contains a control record. The control record contains the dates and times of when inventory management functions last ran and the counts of rack numbers and bin numbers for built-in storage locations.

You can merge the control record from a single CDS and copy it to all target CDSs if you are splitting a CDS. In both cases you must use EDGUTIL with PARM=MEND to correct the record counts after processing is completed.

The control record also contains the CDSID value. If you use this value to identify a CDS, you must ensure that each CDS you create uses a different value. Use EDGUTIL with PARM='UPDATE' after processing is completed and specify the CDSID value, using the SYSIN DD CONTROL statement.

13.4.2.2 Action

Action records contain information about the librarian actions that are outstanding. They include the release actions and any volume movements that are required. Omit these records, as there is a strong chance that they exist in each CDS, and the records are rebuilt by DFSMSrmm during inventory management.

13.4.2.3 Data Set

A data set record exists for each volume on which the data set has been written. Each record is uniquely identified by the VOLSER and physical file number of the data set.

When merging CDSs, refer to the information on volume records in 13.4.2.9, "Volume" on page 518 for an explanation of how to handle duplicate volumes and the data sets on them.

When splitting a CDS, you must copy all data set records to each target CDS and then use the facilities of DFSMSrmm to delete those data set records that are not required. The DFSMSrmm EDGUTIL utility determines which are required by using the volume information you have copied into each target CDS. Use the PARM=MEND option to delete the unnecessary data set records from the CDS.

13.4.2.4 VRS

The VRS records contain the retention and movement policies for your data. If you have multiple CDSs, it is likely that you have the same policies defined in each CDS for some types of data. The policies are identified by data set name and job name, VOLSER, and VOLSER prefix.

When you merge CDSs you must check for and resolve any duplicate policies. If you have a policy for the same data set name and job name combination in more than one CDS, you should check that the retention and movement defined are the same. If they are the same, you can use the policy from any CDS. If they are not the same, you must resolve the conflict before merging the CDSs.

One way to resolve the conflict would be to ensure that one CDS contains the complete set of VRS policies. Identify the CDS with most of the VRSs you require in the merged CDS. Identify the VRSs that are unique in the other CDSs and define them in the one CDS.

When you split a CDS, you copy the complete set of VRS records to all target CDSs. This action may result in having VRSs that are unused in some CDSs, but it does not cause a problem in DFSMSrmm processing. Once DFSMSrmm is started using the new CDS, you can use the DELETEVRS subcommand to delete the unused VRSs.

13.4.2.5 Owner

There is one owner record for each identified user that owns a DFSMSrmm resource such as a volume or VRS. There can also be owner records for users that own no resources. Perhaps you have defined these so that the information can be used within your installation.

The owner record contains detailed information about the user and the count of volumes owned. For users who own volumes DFSMSrmm uses multiple owner records to provide directed retrieval of volume and data set information during search commands. You must consider the owner details and the owner volume records when splitting and merging the CDS.

Whether you are merging or splitting CDSs, you must use a SORT program to select the owner records, remove the additional owner records that contain just volume information, and remove the duplicate owner records. Once the records are loaded into a CDS, use the EDGUTIL utility with the PARM=MEND option to set the correct counts of owned volumes and build the owner volume records for owned volumes.

When merging CDSs, you must check for duplicate owners across the CDSs being merged. Duplicates are not a problem, as they can be handled as long as the owner record identifies the same person or task in each CDS. If the duplicate owners are the same user, determine which CDS has the correct owner details, such as address and contact information. Ensure that at least one CDS has the correct owner details for the users and use the owner details as the primary source of input to the merge. If the duplicate owners are really different users, you must decide how to handle that situation. You may have to assign different user IDs or define a new owner ID and transfer the owned resources to the new owner ID. Your objective is to have no duplicate owners, or to have any duplicates be for the same user in each case.

13.4.2.6 Product

A product record identifies a single product and software level and the volumes associated with that product. Whether you are merging or splitting a CDS the existence of product records is a problem that must be resolved.

When you merge CDSs there must be no duplicate product records. If there are, you must delete the duplicates and redefine the product information after the merge is completed. For each duplicate product that must be deleted, list the volumes for that product. Figure 202 shows how to list the volumes, assuming that all product information is being deleted from a single CDS.

```
RMM SEARCHPRODUCT NUMBER(*) LIMIT(*) NOLIST -  
  CLIST('RMM LISTPRODUCT ')  
EXEC EXEC.RMM LIST
```

Figure 202. Listing Product Information

Figure 203 on page 516 shows how to delete all product information from a single CDS.

```
RMM SEARCHPRODUCT NUMBER(*) LIMIT(*) NOLIST -  
CLIST('RMM DELETEPRODUCT ',' NORELEASE')  
EXEC EXEC.RMM LIST
```

Figure 203. Deleting Product Information

After you merge the CDSs you can add the product information using the RMM CHANGEVOLUME subcommand as shown in Figure 204.

```
RMM CHANGEVOLUME DZ11C0 NUMBER(5695-DF1) FEATCD(A001) -  
LEVEL(V01R03M00)
```

Figure 204. Adding Product Information

When you split a CDS, the easiest way to handle product records is to target all product volumes to a single CDS. If this is not possible, you can either:

- Delete the product information and then add it after the CDS is split
- Individually copy the product records by key to the correct target CDS. You can do this only if all volumes for a product are targeted to the same CDS.

13.4.2.7 Rack Numbers

There is a single rack record for each shelf location you have defined to DFSMSrmm. There must be at least one for each volume defined, but there can be more if you also defined empty rack numbers for use when volumes are added to the CDS. The rack number represents the volume's external VOLSER.

There are three rack number records, each using a unique key to aid retrieval by DFSMSrmm. During the copy operation the three key ranges must be copied, but you only need to consider the rack numbers that you have defined.

When you merge CDSs you must not have any duplicate rack numbers. Any duplicates must be deleted. If the duplicate already contains a volume, you can reassign the volume to another rack number by using the RMM CHANGEVOLUME subcommand. Figure 205 shows two different ways of reassigning a volume to a new rack number:

```
RMM CHANGEVOLUME PP1234 POOL(A*)  
  
RMM CHANGEVOLUME PP1234 RACK(A00001)
```

Figure 205. Assigning a New Rack Number

In the first command DFSMSrmm picks an empty rack in pool A*. In the second example you have selected the empty rack to be used.

Before cleaning up any duplicate rack numbers, refer to 13.4.2.9, "Volume" on page 518 for an explanation of how to handle duplicate volumes.

When you are splitting a CDS you split the rack numbers in the same way that you split the volume ranges into the target CDSs. You decide which volumes are copied to each CDS and ensure that the rack numbers with which they are associated are also copied to the same target CDS. During the copy process you identify the rack numbers using key ranges to include or exclude the appropriate rack numbers. The significant part of the key of rack number

records is the first 16 characters. The key must be specified in hexadecimal, as the second byte is hex zero. When copying ranges of volumes, you can use the COUNT keyword on the REPRO statement; however, you must specify a hexadecimal key for the TOKEY as well as the FROMKEY. The key includes an 8-character media name that matches the VLPOOL media name as well as the rack number. Here is an example of copying the rack numbers for the volumes in the 001000 - 001999 range:

```
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
      FROMKEY(X' C500F3F4F8F0404040F0F0F1F0') -
      TOKEY(X' C500F3F4F8F0404040F0F0F1F9F9F9')
/* COPY racks from E 3480    001000          to E 3480    001999 */
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
      FROMKEY(X' C600F3F4F8F0404040F0F0F1F0') -
      TOKEY(X' C600F3F4F8F0404040F0F0F1F9F9F9')
/* COPY racks from F 3480    001000          to F 3480    001999 */
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
      FROMKEY(X' E400F3F4F8F0404040F0F0F1F0') -
      TOKEY(X' E400F3F4F8F0404040F0F0F1F9F9F9')
/* COPY racks from U 3480    001000          to U 3480    001999 */
```

The first character of the key is either E, F, or U, the media name starts in position 3, and the rack number starts in position 11. Only copy a range of volumes to a single target CDS.

13.4.2.8 Bin Numbers

There is a single bin record for each storage location shelf location you have defined to DFSMSrmm.

There are two different bin number records, each using a unique key to aid retrieval by DFSMSrmm. During the copy operation the two key ranges must be copied, but you only need to consider the bin numbers that you have defined.

When you merge CDSs, you must not have any duplicate bin numbers. Any duplicates must be deleted. If the duplicate already contains a volume, you can reassign the volume to another bin number by using the RMM CHANGEVOLUME subcommand. Figure 206 shows how to reassign a volume to a new bin number.

```
RMM CHANGEVOLUME A00123 BIN(000029)
```

Figure 206. Assigning a New Bin Number

You can also remove the volume from the storage location, free up the bin number, and delete the unused bin number. Figure 207 shows how to delete duplicate bin number X00022 in location VAULT1 of media name CARTS.

```
RMM CHANGEVOLUME A00123 LOCATION(HOME) CMOVE
RMM DELETEBIN X00022 MEDIANAME(CARTS) LOCATION(VAULT1)
```

Figure 207. Freeing and Deleting a Duplicate Bin Number

Next time you run inventory management, the volume is reassigned to the storage location, and another bin number is assigned.

Note: When you need to reassign bin numbers, you should consider how offsite movements are managed in your installation. If you depend on the

volume being in a specific slot in a storage location, you should ensure that any volumes you change are physically moved in the storage location.

When you are splitting a CDS you split the bin numbers according to how you want to handle storage locations in the split environment. If the volumes in the storage locations are being split such that each target CDS contains volumes for each shelf-managed storage location, handling the splitting of bin numbers is very complex. We suggest that you use DFSMSrmm commands to change the location of all volumes and confirm their return to their home location. This approach frees up the in-use bin numbers. Do not copy the bin numbers to the target CDSs; but add empty bin numbers after the split and use inventory management VRSEL and DSTORE processing to assign new bin numbers to the volumes after the split processing is completed.

Targeting all volumes in a single storage location for the same CDS simplifies the copying of bin numbers. During the copy process you identify the bin numbers using key ranges to include or exclude the appropriate bin numbers. The significant part of the key of bin number records is the first 24 characters. The key can be specified in character form. When copying ranges of bin numbers you must specify a TOKEY as well as the FROMKEY because a range of bin numbers can include both empty and in-use bin numbers. The key includes an 8-character media name, which matches the LOCDEF media name as well as the bin number, and the location name. Here is an example of copying the bin numbers for the bin numbers in storage location VLT1, media name CARTS, and the V10001 - V10500 range:

```
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
      FROMKEY(C' RUVLT1    CARTS  V10001') -
      TOKEY(C' RUVLT1    CARTS  V10500')
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
      FROMKEY(C' SUVLT1    CARTS  V10001') -
      TOKEY(C' SUVLT1    CARTS  V10500')
```

The first two characters of the key are either RU or SU, the storage location is in position 3, the media name starts in position 11, and the bin number starts in position 19. Only copy a range of bin numbers to a single target CDS.

13.4.2.9 Volume

The volume record contains detailed information about a volume and its contents. It also identifies the data sets that are on the volume and volume chaining information. Some fields in the records identify the existence of other records in the CDS.

When moving volume records from one CDS to another, you must be sure to move the related records as well. These include the data set records, owner record, rack record, bin records, and product record. If you do not move them, the EDGUTIL utility can identify the missing records and, with the MEND option, create the missing records with one exception—product records cannot be re-created.

When you are merging CDSs, you should not have any duplicate volumes. If you do, you must either skip the duplicate volumes or delete them from the source CDS before starting the merge process. You should understand why there are duplicate volumes and, based on your understanding, select the best approach for dealing with them.

If you have duplicate volumes, there could also be duplicate data set records for those volumes. If you delete the volumes, you also delete the data set records.

When you are splitting a CDS you will most probably split the volumes according to VOLSER ranges. During the copy process you identify the volumes using key ranges to include or exclude the appropriate volumes. The significant part of the key of volume records is the first eight characters. The key must be specified in hexadecimal as the second byte is hex zero. When copying ranges of volumes you can use the COUNT keyword on the REPRO statement, otherwise you must specify a hexadecimal key for the TOKEY as well. Here is an example of copying the volumes in the 001000 - 001999 range:

```
REPRO INFILE(SOURCE) OUTFILE(OUTS1) -  
      FROMKEY(X'E500F0F0F1') COUNT(1000)
```

The first character of the key is V, and the VOLSER starts in position 3. Only copy a range of volumes to a single target CDS.

13.4.3 PARMLIB Options

The PARMLIB options that are affected by the splitting and merging of CDSs are:

- LOCDEF
- VLPOOL
- REJECT

13.4.3.1 LOCDEF

These options identify your locations to DFSMSrmm. When you are merging CDSs, consolidate the LOCDEF options from all CDSs you are merging to ensure that you have no duplicates. If you have duplicates, ensure that they are identified as the same location type and with the same media names.

When you split a CDS, each system that will use one of the target CDSs should have defined to it the locations where the volumes in the target CDS reside. You can simply include all LOCDEFs in each new system, but you have the option of removing those that are not required.

13.4.3.2 VLPOOL

These options identify the ranges of rack numbers and pools of shelf space you have in your library. If your library is not changing, we recommend that you do not change your current VLPOOL options.

If you are merging CDSs and your existing VLPOOL definitions are different for each existing CDS, you must merge the definitions together to cover the merged sets of volumes and rack numbers. Ensure that the operands specified on each VLPOOL are correct when the same pool prefix is defined in more than one source environment.

When you split a CDS we recommend that you take the existing VLPOOL definitions and use them on each of the new CDSs.

13.4.3.3 REJECT

These options identify the ranges of volumes to be rejected for use on the system. The reject specifies the rack number prefix and should match the VLPOOL definitions that you have.

Where you have a tape library shared by multiple systems, define the same VLPOOL definitions to all systems and use the REJECT definitions to prevent systems from using volumes that are for use on other systems.

The actions you take for the REJECT definitions should match how you handle the VLPOOL definitions.

13.4.3.4 Running IDCAMS REPRO

When using IDCAMS REPRO to copy records to the target CDS, you may get messages indicating that records are duplicates or that there are no input records:

- Return code 4 from a REPRO command with this message:

```
IDC11465I DELIMITERS WERE SPECIFIED, BUT NO RECORDS PROCESSED.
```

The only key range for which this should occur is *P* for product records. Expect this message if you do not have any product definitions in the CDS.

- Return code 8 or 12 from a REPRO command with messages:

```
IDC3302I ACTION ERROR ON DFRMM.MASTER.MERGE  
IDC3308I ** DUPLICATE RECORD X'ccXXXXXXXX'
```

You should not get these messages if you have followed the recommendations made above. After four duplicate records IDCAMS stops processing the records, so the copy operation is incomplete. You must resolve the duplicates by taking the actions recommended in Section 13.4.2, “Considerations” on page 513 and then rerun the merge or split job.

13.4.4 Merging the CDS

Follow the steps outlined below to perform the merge of two CDSs. SYS1 is the system on which the merge is being performed, and SYS2 is the second system from which CDS data is being merged.

1. Refer to Section 13.4.2, “Considerations” on page 513 and use the information to analyze the CDSs that are being merged. If any potential problems or duplicate records are identified, you must make the recommended changes before continuing.
2. Stop DFSMSrmm on all systems affected by the merging of the CDSs. You cannot use tapes while you merge CDSs.
3. Back up the CDS on SYS2 and send the backup to SYS1. Run the backup job shown in Figure 208 on page 521. If you have shared DASD between SYS1 and SYS2, you can skip this step.
4. Restore the SYS2 CDS on SYS1. Run the restore job shown in Figure 209 on page 522. If you have shared DASD between SYS1 and SYS2, you can skip this step.
5. Run the merge job shown in Figure 210 on page 522. The job includes the following steps:

BACKS1 Backs up the CDS used for SYS1.

COPYO	Copies the owner records from both source CDSs to temporary files for input to step SORTO.
SORTO	Removes any duplicate owner records found in SYS2 input and deletes the owner volume records.
DEFINE	Deletes and then redefines a target CDS.
COPY	Copies most of the records from both source CDSs. For VRS and product records only one CDS is used as input, and a later step copies from the second CDS. This is done as these are the most likely records to still have duplicates.
COPYK2	Copies the VRS records from SYS2. Expect duplicates to be found if you use the same VRS definitions on both CDSs. You can remove this step if you have already ensured that all VRS definitions were in SYS1.
COPYP2	Copies the product records from SYS2.
MEND	Validates the newly merged CDS and makes any corrections to record relationships. Builds the owner volume records, corrects the count of owned volumes, and corrects the rack number and bin number counts in the control record.
VERIFY	Verifies that the CDS is logically sound and ready to use.

If the merge job is successful, you are ready to use the merged CDS.

- Update SYS1.PARMLIB member EDGRMMxx:

Command Required change

OPTION	Update DSNNAME to identify the new CDS name.
LOCDEF	Update to reflect all locations used in the merged CDS.
VLPOOL	Update to reflect the new set of pools.
REJECT	Update to reflect any changes in VLPOOL definitions.

- Start the DFSMSrmm procedure using the new PARMLIB member.
- Check that the merged information is good by using DFSMSrmm commands and the ISPF dialog to retrieve and display information.
- Execute any commands that are identified to be run in Section 13.4.1, “When to Merge or Split” on page 512.
- Run DFSMSrmm inventory management on the new CDS. During this run DFSMSrmm VRSEL and DSTORE processing identifies volumes to be moved to the correct location. If any moves are identified, use EDGRPTD to produce the picking lists and action any moves that you know are still required.

The inventory management run should execute all inventory management functions including backup.

Figure 208 shows the JCL for backing up the CDS on SYS2.

```
//BACKS2 EXEC PGM=EDGBKUP,PARM=' BACKUP'
//SYSPRINT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//MASTER DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//BACKUP DD DSN=DFRMM.MASTER.BACKUP2(+1),DISP=(,CATLG,DELETE),
// UNIT=SYSDA,SPACE=(6233,(1000,999),RLSE),
// DCB=(MODEL.VB9216,DSORG=PS)
```

Figure 208. JCL to Back Up the CDS on SYS2

Figure 209 on page 522 shows the JCL for restoring the SYS2 CDS on SYS1.

```
//RESTS2 EXEC PGM=EDGBKUP,PARM=' RESTORE'
//SYSPRINT DD SYSOUT=*
//MASTER DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//BACKUP DD DSN=DFRMM.MASTER.BACKUP2(0),DISP=SHR
```

Figure 209. JCL to Restore the SYS2 CDS on SYS1

Figure 210 shows the JCL for executing the merge process for two CDSs.

```
//BACKS1 EXEC PGM=EDGBKUP,PARM=' BACKUP'
//SYSPRINT DD SYSOUT=*
//MASTER DD DSN=DFRMM.MASTER.SYS1,DISP=SHR
//BACKUP DD DSN=DFRMM.MASTER.BACKUP1(+1),DISP=(,CATLG,DELETE),
// UNIT=SYSDA,SPACE=(6233,(1000,999),RLSE),
// DCB=(MODEL.VB9216,DSORG=PS)
//*****
/** COPY THE OWNER RECORDS FROM BOTH SOURCE CDSs TO **
/** A SEQUENTIAL DATASET TO BE USED FOR THE MERGED CDS. **
//*****
//COPY0 EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//INS1 DD DSN=DFRMM.MASTER.SYS1,DISP=SHR
//INS2 DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//OUTS1 DD DSN=&&IN01,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(1024,(5000,5000),RLSE),
// LRECL=2048,RECFM=VB
//OUTS2 DD DSN=&&IN02,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(1024,(5000,5000),RLSE),
// LRECL=2048,RECFM=VB
//SYSIN DD *
REPRO INFILE(INS1) OUTFILE(OUTS1) FROMKEY(0) TOKEY(0)
REPRO INFILE(INS2) OUTFILE(OUTS2) FROMKEY(0) TOKEY(0)
//*****
/** SORT THE OWNER RECORDS FROM BOTH SYSTEMS TO A TEMPORARY **
/** SEQUENTIAL DATASET TO BE PASSED TO A LATER STEP COPYALL. **
/** REMOVE DUPLICATE OWNERS FROM SOURCE S2. **
//*****
//SORT0 EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTIN DD DSN=&&IN01,DISP=(OLD,DELETE)
// DD DSN=&&IN02,DISP=(OLD,DELETE)
//SORTWK01 DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT DD DISP=(NEW,PASS),UNIT=SYSDA,DCB=*.SORTIN,
// SPACE=(1024,(5000,5000),RLSE),DSN=&&SORT0
//SYSOUT DD SYSOUT=*
//SYSIN DD *
SORT FIELDS=(5,9,CH,A)
INCLUDE COND=(5,1,CH,EQ,C'0',AND,14,1,CH,EQ,X'00') BASE OWNERS ONLY
SUM FIELDS=NONE ELIMINATE DUPS
```

Figure 210 (Part 1 of 3). Merge JCL

```

//DEFINE EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//MSTVOL DD DISP=SHR,UNIT=3380,VOL=SER=VSMN01
//SYSIN DD *
DEL DFRMM.MASTER.MERGE
SET MAXCC=0
DEF CL(NAME(DFRMM.MASTER.MERGE) FILE(MSTVOL) -
        VOLUME(VSMN01) KILOBYTES(50000 9000) FSPC(20 20) -
        RECORDSIZE(512 9216) KEYS(56 0) SHR(3 3) RUS) -
DATA(NAME(DFRMM.MASTER.MERGE.DATA) -
      CISZ(10240) ) -
INDEX(NAME(DFRMM.MASTER.MERGE.INDEX) IMBED REPL -
      CISZ(2048) )
//*****
//** TARGET SYSTEM **
//** COPY THE RECORDS FROM BOTH SOURCE CDS TO THE TARGET CDS **
//*****
//COPY EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//INS1 DD DSN=DFRMM.MASTER.SYS1,DISP=SHR
//INS2 DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//TARGET DD DSN=DFRMM.MASTER.MERGE,DISP=SHR
//OWNER DD DSN=&&SORTO,DISP=(OLD,DELETE)
//SYSIN DD *
REPRO INFILE(INS1) OUTFILE(TARGET) FROMKEY(C) TOKEY(C) COUNT(1)
REPRO INFILE(INS1) OUTFILE(TARGET) FROMKEY(D) TOKEY(F)
REPRO INFILE(INS2) OUTFILE(TARGET) FROMKEY(D) TOKEY(F)
REPRO INFILE(INS1) OUTFILE(TARGET) FROMKEY(K) TOKEY(K)
REPRO INFILE(OWNER) OUTFILE(TARGET)
REPRO INFILE(INS1) OUTFILE(TARGET) FROMKEY(P) TOKEY(P)
REPRO INFILE(INS1) OUTFILE(TARGET) FROMKEY(R) TOKEY(V)
REPRO INFILE(INS2) OUTFILE(TARGET) FROMKEY(R) TOKEY(V)
//*****
//** SOURCE CDS 2 - ON TARGET SYSTEM **
//** COPY "K" RECORDS FROM THE SOURCE CDS 2 TO THE TARGET CDS. **
//** RETURN CODE 8 or 12 INDICATES DUPLICATE RECORDS. **
//*****
//COPYK2 EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//INS2 DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//TARGET DD DSN=DFRMM.MASTER.MERGE,DISP=SHR
//SYSIN DD *
REPRO INFILE(INS2) OUTFILE(TARGET) FROMKEY(K) TOKEY(K)

```

Figure 210 (Part 2 of 3). Merge JCL

```

//*****
//**          SOURCE CDS 2 - ON TARGET SYSTEM          **
//** COPY "P" RECORDS FROM THE SOURCE CDS 2 TO THE TARGET CDS. **
//** RETURN CODE 8 or 12 INDICATES DUPLICATE RECORDS. **
//*****
//COPYP2 EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//INS2 DD DSN=DFRMM.MASTER.SYS2,DISP=SHR
//TARGET DD DSN=DFRMM.MASTER.MERGE,DISP=SHR
//SYSIN DD *
REPRO INFILE(INS2) OUTFILE(TARGET) FROMKEY(P) TOKEY(P)
//*****
//**          MERGED TARGET CDS          **
//** RUN EDGUTIL WITH PARM='MEND' TO UPDATE THE NEW MERGED **
//** CDS TO HAVE THE CORRECT COUNTS FOR RACK/BINS. **
//** ALSO SETS OWNER VOLUME COUNTS AND BUILDS OWNER VOLUME **
//** INFORMATION. **
//*****
//MEND EXEC PGM=EDGUTIL,PARM='MEND'
//MASTER DD DISP=SHR,DSN=DFRMM.MASTER.MERGE
//SYSPRINT DD SYSOUT=*
//*****
//**          MERGED TARGET CDS          **
//** RUN EDGUTIL WITH PARM='VERIFY(ALL)' ON THE NEW MERGED **
//** CDS TO VERIFY THE INTEGRITY OF THE NEW CDS. **
//*****
//VERIFY EXEC PGM=EDGUTIL,PARM='VERIFY(ALL)'
//SYSPRINT DD SYSOUT=*
//MASTER DD SUBSYS=(BLSR,'DDNAME=MASTERB')
//MASTERB DD DISP=SHR,DSN=DFRMM.MASTER.MERGE
/* Use of Batch LSR can speed the running of EDGUTIL
/* IF you have not got BLSR setup on your system you must replace
/* the MASTER DD statement with the following one, and delete
/* the MASTERB DD statement
/*MASTER DD DISP=OLD,DSN=DFRMM.MASTER.MERGE,
/* AMP=('BUFND=50,BUFNI=50')
//SYSIN DD DUMMY

```

Figure 210 (Part 3 of 3). Merge JCL

13.4.5 Splitting the CDS

Follow the steps outlined below to split a CDS into two CDSs. SYS1 and SYS2 are the new systems to which CDS data is being split.

1. Refer to Section 13.4.2, "Considerations" on page 513 and use the information to analyze the CDS that is being split. If any potential problems are identified, you must make the recommended changes before continuing. You need to consider how you plan to split the ranges of volumes, rack numbers, and bin numbers. You must decide now using the information in Section 13.4.2, "Considerations" on page 513 and prepare the AMS statements for use in job steps COPYV and COPYB.
2. Stop DFSMSrmm on all systems affected by the splitting of the CDS. You cannot use tapes while you split a CDS.
3. Run the merge job shown in Figure 211 on page 526. The job includes the following steps:

BACKS	Backs up the CDS to be split.
COPYO	Copies the owner records from the source CDS to a temporary file for input to step SORTO.
SORTO	Removes the owner volume records.
DEFINE	Deletes and then redefines the target CDSs.
COPY	Copies those records from the source CDS that must be copied to both target CDSs. The copying of the record ranges that need to be split are performed in later steps.
COPYP2	Copies the product records to SYS2. This assumes that all product volumes are split out to SYS2 CDS. If this is not the case, refer to Section 13.4.2.6, "Product" on page 515 and update the JCL based on the recommended actions.
COPYV	Copies the volume and rack numbers to the two target CDSs. The JCL example shows how you can specify to AMS which records are to be copied. You must update the sample AMS statements to copy the actual ranges you want to have in each target CDS.
COPYB	Copies the bin numbers to the two target CDSs. The JCL example shows how you can specify to AMS which records are to be copied. You must update the sample AMS statements to copy the actual ranges you want to have in each target CDS.
MEND1	Validates the newly created SYS1 CDS and makes any corrections to record relationships. Builds the owner volume records, corrects the count of owned volumes, corrects the rack number and bin number counts in the control record, and deletes any unused data set records.
MEND2	Validates the newly created SYS2 CDS and makes any corrections to record relationships. Builds the owner volume records, corrects the count of owned volumes, corrects the rack number and bin number counts in the control record, and deletes any unused data set records.
VERIFY1	Verifies that the SYS1 CDS is logically sound and ready to use.
VERIFY2	Verifies that the SYS2 CDS is logically sound and ready to use.

If the merge job is successful, you are ready to use the new CDSs.

4. Back up the newly created CDSs for SYS1 and SYS2. Run the backup job shown in Figure 212 on page 531.

5. Update SYS1.PARMLIB member EDGRMMxx on all affected systems:

Command Required change

OPTION	Update DSNNAME to identify the new CDS name.
OPTION	Update SYSID to identify a unique system ID. If you do not specify a value, DFSMSrmm uses the SMF system ID.
OPTION	If you specified CDSID in PARMLIB for the source CDS, ensure that CDSID still matches the value you used on the source CDS.
LOCDEF	Update to reflect the locations used in each CDS.
VLPOOL	Update to reflect the new set of pools. Update the SYSID to match the value specified on the OPTION SYSID keyword. If SYSID is not specified, ensure that the VLPOOL SYSID, if required, matches the SMF system ID of the running system.
REJECT	Update to reflect any changes in VLPOOL definitions.

6. Start the DFSMSrmm procedure on each system, using the new PARMLIB member.

7. Check that the merged information is good by using DFSMSrmm commands and the ISPF dialog to retrieve and display information.

8. Execute any commands that are identified to be run in Section 13.4.1, “When to Merge or Split” on page 512.
9. Run DFSMSrmm inventory management on each of the new CDSs. During this run DFSMSrmm VRSEL and DSTORE processing identifies volumes to be moved to the correct location. If any moves are identified, use EDGRPTD to produce the picking lists and action any moves that you know are still required.

The inventory management run should execute all inventory management functions including backup.

Figure 211 shows the JCL for executing the split process to two target CDSs.

```
//BACKS      EXEC PGM=EDGBKUP,PARM=' BACKUP'
//SYSPRINT   DD SYSOUT=*
//MASTER     DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//BACKUP      DD DSN=DFRMM.MASTER.BACKUP(+1),DISP=(,CATLG,DELETE),
// UNIT=SYSDA,SPACE=(6233,(1000,999),RLSE),
// DCB=(MODEL.VB9216,DSORG=PS)
//*****
//** COPY THE OWNER RECORDS FROM SOURCE CDS TO                **
//** A SEQUENTIAL DATASET TO BE USED FOR THE NEW CDSs          **
//*****
//COPY0      EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT   DD SYSOUT=*
//SOURCE     DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//OUT        DD DSN=&&OUT0,DISP=(,PASS),UNIT=SYSDA,
//           SPACE=(1024,(5000,5000),RLSE),
//           LRECL=2048,RECFM=VB
//SYSIN      DD *
REPRO INFILE(SOURCE) OUTFILE(OUT) FROMKEY(0) TOKEY(0)
//*****
//** SORT THE OWNER RECORDS FROM BOTH SYSTEMS TO A TEMPORARY   **
//** SEQUENTIAL DATASET TO BE PASSED TO A LATER STEP COPYALL.   **
//** REMOVE DUPLICATE OWNERS FROM SOURCE S2.                   **
//*****
//SORT0      EXEC PGM=SORT,REGION=6M
//SYSPRINT   DD SYSOUT=*
//SORTIN     DD DSN=&&OUT0,DISP=(OLD,DELETE)
//SORTWK01   DD UNIT=SYSDA,SPACE=(CYL,(10,10))
//SORTOUT    DD DISP=(NEW,PASS),UNIT=SYSDA,DCB=*.SORTIN,
//           SPACE=(1024,(5000,5000),RLSE),DSN=&&SORT0
//SYSOUT     DD SYSOUT=*
//SYSIN      DD *
SORT FIELDS=(5,9,CH,A)
INCLUDE COND=(5,1,CH,EQ,C'0',AND,14,1,CH,EQ,X'00')  BASE OWNERS ONLY
SUM FIELDS=NONE                                         ELIMINATE DUPS
```

Figure 211 (Part 1 of 6). Split JCL

```

//DEFINE EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYS1VOL DD DISP=SHR,UNIT=3380,VOL=SER=VSMN01
//SYS2VOL DD DISP=SHR,UNIT=3380,VOL=SER=VSMN02
//SYSIN DD *
DEL DFRMM.NEW.SYS1
DEL DFRMM.NEW.SYS2
SET MAXCC=0
DEF CL(NAME(DFRMM.NEW.SYS1) FILE(SYS1VOL) -
      VOLUME(VSMN01) KILOBYTES(50000 9000) FSPC(20 20) -
      RECORDSIZE(512 9216) KEYS(56 0) SHR(3 3) RUS) -
DATA(NAME(DFRMM.NEW.SYS1.DATA) -
      CISZ(10240) ) -
INDEX(NAME(DFRMM.NEW.SYS1.INDEX) IMBED REPL -
      CISZ(2048) )
DEF CL(NAME(DFRMM.NEW.SYS2) FILE(SYS2VOL) -
      VOLUME(VSMN02) KILOBYTES(50000 9000) FSPC(20 20) -
      RECORDSIZE(512 9216) KEYS(56 0) SHR(3 3) RUS) -
DATA(NAME(DFRMM.NEW.SYS2.DATA) -
      CISZ(10240) ) -
INDEX(NAME(DFRMM.NEW.SYS2.INDEX) IMBED REPL -
      CISZ(2048) )

```

Figure 211 (Part 2 of 6). Split JCL

```

//*****
//**          SOURCE SYSTEM                      **
//** COPY THE RECORDS FROM SOURCE TO BOTH TARGET CDSs      **
//*****
//COPY      EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//OUTS1     DD DSN=DFRMM.NEW.SYS1,DISP=SHR
//OUTS2     DD DSN=DFRMM.NEW.SYS2,DISP=SHR
//SOURCE    DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//OWNER     DD DSN=&&SORTO,DISP=(OLD,DELETE)
//SYSIN     DD *
REPRO OUTFILE(OUTS1) INFILE(SOURCE) FROMKEY(C) TOKEY(C) COUNT(1)
REPRO OUTFILE(OUTS2) INFILE(SOURCE) FROMKEY(C) TOKEY(C) COUNT(1)
REPRO OUTFILE(OUTS1) INFILE(SOURCE) FROMKEY(D) TOKEY(D)
REPRO OUTFILE(OUTS2) INFILE(SOURCE) FROMKEY(D) TOKEY(D)
REPRO OUTFILE(OUTS1) INFILE(SOURCE) FROMKEY(K) TOKEY(K)
REPRO OUTFILE(OUTS2) INFILE(SOURCE) FROMKEY(K) TOKEY(K)
REPRO INFILE(OWNER) OUTFILE(OUTS1)
REPRO INFILE(OWNER) OUTFILE(OUTS2)
//*****
//**          SOURCE TO A SINGLE TARGET                **
//** COPY "P" RECORDS FROM THE SOURCE CDS TO THE ONE TARGET  **
//*****
//COPYP2    EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//OUTS2     DD DSN=DFRMM.NEW.SYS2,DISP=SHR
//SOURCE    DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//SYSIN     DD *
REPRO INFILE(SOURCE) OUTFILE(OUTS2) FROMKEY(P) TOKEY(P)

```

Figure 211 (Part 3 of 6). Split JCL


```

//*****
//**          SOURCE TO TWO TARGETS          **
//** COPY "V" RECORDS BY VOLSER RANGE 100000-199999 TO SYS1 **
//** COPY "V" RECORDS BY VOLSER RANGE A00000-A00999 TO SYS2 **
//** COPY RACK RECORDS FOR THIS RANGE OF VOLUMES AS WELL **
//*****
//COPYV EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//OUTS1 DD DSN=DFRMM.NEW.SYS1,DISP=SHR
//OUTS2 DD DSN=DFRMM.NEW.SYS2,DISP=SHR
//SOURCE DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//SYSIN DD *
  REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
    FROMKEY(X' E500F1F0') COUNT(100000)
  /* copy volumes from V 100000 to V 199999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
    FROMKEY(X' C500F3F4F8F0404040F1F0') -
    TOKEY(X' C500F3F4F8F0404040F1F9F9F9F9')
  /* COPY racks from E 3480 100000 to E 3480 199999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
    FROMKEY(X' C600F3F4F8F0404040F1F0') -
    TOKEY(X' C600F3F4F8F0404040F1F9F9F9F9')
  /* COPY racks from F 3480 100000 to F 3480 199999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
    FROMKEY(X' E400F3F4F8F0404040F1F0') -
    TOKEY(X' E400F3F4F8F0404040F1F9F9F9F9')
  /* COPY racks from U 3480 100000 to U 3480 199999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
    FROMKEY(X' V 00C1F0') COUNT(1000)
  /* COPY from V A00000 to V A00999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
    FROMKEY(X' C500C3C1D9E3E2404040C1F0') -
    TOKEY(X' C500C3C1D9E3E2404040C1F0F0F9F9F9')
  /* COPY racks from E CARTS A00000 to E CARTS A00999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
    FROMKEY(X' C600C3C1D9E3E2404040C1F0') -
    TOKEY(X' C600C3C1D9E3E2404040C1F0F0F9F9F9')
  /* COPY racks from F CARTS A00000 to F CARTS A00999 */
  REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
    FROMKEY(X' E400C3C1D9E3E2404040C1F0') -
    TOKEY(X' E400C3C1D9E3E2404040C1F0F0F9F9F9')
  /* COPY racks from U CARTS A00000 to U CARTS A00999 */

```

Figure 211 (Part 4 of 6). Split JCL

```

//*****
//**                      SOURCE TO TWO TARGETS                      **
//** COPY BIN RECORDS FOR STORE VLT1  TO SYS1                      **
//** COPY BIN RECORDS FOR STORE VLT2  TO SYS2                      **
//*****
//COPYB   EXEC PGM=IDCAMS,REGION=5000K
//SYSPRINT DD SYSOUT=*
//OUTS1   DD DSN=DFRMM.NEW.SYS1,DISP=SHR
//OUTS2   DD DSN=DFRMM.NEW.SYS2,DISP=SHR
//SOURCE  DD DSN=DFRMM.MASTER.CDS,DISP=SHR
//SYSIN   DD *
    REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
        FROMKEY(X' D9E4E5D3E3F140404040C3C1D9E3E2404040F0F0F0F0F0') -
        TOKEY(X' D9E4E5D3E3F140404040C3C1D9E3E2404040F0F0F0F9F9F9')
    /* COPY bins from RUVLT1   CARTS   000000 for 1000 bins */
    REPRO INFILE(SOURCE) OUTFILE(OUTS1) -
        FROMKEY(X' E2E4E5D3E3F140404040C3C1D9E3E2404040F0F0F0F0F0') -
        TOKEY(X' E2E4E5D3E3F140404040C3C1D9E3E2404040F0F0F0F9F9F9')
    /* COPY bins from RUVLT1   CARTS   000000 for 1000 bins */
    REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
        FROMKEY(X' D9E4E5D3E3F240404040F3F4F8F0404040F0F0F0F0F0') -
        TOKEY(X' D9E4E5D3E3F240404040F3F4F8F0404040F0F0F1F9F9F9')
    /* COPY bins from RUVLT1   3480   000000 for 2000 bins */
    REPRO INFILE(SOURCE) OUTFILE(OUTS2) -
        FROMKEY(X' E2E4E5D3E3F240404040F3F4F8F0404040F0F0F0F0F0') -
        TOKEY(X' E2E4E5D3E3F240404040F3F4F8F0404040F0F0F1F9F9F9')
    /* COPY bins from RUVLT1   3480   000000 for 2000 bins */
//*****
//**                      EACH TARGET CDS                      **
//** RUN EDGUTIL WITH   PARM='MEND' TO UPDATE THE NEW TARGET      **
//** CDS TO HAVE THE CORRECT COUNTS FOR RACK/BINS.                **
//** ALSO SETS OWNER VOLUME COUNTS AND BUILDS OWNER VOLUME        **
//** INFORMATION, AND DELETES UNUSED DATA SET RECORDS            **
//*****
//MEND1   EXEC PGM=EDGUTIL,PARM='MEND'
//MASTER  DD DISP=SHR,DSN=DFRMM.NEW.SYS1
//SYSPRINT DD SYSOUT=*
//MEND2   EXEC PGM=EDGUTIL,PARM='MEND'
//MASTER  DD DISP=SHR,DSN=DFRMM.NEW.SYS2
//SYSPRINT DD SYSOUT=*

```

Figure 211 (Part 5 of 6). Split JCL

```

//*****
//**          EACH TARGET CDS          **
//** RUN EDGUTIL WITH  PARM=' VERIFY(ALL)' ON THE NEW TARGET **
//**   CDS TO VERIFY THE INTEGRITY OF THE NEW CDS.          **
//*****
//VERIFY1 EXEC PGM=EDGUTIL,PARM=' VERIFY(ALL)'
//SYSPRINT DD SYSOUT=*
//MASTER  DD SUBSYS=(BLSR,' DDNAME=MASTERB')
//MASTERB DD DISP=SHR,DSN=DFRMM.NEW.SYS1
//* Use of Batch LSR can speed the running of EDGUTIL
//* IF you have not got BLSR setup on your system you must replace
//* the MASTER DD statement with the following one, and delete
//* the MASTERB DD statement
//*MASTER DD DISP=OLD,DSN=DFRMM.NEW.SYS1,
//*          AMP=(' BUFND=50,BUFNI=50')
//SYSIN   DD DUMMY
//VERIFY2 EXEC PGM=EDGUTIL,PARM=' VERIFY(ALL)'
//SYSPRINT DD SYSOUT=*
//MASTER  DD SUBSYS=(BLSR,' DDNAME=MASTERB')
//MASTERB DD DISP=SHR,DSN=DFRMM.NEW.SYS2
//* Use of Batch LSR can speed the running of EDGUTIL
//* IF you have not got BLSR setup on your system you must replace
//* the MASTER DD statement with the following one, and delete
//* the MASTERB DD statement
//*MASTER DD DISP=OLD,DSN=DFRMM.NEW.SYS2,
//*          AMP=(' BUFND=50,BUFNI=50')
//SYSIN   DD DUMMY

```

Figure 211 (Part 6 of 6). Split JCL

Figure 212 shows the JCL for backing up the CDSs on SYS1 and SYS2.

```

//BACKCDS  PROC ID=
//BACKUP   EXEC PGM=EDGBKUP,PARM=' BACKUP'
//SYSPRINT DD SYSOUT=*
//MASTER  DD DSN=DFRMM.MASTER.SYS&ID.,DISP=SHR
//BACKUP   DD DSN=DFRMM.MASTER.BACKUP&ID.(+1),DISP=(,CATLG,DELETE),
// UNIT=SYSDA,SPACE=(6233,(1000,999),RLSE),
// DCB=(MODEL.VB9216,DSORG=PS)
// PEND
//BACKS1   EXEC BACKCDS,ID=1
//BACKS2   EXEC BACKCDS,ID=2

```

Figure 212. JCL to Back Up the CDSs on SYS1 and SYS2

13.4.6 Converting to an Existing CDS

When you are converting information to DFSMSrmm and want the target to be an existing CDS, you are really merging databases. We recommend that you follow the normal conversion process (as described in Chapter 9, “Building the DFSMSrmm CDS” on page 403) to the point where the CDS is built and verified, and you have started DFSMSrmm and executed any commands necessary to complete the conversion. At that point you know the conversion is good and that you have a valid and complete CDS. You can now use the process described in Section 13.4.4, “Merging the CDS” on page 520 to merge two DFSMSrmm CDSs.

If you have identified duplicate records, you may be able to change the extract and conversion control statements to avoid those duplicates rather than taking the actions recommended in Section 13.4.2, “Considerations” on page 513.

Chapter 14. Extending DFSMSrmm Reporting

In this chapter we describe how you can use the information that DFSMSrmm creates to generate customized reports that meet your specific tape management needs. You may be able to use the samples described in this chapter to meet some of your reporting requirements.

Objectives of Chapter

- Complete the report information provided with DFSMSrmm
- Provide samples that are ready to use
- Provide samples that match the key reports of your previous tape management system

Audience

- Production controllers
- Tape librarians
- Storage administrators

14.1 Producing Standard Reports

DFSMSrmm offers a series of standard reports that you can produce using the EDGHSKP, EDGRPTD, and EDGAUD utilities. The standard reports and utilities are documented in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932. Refer to that book for a detailed description.

14.1.1 EDGHSKP Reports

With the housekeeping utility, EDGHSKP, you can obtain two standard reports: a Vital Records Retention Report and a Summary Report.

You can also produce an activity report from either a normal run or a trial run of vital record processing.

The DD cards and EXEC PARMs for EDGHSKP are described in 10.1.7.1, “EDGHSKP Procedure” on page 438. Figure 213 shows sample JCL for VRS processing.

```
//HSKP    EXEC PGM=EDGHSKP,  
//        PARM='VRSEL'  
//MESSAGE DD DISP=SHR,DSN=HSKP.MESSAGES  
//REPORT  DD DISP=SHR,DSN=HSKP.VRS.REPORT
```

Figure 213. Sample JCL for EDGHSKP VRS Processing

Figure 214 on page 534 shows sample JCL of VRS trial run processing.

```
//HSCP    EXEC PGM=EDGHSKP,
//          PARM='VRSEL,VERIFY'
//MESSAGE DD DISP=SHR,DSN=HSCP.MESSAGES
//REPORT  DD DISP=SHR,DSN=HSCP.VRS.REPORT
//ACTIVITY DD DISP=SHR,DSN=HSCP.ACTIVITY
```

Figure 214. Sample JCL for Trial Run VRS Processing

The ACTIVITY DD file reports detailed information about changes made to data sets during vital record processing. The ACTIVITY file can be viewed online. To print the ACTIVITY file, use a tool like DFSORT or DFSORT ICETOOL to selectively format and print fields. DFSMSrmm provides a sample job EDGJACTP in SAMPLIB for this.

14.1.1.1 Vital Records Retention Report

During VRSEL processing, DFSMSrmm can create a Vital Records Retention Report (Figure 215) describing data sets and volumes that are being retained and the location of the data sets or volumes.

Removable Media Manager (C) IBM Corporation 1993		VITAL RECORDS RETENTION REPORT -----								PAGE 1 DATE 1997/01/19	
JOB MASK	DATA SET OR VOLUME MASK	OWNER	TYPE	RETN	C	X	DELETE	DLY	COUNT	STNUM	LOCATION
*	MOVE.**	MOVER	DSN	CYCLES	N	N	1999/12/31	0	99999	99999	SHELF
JOB NAME	DATASET NAME	FSEQ	DSEQ	VOLUME	VSEQ	OWNER	CURRENT	REQUIRED	PRTY	RETDTE	
STSGMAMW	MOVE.VOLUME.SOMEWHERE.NORETURN	1	1	987654	1	JC41444	SHELF	LIBDEF3	9000	CYCL/99999	
STSGMAMW	MOVE.VOLUME.SOMEWHERE.NORETURN	1	1	123456	1	JC41444	SHELF	LIBDEF3	9000	CYCL/99999	
TOTALS=								2	2		
STSGMAX1	MOVE.VOLUME.SOMEWHERE.NORETURN	1	1	STG001	1	BCG1234	SHELF	LIBDEF3	9000	CYCL/99999	
STSGMAX1	MOVE.VOLUME.SOMEWHERE.NORETURN	1	1	STG102	1	BCG2222	SHELF	LIBDEF3	9000	CYCL/99999	
TOTALS=								2	2		

Figure 215. Vital Records Retention Report Produced by EDGHSKP

If the VRS SPE is installed, the layout of the Vital Record Retention Report is changed to reflect the new VRS enhancements. Figure 216 on page 535 is an example of the new report layout.

Removable Media Manager (C) IBM Corporation 1993		VITAL RECORDS RETENTION REPORT -----										PAGE TIME 23:11:45 DATE	1 1997/331
JOB MASK	DATA SET OR VOLUME MASK	OWNER	TYPE	RETN	C	X	DELETE	DLY	COUNT	STNUM	LOCATION	RLSE	
	STSGMA.**	STSGMA	DSN	CYCLES	N	N	1999/365	0	3	2	HOME	XI	SI
	DAYS	STSGMA	NEXT	LRDAYS	N	N	1999/365		3	3	HOME		
JOB NAME	DATASET NAME	2ndVRS	2nd NAME	FSEQ	DSEQ	VOLUME	VSEQ	OWNER	CURRENT	REQUIRED	PRTY	RETDAT	RETNAME
	STSGMA.TEST.G0010V00			1	0	115009	1	STSGMA	SHELF	SHELF	5000	CYCL/00003	*
	STSGMA.TEST.G0009V00			1	0	115008	1	STSGMA	SHELF	SHELF	5000	CYCL/00003	*
	STSGMA.TEST.G0008V00			1	0	115007	1	STSGMA	SHELF	SHELF	5000	CYCL/00003	*
	STSGMA.TEST.G0007V00			1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/334	DAYS
	STSGMA.TEST.G0006V00			1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/334	DAYS
	STSGMA.TEST.G0005V00			1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/334	DAYS
	STSGMA.TEST.G0004V00			1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/334	DAYS
	STSGMA.TEST.G0003V00			1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/334	DAYS
NUMBER OF DATA SETS RETAINED (GROUP STORE) =										8	0		
JOB MASK	DATA SET OR VOLUME MASK	OWNER	TYPE	RETN	C	X	DELETE	DLY	COUNT	STNUM	LOCATION	RLSE	
	STSGMA.TEST.MC*.**	STSGMA	DSN	CYCLES	N	Y	1999/365	0	99999	99999	HOME	XI	SI
	UNCATLG	STSGMA	NEXT	XDAYS	N	N	1999/365		3	3	HOME		
JOB NAME	DATASET NAME	2ndVRS	2nd NAME	FSEQ	DSEQ	VOLUME	VSEQ	OWNER	CURRENT	REQUIRED	PRTY	RETDAT	RETNAME
	STSGMA.TEST.MC	MC99000		1	0	115009	1	STSGMA	SHELF	SHELF	5000	WHILECATLG	*
	STSGMA.TEST.MC	MC99000		1	0	115008	1	STSGMA	SHELF	SHELF	5000	WHILECATLG	*
	STSGMA.TEST.MC	MC99000		1	0	115007	1	STSGMA	SHELF	SHELF	5000	WHILECATLG	*
	STSGMA.TEST.MC	MC99000		1	0	115006	1	STSGMA	SHELF	SHELF	5000	1997/332	UNCATLG
	STSGMA.TEST.MC	MC99000		1	0	115005	1	STSGMA	SHELF	SHELF	5000	1997/333	UNCATLG
	STSGMA.TEST.MC	MC99000		1	0	115004	1	STSGMA	SHELF	SHELF	5000	1997/334	UNCATLG
NUMBER OF DATA SETS RETAINED (GROUP STORE) =										6	0		
JOB MASK	DATA SET OR VOLUME MASK	OWNER	TYPE	RETN	C	X	DELETE	DLY	COUNT	STNUM	LOCATION	RLSE	
	STSGMA.AND.**	STSGMA	DSN	BYDAYS	N	N	1999/365	0	3	3	HOME	XI	SI
	WHILEC	STSGMA	AND	CYCLES	Y	N	1999/365		99999				
JOB NAME	DATASET NAME	2ndVRS	2nd NAME	FSEQ	DSEQ	VOLUME	VSEQ	OWNER	CURRENT	REQUIRED	PRTY	RETDAT	RETNAME
	STSGMA.AND.G0002V00			1	0	AND002	1	STSGMA	SHELF	SHELF	5000	CYCL/00003	*
NUMBER OF DATA SETS RETAINED (GROUP STORE) =										1	0		

Figure 216. Vital Records Retention Report with VRS SPE Installed

14.1.1.2 Summary Report

During inventory management processing, DFSMSrmm provides a Summary Report in the MESSAGE DD file. The report contains a summary of EDGHSKP processing.

In addition to the statistics, DFSMSrmm identifies data sets that might be open and puts them in a list in the MESSAGE data set. Figure 217 on page 536 is an example of this output.

```

EDG2404W VOLUME 111020 FOR JOB IT23004A IS OPEN - VOLUME HAS EXPIRATION DATE 16/06/1997
EDG2404W VOLUME 111023 FOR JOB IT23004B IS OPEN - VOLUME HAS EXPIRATION DATE 16/06/1997
EDG2420I TOTAL VOLUMES READ = 1970
EDG2421I TOTAL VOLUMES UPDATED = 330
EDG2422I TOTAL VOLUMES, THIS RUN, KEPT FOR VRS = 50
EDG2423I TOTAL VOLUMES, THIS RUN, ASSIGNED TO STORES = 85
EDG2424I TOTAL VOLUMES, THIS RUN, SET PENDING RELEASE = 126
EDG2425I TOTAL VOLUMES RETURNED TO SCRATCH = 55
EDG2429I MAIN INVENTORY MANAGEMENT UPDATES HAVE COMPLETED SUCCESSFULLY
EDG2307I INVENTORY MANAGEMENT TASK EXPROC COMPLETED SUCCESSFULLY
EDG2307I INVENTORY MANAGEMENT TASK DSTORE COMPLETED SUCCESSFULLY
EDG2307I INVENTORY MANAGEMENT TASK VRSEL COMPLETED SUCCESSFULLY
EDG2307I INVENTORY MANAGEMENT TASK RPTEXT COMPLETED SUCCESSFULLY
EDG6401I MASTER FILE BACKUP SUCCESSFUL

```

Figure 217. EDGHSKP Summary Report

14.1.2 EDGRPTD Reports

The DFSMSrmm EDGRPTD utility produces inventory and movement reports from the extract data set created by using the EDGHSKP utility.

Use EDGRPTD to:

- List the contents of the system-managed libraries, non-system-managed shelf space, and storage locations
- List volumes that should be moved from the library to storage locations
- List volumes that should be moved from storage locations to the library
- List volumes that have different handling requirements

Figure 218 is an example of JCL for Inventory and Movement Reporting using the EDGRPTD utility.

```

//RPTD      EXEC PGM=EDGRPTD,
//          PARM=' SEC' '( INTERNAL USE ONLY)'', DATEFORM(I),LINECOUNT(54)'
//REPTXT    DD DISP=SHR,DSN=HSKP.REPTXT
//SYSPRINT  DD SYSOUT=*
//INSTVOL   DD DISP=SHR,DSN=HSKP.INSTVOL
//INSTBIN   DD DISP=SHR,DSN=HSKP.INSTBIN
//INSTOWN   DD DISP=SHR,DSN=HSKP.INSTOWN
//TOSTRCK   DD DISP=SHR,DSN=HSKP.TOSTRCK
//TOSTOWN   DD DISP=SHR,DSN=HSKP.TOSTOWN
//FMSTBIN   DD DISP=SHR,DSN=HSKP.FMSTBIN
//FMSTOWN   DD DISP=SHR,DSN=HSKP.FMSTOWN
//RDYTOSCR  DD DISP=SHR,DSN=HSKP.RDYTO SCR
//SYSOUT     DD SYSOUT=*

```

Figure 218. Sample of JCL for Inventory and Movement Reporting

The input file in the EDGRPTD run is the REPTXT DD file. The input file is the extract data set, created in the EDGHSKP Inventory Management run with PARM(RPTEXT) specified.

The output files can be allocated as GDSs. You only need to code DD statements for the reports you intend to create.

The DD cards for EDGRPTD are described in 10.1.7.3, “ EDGRPTD Report Utility” on page 440.

Figure 219 on page 537 is an example of an inventory report created by using EDGRPTD to list volumes in a specified location.

INTERNAL USE ONLY														
REMOVABLE MEDIA MANAGER					INVENTORY OF VOLUMES IN LOCATION DISTANT					PAGE				
5695-DF1 (C) Copyright IBM Corporation 1993					-----					DATE 08/22/1997				
VOLUME	RACK	BIN	OWNER	MEDIANAME T	VOLUME	RACK	BIN	OWNER	MEDIANAME T	VOLUME	RACK	BIN	OWNER	MEDIANAME T

A00003	A00003	13	@@@OWNER	0										
A00471	A00471	20	D094746	I										
A01128	A01128	4	D094746	0										
A01658	A01658	5	D094746	0										
A01668	A01668	6	D094746	0										
A01669	A01669	7	D094746	0										
TOTAL NUMBER OF ENTRIES LISTED = 6														

Figure 219. Volume Inventory Report

Figure 220 is a movement report showing volumes to be moved from the LOCAL storage location to library ATL1.

INTERNAL USE ONLY																	
Removable Media Manager					VOLUMES TO BE MOVED FROM LOCATION LOCAL TO LOCATION ATL1					PAGE 1							
5685-086 (C) Copyright IBM Corporation 1993					-----					DATE 08/22/1997							
BIN	VOLUME	RACK	OWNER	MEDIANAME	T	BIN	VOLUME	RACK	OWNER	MEDIANAME	T	BIN	VOLUME	RACK	OWNER	MEDIANAME	T
12	C00123	C00123	GARROTTO		3480 Y												
13	C00125	C00125	GLYN		3480 Y												
17	C00789	C00789	WANGSP		3480 Y												
21	C00801	C00801	GARROTTO		3480 Y												
24	C00946	C00946	KAHIL		3480 Y												
36	C01247	C01246	WANGSP		3480 Y												
44	C01249	C01249	TOMKIN1		3480 Y												
45	C01256	C01256	FLECK		3480 Y												
46	C01257	C01257	MBAKER		3480 Y												

Figure 220. Volume Movement Report

14.1.3 EDGAUD Reports

Use EDGAUD to create security and audit reports using either previously selected and sorted SMF records or raw SMF data. DFSMSrmm produces SMF records when you specify the SMFAUD or SMFSEC installation options in the PARMLIB member EDGRMMxx. DFSMSrmm uses the current subsystem startup option for the SMF record types and the default report option, unless you override them with the EDGAUD EXEC parameters.

Figure 221 on page 538 is a sample of JCL for the EDGAUD utility.

```

//AUDT      EXEC PGM=EDGAUD,
//          PARM=' SMFAUD=(nnn),SMFSEC=(nnn),SEC('' security classification'')'
//SYSPRINT DD SYSOUT=*
//SMFIN     DD DISP=SHR,DSN=BSYSMF.WEEK(0)
//AUDREPT   DD DISP=(NEW,CATLG),UNIT=SYSDA,
//          DSN=HSCP.AUDREPT,
//          SPACE=(TRK,(5,5),RLSE),
//          DCB=(DSORG=PS,RECFM=VBS,LRECL=32000,BLKSIZE=4096)
//SECREPT   DD DISP=(NEW,CATLG),UNIT=SYSDA,
//          DSN=HSCP.SECREPT,
//          SPACE=(TRK,(5,5),RLSE),
//          DCB=(DSORG=PS,RECFM=VBS,LRECL=32000,BLKSIZE=4096)
//SYSOUT    DD SYSOUT=*
//SYSIN     DD *
          SELECT DATE(06/21/1997,06/30/1997) -
          VOLUMES(A101*,B200*) RACK(A101*,B200*) -
          USERS(RMMU001,TAPE001)
/*

```

Figure 221. Sample of JCL for Audit Reporting Using EDGAUD Utility

Following are the DD statements for EDGAUD utility:

- SYSPRINT** Specifies program and information messages. This DD statement is required.
- SMFIN** Specifies the SMF record input data set. This DD statement is required.
- AUDREPT** Specifies to create an audit report in this data set. DFSMSrmm does not produce a report unless you do not specify this DD statement. The report data set record length is 132 characters. This DD statement is optional.
- SECREPT** Specifies to create a security report in this data set. DFSMSrmm does not produce a report unless you do not specify this DD statement. The report data set record length is 132 characters. This DD statement is optional.
- SYSOUT** Specifies an output file for DFSORT messages. To suppress DFSORT messages, define this DD statement as a DUMMY file. This DD statement is required.
- SYSIN** Specifies select statements for the audit report, which you specify with the AUDREPT DD statement. The select statement help you tailor the report. This DD statement is optional.

The format of the audit report selection options that can be supplied for SYSIN are shown in Figure 222:

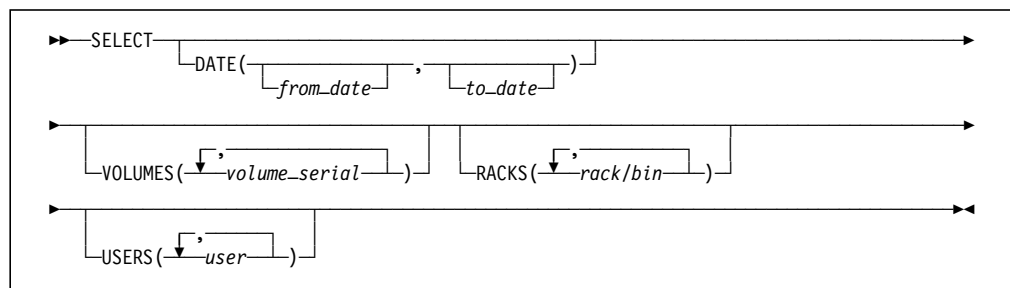


Figure 222. EDGAUD SYSIN Commands

Following is a description of the commands you can specify in the SYSIN DD statement. All SYSIN commands are optional and you can specify them in any order, except for SELECT, which must be specified first if you use any other command.

SELECT	Select is not required, but must be specified if you want it.
DATE(from_date,to_date)	Specify the date range of records to be selected for use in audit reports. The format of the date value depends of the setting of the DATEFORM parameter value.
RACKS(rack/bin)	Specify to limit the report to specific rack number or bin number. You can specify a list of values.
VOLUMES(volume_serial)	Specify to limit the report to specific volumes. You can specify a list of values.
USERS(user)	Specify to include only those changes made by specific users in the report. You can specify a list of users.

You can specify generic volume, rack, or user information. For example, you can specify VOLUMES(ABC*) to request all volumes whose volume serial number starts with 'ABC'.

See the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for a more detailed description about EDGAUD JCL, EXEC PARMS, and SYSIN statements.

Figure 223 is an example of a security report.

REMOVABLE MEDIA MANAGER (C) COPYRIGHT IBM CORPORATION 1993				REPORT OF ACCESSES TO SECURE VOLUMES ----- --				PAGE 1 DATE 02/26/1997			
DATA SET NAME	VOLUME	VSQ	DSQ	MEDIA	ACTION	SECURITY	GROUP	USERID	SYST	DATE	TIME
USERJOY.S1ATL026.D65DM1.BACKUP	002030	1	2	3490	CREATE	SECURE	SYS1	DILE	3090	02/23/1997	16:22:28
USERJOY.S1ATL026.D65DM1.BACKUP	002031	2	2	3490	CREATE	SECURE	SYS1	DILE	3090	02/23/1997	16:28:25
USERJOY.S1ATL026.D65DM1.BACKUP	002033	3	2	3490	CREATE	SECURE	SYS1	DILE	3090	02/23/1997	16:35:02
USERJOY.S1ATL026.USRPCK.BACKUP	002030	1	1	3490	CREATE	SECURE	SYS1	DILE	3090	02/19/1997	14:50:50
USERJOY.S1ATL026.USRPCK.BACKUP	002030	1	1	3490	CREATE	SECURE	SYS1	MIKE	3090	02/19/1997	15:15:11
USERJOY.S1ATL026.USRPCK.BACKUP	002030	1	1	3490	CREATE	SECURE	SYS1	DILE	3090	02/23/1997	16:17:37
RMMU001.RAC005.DS1	A00099	1	1	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/19/1997	11:06:46
RMMU001.RAC005.DS1	123456	1	1	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/19/1997	14:19:12
RMMU001.RAC005.DS1	A04101	1	1	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/22/1997	11:08:05
RMMU001.RAC005.DS1	A04101	1	1	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/22/1997	13:14:28
RMMU001.RAC005.DS2	A00099	1	2	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/19/1997	11:06:48
RMMU001.RAC005.DS2	123456	1	2	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/19/1997	14:19:14
RMMU001.RAC005.DS2	A04101	1	2	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/22/1997	11:08:07
RMMU001.RAC005.DS2	A04101	1	2	3480	CREATE	GENERAL	D65RMM	RMMU001	3090	02/22/1997	13:14:29
CAUDILL.S1VVA09.V1F1	A04201	1	1	3480	CREATE	CLASS11	SYS1	LYONS	3090	02/23/1997	16:48:24
CAUDILL.S1VVA09.V1F1	A04201	1	1	3480	CREATE	CLASS11	SYS1	LYONS	3090	02/23/1997	16:56:46
CAUDILL.S1VVA09.V2F1	A04301	1	1	3480	CREATE	CLASS11	SYS1	LYONS	3090	02/23/1997	16:53:47
CAUDILL.S1VVA09.V2F1	A04301	1	1	3480	CREATE	CLASS11	SYS1	LYONS	3090	02/23/1997	16:57:15
NICHOLSA.DATASET	A02000	1	1	3480	CREATE	SECURE	SYS1	GEORGE	3090	02/19/1997	09:37:02
NICHOLSB.DATASET	A02001	1	1	3480	CREATE	SECURE	SYS1	GEORGE	3090	02/19/1997	09:37:16

Figure 223. Report of Accesses to Secure Volumes

Figure 224 on page 540 shows excerpts from an audit trail report.

REMOVABLE MEDIA MANAGER (C) COPYRIGHT IBM CORPORATION 1993						AUDIT TRAIL REPORT -----				PAGE 1 DATE 07/11/1997				
VOLUME	RACK	BIN	USERID	DATE	TIME	SYSTEM	STATUS	LOCATION	LOAN	LOC	OWNER	EXP DATE	SECURITY	ACTIVITY
111000	111000	000033	DENZEL	16/11/1997	04:00:10	E4E4	MASTER	<REMOTE			RDRHSME	07/11/1997	U	UPDATE
111041	111041	000042	BJK	16/11/1997	04:00:03	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111054	111054	000043	PALMER	16/11/1997	04:00:14	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111056	111056	000044	WRIGHT	16/11/1997	04:00:10	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111089	111089	000048	GILLPAT	16/11/1997	04:00:08	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111113	111113	000121	WHEELER	16/11/1997	04:00:12	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111122	111122	000122	PENDLTN	16/11/1997	04:00:12	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111124	111124	000123	RDRHSME	16/11/1997	04:00:15	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111127	111127	000124	RDRHSME	16/11/1997	04:00:14	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE
111128	111128	000125	RDRHSME	16/11/1997	04:00:07	E4E4	MASTER	REMOTE			RDRHSME	07/11/1997	U	UPDATE

Figure 224. Audit Trail Report

Appendix D, "Sample Exits and Reports" on page 603 contains sample JCL for producing reports using EDGAUD.

14.2 DFSMSrmm Supplied Reports

DFSMSrmm provides REXX execs and sample JCL that can be used as-is to create reports. You can copy these reports and use them to create reports that are tailored for your installation. Documentation about the reports is available in the EDGDOC member in SYS1.SAMPLIB.

The following REXX execs are available to create an extended extract data set. The REXX execs are in SYS1.SEDGEXE1.

EDGRRPTM	REXX Exec to create an extended extract data set only for multiple data set/multiple volume reporting.
EDGRRPTN	REXX Exec to add the volume count to the extended extract data set for the entire inventory.
EDGRRPTR	REXX Exec to create an extended report extract data set.
EDGRRPTE	REXX Exec to create the reports.

Appendix D, "Sample Exits and Reports" on page 603 contains sample JCL to create reports using the EDGJRPT job. For a detailed description of how to customize the EDGJRPT JCL see the EDGDOC member in SAMPLIB.

Report Name	Description
REPORT01	Pull list for scratch tapes sorted by volume serial number
REPORT02	Pull list for scratch tapes sorted by data set name
REPORT03	Inventory list by volume serial number
REPORT04	Inventory list by data set name
REPORT05	Inventory of data sets including number of KB used
REPORT06	Inventory of volume serial number by location
REPORT07	Inventory of data set names by location
REPORT08	Inventory of bin numbers by location
REPORT09	List of all data set names at loan locations
REPORT10	List of all volume serial numbers at loan locations
REPORT11	List of all multiple volume, multiple data sets
REPORT12	Movement report including the first data set name of the volume
REPORT13	Movement report sorted by storage location bin number
REPORT14	Movement report sorted by volume serial number
REPORT15	Inventory list sorted by volume serial number including volume count

In the following section we show samples of REPORT01-14.

This section contains samples of the DFSMSrmm supplied reports, using the EDGJRPT JCL.

14.2.1 REPORT01

Figure 225 shows sample output from the REPORT01 report; pull list for scratch tapes sorted by volume serial number.

DFSMSrmm IBM INTERNAL USE ONLY		Scratch Tapes by Volume Serial Number										PAGE - 00001
EDGRPT01 -----												DATE - 97286
Volume	Data Set Name	Vol-Seq.	DSN-Seq.	Create Date	Org. Exp. Date	V F	LBL Typ	Media Type	Rec. Fmt	Home Location	S Name	Sum. Error
M00000		1		02/04/1996			SL *		*	SHELF		0
M00001		2		02/04/1996			SL *		*	SHELF		0
M00002		3		02/04/1996			SL *		*	SHELF		0
M00003		1		02/04/1996			SL *		*	SHELF		0
M00004		1		02/04/1996			SL *		*	SHELF		0
M00005		1		02/04/1996			SL *		*	SHELF		0
M00006		1		02/04/1996			SL *		*	SHELF		0
M00007		1		02/04/1996			SL *		*	SHELF		0
M00008		1		02/04/1996			SL *		*	SHELF		0
M00009		1		02/04/1996			SL *		*	SHELF		0
M00010		1		02/04/1996			SL *		*	SHELF		0
M00011		1		02/04/1996			SL *		*	SHELF		0
M00012		1		02/04/1996			SL *		*	SHELF		0
M00013		1		02/04/1996			SL *		*	SHELF		0
M00014		1		02/04/1996			SL *		*	SHELF		0
NR1154	SMPMCS	1	1	27/08/1992			SL *		*	SHELF		0
NR1302		1		23/07/1993			SL *		*	SHELF		0
SC0082		3		27/08/1992			SL CST		*	SHELF		0
SC0095	SYS1.DUMP3480.MXXSY0.DAY280.IBBT	1	1	14/07/1994			SL CST		*	SHELF		10
SC0096	SYS1.DUMP3480.MXXSY0.DAY280.IBBT	1	1	27/08/1992			SL CST		*	SHELF		0
SC0097	SYS1.DUMP3480.MXXSY0.DAY280.IBBT	2	1	27/08/1992			SL CST		*	SHELF		0
SC0098	SYS1.DUMP3480.MXXSY0.DAY280.IBBT	3	1	27/08/1992			SL CST		*	SHELF		0
SC0099	SYS1.DUMP3480.MXXSY0.DAY280.IBBT	4	1	27/08/1992			SL CST		*	SHELF		0
SC3011	HBAC.DMP.BUILD.VMBLD08.D97265.T431619	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3014	HBAC.DMP.REPOSIT.VMREP18.D97259.T240821	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3015	HBAC.DMP.REPOSIT.VMREP20.D97259.T062121	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3022	HBAC.DMP.REPOSIT.VMREP39.D97259.T350422	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3025	HBAC.DMP.REPOSIT.VMREP15.D97259.T215320	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3037	HBAC.DMP.REPOSIT.VMREP14.D97259.T294020	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3049	HBAC.DMP.REPOSIT.VMREP36.D97259.T154320	2	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3057	HBAC.DMP.REPOSIT.VMREP48.D97252.T102723	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3059	HBAC.DMP.REPOSIT.VMREP36.D97259.T154320	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3063	HBAC.BACKTAPE.DATASET	1	1	28/02/1997	31/12/1999		SL ECCST		*	SHELF		0
SC3084	HBAC.DMP.REPOSIT.VMREP16.D97259.T215620	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3087	HMIG.HMIGTAPE.DATASET	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3089	HMIG.HMIGTAPE.DATASET	1	1	28/02/1997	31/12/1999		SL ECCST		*	SHELF		4
SC3098	SSC.VITALREC.SYSTEM.MZBSP1.G0092V00	1	1	28/02/1997			SL CST		*	SHELF		1
SC3100	BSYOPC.OPCPR131.SAVE.DSN.G0024V00	1	1	28/02/1997	09/10/1997		SL CST		*	SHELF		0
SC3126	HBAC.DMP.REPOSIT.VMREP37.D97259.T220121	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3137	HBAC.BACKTAPE.DATASET	2	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3138	HMIG.HMIGTAPE.DATASET	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		1
SC3146	HBAC.DMP.BUILD.VMBLD10.D97265.T423219	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3150	HBAC.DMP.REPOSIT.VMREP17.D97259.T500521	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3161	HMIG.HMIGTAPE.DATASET	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3167	SSC.VITALREC.SYSTEM.MZBSP1.G0092V00	2	1	28/02/1997			SL CST		*	SHELF		0
SC3207	HBAC.DMP.BUILD.VEMEA05.D97265.T175321	2	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3208	HBAC.DMP.REPOSIT.VMREP37.D97259.T220121	2	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3211	HBAC.DMP.REPOSIT.VMREP19.D97259.T571721	1	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0
SC3217	SSC.VITALREC.SYSTEM.MZBSP1.G0092V00	3	1	28/02/1997			SL CST		*	SHELF		0
SC3219	HBAC.DMP.REPOSIT.VMREP39.D97259.T350422	2	1	28/02/1997	31/12/1999		SL CST		*	SHELF		0

Figure 225. REPORT01 Output: Pull List for Scratch Tapes Sorted by Volume Serial

14.2.2 REPORT02

Figure 226 shows sample output from the REPORT02 report; pull list for scratch tapes sorted by data set name.

DFSMSrmm IBM INTERNAL USE ONLY		Scratch Tapes by Data Set Name										PAGE - 00001
EDGRPT02	-----	Vol-	DSN-	Create	Org. Exp.	V LBL	Media	Rec.	Home	S	Location	Sum.
Serial	Data Set Name	Seq.	Seq.	Date	Date	F Typ	Type	Fmt	Location	S	Name	Error
SR0090		1		1997/191		SL	CST	*	SHELF			0
SR0091		1		1997/191		SL	CST	*	SHELF			0
SR0092		1		1997/191		SL	CST	*	SHELF			0
SR0093		1		1997/191		SL	CST	*	SHELF			0
SR0094		1		1997/191		SL	CST	*	SHELF			0
SR0095		1		1997/191		SL	CST	*	SHELF			0
SR0096		1		1997/191		SL	CST	*	SHELF			0
SR0097		1		1997/191		SL	CST	*	SHELF			0
SR0098		1		1997/191		SL	CST	*	SHELF			0
SR0099		1		1997/191		SL	CST	*	SHELF			0
T00000		1		1994/245		SL	*	*	SHELF			0
T00001		1		1994/245		SL	*	*	SHELF			0
T00002		1		1994/245		SL	*	*	SHELF			0
SC4421	BSYOPC.OPCPR131.SAVE.AD.G0026V00	1	1	1997/260	1997/313	SL	CST	*	SHELF			0
SC3445	EMEGZBD.EGZB.D1002332.OS#120.TEMP	1	1	1997/059		SL	CST	*	SHELF			0
SC3446	EMEGZBD.EGZB.D2002332.OS#120.TEMP	1	1	1997/059		SL	CST	*	SHELF			0
SC3027	HBAC.BACKTAPE.DATASET	2	1	1997/059	1999/365	SL	ECCST	*	SHELF			0
SC3361	HBAC.BACKTAPE.DATASET	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3487	HBAC.BACKTAPE.DATASET	2	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3491	HBAC.BACKTAPE.DATASET	2	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3625	HBAC.BACKTAPE.DATASET	2	1	1997/059	1999/365	SL	ECCST	*	SHELF			0
SC3626	HBAC.BACKTAPE.DATASET	3	1	1997/059	1999/365	SL	ECCST	*	SHELF			0
SC3634	HBAC.BACKTAPE.DATASET	3	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3637	HBAC.BACKTAPE.DATASET	2	1	1997/059	1999/365	SL	ECCST	*	SHELF			0
SC4043	HBAC.BACKTAPE.DATASET	1	1	1997/164	1999/365	SL	ECCST	*	SHELF			0
SC4054	HBAC.BACKTAPE.DATASET	2	1	1997/164	1999/365	SL	CST	*	SHELF			0
SC4159	HBAC.BACKTAPE.DATASET	3	1	1997/164	1999/365	SL	CST	*	SHELF			0
SC4192	HBAC.BACKTAPE.DATASET	1	1	1997/164	1999/365	SL	CST	*	SHELF			0
SC4259	HBAC.BACKTAPE.DATASET	1	1	1997/199	1999/365	SL	CST	*	SHELF			0
SC4499	HBAC.BACKTAPE.DATASET	2	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC3389	HBAC.DMP.BUILD.VEMEA00.D97293.T133421	2	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3353	HBAC.DMP.BUILD.VEMEA02.D97293.T304621	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3429	HBAC.DMP.BUILD.VEMEA04.D97293.T395821	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3423	HBAC.DMP.BUILD.VEMEA05.D97293.T000422	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3444	HBAC.DMP.BUILD.VEMEA05.D97293.T000422	2	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3461	HBAC.DMP.BUILD.VEMEA06.D97293.T440522	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3473	HBAC.DMP.BUILD.VEMEA07.D97293.T442122	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3300	HBAC.DMP.BUILD.VEMEA10.D97293.T483218	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3457	HBAC.DMP.BUILD.VMBLD01.D97293.T483218	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3401	HBAC.DMP.BUILD.VMBLD02.D97293.T483218	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3459	HBAC.DMP.BUILD.VMBLD04.D97293.T034718	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC4460	HBAC.DMP.BUILD.VMBLD06.D97286.T485818	1	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC4462	HBAC.DMP.BUILD.VMBLD12.D97286.T375119	1	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC4464	HBAC.DMP.BUILD.VMBLD13.D97286.T480120	1	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC4463	HBAC.DMP.BUILD.VMBLD14.D97286.T330420	1	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC3297	HBAC.DMP.BUILD.VMBLD14.D97293.T044420	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC4466	HBAC.DMP.BUILD.VMBLD15.D97286.T561220	1	1	1997/260	1999/365	SL	CST	*	SHELF			0
SC3312	HBAC.DMP.BUILD.VMBLD20.D97293.T342621	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3501	HBAC.DMP.REPOSIT.VMREP01.D97294.T343018	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3498	HBAC.DMP.REPOSIT.VMREP02.D97294.T343018	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3506	HBAC.DMP.REPOSIT.VMREP03.D97294.T214618	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3511	HBAC.DMP.REPOSIT.VMREP04.D97294.T155218	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3530	HBAC.DMP.REPOSIT.VMREP05.D97294.T585918	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3542	HBAC.DMP.REPOSIT.VMREP06.D97294.T010519	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3541	HBAC.DMP.REPOSIT.VMREP07.D97294.T471619	1	1	1997/059	1999/365	SL	CST	*	SHELF			0
SC3544	HBAC.DMP.REPOSIT.VMREP08.D97294.T353519	1	1	1997/059	1999/365	SL	CST	*	SHELF			0

Figure 226. REPORT02 Output: Pull List for Scratch Tapes by Data Set Name

14.2.3 REPORT03

Figure 227 shows sample output from the REPORT03 report; inventory list sorted by volume serial number.

DFSMSrmm IBM INTERNAL USE ONLY		Inventory List by Volume Serial Number								PAGE -	00001
EDGRPT03 -----										DATE -	97286
Volume	Serial	Data Set Name	Vol-Seq.	DSN-Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Volume Ref. Date	LBL	Rec. V V Location
Serial	Data Set Name										
ABJ001			1					03/11/1997		SL	* M N
ACQ32A			1					03/11/1997		SL	* M N
ADB100	SMPMCS		1	1		28/08/1995	134142	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F1		1	2		28/08/1995	141233	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F2		1	3		28/08/1995	141242	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F3		1	4		28/08/1995	141252	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F4		1	5		28/08/1995	141302	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F5		1	6		28/08/1995	141312	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F6		1	7		28/08/1995	141322	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F7		1	8		28/08/1995	141336	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F8		1	9		28/08/1995	141345	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F9		1	10		28/08/1995	141355	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F10		1	11		28/08/1995	141405	22/08/1997	28/08/1995	SL	18TR M N
ADB100	IBM.H0IH110.F11		1	12		28/08/1995	141416	22/08/1997	28/08/1995	SL	18TR M N
AD2502			1					08/11/1997		SL	* M N
AD2522	SMPMCS		1	1		30/07/1992	200559	08/11/1997		SL	* M N
AD2524	SMPMCS		1	1		30/07/1992	200701	08/11/1997		SL	* M N
AD2526			1					08/11/1997		SL	* M N
APE211			1					03/11/1997		SL	* M N
AQ1102			1					03/11/1997		SL	* M N
ASAP16	ASAP.SAMPLE.MONITOR		1	1		25/08/1992	082701	08/11/1997		SL	* M N
AU1103			1					03/11/1997		SL	* M N
AZ1200			1					03/11/1997		SL	* M N
AZ1230			1					03/11/1997		SL	* M N
A01200			1					03/11/1997		SL	* M N
A01230			1					03/11/1997		SL	* M N
BBF200			1					03/11/1997		SL	* M N
BBF230			1					03/11/1997		SL	* M N
BB5510	SMPMCS		1	1		28/07/1994	103509	27/10/1998	28/07/1994	SL	18TR M N
BB5511	IBM.H0M1120.F2		2	1		28/07/1994	104111	27/10/1998	28/07/1994	SL	18TR M N
BB5511	IBM.J0M12N0.F1		2	2		28/07/1994	104134	27/10/1998	28/07/1994	SL	18TR M N
BB5511	IBM.J0M12N0.F2		2	3		28/07/1994	104146	27/10/1998	28/07/1994	SL	18TR M N
BB5520			1					27/10/1998		SL	* M N
BB5521			1					27/10/1998		SL	* M N
BB5522	SMPMCS		1	1	REC522BA	19/10/1995	104023	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.HBB5520.F1		1	2	REC522BA	19/10/1995	104458	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.HBB5520.F2		1	3	REC522BA	19/10/1995	104511	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.HBB5520.F3		1	4	REC522BA	19/10/1995	104527	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.HBB5520.F4		1	5	REC522BA	19/10/1995	104747	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.JBB5522.F1		1	6	REC522BA	19/10/1995	104955	27/10/1998	19/10/1995	SL	18TR M N
BB5522	IBM.JBB5522.F2		1	7	REC522BA	19/10/1995	105015	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JBB5522.F2		2	1	REC522BA	19/10/1995	105025	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JBB52N0.F1		2	2	REC522BA	19/10/1995	105035	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JBB52N0.F2		2	3	REC522BA	19/10/1995	105052	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JBB55N2.F1		2	4	REC522BA	19/10/1995	105105	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.HCSH521.F1		2	5	REC522BA	19/10/1995	105116	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.HCSH521.F2		2	6	REC522BA	19/10/1995	105127	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.HCSH521.F3		2	7	REC522BA	19/10/1995	105141	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JCSH523.F1		2	8	REC522BA	19/10/1995	105207	27/10/1998	19/10/1995	SL	18TR M N
BB5523	IBM.JCSH523.F2		2	9	REC522BA	19/10/1995	105220	27/10/1998	19/10/1995	SL	18TR M N

Figure 227. REPORT03 Output: Inventory List Sorted by Volume Serial Number

14.2.4 REPORT04

Figure 228 on page 544 shows sample output from the REPORT04 report; inventory list sorted by data set name.

DFSMSrmm IBM INTERNAL USE ONLY		Inventory List by Data Set Name										PAGE - 00001			
EDGRPT04 -----		Volume	Vol-	DSN-	Creating	Create	Create	Expiration	Volume			DATE - 97286	Rec.	V V	Location
Data Set Name	Serial	Seq.	Seq.	Jobname	Date	Time	Date	Date	Ref. Date	LBL		Fmt	S	R	Name
ADDONS.CNTL	SD2573	1	6	POOLSD	20/11/1996	134126	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.EXEC	SD2234	1	3	POOLSD	14/12/1995	132803	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.EXEC	SD2573	1	3	POOLSD	20/11/1996	134058	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.EXECFB	SD2234	1	4	POOLSD	14/12/1995	132809	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.EXECFB	SD2573	1	4	POOLSD	20/11/1996	134106	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.INITVARS	SD2234	1	6	POOLSD	14/12/1995	132821	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.INITVARS	SD2573	1	10	POOLSD	20/11/1996	134152	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.LOAD	SD2573	1	7	POOLSD	20/11/1996	134133	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.MSGS	SD2234	1	5	POOLSD	14/12/1995	132817	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.MSGS	SD2573	1	5	POOLSD	20/11/1996	134121	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.OBJ	SD2573	1	8	POOLSD	20/11/1996	134139	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.PANELS	SD2234	1	1	POOLSD	14/12/1995	132752	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.PANELS	SD2573	1	1	POOLSD	20/11/1996	134045	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.SKELS	SD2234	1	2	POOLSD	14/12/1995	132759	13/01/1998	14/12/1995	SL	18TR	M N	GDF			
ADDONS.SKELS	SD2573	1	2	POOLSD	20/11/1996	134052	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ADDONS.SOURCE	SD2573	1	9	POOLSD	20/11/1996	134144	21/12/1998	20/11/1996	SL	18TR	M N	LIDL			
ASAP.SAMPLE.MONITOR	ASAP16	1	1		25/08/1992	082701	08/11/1997		SL	*		M N			
AUDITFMT.SRC	HQK642	1	7		01/03/1995	073854	28/02/1999	01/03/1995	SL	18TR	M N				
A0D204MS.BOOK	HQK642	1	9		01/03/1995	073901	28/02/1999	01/03/1995	SL	18TR	M N				
BADER.TLMS.RMF	68059C	1	1		27/03/1995	153951	10/04/1998	27/03/1995	SL	18TR	U N	REMOTE			
BOOKENG.AGREEMNT	HQK642	1	8		01/03/1995	073857	28/02/1999	01/03/1995	SL	18TR	M N				
BSYMS.DATSICTH.BLD000.MVSSSC.D930719A	SC0022	1	1		19/07/1993	145431	03/01/2021	19/07/1993	SL	*		U N			
BSYOPC.OPCPR131.SAVE.AD.G0022V00	SC3214	1	1	SAVEOPC1	12/09/1997	180115	12/10/1997	12/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.AD.G0023V00	SC3181	1	1	SAVEOPC1	19/09/1997	180052	19/10/1997	19/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.AD.G0024V00	SC3999	1	1	SAVEOPC1	26/09/1997	180057	26/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.AD.G0025V00	SC4373	1	1	SAVEOPC1	06/10/1997	180117	05/11/1997	06/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.AD.G0026V00	SC4421	1	1	SAVEOPC1	10/10/1997	180125	09/11/1997	10/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP1.G0022V00	SC3214	1	2	SAVEOPC1	12/09/1997	180119	12/10/1997	12/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP1.G0023V00	SC3181	1	2	SAVEOPC1	19/09/1997	180057	19/10/1997	19/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP1.G0024V00	SC3999	1	2	SAVEOPC1	26/09/1997	180101	26/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP1.G0025V00	SC4373	1	2	SAVEOPC1	06/10/1997	180121	05/11/1997	06/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP1.G0026V00	SC4421	1	2	SAVEOPC1	10/10/1997	180129	09/11/1997	10/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP2.G0022V00	SC3214	1	3	SAVEOPC1	12/09/1997	180124	12/10/1997	12/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP2.G0023V00	SC3181	1	3	SAVEOPC1	19/09/1997	180104	19/10/1997	19/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP2.G0024V00	SC3999	1	3	SAVEOPC1	26/09/1997	180106	26/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP2.G0025V00	SC4373	1	3	SAVEOPC1	06/10/1997	180126	05/11/1997	06/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CP2.G0026V00	SC4421	1	3	SAVEOPC1	10/10/1997	180135	09/11/1997	10/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CX.G0022V00	SC3214	1	4	SAVEOPC1	12/09/1997	180128	12/10/1997	12/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CX.G0023V00	SC3181	1	4	SAVEOPC1	19/09/1997	180111	19/10/1997	19/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CX.G0024V00	SC3999	1	4	SAVEOPC1	26/09/1997	180112	26/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CX.G0025V00	SC4373	1	4	SAVEOPC1	06/10/1997	180131	05/11/1997	06/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.CX.G0026V00	SC4421	1	4	SAVEOPC1	10/10/1997	180141	09/11/1997	10/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.DSN.G0024V00	SC3100	1	1	SAVEOPC2	19/09/1997	180316		19/09/1997	SL	*		S N			
BSYOPC.OPCPR131.SAVE.DSN.G0025V00	SC4011	1	1	SAVEOPC2	26/09/1997	180244	16/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.DSN.G0026V00	SC4377	1	1	SAVEOPC2	06/10/1997	180300	26/10/1997	06/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.DSN.G0027V00	SC4318	1	1	SAVEOPC2	10/10/1997	180313	30/10/1997	10/10/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.LB.G0022V00	SC3214	1	5	SAVEOPC1	12/09/1997	180133	12/10/1997	12/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.LB.G0023V00	SC3181	1	5	SAVEOPC1	19/09/1997	180118	19/10/1997	19/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.LB.G0024V00	SC3999	1	5	SAVEOPC1	26/09/1997	180117	26/10/1997	26/09/1997	SL	36TR	M N				
BSYOPC.OPCPR131.SAVE.LB.G0025V00	SC4373	1	5	SAVEOPC1	06/10/1997	180137	05/11/1997	06/10/1997	SL	36TR	M N				

Figure 228. REPORT04 Output: Inventory List Sorted by Data Set Name

14.2.5 REPORT05

Figure 229 on page 545 shows sample output from the REPORT05 report; inventory of data sets including number of KB used.

DFSMSrmm IBM INTERNAL USE ONLY		Inventory List by Data Set Name incl. used KB										PAGE - 00001		
EDGRPT05 -----												DATE - 97286		
Data Set Name	Volume Serial	Vol-Seq.	DSN-Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Volume Ref. Date	LBL	Rec. Fmt	V S	Kilobytes used		
ADDONS.CNTL	SD2573	1	6	POOLSD	20/11/1996	134126	21/12/1998	20/11/1996	SL	18TR	M	1993		
ADDONS.EXEC	SD2234	1	3	POOLSD	14/12/1995	132803	13/01/1998	14/12/1995	SL	18TR	M	2042		
ADDONS.EXEC	SD2573	1	3	POOLSD	20/11/1996	134058	21/12/1998	20/11/1996	SL	18TR	M	3602		
ADDONS.EXECFB	SD2234	1	4	POOLSD	14/12/1995	132809	13/01/1998	14/12/1995	SL	18TR	M	1852		
ADDONS.EXECFB	SD2573	1	4	POOLSD	20/11/1996	134106	21/12/1998	20/11/1996	SL	18TR	M	4115		
ADDONS.INITVARS	SD2234	1	6	POOLSD	14/12/1995	132821	13/01/1998	14/12/1995	SL	18TR	M	27		
ADDONS.INITVARS	SD2573	1	10	POOLSD	20/11/1996	134152	21/12/1998	20/11/1996	SL	18TR	M	54		
ADDONS.LOAD	SD2573	1	7	POOLSD	20/11/1996	134133	21/12/1998	20/11/1996	SL	18TR	M	337		
ADDONS.MSGS	SD2234	1	5	POOLSD	14/12/1995	132817	13/01/1998	14/12/1995	SL	18TR	M	412		
ADDONS.MSGS	SD2573	1	5	POOLSD	20/11/1996	134121	21/12/1998	20/11/1996	SL	18TR	M	435		
ADDONS.OBJ	SD2573	1	8	POOLSD	20/11/1996	134139	21/12/1998	20/11/1996	SL	18TR	M	368		
ADDONS.PANELS	SD2234	1	1	POOLSD	14/12/1995	132752	13/01/1998	14/12/1995	SL	18TR	M	2084		
ADDONS.PANELS	SD2573	1	1	POOLSD	20/11/1996	134045	21/12/1998	20/11/1996	SL	18TR	M	3184		
ADDONS.SKELS	SD2234	1	2	POOLSD	14/12/1995	132759	13/01/1998	14/12/1995	SL	18TR	M	733		
ADDONS.SKELS	SD2573	1	2	POOLSD	20/11/1996	134052	21/12/1998	20/11/1996	SL	18TR	M	1489		
ADDONS.SOURCE	SD2573	1	9	POOLSD	20/11/1996	134144	21/12/1998	20/11/1996	SL	18TR	M	3596		
ASAP.SAMPLE.MONITOR	ASAP16	1	1		25/08/1992	082701	08/11/1997		SL	*	M	1131		
AUDITFMT.SRC	HQK642	1	7		01/03/1995	073854	28/02/1999	01/03/1995	SL	18TR	M	78		
A0D204MS.BOOK	HQK642	1	9		01/03/1995	073901	28/02/1999	01/03/1995	SL	18TR	M	476		
BADER.TLMS.RMF	68059C	1	1		27/03/1995	153951	10/04/1998	27/03/1995	SL	18TR	U	34		
BOOKENG.AGREEMNT	HQK642	1	8		01/03/1995	073857	28/02/1999	01/03/1995	SL	18TR	M	31		
BSYMS.DATSICTH.BLD000.MVSSSC.D930719A	SC0022	1	1		19/07/1993	145431	03/01/2021	19/07/1993	SL	*	U	351040		
BSYOPC.OPCPR131.SAVE.AD.G0022V00	SC3214	1	1	SAVEOPC1	12/09/1997	180115	12/10/1997	12/09/1997	SL	36TR	M	63		
BSYOPC.OPCPR131.SAVE.AD.G0023V00	SC3181	1	1	SAVEOPC1	19/09/1997	180052	19/10/1997	19/09/1997	SL	36TR	M	63		
BSYOPC.OPCPR131.SAVE.AD.G0024V00	SC3999	1	1	SAVEOPC1	26/09/1997	180057	26/10/1997	26/09/1997	SL	36TR	M	63		
BSYOPC.OPCPR131.SAVE.AD.G0025V00	SC4373	1	1	SAVEOPC1	06/10/1997	180117	05/11/1997	06/10/1997	SL	36TR	M	63		
BSYOPC.OPCPR131.SAVE.AD.G0026V00	SC4421	1	1	SAVEOPC1	10/10/1997	180125	09/11/1997	10/10/1997	SL	36TR	M	63		
BSYOPC.OPCPR131.SAVE.CP1.G0022V00	SC3214	1	2	SAVEOPC1	12/09/1997	180119	12/10/1997	12/09/1997	SL	36TR	M	190		
BSYOPC.OPCPR131.SAVE.CP1.G0023V00	SC3181	1	2	SAVEOPC1	19/09/1997	180057	19/10/1997	19/09/1997	SL	36TR	M	222		
BSYOPC.OPCPR131.SAVE.CP1.G0024V00	SC3999	1	2	SAVEOPC1	26/09/1997	180101	26/10/1997	26/09/1997	SL	36TR	M	285		
BSYOPC.OPCPR131.SAVE.CP1.G0025V00	SC4373	1	2	SAVEOPC1	06/10/1997	180121	05/11/1997	06/10/1997	SL	36TR	M	253		
BSYOPC.OPCPR131.SAVE.CP1.G0026V00	SC4421	1	2	SAVEOPC1	10/10/1997	180129	09/11/1997	10/10/1997	SL	36TR	M	349		
BSYOPC.OPCPR131.SAVE.CP2.G0022V00	SC3214	1	3	SAVEOPC1	12/09/1997	180124	12/10/1997	12/09/1997	SL	36TR	M	190		
BSYOPC.OPCPR131.SAVE.CP2.G0023V00	SC3181	1	3	SAVEOPC1	19/09/1997	180104	19/10/1997	19/09/1997	SL	36TR	M	222		
BSYOPC.OPCPR131.SAVE.CP2.G0024V00	SC3999	1	3	SAVEOPC1	26/09/1997	180106	26/10/1997	26/09/1997	SL	36TR	M	285		
BSYOPC.OPCPR131.SAVE.CP2.G0025V00	SC4373	1	3	SAVEOPC1	06/10/1997	180126	05/11/1997	06/10/1997	SL	36TR	M	253		
BSYOPC.OPCPR131.SAVE.CP2.G0026V00	SC4421	1	3	SAVEOPC1	10/10/1997	180135	09/11/1997	10/10/1997	SL	36TR	M	349		
BSYOPC.OPCPR131.SAVE.CX.G0022V00	SC3214	1	4	SAVEOPC1	12/09/1997	180128	12/10/1997	12/09/1997	SL	36TR	M	31		
BSYOPC.OPCPR131.SAVE.CX.G0023V00	SC3181	1	4	SAVEOPC1	19/09/1997	180111	19/10/1997	19/09/1997	SL	36TR	M	31		
BSYOPC.OPCPR131.SAVE.CX.G0024V00	SC3999	1	4	SAVEOPC1	26/09/1997	180112	26/10/1997	26/09/1997	SL	36TR	M	31		
BSYOPC.OPCPR131.SAVE.CX.G0025V00	SC4373	1	4	SAVEOPC1	06/10/1997	180131	05/11/1997	06/10/1997	SL	36TR	M	31		
BSYOPC.OPCPR131.SAVE.CX.G0026V00	SC4421	1	4	SAVEOPC1	10/10/1997	180141	09/11/1997	10/10/1997	SL	36TR	M	31		
BSYOPC.OPCPR131.SAVE.DSN.G0024V00	SC3100	1	1	SAVEOPC2	19/09/1997	180316		19/09/1997	SL	*	S	22089		
BSYOPC.OPCPR131.SAVE.DSN.G0025V00	SC4011	1	1	SAVEOPC2	26/09/1997	180244	16/10/1997	26/09/1997	SL	36TR	M	21264		
BSYOPC.OPCPR131.SAVE.DSN.G0026V00	SC4377	1	1	SAVEOPC2	06/10/1997	180300	26/10/1997	06/10/1997	SL	36TR	M	21232		
BSYOPC.OPCPR131.SAVE.DSN.G0027V00	SC4318	1	1	SAVEOPC2	10/10/1997	180313	30/10/1997	10/10/1997	SL	36TR	M	21391		
BSYOPC.OPCPR131.SAVE.LB.G0022V00	SC3214	1	5	SAVEOPC1	12/09/1997	180133	12/10/1997	12/09/1997	SL	36TR	M	1142		
BSYOPC.OPCPR131.SAVE.LB.G0023V00	SC3181	1	5	SAVEOPC1	19/09/1997	180118	19/10/1997	19/09/1997	SL	36TR	M	1142		
BSYOPC.OPCPR131.SAVE.LB.G0024V00	SC3999	1	5	SAVEOPC1	26/09/1997	180117	26/10/1997	26/09/1997	SL	36TR	M	253		
BSYOPC.OPCPR131.SAVE.LB.G0025V00	SC4373	1	5	SAVEOPC1	06/10/1997	180137	05/11/1997	06/10/1997	SL	36TR	M	253		

Figure 229. REPORT05 Output: Inventory of Data Sets Including KB Used

14.2.6 REPORT06

Figure 230 on page 546 shows sample output from the REPORT06 report; inventory of volume serial number by location.

DFSMSrmm IBM INTERNAL USE ONLY		Inventory of Volumes in Storage Location DISTANT								PAGE - 00001
EDGRPT06 -----										DATE - 97316
Volume		BIN	Creating	Vol-	DSN-	Create	Create	Expiration	Date	Rec. V
Serial	Data Set Name	Number	Jobname	Seq.	Seq.	Date	Time	Date	stored	LBL Fmt S

SC3354	SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	000030	DMZBPP0#	1	1	1997/310	222851	1997/315	1997/315	SL 36TR M
SC3357	SSC.VITALREC.BUILD.DB.G0172V00	000031	VRDB	1	1	1997/311	130501	1997/316	1997/315	SL 36TR M
SC3362	SSC.VITALREC.BUILD.PP.G0165V00	000032	VRPP	1	1	1997/311	134953	1997/316	1997/315	SL 36TR M
SC3365	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000033	VRWWZ38#	1	1	1997/311	172236	1997/316	1997/315	SL 36TR M
SC3368	SSC.VITALREC.BUILD.WWP004.G0041V00	000034	VRWWP004	1	1	1997/311	163655	1997/316	1997/315	SL 36TR M
SC3371	SSC.VITALREC.BUILD.NET.G0164V00	000035	VRNET	2	1	1997/311	140201	1997/316	1997/315	SL 36TR M
SC3377	SSC.VITALREC.BUILD.ESA52.G0080V00	000025	VRSA52L	2	1	1997/119	101517	1999/119	1997/126	SL 36TR M
SC3382	SSC.VITALREC.BUILD.NET.G0164V00	000036	VRNET	3	1	1997/311	141819	1997/316	1997/315	SL 36TR M
SC3386	SSC.VITALREC.BUILD.NET.G0164V00	000037	VRNET	1	1	1997/311	134339	1997/316	1997/315	SL 36TR M
SC3391	SSC.VITALREC.BUILD.ES52.G0031V00	000038	VRSA52	1	1	1997/311	143509	1997/316	1997/315	SL 36TR M
SC3393	SSC.VITALREC.BUILD.ES52.G0031V00	000039	VRSA52	2	1	1997/311	144903	1997/316	1997/315	SL 36TR M
SC3394	SSC.VITALREC.BUILD.PP.G0165V00	000040	VRPP	2	1	1997/311	141145	1997/316	1997/315	SL 36TR M
SC3407	SSC.VITALREC.BUILD.PP.G0165V00	000041	VRPP	3	1	1997/311	142639	1997/316	1997/315	SL 36TR M
SC3415	SSC.VITALREC.BUILD.PP.G0165V00	000042	VRPP	4	1	1997/311	144048	1997/316	1997/315	SL 36TR M
SC3417	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000043	VRWWZ38#	2	1	1997/311	180657	1997/316	1997/315	SL 36TR M
SC3421	SSC.VITALREC.BUILD.ES52.G0031V00	000044	VRSA52	5	1	1997/311	153214	1997/316	1997/315	SL 36TR M
SC3427	SSC.VITALREC.BUILD.PP.G0165V00	000045	VRPP	5	1	1997/311	150122	1997/316	1997/315	SL 36TR M
SC3433	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000046	VRWWZ38#	3	1	1997/311	182520	1997/316	1997/315	SL 36TR M
SC3452	SSC.VITALREC.BUILD.ES52.G0031V00	000047	VRSA52	3	1	1997/311	150311	1997/316	1997/315	SL 36TR M
SC3462	SSC.VITALREC.BUILD.ES52.G0031V00	000048	VRSA52	4	1	1997/311	151656	1997/316	1997/315	SL 36TR M
SC3463	SSC.VITALREC.BUILD.WWP115.G0041V00	000049	VRWWP115	2	1	1997/311	170137	1997/316	1997/315	SL 36TR M
SC3467	SSC.VITALREC.BUILD.ES52.G0031V00	000050	VRSA52	6	1	1997/311	154353	1997/316	1997/315	SL 36TR M
SC3476	SSC.VITALREC.BUILD.WWP115.G0041V00	000052	VRWWP115	1	1	1997/311	164732	1997/316	1997/315	SL 36TR M
SC3483	SSC.VITALREC.BUILD.ESAS.G0134V00	000028	VRSA5L	2	1	1997/119	105757	1999/119	1997/126	SL 36TR M
SC3484	SSC.VITALREC.BUILD.ESAS.G0134V00	000029	VRSA5L	1	1	1997/119	103831	1999/119	1997/126	SL 36TR M
SC3490	SSC.VITALREC.BUILD.WWC150.G0041V00	000053	VRWWC150	2	1	1997/311	162546	1997/316	1997/315	SL 36TR M
SC3510	SSC.VITALREC.BUILD.WWC150.G0041V00	000054	VRWWC150	1	1	1997/311	161143	1997/316	1997/315	SL 36TR M
SC3568	SSC.VITALREC.BUILD.MVSSMP.G0163V00	000055	VRMVSSMP	1	1	1997/311	155507	1997/316	1997/315	SL 36TR M
SC3621	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000056	VRWWZ38#	4	1	1997/311	183740	1997/316	1997/315	SL 36TR M
SC3799	SSC.VITALREC.BUILD.ESA52.G0080V00	000051	VRSA52L	1	1	1997/119	095924	1999/119	1997/126	SL 36TR M
SC3926	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000058	VRWWZ38#	5	1	1997/311	185336	1997/316	1997/315	SL 36TR M
SC3928	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000059	VRWWZ38#	6	1	1997/311	192122	1997/316	1997/315	SL 36TR M
SC3930	SSC.VITALREC.BUILD.WWZ038.G0043V00	000060	VRWWZ038	2	1	1997/311	230837	1997/316	1997/315	SL 36TR M
SC3932	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000061	VRWWZ38#	7	1	1997/311	195327	1997/316	1997/315	SL 36TR M
SC3933	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000062	VRWWZ38#	9	1	1997/311	205912	1997/316	1997/315	SL 36TR M
SC3935	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000063	VRWWZ38#	8	1	1997/311	203412	1997/316	1997/315	SL 36TR M
SC3939	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000064	VRWWZ38#	10	1	1997/311	212727	1997/316	1997/315	SL 36TR M
SC3943	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000065	VRWWZ38#	11	1	1997/311	214047	1997/316	1997/315	SL 36TR M
SC3947	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000066	VRWWZ38#	12	1	1997/311	221035	1997/316	1997/315	SL 36TR M
SC3949	SSC.VITALREC.BUILD.WWZ38#.G0040V00	000067	VRWWZ38#	13	1	1997/311	222530	1997/316	1997/315	SL 36TR M
SC3960	SSC.VITALREC.BUILD.WWZ038.G0043V00	000068	VRWWZ038	1	1	1997/311	224548	1997/316	1997/315	SL 36TR M
SC3970	SSC.VITALREC.BUILD.WWZ038.G0043V00	000069	VRWWZ038	5	1	1997/311	234856	1997/317	1997/315	SL 36TR M
SC3974	SSC.VITALREC.BUILD.WWZ038.G0043V00	000070	VRWWZ038	3	1	1997/311	232230	1997/316	1997/315	SL 36TR M
SC3982	SSC.VITALREC.BUILD.WWZ038.G0043V00	000071	VRWWZ038	4	1	1997/311	233408	1997/316	1997/315	SL 36TR M
End of Report. 44 Entries listed										
DFSMSrmm IBM INTERNAL USE ONLY		Inventory of Volumes in Storage Location REMOTE								PAGE - 00001
EDGRPT06 -----										DATE - 97316
Volume		BIN	Creating	Vol-	DSN-	Create	Create	Expiration	Date	Rec. V
Serial	Data Set Name	Number	Jobname	Seq.	Seq.	Date	Time	Date	stored	LBL Fmt S

68059C	BADER.TLMS.RMF	000002		1	1	1995/086	153951	1998/100	1995/142	SL 18TR M
End of Report. 1 Entries listed										

Figure 230. REPORT06 Output: Inventory of Volume Serial Number by Location

14.2.7 REPORT07

Figure 231 on page 547 shows sample output from the REPORT07 report; inventory of data set names by location.

DFSMSrmm IBM INTERNAL USE ONLY		Inventory of Data Set Names in Storage Location DISTANT							PAGE - 00001	
EDGRPT07 -----									DATE - 97316	
Data Set Name	Volume Serial	BIN Number	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Date stored	Rec. V LBL Fmt S
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3365	000033	1	1	VRWWZ38#	1997/311	172236	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3417	000043	2	1	VRWWZ38#	1997/311	180657	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3433	000046	3	1	VRWWZ38#	1997/311	182520	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3621	000056	4	1	VRWWZ38#	1997/311	183740	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3926	000058	5	1	VRWWZ38#	1997/311	185336	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3928	000059	6	1	VRWWZ38#	1997/311	192122	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3932	000061	7	1	VRWWZ38#	1997/311	195327	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3935	000063	8	1	VRWWZ38#	1997/311	203412	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3933	000062	9	1	VRWWZ38#	1997/311	205912	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3939	000064	10	1	VRWWZ38#	1997/311	212727	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3943	000065	11	1	VRWWZ38#	1997/311	214047	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3947	000066	12	1	VRWWZ38#	1997/311	221035	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3949	000067	13	1	VRWWZ38#	1997/311	222530	1997/316	1997/315	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3246	000093	1	1	VRW3897A	1997/305	014101	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3247	000095	2	1	VRW3897A	1997/305	020946	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3251	000097	3	1	VRW3897A	1997/305	022128	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3253	000098	4	1	VRW3897A	1997/305	023642	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3268	000106	5	1	VRW3897A	1997/305	024739	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3273	000108	6	1	VRW3897A	1997/305	030007	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3272	000107	7	1	VRW3897A	1997/305	031252	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3274	000109	8	1	VRW3897A	1997/305	032418	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3258	000104	9	1	VRW3897A	1997/305	033547	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3202	000079	10	1	VRW3897A	1997/305	034612	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3285	000111	11	1	VRW3897A	1997/305	035901	1997/310	1997/308	SL 36TR M
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3277	000110	12	1	VRW3897A	1997/305	041151	1997/310	1997/308	SL 36TR M
SSC.VITALREC.FILTER.SELECT.G0160V00	SC3260	000105	1	1	VRSELECT	1997/305	042538	1997/310	1997/308	SL 36TR M
SSC.VITALREC.MASTER.JCL.G0160V00	SC3184	000006	1	1	VRMASTER	1997/305	044653	1997/310	1997/308	SL 36TR M
SSC.VITALREC.SYSTEM.MZASH1.G0116V00	SC3331	000023	1	1	DMZASH1#	1997/310	220354	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZASH2.G0113V00	SC3336	000024	1	1	DMZASH1#	1997/310	221621	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBMC1.G0115V00	SC3286	000017	1	1	DMZBMC1#	1997/310	210250	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBPG1.G0115V00	SC3287	000018	1	1	DMZBPG1#	1997/310	210441	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBPG2.G0115V00	SC3288	000019	1	1	DMZBPG1#	1997/310	211529	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	SC3354	000030	1	1	DMZBPP0#	1997/310	222851	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBPP2.G0098V00	SC3338	000026	1	1	DMZBPP0#	1997/310	224203	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3087	000010	1	1	DMZBSP0#	1997/310	200128	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3100	000012	2	1	DMZBSP0#	1997/310	201405	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3098	000011	3	1	DMZBSP0#	1997/310	202311	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3219	000013	1	1	DMZBSP0#	1997/310	202510	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3228	000014	2	1	DMZBSP0#	1997/310	204510	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3252	000015	3	1	DMZBSP0#	1997/310	205955	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSY1.G0116V00	SC3316	000021	1	1	DMZBSY1#	1997/310	212431	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSY1.G0116V00	SC3326	000022	2	1	DMZBSY1#	1997/310	214803	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSY2.G0114V00	SC3276	000016	1	1	DMZBSY1#	1997/310	215102	1997/315	1997/315	SL 36TR M
SSC.VITALREC.SYSTEM.MZBSY2.G0114V00	SC3309	000020	2	1	DMZBSY1#	1997/310	220059	1997/315	1997/315	SL 36TR M
End of Report. 44 Entries listed										
DFSMSrmm IBM INTERNAL USE ONLY		Inventory of Data Set Names in Storage Location REMOTE							PAGE - 00001	
EDGRPT07 -----									DATE - 97316	
Data Set Name	Volume Serial	BIN Number	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Date stored	Rec. V LBL Fmt S
BADER.TLMS.RMF	68059C	000002	1	1		1995/086	153951	1998/100	1995/142	SL 18TR M
End of Report. 1 Entries listed										

Figure 231. REPORT07 Output: Inventory of Data Set Names by Location

14.2.8 REPORT08

Figure 232 on page 548 shows sample output from the REPORT08 report; inventory of bin numbers by location.

DFSMSrmm EDGRPT08		IBM INTERNAL USE ONLY		Inventory of BIN numbers in Storage Location DISTANT						PAGE - 00001	
										DATE - 97316	
BIN		Volume	Vol-	DSN-	Creating	Create	Create	Expiration	Date	Rec. V	
Number	Data Set Name	Serial	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL Fmt S	
000006	SSC.VITALREC.MASTER.JCL.G0160V00	SC3184	1	1	VRMASTER	1997/305	044653	1997/310	1997/308	SL 36TR M	
000010	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3087	1	1	DMZBSP0#	1997/310	200128	1997/315	1997/315	SL 36TR M	
000011	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3098	3	1	DMZBSP0#	1997/310	202311	1997/315	1997/315	SL 36TR M	
000012	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3100	2	1	DMZBSP0#	1997/310	201405	1997/315	1997/315	SL 36TR M	
000013	SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3219	1	1	DMZBSP0#	1997/310	202510	1997/315	1997/315	SL 36TR M	
000014	SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3228	2	1	DMZBSP0#	1997/310	204510	1997/315	1997/315	SL 36TR M	
000015	SSC.VITALREC.SYSTEM.MZBSP2.G0099V00	SC3252	3	1	DMZBSP0#	1997/310	205955	1997/315	1997/315	SL 36TR M	
000016	SSC.VITALREC.SYSTEM.MZBSY2.G0114V00	SC3276	1	1	DMZBSY1#	1997/310	215102	1997/315	1997/315	SL 36TR M	
000017	SSC.VITALREC.SYSTEM.MZBMC1.G0115V00	SC3286	1	1	DMZBMC1#	1997/310	210250	1997/315	1997/315	SL 36TR M	
000018	SSC.VITALREC.SYSTEM.MZBPG1.G0115V00	SC3287	1	1	DMZBPG1#	1997/310	210441	1997/315	1997/315	SL 36TR M	
000019	SSC.VITALREC.SYSTEM.MZBPG2.G0115V00	SC3288	1	1	DMZBPG1#	1997/310	211529	1997/315	1997/315	SL 36TR M	
000020	SSC.VITALREC.SYSTEM.MZBSY2.G0114V00	SC3309	2	1	DMZBSY1#	1997/310	220059	1997/315	1997/315	SL 36TR M	
000021	SSC.VITALREC.SYSTEM.MZBSY1.G0116V00	SC3316	1	1	DMZBSY1#	1997/310	212431	1997/315	1997/315	SL 36TR M	
000022	SSC.VITALREC.SYSTEM.MZBSY1.G0116V00	SC3326	2	1	DMZBSY1#	1997/310	214803	1997/315	1997/315	SL 36TR M	
000023	SSC.VITALREC.SYSTEM.MZASH1.G0116V00	SC3331	1	1	DMZASH1#	1997/310	220354	1997/315	1997/315	SL 36TR M	
000024	SSC.VITALREC.SYSTEM.MZASH2.G0113V00	SC3336	1	1	DMZASH1#	1997/310	221621	1997/315	1997/315	SL 36TR M	
000025	SSC.VITALREC.BUILD.ESA52.G0080V00	SC3377	2	1	VRESA52L	1997/119	101517	1999/119	1997/126	SL 36TR M	
000026	SSC.VITALREC.SYSTEM.MZBPP2.G0098V00	SC3338	1	1	DMZBPP0#	1997/310	224203	1997/315	1997/315	SL 36TR M	
000027	SSC.VITALREC.BUILD.DB.G0172V00	SC3348	2	1	VRDB	1997/311	132021	1997/316	1997/315	SL 36TR M	
000028	SSC.VITALREC.BUILD.ESAS.G0134V00	SC3483	2	1	VRESASL	1997/119	105757	1999/119	1997/126	SL 36TR M	
000029	SSC.VITALREC.BUILD.ESAS.G0134V00	SC3484	1	1	VRESASL	1997/119	103831	1999/119	1997/126	SL 36TR M	
000030	SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	SC3354	1	1	DMZBPP0#	1997/310	222851	1997/315	1997/315	SL 36TR M	
000031	SSC.VITALREC.BUILD.DB.G0172V00	SC3357	1	1	VRDB	1997/311	130501	1997/316	1997/315	SL 36TR M	
000032	SSC.VITALREC.BUILD.PP.G0165V00	SC3362	1	1	VRPP	1997/311	134953	1997/316	1997/315	SL 36TR M	
000033	SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3365	1	1	VRWWZ38#	1997/311	172236	1997/316	1997/315	SL 36TR M	
000034	SSC.VITALREC.BUILD.WWP004.G0041V00	SC3368	1	1	VRWWP004	1997/311	163655	1997/316	1997/315	SL 36TR M	
000035	SSC.VITALREC.BUILD.NET.G0164V00	SC3371	2	1	VRNET	1997/311	140201	1997/316	1997/315	SL 36TR M	
000036	SSC.VITALREC.BUILD.NET.G0164V00	SC3382	3	1	VRNET	1997/311	141819	1997/316	1997/315	SL 36TR M	
000037	SSC.VITALREC.BUILD.NET.G0164V00	SC3386	1	1	VRNET	1997/311	134339	1997/316	1997/315	SL 36TR M	
000038	SSC.VITALREC.BUILD.ES52.G0031V00	SC3391	1	1	VRES52	1997/311	143509	1997/316	1997/315	SL 36TR M	
000039	SSC.VITALREC.BUILD.ES52.G0031V00	SC3393	2	1	VRES52	1997/311	144903	1997/316	1997/315	SL 36TR M	
000040	SSC.VITALREC.BUILD.PP.G0165V00	SC3394	2	1	VRPP	1997/311	141145	1997/316	1997/315	SL 36TR M	
000041	SSC.VITALREC.BUILD.PP.G0165V00	SC3407	3	1	VRPP	1997/311	142639	1997/316	1997/315	SL 36TR M	
000042	SSC.VITALREC.BUILD.PP.G0165V00	SC3415	4	1	VRPP	1997/311	144048	1997/316	1997/315	SL 36TR M	
000043	SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3417	2	1	VRWWZ38#	1997/311	180657	1997/316	1997/315	SL 36TR M	
000044	SSC.VITALREC.BUILD.ES52.G0031V00	SC3421	5	1	VRES52	1997/311	153214	1997/316	1997/315	SL 36TR M	
000045	SSC.VITALREC.BUILD.PP.G0165V00	SC3427	5	1	VRPP	1997/311	150122	1997/316	1997/315	SL 36TR M	
000046	SSC.VITALREC.BUILD.WWZ38#.G0040V00	SC3433	3	1	VRWWZ38#	1997/311	182520	1997/316	1997/315	SL 36TR M	
000047	SSC.VITALREC.BUILD.ES52.G0031V00	SC3452	3	1	VRES52	1997/311	150311	1997/316	1997/315	SL 36TR M	
000048	SSC.VITALREC.BUILD.ES52.G0031V00	SC3462	4	1	VRES52	1997/311	151656	1997/316	1997/315	SL 36TR M	
000049	SSC.VITALREC.BUILD.WWP115.G0041V00	SC3463	2	1	VRWWP115	1997/311	170137	1997/316	1997/315	SL 36TR M	
000050	SSC.VITALREC.BUILD.ES52.G0031V00	SC3467	6	1	VRES52	1997/311	154353	1997/316	1997/315	SL 36TR M	
000051	SSC.VITALREC.BUILD.ESA52.G0080V00	SC3799	1	1	VRESA52L	1997/119	095924	1999/119	1997/126	SL 36TR M	
000052	SSC.VITALREC.BUILD.WWP115.G0041V00	SC3476	1	1	VRWWP115	1997/311	164732	1997/316	1997/315	SL 36TR M	
End of Report.				44 Entries listed							

DFSMSrmm EDGRPT08		IBM INTERNAL USE ONLY		Inventory of BIN numbers in Storage Location REMOTE						PAGE - 00001	
										DATE - 97316	
BIN		Volume	Vol-	DSN-	Creating	Create	Create	Expiration	Date	Rec. V	
Number	Data Set Name	Serial	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL Fmt S	
000002	BADER.TLMS.RMF	68059C	1	1		1995/086	153951	1998/100	1995/142	SL 18TR M	
End of Report.				1 Entries listed							

Figure 232. REPORT08 Output: Inventory of Bin Numbers by Location

14.2.9 REPORT09

Figure 233 on page 549 shows sample output from the REPORT09 report; list of all data set names at loan locations.

DFSMSrmm EDGRPT09	IBM INTERNAL USE ONLY -----	Inventory of Data Set Names in Loan Location EXPORT									PAGE - 00001 DATE - 97286
Data Set Name	Volume Serial	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Volume Ref. Date	LBL	Rec. V V Fmt S R	
EMDISDU.DUMP	SC3915	1	1	P120TRD1	04/07/1997	181712	09/07/1997	04/07/1997	SL	36TR M N	
EMDISDU.DUMP	SC3880	1	1	PI96A1D1	15/05/1997	153452	22/06/1997	15/05/1997	SL	* M N	
EMDISDU.DUMP	SC3881	2	1	PI96A1D1	15/05/1997	154356	22/06/1997	15/05/1997	SL	* M N	
SCHLUM.TAPE.RESTORE.JCL	SD1722	1	1	POOLSD	13/01/1997	154733	14/01/1998	13/01/1997	SL	18TR M N	
SIE.SOC4.DUMP31	SC0081	1	1	CICSDUMP	03/07/1997	105241	08/07/1997	03/07/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0079	1	1	IBBTRES	07/10/1997	111007	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0080	2	1	IBBTRES	07/10/1997	111332	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0084	3	1	IBBTRES	07/10/1997	111701	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0086	4	1	IBBTRES	07/10/1997	112026	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0088	5	1	IBBTRES	07/10/1997	112353	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0089	6	1	IBBTRES	07/10/1997	112757	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0090	7	1	IBBTRES	07/10/1997	113119	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0091	8	1	IBBTRES	07/10/1997	113438	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0093	9	1	IBBTRES	07/10/1997	113813	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3480.MXXSYO.DAY280.IBBT	SC0000	10	1	IBBTRES	07/10/1997	125209	12/10/1997	07/10/1997	SL	18TR M N	
SYS1.DUMP3490.MXXSYO.DAY280.IBBT	SC3170	1	1	IBBTRES	07/10/1997	151914	12/10/1997	07/10/1997	SL	36TR M N	
SYS1.DUMP3490.MXXSYO.DAY280.IBBT	SC3177	2	1	IBBTRES	07/10/1997	153924	12/10/1997	07/10/1997	SL	36TR M N	
XXXXX.RMM.ADDONS.EXEC	SD1722	1	2	POOLSD	13/01/1997	154737	14/01/1998	13/01/1997	SL	18TR M N	
XXXXX.RMM.ADDONS.INITVARS	SD1722	1	6	POOLSD	13/01/1997	154803	14/01/1998	13/01/1997	SL	18TR M N	
XXXXX.RMM.ADDONS.MSGS	SD1722	1	3	POOLSD	13/01/1997	154745	14/01/1998	13/01/1997	SL	18TR M N	
XXXXX.RMM.ADDONS.PANELS	SD1722	1	4	POOLSD	13/01/1997	154750	14/01/1998	13/01/1997	SL	18TR M N	
XXXXX.RMM.ADDONS.SKELS	SD1722	1	5	POOLSD	13/01/1997	154757	14/01/1998	13/01/1997	SL	18TR M N	
End of Report. 22 Entries listed											

Figure 233. REPORT09 Output: List of All Data Sets at Loan Locations

14.2.10 REPORT10

Figure 234 shows sample output from the REPORT10 report; list of all volume serial numbers at loan locations.

DFSMSrmm	IBM INTERNAL USE ONLY		Inventory of Volumes in Loan Location EXPORT								PAGE - 00001	
EDGRPT10											DATE - 97286	
Volume			Vol-	DSN-	Creating	Create	Create	Expiration	Volume		Rec. V	
Serial	Data Set Name		Seq.	Seq.	Jobname	Date	Time	Date	Ref. Date	LBL	Fmt S R	
SC0000	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		10	1	IBBTRES	07/10/1997	125209	12/10/1997	07/10/1997	SL	18TR M N	
SC0079	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		1	1	IBBTRES	07/10/1997	111007	12/10/1997	07/10/1997	SL	18TR M N	
SC0080	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		2	1	IBBTRES	07/10/1997	111332	12/10/1997	07/10/1997	SL	18TR M N	
SC0081	SIE.SOC4.DUMP31		1	1	CICSDUMP	03/07/1997	105241	08/07/1997	03/07/1997	SL	18TR M N	
SC0084	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		3	1	IBBTRES	07/10/1997	111701	12/10/1997	07/10/1997	SL	18TR M N	
SC0086	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		4	1	IBBTRES	07/10/1997	112026	12/10/1997	07/10/1997	SL	18TR M N	
SC0088	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		5	1	IBBTRES	07/10/1997	112353	12/10/1997	07/10/1997	SL	18TR M N	
SC0089	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		6	1	IBBTRES	07/10/1997	112757	12/10/1997	07/10/1997	SL	18TR M N	
SC0090	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		7	1	IBBTRES	07/10/1997	113119	12/10/1997	07/10/1997	SL	18TR M N	
SC0091	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		8	1	IBBTRES	07/10/1997	113438	12/10/1997	07/10/1997	SL	18TR M N	
SC0093	SYS1.DUMP3480.MXXSYO.DAY280.IBBT		9	1	IBBTRES	07/10/1997	113813	12/10/1997	07/10/1997	SL	18TR M N	
SC3170	SYS1.DUMP3490.MXXSYO.DAY280.IBBT		1	1	IBBTRES	07/10/1997	151914	12/10/1997	07/10/1997	SL	36TR M N	
SC3177	SYS1.DUMP3490.MXXSYO.DAY280.IBBT		2	1	IBBTRES	07/10/1997	153924	12/10/1997	07/10/1997	SL	36TR M N	
SC3880	EMDISDU.DUMP		1	1	PI96A1D1	15/05/1997	153452	22/06/1997	15/05/1997	SL	* M N	
SC3881	EMDISDU.DUMP		2	1	PI96A1D1	15/05/1997	154356	22/06/1997	15/05/1997	SL	* M N	
SC3915	EMDISDU.DUMP		1	1	P120TRD1	04/07/1997	181712	09/07/1997	04/07/1997	SL	36TR M N	
SD1722	SCHLUM.TAPE.RESTORE.JCL		1	1	POOLSD	13/01/1997	154733	14/01/1998	13/01/1997	SL	18TR M N	
SD1722	XXXXX.RMM.ADDONS.EXEC		1	2	POOLSD	13/01/1997	154737	14/01/1998	13/01/1997	SL	18TR M N	
SD1722	XXXXX.RMM.ADDONS.MSGS		1	3	POOLSD	13/01/1997	154745	14/01/1998	13/01/1997	SL	18TR M N	
SD1722	XXXXX.RMM.ADDONS.PANELS		1	4	POOLSD	13/01/1997	154750	14/01/1998	13/01/1997	SL	18TR M N	
SD1722	XXXXX.RMM.ADDONS.SKELS		1	5	POOLSD	13/01/1997	154757	14/01/1998	13/01/1997	SL	18TR M N	
SD1722	XXXXX.RMM.ADDONS.INITVARS		1	6	POOLSD	13/01/1997	154803	14/01/1998	13/01/1997	SL	18TR M N	
End of Report. 22 Entries listed												

Figure 234. REPORT10 Output: List of All Serial Numbers at Loan Locations

14.2.11 REPORT11

Figure 235 shows sample output from the REPORT11 report; list of all multiple volume, multiple data sets.

DFSMSrmm		IBM INTERNAL USE ONLY				Multi-Volume/Multi-Data Set Report						PAGE - 00001	
EDGRPT11		-----										DATE - 97286	
Volume	Vol-	DSN-				Expiration	First	Prev.	Next	Create	Creating	Create	Create
Serial	Seq.	Seq.	Data	Set	Name	Date	Volser	Volser	Volser	Userid	Jobname	Date	Time
HD0311	1					27/10/1998	HD0311		HD0313				
Used kilobytes for volume HD0311							0						
HD0313	2	154	IBM.HTCP310.F6			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	132131
HD0313	2	155	IBM.HTCP310.F7			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134151
HD0313	2	156	IBM.HTCP310.F8			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134210
HD0313	2	157	IBM.HTCP310.F9			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134220
HD0313	2	158	IBM.HTCP310.F10			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134311
HD0313	2	159	IBM.HTCP310.F11			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134320
HD0313	2	160	IBM.HTCP310.F12			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134331
HD0313	2	161	IBM.HTCP310.F13			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134341
HD0313	2	162	IBM.HTCP310.F14			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134354
HD0313	2	163	IBM.HTCP310.F15			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134409
HD0313	2	164	IBM.HTCP310.F16			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134419
HD0313	2	165	IBM.HTCP310.F17			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134433
HD0313	2	166	IBM.HTCP310.F18			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134448
HD0313	2	167	IBM.HTCP310.F19			27/10/1998	HD0311	HD0311		EGE	RCVPD03	16/01/1996	134458
Used kilobytes for volume HD0313							141429						
HD0577	1					27/10/1998	HD0577		HD0578				
Used kilobytes for volume HD0577							0						
HD0578	2	338	IBM.JIF4310.F2			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	203830
HD0578	2	339	IBM.JIF4310.F3			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	205859
HD0578	2	340	IBM.JIF4310.F4			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	205942
HD0578	2	341	IBM.JIF4310.F5			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210001
HD0578	2	342	IBM.JIF4370.F1			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210015
HD0578	2	343	IBM.JIF4370.F2			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210028
HD0578	2	344	IBM.JIF4370.F3			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210044
HD0578	2	345	IBM.JIF4370.F4			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210057
HD0578	2	346	IBM.JIF4370.F5			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210123
HD0578	2	347	IBM.JIF4370.F6			27/10/1998	HD0577	HD0577	HD0579	EGE	RCVPD01	16/07/1996	210138
Used kilobytes for volume HD0578							191978						
HD0579	3	397	IBM.JMWL650.F4			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	211055
HD0579	3	398	IBM.JOM14N0.F1			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	212101
HD0579	3	399	IBM.JOM14N0.F2			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	212118
HD0579	3	400	IBM.JOT14N0.F1			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	212130
HD0579	3	401	IBM.JOT14N0.F2			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	212145
HD0579	3	402	IBM.JOT14N0.F3			27/10/1998	HD0577	HD0578		EGE	RCVPD01	16/07/1996	212157
Used kilobytes for volume HD0579							26389						
HD0580	1					27/10/1998	HD0580		HD0626				
Used kilobytes for volume HD0580							0						
HD0626	2	494	IBM.JOHL15B.F1			27/10/1998	HD0580	HD0580		EGE	RCVPD01	16/07/1996	213153
HD0626	2	495	IBM.JOHL152.F1			27/10/1998	HD0580	HD0580		EGE	RCVPD01	16/07/1996	213407
HD0626	2	496	IBM.JOHL153.F1			27/10/1998	HD0580	HD0580		EGE	RCVPD01	16/07/1996	213436
HD0626	2	497	IBM.JOHL154.F1			27/10/1998	HD0580	HD0580		EGE	RCVPD01	16/07/1996	213508
Used kilobytes for volume HD0626							118497						
HC0004	1					30/09/1999	HC0004		HC0006				
Used kilobytes for volume HC0004							0						
HC0006	2	5	SMPMCS			30/09/1999	HC0004	HC0004	HC0008	EGE	RCVPD0GA	05/09/1996	150951

DFSMSrmm	IBM INTERNAL USE ONLY	Movement report by Data Set Names										PAGE -	00001
EDGRPT12	-----	from location SHELF					to location DISTANT			DATE -	97316		
Data Set Name	Volume Serial	BIN Number	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Date stored	LBL	Rec. V Fmt S		
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3261	000002	1	1	VRW3897A	1997/316	095752	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3305	000004	2	1	VRW3897A	1997/316	101636	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3420	000080	3	1	VRW3897A	1997/316	102949	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3327	000007	4	1	VRW3897A	1997/316	104816	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3345	000008	5	1	VRW3897A	1997/316	110018	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3418	000078	6	1	VRW3897A	1997/316	111441	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3052	000001	7	1	VRW3897A	1997/316	112627	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC4408	000102	8	1	VRW3897A	1997/316	113949	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3351	000009	9	1	VRW3897A	1997/316	115141	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3392	000076	10	1	VRW3897A	1997/316	120530	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3378	000057	11	1	VRW3897A	1997/316	121858	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0018V00	SC3404	000077	12	1	VRW3897A	1997/316	123033	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.FILTER.SELECT.G0161V00	SC3281	000003	1	1	VRSELECT	1997/316	124107	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.MASTER.JCL.G0161V00	SC3317	000005	1	1	VRMASTER	1997/316	125851	1997/321	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZASH1.G0117V00	SC3866	000096	1	1	DMZASH1#	1997/317	220323	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZASH2.G0114V00	SC3896	000100	1	1	DMZASH1#	1997/317	221753	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBMC1.G0116V00	SC3807	000087	1	1	DMZBMC1#	1997/317	205221	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPG1.G0116V00	SC3818	000088	1	1	DMZBPG1#	1997/317	205358	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPG2.G0116V00	SC3825	000089	1	1	DMZBPG1#	1997/317	210036	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPP1.G0099V00	SC3888	000099	1	1	DMZBPP0#	1997/317	223108	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPP2.G0099V00	SC3901	000101	1	1	DMZBPP0#	1997/317	224356	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3733	000081	1	1	DMZBSP0#	1997/317	200156	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3742	000082	2	1	DMZBSP0#	1997/317	201535	1997/322	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3753	000083	3	1	DMZBSP0#	1997/317	202608	1997/322	1997/318	SL	36TR M		
End of Report. 24 Entries listed													

DFSMSrmm	IBM INTERNAL USE ONLY	Movement report by Data Set Names										PAGE -	00001
EDGRPT12	-----	from location DISTANT					to location SHELF			DATE -	97316		
Data Set Name	Volume Serial	BIN Number	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Expiration Date	Date stored	LBL	Rec. V Fmt S		
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3246	000093*	1	1	VRW3897A	1997/305	014101	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3247	000095*	2	1	VRW3897A	1997/305	020946	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3251	000097*	3	1	VRW3897A	1997/305	022128	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3253	000098*	4	1	VRW3897A	1997/305	023642	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3268	000106*	5	1	VRW3897A	1997/305	024739	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3273	000108*	6	1	VRW3897A	1997/305	030007	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3272	000107*	7	1	VRW3897A	1997/305	031252	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3274	000109*	8	1	VRW3897A	1997/305	032418	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3258	000104*	9	1	VRW3897A	1997/305	033547	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3202	000079*	10	1	VRW3897A	1997/305	034612	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3285	000111*	11	1	VRW3897A	1997/305	035901	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.BUILD.W3897A.G0017V00	SC3277	000110*	12	1	VRW3897A	1997/305	041151	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.FILTER.SELECT.G0160V00	SC3260	000105*	1	1	VRSELECT	1997/305	042538	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.MASTER.JCL.G0160V00	SC3184	000006*	1	1	VRMASTER	1997/305	044653	1997/310	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZASH1.G0116V00	SC3331	000023*	1	1	DMZASH1#	1997/310	220354	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZASH2.G0113V00	SC3336	000024*	1	1	DMZASH1#	1997/310	221621	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBMC1.G0115V00	SC3286	000017*	1	1	DMZBMC1#	1997/310	210250	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPG1.G0115V00	SC3287	000018*	1	1	DMZBPG1#	1997/310	210441	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPG2.G0115V00	SC3288	000019*	1	1	DMZBPG1#	1997/310	211529	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	SC3354	000030*	1	1	DMZBPP0#	1997/310	222851	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBPP2.G0098V00	SC3338	000026*	1	1	DMZBPP0#	1997/310	224203	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3087	000010*	1	1	DMZBSP0#	1997/310	200128	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3100	000012*	2	1	DMZBSP0#	1997/310	201405	1997/315	1997/318	SL	36TR M		
SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3098	000011*	3	1	DMZBSP0#	1997/310	202311	1997/315	1997/318	SL	36TR M		
End of Report. 24 Entries listed													

Figure 236. REPORT12 Output: Movement Report With First Data Set Name

Note: When the bin number ends with an asterisk, it indicates that the volume is in movement from the bin, but movement is not confirmed.

14.2.13 REPORT13

Figure 237 on page 552 shows sample output from the REPORT13 report; movement report sorted by storage location bin number.

DFSMSrmm IBM INTERNAL USE ONLY		Movement report by BIN number								PAGE - 00001
EDGRPT13 -----		from location SHELF				to location DISTANT				DATE - 97316
BIN		Volume	Vol-	DSN-	Creating	Create	Create	Expiration	Date	Rec. V
Number	Data Set Name	Serial	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL Fmt S

000001	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3052	7	1	VRW3897A	1997/059	112627	1997/321	1997/318	SL 36TR M
000002	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3261	1	1	VRW3897A	1997/059	095752	1997/321	1997/318	SL 36TR M
000003	SSC.VITALREC.FILTER.SELECT.G0161V00	SC3281	1	1	VRSELECT	1997/059	124107	1997/321	1997/318	SL 36TR M
000004	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3305	2	1	VRW3897A	1997/059	101636	1997/321	1997/318	SL 36TR M
000005	SSC.VITALREC.MASTER.JCL.G0161V00	SC3317	1	1	VRMASTER	1997/059	125851	1997/321	1997/318	SL 36TR M
000007	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3327	4	1	VRW3897A	1997/059	104816	1997/321	1997/318	SL 36TR M
000008	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3345	5	1	VRW3897A	1997/059	110018	1997/321	1997/318	SL 36TR M
000009	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3351	9	1	VRW3897A	1997/059	115141	1997/321	1997/318	SL 36TR M
000057	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3378	11	1	VRW3897A	1997/059	121858	1997/321	1997/318	SL 36TR M
000076	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3392	10	1	VRW3897A	1997/059	120530	1997/321	1997/318	SL 36TR M
000077	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3404	12	1	VRW3897A	1997/059	123033	1997/321	1997/318	SL 36TR M
000078	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3418	6	1	VRW3897A	1997/059	111441	1997/321	1997/318	SL 36TR M
000080	SSC.VITALREC.BUILD.W3897A.G0018V00	SC3420	3	1	VRW3897A	1997/059	102949	1997/321	1997/318	SL 36TR M
000081	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3733	1	1	DMZBSP0#	1997/059	200156	1997/322	1997/318	SL 36TR M
000082	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3742	2	1	DMZBSP0#	1997/059	201535	1997/322	1997/318	SL 36TR M
000083	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	SC3753	3	1	DMZBSP0#	1997/059	202608	1997/322	1997/318	SL 36TR M
000087	SSC.VITALREC.SYSTEM.MZBMC1.G0116V00	SC3807	1	1	DMZBMC1#	1997/059	205221	1997/322	1997/318	SL 36TR M
000088	SSC.VITALREC.SYSTEM.MZBPG1.G0116V00	SC3818	1	1	DMZBPG1#	1997/059	205358	1997/322	1997/318	SL 36TR M
000089	SSC.VITALREC.SYSTEM.MZBPG2.G0116V00	SC3825	1	1	DMZBPG1#	1997/059	210036	1997/322	1997/318	SL 36TR M
000096	SSC.VITALREC.SYSTEM.MZASH1.G0117V00	SC3866	1	1	DMZASH1#	1997/059	220323	1997/322	1997/318	SL 36TR M
000099	SSC.VITALREC.SYSTEM.MZBPP1.G0099V00	SC3888	1	1	DMZBPP0#	1997/059	223108	1997/322	1997/318	SL 36TR M
000100	SSC.VITALREC.SYSTEM.MZASH2.G0114V00	SC3896	1	1	DMZASH1#	1997/059	221753	1997/322	1997/318	SL 36TR M
000101	SSC.VITALREC.SYSTEM.MZBPP2.G0099V00	SC3901	1	1	DMZBPP0#	1997/059	224356	1997/322	1997/318	SL 36TR M
000102	SSC.VITALREC.BUILD.W3897A.G0018V00	SC4408	8	1	VRW3897A	1997/260	113949	1997/321	1997/318	SL 36TR M
End of Report. 24 Entries listed										

DFSMSrmm IBM INTERNAL USE ONLY		Movement report by BIN number								PAGE - 00001
EDGRPT13 -----		from location DISTANT				to location SHELF				DATE - 97316
BIN		Volume	Vol-	DSN-	Creating	Create	Create	Expiration	Date	Rec. V
Number	Data Set Name	Serial	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL Fmt S

000006*	SSC.VITALREC.MASTER.JCL.G0160V00	SC3184	1	1	VRMASTER	1997/059	044653	1997/310	1997/318	SL 36TR M
000010*	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3087	1	1	DMZBSP0#	1997/059	200128	1997/315	1997/318	SL 36TR M
000011*	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3098	3	1	DMZBSP0#	1997/059	202311	1997/315	1997/318	SL 36TR M
000012*	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	SC3100	2	1	DMZBSP0#	1997/059	201405	1997/315	1997/318	SL 36TR M
000017*	SSC.VITALREC.SYSTEM.MZBMC1.G0115V00	SC3286	1	1	DMZBMC1#	1997/059	210250	1997/315	1997/318	SL 36TR M
000018*	SSC.VITALREC.SYSTEM.MZBPG1.G0115V00	SC3287	1	1	DMZBPG1#	1997/059	210441	1997/315	1997/318	SL 36TR M
000019*	SSC.VITALREC.SYSTEM.MZBPG2.G0115V00	SC3288	1	1	DMZBPG1#	1997/059	211529	1997/315	1997/318	SL 36TR M
000023*	SSC.VITALREC.SYSTEM.MZASH1.G0116V00	SC3331	1	1	DMZASH1#	1997/059	220354	1997/315	1997/318	SL 36TR M
000024*	SSC.VITALREC.SYSTEM.MZASH2.G0113V00	SC3336	1	1	DMZASH1#	1997/059	221621	1997/315	1997/318	SL 36TR M
000026*	SSC.VITALREC.SYSTEM.MZBPP2.G0098V00	SC3338	1	1	DMZBPP0#	1997/059	224203	1997/315	1997/318	SL 36TR M
000030*	SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	SC3354	1	1	DMZBPP0#	1997/059	222851	1997/315	1997/318	SL 36TR M
000079*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3202	10	1	VRW3897A	1997/059	034612	1997/310	1997/318	SL 36TR M
000093*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3246	1	1	VRW3897A	1997/059	014101	1997/310	1997/318	SL 36TR M
000095*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3247	2	1	VRW3897A	1997/059	020946	1997/310	1997/318	SL 36TR M
000097*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3251	3	1	VRW3897A	1997/059	022128	1997/310	1997/318	SL 36TR M
000098*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3253	4	1	VRW3897A	1997/059	023642	1997/310	1997/318	SL 36TR M
000104*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3258	9	1	VRW3897A	1997/059	033547	1997/310	1997/318	SL 36TR M
000105*	SSC.VITALREC.FILTER.SELECT.G0160V00	SC3260	1	1	VRSELECT	1997/059	042538	1997/310	1997/318	SL 36TR M
000106*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3268	5	1	VRW3897A	1997/059	024739	1997/310	1997/318	SL 36TR M
000107*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3272	7	1	VRW3897A	1997/059	031252	1997/310	1997/318	SL 36TR M
000108*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3273	6	1	VRW3897A	1997/059	030007	1997/310	1997/318	SL 36TR M
000109*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3274	8	1	VRW3897A	1997/059	032418	1997/310	1997/318	SL 36TR M
000110*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3277	12	1	VRW3897A	1997/059	041151	1997/310	1997/318	SL 36TR M
000111*	SSC.VITALREC.BUILD.W3897A.G0017V00	SC3285	11	1	VRW3897A	1997/059	035901	1997/310	1997/318	SL 36TR M
End of Report. 24 Entries listed										

Figure 237. REPORT13 Output: Movement Report by Storage Location Bin

Note: When the bin number ends with an asterisk, it indicates that the volume is in movement from the bin, but movement is not confirmed.

14.2.14 REPORT14

Figure 238 on page 553 shows sample output from the REPORT14 report; movement report sorted by volume serial number.

DFSM\$rm IBM INTERNAL USE ONLY		Movement report by Volume Serial Number										PAGE - 00001	
EDGRPT14 -----		from location SHELF to location DISTANT										DATE - 97316	
Volume		BIN	Vol-	DSN-	Creating	Create	Create	Expiration	Date			Rec. V	
Serial	Data Set Name	Number	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL	Fmt	S	
-----	-----	-----	Seq.	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
SC3052	SSC.VITALREC.BUILD.W3897A.G0018V00	000001	7	1	VRW3897A	1997/059	112627	1997/321	1997/318	SL	36TR	M	
SC3261	SSC.VITALREC.BUILD.W3897A.G0018V00	000002	1	1	VRW3897A	1997/059	095752	1997/321	1997/318	SL	36TR	M	
SC3281	SSC.VITALREC.FILTER.SELECT.G0161V00	000003	1	1	VRSELECT	1997/059	124107	1997/321	1997/318	SL	36TR	M	
SC3305	SSC.VITALREC.BUILD.W3897A.G0018V00	000004	2	1	VRW3897A	1997/059	101636	1997/321	1997/318	SL	36TR	M	
SC3317	SSC.VITALREC.MASTER.JCL.G0161V00	000005	1	1	VRMASTER	1997/059	125851	1997/321	1997/318	SL	36TR	M	
SC3327	SSC.VITALREC.BUILD.W3897A.G0018V00	000007	4	1	VRW3897A	1997/059	104816	1997/321	1997/318	SL	36TR	M	
SC3345	SSC.VITALREC.BUILD.W3897A.G0018V00	000008	5	1	VRW3897A	1997/059	110018	1997/321	1997/318	SL	36TR	M	
SC3351	SSC.VITALREC.BUILD.W3897A.G0018V00	000009	9	1	VRW3897A	1997/059	115141	1997/321	1997/318	SL	36TR	M	
SC3378	SSC.VITALREC.BUILD.W3897A.G0018V00	000057	11	1	VRW3897A	1997/059	121858	1997/321	1997/318	SL	36TR	M	
SC3392	SSC.VITALREC.BUILD.W3897A.G0018V00	000076	10	1	VRW3897A	1997/059	120530	1997/321	1997/318	SL	36TR	M	
SC3404	SSC.VITALREC.BUILD.W3897A.G0018V00	000077	12	1	VRW3897A	1997/059	123033	1997/321	1997/318	SL	36TR	M	
SC3418	SSC.VITALREC.BUILD.W3897A.G0018V00	000078	6	1	VRW3897A	1997/059	111441	1997/321	1997/318	SL	36TR	M	
SC3420	SSC.VITALREC.BUILD.W3897A.G0018V00	000080	3	1	VRW3897A	1997/059	102949	1997/321	1997/318	SL	36TR	M	
SC3733	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	000081	1	1	DMZBSP0#	1997/059	200156	1997/322	1997/318	SL	36TR	M	
SC3742	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	000082	2	1	DMZBSP0#	1997/059	201535	1997/322	1997/318	SL	36TR	M	
SC3753	SSC.VITALREC.SYSTEM.MZBSP1.G0100V00	000083	3	1	DMZBSP0#	1997/059	202608	1997/322	1997/318	SL	36TR	M	
SC3807	SSC.VITALREC.SYSTEM.MZBMC1.G0116V00	000087	1	1	DMZBMC1#	1997/059	205221	1997/322	1997/318	SL	36TR	M	
SC3818	SSC.VITALREC.SYSTEM.MZBPG1.G0116V00	000088	1	1	DMZBPG1#	1997/059	205358	1997/322	1997/318	SL	36TR	M	
SC3825	SSC.VITALREC.SYSTEM.MZBPG2.G0116V00	000089	1	1	DMZBPG1#	1997/059	210036	1997/322	1997/318	SL	36TR	M	
SC3866	SSC.VITALREC.SYSTEM.MZASH1.G0117V00	000096	1	1	DMZASH1#	1997/059	220323	1997/322	1997/318	SL	36TR	M	
SC3888	SSC.VITALREC.SYSTEM.MZBPP1.G0099V00	000099	1	1	DMZBPP0#	1997/059	223108	1997/322	1997/318	SL	36TR	M	
SC3896	SSC.VITALREC.SYSTEM.MZASH2.G0114V00	000100	1	1	DMZASH1#	1997/059	221753	1997/322	1997/318	SL	36TR	M	
SC3901	SSC.VITALREC.SYSTEM.MZBPP2.G0099V00	000101	1	1	DMZBPP0#	1997/059	224356	1997/322	1997/318	SL	36TR	M	
SC4408	SSC.VITALREC.BUILD.W3897A.G0018V00	000102	8	1	VRW3897A	1997/260	113949	1997/321	1997/318	SL	36TR	M	
End of Report. 24 Entries listed													

DFSM\$rm IBM INTERNAL USE ONLY		Movement report by Volume Serial Number										PAGE - 00001	
EDGRPT14 -----		from location DISTANT to location SHELF										DATE - 97316	
Volume		BIN	Vol-	DSN-	Creating	Create	Create	Expiration	Date			Rec. V	
Serial	Data Set Name	Number	Seq.	Seq.	Jobname	Date	Time	Date	stored	LBL	Fmt	S	
-----	-----	-----	Seq.	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
SC3087	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	000010*	3	1	DMZBSP0#	1997/059	200128	1997/315	1997/318	SL	36TR	M	
SC3098	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	000011*	1	1	DMZBSP0#	1997/059	202311	1997/315	1997/318	SL	36TR	M	
SC3100	SSC.VITALREC.SYSTEM.MZBSP1.G0099V00	000012*	2	1	DMZBSP0#	1997/059	201405	1997/315	1997/318	SL	36TR	M	
SC3184	SSC.VITALREC.MASTER.JCL.G0160V00	000006*	1	1	VRMASTER	1997/059	044653	1997/310	1997/318	SL	36TR	M	
SC3202	SSC.VITALREC.BUILD.W3897A.G0017V00	000079*	10	1	VRW3897A	1997/059	034612	1997/310	1997/318	SL	36TR	M	
SC3246	SSC.VITALREC.BUILD.W3897A.G0017V00	000093*	1	1	VRW3897A	1997/059	014101	1997/310	1997/318	SL	36TR	M	
SC3247	SSC.VITALREC.BUILD.W3897A.G0017V00	000095*	2	1	VRW3897A	1997/059	020946	1997/310	1997/318	SL	36TR	M	
SC3251	SSC.VITALREC.BUILD.W3897A.G0017V00	000097*	3	1	VRW3897A	1997/059	022128	1997/310	1997/318	SL	36TR	M	
SC3253	SSC.VITALREC.BUILD.W3897A.G0017V00	000098*	4	1	VRW3897A	1997/059	023642	1997/310	1997/318	SL	36TR	M	
SC3258	SSC.VITALREC.BUILD.W3897A.G0017V00	000104*	9	1	VRW3897A	1997/059	033547	1997/310	1997/318	SL	36TR	M	
SC3260	SSC.VITALREC.FILTER.SELECT.G0160V00	000105*	1	1	VRSELECT	1997/059	042538	1997/310	1997/318	SL	36TR	M	
SC3268	SSC.VITALREC.BUILD.W3897A.G0017V00	000106*	5	1	VRW3897A	1997/059	024739	1997/310	1997/318	SL	36TR	M	
SC3272	SSC.VITALREC.BUILD.W3897A.G0017V00	000107*	7	1	VRW3897A	1997/059	031252	1997/310	1997/318	SL	36TR	M	
SC3273	SSC.VITALREC.BUILD.W3897A.G0017V00	000108*	6	1	VRW3897A	1997/059	030007	1997/310	1997/318	SL	36TR	M	
SC3274	SSC.VITALREC.BUILD.W3897A.G0017V00	000109*	8	1	VRW3897A	1997/059	032418	1997/310	1997/318	SL	36TR	M	
SC3277	SSC.VITALREC.BUILD.W3897A.G0017V00	000110*	12	1	VRW3897A	1997/059	041151	1997/310	1997/318	SL	36TR	M	
SC3285	SSC.VITALREC.BUILD.W3897A.G0017V00	000111*	11	1	VRW3897A	1997/059	035901	1997/310	1997/318	SL	36TR	M	
SC3286	SSC.VITALREC.SYSTEM.MZBMC1.G0115V00	000017*	1	1	DMZBMC1#	1997/059	210250	1997/315	1997/318	SL	36TR	M	
SC3287	SSC.VITALREC.SYSTEM.MZBPG1.G0115V00	000018*	1	1	DMZBPG1#	1997/059	210441	1997/315	1997/318	SL	36TR	M	
SC3288	SSC.VITALREC.SYSTEM.MZBPG2.G0115V00	000019*	1	1	DMZBPG1#	1997/059	211529	1997/315	1997/318	SL	36TR	M	
SC3331	SSC.VITALREC.SYSTEM.MZASH1.G0116V00	000023*	1	1	DMZASH1#	1997/059	220354	1997/315	1997/318	SL	36TR	M	
SC3336	SSC.VITALREC.SYSTEM.MZASH2.G0113V00	000024*	1	1	DMZASH1#	1997/059	221621	1997/315	1997/318	SL	36TR	M	
SC3338	SSC.VITALREC.SYSTEM.MZBPP2.G0098V00	000026*	1	1	DMZBPP0#	1997/059	224203	1997/315	1997/318	SL	36TR	M	
SC3354	SSC.VITALREC.SYSTEM.MZBPP1.G0098V00	000030*	1	1	DMZBPP0#	1997/059	222851	1997/315	1997/318	SL	36TR	M	
End of Report. 24 Entries listed													

Figure 238. REPORT14 Output: Movement Report Sorted by Volume Serial Number

Note: When the bin number ends with an asterisk, it indicates that the volume is in movement from the bin, but movement is not confirmed.

14.2.15 REPORT15

Figure 239 on page 554 shows sample output from the REPORT15 report; inventory list sorted by volume serial number including volume count.

DFSMSrmm INTERNAL USE ONLY		Inventory List by Volume Serial							PAGE - 00001
EDGRPT15 -----									DATE - 98104
Volume		Vol-	Vol-	DSN-	Create	Create	Vol	Location	BIN
Serial	Data Set Name	Seq.	Cnt.	Seq.	Date	Time	Scr	Name	Number

AD2522	SMPMCS	1	1	1	30/07/1992	200559	NO		*
AD2524	SMPMCS	1	1	1	30/07/1992	200701	NO		*
ASAP16	ASAP.SAMPLE.MONITOR	1	1	1	25/08/1992	082701	NO		*
BB5510	SMPMCS	1	1	1	28/07/1994	103509	NO		*
BB5511	IBM.HOM1120.F2	2	2	1	28/07/1994	104111	NO		*
BB5511	IBM.JOM12NO.F1	2	2	2	28/07/1994	104134	NO		*
BB5511	IBM.JOM12NO.F2	2	2	3	28/07/1994	104146	NO		*
BB5522	SMPMCS	1	1	1	19/10/1995	104023	NO		*
BB5522	IBM.HBB5520.F1	1	1	2	19/10/1995	104458	NO		*
BB5522	IBM.HBB5520.F2	1	1	3	19/10/1995	104511	NO		*
BB5522	IBM.HBB5520.F3	1	1	4	19/10/1995	104527	NO		*
BB5522	IBM.HBB5520.F4	1	1	5	19/10/1995	104747	NO		*
BB5522	IBM.JBB5522.F1	1	2	6	19/10/1995	104955	NO		*
BB5522	IBM.JBB5522.F2	1	2	7	19/10/1995	105015	NO		*
BB5523	IBM.JBB5522.F2	2	2	1	19/10/1995	105025	NO		*
BB5523	IBM.JBB52NO.F1	2	2	2	19/10/1995	105035	NO		*
BB5523	IBM.JBB52NO.F2	2	2	3	19/10/1995	105052	NO		*
BB5523	IBM.JBB55N2.F1	2	2	4	19/10/1995	105105	NO		*
BB5523	IBM.HCSH521.F1	2	2	5	19/10/1995	105116	NO		*
BB5523	IBM.HCSH521.F2	2	2	6	19/10/1995	105127	NO		*
BB5523	IBM.HCSH521.F3	2	2	7	19/10/1995	105141	NO		*
BB5523	IBM.JCSH523.F1	2	2	8	19/10/1995	105207	NO		*
BB5523	IBM.JCSH523.F2	2	2	9	19/10/1995	105220	NO		*
DFSMSrmm INTERNAL USE ONLY		Inventory List by Volume Serial							PAGE - 00117
EDGRPT15 -----									DATE - 98104
Volume		Vol-	Vol-	DSN-	Create	Create	Vol	Location	BIN
Serial	Data Set Name	Seq.	Cnt.	Seq.	Date	Time	Scr	Name	Number

VT4201	HVT4201.F2	1	1	3	27/07/1994	143455	NO		*
VT4201	HVT4201.F3	1	1	4	27/07/1994	143523	NO		*
VT4201	HVT4201.F4	1	1	5	27/07/1994	143612	NO		*
VT4201	HVT4201.F5	1	1	6	27/07/1994	143623	NO		*
VT4202	SMPMCS	1	1	1	27/07/1994	143640	NO		*
VT4202	JVT4202.F1	1	1	2	27/07/1994	145021	NO		*
VT4202	JVT4202.F2	1	1	3	27/07/1994	145036	NO		*
VT4202	JVT4202.F3	1	1	4	27/07/1994	145053	NO		*
VT4202	JVT4202.F4	1	1	5	27/07/1994	145113	NO		*
VT4202	JVT4202.F5	1	1	6	27/07/1994	145130	NO		*
VT4301	SMPMCS	1	1	1	10/08/1995	155519	NO		*
VT4301	IBM.HVT4301.F1	1	1	2	10/08/1995	155605	NO		*
VT4301	IBM.HVT4301.F2	1	1	3	10/08/1995	155636	NO		*
VT4301	IBM.HVT4301.F3	1	1	4	10/08/1995	155742	NO		*
VT4301	IBM.HVT4301.F4	1	1	5	10/08/1995	155756	NO		*
VT4301	IBM.HVT4301.F5	1	1	6	10/08/1995	155814	NO		*
VT4301	IBM.HVT4301.F6	1	1	7	10/08/1995	155826	NO		*
VT4301	IBM.HVT4301.F7	1	1	8	10/08/1995	155836	NO		*
VT4301	IBM.HVT4301.F8	1	1	9	10/08/1995	155848	NO		*
VT4302	SMPMCS	1	1	1	03/11/1995	130702	NO		*
VT4302	IBM.JVT4302.F1	1	1	2	03/11/1995	130714	NO		*
VT4302	IBM.JVT4302.F2	1	1	3	03/11/1995	130727	NO		*
VT4302	IBM.JVT4302.F3	1	1	4	03/11/1995	130741	NO		*
WB1102		1	1				NO		*
		End of Report. 6239 Entries listed							

Figure 239. REPORT15 Output: Inventory List by Volume Serial Number

14.3 Producing Customized Reports from DFSMSrmm Extract Data

If you cannot produce the information you want either from the DFSMSrmm ISPF dialog or TSO search commands, or from the reports provided with DFSMSrmm, you have the option of producing customized reports. Customized reports can be produced from the DFSMSrmm report extract file created by a EDGHSKP utility run.

The extract file is an extracted point-in-time copy of information in the DFSMSrmm CDS. It is a sequential file containing:

- EDGRDTEXT data set records

- EDGRKEXT VRS records
- EDGROEXT owner records
- EDGRPEXT software product records
- EDGRREXT rack records
- EDGRSEXT storage location records
- EDGRVEXT volume records.

These records are mapped in *DFSMSrmm Implementation and Customization Guide*, SC26-4932. You can use reporting programs such as SAS, SLR, or DFSORT and REXX to obtain useful reports using the extract file as input.

To obtain an extract file from the DFSMSrmm database you can use the sample JCL in Figure 240.

```
//HSKP    EXEC PGM=EDGHSKP,
//          PARM='RPTTEXT'
//MESSAGE DD DISP=SHR,DSN=HSKP.MESSAGES
//RPTTEXT DD DISP=SHR,DSN=HSKP.RPTTEXT
```

Figure 240. EDGHSKP JCL for Extract Processing

If you specify RPTTEXT as the PARM of the EDGHSKP utility, you obtain, in the data set referenced by the RPTTEXT DD statement, a sequential extracted copy of information in the DFSMSrmm CDS. Appendix D, “Sample Exits and Reports” on page 603 contains examples of customized reports, where the DFSMSrmm extract data set is used as input. Also see the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for examples of how to use the DFSORT utility ICETOOL to produce customized reports.

14.4 Producing Reports Using Other Reporting Products

The flexibility of DFSMSrmm reporting derives from a complete set of commands and the sequential extract data set, which most reporting programs can easily use as input.

It is very simple to obtain useful DFSMSrmm reports using DB2 and QMF. The use of SQL is a very good way to get flexible reporting.

Do not use DFSMSrmm commands to gather the input. Instead, use the report extract data set, which you can produce quickly by using the EDGHSKP utility.

You will need the record layouts of the extract data set (see *DFSMSrmm Implementation and Customization Guide*, SC26-4932) to load a relational database and for query-based reporting. Once you have loaded the relational database, you have the option of running some predefined reports or individual queries.

With REXX, you can directly execute DFSMSrmm commands and build customized reports, or you can use the information from the DFSMSrmm report extract data set as input to building your own reports.

The SAS program can simply read the extract file and generate a set of reports properly customized to meet your needs.

The choice of reporting method is up to you; it depends on your environment and the results you want to obtain.

14.4.1 Scratch Forecast Using Retention Date

The DFSMSrmm retention date is included in the extract data set and volume records and in the EDGHSKP VRS report. This field has different meanings, depending on the VRS retention type (see Table 106 and Table 107 on page 557).

<i>Table 106 (Page 1 of 2). DFSMSrmm Retention Date Calculation by COUNT from 1 - 99998</i>		
Retention Type	Retention Date	Calendar Date Format
CYCLES	Special cycles date format ¹	CYCL/ccccc
DAYS	COUNT plus the create date ¹	Date format specified by your installation
XD	COUNT plus the date ^{1,5}	Date format specified by your installation
LASTREF	COUNT plus the date last referenced ¹	Date format specified by your installation
CYCLES + WC	Special catalog date format ³	WHILECATLG
DAYS + WC	Special catalog date format ¹	Date format specified by your installation
LASTREF + WC	Special catalog date format	Date format specified by your installation
CYCLES + UEX	Volume expiration date ¹	Date format specified by your installation
DAYS + UEX	Lower of volume expiration date and COUNT plus create date ¹	Date format specified by your installation
LASTREF + UEX	Lower of volume expiration date and COUNT plus date last referenced ¹	Date format specified by your installation
CYCLES + WC + UEX	Volume expiration date ¹	Date format specified by your installation
DAYS + WC + UEX	Lower of volume expiration date and COUNT plus create date ¹	Date format specified by your installation
LASTREF + WC + UEX	Lower of volume expiration date and COUNT plus date last referenced ¹	Date format specified by your installation
(CYCLES + UEX) and (MV/MC = WC) ²	Special catalog date format ³	WHILECATLG
(DAYS + UEX) and (MV/MC = WC) ²	Count plus the create date ¹	Date format specified by your installation
(LASTREF + UEX) and (MV/MC = WC) ²	Count plus the last reference date ¹	Date format specified by your installation
(DSN = UEX) and (MV/MC) ²	Calculates two dates. One date using primary DSN VRS. One date using secondary MV or MC vital record specification. Uses the lowest of the two dates. ^{1,4}	As determined by the VRS retention options

Table 106 (Page 2 of 2). DFSMSrmm Retention Date Calculation by COUNT from 1 - 99998

Retention Type	Retention Date	Calendar Date Format
¹The VRS deletion date is used as long as it is not 1999/365, and is lower than the retention date calculated. The deletion date used is the earlier of the current retention VRS and the firstt VRS in the VRS chain. ²Management value specifies WHILECATALOG and VRSEL(OLD) is in use, and the data set is cataloged. ³The VRS deletion date is used as long as it is not 1999/365, and is lower than the retention date calculated. If a data set is not cataloged and is retained using the parmlib CATRETPD operand, DFSMSrmm sets the CATRETPD retention date. ⁴Parmlib option VRSEL(NEW) must be in use. The retention date format can be any of the special formats or a date. ⁵The retention date is determined using the date VRSEL determined that the VRS subchain started to retain the data set. Legend • LASTREF = LASTREFERENCE LASTREF date is the last referenced date in the data set record and is checked each time DFSMSrmm vital record processing is run. • WC = WHILECATALOG • UEX = UNTILEXPIRED • MV = Management value • MC = Management class • CYCLES = CYCLES and BYDAYSCYCLE retention types • DSN = matching data set name VRS • XD = EXTRADAYS		

Table 107 shows how DFSMSrmm calculates retention date when you specify a COUNT(99999) on the DFSMSrmm ADDVRS subcommand. The COUNT(99999) indicates that DFSMSrmm retains all cycles of a data set.

Table 107 (Page 1 of 2). Retention Date Calculation by COUNT(99999)

Retention Type	Retention Date	Calendar Date Format
CYCLES	VRS delete date or cycles date format ¹	Date format specified by your installation
DAYS	VRS delete date	Date format specified by your installation
XD	VRS delete date	Date format specified by your installation
LASTREF	VRS delete date	Date format specified by your installation
CYCLES + WC	VRS delete date or catalog date format ²	Date format specified by your installation or WHILECATALOG
DAYS + WC	VRS delete date or catalog date format ²	Date format specified by your installation or WHILECATALOG
LASTREF + WC	VRS delete date or catalog date format ²	Date format specified by your installation or WHILECATALOG
CYCLES + UEX	Volume expiration date ³	Date format specified by your installation
DAYS + UEX	Volume expiration date ³	Date format specified by your installation

Table 107 (Page 2 of 2). Retention Date Calculation by COUNT(99999)

Retention Type	Retention Date	Calendar Date Format
LASTREF + UEX	Volume expiration date ³	Date format specified by your installation
CYCLES + WC + UEX	Volume expiration date ³	Date format specified by your installation
DAYS + WC + UEX	Volume expiration date ³	Date format specified by your installation
LASTREF + WC + UEX	Volume expiration date ³	Date format specified by your installation
(CYCLES + UEX) and (MV/MC = WC) ⁴	Data set VRS delete date OR catalog date format ²	Date format specified by your installation or WHILECATALOG
(DAYS + UEX) and (MV/MC = WC) ⁴	Data set VRS delete date OR catalog date format ²	Date format specified by your installation or WHILECATALOG
(LASTREF + UEX) and (MV/MC = WC) ⁴	Data set VRS delete date OR catalog date format ²	Date format specified by your installation or WHILECATALOG
(DSN = UEX) and (MV/MC)	Calculates two dates: one date using DSN VRS; and one date using MV or MC VRS. Uses the lowest of the two dates. ^{2,5}	As determined by the VRS retention options.

¹The VRS deletion date is used as long as it is not 1999/365. The special cycles date format is used when the VRS deletion date is 1999/365. The deletion date used is the earlier of the current retention VRS and the 1st VRS in the VRS chain.

²The VRS deletion date is used as long as it is not 1999/365. The special catalog date format is used when the VRS deletion date is 1999/365. If the data set is not cataloged and CATRETPD retains the data set, DFSMSrmm uses the CATRETPD retention date. The deletion date used is the earlier of the current retention VRS in the VRS chain.

³The VRS deletion date is used as long as it is not 1999/365. The deletion date used is the earlier of the current retention VRS and the 1st VRS in the VRS chain.

⁴Management value or management class specifies WHILECATALOG and VRSEL(OLD), and the data set is cataloged.

⁵PARMLIB option VRSEL(NEW) must be in use. To determine the retention date use both VRSs and this table.

Legend

- LASTREF = LASTREFERENCE
LASTREF is the last referenced date in the data set record and is checked each time DFSMSrmm vital record processing is run.
- WC = WHILECATALOG
- UEX = UNTILEXPIRED
- MV = Management value
- MC = Management class
- CYCLES = CYCLES and BYDAYSCYCLE retention types
- DSN = matching data set name VRS
- XD = EXTRADAYS

The volume retention date is the highest of all of the data sets on the volume.
The order is:

1. 1999/366
2. 1999/365
3. CYCL/cccc, where ccccc is the COUNT value for cycles used in the VRS
4. WHILECATLG
5. A date in the format selected by your installation

6. CATRETPD

If you do not define a retention policy for a data set or volume, DFSMSrmm sets an expiration date for each volume defined to it, using one of the following:

- User-specified JCL expiration date for a data set on the volume, not to exceed the maximum retention period (MAXRETPD) set by your installation in PARMLIB member EDGRMMxx
- Default retention period for data sets and volumes defined by your installation, if no retention period or expiration date was set in user-specified JCL for a data set on the volume
- Expiration date or retention period specified by a user for a volume when manually requesting a scratch volume or manually adding or changing information about the volume to DFSMSrmm. This value cannot exceed the maximum retention period (MAXRETPD) set by your installation in PARMLIB member EDGRMMxx

The new cycles and catalog date formats are displayed on the LISTDATASET and LISTVOLUME command output:

Retention Date	Meaning
WHILECATLG	Data set is retained by a VRS that specifies the WHILECATALOG option.
CYCL/cccc	Data set is retained by CYCLES.

Note: If a volume is not retained by vital records processing, its retention date is equal to its expiration date. The retention date of a volume is set to the expiration date when the volume is opened for output.

Appendix A. Security Considerations and Implementation

This appendix describes how you can protect DFSMSrmm functions using RACF.

Objectives of Chapter

- Describe the different DFSMSrmm users and the functions they typically use.
- Describe the RACF FACILITY class profiles used to protect DFSMSrmm resources
- Define DFSMSrmm resources to RACF

Audience

- System programmers
- Security administrator

This appendix provides you with the information you need to implement your installation security policies in DFSMSrmm using the RACF FACILITY class profiles. To protect DFSMSrmm functions, you need to use an external security product, such as RACF. You can use the information provided here and the information provided in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932, for authorizing users and ensuring security.

A.1 Authorize Users and Ensure Security

In our examples we use six different user groups, which we will define here:

- General User
- Storage Administrator
- System Programmer
- Librarian
- Inventory Management Functions
- Operator

A.1.1 General User

The *General User*, or *End User*, is a person outside the computing center. The General User needs only limited access to DFSMSrmm functions. They might want to manage volumes they own and request information about resources defined to DFSMSrmm.

The General User can:

- Manually request a scratch volume by using the TSO subcommand GETVOLUME
- Display all DFSMSrmm CDS details except VRSS control information
- Search for data sets, software products, shelf locations, and volumes
- Update their owner ID information
- Release an owned volume

A.1.2 Storage Administrator

The *Storage Administrator* is responsible for specifying the policies to retain and move data sets and volumes. These policies are specified in VRSs and processed during DFSMSrmm inventory management.

The Storage Administrator can:

- Define retention and storage policies for data sets and volumes
- Change information about any volume they own, their owner ID, and any VRS information
- Manually request a scratch volume by using the TSO subcommand GETVOLUME
- Create a list of software products and display information about any resource defined to DFSMSrmm
- Update their owner ID information
- Release an owned volume

A.1.3 Librarian

The *Librarian* creates scratch pull lists, moves the tapes between the built-in locations (LOCAL, DISTANT, REMOTE, SHELF), installation defined storage locations, and system managed libraries. The Librarian takes the inventory, expands or reduces volume pools, and adds and deletes the external tapes or cartridges to the DFSMSrmm CDS. They can add, change, list, and delete all information in the DFSMSrmm CDS except the VRSs.

The *Librarian* can:

- Display control information and installation options and rules using the TSO subcommand LISTCONTROL
- Display all DFSMSrmm CDS details
- Request scratch volumes for themselves and other users
- Release volumes
- Add information about volumes, data sets, shelf locations, owner IDs, and software products
- Update volume, data set, owner, and software product details.
- Delete information about volumes, data sets, shelf locations, owner IDs, and software products
- Initialize and erase volumes

A.1.4 System Programmer

The *System Programmer* is responsible for the DFSMSrmm installation and implementation, and provides support for DFSMSrmm in the computing center.

The System Programmer can:

- Display control information and installation options and rules using the TSO subcommand LISTCONTROL
- Display all DFSMSrmm CDS details
- Request scratch volumes for themselves and other users
- Release volumes
- Add information about volumes, data sets, shelf locations, owner IDs, and software products
- Update volume, data set, owner, and software product details.
- Delete information about volumes, data sets, shelf locations, owner IDs, and software products

- Initialize and erase volumes
- Remove DFSMSrmm from the system. This means stop the DFSMSrmm subsystem, the started task, and disable the subsystem interface.

A.1.5 Inventory Management Function

The *Inventory Management Function* is responsible for the daily and weekly DFSMSrmm inventory management.

The *Inventory Management Function* can:

- Process vital records
- Perform expiration processing
- Perform storage location management
- Create an extract data set
- Create volume movement reports
- Back up the DFSMSrmm CDS
- Back up the DFSMSrmm journal and clear the journal

A.1.6 Operator

The *Operator* handles DFSMSrmm system requests. They take care of tape volumes requests outside an IBM automated tape library data server or any other robot system, display DFSMSrmm information, and add and delete information about volumes, data sets, shelf locations, owner IDs, and software products,

The *Operator* can:

- Display control information and installation options and rules using the TSO subcommand LISTCONTROL
- Manually request a scratch volume by using the TSO subcommand GETVOLUME
- Display all DFSMSrmm CDS details except VRs and control information
- Search for data sets, software products, shelf locations, and volumes
- Update owner ID details
- Release an owned volume
- Respond to console mount messages
- Manually erase and initialize tapes

After you cut over to production, you should remove the the Operator from the access list for the resource STGADMIN.EDG.RESET.SSI.

A.2 Protecting DFSMSrmm Resources

Define RACF FACILITY class profiles to control access to DFSMSrmm functions protected by the DFSMSrmm resource. The DFSMSrmm resources you protect with RACF profiles in the FACILITY class each have an entity name prefixed with STGADMIN.EDG. Table 108 on page 564 lists the DFSMSrmm resources.

If you do not protect DFSMSrmm resources with RACF, an equivalent security product, or you do not have a security product in place, DFSMSrmm provides control for some resources as defined in the *DFSMSrmm Implementation and Customization Guide*, SC26-4932.

<i>Table 108. RACF FACILITY Class Profiles for DFSMSrmm Functions</i>	
DFSMSrmm Resource	Function Protected
STGADMIN.EDG.FORCE	Use of the RMM FORCE parameter that can be used in the AD DD DATASET, CHANGEDATASET, CHANGEVOLUME and DELETEDATASET subcommand.
STGADMIN.EDG.HOUSEKEEP	Use of DFSMSrmm inventory management functions.
STGADMIN.EDG.IGNORE.TAPE.volser	Use of duplicate volume serial numbers and ignored volumes.
STGADMIN.EDG.LABEL.volser	Creation of standard tape labels.
STGADMIN.EDG.LISTCONTROL	Use of the RMM LISTCONTROL subcommand to display DFSMSrmm CDS control record information and EDGRMMxx PARMLIB settings.
STGADMIN.EDG.MASTER	Access to information in the DFSMSrmm CDS.
STGADMIN.EDG.NOLABEL.volser	Creation of tapes without labels.
STGADMIN.EDG.OPERATOR	Use of the initialize and erase function.
STGADMIN.EDG.OWNER.owner	Access to owned resources. DFSMSrmm checks this entity only if the command issuer is not the owner of the resource or does not have CONTROL access to STDADMIN.EDG.MASTER.
STGADMIN.EDG.RELEASE	Use of the RMM DELETEVOLUME RELEASE subcommand to process any release actions specified for a volume.
STGADMIN.EDG.RESET.SSI	Use of the RESET facility for removing DFSMSrmm from the system.
STGADMIN.EDG.VRS	Use of the RMM LISTVRS or SEARCHVRS subcommands to obtain information about VRSs. Use of the RMM ADDVRS and DELETEVRS subcommands to define or remove vital record specifications.
STGADMIN.EDG.INERS.WRONGLABEL	Processing for volume mounted with the wrong label using the EDGINERS initialization process.

A.2.1 DFSMSrmm Resource Access Levels

Table 109 provides suggestions for authorizing different types of users. Implementing security authorization to DFSMSrmm resources depends on your organization's structure, administrative procedures, and security requirements. For example, in Table 109, we suggest that the general user be given ALTER access to the STGADMIN.EDG.LABEL.volser and STGADMIN.EDG.NOLABEL.volser resources. This allows the relabelling of tapes to SL and the use of NL for non-specific mounts. You may choose not to give ALTER access to all general users, and only provide access to those users who have a need to create BLP or NL tapes.

Table 109 (Page 1 of 2). Suggested DFSMSrmm Resource Access

RACF FACILITY Class Resource	General User	Storage Adminis- trator	System Pro- grammer	Librarian	Inventory Manage- ment Functions	Operator
STGADMIN.EDG.FORCE	-	-	U	U	-	-

Table 109 (Page 2 of 2). Suggested DFSMSrmm Resource Access

RACF FACILITY Class Resource	General User	Storage Administrator	System Programmer	Librarian	Inventory Management Functions	Operator
STGADMIN.EDG.HOUSKEEP	-	-	-	-	R	-
STGADMIN.EDG.IGNORE.TAPE.volser	-	-	-	-	-	-
STGADMIN.EDG.LABEL.volser	A	-	-	-	-	-
STGADMIN.EDG.LISTCONTROL	C	C	C	C	-	C
STGADMIN.EDG.MASTER	R	C	C	C	-	R
STGADMIN.EDG.NOLABEL.volser	A	-	-	-	-	-
STGADMIN.EDG.OPERATOR	-	-	U	U	-	U
STGADMIN.EDG.OWNER.userid	-	U	-	-	-	-
STGADMIN.EDG.RELEASE	R	-	-	-	-	-
STGADMIN.EDG.RESET.SSI	-	-	-	-	-	A
STGADMIN.EDG.VRS	R	C	C	C	-	-
STGADMIN.EDG.INERS.WRONGLABEL	-	-	-	U	-	-
STGADMIN.ADR.DUMP.CNCURRNT	-	-	-	-	R	-
Access: R-Read, U-Update, C-Control, A-Alter						

A.3 Security Implementation Steps

In this section we take you through the process of using RACF to implement security for DFSMSrmm resources and functions in your installation.

The first step is to create a RACF user ID for DFSMSrmm and define the DFSMSrmm started task to RACF. The second step is to create RACF groups, by function, and connect user IDs to these groups. Next, we define all DFSMSrmm resources in the RACF FACILITY class and permit the function groups to the resources. After this we will define all the required DFSMSrmm data set profiles. Finally, we will authorize other products and procedures to DFSMSrmm.

The following list shows all the activities that should be done before you can start DFSMSrmm:

- Assign a RACF user ID to DFSMSrmm
- Define RACF groups for tape administration
- Define DFSMSrmm resources to RACF
 - STGADMIN.EDG.FORCE
 - STGADMIN.EDG.HOUSEKEEP
 - STGADMIN.EDG.IGNORE.TAPE.volser
 - STGADMIN.EDG.LABEL.volser
 - STGADMIN.EDG.LISTCONTROL
 - STGADMIN.EDG.MASTER
 - STGADMIN.EDG.NOLABEL.volser
 - STGADMIN.EDG.OPERATOR
 - STGADMIN.EDG.OWNER.userid
 - STGADMIN.EDG.RELEASE

- STGADMIN.EDG.RESET.SSI
- STGADMIN.EDG.VRS
- STGADMIN.EDG.INERS.WRONG.LABEL
- Generic DFSMSrmm data set profile
- Data set profiles for CDS, journal, and message data sets
- Data set profiles for housekeeping data sets
- Data set profiles for audit and report data sets
- Define the STGADMIN.ADR.DUMP.CNCURRNT resource

Note: You have to define the *STGADMIN.ADR.DUMP.CNCURRNT* resource only when you use DFSMSdss with concurrent copy.

A.3.1 Assigning DFSMSrmm a RACF User ID

You can assign a RACF user ID that matches the name of the DFSMSrmm procedure you created in 3.1.3, “Update the Procedure Library” on page 39, but any installation-selected RACF user ID is acceptable. As data sets are created for use by the DFSMSrmm procedure, add the DFSMSrmm RACF user ID to the access list for the data sets.

For a started procedure to access any of the system’s resources, a user ID must be associated with that task. The user ID should have all the proper authorizations for accessing the system resources, therefore it should be assigned the RACF OPERATIONS attribute. We recommend defining a new user ID for each started procedure rather than using the default started task user ID (for example, STCUSER). In our system we associated userid *DFRMM* with the DFSMSrmm started procedure. Following is the RACF command we used to define the DFSMSrmm user ID:

```
ADDUSER DFRMM OPERATIONS DFLTGRP(SYS1) NAME('DFSMSrmm Userid')
```

Note: If you do not specify the DFLTGRP parameter on the ADDUSER command, the current connect group of the user issuing the ADDUSER command is used as the default.

A.3.2 Identifying Started Procedures to RACF

Before RACF 2.1, the only way to associate a started procedure with a RACF user ID was by coding the RACF started procedures table, ICHRIN03. With RACF 2.1, assigning RACF identities to started procedures has been greatly simplified by the introduction of the RACF STARTED class. You can add or modify security definitions for new and existing started procedures by issuing the RDEFINE and RALTER commands.

Even though the use of the RACF STARTED class is the preferred way of identifying started procedures to RACF, it is still mandatory to have the ICHRIN03 module. RACF cannot be initialized if ICHRIN03 is not present in the system. A dummy ICHRIN03 is shipped and installed with RACF.

A.3.2.1 RACF STARTED Class

The STARTED class allows you to assign RACF identities to started procedures dynamically using the RDEFINE and RALTER commands. Resource names in the STARTED class have the format *membername.jobname*. You assign identities such as the RACF user ID and group ID, using fields in the STDATA segment. You can define the started procedure resource, using either a generic profile name or a discrete profile name. A RACF generic profile describes one or more data sets that have a similar name structure. A RACF discrete profile describes

a specific data set on a specific volume. In our system we created generic profiles for started procedures. Issue the following RACF commands to assign RACF identities to the DFSMSrmm started procedure:

```
RDEFINE STARTED (DFRMM.*) UACC(NONE) STDATA(USER(DFRMM) GROUP(SYS1))
SETROPTS RACLIST(STARTED) REFRESH
SETROPTS GENERIC(STARTED) REFRESH
```

After you have added profiles to the RACF STARTED class, refresh the in-storage profiles, using the SETROPTS REFRESH command. The SETROPTS GENERIC command is needed only when you define generic profiles.

Note: If you plan to use the EDGLABEL, EDGXPROC, or the EDGBKUP procedure, you must define the procedures in the RACF STARTED class or in ICHRIN03.

For more information on defining procedures to the RACF STARTED class refer to the *OS/390 Security Server (RACF) Security Administrator's Guide*, SC28-1915.

Figure 241 shows a batch job that can be used to add the DFSMSrmm user ID in the RACF STARTED class.

```
//RACFRMM JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
//* Note
//* ====
//* To execute this job you need the RACF SPECIAL attribute
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* CREATE AN ENTRY IN THE DYNAMIC STARTED PROCEDURE TABLE FOR DFRMM */
/* ***** */
RDEFINE STARTED DFRMM.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
```

Figure 241. JCL to Add DFSMSrmm User ID in RACF STARTED Class

A.3.2.2 ICHRIN03 Started Procedures Table

We do not recommend using the RACF started procedures table (ICHRIN03). The preferred way of adding started procedure users to the RACF database is by using the STARTED class.

The sample JCL in Figure 242 on page 568 shows how to identify the DFSMSrmm started procedure to RACF by assembling ICHRIN03.


```

//RACFGRP JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
/* Note
/* ====
/* To execute this job you need the RACF SPECIAL attribute
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* DEFINE RACF ACCESS GROUP FOR ADMINISTRATOR */
/* ***** */
ADDGROUP EDGADMIN -
      SUBGROUP(SYS1) OWNER(SYS1)
      CONNECT WOODY GROUP(EDGADMIN) AUTHORITY(USE)
      CONNECT KUEHN GROUP(EDGADMIN) AUTHORITY(USE)
      CONNECT GOHR GROUP(EDGADMIN) AUTHORITY(USE)
      CONNECT ETZ GROUP(EDGADMIN) AUTHORITY(USE)
      CONNECT BENGT GROUP(EDGADMIN) AUTHORITY(USE)
      CONNECT NELSON GROUP(EDGADMIN) AUTHORITY(USE)
/* ***** */
/* DEFINE RACF ACCESS GROUP FOR SYSTEM PROGRAMMER */
/* ***** */
ADDGROUP EDGSYSPG -
      SUBGROUP(SYS1) OWNER(SYS1)
      CONNECT SCHLUM GROUP(EDGSYSPG) AUTHORITY(USE)
      CONNECT SIEGEL GROUP(EDGSYSPG) AUTHORITY(USE)
/* ***** */
/* DEFINE RACF ACCESS GROUP FOR LIBRARIAN */
/* ***** */
ADDGROUP EDGLIB -
      SUBGROUP(SYS1) OWNER(SYS1)
      CONNECT JOY GROUP(EDGLIB) AUTHORITY(USE)
      CONNECT NADINE GROUP(EDGLIB) AUTHORITY(USE)
      CONNECT ZERBINI GROUP(EDGLIB) AUTHORITY(USE)
      CONNECT STRAUS GROUP(EDGLIB) AUTHORITY(USE)
      CONNECT SEYFERT GROUP(EDGLIB) AUTHORITY(USE)
/* ***** */
/* DEFINE RACF ACCESS GROUP FOR INVENTORY MANAGEMENT FUNCTION */
/* ***** */
ADDGROUP EDGJOB -
      SUBGROUP(SYS1) OWNER(SYS1)
      CONNECT WILDEN GROUP(EDGJOB) AUTHORITY(USE)
      CONNECT HENICKE GROUP(EDGJOB) AUTHORITY(USE)
      CONNECT KEIL GROUP(EDGJOB) AUTHORITY(USE)
      CONNECT BRIAN GROUP(EDGJOB) AUTHORITY(USE)

```

Figure 243 (Part 1 of 2). JCL to Create RACF Groups and Connect User IDs

```

/* ***** */
/*  DEFINE RACF ACCESS GROUP FOR OPERATOR  */
/* ***** */
ADDGROUP EDGOPER -
        SUBGROUP(SYS1) OWNER(SYS1)
        CONNECT LOVELACE GROUP(EDGOPER) AUTHORITY(USE)
        CONNECT TAUBER  GROUP(EDGOPER) AUTHORITY(USE)
        CONNECT ZINK    GROUP(EDGOPER) AUTHORITY(USE)

```

Figure 243 (Part 2 of 2). JCL to Create RACF Groups and Connect User IDs

A.3.4 RACF Data Set Profiles

To protect the DFSMSrmm data sets, we define discrete profiles for the following data sets, and then permit the appropriate RACF groups to the discrete data set profile:

- DFSMSrmm control data set
- DFSMSrmm journal data set
- DFSMSrmm message data set
- DFSMSrmm housekeeping data sets
- DFSMSrmm report data sets
- DFSMSrmm security report data sets

To protect all other DFSMSrmm data sets define a generic data set profile using the high level qualifier of your DFSMSrmm data sets.

Note: Refer to the *OS/390 Security Server (RACF) Command Language Reference*, SC28-1918, for enhanced generic naming considerations. The enhanced generic naming option applies only to data sets and allows you to use double asterisk (**) in the the DATASET class. It also changes the meaning of the single asterisk (*) at the end of a profile name.

Figure 244 on page 571 shows a batch job to define the DFSMSrmm data set profiles and permit the appropriate RACF groups and DFSMSrmm procedure to the data set profiles.

```

//RACFDSN JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
//* Note
//* ====
//* To execute this job you need the RACF SPECIAL attribute
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* DEFINE RACF GROUP */
/* ***** */
ADDGROUP RMM -
      SUBGROUP(SYS1) OWNER(SYS1)
/* ***** */
/* DEFINE GLOBAL DATA SET PROFILE */
/* ***** */
ADDSD 'RMM.*' UACC(NONE)
      PERMIT 'RMM.*' -
            ID(EDGADMIN) ACC(NONE)
      PERMIT 'RMM.*' -
            ID(EDGLIB) ACC(NONE)
      PERMIT 'RMM.*' -
            ID(EDGJOB) ACC(NONE)
      PERMIT 'RMM.*' -
            ID(EDGOPER) ACC(NONE)
      PERMIT 'RMM.*' -
            ID(EDGSYSPG) ACC(NONE)
/* ***** */
/* DEFINE CONTROL DATA SET DATA SET PROFILE */
/* ***** */
ADDSD 'RMM.PROD.CDS' UACC(NONE) GENERIC
      PERMIT 'RMM.PROD.CDS' -
            ID(EDGADMIN) ACC(READ)
      PERMIT 'RMM.PROD.CDS' -
            ID(EDGLIB) ACC(READ)
      PERMIT 'RMM.PROD.CDS' -
            ID(EDGSYSPG) ACC(ALTER)
/* ***** */
/* DEFINE JOURNAL DATA SET DATA SET PROFILE */
/* ***** */
ADDSD 'RMM.PROD.JRNL' UACC(NONE) GENERIC
      PERMIT 'RMM.PROD.JRNL' -
            ID(EDGADMIN) ACC(READ)
      PERMIT 'RMM.PROD.JRNL' -
            ID(EDGLIB) ACC(READ)
      PERMIT 'RMM.PROD.JRNL' -
            ID(EDGSYSPG) ACC(ALTER)

```

Figure 244 (Part 1 of 2). JCL to Permit Access to DFSMSrmm Data Set Profiles

```

/* ***** */
/*  DEFINE MESSAGE DATA SET DATA SET PROFILE  */
/* ***** */
ADDSD 'RMM.PROD.MSGS' UACC(NONE) GENERIC
PERMIT 'RMM.PROD.MSGS' -
      ID(EDGADMIN) ACC(UPDATE)
PERMIT 'RMM.PROD.MSGS' -
      ID(EDGLIB) ACC(UPDATE)
PERMIT 'RMM.PROD.MSGS' -
      ID(EDGSYSPG) ACC(ALTER)
/* ***** */
/*  DEFINE HOUSEKEEP DATA SET DATA SET PROFILE  */
/* ***** */
ADDSD 'RMM.HSKP.*' UACC(NONE)
PERMIT 'RMM.HSKP.*' -
      ID(EDGSYSPG) ACC(ALTER)
PERMIT 'RMM.HSKP.*' -
      ID(EDGLIB) ACC(ALTER)
PERMIT 'RMM.HSKP.*' -
      ID(EDGJOB) ACC(ALTER)
PERMIT 'RMM.HSKP.*' -
      ID(EDGOPER) ACC(ALTER)
/* ***** */
/*  DEFINE REPORT DATA SET DATA SET PROFILE  */
/* ***** */
ADDSD 'RMM.REPORT.*' UACC(NONE)
PERMIT 'RMM.REPORT.*' -
      ID(EDGSYSPG) ACC(ALTER)
PERMIT 'RMM.REPORT.*' -
      ID(EDGLIB) ACC(ALTER)
PERMIT 'RMM.REPORT.*' -
      ID(EDGJOB) ACC(ALTER)
PERMIT 'RMM.REPORT.*' -
      ID(EDGOPER) ACC(ALTER)
/* ***** */
/*  DEFINE SECURITY DATA SET DATA SET PROFILE  */
/* ***** */
ADDSD 'RMM.SECURITY.*' UACC(NONE)
PERMIT 'RMM.SECURITY.*' -
      ID(EDGSYSPG) ACC(ALTER)
PERMIT 'RMM.SECURITY.*' -
      ID(EDGJOB) ACC(ALTER)
PERMIT 'RMM.SECURITY.*' -
      ID(AUDITOR) ACC(READ)
/* ***** */
/* PERMIT DFRMM TO THE CONTROL DATA SET, JOURNAL DATA SET,  */
/* EXTRACT DATA SET, ACTIVITY LOG AND REPORT DATA SET */
/* ***** */
PERMIT 'RMM.MASTER.CDS' -
      ID(DFRMM) ACC(CONTROL)
PERMIT 'RMM.MASTER.JRNL' -
      ID(DFRMM) ACC(CONTROL)
PERMIT 'RMM.HSKP*' -
      ID(DFRMM) ACC(UPDATE)

```

Figure 244 (Part 2 of 2). JCL to Permit Access to DFSMSrmm Data Set Profiles

A.3.5 RACF Resource Profiles

The following DFSMSrmm resources, described in Table 108 on page 564, require RACF profiles in the FACILITY class:

- STGADMIN.EDG.FORCE
- STGADMIN.EDG.HOUSEKEEP
- STGADMIN.EDG.IGNORE.TAPE.volser
- STGADMIN.EDG.LABEL.volser
- STGADMIN.EDG.LISTCONTROL
- STGADMIN.EDG.MASTER
- STGADMIN.EDG.NOLABEL.volser
- STGADMIN.EDG.OPERATOR
- STGADMIN.EDG.OWNER.userid
- STGADMIN.EDG.RELEASE
- STGADMIN.EDG.RESET.SSI
- STGADMIN.EDG.VRS
- STGADMIN.EDG.INERS.WRONGLABEL

Figure 245 shows a batch job to define the resource profiles in the RACF FACILITY class.

```
//RACFRDEF JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
/* *****
/* Note
/* ====
/* To execute this job you need the RACF SPECIAL attribute
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* DEFINE RESOURCE STGADMIN.EDG.FORCE */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.FORCE -
          UACC(NONE) OWNER(SYS1)
          RALT FACILITY STGADMIN.EDG.FORCE -
          GLOBALAUDIT(ALL(UPDATE))
          PERMIT STGADMIN.EDG.FORCE -
          ID(EDGSYSPG) ACC(READ) CLASS(FACILITY)
          PERMIT STGADMIN.EDG.FORCE -
          ID(EDGLIB) ACC(READ) CLASS(FACILITY)
```

Figure 245 (Part 1 of 4). JCL to Define DFSMSrmm Resources in the FACILITY Class

```

/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.HOUSEKEEP                               */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.HOUSEKEEP -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.HOUSEKEEP -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.HOUSEKEEP -
        ID(EDGSYSPG) ACC(READ) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.HOUSEKEEP -
        ID(EDGLIB ) ACC(READ) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.HOUSEKEEP -
        ID(EDGJOB  ) ACC(READ) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.IGNORE.TAPE.*                           */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.IGNORE.TAPE.VOLSER.* -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.IGNORE.TAPE.VOLSER.* -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.IGNORE.TAPE.VOLSER.* -
        ID(EDGSYSPG) ACC(READ) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.IGNORE.TAPE.VOLSER.* -
        ID(EDGJOB  ) ACC(UPDATE) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.LABEL.*                                 */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.LABEL.* -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.LABEL.* -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.LABEL.* -
        ID(EDGLIB) ACC(ALTER) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.LABEL.* -
        ID(EDGSYSPG) ACC(ALTER) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.LABEL.* -
        ID(EDGSYSPG) ACC(ALTER) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.LISTCONTROL                             */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.LISTCONTROL -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.LISTCONTROL -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.LISTCONTROL -
        ID(EDGADMIN) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.LISTCONTROL -
        ID(EDGSYSPG) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.LISTCONTROL -
        ID(EDGLIB ) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.LISTCONTROL -
        ID(EDGJOB ) ACC(CONTROL) CLASS(FACILITY)

```

Figure 245 (Part 2 of 4). JCL to Define DFSMSrmm Resources in the FACILITY Class

```

/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.MASTER */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.MASTER -
    UACC(READ) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.MASTER -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.MASTER -
        ID(EDGADMIN) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.MASTER -
        ID(EDGSYSPG) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.MASTER -
        ID(EDGLIB) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.MASTER -
        ID(EDGJOB) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.MASTER -
        ID(EDGOPER) ACC(CONTROL) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.NOLABEL.* */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.NOLABEL.* -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.NOLABEL.* -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.NOLABEL.* -
        ID(EDGADMIN) ACC(ALTER) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.NOLABEL.* -
        ID(EDGLIB ) ACC(ALTER) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.NOLABEL.* -
        ID(EDGOPER ) ACC(ALTER) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.OPERATOR */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.OPERATOR -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.OPERATOR -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.OPERATOR -
        ID(EDGSYSPG) ACC(UPDATE) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.OPERATOR -
        ID(EDGLIB) ACC(UPDATE) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.OPERATOR -
        ID(EDGOPER) ACC(UPDATE) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.OWNER.* */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.OWNER.* -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.OWNER.* -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.OWNER.* -
        ID(EDGLIB) ACC(UPDATE) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.OWNER.* -
        ID(EDGADMIN) ACC(UPDATE) CLASS(FACILITY)

```

Figure 245 (Part 3 of 4). JCL to Define DFSMSrmm Resources in the FACILITY Class

```

/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.RELEASE */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.RELEASE -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.RELEASE -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.RELEASE -
        ID(DFHSM) ACC(READ) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.RESET.SSI */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.RESET.SSI -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.RESET.SSI -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.RESET.SSI -
        ID(EDGSYSPG) ACC(ALTER) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.RESET.SSI -
        ID(EDGOPER) ACC(ALTER) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.VRS */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.VRS -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.VRS -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.VRS -
        ID(EDGADMIN) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.VRS -
        ID(EDGSYSPG) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.VRS -
        ID(EDGLIB) ACC(CONTROL) CLASS(FACILITY)
/* ***** */
/*  DEFINE RESOURCE STGADMIN.EDG.INERS.WRONGLABEL */
/* ***** */
RDEFINE FACILITY STGADMIN.EDG.INERS.WRONG.LABEL -
    UACC(NONE) OWNER(SYS1)
    RALT FACILITY STGADMIN.EDG.INERS.WRONG.LABEL -
        GLOBALAUDIT(ALL(UPDATE))
    PERMIT STGADMIN.EDG.INERS.WRONG.LABEL -
        ID(EDGOPER) ACC(UPDATE) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.INERS.WRONG.LABEL -
        ID(EDGSYSPG) ACC(CONTROL) CLASS(FACILITY)
    PERMIT STGADMIN.EDG.INERS.WRONG.LABEL -
        ID(EDGLIB) ACC(CONTROL) CLASS(FACILITY)

```

Figure 245 (Part 4 of 4). JCL to Define DFSMSrmm Resources in the FACILITY Class

A.3.6 Removing DFSMSrmm from the System

When you first install and start to use DFSMSrmm, it is likely that you will need to stop or even remove DFSMSrmm from your system. Although you can easily stop DFSMSrmm, if you have EDGSSSI in the subsystem name table DFRM entry, or have been using a running mode that supports tape recording, DFSMSrmm prevents all tape usage until it is restarted. To allow tapes to be used, you must restart DFSMSrmm, or remove it from the system.

The RACF FACILITY class profile STGADMIN.EDG.RESET.SSI controls the use of the RESET facility for removing DFSMSrmm from the system. Be sure to define this security profile and authorize the DFSMSrmm started task to reset the subsystem interface. Grant the DFSMSrmm started task procedure RACF user ID ALTER access authority to STGADMIN.EDG.RESET.SSI so you can shut down and remove the DFSMSrmm subsystem using the commands shown in Figure 246.

Note: You can use the RESET facility without defining this profile when you do not have a security product installed.

```
S DFRMM,OPT=RESET
P DFRMM
```

Figure 246. Disabling the DFSMSrmm Subsystem Interface

After you cut over to production, you should remove the the DFSMSrmm started task procedure from the access list for the resource STGADMIN.EDG.RESET.SSI.

A.3.7 Protecting DFSMSdss Resources

You can use EDGHSKP or EDGBKUP with DFSMSdss DUMP, or the access method services (AMS) REPRO command, to create a backup copy of your DFSMSrmm CDS and journal data set. To use DFSMSdss concurrent copy, you must define or update the following profile in the RACF FACILITY class:

- STGADMIN.ADR.DUMP.CNCURRNT

Figure 247 shows a batch job to define the resource in the RACF FACILITY class.

```
//RACRDEF JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
//* Note
//* ===
//* To execute this job you need the RACF SPECIAL attribute
//* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//SYSTSIN DD *
/* ***** */
/* DEFINE RESOURCE STGADMIN.EDG.FORCE */
/* ***** */
RDEFINE FACILITY STGADMIN.ADR.DUMP.CNCURRNT -
          UACC(NONE) OWNER(SYS1 )
          PERMIT STGADMIN.ADR.DUMP.CNCURRNT-
          ID(EDGADMIN) ACC(READ) CLASS(FACILITY)
```

Figure 247. JCL to Define DFSMSdss RACF FACILITY Class Profile

A.3.8 Authorizing DFSMShsm to DFSMSrmm Resources

DFSMShsm can provide enhanced management functions for the tape volumes that DFSMShsm uses for each of its tape functions. DFSMSrmm treats DFSMShsm like any other tape user, not like a tape manager.

To run DFSMShsm with DFSMSrmm, DFSMS/MVS Hierarchical Storage Manager procedure names must be defined to RACF. Add the name of the DFSMShsm procedure to the RACF STARTED class or the started procedure table ICHRIN03. Use a DFSMShsm user ID other than the default user ID. Refer to *DFSMS/MVS DFSMShsm Storage Administrators Guide*, SH21-1076, for information about defining the DFSMShsm and ABARS procedure names.

Before you can use DFSMSrmm with DFSMShsm, you must authorize DFSMShsm to the following resources:

```
STGADMIN.EDG.MASTER
STGADMIN.EDG.RELEASE
STGADMIN.EDG.OWNER.hsmid
```

If you have multiple DFSMShsm user IDs, for example, in a multi-system environment, and any DFSMShsm ID can return tapes to scratch status or to the DFSMShsm tape pool, you must authorize each DFSMShsm user ID. Define a resource profile STGADMIN.EDG.OWNER.hsmid for each DFSMShsm user ID, and give the other DFSMShsm user IDs UPDATE access to it.

When using DFSMSrmm with DFSMShsm, DFSMShsm needs access to the following resources to use scratch tapes.

STGADMIN.EDG.MASTER	READ
STGADMIN.EDG.RELEASE	READ
STGADMIN.EDG.OWNER.hsmid	UPDATE

DFSMShsm needs access to the following resources to use a DFSMShsm-managed scratch tape pool:

STGADMIN.EDG.MASTER	UPDATE
STGADMIN.EDG.OWNER.hsmid	UPDATE

Figure 248 on page 579 shows a batch job to add the name of the DFSMShsm procedure in the STARTED class, and permits the DFSMShsm user ID to the required DFSMSrmm RACF FACILITY class profiles.

```

//RACFHSM JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* ****
/* Note
/* ====
/* To execute this job you need the RACF SPECIAL attribute
/* ****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
/* **** */
/* Create an entry in the dynamic STARTED procedure table for DFHSM */
/* **** */
RDEFINE STARTED DFHSM.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
/* **** */
/* Permit DFHSM to STGADMIN.EDG.nnnnn resources */
/* **** */
PERMIT STGADMIN.EDG.MASTER -
      ID(DFHSM) ACC(READ) CLASS(FACILITY)
PERMIT STGADMIN.EDG.RELEASE -
      ID(DFHSM) ACC(READ) CLASS(FACILITY)
PERMIT STGADMIN.EDG.OWNER.DFHSM1
      ID(DFHSM) ACC(UPDATE) CLASS(FACILITY)

```

Figure 248. JCL to Define DFSMSHsm to DFSMSrmm Resources

A.3.9 Authorizing ABARS to DFSMSrmm Resources

To run ABARS with DFSMSrmm, the ABARS procedure names must be defined to RACF. Add the name of the ABARS procedure to the RACF STARTED class or to the started procedure table ICHRIN03. Refer to the *DFSMS/MVS DFSMSHsm Storage Administrators Guide*, SH21-1076, for information about defining the DFSMSHsm and ABARS procedure names to RACF.

To use DFSMSrmm with ABARS, you must authorize ABARS to the following resources:

```

STGADMIN.EDG.MASTER
STGADMIN.EDG.RELEASE
STGADMIN.EDG.OWNER.abarsid

```

If you have multiple ABARS user IDs, for example, in a multi-system environment, and any ABARS ID can return tapes to scratch status, you must authorize each ABARS user ID. Define a RACF resource profile STGADMIN.EDG.OWNER.abarsid for each ABARS user ID and give the other ABARS user IDs UPDATE access to it. This allows one ABARS to release the tapes initially obtained from scratch by another ABARS procedure.

The ABARS user ID needs access to the following resources to use DFSMSrmm with ABARS:

STGADMIN.EDG.MASTER	READ
STGADMIN.EDG.RELEASE	READ
STGADMIN.EDG.OWNER.abarsid	UPDATE

Figure 249 shows a batch job to add the name of the ABARS procedure in the STARTED class and permit the ABARS user ID to the required DFSMSrmm RACF FACILITY class resources.

```
//RACFABAR JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
/* *****
/* Note
/* ====
/* To execute this job you need the RACF SPECIAL attribute
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSSTSIN DD *
/* ***** */
/* Create an entry in the dynamic STARTED procedure table for ABARS */
/* ***** */
RDEFINE STARTED ABARS.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
/* ***** */
/* Permit ABARS to STGADMIN.EDG.nnnnn resources */
/* ***** */
PERMIT STGADMIN.EDG.MASTER -
      ID(ABARS) ACC(READ) CLASS(FACILITY)
PERMIT STGADMIN.EDG.RELEASE -
      ID(ABARS) ACC(READ) CLASS(FACILITY)
PERMIT STGADMIN.EDG.OWNER.ABARS1
      ID(ABARS) ACC(UPDATE) CLASS(FACILITY)
```

Figure 249. JCL to Define ABARS to DFSMSrmm Resources

A.3.10 Authorizing EDGBKUP to DFSMSrmm Resources

You use the OPTION JOURNALFULL(xx) command in PARMLIB member EDGRMMnn, to define a percentage full threshold for the journal data set. When DFSMSrmm detects that the journal has reached this threshold, the procedure specified on the BACKUPPROC operand is started automatically. In this book we use the name EDGBKUP for the backup procedure name.

The EDGBKUP procedure requires access to the following resources to use the backup housekeeping function:

STGADMIN.EDG.HOUSEKEEP	READ
-------------------------------	------

Figure 250 on page 581 shows a batch job to add the EDGBKUP procedure in the STARTED class and permit it to the required RACF FACILITY class resources.

```

//RACFBKUP JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* *****
/* Note
/* ====
/* To execute this job you need the RACF SPECIAL attribute
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD  SYSOUT=*
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD   *
/* ***** */
/* Create an entry in the STARTED procedure table for EDGBKUP */
/* ***** */
RDEFINE STARTED EDGBKUP.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC      ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
/* ***** */
/* Permit EDGHSKP to STGADMIN.EDG.nnnnn resources */
/* ***** */
PERMIT STGADMIN.EDG.HOUSEKEEP -
      ID(EDGBKUP) ACC(READ) CLASS(FACILITY)
PERMIT 'RMM.PROD.MSGS' -
      ID(EDGBKUP) ACC(ALTER)
PERMIT 'RMM.HSKP.*' -
      ID(EDGBKUP) ACC(ALTER)
PERMIT STGADMIN.ADR.DUMP.CNCURRNT -
      ID(EDGBKUP) ACC(READ) CLASS(FACILITY)

```

Figure 250. JCL to Define EDGBKUP to DFSMSrmm Resources

A.3.11 Authorizing EDGXPROC to DFSMSrmm Resources

You use the OPTION SCRATCHPROC command in PARMLIB member EDGRMMnn, to specify the name of the procedure that you want to be started automatically to replenish scratch volumes in an automated tape library data server (IBM 3494 or IBM 3495). In this book we use the default procedure name of EDGXPROC by specifying SCRATCHPROC(EDGXPROC) in EDGRMMnn. A sample EDGXPROC procedure is shipped with DFSMSrmm.

When a manual or automatic tape library dataserver detects a low-on-scratch condition, where more scratch volumes are needed, OAM issues a write-to-operator message. DFSMSrmm intercepts the message and starts the procedure you specify with SCRATCHPROC value in PARMLIB.

The EDGXPROC procedure requires access to the following resources to use the expiration processing housekeeping function:

STGADMIN.EDG.HOUSEKEEP **READ**

Figure 251 on page 582 shows a batch job to add the EDGXPROC procedure in the STARTED class and permit it to the required RACF FACILITY class resources.

```

//RACFXPRC JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* ****
//* Note
//* ====
//* To execute this job you need the RACF SPECIAL attribute
//* ****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
/* **** */
/* Create an entry in the dynamic STARTED procedure table for ABARS */
/* **** */
RDEFINE STARTED EDGXPROC.* -
          STDATA(USER(=MEMBER ) -
                  GROUP(STC ) -
                  PRIVELEGED(YES) -
                  TRACE(YES) -
                  TRUSTED(YES) )
/* **** */
/* Permit EDGXPROC to STGADMIN.EDG.nnnnn resources */
/* **** */
PERMIT STGADMIN.EDG.HOUSEKEEP -
      ID(EDGXPROC) ACC(READ) CLASS(FACILITY)
PERMIT 'RMM.PROD.MSGS' -
      ID(EDGXPROC) ACC(ALTER)
PERMIT 'RMM.HSKP.*' -
      ID(EDGXPROC) ACC(ALTER)

```

Figure 251. JCL to Define EDGXPROC to DFSMSrmm Resources

A.3.12 Authorizing EDGLABEL to DFSMSrmm Resources

You can run the EDGINERS utility as an operator-started procedure, so that the operator or librarian can make requests for tape labeling and erasing. The name of the procedure can be any valid alphanumeric procedure name from 1 to 8 characters. In this book we use EDGLABEL as the procedure name.

To automatically initialize or erase tapes, all volumes must be defined in the DFSMSrmm CDS, and the requested action must be set. In the automatic processing mode, EDGINERS requests that DFSMSrmm scan its CDS for the first 10 volumes waiting to be initialized or erased. If no volumes are found, the procedure ends without an error.

Note: DFSMSrmm erases volumes using the hardware security erase feature, when supported.

The EDGLABEL procedure requires access to the following resources to use the expiration processing housekeeping function:

STGADMIN.EDG.OPERATOR UPDATE

Figure 252 on page 583 shows a batch job to add the EDGLABEL procedure in the STARTED class and permit it to the required RACF FACILITY class resources.

```

//RACFLBL JOB ,RMM,NOTIFY=SCHLUM,
//      MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
/* *****
/* Note
/* ===
/* To execute this job you need the RACF SPECIAL attribute
/* *****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD  SYSOUT=*
//SYSTSPRT DD  SYSOUT=*
//SYSTSIN DD   *
/* ***** */
/* Create an entry in the dynamic STARTED procedure table for LABEL */
/* ***** */
RDEFINE STARTED EDGLABEL.* -
      STDATA(USER(=MEMBER ) -
              GROUP(STC      ) -
              PRIVELEGED(YES) -
              TRACE(YES) -
              TRUSTED(YES) )
/* ***** */
/* Permit EDGLABEL to STGADMIN.EDG.nnnnn resources */
/* ***** */
PERMIT STGADMIN.EDG.OPERATOR -
      ID(EDGLABEL) ACC(READ) CLASS(FACILITY)

```

Figure 252. JCL to Define EDGLABEL to DFSMSrmm Resources

A.3.13 Final RACF Activities

After you have defined the DFSMSrmm RACF function groups, data set profiles, and resources in RACF FACILITY class, and assigned DFSMSrmm a user ID, you must refresh the RACF in-storage profiles.

Figure 253 on page 584 shows a batch job to REFRESH the RACF in-storage profiles.

```

//RACREFR JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//* ****
//* Caution
//* =====
//* To execute this job you need the RACF "SPECIAL" attribute
//* ****
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
/* **** */
/* FINAL ACTIVITIES */
/* **** */
    SETROPTS GENERIC(FACILITY) REFRESH
    SETROPTS RACLIST(FACILITY) REFRESH
    SETROPTS GENERIC(DATASET) REFRESH
    SETROPTS GENERIC(STARTED) REFRESH
    SETROPTS LIST
/*
//

```

Figure 253. JCL to REFRESH RACF In-Storage Profiles

Appendix B. Conversion Checklist

Use the checklist in Table 110 to track the activities to be performed to achieve a successful conversion.

<i>Table 110 (Page 1 of 4). Conversion Checklist</i>				
Task	Person	Plan Date	Actual Date	Elapsed Time
Education - learn DFSMSrmm				
Build a plan				
Assign responsibilities				
<i>Milestone 1A - INSTALLING THE DFSMSrmm PRODUCT</i>				
Install DFSMSrmm product through SMP/E				
Run DFRMM IVPs				
Authorize DFSMSrmm commands and utilities				
Define DFSMSrmm PARMLIB member EDGRMMxx				
CREATE DFSMSrmm initialization parameters (EDGRMM00)				
DFSMSrmm serialization (through GRS)				
Update SCHEDxx member to provide PARMLIB serialization				
Update DFSMS/MVS licensed components				
Update SYS1.PROCLIB member for starting DFRMM				
Turn off ARCTVEXT exit ¹				
If using ABARS, consider using ARCTDEXT				
Assign RACF UID to DFRMM				
Verify DFSMSrmm initialization parameters (EDGRMM00)				
JES3 considerations				
Authorize appropriate DFSMSrmm users using RACF facility classes				
Update PARMLIB member IKJTSOxx to authorize DFSMSrmm commands and utilities				
Update PARMLIB member SMFPRMxx if you want security or audit records				

<i>Table 110 (Page 2 of 4). Conversion Checklist</i>				
Task	Person	Plan Date	Actual Date	Elapsed Time
Add DFRMM to ISPF (to execute DFSMSrmm functions under ISPF)				
Enable the DFSMSrmm subsystem interface PARMLIB member IEFSSNxx				
<i>Milestone 1B - DATABASE CONVERSION</i>				
Vendor database cleanup				
Run TMSDATA to get VOLS and DSNBs for TMS/CA-1				
Run data extract tools				
Run EDGCNVT				
Postprocess lib and owners				
Sort and/or merge sequential CNVT output				
Allocate DFSMSrmm CDS and run REPRO for data records				
Verify new DFSMSrmm CDS				
Vendor reports analysis				
Develop new reports under DFSMSrmm				
<i>Milestone 1C - DFSMSrmm COMES UP IN RECORD MODE</i>				
Halt or quiesce all tape activity				
Shut down vendor product				
Run vendor utilities to clean up data base				
Run TMSDATA to dump TMC VOLS and DSNBs				
Run extract tools				
Run EDGCNVT				
Enable your DFSMSrmm user exits				
Refresh LLA				
Start DFSMSrmm with MODE(R)				
Start vendor product				
<i>Milestone 2A - DETERMINE EXITS NEEDED FOR DFSMSrmm</i>				
CBRUXCUA				
CBRUXEJC				
CBRUXENT				
OMODVOL1				
EDGUX100				

<i>Table 110 (Page 3 of 4). Conversion Checklist</i>				
Task	Person	Plan Date	Actual Date	Elapsed Time
IGXMSGEX				
DMS/OS USERS ONLY				
<i>Milestone 2B - RECORD MODE</i>				
Ensure all previous tasks completed				
Run IDCAMS CREATE to build CDS and allocate journal				
Shut down TMS temporarily (Sunday test shot)				
Use vendor utilities to clean up data base				
Recollect the data from the database				
Rerun all EXTRACTION and CONVERT programs				
Run EDGCVOL, final sort				
Run IDCAMS REPRO for data from final sort into the CDS built above				
Run EDGUTIL with PARM='CREATE'				
RUN EDGUTIL with PARM='VERIFY'				
Ensure that DFSMSrmm exits are in LINKLIB and LLA refreshed				
If VERIFY is clean, start up DFRMM(OPMODE R) or modify DFSMSrmm with updated PARMLIB member				
Restart CA-1 or open tape activity (S TMSINIT,PARM=SOFTIPL and then response U).				
Run test cases to test all functions				
Analyze any problems encountered				
<i>Milestone 3 - WARNING MODE</i>				
Validate all activity in record mode				
Run EDGUTIL,PARM='VERIFY(ALL)'				
Run EDGHSKP,PARM='BACKUP'				
Run VRSEL (vital record processing)				
RUN EDGHSKP,PARM='DSTORE'				

Table 110 (Page 4 of 4). Conversion Checklist

Task	Person	Plan Date	Actual Date	Elapsed Time
RUN EDGHSKP,PARM='EXPROC'				
Review output from all above				
Stop all tape activity				
Redo TMS data collection				
Rerun EXTRACTION and CONVERT tools				
Reload CDS and bring DFSMSrmm in warning mode				
Restart TMS				
Reenable tape activity				
Generate EDGAUD (security and audit report)				
Generate EDGRPTD (movement and inventory report)				
Generate scratch lists through ICETOOL, SAS, REXX, or similar tool				
Generate volumes pending release (ICETOOL, SAS, REXX)				
Modify exits and user modifications as appropriate				
Resubmit all DFSMSrmm tests cases				
<i>Milestone 4 - DFSMSrmm IN PROTECT MODE</i>				
Prepare for cutover to DFSMSrmm				
Present user education				
Verify operations staff comfort level with DFSMSrmm reports and facilities				
Plan for cutover timing (vendor product to be removed)				
Vendor product disabled				
If you use RACF FACILITY class, remove profile STGADMIN.EDG.RESET.SSI access				
Control/Check use of the resources STGADMIN.EDG.*				
Check the PARMLIB options UNCATALOG / TPRACF / EXPDTCHECK				
¹ Code DFSMSrmm exit (ARCTVEXT) if you have a release prior to DFSMS/MVS Version 1 Release 4.0. DFSMSrmm uses the DFHSM tape volume exit (ARCTVEXT) to manage DFHSM tapes. During parallel testing both CA-1 and DFSMSrmm code is called. A modified version (that calls both TMS and DFSMSrmm) must be built.				

Appendix C. Conversion Record Layouts

The IBM-written EDGCNVT program requires a sequential file as input and an optional file that includes customer-written policy and rules statements. This appendix provides details of the required record layouts for the sequential input file. There are five different records that describe input to EDGCNVT:

EDGCDREC	Maps the data set details record
EDGCEREC	Maps the empty shelf details record
EDGCKREC	Maps the policy details record
EDGCLREC	Maps the volume and first data set details record
EDGCOREC	Maps the owner details record

C.1 Record Layout Description

The description of the different five input records for the EDGCNVT program:

Record Type	Description
L	Library. An L-Record contains details of a single volume and the first data set on that volume.
D	Data set name. A D-Record contains details of a single data set. One D-Record is required for each additional data set on a volume. If you supply a D-Record for a file, all previous files on the same volume must be specified in D-Records that precede the record. The L- and D-Records have the following sequence requirements: • Library (L) records for a multivolume set of volumes must appear in ascending volume sequence order, interrupted only by: – Data set name (D) records, in ascending file sequence order, after each L-Record that is created for the volumes from which the data set originates. • The count of D-Records that follow after a nonscratch L-record must be one less than the count of data sets in the LVDSNCNT field of that L-Record, because each L-Record contains the first file's data set information.
E	Empty rack and bin. The E-Record contains details of a single, empty shelf location. It can describe either a rack number or a bin number.
O	Owner. The O-Record contains details of a single owner ID. The owner may own resources that are being converted, but it need not own any resources.
K	VRS. The K-Record contains details of a single policy definition.

C.2 EDGCDREC Record Layout

RMM CONVERSION DATA SET RECORD MAPPING

DFSMS/MVS 1.4.0 DFSMSrmm

PROPRIETARY V3 STATEMENT

Licensed Materials - Property Of IBM

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END PROPRIETARY V3 STATEMENT

FUNCTION : EDGCDREC MAPS THE CONVERSION DATA SET RECORD

LOCATION : N/A

METHOD OF ACCESS:

ASSEMBLER - USING ON RMMDREC

LA Rx,DRECAREA

USING RMMDREC,Rx

MVI DSNRID,C'D' Set D record type

LISTING : ASSEMBLER - SPECIFY LIST=YES ON THE MACRO

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
START OF RMMDREC					
0	(0)	STRUCTURE	192	CDREC	ZERO
0	(0)	CHARACTER		DREC	BOUNDARY ALIGN
0	(0)	SIGNED	4	DRECRDW	RESERVED FOR RDW
4	(4)	CHARACTER	1	DSNRID	(REQUIRED) DATA SET NAME RECORD ID
		11.. .1..		DSNRTYPE	'D' - DATASET NAME RECORD TYPE
5	(5)	CHARACTER	4	DDCRDATE	(REQUIRED) DSN CREATE DATE - YYYYDDDD
9	(9)	CHARACTER	4	DDCRTIME	DSN CREATE TIME - HHMMSSST
13	(D)	CHARACTER	44	DSNNAME	(REQUIRED) DATA SET NAME
57	(39)	BITSTRING	1	DSNTRTCH	(REQUIRED) RECORDING TECHNIQUE
					TRTCH
		1111		*	RESERVED
		 1...		DSNTIDRC	DATA is compacted
		1..		DSBTNCOM	DATA is not compacted
		11		*	RESERVED
58	(3A)	SIGNED	2	DSNFSEQ	(REQUIRED) FILE SEQ NUMBER ON VOLUME
60	(3C)	CHARACTER	6	DSNVOL	(REQUIRED) VOLSER HOLDING DATA SET
66	(42)	SIGNED	2	DSNVOLSQ	(REQUIRED) VOLSEQ WITHIN SET OF VOLS
68	(44)	CHARACTER	4	DSNUNITA	UNIT WHERE DS WAS WRITTEN
72	(48)	CHARACTER	4	DSNRECFM	(REQUIRED) DS RECORD FORMAT
76	(4C)	SIGNED	4	DSNLRECL	(REQUIRED) DS LOGICAL RECORD LENGTH
80	(50)	SIGNED	4	DSNBLKSZ	(REQUIRED) DS BLOCK SIZE
84	(54)	SIGNED	4	DSNBLKCT	(REQUIRED) DS BLOCK COUNT
88	(58)	SIGNED	4	DSNKBYS	(REQUIRED) DS SIZE IN KBYTES
92	(5C)	CHARACTER	8	DSNOWNER	(REQUIRED) DS OWNER NAME
100	(64)	SIGNED	2	DSNLFSEQ	(REQUIRED) LOGICAL FILE SEQ NUMBER
102	(66)	SIGNED	2	DSNSCLV	DATA SET SECURITY LEVEL
104	(68)	CHARACTER	4	DSLRLDATE	(REQUIRED) DATE LAST READ YYYYDDD
108	(6C)	CHARACTER	4	DSLWDATE	(REQUIRED) DATE LAST WRITTEN
					YYYYDDD
112	(70)	CHARACTER	8	DSVRSVAL	VRS MANAGEMENT VALUE
120	(78)	BITSTRING	1	DSFLAG	DSN FLAG BYTE
		1...		DSFLGCAT	DATA SET IS CATALOGED
		.111		*	RESERVED
		 1...		DSFLGABD	ABEND IN PROCESS WHEN DATA SET
					CLOSED
		111		*	RESERVED
121	(79)	CHARACTER	3	*	RESERVED
124	(7C)	CHARACTER	8	DDCRSID	CREATING SYSTEM ID

Offsets					
Dec	Hex	Type	Len	Name (Dim)	Description
132	(84)	CHARACTER	8	DSCJOB	DATA SET CREATING JOBNAME
140	(8C)	CHARACTER	8	DSNCSTEP	(COND) DSN1 Creating step name
148	(94)	CHARACTER	8	DSNCDDNM	(COND) DSN1 Creating DD name
156	(9C)	CHARACTER	4	DSRETDTE	(COND) DATA SET RETENTION DATE
160	(A0)	SIGNED	4	* (8)	RESERVED
192	(C0)	CHARACTER		CDRECEND	END OF CDREC

C.2.1.1 Constants

Len	Type	Value	Name	Description
1	CHARACTER	D	DSNRTYPE	'D' - DATASET RECORD TYPE

C.2.1.2 Cross Reference

Name	Hex Offset	Hex Value	Level
CDREC	0		1
CDRECEND	C0		2
DDCRDATE	5		2
DDCRSID	7C		2
DDCRTIME	9		2
DREC	0		2
DRECRDW	0		2
DSBTNCOM	39	40	3
DSCJOB	84		2
DSFLAG	78		2
DSFLGABD	78	08	3
DSFLGCAT	78	80	3
DSLRLDATE	68		2
DSLWDATE	6C		2
DSNBLKCT	54		2
DSNBLKSZ	50		2
DSNCDDNM	94		2
DSNCSTEP	8C		2
DSNFSEQ	3A		2
DSNKBYTS	58		2
DSNLFSEQ	64		2
DSNLRECL	4C		2
DSNNAME	D		2
DSNOWNER	5C		2
DSNRECFM	48		2
DSNRID	4		2
DSNSCLV	66		2
DSNTIDRC	39	80	3
DSNTRTCH	39		2
DSNUNITA	44		2
DSNVOL	3C		2
DSNVOLSQ	42		2
DSRETDTE	9C		2
DSVRSVAL	70		2

C.3 EDGCEREC Record Layout

RMM CONVERSION EMPTY SHELF RECORD MAPPING

DFSMS/MVS 1.4.0 DFSMSrmm

PROPRIETARY V3 STATEMENT

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FUNCTION : EDGCEREC MAPS THE CONVERSION EMPTY SHELF RECORD

LOCATION : N/A

METHOD OF ACCESS:

ASSEMBLER - USING ON RMMEREC

LA Rx,ERECAREA

USING RMMEREC,Rx

MVI ESRID,C'E' Set E record type

LISTING : ASSEMBLER - SPECIFY LIST=YES ON THE MACRO

Offsets					
Dec	Hex	Type	Len	Name (Dim)	Description
START OF RMMEREC					
0	(0)	STRUCTURE	68	CEREC	0
0	(0)	CHARACTER		EREC	BOUNDARY ALIGN
0	(0)	SIGNED	4	ERECRDW	RESERVED FOR RDW
4	(4)	CHARACTER	1	ESRID	(REQUIRED) EMPTY RACK RECORD ID
		11.. .1.1		ESRTYPE	'E' - EMPTY RACK RECORD TYPE
5	(5)	CHARACTER	1	ESTYPE	(REQUIRED) SHELF TYPE
		11..1 ..1.		EBSHELF	'B' - BIN TYPE LOCATION RECCORD
		11.1 1..1		ERSHELF	'R' - RACK TYPE SHELF RECCORD
6	(6)	CHARACTER	8	ESLOCNAM	(REQUIRED) NAME OF ASSOCIATED LIBRY
14	(E)	CHARACTER	6	ESSTART	(REQUIRED) FIRST OF EMPTY SHELVES
20	(14)	CHARACTER	6	ESCOUNT	(REQUIRED) N-1 OF CONTIGUOUS SHELVES
26	(1A)	CHARACTER	8	EVMEDIA	(REQUIRED) MEDIA NAME
34	(22)	SIGNED	2	*	RESERVED
36	(24)	SIGNED	4	* (8)	RESERVED
68	(44)	CHARACTER		CNVEEND	END OF CNVEREC

C.3.1.1 Constants

Len	Type	Value	Name	Description
1	CHARACTER	E	ESRTYPE	'E' - EMPTY RACK RECORD
1	CHARACTER	R	ERSHELF	'R' - RACK TYPE SHELF
1	CHARACTER	B	EBSHELF	

C.3.1.2 Cross Reference

Name	Hex Offset	Hex Value	Level
CEREC	0		1
CNVEEND	44		2
EREC	0		2
ERECRDW	0		2
ESCOUNT	14		2
ESLOCNAM	6		2
ESRID	4		2
ESSTART	E		2
ESTYPE	5		2
EVMEDIA	1A		2

C.4 EDGCKREC Record Layout

RMM CONVERSION VRS RECORD MAPPING

DFSMS/MVS 1.4.0 DFSMSrmm

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END PROPRIETARY V3 STATEMENT

FUNCTION : EDGCKREC MAPS THE CONVERSION VRS RECORD

LOCATION : N/A

METHOD OF ACCESS:

ASSEMBLER - USING ON RMMKREC

LA Rx,KRECAREA

USING RMMKREC,Rx

MVI VRSRID,C'K' Set K record type

LISTING : ASSEMBLER - SPECIFY LIST=YES ON THE MACRO

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
START OF RMMKREC					
0	(0)	STRUCTURE	184	CKREC	0
0	(0)	CHARACTER		KREC	BOUNDARY ALIGN
0	(0)	SIGNED	4	KRECRDW	RESERVED FOR RDW
4	(4)	CHARACTER	1	VRSRID	(REQUIRED) VRS RECORD ID
		11.1 ..1.		VRSRTYPE	'K' - VITAL RECORDS RECORD TYPE
5	(5)	CHARACTER	4	VCRDATE	VRS CREATE DATE - YYYYDDD
9	(9)	CHARACTER	4	VCRTIME	VRS CREATE TIME - HHMSST
13	(D)	CHARACTER	1	VTYPEID	(REQUIRED) VRS TYPE
1		11.. .1..	D	VTYPED	'D' - DATASET TYPE VRS
1		111. .1.1	N	VTYPEN	'N' - NAME TYPE VRS
1		111. .1.1	V	VTYPEV	'V' - VOLUME TYPE VRS
14	(E)	CHARACTER	44	VDSNAME	(COND) DATASET NAME
58	(3A)	BITSTRING	1	VDSNTYPE	(COND) FORMAT OF DATA SET NAME
		1...		VDSNGDG	GDG FORMAT DSN
		.1..		*	
		..1.		VDSNSTD	STD FORMAT DSN
		...1 1111		*	
59	(3B)	BITSTRING	1	VRETN	VRS RETENTION TYPE
		1...		VRETNCYC	RETAIN BY CYCLES
		.1..		VRETNELP	RETAIN BY ELAPSED DAYS
		..1.		VRETNLRF	RETAIN BY DAYS SINCE REF

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
		...1		VRETNCAT	RETAIN WHILST CATALOGUED
		 1...		VRETNUEX	RETAIN UNTIL EXPIERED
		1..		VRETNXD	EXTRADAYS
		1.		VRETNBDC	BYDAYSCYCLE
		1		*	
60	(3C)	CHARACTER	6	VVOLSER	(COND) VOLUME SERIAL NUMBER
66	(42)	SIGNED	2	VDELAY	(REQUIRED) DAYS DELAY BEFORE MOVEMNT
68	(44)	SIGNED	4	VRCOUNT	(REQUIRED) VITAL RECORDS NUMBER. NUMBER OF CYCLES OR DAYS FOR VOLUME TO BE KEPT.
72	(48)	CHARACTER	8	VRSNAME	(COND) VRS NAME
80	(50)	SIGNED	4	VSTRKEEP	(REQUIRED) STORE KEEP NUMBER. NUMBER OF CYCLES OR DAYS OF VOLUME TO BE KEPT IN STORE
84	(54)	CHARACTER	8	VOWNER	(REQUIRED) VRS OWNER ID
92	(5C)	CHARACTER	4	VDELDATE	AUTO-DELETE DATE -YYYYDDD
96	(60)	CHARACTER	30	VDESC	VRS DESCRIPTION
126	(7E)	CHARACTER	8	VLOCNAM	(REQUIRED) LOCATION NAME
134	(86)	CHARACTER	8	VRSNEXT	(COND) NAME OF CHAINED VRS
142	(8E)	CHARACTER	8	VRSJOB	(COND) NAME OF CREATING JOB
150	(96)	BITSTRING	1	VRSOPTS	OPTIONS BYTE
		1...		VANDVRS	VRSNEXT IS ANDVRS()
		.1..		VRELIXD	EXPIRY DATE IGNORE
		..1.		VRELSI	SCRATCH IMMEDIATE
		...1 1111		*	RESERVED
151	(97)	CHARACTER	1	*	RESERVED
152	(98)	SIGNED	4	* (8)	RESERVED
184	(B8)	CHARACTER		CKRECEND	END OF CNVKREC

C.4.1.1 Constants

Len	Type	Value	Name	Description
1	CHARACTER	K	VRSRTYPE	'K' - VRS RECORD TYPE
1	CHARACTER	D	VTYPED	'D' - DATASET TYPE VRS
1	CHARACTER	N	VTYPEN	'N' - NAME TYPE VRS
1	CHARACTER	V	VTYPEV	'V' - VOLUME TYPE VRS

C.4.1.2 Cross Reference

Name	Hex Offset	Hex Value	Level
CKREC	0		1
CKRECEND	B8		2
KREC	0		2
KRECRDW	0		2
VANDVRS	96	80	3
VCRDATE	5		2
VCRTIME	9		2
VDELAY	42		2
VDELDATE	5C		2
VDESC	60		2
VDSNAME	E		2
VDSNGDG	3A	80	3
VDSNSTD	3A	10	3
VDSNTYPE	3A		2
VLOCNAM	7E		2
VOWNER	54		2
VRCOUNT	44		2
VRELIXD	96	40	3
VRELSI	96	20	3
VRETN	3B		2
VRETNBDC	3B	02	3
VRETNCAT	3B	10	3

Name	Hex Offset	Hex Value	Level
VRETNCYC	3B	80	3
VRETNELP	3B	40	3
VRETNLRF	3B	20	3
VRETNUEX	3B	08	3
VRETNXD	3B	04	3
VRSJOB	8E		2
VRSNAME	48		2
VRSNEXT	86		2
VRSOPTS	96		2
VRSRID	4		2
VSTRKEEP	50		2
VTYPERID	D		2
VVOLSER	3C		2

C.5 EDGCLREC Record Layout

RMM CONVERSION LIBRARY RECORD MAPPING

DFSMS/MVS 1.4.0 DFSMSrmm

PROPRIETARY V3 STATEMENT

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END PROPRIETARY V3 STATEMENT

FUNCTION : EDGCLREC MAPS THE CONVERSION LIBRARY RECORD

LOCATION : N/A

METHOD OF ACCESS:

ASSEMBLER - USING ON RMMLREC

LA Rx,LRECAREA

USING RMMLREC,Rx

MVI LIBRID,C'L' Set L record type

LISTING : ASSEMBLER - SPECIFY LIST=YES ON THE MACRO

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
START OF RMMLREC					
0	(0)	STRUCTURE	580	CLREC	0
0	(0)	CHARACTER		LREC	RESERVED FOR INFORMATION
0	(0)	SIGNED	4	LRECRDW	RECORD DESCRIPTOR WORD
4	(4)	CHARACTER	1	LIBRID	(REQUIRED) LIBRARY RECORD ID
		11.1 ..11		LVTYPE	'L' - LIBRARY RECORD TYPE
5	(5)	CHARACTER	6	LVOLSER	(REQUIRED) VOLUME SERIAL NUMBER
11	(B)	CHARACTER	8	LVOWNER	(REQUIRED) VOLUME OWNER
19	(13)	CHARACTER	8	LVOUID	VOLUME NOTIFY USERID
27	(1B)	CHARACTER	8	LVONODE	VOLUME NOTIFY NODE
35	(23)	CHARACTER	1	LRESRVA	RESERVED
36	(24)	SIGNED	2	LVSECLEV	VOLUME SECURITY LEVEL
38	(26)	SIGNED	2	LVSEQNO	(REQUIRED) VOLUME SEQUENCE NUMBER
40	(28)	CHARACTER	4	LVCRCDATE	(REQUIRED) VOLUME CREATE DATE - YYYYDDD
44	(2C)	CHARACTER	4	LVCRTIME	VOLUME CREATE TIME - HHMMSS
48	(30)	CHARACTER	8	LVCRCUID	VOLUME CREATE USERID
56	(38)	CHARACTER	8	LVCRCJOB	VOLUME CREATE JOBNAME
64	(40)	BITSTRING	4	LVUCBTYP	CUPY OF UCBTYP
68	(44)	BITSTRING	4	LVTDSI	(COND) TAPE MEDIA TYPE INFORMATION
68	(44)	BITSTRING	1	LVTDSRC	TAPE RECORDING FORMAT.
		1111 11..		*	ONE OF: X'00' - NOT 3480

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
69	(45)	1.	1	LVM36TRK	X'02' - 36 TRACK
		1		LVM18TRK	X'01' - 18 TRACK
		1111 11..		LVTDSY	TAPE MEDIA TYPE. ONE OF:
		1.		*	X'00' - NOT 3480
70	(46)	1.	1	LV MENHCP	X'02' - ENHANCED CAPACITY
		1		LVMSTDCP	X'01' - STANDARD CAPACITY
		1111 11..		LVTDSCM	TAPE COMPACTION TECHNIQUE
		1.		*	ONE OF: X'00' - UNKNOWN
71	(47)	1.	1	LV MIDRC	X'02' - IDRC
		1		LV MNCOMP	X'01' - NOT COMPACT
		1111 11..		LVTDSSA	TAPE SPECIAL ATTRIBUTES.
		1		*	ONE OF: X'00' - NONE
72	(48)	SIGNED	4	LVM18TRD	X'01' - 18 TRACK READ ONL
76	(4C)	CHARACTER	4	LVTUSE	TAPE USAGE IN K-BYTES
80	(50)	CHARACTER	4	LVEXDATO	(REQUIRED) JCL VOL EXPIRATION DATE
84	(54)	CHARACTER	4	LVEXDATE	(REQUIRED) VOLUME EXPIRATION DATE
88	(58)	SIGNED	2	LVRETDTE	(COND) VOLUME RETENTION DATE
90	(5A)	SIGNED	2	LVDSNCNT	(REQUIRED) COUNT OF DS W/ ORIG ONVOL
92	(5C)	SIGNED	2	LVTRERR	TEMPORARY READ ERRORS
94	(5E)	SIGNED	2	LVTWERR	TEMPORARY WRITE ERRORS
96	(60)	SIGNED	2	LVPRERR	PERMANENT READ ERRORS
98	(62)	SIGNED	2	LV PWERR	PERMANENT WRITE ERRORS
99	(63)	BITSTRING	1	LVRDEN	COPY OF JFCBDEN
100	(64)	CHARACTER	1	LBPIIND	(REQUIRED) BPI INDICATOR
		1111 ..11	1	DSNRTYPE	'3' - 1600BPI
		1111 ..1.		DSNRTYPE	'4' - 6250BPI
		1111 1..1		DSNRTYPE	'9' - 3480
		11.. ..11		DSNRTYPE	'C' - 3480 COMPACT (IDRC)
		..1.1. 11..		DSNRTYPE	'*' - UNDEFINED
		1...		LVFLAGA	(REQUIRED) FLAGS 'A' - STATUS
		..1.		LVMSTFLG	VOLUME IS MASTER
		..1.		LVRLSFLG	VOLUME PENDING RELEASE
		...1		LVVRFLG	VITAL RECORD - DO NOT RELEASE
		 1...		LVASSFLG	USER TAPE (ASSIGNED BY LIB)
101	(65)	 1..	1	LV LONFLG	TAPE IS ON LOAN
		1.		LVOPNFLG	TAPE OPENED AND NOT YET CLOSED
		1.		LVSCRFLG	VOLUME IS SCRATCH
		1		LVOCEFLG	VOLUME RECORDED BY O/C/EOV
		1...		LVFLGAX	(REQUIRED) FLAGS 'A' - STATUS
		..1.		LVGVCFLG	EXTENSION
		..1.		LVXINFLG	SCRATCH VOL CLAIMED VIA GETVOL
		..1.		LVINIFLG	SCRATCH VOL NEVER INITIALIZED
		...1		LVENTFLG	SCRATCH VOL W/INIT ACTION PEND
		 1...		LVFABEND	SCRATCH VOL WAIT TO ENTER ATL
102	(66)	1.	1	LVFOCEAB	ABEND IN PROCESS WHEN DS
		1...		LVFLAGB	CLOSED@02A
		..1.		LVDEFRET	ABEND PROBABLY IN O/C/EOV
		 1..		*	(REQUIRED) FLAGS ARE 0 OR 1
		...1		LVNLTAPE	DEF. RETEN. PERIOD USED
		 1...		LVALTAPE	LABEL TYPE IS NL
		1.		LVSLTAPE	LABEL TYPE IS AL
		1.		*	LABEL TYPE IS SL
		1.		LVBLTAPE	TAPE LAST WRITTEN USING BLP
		1		LVULTAPE	SL OR AL W/USER LABELS
103	(67)	1...	1	LVFLAGC	(REQUIRED) FLAGS 'C' - RELEASE ACTIONS
		..1.		LVRETSCR	RETRN TO SCRATCH POOL - DEFLT
		..1.		LVREPREL	REPLACE TAPE ON RELEASE
		...1		LVREINIT	REINITIALISE
		 1...		LVDEGAUS	DEGAUS/SECURITY ERASE
		1.		LVROWNER	RETURN TO OWNER
104	(68)	1.	1	LVNOWNER	NOTIFY OWNER
		1...		LVFLAGD	(REQUIRED) FLAGS 'D' - ACCESS
		..1.		LVOREAD	OWNER MAY READ VOLUME
		..1.		LVOUPD	OWNER MAY UPDATE VOLUME
		...1		LVPROTR	OWNER MAY ALTER VOLUME
					READ-ONLY PROTECTION

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
		 1...		LVPROTU	UPDATE PROTECTION
		1..		LVMVSUSE	MAY BE USED ON MVS SYSTEMS
		1.		LVVMUSE	MAY BE USED ON VM SYSTEMS
105	(69)	CHARACTER	8	LVMEDIA	MEDIA NAME
113	(71)	CHARACTER	4	LVUNITAD	UNIT ADDRESS USED
117	(75)	CHARACTER	4	LVLSTUAD	LAST UNIT ADDRESS USED
121	(79)	CHARACTER	8	LVLONLOC	LOAN LOCATION
129	(81)	CHARACTER	4	LVLRRDAT	DATE VOLUME LAST READ (YYYYDDD)@02A
133	(85)	CHARACTER	4	LVLWTDAT	DATE VOLUME LAST WRIT (YYYYDDD)@02A
137	(89)	CHARACTER	4	LVASDATE	ASSIGN FROM SCR DATE (YYYYDDD)
141	(8D)	CHARACTER	4	LVASTIME	ASSIGN FROM SCR TIME (HHMMSS)
145	(91)	BITSTRING	1	LVLTYTYP	COPY OF JFCBLTYP
146	(92)	BITSTRING	1	LVTRTCH	(REQUIRED) FROM JFCTRCH - IDRC SUPPORT
		1111		*	
		 1...		LVTCOMP	DSN USED 3480 IDRC
		1..		LVTNCOMP	NO COMPACTION
147	(93)	CHARACTER	1	LRESRVB	RESERVED
148	(94)	CHARACTER	8	LVLOCNAM	(REQUIRED) CURRENT LOCATION OF VOLUME
156	(9C)	CHARACTER	8	LVDEST	DESTINATION NAME
164	(A4)	CHARACTER	8	LVHLOC	HOME LOCATION NAME
172	(AC)	CHARACTER	8	LVSGNAME	STORAGE GROUP NAME
180	(B4)	CHARACTER	40	LACCINF	ACCOUNTING INFORMATION
220	(DC)	CHARACTER	30	LDESC	USER DESCRIPTION
250	(FA)	CHARACTER	96	LAUTHIDS	AUTHORISED USER IDS
250	(FA)	CHARACTER	8	LAUTHID1	1ST USERID SLOT
258	(102)	CHARACTER	8	LAUTHID2	2ND USERID SLOT
266	(10A)	CHARACTER	8	LAUTHID3	3RD USERID SLOT
274	(112)	CHARACTER	8	LAUTHID4	4TH USERID SLOT
282	(11A)	CHARACTER	8	LAUTHID5	5TH USERID SLOT
290	(122)	CHARACTER	8	LAUTHID6	6TH USERID SLOT
298	(12A)	CHARACTER	8	LAUTHID7	7TH USERID SLOT
306	(132)	CHARACTER	8	LAUTHID8	8TH USERID SLOT
314	(13A)	CHARACTER	8	LAUTHID9	9TH USERID SLOT
322	(142)	CHARACTER	8	LAUTHIDA	10TH USERID SLOT
330	(14A)	CHARACTER	8	LAUTHIDB	11TH USERID SLOT
338	(152)	CHARACTER	8	LAUTHIDC	12TH USERID SLOT
346	(15A)	CHARACTER	6	LRACKNO	(REQUIRED) RACK NUMBER
352	(160)	CHARACTER	6	LRSEBIN	(COND) REMOTE STORE BIN NUMBER
358	(166)	CHARACTER	4	LRSDATE	(COND) REMOTE STORE DATE (YYYYDDD)
362	(16A)	CHARACTER	8	LRBINMDN	(COND) BIN MEDIA NAME
370	(172)	SIGNED	2	LLABNO1	(REQUIRED) LABEL NUMBER OF FIRST FILE
372	(174)	CHARACTER	44	LDS1NAM	NAME OF 1ST DATASET ON VOLUME
416	(1A0)	CHARACTER	8	LDS1OWNR	DATASET OWNER
424	(1A8)	SIGNED	2	LDS1SEQ	(COND) SEQ NO OF DATA SET ONE
426	(1AA)	CHARACTER	4	LDS1RECF	(COND) FIRST DATA SET'S RECFM
430	(1AE)	CHARACTER	2	*	ALIGNMENT
432	(1B0)	SIGNED	4	LDS1LREC	(COND) FIRST DATA SET'S LRECL
436	(1B4)	SIGNED	4	LDS1BSIZ	(COND) FIRST DATA SET'S BLKSZ
440	(1B8)	SIGNED	4	LDS1BLKS	(COND) FIRST DATA SET'S BLK COUNT
444	(1BC)	SIGNED	4	LDS1KBYT	(COND) FIRST DATA SET'S K-BYTES
448	(1C0)	SIGNED	2	LDS1SCLV	FIRST DATA SET SECURITY LEVEL
450	(1C2)	CHARACTER	4	LDCRDATE	(REQUIRED) DSN1 CREATE DATE YYYYDDD
454	(1C6)	CHARACTER	4	LDCRTIME	DSN1 CREATE TIME - HHMMSS
458	(1CA)	CHARACTER	4	LDLRDATE	(COND) DSN1 DATE LAST READ - YYYYDDD
462	(1CE)	CHARACTER	4	LDLWDATE	(COND) DSN1 DATE LAST WRIT - YYYYDDD
466	(1D2)	BITSTRING	1	LDS1FLAG	DSN1 FLAG BYTE
		1...		LDS1FCAT	DATA SET IS CATALOGED
		.111		*	RESERVED
		 1...		LDS1FABD	ABEND IN PROCESS WHEN DS CLOS
467	(1D3)	CHARACTER	8	LDVRSVAL	(COND) VRS MANAGEMENT VALUE
475	(1DB)	CHARACTER	8	LDS1STEP	(COND) DSN1 Creating step name
483	(1E3)	CHARACTER	8	LDS1DDNM	(COND) DSN1 Creating DD name
491	(1EB)	CHARACTER	4	LDRETDTE	(COND) DATA SET RETENTION DATE

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
495	(1EF)	CHARACTER	44	LDSXNAME	(COND) NAME OF LAST DATASET ON VOL
RESERVED (OF)					
539	(21B)	CHARACTER	8	LVCRSID	CREATING SYSTEM ID
547	(223)	BITSTRING	1	*	RESERVED
548	(224)	SIGNED	4	* (8)	RESERVED
580	(244)	CHARACTER		CLRECEND	END OF CNVLREC

C.5.1.1 Constants

Len	Type	Value	Name	Description
1	CHARACTER	L	LVTYP	'L' - LIBRARY RECORD TYPE
1	CHARACTER	3	BPI1600	'3' - 1600BPI
1	CHARACTER	4	BPI6250	'4' - 6250BPI
1	CHARACTER	9	BPI3480	'9' - 3480
1	CHARACTER	C	BPI348C	'C' - 3480 COMPACT (IDRC)
1	CHARACTER	*	BPIUNDEF	'*' - UNDEFINED

C.5.1.2 Cross Reference

Name	Hex Offset	Hex Value	Level
CLREC	0		1
CLRECEND	244		2
LACCINF	B4		2
LAUTHIDA	142		3
LAUTHIDB	14A		3
LAUTHIDC	152		3
LAUTHIDS	FA		2
LAUTHID1	FA		3
LAUTHID2	102		3
LAUTHID3	10A		3
LAUTHID4	112		3
LAUTHID5	11A		3
LAUTHID6	122		3
LAUTHID7	12A		3
LAUTHID8	132		3
LAUTHID9	13A		3
LBPIIND	63		2
LDCRDATE	1C2		2
LDCRTIME	1C6		2
LDESC	DC		2
LDLRDATE	1CA		2
LDLWDATE	1CE		2
LDRETDTE	1EB		2
LDSXNAME	1EF		2
LDS1BLKS	1B8		2
LDS1BSIZ	1B4		2
LDS1DDNM	1E3		2
LDS1FABD	1D2	08	3
LDS1FCAT	1D2	80	3
LDS1FLAG	1D2		2
LDS1KBYT	1BC		2
LDS1LREC	1B0		2
LDS1NAM	174		2
LDS1OWNR	1A0		2
LDS1RECF	1AA		2
LDS1SCLV	1C0		2
LDS1SEQ	1A8		2
LDS1STEP	1DB		2
LDVRSVAL	1D3		2
LIBRID	4		2
LLABNO1	172		2
LRACKNO	15A		2

Name	Hex Offset	Hex Value	Level
LRBINMDN	16A		2
LREC	0		2
LRECRDW	0		2
LRESRVA	23		2
LRESRVB	93		2
LRSBIN	160		2
LRSDATE	166		2
LVALTAPE	66	10	3
LVASDATE	89		2
LVASSFLG	64	10	3
LVASTIME	8D		2
LVBLTAPE	66	02	3
LVCRCRDATE	28		2
LVCRCRJOB	38		2
LVCRCRSID	21B		2
LVCRCRTIME	2C		2
LVCRCRUID	30		2
LVDEFRET	66	80	3
LVDEGAUS	67	10	3
LVDEST	9C		2
LVDSNCNT	58		2
LVENTFLG	65	10	3
LVEXDATE	50		2
LVEXDATO	4C		2
LVFABEND	65	08	3
LVFLAGA	64		2
LVFLAGB	66		2
LVFLAGC	67		2
LVFLAGD	68		2
LVFLAGX	65		2
LVFOCEAB	65	04	3
LVGVCFLG	65	80	3
LVHLOC	A4		2
LVINIFLG	65	20	3
LVLOCNAM	94		2
LVLONFLG	64	08	3
LVLONLOC	79		2
LVLRDDAT	81		2
LVLSTUAD	75		2
LVLSTYP	91		2
LVLWTDAT	85		2
LVMEDIA	69		2
LVMENHCP	45	02	4
LVMIDRC	46	02	4
LVMNCOMP	46	01	4
LVMSTDCP	45	01	4
LVMSTFLG	64	80	3
LVMVSUSE	68	04	3
LVM18TRD	47	01	4
LVM18TRK	44	01	4
LVM36TRK	44	02	4
LVNLTAPE	66	20	3
LVNOWNER	67	04	3
LVOALT	68	20	3
LVOCEFLG	64	01	3
LVOLSER	5		2
LVONODE	1B		2
LVOPNFLG	64	04	3
LVOREAD	68	80	3
LVOUID	13		2
LVOUNPD	68	40	3
LVOWNER	B		2
LVPRERR	5E		2
LVPROTR	68	10	3
LVPROTU	68	08	3
LVPWERR	60		2
LVRDEN	62		2
LVREINIT	67	20	3

Name	Hex Offset	Hex Value	Level
LVREPREL	67	40	3
LVRETDTE	54		2
LVRETSCR	67	80	3
LVRLSFLG	64	40	3
LVROWNER	67	08	3
LVSCRFLG	64	02	3
LVSECLEV	24		2
LVSEQNO	26		2
LVSGNAME	AC		2
LVSLTAPE	66	08	3
LVTCOMP	92	08	3
LVTDSCM	46		3
LVTDSI	44		2
LVTDSRC	44		3
LVTDSSA	47		3
LVTDSTY	45		3
LVTNCOMP	92	04	3
LVTRERR	5A		2
LVRTTCH	92		2
LVTUSE	48		2
LVTWERR	5C		2
LVUCBTYP	40		2
LVULTAPE	66	01	3
LVUNITAD	71		2
LVMUSE	68	02	3
LVVRFLG	64	20	3
LVXINFLG	65	40	3

C.6 EDGCOREC Record Layout

RMM CONVERSION OWNER RECORD MAPPING

DFSMS/MVS 1.4.0 DFSMSrmm

PROPRIETARY V3 STATEMENT

Licensed Materials - Property Of IBM

"Restricted Materials of IBM"

5695-DF1 / 5645-001

(C) COPYRIGHT 1993,1996 IBM CORP.

END PROPRIETARY V3 STATEMENT

FUNCTION : EDGCOREC MAPS THE CONVERSION OWNER RECORD

LOCATION : N/A

METHOD OF ACCESS:

ASSEMBLER - USING ON RMMOREC

LA Rx,ORECAREA

USING RMMOREC,Rx

MVI OWNERID,C'O' Set O record type

LISTING : ASSEMBLER - SPECIFY LIST=YES ON THE MACRO

Offsets		Type	Len	Name (Dim)	Description
Dec	Hex				
START OF RMMOREC					
0	(0)	STRUCTURE	292	COREC	0
0	(0)	CHARACTER		OREC	COREC INFORMATION
0	(0)	SIGNED	4	ORECRDW	RDW
4	(4)	CHARACTER	1	OWNERRID	OWNER'S NAME RECORD ID
		11.1 .11.		OWNRRTYPE	'O' - OWNER RECORD TYPE
5	(5)	CHARACTER	8	OWNER	OWNER'S NAME
13	(D)	CHARACTER	20	OWNSNAME	OWNER'S SURNAME

Offsets					
Dec	Hex	Type	Len	Name (Dim)	Description
33	(21)	CHARACTER	20	OWNFNAME	OWNER'S FIRST NAME
53	(35)	CHARACTER	40	OWNDEPT	(REQUIRED) OWNER'S DEPARTMENT
93	(5D)	CHARACTER	40	OWNADDR1	OWNER'S 1ST ADDRESS LINE
133	(85)	CHARACTER	40	OWNADDR2	OWNER'S 2ND ADDRESS LINE
173	(AD)	CHARACTER	40	OWNADDR3	OWNER'S 3RD ADDRESS LINE
213	(D5)	CHARACTER	8	OWNINTEL	OWNER'S INTERNAL TELE NO.
221	(DD)	CHARACTER	20	OWNEXTTEL	OWNER'S EXTERNAL TELE NO.
241	(F1)	CHARACTER	8	OWNUID	OWNER'S USERID
249	(F9)	CHARACTER	8	OWNNODE	OWNER'S NODE NAME
257	(101)	CHARACTER	3	OWNPAD1	PADDING
260	(104)	SIGNED	4	* (8)	RESERVED
292	(124)	CHARACTER		CNVOEND	END OF CNVOREC

C.6.1.1 Constants

Len	Type	Value	Name	Description
1	CHARACTER	O	OWNRTYPE	'O' - OWNER RECORD TYPE

C.6.1.2 Cross Reference

Name	Hex Offset	Hex Value	Level
CNVOEND	124		2
COREC	0		1
OREC	0		2
ORECRDW	0		2
OWNADDR1	5D		2
OWNADDR2	85		2
OWNADDR3	AD		2
OWNDEPT	35		2
OWNER	5		2
OWNERRID	4		2
OWNEXTTEL	DD		2
OWNFNAME	21		2
OWNINTEL	D5		2
OWNNODE	F9		2
OWNPAD1	101		2
OWNSNAME	D		2
OWNUID	F1		2

Appendix D. Sample Exits and Reports

You can use the samples in this appendix to perform the function described, or you can modify the samples to meet your specific needs.

This appendix contains the samples listed in Table 111.

Table 111. Sample Exits and Reports		
Name	Type	Reference
ARCTVEXT	Exit	D.1.1, "Sample DFSMSHsm ARCTVEXT Router Exit"
CBRUXENT	Exit	D.1.2, "Sample CA-1 - DFSMSrmm Merged ARCTVEXT"
BTLSRMM	JCL	D.2.1, "Synchronize BTLS and DFSMSrmm CDS Status"
EDGHSKP	JCL	D.2.2, "EDGHSKP Inventory Management example JCL"
EDGJRPT	JCL	D.2.3, "Sample JCL for Reports"
RMMCVB	Report	D.3.1.1, "Sample CHANGEVOLUME Command: Job RMMCVB"
RMMAUDIB	Report	D.3.2.1, "Archiving Audit Reports: Job RMMAUDIB"
RMMAUD2B	Report	D.3.2.2, "Archiving Audit Reports: Job RMMAUD2B"
RMMRECI1	Report	D.3.3.1, "Listing Information for Recovery: Job RMMRECI1"
RMMRECVL	Report	D.3.3.2, "Recovery Lost Information: Job RMMRECVL"
RMMSMF	Report	D.3.4.1, "Listing SMF DFSMSrmm Volume Records: Job RMMSMF"
SMFPRNT	Report	D.3.4.2, "List Number of SMF DFSMSrmm Records: Job SMFPRNT"
BARCMDS	Report	D.3.5.1, "Create ADDVOLUME Commands: Job BARCMDS"
BARCOMP	Report	D.3.5.2, "Audit Using Barcode Scanner: Job BARCOMP"

D.1 ARCTVEXT and CBRUXENT Sample Exits

D.1.1 Sample DFSMSHsm ARCTVEXT Router Exit

The first part contains samples of the ARCTVEXT exit that interfaces with DFSMSHsm for tape retention.

Figure 254 on page 604 shows the ARCTVEXT router exit.

```

ARCTVEXT TITLE 'ARCTVEXT - ARCTVEXT Router Exit'
      SPACE 1
ARCTVEXT AMODE 24
ARCTVEXT RMODE 24
*****
*
*  MODULE-NAME      = ARCTVEXT
*  DESCRIPTIVE NAME = ARCTVEXT - ARCTVEXT Router Exit
*  COPYRIGHT        = (C) Copyright IBM Corp 1994
*  AUTHOR           = Dieter Wellerdiek
*  FUNCTION = This module is a router module. This means, it routs
*              the same information to two other modules.
*              These module are:
*              - The CA1 ARCTVEXT and
*              - the RMM ARCTVEXT.
*
*              This allows to run RMM and CA1 in parallel.
*
*  MIGRATION OF THE ORIGINAL RMM AND CA1 ARCTVEXT EXIT=
*
*              When you want to use this router, you must make
*              some changes in the original module
*              delivered by RMM and CA1.
*
*              Change all name related statements in the exit from
*              ARCTVEXT to RMMTVEXT/CA1TVEXT. These statements are:
*              - CSECT statement (RMM and CA1)
*              - USING statement (RMM)
*              - TITLE statement (RMM)
*              - AMODE statement (RMM)
*              - RMODE statement (RMM)
*              - END statement (RMM)
*
*              Link all modules together to one module.
*  ENTRY-REGISTERS = 0 - Irrelevant
*                  1 - PTR to parameter list
*                  13 - Caller's save area
*                  14 - Return address
*                  15 - Entry point address of ARCTVEXT
*

```

Figure 254 (Part 1 of 5). Sample ARCTVEXT Router Exit

```

* INPUT =      1      - None at the moment                                *
*                                                                 *
* EXIT =      Returns to the caller with 'BSM 0,R14'                    *
*                                                                 *
* RETURN CODES = 0      No error occurred                                *
*                                                                 *
*-----*
*                                                                 *
* MODULE TYPE = MODULE                                                  *
*   PROCESSOR = Assembler H                                            *
*   ATTRIBUTES = reentrant AC(0)                                        *
*                                                                 *
*****
EJECT
*****
*                                                                 *
* Equates and Macros                                                    *
*                                                                 *
*****
*-----*
* General purpose register's                                           *
*-----*
R0      EQU    0
R1      EQU    1
R2      EQU    2
R3      EQU    3
R4      EQU    4
R5      EQU    5
R6      EQU    6
R7      EQU    7
R8      EQU    8
R9      EQU    9
R10     EQU    10
R11     EQU    11
R12     EQU    12
R13     EQU    13
R14     EQU    14
R15     EQU    15
        SPACE 1

```

Figure 254 (Part 2 of 5). Sample ARCTVEXT Router Exit

```

                TITLE 'ARCTVEXT - Workarea'
*****
*
* Work area - getmained      (first pointed by R11, then R13)
*
*****
RENTAREA DSECT
RENTEYE1 DC    CL8' ARCTVEXT'      Eyecatcher 1
RENTEYE2 DC    CL8' WORKAREA'      Eyecatcher 2
        SPACE 1
WORK      DS    0F                  Start (offset 0 - R13)
SAVEAREA  DS    18F                Program save area
CALLSAVE  DS    18F                Temp. savearea
        SPACE 1
RTNCODE   DS    F                  Return code field
        SPACE 1
*-----*
* End of the work area
*-----*
RENTLEN   EQU   *-RENTAREA
                TITLE 'ARCTVEXT - Main module'
*****
*
* Startup Processing
*
*****
ARCTVEXT CSECT
        USING *,R15
        B      INIT010              Branch around constant
        DC     CL8' ARCTVEXT'        Module name
VERSION   DC     CL8' V01.00'        Version
        DC     CL8'&SYSDATE'         Date
        DC     CL6' &SYSTIME'        Time
INIT010   DS     0H
        DROP   R15                  Temp. base
        STM    R14,R12,12(R13)       Save all registers
        LR     R12,R15               Set up program base register
        USING  ARCTVEXT,R12          Inform assembler
        SPACE 1

```

Figure 254 (Part 3 of 5). Sample ARCTVEXT Router Exit

```

*-----*
* Getmain work area and initialize it                                     *
*-----*
        STORAGE OBTAIN,          Get a work area that's reentrant      x
        LENGTH=RENTLEN,         Work area size                        x
        COND=NO,                 Unconditional req - no RC info        x
        LOC=BELOW,               Getmain area anywhere                 x
        RELATED=' ARCTVEXT-WORKAREA'
        USING RENTAREA,R1        Set temp. addressing
        MVC  RENTYE1,=C' ARCTVEXT' Set eyecatcher 1 (x'-10' R13)
        MVC  RENTYE2,=C' WORKAREA' Set eyecatcher 2 (x'-08' R13)
        ST   R13,SAVEAREA+4      Backward pointer
        LA   R0,SAVEAREA         Get savearea address
        ST   R0,8(R13)           Forward Pointer
        LA   R13,SAVEAREA        Get savearea address
        DROP R1
        USING WORK,R13          Set addressability for save and work
        SPACE 1
*****
*
* Main Processing                                                         *
*
*****
        STM  R0,R12,CALLSAVE     Save all work register
        SPACE 1
        L    R15,RMM             Get the RMM exit address
        L    R14,SAVEAREA+4      Get address of lower savearea
        LM   R0,R12,20(R14)      Restore registers of entry
        BALR R14,R15             Call the RMM exit
        LM   R0,R12,CALLSAVE     Restore registers
        SPACE 1
        L    R15,CA1            Get the CA1 exit address
        L    R14,SAVEAREA+4      Get address of lower savearea
        LM   R0,R12,20(R14)      Restore registers of entry
        BALR R14,R15            Call the CA1 exit
        LM   R0,R12,CALLSAVE     Restore registers
        SPACE 1
*****
*
* Return Processing                                                       *
*
*****
        RETURN   DS    OH

```

Figure 254 (Part 4 of 5). Sample ARCTVEXT Router Exit

```

*-----*
* Free work area and save return code                                     *
*-----*
FASTEND DS    OH
        L      R3,=F'0'          Load and save return code in R3
        LR     R1,R13            Copy R13 to R1 for STORAGE RELEASE
        S      R1,=F'16'        Get real work area address
        L      R13,4(R13)        Restore previous savearea
        STORAGE RELEASE,        Free work area                                X
        LENGTH=RENTLEN,          X
        ADDR=(1),                X
        RELATED=' ARCTVEXT-WORKAREA'
        SPACE 1

*-----*
* Return                                                                *
*-----*
        LM     R0,R12,20(R13)    Restore registers
        L      R14,12(R13)       Restore return address
        BSM    0,R14             Return to caller
        EJECT

*****
*
* Data area's
*
*****
RMM     DC     V(RMMTVEXT)        Address of the RMM exit
CA1     DC     V(CA1TVEXT)       Address of the CA1 exit
        SPACE 1

*-----*
* Literal's
*-----*
        LTORG
        SPACE 1
        END    ARCTVEXT          This is the end

```

Figure 254 (Part 5 of 5). Sample ARCTVEXT Router Exit

D.1.2 Sample CA-1 - DFSMSrmm Merged ARCTVEXT

Figure 255 on page 609 shows the CA-1 - DFSMSrmm merged ARCTVEXT.


```

ARCTVEXT TITLE 'ARCTVEXT MODULE - PROLOGUE'
*****
** DFSMS/MVS 1.4.0 DFSMSrmm                                     *
**                                                             *
** CONTAINS "RESTRICTED MATERIALS OF IBM"                       *
** MODIFIED SPECIFICALLY FOR CUSTOMER MIGRATIONS TO DFRMM AND  *
** DFSMSRMM FROM CA-1.                                         *
** DEVELOPED/TESTED BY THE SOFTWARE MIGRATION PROJECT OFFICE (SMPO) *
** 5695-DF1 (C) COPYRIGHT IBM CORP. 1979, 1993.               *
** ALL RIGHTS RESERVED.                                         *
** SEE COPYRIGHT INSTRUCTIONS                                   *
** LICENSED MATERIALS - PROPERTY OF IBM                         *
**                                                             *
** 1/10/94 - ADD LOGIC TO CA-1 AND RMM INTERFACES IN ORDER TO RUN *
**          BOTH CA-1 AND RMM IN PARALLEL MODE. GLH            *
**                                                             *
*****
**
** MODULE NAME = ARCTVEXT                                       *
**                                                             *
** DESCRIPTIVE NAME = DFHSM TAPE VOLUME EXIT                   *
**                                                             *
** STATUS = OS/VS2  FMID (JDZ1150)                             *
**                                                             *
** FUNCTION = TO INVOKE EDGDFHSM TO TELL RMM OF VOLUMES        *
**          CHANGING STATUS                                     *
**                                                             *
*****
EJECT
*****
**
** DEPENDENCIES = NONE                                         *
** RESTRICTIONS = NONE                                         *
** REGISTER CONVENTIONS = R0 = WORK REGISTER                   *
**                               R1 = PARAMETER LIST             *
**                               R2 = WORK REGISTER              *
**                               R3 = WORK REGISTER              *
**                               R4 = WORK REGISTER              *
**                               R5 = WORK REGISTER              *
**                               R6 = WORK REGISTER              *
**                               R7 = WORK REGISTER              *
**                               R8 = WORK REGISTER              *
**                               R9 = WORK REGISTER              *
**                               R10 = ADDRESS OF MY WORK AREA   *
**                               R11 = WORK REGISTER              @005 *
**                               R12 = PROGRAM BASE               *
**                               R13 = SAVE AREA                 *
**                               R14 = LINK REGISTER              *
**                               R15 = CALLED PROGRAMS/RETURN CODE *

```

Figure 255 (Part 1 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

```

*****
EJECT
*****
*
* ATTRIBUTES = REENTRANT, RMODE 24, AMODE 24
*             APF AUTHORIZED
*
* ENTRY POINTS = ARCTVEXT -- MAIN ENTRY POINT
*
* LINKAGE = FROM DFHSM
*
* INPUT = R0-R10 = ZERO
*         R11 =
*         R12 = PROGRAM ADDRESSABILITY
*         R13 = ZERO
*         R14 = ZERO
*         R15 = ADDRESS OF ENTRY POINT
*
* OUTPUT = LINK TO EDGDFHSM WITH PARAMETER LIST
*
* RETURN CODES - NONE
*
* EXTERNAL REFERENCES = SEE BELOW
*
* ROUTINES = NONE
*
* DATA AREAS = NONE
*
* MACROS = MVS - GETMAIN, FREEMAIN, LINK, RETURN
*
*             RMM - NONE
*
*****
EJECT
*****
*
* CHANGE ACTIVITY
*
* @210      CREATED
* 23/MAY/91 001/177284 CHANGE TO AMODE/RMODE 24      AS @001*
* 06/09/91 DSP PRODUCT NAME CHANGE                  @RMM
* ,AS,100192,MERGE TLCSV1 LINK                      @002 *
* 265034 ,WD,210492,REMOVE EDGENTR MACRO             @003 *
* 265115 ,MC,200592,CORRECT ADDRESSABILITY           @004 *
* ,WD,110992,CORRECT COPYRIGHT                      @005 *
* 268557 ,MC,150393,Getmain/freemain corrections     @006 *
*
*****

```

Figure 255 (Part 2 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

```

SPACE 1
ARCTVEXT CSECT
ARCTVEXT AMODE 24 @001
ARCTVEXT RMODE 24 @001
PRINT GEN
TITLE 'ARCTVEXT MODULE MAINLINE PROGRAM'
USING ARCTVEXT,R15 DECLARE ADDRESSABILITY @003
SPACE 2 @003
R15 EQU 15 REGISTER 15 IS E.P. @003
R14 EQU 14 REGISTER 14 RETURN ADDRESS @003
R13 EQU 13 REGISTER 13 SAVE AREA @003
R12 EQU 12 REGISTER 12 @003
R11 EQU 11 REGISTER 11 @003
R10 EQU 10 REGISTER 10 @003
R9 EQU 9 REGISTER 9 @003
R8 EQU 8 REGISTER 8 @003
R7 EQU 7 REGISTER 7 @003
R6 EQU 6 REGISTER 6 @003
R5 EQU 5 REGISTER 5 @003
R4 EQU 4 REGISTER 4 @003
R3 EQU 3 REGISTER 3 @003
R2 EQU 2 REGISTER 2 @003
R1 EQU 1 REGISTER 1 @003
R0 EQU 0 REGISTER 0 @003
SPACE 3 @003
MODID , SET IDENTIFICATION TEXT @003
SPACE 2 @003
STM R14,R12,12(R13) SAVE CALLERS REGISTERS @003
LR R12,R15 SET R12 AS PROGRAM BASE @003
DROP R15 RESET ADDRESSABILITY @004
USING ARCTVEXT,R12 DECLARE ADDRESSABILITY @003
EJECT
*****
*
* ARCTVEXT MAINLINE PROGRAM
*
*****
SPACE 1
*****
* GETMAIN WORK AREA
*****
SPACE 1
GETMAIN1 DS OH
USING PARMLIST,R4 USE DSECT
LR R4,R1 GET PARMLIST ADDRESS
L R1,RC LOAD RC ADDRESS
SR R0,R0 PUT 0 IN R0
ST R0,0(,R1) STORE IN RC FIELD
LA R0,GMAREAL LENGTH OF AREA TO GETMAIN
SPACE 1
GETMAIN RC,LV=(0),SP=0 GETMAIN @006
LTR R15,R15 SUCCESSFUL?
BZ GETMAIN2 YES - CONTINUE
WTO 'ARCTVEXT - GETMAIN ERROR',ROUTCDE=13 ERROR MESSAGE
B ERROR GO RETURN
SPACE 1

```

Figure 255 (Part 3 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

```

GETMAIN2 DS    OH
          ST    R13,4(,R1)          LINK CALLER'S SAVE AREA
          ST    R1,8(,R13)          LINK NEW SAVE AREA
          LR     R13,R1             USE NEW SAVE AREA
          USING GMAREA,R13         USE DSECT FOR WORK AREA
          SPACE 1
          LR     R1,R4              load parameter register
          LINK EP=TMSARCTV,ERRET=TMSERROR LINK TO CA1 5.0 EXIT
*         LINK EP=TMSTVEXT,ERRET=TMSERROR LINK TO CA1 4.9 EXIT
          B      RMMHSM             GO TO RMM PROCESS
TMSERROR WTO 'ARCTVEXT - Error calling CA1 HSM exit'
          B      FREEMAIN           Free storage
          SPACE 1
          EJECT
*****
* COPY FIRST VOLUME DATA TO EDGDFHSM PARM LIST
* AND SET REGISTERS TO TEST FOR ALTVOL PARAMETER
*****
          SPACE 1
RMMHSM   DS    OH
          LA     R6,ALTCHK          SET ADDRESS TO CHECK ALTVOL
          L      R5,DATA            GET A(DATA AREA)
NEXTVOL  MVC    RMMDATA,0(R5)      COPY THE TAPE VOLUME AND FLAGS @RMM
          EJECT
*****
* INVOKE EDGDFHSM VIA LINK TO TELL RMM STATUS
*****
          SPACE 1
          LA     R1,RMMPARM         SETUP PARMLIST @RMM
          LA     R14,RMMDATA        POINT TO VOLSER @RMM
          ST     R14,0(,R1)         PUT ADDRESS INTO PARMLIST
          OI     0(R1),X'80'        INDICATE LAST PARAMETER
          LINK EP=EDGDFHSM,ERRET=SETEXP INVOKE EDGDFHSM IF INSTALLED
          EJECT
*****
* INVOKE TLCDFHSM VIA LINK TO TELL TLCS STATUS
*
* TO CREATE AN ARCTVEXT THAT LINKS TO BOTH RMM AND TLCSV1 THEN
* DO THE FOLLOWING
*
* 1. REPLACE THE FIRST LINK STATEMENT WITH
*
* LINK EP=EDGDFHSM,ERRET=TLCSV1 INVOKE EDGDFHSM IF INSTALLED
*
*
* 2. UNCOMMENT THE STATEMENTS BETWEEN LABELS TLCSV1 AND SETEXP
*
*****
*
* SPACE 1
*TLCSV1 LA     R1,RMMPARM         SETUP PARMLIST @002
*       LA     R14,RMMDATA        POINT TO VOLSER @002
*       ST     R14,0(,R1)         PUT ADDRESS INTO PARMLIST @002
*       OI     0(R1),X'80'        INDICATE LAST PARAMETER @002
*       LINK EP=TLCDFHSM,ERRET=SETEXP INVOKE TLCDFHSM IF INSTALLED
*       EJECT

```

Figure 255 (Part 4 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

```

*****
*   CHECK FOR EXISTENCE OF A COPY VOLUME                               *
*****

      SPACE 1
SETEXP  BR   R6                                BRANCH TO ALTCHK OR FREEMAIN..
ALTCHK  DS   OH
        LA   R6,RC                            POINT TO RC ADDRESS
        TM   0(R6),X'80'                      IS HIGH ORDER BIT ON
        BO   FREEMAIN                        ALL DONE
        LA   R6,FREEMAIN                     ENSURE WE STOP AFTER NEXT LINK
        L    R5,ALTVOL                      POINT TO COPY VOLUME
        B    NEXTVOL                         DO COPY AS WELL
      EJECT

*****
*   FREEMAIN THE GOTTEN WORK AREA                                       *
*****

      SPACE 1
FREEMAIN DS   OH
        LR   R1,R13                          A(GETMAINED AREA)
        L    R13,SAVEAREA+4                  LOAD A(CALLER'S SAVE AREA)
        LA   R0,GMAREAL                      SIZE OF GETMAINED AREA
        FREEMAIN RC,A=(1),LV=(0)             FREE THE AREA @006
      SPACE 1
      LTR   R15,R15                          SUCCESSFUL?
      BZ    RETURN                          YES - SKIP ERROR MESSAGE
      WTO   'ARCTVEXT - FREEMAIN ERROR'      ERROR MESSAGE
ERROR    DS   OH
        L    R1,RC                          LOAD RC ADDRESS
        LA   R0,4                            PUT 4 IN R0
        ST   R0,0(R1)                        STORE IN RC FIELD
      EJECT

*****
*   RETURN TO DFHSM WITH PROPER RETURN CODE SET IN RC                 *
*****

      SPACE 1
RETURN   DS   OH
        RETURN (14,12),RC=0                 RETURN WITH R15=0
      SPACE 1
        DROP R4                            RELEASE PARMLIST
        DROP R13                           RELEASE GMAREA
        DROP R12                           RELEASE BASE
      EJECT 1
      LTORG ,
      EJECT

*****
*   DSECT DEFINITIONS                                                 *
*****

      SPACE 1

```

Figure 255 (Part 5 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

```

*****
*   WORK AREA                                     *
*****
          SPACE 1
GMAREA   DSECT
SAVEAREA DS    18F          THIS PROGRAM'S SAVE AREA
RMMPARM  DS     F           EDGDFHSM CALL PARAMETER LIST  @RMM
RMMDATA  DS    CL8          DATA AREA AS PASSED BY DFHSM TO @RMM
          SPACE 1
GMAREAL  EQU    *-GMAREA    LENGTH OF WORK AREA
          EJECT
*****
*   THIS IS A VARIABLE LENGTH PARM. THE HIGH ORDER BIT IS SET ON   *
*   IN EITHER RC OR ALTVOL POINTER.                                *
*****
          SPACE 1
PARMLIST DSECT
DATA     DS     F           POINTER TO DATA AREA
RC       DS     F           POINTER TO RETURN CODE
ALTVOL   DS     F           POINTER TO DATA AREA COPY VOLUME
          SPACE 1
          SPACE 1
          END   ARCTVEXT

```

Figure 255 (Part 6 of 6). Sample CA-1 - DFSMSrmm Merged ARCTVEXT

D.1.3 Sample TLMS - DFSMSrmm Merged ARCTVEXT

Figure 256 on page 615 shows the TLMS - DFSMSrmm merged ARCTVEXT.

```

*****
*
* Both CA-DYNAM/TLMS and DFSMSrmm supplied ARCTVEXT exits are
* included;
* First CA-DYNAM/TLMS and then with DFSMSrmm exit included, but
* modified to use RMMTVEXT as the CSECT name.
* The CA-DYNAM/TLMS exit is modified to restore registers and
* pass control to the DFSMSrmm exit.
*
*****
      TITLE 'CA-DYNAM/TLMS/DFSMSHsm SAMPLE ARCTVEXT'
ARCTVEXT CSECT
*****
*
* TLMShsmX - TLMShsmX PROVIDES AN INTERFACE BETWEEN DFSMSHsm AND
* CA-DYNAM/TLMS. WHEN DFSMSHsm HAS DETERMINED THAT IT NO LONGER
* REQUIRES A TAPE (OR TAPES) IT CALLS THIS EXIT WITH A PARAMETER
* LIST ADDRESSED BY REGISTER 1 CONTAINING THE VOLSER(S) TO BE
* RETURNED TO THE CA-DYNAM/TLMS SCRATCH POOL. TLMShsmX CREATES THE
* UPDATE TRANSACTION AND CALL TLMSPUT TO PASS THE TRANSACTION TO THE
* ONLINE RECORDER. TAPES CREATED BY DFSMSHsm SHOULD BE CONTROLLED WITH*
* AN RMF ENTRY THAT PLACES ALL TAPES UNDER MANUAL CONTROL.
*
* THE TRANSACTION (VMFTRAN) IS BUILT INTERNALLY RATHER THAN CALL
* TLMSCMND, BECAUSE OF IT'S EXCESSIVE SIZE.
*
*
* ENTRY: ARCTVEXT
*
* INPUTS:
*
*      R1          POINTS TO THE PARMLIST
*      R13         POINTS TO THE CALLER'S SAVEAREA
*      R14         RETURN ADDRESS
*      R15         ENTRY POINT ADDRESS
*
*
* PARMLIST:
*
*      A          ADDRESS OF TAPE VOLSER
*      A          ADDRESS OF FULLWORD RETURN CODE
*      A          ADDRESS OF ALTERNATE VOLSER
*                (HSM 2.3.1 AND UP)
*
* OUTPUTS:
*
*      R0-R14     CALLER'S REGISTERS
*      R15        RETURN CODE
*

```

Figure 256 (Part 1 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

* RETURN CODES:
*
*      0      FUNCTION SUCCESSFUL
*      4      ERROR DETECTED ( NEVER CALL AGAIN)
*
* MESSAGES:
*
*      TLMShsm1 -- LOAD FAILED FOR PROGRAM XXXXXXXX
*      TLMShsm2 -- CRITICAL ERROR - EXIT DISABLED
*
* ENVIRONMENT:
*
*      1. THIS MODULE IS REENTRANT
*
* CALLED BY:
*
*      1. DFHSM      - TO RETURN A TAPE VOLUME TO A TAPE
*                     MANAGEMENT SYSTEM.
*
* CALLS TO:
*
*      1. TLMSPUT    - TO ISSUE THE QUEUE MANAGER SVC TO PASS
*                     THE UPDATE TRANSACTION TO THE TLMS ONLINE
*                     RECORDER.
*
*****
EJECT
R0      EQU 0      WORK REGISTER
R1      EQU 1      WORK REGISTER
R2      EQU 2      WORK REGISTER
R3      EQU 3      ADDRESS OF WORK AREA
R4      EQU 4      ADDRESS OF PASSED PARMLIST
R5      EQU 5      NOT USED
R6      EQU 6      NOT USED
R7      EQU 7      NOT USED
R8      EQU 8      NOT USED
R9      EQU 9      NOT USED
R10     EQU 10     NOT USED
R11     EQU 11     NOT USED
R12     EQU 12     PROGRAM BASE
R13     EQU 13     ADDRESS OF SAVEAREA
R14     EQU 14     LINKAGE REGISTER
R15     EQU 15     LINKAGE REGISTER
EJECT
SAVE (14,12)      SAVE CALLER REGISTERS
LR R12,R15        GET ENTRY ADDRESS IN R12
USING ARCTVEXT,R12 SET UP ADDRESSABILITY FOR
B INITCODE        BRANCH AROUND IDENTIFIER
DC C'TLMShsmX -'   PROGRAM ID
DC C'(C) COPYRIGHT COMPUTER ASSOCIATES INTERNATIONAL 1988'
DC C'&SYSDATE'     DATE ASSEMBLED
DC C'&SYSTIME'     TIME ASSEMBLED

```

Figure 256 (Part 2 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

INITCODE	DS	OH	
	LR	R4,R1	SAVE ADDRESS OF INPUT PARM
	LA	R0,LWORK	ADDRESS OF STORAGE NEEDED
	GETMAIN	RC,LV=(0)	OBTAIN WORKAREA
	LR	R3,R1	ADDRESS GETMAINED AREA
	SLR	R0,R0	CLEAR FOR MVCL
	LR	R1,R0	CLEAR AGAIN
	LR	R14,R3	ADDRESS GETMAINED AREA
	LA	R15,LWORK	LENGTH OF AREA
	MVCL	R14,R0	CLEAR THE AREA
	USING	WORKAREA,R3	TELL THE ASSEMBLER
	LA	R2,SAVEAREA	GET ADDRESS OF THE SAVE AREA
	ST	R2,8(,R13)	FORWARD CHAIN SAVE AREA POINTER
	ST	R13,4(,R2)	BACK CHAIN SAVE AREAS
	LR	R13,R2	GET THE ADDRESS OF THE SAVEAREA
	L	R2,0(,R4)	GET ADDRESS OF VOLSER
	TM	6(R2),X'80'	IS THIS TAPE BEING PURGED??
	BZ	CHKKALT	NO, CHECK FOR ALT VOL..
	BAL	R14,SCRTAPE	GO ISSUE THE SCRATCH TRAN
CHKKALT	DS	OH	
	TM	4(R4),X'80'	IS THERE ALT VOLSER??
	BO	RC0	NO, GO SET RETURN CODE
	L	R2,8(,R4)	GET ADDRESS OF LAST PARM
	TM	6(R2),X'80'	IS VOLUME BEING PURGED?
	BZ	RC0	NO, GO SET RETURN CODE
	BAL	R14,SCRTAPE	GO ISSUE THE SCRATCH TRAN
	B	RC0	GO RETURN
SCRTAPE	TITLE	'BUILD THE TRANSACTION AND CALL TLMSPUT'	
	DS	OH	
	ST	R14,XSCRTAPE	SAVE THE RETURN ADDRESS
	TM	PRFLAG,PGMLODED	HAVE WE LOADED TLMSPUT?
	BO	SKIPLOAD	YES, SKIP THE LOAD
	LOAD	EPLOC=TLMSPUT,ERRET=LOADFAIL	LOAD THE PROGRAM
	ST	R0,PUTADDR	STORE THE ADDRESS
	OI	PRFLAG,PGMLODED	INDICATE PROGRAM LOADED
	MVC	ENQPL(ENQ1L),ENQ1	MOVE IN THE PARM LIST
	LA	R1,EXITNAME	GET ADDRESS OF THE QNAME
	ST	R1,ENQPL+4	STORE IT IN THE PARMLIST
	LA	R1,TLMSPUT	GET ADDRESS OF THE RNAME
	ST	R1,ENQPL+8	STORE IT IN THE PARMLIST
	ENQ	,MF=(E,ENQPL)	

Figure 256 (Part 3 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

SKIPLOAD DS      OH
*****
* THE FOLLOWING CODE IS POSITIONALLY DEPENDENT AND MUST NOT BE      *
* ALTERED.  FAILURE TO ADHERE TO THIS WARNING MAY RESULT IN SOC7    *
* ABENDS IN THE CA-DYNAM/TLMS ONLINE RECORDER.                      *
*****

        MVI     VMFTRAN,X'FF'          SET UP NULL FIELDS
        MVC     VMFTRAN+1(TRLENGTH-1),VMFTRAN  CLEAR TO X'FF'S
        MVC     TRRECLN,=AL2(TRLENGTH)  MOVE IN TRANSACTION LENGTH
        MVI     TRTYPE,C'1'            MOVE IN TYPE=UPDATE
        MVI     TRMTYPE,C'5'           MOVE IN MESSAGE=BATCH
        MVC     TRVSN(6),0(R2)         MOVE IN VSN
        MVI     TRSRVSCR,C'2'         MOVE IN IN-SERVICE/SCRATCH
        MVC     TRCSTEP(8),EXITNAME    INDICATE XACTION FROM ARCTVEXT
        TIME    DEC                   GET THE TIME IS DECIMAL
        STCM    R1,7,TRSCRDT          STORE IT IN THE TRANSACTION
        LA      R1,VMFTRAN            GET ADDRESS OF TRANS
        ST      R1,PUTPLIST           STORE IT IN THE PLIST
        LA      R1,RETURNCD           GET ADDRESS OF RETURN CODE
        ST      R1,PUTPLIST+4         STORE IT IN THE PLIST
        OI      PUTPLIST+4,X'80'      INDICATE LAST ENTRY
        LA      R1,PUTPLIST           INDICATE LAST ENTRY
        L       R15,PUTADDR           GET ADDRESS OF THE PUT ROUTINE
        BALR    R14,R15              GO TO PUT ROUTINE
*****
*                               END OF POSITIONALLY DEPENDENT CODE.    *
*****

        L       R14,XSCRTAPE          RESTORE RETURN ADDRESS
        BR      R14                   RETURN
        TITLE   'RETURN TO CALLING MODULE'
RCO      DS      OH
        TM      PRFLAG,PGMLODED       DID WE LOAD THE PGM?
        BZ      RETURN                NO, THEN DONT DELETE
        DELETE  EPLOC=TLMSPUT
        MVC     DEQPL(DEQ1L),DEQ1     MOVE IN THE PARM LIST
        LA      R1,EXITNAME           GET ADDRESS OF THE QNAME
        ST      R1,DEQPL+4            STORE IT IN THE PARMLIST
        LA      R1,TLMSPUT            GET ADDRESS OF THE RNAME
        ST      R1,DEQPL+8            STORE IT IN THE PARMLIST
        DEQ     ,MF=(E,DEQPL)
        XC      PUTADDR,PUTADDR       CLEAR PROGRAM ADDRESS
        NI      PRFLAG,X'FF'-PGMLODED INDICATE PGM NOT LOADED
        SLR     R15,R15               CLEAR RETURN CODE
        B       RETURN
LOADFAIL DS      OH
        MVC     WTOPL(WTO1L),WTO1     MOVE IN WTO PARMLIST
        MVC     WTOPL+8(80),TLMShsm1 MOVE IN THE MESSAGE
        MVC     WTOPL+45(8),TLMSPUT   MOVE IN THE PROGRAM ID
        WTO     MF=(E,WTOPL)
        MVC     WTOPL(WTO1L),WTO1     MOVE IN WTO PARMLIST
        MVC     WTOPL+8(80),TLMShsm2 MOVE IN THE MESSAGE
        WTO     MF=(E,WTOPL)
        LA      R15,4                 SET NONZERO RETURN CODE
        B       RETURN

```

Figure 256 (Part 4 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

RETURN  DS   OH
        L     R2,4(,R4)           GET ADDRESS OF RC AREA
        ST    R15,0(,R2)          STORE THE RETURN CODE
        L     R13,4(,R13)         GET SAVE AREA ADDRESS
        LA    R0,LWORK            GET WORKAREA LENGTH
        LR    R1,R3               GET WA ADDRESS
        FREEMAIN R,LV=(0),A=(1)   FREE THE WORKAREA
        SR    R15,R15             FORCE RC=0
        RETURN (14,12),,RC=(15)  RETURN
*****
* RMM ENTRANCE CODE
* RESTORE ALL PARMS AND REGS AT ENTRANCE TO TLMS EXIT
*****
        L     R15,RMMEXT          ADDR OF TMM EXIT
        L     R14,R12(0,R13)      RESTORE REG 14
        LM    R0,R12,20(R13)      RESTORE REGS
        BR    R15                 ENTER RMM EXIT
        TITLE ' PATCH AREA      '
        PRINT DATA
PATCH  DC    64S(*)              PATCH AREA
        PRINT NODATA
        TITLE ' PROGRAM CONSTANTS'
WT01   WTO    '
                                           ',ROUTCDE=(2,11),DESC=2,MF=L
WT01L   EQU   *-WT01
ENQ1    ENQ   (EXITNAME,TLMSPUT,E,7,STEP),MF=L
ENQ1L   EQU   *-ENQ1
DEQ1    DEQ   (EXITNAME,TLMSPUT,7,STEP),MF=L
DEQ1L   EQU   *-DEQ1
RMMEXT  DC    V(RMMTVEXT)
EXITNAME DC   CL8' TLMSHSMX'
TLMSPUT  DC   CL8' TLMSPUT '
TLMSHSM1 DC   CL80' TLMSHSM1  -- LOAD FAILED FOR PROGRAM XXXXXXXX'
TLMSHSM2 DC   CL80' TLMSHSM2  -- CRITICAL ERROR - EXIT DISABLED '
        LTORG
        TITLE ' WORK AREA DSECT '
WORKAREA DSECT
PUTADDR  DS   F
PUTPLIST DS   2F
RETURNCD DS   F
XSCRTAPE DS   F
        COPY  VMFTRAN
PRFLAG   DS   XL1
PGMLODED EQU  X'80'
PLISTS   DS   OD
ENQPL    DS   XL(ENQ1L)
DEQPL    DS   XL(DEQ1L)
WTOPL    DS   XL(WT01L)
SAVEAREA DS   OD                SAVE AREA
PUTWORK   DS   XL512            SAVE/WORK AREA FOR TLMSPUT
LWORK    EQU   *-WORKAREA
END

```

Figure 256 (Part 5 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

RMMTEXT TITLE 'ARCTVEXT MODULE - PROLOGUE'
*****
** DFSMS/MVS 1.1.0 DFSMSrmm                                     *
**                                                             *
** CONTAINS "RESTRICTED MATERIALS OF IBM"                       *
** 5695-DF1 (C) COPYRIGHT IBM CORP. 1979, 1993.                 *
** ALL RIGHTS RESERVED.                                         *
** SEE COPYRIGHT INSTRUCTIONS                                   *
** LICENSED MATERIALS - PROPERTY OF IBM                         *
*****
*                                                             *
*   MODULE NAME = ARCTVEXT                                       *
*                                                             *
*   DESCRIPTIVE NAME = DFHSM TAPE VOLUME EXIT                   *
*                                                             *
*   STATUS = OS/VS2  FMID (JDZ1150)                             *
*                                                             *
*   FUNCTION = TO INVOKE EDGDFHSM TO TELL RMM OF VOLUMES        *
*               CHANGING STATUS                                  *
*                                                             *
*****
EJECT
*****
*                                                             *
*   DEPENDENCIES = NONE                                         *
*                                                             *
*   RESTRICTIONS = NONE                                         *
*                                                             *
*   REGISTER CONVENTIONS = R0 = WORK REGISTER                   *
*                           R1 = PARAMETER LIST                 *
*                           R2 = WORK REGISTER                   *
*                           R3 = WORK REGISTER                   *
*                           R4 = WORK REGISTER                   *
*                           R5 = WORK REGISTER                   *
*                           R6 = WORK REGISTER                   *
*                           R7 = WORK REGISTER                   *
*                           R8 = WORK REGISTER                   *
*                           R9 = WORK REGISTER                   *
*                           R10 = ADDRESS OF MY WORK AREA       *
*                           R11 = WORK REGISTER                  *
*                           R12 = PROGRAM BASE                   *
*                           R13 = SAVE AREA                      *
*                           R14 = LINK REGISTER                  *
*                           R15 = CALLED PROGRAMS/RETURN CODE   *
*                                                             *
*****
EJECT

```

Figure 256 (Part 6 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
*
*   ATTRIBUTES = REENTRANT, RMODE 24, AMODE 24
*               APF AUTHORIZED
*
*   ENTRY POINTS = ARCTVEXT -- MAIN ENTRY POINT
*
*   LINKAGE = FROM DFHSM
*
*   INPUT = R0-R10 = ZERO
*           R11  =
*           R12  = PROGRAM ADDRESSABILITY
*           R13  = ZERO
*           R14  = ZERO
*           R15  = ADDRESS OF ENTRY POINT
*
*   OUTPUT = LINK TO EDGDFHSM WITH PARAMETER LIST
*
*   RETURN CODES - NONE
*
*   EXTERNAL REFERENCES = SEE BELOW
*
*   ROUTINES = NONE
*
*   DATA AREAS = NONE
*
*   MACROS = MVS - GETMAIN, FREEMAIN, LINK, RETURN
*
*           RMM - NONE
*
*****

      EJECT
      SPACE 1
RMMTVEXT CSECT
RMMTVEXT AMODE 24
RMMTVEXT RMODE 24
      PRINT GEN
      TITLE 'ARCTVEXT MODULE MAINLINE PROGRAM'
      USING RMMTVEXT,R15          DECLARE ADDRESSABILITY
      SPACE 2
R15     EQU    15          REGISTER 15 IS E.P.
R14     EQU    14          REGISTER 14 RETURN ADDRESS
R13     EQU    13          REGISTER 13 SAVE AREA
R12     EQU    12          REGISTER 12
R11     EQU    11          REGISTER 11
R10     EQU    10          REGISTER 10
R9      EQU    9           REGISTER 9
R8      EQU    8           REGISTER 8
R7      EQU    7           REGISTER 7

```

Figure 256 (Part 7 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

R6      EQU    6          REGISTER 6
R5      EQU    5          REGISTER 5
R4      EQU    4          REGISTER 4
R3      EQU    3          REGISTER 3
R2      EQU    2          REGISTER 2
R1      EQU    1          REGISTER 1
R0      EQU    0          REGISTER 0
      SPACE 3
      MODID ,           SET IDENTIFICATION TEXT
      SPACE 2
      STM  R14,R12,12(R13)  SAVE CALLERS REGISTERS
      LR   R12,R15          SET R12 AS PROGRAM BASE
      DROP R15              RESET ADDRESSABILITY
      USING RMMTEXT,R12      DECLARE ADDRESSABILITY
      EJECT
*****
*
*          ARCTVEXT MAINLINE PROGRAM
*
*****
      SPACE 1
*****
*  GETMAIN WORK AREA
*****
      SPACE 1
GETMAIN1 DS    0H
      USING PARMLIST,R4      USE DSECT
      LR   R4,R1             GET PARMLIST ADDRESS
      L    R1,RC             LOAD RC ADDRESS
      SR   R0,R0             PUT 0 IN R0
      ST   R0,0(,R1)         STORE IN RC FIELD
      LA   R0,GMAREAL        LENGTH OF AREA TO GETMAIN
      SPACE 1
      GETMAIN RU,LV=(0),SP=0  GETMAIN
      LTR  R15,R15           SUCCESSFUL?
      BZ   GETMAIN2          YES - CONTINUE
      WTO  'ARCTVEXT - GETMAIN ERROR',ROUTCDE=13  ERROR MESSAGE
      B    ERROR             GO RETURN
      SPACE 1
GETMAIN2 DS    0H
      ST   R13,4(,R1)        LINK CALLER'S SAVE AREA
      ST   R1,8(,R13)        LINK NEW SAVE AREA
      LR   R13,R1            USE NEW SAVE AREA
      USING GMAREA,R13       USE DSECT FOR WORK AREA
      EJECT

```

Figure 256 (Part 8 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
* COPY FIRST VOLUME DATA TO EDGDFHSM PARM LIST *
* AND SET REGISTERS TO TEST FOR ALTVOL PARAMETER *
*****
        SPACE 1
        LA    R6,ALTCHECK          SET ADDRESS TO CHECK ALTVOL
        L     R5,DATA              GET A(DATA AREA)
NEXTVOL MVC  RMMDATA,0(R5)        COPY THE TAPE VOLUME AND FLAGS
        EJECT
*****
* INVOKE EDGDFHSM VIA LINK TO TELL RMM STATUS *
*****
        SPACE 1
        LA    R1,RMPPARM          SETUP PARMLIST
        LA    R14,RMMDATA         POINT TO VOLSER
        ST    R14,0(R1)          PUT ADDRESS INTO PARMLIST
        OI    0(R1),X'80'        INDICATE LAST PARAMETER
        LINK  EP=EDGDFHSM,ERRET=SETEXP INVOKE EDGDFHSM IF INSTALLED
        EJECT
*****
* CHECK FOR EXISTENCE OF A COPY VOLUME *
*****
        SPACE 1
SETEXP BR    R6                  BRANCH TO ALTCHECK OR FREEMAIN..
ALTCHECK DS  OH
        LA    R6,RC              POINT TO RC ADDRESS
        TM    0(R6),X'80'        IS HIGH ORDER BIT ON
        BO    FREEMAIN          ALL DONE
        LA    R6,FREEMAIN        ENSURE WE STOP AFTER NEXT LINK
        L     R5,ALTVOL          POINT TO COPY VOLUME
        B     NEXTVOL            DO COPY AS WELL
        EJECT
*****
* FREEMAIN THE GOTTEN WORK AREA *
*****
        SPACE 1
FREEMAIN DS  OH
        LR    R1,R13             A(GETMAINED AREA)
        L     R13,SAVEAREA+4     LOAD A(CALLER'S SAVE AREA)
        LA    R0,GMAREAL         SIZE OF GETMAINED AREA
        FREEMAIN RU,A=(1),LV=(0) FREE THE AREA
        SPACE 1
        LTR   R15,R15            SUCCESSFUL?
        BZ    RETURN             YES - SKIP ERROR MESSAGE
        WTO   'ARCTVEXT - FREEMAIN ERROR' ERROR MESSAGE
ERROR DS    OH
        L     R1,RC              LOAD RC ADDRESS
        LA    R0,4               PUT 4 IN R0
        ST    R0,0(R1)          STORE IN RC FIELD
        EJECT

```

Figure 256 (Part 9 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
*   RETURN TO DFHSM WITH PROPER RETURN CODE SET IN RC   *
*****

      SPACE 1
RETURN  DS    OH
      RETURN (14,12),RC=0          RETURN WITH R15=0
      SPACE 1
      DROP  R4                      RELEASE PARMLIST
      DROP  R13                     RELEASE GMAREA
      DROP  R12                     RELEASE BASE
      EJECT 1
      LTORG ,
      EJECT

*****
*   DSECT DEFINITIONS                                   *
*****
      SPACE 1
*****
*   WORK AREA                                           *
*****
      SPACE 1
GMAREA  DSECT
SAVEAREA DS    18F                THIS PROGRAM'S SAVE AREA
RMMPARM DS     F                  EDGDFHSM CALL PARAMETER LIST
RMMDATA DS    CL8                 DATA AREA AS PASSED BY DFHSM TO RMM
      SPACE 1
GMAREAL EQU  *-GMAREA             LENGTH OF WORK AREA
      EJECT

*****
*   THIS IS A VARIABLE LENGTH PARM. THE HIGH ORDER BIT IS SET ON   *
*   IN EITHER RC OR ALTVOL POINTER.                                *
*****
      SPACE 1
PARMLIST DSECT
DATA     DS     F                POINTER TO DATA AREA
RC        DS     F                POINTER TO RETURN CODE
ALTVOL    DS     F                POINTER TO DATA AREA COPY VOLUME
      SPACE 1
      SPACE 1
      END    RMMTVEXT

```

Figure 256 (Part 10 of 10). Sample TLMS - DFSMSrmm Merged ARCTVEXT

D.1.4 CBRUXENT Sample Exit

Figure 257 on page 625 shows a merged CA-1 - DFSMSrmm version of the exit. It contains a sample of the CBRUXENT exit that interfaces with the IBM 3494 and 3495 Automated Tape Library Dataservers.


```

CBRUXENT TITLE 'SAMPLE CART. ENTRY INSTALLATION EXIT W/CA-TAPE-MGMT'
CBRUXENT START 0
      SPACE 2
*****
*
*  MODULE NAME: CBRUXENT
*
*  DESCRIPTIVE NAME: SAMPLE CARTRIDGE ENTRY INSTALLATION EXIT
*
*  FUNCTION:
*    MODULE CBRUXENT IS INVOKED EACH TIME A CARTRIDGE IS ENTERED
*    INTO A TAPE LIBRARY. CBRUXENT MAY REFUSE TO ALLOW THE
*    CARTRIDGE TO BE ENTERED, OR UPDATE CERTAIN FILEDS IN THE TAPE
*    VOLUME RECORD, OR APPROVE THE CARTRIDGE ENTRY WITHOUT CHANGE,
*    OR LEAVE THE CARTRIDGE FOR PROCESSING BY SOME OTHER SYSPLEX.
*
*
*  NOTES:
*
*    DEPENDENCIES:          MVS/SP VERSION 4.3.0
*                          DFSMS/MVS 1.1.0
*
*    CHARACTER CODE:        EBCDIC
*
*    RESTRICTIONS:          NONE
*
*    MODULE TYPE:           CONTROL SECTION
*
*    PROCESSOR:             ASSEMBLER H
*
*    ATTRIBUTES:
*      LOCATION:            JOB PACK AREA
*      STATE:               PROBLEM
*      AMODE:               31 - MAY BE 24
*      RMODE:               ANY - MAY BE 24
*      KEY:                 5 (DATA MANAGEMENT)
*      MODE:                TASK
*      SERIALIZATION:       UNLOCKED
*      AUTHORIZATION:       APF AUTHORIZED
*
*    LINKAGE:               STANDARD LINKAGE CONVENTIONS
*
*    CALLING SEQUENCE:
*      CBRUXENT IS INVOKED IN THE OAM ADDRESS SPACE USING THE MVS
*      LINK MACRO.
*
*    INPUT:
*      AT ENTRY TO CBRUXENT, REGISTER 1 CONTAINS THE ADDRESS OF
*      THE CARTRIDGE ENTRY INSTALLATION EXIT PARAMETER LIST.

```

Figure 257 (Part 1 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*      OUTPUT:
*      A RETURN CODE IS PLACED IN REGISTER 15:
*      CODE  MEANING
*      0      ENTER THE CARTRIDGE INTO THE LIBRARY.  THE TAPE
*             VOLUME RECORD IS UNCHANGED.
*      4      ENTER THE CARTRIDGE INTO THE LIBRARY.  ONE OR MORE
*             FIELDS IN THE TAPE VOLUME RECORD HAVE BEEN CHANGED.
*      8      DO NOT ALLOW THE CARTRIDGE TO BE ENTERED INTO THE
*             LIBRARY.  EJECT THE CARTRIDGE IMMEDIATELY.
*      12     IGNORE THE REQUEST TO ENTER THE CARTRIDGE INTO THE
*             LIBRARY.  LEAVE THE CARTRIDGE IN THE INSERT
*             CATEGORY FOR PROCESSING BY ANOTHER SYSPLEX
*      16     DO NOT INVOKE THE CARTRIDGE ENTRY INSTALLATION EXIT
*             AGAIN.  ALLOW ALL CARTRIDGES TO BE ENTERED WITHOUT
*             CHANGES TO THEIR TAPE VOLUME RECORDS.
*
*      EXTERNAL REFERENCES:
*      TLMSQSTS:                QUERY TAPE STATUS
*
*      ROUTINES:  NONE
*
*      CONTROL BLOCKS:
*      CBRUXEPL - CARTRIDGE ENTRY INSTALLATION EXIT
*                PARAMETER LIST                                -R/W
*
*      EXECUTABLE MACROS:
*      RETURN
*      SAVE
*      GETMAIN
*      FREEMAIN
*      CATAPE
*
*      MESSAGES:  NONE
*
*      ABEND CODES: NONE
*
*      1/10/94 - ADD IBM LOGIC BACK INTO CA-1 CBRUXENT MODULE -
*****
** DFSMS/MVS 1.1.0 DFSMSrmm
**
** CONTAINS "RESTRICTED MATERIALS OF IBM SOFTWARE MIGRATION SERVICES
** 5695-DF1 (C) COPYRIGHT IBM CORP. 1994.
** ALL RIGHTS RESERVED.
** SEE COPYRIGHT INSTRUCTIONS
** LICENSED MATERIALS - PROPERTY OF IBM-USED BY SOFTWARE OFFERINGS
*****
      TITLE '                CARTRIDGE ENTRY INST PARAMETER LIST'
      CBRUXEPL ,
      TITLE '                STANDARD REGISTER DEFINITIONS'
R0      EQU  0          WORK REGISTER
R1      EQU  1          ADR PARM LIST
R2      EQU  2
R3      EQU  3
R4      EQU  4
R5      EQU  5
R6      EQU  6

```

Figure 257 (Part 2 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

R7      EQU 7
R8      EQU 8
R9      EQU 9          RETURN CODE WORK REGISTER
R10     EQU 10
R11     EQU 11         UXEPL BASE REGISTER
R12     EQU 12         CBRUXENT BASE REGISTER
R13     EQU 13         ADR OF SAVE AREA
R14     EQU 14         ADR OF RETURN POINT
R15     EQU 15         ADR OF ENTRY POINT/RETURN CODE
        TITLE '        SAMPLE CART-ENTRY EXIT WITH CA-TAPE-MGMT'
CBRUXENT CSECT ,
CBRUXENT AMODE 31
CBRUXENT RMODE ANY
        SAVE (14,12),,,'CBRUXENT&SYSDATE'
        LR R12,R15          ESTABLISH PGM BASE
        USING CBRUXENT,R12
        USING WSTO,R13
        LR R11,R1          ESTABLISH BASE FOR PARMLIST
        SPACE 1
        L R0,STORPARM
        GETMAIN R,LV=(0)
        SPACE 1
        ST R13,4(,R1)      *****CXNNNNNN
        LR R13,R1          *                  CXNNNNNN
        LA R0,SAVE          *          DO          *
        ST R0,8(,R13)      *  SAVEAREA CHAINING  *
        LR R13,R0          *****
        USING UXEPL,R11
        EJECT
*****
*          MAINLINE LOGIC          *
*****
        MVC WKVOLSER,UXEVOLSR
ASKCA   CATAPE QUERY_STATUS,VOLSER=WKVOLSER,MF=(E,LQUERY),      +
        SCRATCH=RC0,      +
        PROTECT=RC4,      +
        FOREIGN=RC16,      +
        OUTSERV=RC12,      +
        ERROR=RC20
        EJECT
RCO     DS OH
        MVI UXEUSEA,UXESCRT      SET VOLUME TO SCRATCH
        LA R15,4                ALLOW ENTRY WITH CHG FIELDS
*----- THIS COMPARE IS PROBABLY NOT NEEDED
        CLC UXEGROUP,BLANK      IF GROUP-CODE = BLANKS
        BNE EXIT                THEN EXIT
*-----
        MVC UXEGROUP,SCRATCH    ELSE FORCE TO SCRATCH
        B EXIT
        SPACE 2

```

Figure 257 (Part 3 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

RC4	DS	OH	
	MVI	UXEUSEA,UXEPRIV	SET VOLUME TO PRIVATE
	LA	R15,4	ALLOW ENTRY WITH CHG FIELDS
	CLC	UXEGROUP,SCRTCH	IF GROUP-CODE $\neq$ SCRATCH
	BNE	EXIT	THEN EXIT
	MVC	UXEGROUP,BLANK	ELSE FORCE TO BLANKS
	B	EXIT	
	SPACE	2	
RC8	DS	OH	
	MVI	UXEUSEA,UXEPRIV	SET VOLUME TO PRIVATE
	LA	R15,4	ALLOW ENTRY WITH CHG FIELDS
	CLC	UXEGROUP,SCRTCH	IF GROUP-CODE $\neq$ SCRATCH
	BNE	EXIT	THEN EXIT
	MVC	UXEGROUP,BLANK	ELSE FORCE TO BLANKS
	B	EXIT	
	SPACE	2	
RC12	DS	OH	
	LA	R15,8	DO NOT ALLOW ENTRY
*			EJECT TAPE IMMEDIATELY
	B	EXIT	
	SPACE	2	
RC16	DS	OH	
	LA	R15,12	IGNORE,
*			LEAVE FOR SOMEONE ELSE
	B	EXIT	
	SPACE	2	
RC20	DS	OH	
	LA	R15,16	DO NOT INVOKE AGAIN
	B	EXIT	
	EJECT		
EXIT	DS	OH	
	L	R0,STORPARM	
	LR	R1,R13	
	L	R13,SAVE+4	R13 --> OLD SA
	FREEMAIN	R,A=(1),LV=(0)	
	EJECT		
	LR	R5,R15	SAVE CA-1'S RETURN CODE ACROSS IBM CODE

Figure 257 (Part 4 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
*                                                                 *
*   CBRUXENT MAINLINE PROGRAM FROM VANILLA CBRUXENT             *
*                                                                 *
*****
      SPACE 1
*****
*   GETMAIN WORK AREA                                           *
*****
      SPACE 1
GETMAIN1 DS    OH
          BASR  R12,0              SET R12 AS PROGRAM BASE
          USING *,R12
          LR    R4,R1              SAVE INPUT PARAMETER ADDRESS
          LA    R0,GMAREAL         LENGTH OF AREA TO GETMAIN
      SPACE 1
          GETMAIN RU,LV=(0),SP=230 @003
      SPACE 1
          ST    R13,4(,R1)         LINK CALLER'S SAVE AREA
          ST    R1,8(,R13)         LINK NEW SAVE AREA
          LR    R13,R1             USE NEW SAVE AREA
          USING GMAREA,R13         USE DSECT FOR WORK AREA
          EJECT
*****
*   INITIALISE PARAMETER LIST FOR EDGLCSUX                       *
*****
          MVC   LCSUPWA(LCSUP_LEN#),LCSUP_HDR @002
          LA    R11,LCSUPWA        AREA WITHIN OUR WORKAREA
          USING LCSUP_HDR,R11      SET ADDRESSABILITY FOR LCSUP
          ST    R4,LCSUP_LCSPL     STORE ENTRY PLIST ADDRESS
*****
*   SET THE APPROPRIATE FUNCTION FLAG ACCORDING TO OUR EXIT NAME @004*
*****
          OI    LCSUP_FUNCTION,LCSUP_ENT
          USING UXEPL,R4           ADDRESS PARAMETER LIST
* EXAMINE/MODIFY PARMS
          DROP  R4                 RESET ADDRESSABILITY
      SPACE 1
LINKCSUX LR    R1,R11             PARAMETER LIST FOR EDGLCSUX
          LINK  EP=EDGLCSUX,ERRET=ERROR
      SPACE 1
          CH    R15,=H'12'         ENVIRONMENT ERROR?
          LA    R15,UXEFAIL        ASSUME YES - SET FAIL REQUEST @004
          BE    FREEMAIN           IF ERROR, EXIT WITH FAILURE
          L     R15,LCSUP_LCSRC    OTHERWISE, TAKE LCSUX RETURNED CODE
          B     FREEMAIN           RETURN TO CALLER

```

Figure 257 (Part 5 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
*   HANDLE FAILURE TO LINK TO EDGLCSUX                               *
*****
ERROR   DS      OH                      HANDLE LINK ERROR
        WTO     'LINK TO EDGLCSUX FAILED'
        LA      R15,UXEFAIL             ASSUME YES - SET FAIL REQUEST  @004
        EJECT
*****
*   FREEMAIN THE WORK AREA                                           *
*****
        SPACE 1
FREEMAIN LR      R10,R15                SAVE RETURN CODE ACROSS FREEMAIN
        LR      R1,R13                  A(GETMAINED AREA)
        L       R13,SAVEAREA+4         LOAD A(CALLER'S SAVE AREA)
        LA      R0,GMAREAL              SIZE OF GETMAINED AREA
        FREEMAIN RU,A=(1),LV=(0),SP=230  FREE THE AREA                    @003
        SPACE 1
*****
*   RETURN TO LCS WITH RETURN CODE SET BY EDGLCSUX                  *
*****
        SPACE 1
RETURN  DS      OH
        LR      R15,R5  RESTORE CA-1'S RETURN CODE AFTER IBM CODE
        RETURN  (14,12),RC=(15)
        SPACE 3
        LTORG ,
        DS      OF                      Realign for CA-1 areas
STORPARM DC     AL1(230),AL3(4096)      4K FROM SUB-POOL 230
BLANK    DC     CL8' '                  CONSTANT BLANK
SCRATCH  DC     CL8' *SCRATCH*'         CONSTANT FOR SRATCH
        SPACE 2
*****
*   TEMPLATE FOR EDGLCSUP PARAMETER LIST                            *
*****
        EDGLCSUP DSECT=NO
        EJECT
*****
*   DSECT DEFINITIONS                                               *
*****
*       CBRUXEPL ,                      duplicate
*       SPACE 1                        duplicate

```

Figure 257 (Part 6 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

```

*****
*   WORK AREA                                     *
*****
          SPACE 1
GMAREA   DSECT
SAVEAREA DS    18F                THIS PROGRAM'S SAVE AREA
LCSUPWA  DS    XL(LCSUP_LEN#)     RESERVE LCSUP WORK AREA
GMAREAL  EQU    *-GMAREA          LENGTH OF WORK AREA
*----- POINTED TO BY R13
WSTO     DSECT
SAVE     DC    18F'0'
WKVOLSER DC    CL6' '
LQUERY   CATAPE QUERY_STATUS,VOLSER=WKVOLSER,MF=L
          END    CBRUXENT

```

Figure 257 (Part 7 of 7). Sample TLMS - DFSMSrmm Merged ARCTVEXT

D.2 Sample Jobs

This section contains sample execution JCL for executing some useful DFSMSrmm functions.

D.2.1 Synchronize BTLS and DFSMSrmm CDS Status

This example assumes the following DFSMSrmm VLPOOL definitions using two BTLS-managed scratch pools:

```

VLPOOL PREFIX(10*) TYPE(S) MEDIANAME(CARTS) NAME(SCRTCH1) -
    DESCRIPTION('BTLS pool 1') EXPDTCHECK(N)
VLPOOL PREFIX(2*) TYPE(S) MEDIANAME(3590) NAME(SCRTCH7) -
    DESCRIPTION('BTLS pool 7') EXPDTCHECK(N)
VLPOOL PREFIX(*) TYPE(S) MEDIANAME(TAPE) -
    DESCRIPTION('default pool') EXPDTCHECK(N)

```

The sample JCL in Figure 258 shows how to use DFSMSrmm search commands to build lists of volumes by scratch pools and status and use the volume lists as input to the IDCAMS LIBRARY command to update BTLS volume status.

```

//SYNCBTLS PROC
//TMP      EXEC PGM=IKJEFT01
//SYSTSPRT DD SYSOUT=*
//*
//AMS      EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//LIBIN    DD DISP=SHR,DSN=userid.EXEC.RMM.CLIST
// PEND

```

Figure 258 (Part 1 of 2). BTLS Status Synchronization

```

//S1 EXEC SYNCHBTLS
//TMP.SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(10*) OWNER(*) LIMIT(*) STATUS(SCRATCH) -
  NOLIST CLIST
/*
//AMS.SYSIN DD *
  LIBRARY SETCAT UNIT(123) CATEGORY(SCRTCH1)
/*
//S7 EXEC SYNCHBTLS
//TMP.SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(2*) OWNER(*) LIMIT(*) STATUS(SCRATCH) -
  NOLIST CLIST
/*
//AMS.SYSIN DD *
  LIBRARY SETCAT UNIT(123) CATEGORY(SCRTCH7)
/*
//SM EXEC SYNCHBTLS
//TMP.SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) OWNER(*) LIMIT(*) STATUS(MASTER) -
  LOCATION(SHELF) INTRANSIT(N) NOLIST CLIST
/*
//AMS.SYSIN DD *
  LIBRARY SETCAT UNIT(123) CATEGORY(PRIVATE)
/*
//EJECT EXEC SYNCHBTLS
//TMP.SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) OWNER(*) LIMIT(*) DEST(VAULT1) -
  LOCATION(SHELF) INTRANSIT(Y) NOLIST CLIST
/*
//AMS.SYSIN DD *
  LIBRARY SETCAT UNIT(123) CATEGORY(XEJECTB)
/*

```

Figure 258 (Part 2 of 2). BTLS Status Synchronization

By customizing the DFSMSrmm command, you can select different volumes if necessary, so that BTLS processes the selected group of volumes to the category you want. Similarly, you can use DFSMSrmm commands to build eject lists.

D.2.2 EDGHSKP Inventory Management example JCL

Figure 259 on page 633 shows an example of JCL for EDGHSKP Inventory Management.


```

//LCLHSKP JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//*
//* *****
//* 5706-110 (C) COPYRIGHT IBM CORP. 1992, 1993 *
//* 5706-110 (C) COPYRIGHT IBM DEUTSCHLAND *
//*          SYSTEME UND NETZE GMBH 1993, 199x *
//* ----- *
//* *
//* IBM Deutschland Systeme und Netze GmbH *
//* SN Internationale Projekte (Dept. 3331) *
//* Am Keltenwald 1 *
//* 71139 Ehningen *
//* *
//* ***** *
//STEP01 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) ACTION(SCRATCH) LIMIT(*) OWNER(*)
/*
//* *****
//STEP02 EXEC PGM=EDGHSKP,
// PARM='VRSEL,VERIFY,EXPROC,DSTORE,BACKUP(DSS),RPTEXT,DATEFORM(J)'
//SYSPRINT DD SYSOUT=*
//MESSAGE DD DSN=RMM.HSKP.MESSAGE.DSET,DISP=SHR
//REPORT DD DSN=RMM.HSKP.VRSEL(0),DISP=SHR
//ACTIVITY DD DSN=RMM.HSKP.ACTIVITY(0),DISP=SHR
//REPTXT DD DSN=RMM.HSKP.EXTRACT(0),DISP=SHR
//BACKUP DD DSN=RMM.HSKP.BACKUP(+1),DISP=(,CATLG,DELETE),
//          SPACE=(TRK,(150,10),RLSE),
//          DCB=(GDG,LRECL=9000,BLKSIZE=23476,RECFM=VB)
//JRNLBKUP DD DSN=RMM.HSKP.JRNLBKUP(+1),DISP=(,CATLG,DELETE),
//          SPACE=(TRK,(30,10),RLSE),
//          DCB=(GDG,LRECL=9000,BLKSIZE=23476,RECFM=VB)
//* *****
//STEP03 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.HSKP.MESSAGE.DSET,DISP=SHR
//SYSUT2 DD SYSOUT=*
//SYSIN DD DUMMY
//* *****
//STEP04 EXEC PGM=IEFBR14
//DD1 DD DSN=RMM.HSKP.VRSEL(+1),DISP=(,CATLG,DELETE),
//          SPACE=(TRK,(5,1)),DCB=GDG,
//DD2 DD DSN=RMM.HSKP.ACTIVITY(+1),DISP=(,CATLG,DELETE),
//          SPACE=(TRK,(5,1)),DCB=GDG,

```

Figure 259 (Part 1 of 2). Example EDGHSKP Inventory Management JCL

```

/*          *****
//STEP05 EXEC PGM=IEFBRI4
//DD2 DD DSN=RMM.HSKP.EXTRACT(+1),DISP=(,CATLG,DELETE),
// SPACE=(TRK,(150,10)),
// DCB=(GDG,LRECL=719,BLKSIZE=23476,RECFM=VB)
/*          *****
//STEP06 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD DSN=RMM.HSKP.CONTROL(+1),DISP=(,CATLG,DELETE),
// SPACE=(TRK,(1,1),RLSE),
// DCB=(GDG,LRECL=137,BLKSIZE=23476,RECFM=VBA)
//SYSTSIN DD *
RMM LISTCONTROL ALL
/*
/*          *****
//STEP07 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM SEARCHVOLUME VOLUME(*) ACTION(ALL) LIMIT(*) OWNER(*)
/*
/*          *****
//STEP12 EXEC PGM=IKJEFT01
//SYSPRINT DD SYSOUT=*
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
RMM CHANGEVOLUME * CONFIRMMOVE(ALL,ALL)
/*
/*          *****
//STEP13 EXEC PGM=EDGHSKP,PARM='DSTORE'
//SYSPRINT DD SYSOUT=*
//MESSAGE DD DSN=RMM.HSKP.MESSAGE.DSET,
// DISP=SHR
/*          *****
//STEP14 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.HSKP.MESSAGE.DSET,DISP=SHR
//SYSUT2 DD SYSOUT=*
//SYSIN DD DUMMY
/*          *****

```

Figure 259 (Part 2 of 2). Example EDGHSKP Inventory Management JCL

D.2.3 Sample JCL for Reports

Figure 260 on page 635 shows sample JCL to create DFSMSrmm supplied reports.

```

//LCLRPTE JOB ,RMM,NOTIFY=SCHLUM,
//          MSGCLASS=H,CLASS=A,MSGLEVEL=(1,1),REGION=6M
//*
//* *****
//* 5706-110 (C) COPYRIGHT IBM CORP. 1992, 1993 *
//* 5706-110 (C) COPYRIGHT IBM DEUTSCHLAND      *
//*                               SYSTEME UND NETZE GMBH 1993, 199x *
//* ----- *
//* *
//* IBM Deutschland Systeme und Netze GmbH      *
//* SN Internationale Projekte (Dept. 3331)      *
//* Am Keitenwald 1                             *
//* 71139 Ehningen                             *
//* *
//* *****
//STEP01 EXEC PGM=EDGHSKP,PARM='RPTTEXT'
//SYSPRINT DD SYSOUT=*
//MESSAGE DD DSN=RMM.HSKP.MESSAGE.DSET,DISP=SHR
//REPTXT DD DSN=RMM.HSKP.EXTRACT(0),DISP=SHR
//* *****
//STEP02 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=RMM.HSKP.MESSAGE.DSET,DISP=SHR
//SYSUT2 DD SYSOUT=*
//SYSIN DD DUMMY
//* *****
//STEP03 EXEC PGM=IEFB14
//DD2 DD DSN=RMM.HSKP.EXTRACT(+1),DISP=(,CATLG,DELETE),
//      SPACE=(TRK,(150,10)),
//      DCB=(GDG,LRECL=573,BLKSIZE=23476,RECFM=VB)
//* *****
//SORTVOL EXEC PGM=SORT,REGION=6M
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.EXTRACT(0)
//SORTOUT DD DSN=&SORTVOL,DISP=(,PASS,DELETE),
//           SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//SYSIN DD *
//          SORT FIELDS=(9,6,CH,A)
//          OMIT COND=(5,4,CH,NE,C' V ' )
//*

```

Figure 260 (Part 1 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

/*          *****
//SORTDSN EXEC PGM=SORT,REGION=6M
//SYSPRINT DD  SYSOUT=*
//SORTWK01 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN  DD  DISP=SHR,DSN=RMM.HSKP.EXTRACT(0)
//SORTOUT DD  DSN=&SORTDSN,DISP=(,PASS,DELETE),
//          SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT   DD  SYSOUT=*
//SYSIN    DD  *
          SORT FIELDS=(109,6,CH,A)
          OMIT COND=(5,4,CH,NE,C'D  ')
/*
//          *****
//EXT EXTR EXEC PGM=IKJEFT01,DYNAMNBR=99,REGION=4096K
//SYSTSPRT DD  SYSOUT=*
//SYSPRINT DD  SYSOUT=*
//DSN      DD  DSN=&SORTDSN,DISP=(OLD,PASS)
//VOL      DD  DSN=&SORTVOL,DISP=(OLD,PASS)
//EXT EXTR DD  DSN=RMM.HSKP.LCLEXTR(+1),DISP=(,CATLG),
//          SPACE=(TRK,(200,20),RLSE),
//          DCB=(GDG,LRECL=1200,BLKSIZE=6000,RECFM=FB)
//SYSTSIN  DD  *
          EX 'SYS1.SEDGEX1(EDGRRPTR)'
/*

```

Figure 260 (Part 2 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

/*          *****
//SORT01  EXEC PGM=SORT,REGION=6M
/* *****
/* * CREATING A PULL LIST FOR SCRATCH TAPES INCL. DATA SET NAMES *
/* *****
//SYSPRINT DD  SYSOUT=*
//SORTWK01 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN  DD  DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD  DSN=&TEMP01,DISP=(,PASS,DELETE),
//          SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT  DD  SYSOUT=*
/*          *****
/*          * sorted by volser *
/*          * ----- *
/*          * include only status = SCRATCH and data set sequence LT 2*
/*          * ----- *
//SYSIN  DD  *
          SORT FIELDS=(5,6,CH,A)
          INCLUDE COND=(318,8,CH,EQ,C' SCRATCH ',
                        AND,911,4,CH,LT,C' 2')
/*
//SORT02  EXEC PGM=SORT,REGION=6M
/* *****
/* * CREATING A PULL LIST FOR SCRATCH TAPES INCL. DATA SET NAMES *
/* *****
//SYSPRINT DD  SYSOUT=*
//SORTWK01 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD  SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN  DD  DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD  DSN=&TEMP02,DISP=(,PASS,DELETE),
//          SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT  DD  SYSOUT=*
/*          *****
/*          * sorted by data set name and volume serial number *
/*          * ----- *
/*          * include only status = SCRATCH and data set sequence LT 2*
/*          * ----- *
//SYSIN  DD  *
          SORT FIELDS=(805,44,CH,A,5,6,CH,A)
          INCLUDE COND=(318,8,CH,EQ,C' SCRATCH ',
                        AND,911,4,CH,LT,C' 2')
/*

```

Figure 260 (Part 3 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT03 EXEC PGM=SORT,REGION=6M
//* *****
//* * CREATING AN INVENTORY LIST SORTED BY VOLUME SERIAL NUMBERS *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMPO3,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by volser and physical data set sequence number *
//* * ----- *
//* * include all records *
//* *****
//SYSIN DD *
SORT FIELDS=(5,6,CH,A,911,4,CH,A)
/*
//SORT04 EXEC PGM=SORT,REGION=6M
//* *****
//* * CREATING AN INVENTORY LIST SORTED BY DATA SET NAME *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMPO4,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by data set name, create date and create time *
//* * ----- *
//* * exclude all volumes without any data set name (blank) *
//* *****
//SYSIN DD *
SORT FIELDS=(805,44,CH,A,849,16,CH,A)
OMIT COND=(805,44,CH,EQ,C' ')
/*

```

Figure 260 (Part 4 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT05 EXEC PGM=SORT,REGION=6M
//* *****
//* * INVENTORY OF DATA SETS INCLUDING USED KILOBYTES *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP05,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by data set name, create date and create time *
//* * ----- *
//* * exclude all volumes without any data set name (blank) *
//* * ----- *
//SYSIN DD *
        SORT FIELDS=(805,44,CH,A,849,16,CH,A)
        INCLUDE COND=(805,44,CH,NE,C' ')
/*
//SORT06 EXEC PGM=SORT,REGION=6M
//* *****
//* *CREATING AN INVENTORY LIST OF ALL STORAGE LOCATION WITHOUT SHELF*
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP06,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by data set name, create date and create time *
//* * ----- *
//* * include only volumes in a storage location *
//* * ----- *
//SYSIN DD *
        SORT FIELDS=(152,8,CH,A,5,6,CH,A)
        INCLUDE COND=(579,1,CH,EQ,C' S',
                        OR,579,1,CH,EQ,C' U')
/*

```

Figure 260 (Part 5 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT07 EXEC PGM=SORT,REGION=6M
/* *****
/* * Inventory of Data Set Names in Storage Locations *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP07,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * sorted by store ID, data set name and cr-date and time *
/* * ----- *
/* * include only volumes in a storage location *
/* *****
//SYSIN DD *
SORT FIELDS=(152,8,CH,A,805,44,CH,A,849,16,CH,A)
INCLUDE COND=(579,1,CH,EQ,C' S',
OR,579,1,CH,EQ,C' U')
/*
//SORT08 EXEC PGM=SORT,REGION=6M
/* *****
/* * Inventory of BIN by location incl. data set name *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP08,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * sorted by store ID, bin and data set sequence *
/* * ----- *
/* * include only volumes in a storage location *
/* *****
//SYSIN DD *
SORT FIELDS=(152,8,CH,A,168,6,CH,A,180,10,CH,A,805,44,CH,A)
INCLUDE COND=(579,1,CH,EQ,C' S',
OR,579,1,CH,EQ,C' U')
/*

```

Figure 260 (Part 6 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports


```

//SORT09 EXEC PGM=SORT,REGION=6M
//* *****
//* * Inventory of Data Set Names in LOAN Locations *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP09,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by loan location, data set name and cr-date/-time*
//* * ----- *
//* * include only volumes ON LOAN *
//* *****
//SYSIN DD *
SORT FIELDS=(200,8,CH,A,805,44,CH,A,849,16,CH,A)
OMIT COND=(200,8,CH,EQ,C' ')
/*
//SORT10 EXEC PGM=SORT,REGION=6M
//* *****
//* * Inventory of Volumes in LOAN Locations *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP10,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by loan location, volser and data set sequence *
//* * ----- *
//* * include only volumes ON LOAN *
//* *****
//SYSIN DD *
SORT FIELDS=(200,8,CH,A,5,6,CH,A,911,4,CH,A)
OMIT COND=(200,8,CH,EQ,C' ')
/*

```

Figure 260 (Part 7 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT11A EXEC PGM=SORT,REGION=6M
/* *****
/* * Multit volume / multi data set *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP11A,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * sorted by multi dset multi volume id, volume sequence, *
/* * volume serial number and data set sequence *
/* * ----- *
/* * include only multi volumes and/or multi file volumes *
/* *****
//SYSIN DD *
        SORT FIELDS=(29,8,CH,A,236,16,CH,A,314,4,CH,A,911,4,CH,A)
        INCLUDE COND=(130,4,CH,GT,C' 1',
                        OR,11,12,CH,NE,C' ')
/*
//EXTEXTRM EXEC PGM=IKJEFT01,DYNAMNBR=99,REGION=4096K
/* *****
/* * expand extended extract file *
/* *****
//SYSTSPRT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//EXTEXTRI DD DSN=&TEMP11A,DISP=(OLD,DELETE)
//EXTEXTRO DD DSN=&LCLEXTRM,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),
// DCB=(RECFM=FB,LRECL=1244,BLKSIZE=22392)
//SYSTSIN DD *
        EX 'SYS1.SEDGEX1(EDGRRPTM)'
/*

```

Figure 260 (Part 8 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT11B EXEC PGM=SORT,REGION=6M
//* *****
//* * Multit volume / multi data set *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=(OLD,DELETE),DSN=&LCLEXTRM
//SORTOUT DD DSN=&TEMP11B,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by first file on first volume, multi dset multi *
//* * volume ID, al number and data set sequence *
//* * ----- *
//* * include all records selected in step STEP11A *
//* *****
//SYSIN DD *
SORT FIELDS=(1201,44,CH,A,29,8,CH,A,314,4,CH,A,911,4,CH,A)
/*
//SORT12 EXEC PGM=SORT,REGION=6M
//* *****
//* * CREATING A MOVEMENT LIST INCLUDE THE FIRST FILE ON VOLUME *
//* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP12,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
//* * sorted by next location name, location name, data set *
//* * name, create date and create time *
//* * ----- *
//* * exclude all volumes without any destination (blank) *
//* *****
//SYSIN DD *
SORT FIELDS=(160,8,CH,A,152,8,CH,A,805,44,CH,A,849,16,CH,A)
INCLUDE COND=(160,8,CH,NE,C' ')
/*

```

Figure 260 (Part 9 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT13 EXEC PGM=SORT,REGION=6M
/* *****
/* * CREATING A MOVEMENT LIST SORTED BY STORAGE LOCATION BIN NUMBER *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP13,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * sorted by next location name, location name, BIN number *
/* * ----- *
/* * exclude all volumes without any destination (blank) and *
/* * data set sequence greather one *
/* *****
//SYSIN DD *
SORT FIELDS=(160,8,CH,A,152,8,CH,A,168,6,CH,A,174,6,CH,A)
INCLUDE COND=(160,8,CH,NE,C' ',
AND,911,4,CH,LT,C' 2')
/*
//SORT14 EXEC PGM=SORT,REGION=6M
/* *****
/* * CREATING A MOVEMENT LIST INCLUDE THE FIRST FILE ON VOLUME *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(10,10)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP14,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * sorted by next location name, location name, VOLSER *
/* * ----- *
/* * exclude all volumes without any destination (blank) and *
/* * data set sequence greather one *
/* *****
//SYSIN DD *
SORT FIELDS=(160,8,CH,A,152,8,CH,A,5,6,CH,A)
INCLUDE COND=(160,8,CH,NE,C' ',
AND,911,4,CH,LT,C' 2')
/*

```

Figure 260 (Part 10 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT15A EXEC PGM=SORT,REGION=6M
/* *****
/* * VOLUME / DATA SET INVENTORY LIST *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTIN DD DISP=SHR,DSN=RMM.HSKP.LCLEXTR(+1)
//SORTOUT DD DSN=&TEMP15A,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
/* *****
/* * SORTED BY DATA SET NAME, MULTI DSET MULTI VOLUME ID, *
/* * VOLUME SEQUENCE, DATA SET SEQUENCE *
/* *****
//SYSIN DD *
SORT FIELDS=(805,44,CH,A,29,8,CH,A,236,16,CH,D,314,4,CH,D,911,4,CH,D)
/*
//LCLEXTRN EXEC PGM=IKJEFT01,DYNAMNBR=99,REGION=4096K
/* *****
/* * EXPAND EXTENDED EXTRACT FILE ADD VOLUME COUNT *
/* *****
//SYSTSPRT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
//EXTTEXTRI DD DSN=&TEMP15A,DISP=(OLD,DELETE)
//EXTTEXTRO DD DSN=&LCLEXTRN,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),
// DCB=(RECFM=FB,LRECL=1244,BLKSIZE=22392)
//SYSTSIN DD *
EX 'SYS1.SEDGEX1(EDGRRPTN)'
/*

```

Figure 260 (Part 11 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//SORT15B EXEC PGM=SORT,REGION=6M
//* *****
/* * VOLUME / DATA SET INVENTORY LIST *
/* *****
//SYSPRINT DD SYSOUT=*
//SORTWK01 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTWK04 DD SPACE=(CYL,(11,11)),UNIT=SYSDA
//SORTIN DD DISP=(OLD,DELETE),DSN=&LCLEXTRN
//SORTOUT DD DSN=&TEMP15B,DISP=(,PASS,DELETE),
// SPACE=(TRK,(200,20),RLSE),DCB=*.SORTIN
//SYSOUT DD SYSOUT=*
//* *****
/* * SORTED BY VOLUME ID, DATA SET SEQUENCE *
/* *****
//SYSIN DD *
SORT FIELDS=(5,6,CH,A,911,4,CH,A)
/*
//EXTRPDT EXEC PGM=IKJEFT01,DYNAMNBR=99,REGION=4096K
//OUTDDQ OUTPUT FORMDEF=A10111, ** PRINT ON BOTH SIDES
// PAGEDEF=D4Q8, ** TEXT WILL BE ROTATED 90 DEGREES
// ** AND PRINTED WITH 8 LINES P. INCH
// CHARS=GT12, ** FONT
// DEST=STUTVS.PRT3935
//SYSTSPRT DD SYSOUT=*
//SYSPRINT DD SYSOUT=*
/* ***** ALLOCATION FOR REPORT01 *****
//SORT01 DD DSN=&TEMP01,DISP=(OLD,PASS)
//REPORT01 DD DISP=(,CATLG,DELETE),
// DSN=RMM.REPORT.EXTENDED.REPORT01(+1),
// SPACE=(TRK,(150,10)),
// DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/* ***** ALLOCATION FOR REPORT02 *****
//SORT02 DD DSN=&TEMP02,DISP=(OLD,PASS)
//REPORT02 DD DISP=(,CATLG,DELETE),
// DSN=RMM.REPORT.EXTENDED.REPORT02(+1),
// SPACE=(TRK,(150,10)),
// DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/* ***** ALLOCATION FOR REPORT03 *****
//SORT03 DD DSN=&TEMP03,DISP=(OLD,PASS)
//REPORT03 DD DISP=(,CATLG,DELETE),
// DSN=RMM.REPORT.EXTENDED.REPORT03(+1),
// SPACE=(TRK,(150,10)),
// DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/* ***** ALLOCATION FOR REPORT04 *****
//SORT04 DD DSN=&TEMP04,DISP=(OLD,PASS)
//REPORT04 DD DISP=(,CATLG,DELETE),
// DSN=RMM.REPORT.EXTENDED.REPORT04(+1),
// SPACE=(TRK,(150,10)),
// DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)

```

Figure 260 (Part 12 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

//*          ***** ALLOCATION FOR REPORT05 *****
//SORT05 DD DSN=&TEMP05,DISP=(OLD,PASS)
//REPORT05 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT05(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT06 *****
//SORT06 DD DSN=&TEMP06,DISP=(OLD,PASS)
//REPORT06 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT06(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT07 *****
//SORT07 DD DSN=&TEMP07,DISP=(OLD,PASS)
//REPORT07 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT07(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT08 *****
//SORT08 DD DSN=&TEMP08,DISP=(OLD,PASS)
//REPORT08 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT08(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT09 *****
//SORT09 DD DSN=&TEMP09,DISP=(OLD,PASS)
//REPORT09 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT09(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT10 *****
//SORT10 DD DSN=&TEMP10,DISP=(OLD,PASS)
//REPORT10 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT10(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT11 *****
//SORT11 DD DSN=&TEMP11B,DISP=(OLD,PASS)
//REPORT11 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT11(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
//*          ***** ALLOCATION FOR REPORT12 *****
//SORT12 DD DSN=&TEMP12,DISP=(OLD,PASS)
//REPORT12 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT12(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)

```

Figure 260 (Part 13 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

```

/*          ***** ALLOCATION FOR REPORT13 *****
//SORT13 DD DSN=&TEMP13,DISP=(OLD,PASS)
//REPORT13 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT13(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/*          ***** ALLOCATION FOR REPORT14 *****
//SORT14 DD DSN=&TEMP14,DISP=(OLD,PASS)
//REPORT14 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT14(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/*          ***** ALLOCATION FOR REPORT14 *****
//SORT15 DD DSN=&TEMP15,DISP=(OLD,PASS)
//REPORT15 DD DISP=(,CATLG,DELETE),
//          DSN=RMM.REPORT.EXTENDED.REPORT15(+1),
//          SPACE=(TRK,(150,10)),
//          DCB=(GDG,LRECL=133,BLKSIZE=23476,RECFM=VB)
/*          ***** ALLOCATION FOR REPORT14 *****
//SYSTSIN DD *
EX 'SYS1.SEDGEX1(EDGRRPTE)' -
'054 Y Y Y Y Y Y Y Y Y Y Y Y +
Y Y Y IBM INTERNAL USE ONLY'
/*

```

Figure 260 (Part 14 of 14). Sample REPORT JCL to Create DFSMSrmm Supplied Reports

D.3 Sample Customized Reports from Extract File using DFSORT

This section contains sample execution JCL for the extract file reports.

D.3.1.1 Sample CHANGEVOLUME Command: Job RMMCVB

Figure 261 on page 649 is a job that produces DFSMSrmm CHANGEVOLUME commands for volumes in a store.


```

//RMMRCVB JOB (SOFTOD,XXXX,XXX,XXX),'RMM CV BIN',
//      MSGCLASS=H,MSGLEVEL=(1,1),NOTIFY=STSGWD,
//      CLASS=S,REGION=1M,TIME=(5,0),USER=STSGWD
//*****
/* BUILD 'RMM CV volser LOCATION(store) BIN(bin number)
/*
/* To select the store name to use you must edit the
/* SORT INCLUDE statement and specify the correct location name
/* for offset 156.
/* OUTPUT:
/*      RMMCVB DD STATEMENT - commands sorted by bin number
/*      RMMCVBS DD STATEMENT - COUNT BY store name
//*****
//CLEAN EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
      DELETE 'STSGWD.CMDS.CVB'
      DELETE 'STSGWD.REPT.CVBS'
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=* ICETOOL MESSAGES
//DFSMSG DD SYSOUT=* DFSORT MESSAGES
//TOOLIN DD * CONTROL STATEMENTS
      COPY FROM(EXTRACT) TO(VOLS) USING(VOLS)
      SORT FROM(VOLS) TO(RMMCVB) USING(VOLF)
      OCCUR FROM(VOLS) LIST(RMMCVBS) -
        TITLE(' VOLUME COUNTS IN STORE') DATE TIME PAGE -
        BLANK -
        HEADER(' LOCATION') ON(156,8,CH) -
        HEADER(' COUNT') ON(VLCNT)
//EXTRACT DD DSN=RMM.RPTEXT,DISP=SHR
//VOLS DD DSN=&&TEMP,REFDD=*.EXTRACT,SPACE=(CYL,(10,10))
//VOLSCNTL DD * DFSORT STATEMENTS - SELECT STORE VOLUMES
      INCLUDE COND=(5,1,CH,EQ,C'V',AND,
        (156,8,CH,EQ,C'REMOTE ',OR,156,8,CH,EQ,C'LOCAL ',OR,
        156,8,CH,EQ,C'DISTANT '), store names
        AND,164,8,CH,EQ,C' ') no destination
      OPTION VLSHRT
//VOLFCNTL DD * DFSORT STATEMENTS - SORT AND REFORMAT
* sort on bin number field
      SORT FIELDS=(172,6,CH,A) RVSTBIN

```

Figure 261 (Part 1 of 2). Sample RMMCVB Execution JCL

```

*
* The following OUTREC reformats the input records based
* on the record layout
* described in MACRO EDGRVEXT - and shown in DFSMSrmm Installation&
* Customization manual Appendix.
* All offsets into the record must have 5 added to them
* because of the RDW (variable length record) and that DFSORT
* uses the position rather than the offset.
*
  OUTREC FIELDS=(1,4,          RDW
                  C'RMM CV ',  start command
                  9,6,          RRVOLSER
                  C' LOCATION(',
                  156,8,         RVSTORID
                  C') BIN(',
                  172,6,         RVSTBIN
                  C') ')
//RMMCVB  DD DSN=STSGWD.CMDB,DISP=(,CATLG,DELETE),
//  LRECL=121,DSORG=PS,RECFM=VB,AVGREC=K,SPACE=(121,(9,9),RLSE)
//RMMCVBS DD DSN=STSGWD.REPT.CVBS,DISP=(,CATLG,DELETE),
//  LRECL=121,DSORG=PS,RECFM=FB,AVGREC=K,SPACE=(121,(9,9),RLSE)

```

Figure 261 (Part 2 of 2). Sample RMMCVB Execution JCL

D.3.2 Audit Reports

The jobs presented below can be used to create audit reports.

D.3.2.1 Archiving Audit Reports: Job RMMAUDIB

Figure 262 on page 651 shows a job that creates a weekly archive from daily audit reports.

```

//RMAUDIT JOB ,
//      MSGCLASS=H,MSGLEVEL=(1,1),
//      REGION=4096K,USER=&SYSUID,NOTIFY=&SYSUID,TIME=10
//*****
//*      RUN ONCE A WEEK
//* AUDIT DATA IS SORTED BY VOLUME THEN DATE SO THAT ACTIONS AGAINST
//* A VOLUME CAN BE TRACED FROM TAPE CREATION UNTIL TAPE DELETION
//*      REMEMBER TO CREATE THE 3 GDGS FOR THE WEEKLY CONSOLIDATION REPORT
//*
//*      RMAUD2 WILL ARCHIVE WEEKLY REPORTS INTO A MONTHLY ARCHIVE
//*      AUDIT DATA WILL NOT BE SAVED MORE THAN ONE YEAR.
//*****
//SORTDSP EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=* ICETOOL MESSAGES
//DFSMSG DD SYSOUT=* DFSORT MESSAGES
//TOOLIN DD *          CONTROL STATEMENTS
*****
* SEPARATE THE 3 AUDIT REPORTS AND SORT TO CREATE TAPE HISTORY TRAIL
*      VOLUME REPORT, RACK REPORT, USER REPORT
*****
      SORT FROM(COLLECT) TO(VOEXT) USING(VEXT)
      SORT FROM(COLLECT) TO(RACKEXT) USING(REXT)
      SORT FROM(COLLECT) TO(USEREXT) USING(UEXT)
*****
* DISPLAY THE VOLUME AUDIT REPORT
*****
      DISPLAY FROM(VOEXT) LIST(VREPT) -
          TITLE('RMM VOLUME AUDIT REPORT CONSOLIDATION') DATE TIME PAGE -
          BLANK -
* MAX LENGTH OF OUTPUT RECORD IS 121 INCLUDING 3 SPACES BETWEEN COLUMNS
* MAX # OF HEADERS IS 10
      HEADER(' VOLUME RACK-#') ON(2,13,CH) -
      HEADER(' OWNER  ') ON(87,8,CH) -
      HEADER(' ACTIVITY') ON(116,8,CH) -
      HEADER(' USERID ') ON(23,8,CH) -
      HEADER(' DATE      TIME  SYS ') ON(32,24,CH) -
      HEADER(' EXP-DATE SEC') ON(96,14,CH) -
      HEADER(' STATUS LOCATION') ON(61,16,CH) -
      HEADER(' LOAN-LOC') ON(78,8,CH)
* FIELDS IGNORED (WOULD PUT VREPT OVER THE 121 RECORD LENGTH)
*      HEADER(' BIN--# ') ON(16,8,CH) -

```

Figure 262 (Part 1 of 3). Sample RMAUDIB Execution JCL

```

*
*****
*  DISPLAY THE RACK AUDIT REPORT
*****
  DISPLAY FROM(RACKEXT) LIST(RREPT) -
    TITLE('RMM RACK AUDIT REPORT CONSOLIDATION') DATE TIME PAGE -
    BLANK -
* MAX LENGTH OF OUTPUT RECORD IS 121 INCLUDING 3 SPACES BETWEEN COLUMNS
* MAX # OF HEADERS IS 10
  HEADER(' RACK/BIN VOLUME') ON(2,15,CH) -
  HEADER(' OWNER  ') ON(87,8,CH) -
  HEADER(' ACTIVITY') ON(116,8,CH) -
  HEADER(' USERID ') ON(23,8,CH) -
  HEADER(' DATE      TIME  SYS ') ON(32,24,CH) -
  HEADER(' EXP-DATE  SEC') ON(96,14,CH) -
  HEADER(' STATUS LOCATION') ON(61,16,CH) -
  HEADER(' LOAN-L') ON(78,6,CH)
* HAD TO CUT LOAN LOCATION BY 2 TO FIT IN RECORD OF 121 (8 CHAR NOT 6)
* FIELDS IGNORED (WOULD PUT RREPT OVER THE 121 RECORD LENGTH)
*
*****
*  DISPLAY THE USER AUDIT REPORT
*****
  DISPLAY FROM(USEREXT) LIST(UREPT) -
    TITLE('RMM USER AUDIT REPORT CONSOLIDATION') DATE TIME PAGE -
    BLANK -
* MAX LENGTH OF OUTPUT RECORD IS 121 INCLUDING 3 SPACES BETWEEN COLUMNS
* MAX # OF HEADERS IS 10
  HEADER(' USERID') ON(2,8,CH) -
  HEADER(' VOLUME RACK-#') ON(11,13,CH) -
  HEADER(' ACTIVITY') ON(116,8,CH) -
  HEADER(' OWNER  ') ON(87,8,CH) -
  HEADER(' DATE      TIME  SYS ') ON(32,24,CH) -
  HEADER(' EXP-DATE  SEC') ON(96,14,CH) -
  HEADER(' STATUS LOCATION') ON(61,16,CH) -
  HEADER(' LOAN-LOC') ON(78,8,CH)
* FIELDS IGNORED (WOULD PUT UREPT OVER THE 121 RECORD LENGTH)
*   HEADER(' BIN--# ') ON(15,8,CH) -
*
//COLLECT DD DSN=DFRMM.REPORT.AUDIT(0),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-1),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-2),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-3),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-4),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-5),DISP=SHR
//        DD DSN=DFRMM.REPORT.AUDIT(-6),DISP=SHR
//VOLEXT  DD DSN=&&VEXT,REFDD=*.COLLECT,SPACE=(CYL,(100,10))
//RACKEXT DD DSN=&&REXT,REFDD=*.COLLECT,SPACE=(CYL,(100,10))
//USEREXT DD DSN=&&UEXT,REFDD=*.COLLECT,SPACE=(CYL,(100,10))

```

Figure 262 (Part 2 of 3). Sample RMMAUDIB Execution JCL

```

//VEXTCNTL DD *          DFSORT STATEMENTS - SORT ON VOLUME
OPTION VLSHRT
INCLUDE COND=(2,1,CH,NE,C' ',AND,    VALID VOLUME RECORD
              7,1,CH,NE,C' ',AND,
              10,1,CH,NE,C' ',AND,
              15,1,CH,EQ,C' ',AND,
              22,1,CH,EQ,C' ',AND,
              42,1,CH,EQ,C' ')
SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A,43,8,CH,A)
/*  SORT BY VOLUME THEN DATE THEN TIME SO CAN READ HISTORY OF TAPE.
/*      SORT BY VOLUME 2,6 THEN BY DATE 38,4 THEN 32,5
/*      AND FINALLY BY TIME 43,8
/*      ALTERNATE SORT BY RACK 9,6
/*      ALTERNATE SORT BY OWNER 87,8
//REXTCNTL DD *          DFSORT STATEMENTS - SORT ON RACK
OPTION VLSHRT
INCLUDE COND=(2,1,CH,NE,C' ',AND,    VALID RACK RECORD
              7,1,CH,NE,C' ',AND,
              10,1,CH,EQ,C' ',AND,
              15,1,CH,NE,C' ',AND,
              22,1,CH,EQ,C' ',AND,
              42,1,CH,EQ,C' ')
SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A,43,8,CH,A)
/*  SORT BY RACK THEN DATE THEN TIME SO CAN READ HISTORY OF TAPE.
/*      SORT BY RACK 2,6 THEN BY DATE 38,4 THEN 32,5
/*      AND FINALLY BY TIME 43,8
/*      ALTERNATE SORT BY VOLUME 11,6
/*      ALTERNATE SORT BY OWNER 87,8
//UEXTCNTL DD *          DFSORT STATEMENTS - SORT ON USER
OPTION VLSHRT
INCLUDE COND=(2,1,CH,NE,C' ',AND,    VALID USER RECORD
              7,1,CH,NE,C' ',AND,
              10,1,CH,EQ,C' ',AND,
              15,1,CH,NE,C' ',AND,
              22,1,CH,NE,C' ',AND,
              42,1,CH,EQ,C' ')
SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A,43,8,CH,A)
/*  SORT BY RACK THEN DATE THEN TIME SO CAN READ HISTORY OF TAPE.
/*      SORT BY USER 2,6 THEN BY DATE 38,4 THEN 32,5
/*      AND FINALLY BY TIME 43,8
/*      ALTERNATE SORT BY VOLUME 11,6
/*      ALTERNATE SORT BY RACK 18,6
//VREPT   DD DSN=DFRMM.REPT.AUDVOLS.WEEK(+1),DISP=(,CATLG,DELETE),
//  LRECL=132,DSORG=PS,RECFM=FA,SPACE=(CYL,(100,10),RLSE),UNIT=SYSDA
//RREPT   DD DSN=DFRMM.REPT.AUDRACK.WEEK(+1),DISP=(,CATLG,DELETE),
//  LRECL=132,DSORG=PS,RECFM=FA,SPACE=(CYL,(100,10),RLSE),UNIT=SYSDA
//UREPT   DD DSN=DFRMM.REPT.AUDUSER.WEEK(+1),DISP=(,CATLG,DELETE),
//  LRECL=132,DSORG=PS,RECFM=FA,SPACE=(CYL,(100,10),RLSE),UNIT=SYSDA
/*

```

Figure 262 (Part 3 of 3). Sample RMMAUDIB Execution JCL

D.3.2.2 Archiving Audit Reports: Job RMMAUD2B

Figure 263 shows a job that creates a monthly archive from weekly audit reports.

```
//RMMAUD2 JOB ,
//      MSGCLASS=H,MSGLEVEL=(1,1),
//      REGION=4096K,USER=&SYSUID,NOTIFY=&SYSUID,TIME=10
//*****
//* RUN ONCE PER MONTH
//* AUDIT DATA IS SORTED BY VOLUME THEN DATE SO THAT ACTIONS AGAINST
//* A VOLUME CAN BE TRACED FROM TAPE CREATION UNTIL TAPE DELETION
//* REMEMBER TO CREATE THE 3 GDGS FOR THE MONTHLY CONSOLIDATION REPORT
//*
//* RMMAUDIT WILL ARCHIVE DAILY REPORTS INTO A WEEKLY ARCHIVE
//* AUDIT DATA WILL NOT BE SAVED MORE THAN ONE YEAR.
//*****
//*****
//* CREATE MONTHLY MEMBER FROM WEEKLY REPORTS
//*****
//VSORT EXEC PGM=SORT
//SYSOUT DD SYSOUT=*
//SORTIN DD DSN=DFRMM.REPT.AUDVOLS.WEEK(0),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDVOLS.WEEK(-1),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDVOLS.WEEK(-2),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDVOLS.WEEK(-3),DISP=SHR
//SORTOUT DD DSN=DFRMM.REPT.AUDVOLS.MTH(+1),DISP=(,CATLG,DELETE),
//      RECFM=FBA,LRECL=121,DSORG=PS,
//      SPACE=(CYL,(300,100),RLSE),UNIT=SYSDA
//SORTWK01 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SYSIN DD *
//      OPTION VLSHRT
//      SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A)
//* SORT BY VOLUME 2,6 THEN BY DATE 38,4 THEN 32,5
//* TO GET HISTORY OF EACH TAPE FOR ONE MONTH
//*****
//* CREATE MONTHLY MEMBER FROM WEEKLY REPORTS
//*****
//RSORT EXEC PGM=SORT
//SYSOUT DD SYSOUT=*
//SORTIN DD DSN=DFRMM.REPT.AUDRACK.WEEK(0),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDRACK.WEEK(-1),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDRACK.WEEK(-2),DISP=SHR
//      DD DSN=DFRMM.REPT.AUDRACK.WEEK(-3),DISP=SHR
//SORTOUT DD DSN=DFRMM.REPT.AUDRACK.MTH(+1),DISP=(,CATLG,DELETE),
//      RECFM=FBA,LRECL=121,DSORG=PS,
//      SPACE=(CYL,(300,100),RLSE),UNIT=SYSDA
//SORTWK01 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
```

Figure 263 (Part 1 of 2). Sample RMMAUD2B Execution JCL

```

//SYSIN DD *
OPTION VLSHRT
SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A)
/* SORT BY RACK 2,6 THEN BY DATE 38,4 THEN 32,5
/* TO GET HISTORY OF EACH RACK FOR ONE MONTH
//*****
/* CREATE MONTHLY MEMBER FROM WEEKLY REPORTS
//*****
//USORT EXEC PGM=SORT
//SYSOUT DD SYSOUT=*
//SORTIN DD DSN=DFRMM.REPT.AUDUSER.WEEK(0),DISP=SHR
// DD DSN=DFRMM.REPT.AUDUSER.WEEK(-1),DISP=SHR
// DD DSN=DFRMM.REPT.AUDUSER.WEEK(-2),DISP=SHR
// DD DSN=DFRMM.REPT.AUDUSER.WEEK(-3),DISP=SHR
//SORTOUT DD DSN=DFRMM.REPT.AUDUSER.MTH(+1),DISP=(,CATLG,DELETE),
// RECFM=FBA,LRECL=121,DSORG=PS,
// SPACE=(CYL,(300,100),RLSE),UNIT=SYSDA
//SORTWK01 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK02 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SORTWK03 DD SPACE=(CYL,(100,,1)),UNIT=SYSDA
//SYSIN DD *
OPTION VLSHRT
SORT FIELDS=(2,6,CH,A,38,4,CH,A,32,5,CH,A)
/* SORT BY RACK 2,6 THEN BY DATE 38,4 THEN 32,5
/* TO GET HISTORY OF EACH RACK FOR ONE MONTH

```

Figure 263 (Part 2 of 2). Sample RMMAUD2B Execution JCL

D.3.3 Recovery Reports

The jobs presented below can be used to create reports useful for recovery purposes.

D.3.3.1 Listing Information for Recovery: Job RMMRECI1

Figure 264 on page 656 shows a job that lists information for identified tapes for a recovery.

```

//RMMRECI1 JOB ,
//          MSGLEVEL=(1,1),MSGCLASS=H,
//          NOTIFY=&SYSUID,USER=&SYSUID,
//          CLASS=A,TIME=5,REGION=4000K
//*****
/* USES OLD EXTRACT FILE WHICH CONTAINS ALL INFORMATION ON LOST TAPES
/* INPUT:
/* IN1 DD STATEMENT - LIST OF TAPE VOLUMES TO BE RECOVERED
/* IN2 DD STATEMENT - OLD RMM EXTRACT FILE THAT HAD TAPES BEFORE LOSS
/* OUTPUT:
/* FINAL DD STATEMENT - LIST OF RMM VOLUME INFORMATION
//*****
//CLEAN EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE 'SILVER.RMM.RECOVER2.DATA'
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=*
//IN1 DD DSN=SILVER.CLIST.CLIST(LOSTTAPE),DISP=SHR
//IN2 DD DSN=DFRMM.MASTER.EXTRACT,DISP=SHR
//OUT1 DD DSN=&01,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//OUT2 DD DSN=&02,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
/* MAKE SURE FILE WITH ALL FIELDS TO BE SHOWN IS FIRST
//CONCAT DD DSN=*.OUT2,VOL=REF=*.OUT2,DISP=(OLD,PASS)
// DD DSN=*.OUT1,VOL=REF=*.OUT1,DISP=(OLD,PASS)
//OUT3 DD DSN=&03,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//FINAL DD DSN=SILVER.RMM.RECOVER2.DATA,DISP=(,CATLG),
// LRECL=569,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(569,(1,1),RLSE)
//DFSMSG DD SYSOUT=*
//TOOLIN DD *
MODE CONTINUE
* CREATE NEW DATA SETS WITH FIELD IN SAME COLUMNS
* MOVE 6,6,CH FIELD FROM FILE1 TO POSITION 5
COPY FROM(IN1) TO(OUT1) USING(CPY1)
* MOVE 9,6,CH FIELD FROM FILE2 TO POSITION 5
* MOVE 16,500,CH FIELD FROM FILE2 TO POSITION 11
COPY FROM(IN2) TO(OUT2) USING(CPY2)
* SELECT ONLY THE RECORDS WITH DUPLICATE FIELDS
SELECT FROM(CONCAT) TO(OUT3) ON(5,6,CH) ALLDUPS
* KEEP ONLY THE FIRST OF THE DUPLICATE RECORDS
SELECT FROM(OUT3) TO(FINAL) ON(5,6,CH) FIRST

```

Figure 264 (Part 1 of 2). Sample RMMRECI1 Execution JCL

```

//CPY1CNTL DD *
OUTREC FIELDS=(1,4,6,6)
//CPY2CNTL DD *
INCLUDE COND=(5,1,CH,EQ,C'V')
OUTREC FIELDS=(1,4,9,6,16,500)

```

Figure 264 (Part 2 of 2). Sample RMMRECI1 Execution JCL

D.3.3.2 Recovery Lost Information: Job RMMRECVL

Figure 265 shows a job that creates ADDVOLUME commands for identified lost tapes to recover.

```
//RMMRECV5 JOB ,
// MSGCLASS=H,USER=&SYSUID,MSGLEVEL=(1,1),TIME=5,NOTIFY=&SYSUID,CLASS=A
//*****
/* USES OLD EXTRACT FILE WHICH CONTAINS ALL THE INFORMATION
/* ABOUT THE LOST TAPES.
/* INPUT:
/* IN1 DD STATEMENT - LOST TAPE FILE - CONTAINS A LIST OF THE RACK
/* NUMBERS OF THE TAPES TO BE RECOVERED
/* MUST BE A VB DATASET (CLIST) RACK NUMBERS STARTING
/* IN COLUMN 2 (DATA) + 4 (LENGTH) = 6 (ACTUAL)
/* IN2 DD STATEMENT - OLD RMM EXTRACT FILE - CONTAINS INFORMATION ABOUT
/* TAPES BEFORE LOSS (CREATED WITH AMERICAN DATES)
/* OUTPUT:
/* COMMANDS DD STATEMENT - CLIST OF CREATED RMM ADDVOLUME COMMANDS
//*****
//ASMAM35 EXEC PGM=IFOX00
//SYSPRINT DD SYSOUT=*
//SYSLIB DD DSN=SYS1.MACLIB,DISP=SHR
//SYSUT1 DD UNIT=SYSDA,SPACE=(1700,(600,100))
//SYSUT2 DD UNIT=SYSDA,SPACE=(1700,(600,100))
//SYSUT3 DD UNIT=SYSDA,SPACE=(1700,(600,100))
//SYPUNCH DD DSN=&XE35OBJ(E35FILL),DISP=(,PASS),
// UNIT=SYSDA,SPACE=(80,(200,50,1))
//ASM.SYSIN DD *
E35FILL CSECT
* E35 EXIT TO COMPLETE THE ADDVOLUME COMMANDS.
* SEE COMMENTS IN CMDTCNTL BELOW FOR DETAILS.
R0 EQU 0
R1 EQU 1
R2 EQU 2
R3 EQU 3
R4 EQU 4
R5 EQU 5
R6 EQU 6
R7 EQU 7
R8 EQU 8
R9 EQU 9
R10 EQU 10
R11 EQU 11
R12 EQU 12
R13 EQU 13
R14 EQU 14
R15 EQU 15
```

Figure 265 (Part 1 of 6). Sample RMMRECVL Execution JCL

```

* OFFSETS TO FIELDS IN RECORD PASSED TO E35 TO BE OPERATED UPON
DENS      EQU    81          DENSITY: **** -> *
DENSTARG  EQU    81          TARGET FOR DENSITY
USEF1     EQU    91          USE FLAG 1:  Y -> MVS
USEF2     EQU    92          USE FLAG 2:  Y -> VM
USETARG   EQU    91          TARGET FOR USE STRING
EXDTIN    EQU    203         EXPDATE YYYY/MMDD
*          OUT ->  YYYY/DDD
OWNAF     EQU    226         OWNERACCESS FLAG
*          R -> READ
*          U -> UPDATE
*          A -> ALTER
OWNATARG  EQU    226         TARGET FOR OWNERACCESS STRING
SEC       EQU    243         SECURITY LEVEL: BLANK -> NONE
SECTARG   EQU    243         TARGET FOR SECURITY LEVEL
ASDTIN    EQU    276         ASDATE YYYY/MMDD
*          OUT ->  YYYY/DDD
ACTN      EQU    316         RELEASEACTION STRING
*          OWNER  -> RETURN
*          SCRATCH -> SCRATCH
ACTNF1    EQU    324         RELEASEACTION FLAG 1:  Y -> REPLACE
ACTNF2    EQU    325         RELEASEACTION FLAG 2:  Y -> INIT
ACTNF3    EQU    326         RELEASEACTION FLAG 3:  Y -> ERASE
ACTNF4    EQU    327         RELEASEACTION FLAG 4:  Y -> NOTIFY
ACTNTARG  EQU    316         TARGET FOR RELEASEACTION STRING
          STM    R14,R12,12(R13)  SAVE ALL REGS EXCPT 13
          BALR   R12,R0          SET BASE REG
          USING *,R12          SHOW BASE REG
          ST     R13,SAVE35+4    SAVE BACKWARD POINTER
          LA     R14,SAVE35      SET FORWARD POINTER IN CALLER
          ST     R14,8(R13)      SET SAVE AREA
          LR     R13,R14        SET OUR SAVE AREA
          L      R2,0(,R1)      GET ADDRESS OF RECORD
          LTR    R2,R2          IF EOF,
          BZ     EOF           PASS BACK RC=8 (DO NOT RETURN)
* TEST AND SET 'DENSITY' FIELD
          CLC    DENS(4,R2),=CL4' ****'  IF NOT FOUR '*' S,
          BNE    EXPDT          LEAVE AS IS
          MVC    DENSTARG(4,R2),=CL4' '*'  CHANGE TO ONE *
* CONVERT 'YYYY/MMDD' TO 'YYYY/DDD' (JULIAN) FOR 'EXPDATE' FIELD
EXPDT     LA     R6,EXDTIN(,R2)  GET POINTER TO  'YYYY/MMDD'
          BAL    R10,CVTDATE     GO CONVERT IT TO 'YYYY/DDD'
* TEST AND SET 'USE' FIELD
          LA     R3,WORK         POINT TO START OF WORK AREA
          MVC    0(7,R3),BLANKS  INITIALIZE TO BLANKS
          CLI    USEF1(R2),C' Y'  IF MVS FLAG NOT SET,
          BNE    USECKVM        DO NOT SET FOR MVS
          MVC    0(3,R3),=CL3' MVS,' SET FOR MVS
          LA     R3,4(,R3)      POINT TO NEXT LOCATION

```

Figure 265 (Part 2 of 6). Sample RMMRECVL Execution JCL

```

USECKVM  CLI  USEF2(R2),C' Y'      IF VM FLAG NOT SET,
        BNE  USEDONE              DO NOT SET FOR VM
        MVC  0(2,R3),=CL2' VM'    SET FOR VM
USEDONE  MVC  USETARG(6,R2),WORK  MOVE REPLACEMENT TO RECORD
* TEST AND SET 'OWNERACCESS' FIELD
        LA   R3,WORK              POINT TO START OF WORK AREA
        MVC  0(6,R3),BLANKS       INITIALIZE TO BLANKS
        CLI  OWNAF(R2),C' R'      IF READ FLAG NOT SET,
        BNE  OWNACKU              DO NOT SET FOR READ
        MVC  0(4,R3),=CL4' READ)' SET FOR READ
        B    OWNADONE              ALL DONE
OWNACKU  CLI  OWNAF(R2),C' U'      IF UPDATE FLAG NOT SET,
        BNE  OWNACKA              DO NOT SET FOR UPDATE
        MVC  0(6,R3),=CL6' UPDATE)' SET FOR UPDATE
        B    OWNADONE              ALL DONE
OWNACKA  CLI  OWNAF(R2),C' A'      IF ALTER FLAG NOT SET,
        BNE  OWNADONE              DO NOT SET FOR ALTER
        MVC  0(5,R3),=CL5' ALTER)' SET FOR ALTER
        B    OWNADONE              ALL DONE
OWNADONE MVC  OWNATARG(6,R2),WORK  MOVE REPLACEMENT TO RECORD
* TEST AND SET 'SECLEVEL' FIELD
        CLI  SEC(R2),C' '         IF NOT BLANK,
        BNE  ASDT                 LEAVE AS IS
        MVC  SECTARG(4,R2),=CL4' NONE' CHANGE TO NONE
* CONVERT 'YYYY/MMDD' TO 'YYYY/DDD' (JULIAN) FOR 'ASDATE' FIELD
ASDT     LA   R6,ASDTIN(,R2)      GET POINTER TO 'YYYY/MMDD'
        BAL  R10,CVTDATE          GO CONVERT IT TO 'YYYY/DDD'
* TEST AND SET 'RELEASEACTION' FIELDS
RELCK    LA   R3,WORK              POINT TO START OF WORK AREA
        MVC  0(33,R3),BLANKS      INITIALIZE TO BLANKS
        MVC  0(7,R3),ACTN(R2)     MOVE IN ORIGINAL
        CLC  ACTN(5,R2),=CL5' OWNER' IF NOT OWNER,
        BNE  ACTCKSC              LEAVE ORIGINAL
        MVC  0(6,R3),=CL6' RETURN' SET FOR RETURN
ACTCKSC  LA   R3,8(,R3)            POINT TO NEXT LOCATION
ACTCKREP CLI  ACTNF1(R2),C' Y'     IF REPLACE FLAG NOT SET,
        BNE  ACTCKIN              DO NOT SET FOR REPLACE
        MVC  0(7,R3),=CL7' REPLACE' SET FOR REPLACE
        LA   R3,8(,R3)            POINT TO NEXT LOCATION
ACTCKIN  CLI  ACTNF2(R2),C' Y'     IF INIT FLAG NOT SET,
        BNE  ACTCKER              DO NOT SET FOR INIT
        MVC  0(4,R3),=CL4' INIT'  SET FOR INIT
        LA   R3,5(,R3)            POINT TO NEXT LOCATION
ACTCKER  CLI  ACTNF3(R2),C' Y'     IF ERASE FLAG NOT SET,
        BNE  ACTCKNT              DO NOT SET FOR ERASE
        MVC  0(5,R3),=CL5' ERASE' SET FOR ERASE
        LA   R3,6(,R3)            POINT TO NEXT LOCATION
ACTCKNT  CLI  ACTNF4(R2),C' Y'     IF NOTIFY FLAG NOT SET,
        BNE  ACTDONE              DO NOT SET FOR NOTIFY
        MVC  0(6,R3),=CL6' NOTIFY' SET FOR NOTIFY
ACTDONE  MVC  ACTNTARG(33,R2),WORK MOVE REPLACEMENT TO RECORD

```

Figure 265 (Part 3 of 6). Sample RMMRECVL Execution JCL

```

* PASS BACK CHANGED RECORD
CHANGE LR R1,R2 PASS BACK RECORD
SLR R15,R15 PASS BACK RC=0 (ACCEPT)
B GOBACK
EOF LA R15,8 INDICATE DO NOT RETURN
*
GOBACK L R13,4(,R13)
L R14,12(,R13)
LM R2,R12,28(R13)
BR R14
* SUBROUTINE TO CONVERT 'YYYY/MMDD' TO 'YYYY/DDD' (JULIAN)
* INPUT - R6 = POINTER TO 'YYYY/MMDD'
* OUTPUT - 'YYYY/DDD' REPLACEMENT IN RECORD
DTYYYY EQU 0 YYY
DTMM EQU 5 MM
DTDD EQU 7 DD
DTOUT EQU 5 DDD (JULIAN)
CVTDATE SLR R3,R3 INIT LEAP YEAR DAY TO ZERO
PACK CVTWA,DTMM(2,R6) PACK MM
CVB R4,CVTWA CONVERT MM TO BINARY
C R4,=F'2' IF MM IS 1 OR 2,
BNH DAYCNV SKIP LEAP YEAR TEST
PACK CVTWA,DTYYYY(4,R6) PACK YYYY
CVB R9,CVTWA CONVERT YYYY TO BINARY
SLR R8,R8 PREPARE FOR DIVIDE
D R8,=F'4' DIVIDE YYYY BY 4
LTR R8,R8 IF NOT MODULO 4,
BNZ DAYCNV THEN NOT LEAP YEAR
LA R3,1 SET EXTRA DAY FOR LEAP YEAR
DAYCNV PACK CVTWA,DTDD(2,R6) PACK DD
CVB R5,CVTWA CONVERT DD TO BINARY
AR R5,R3 ADD LEAP YEAR DAY IF ANY
BCTR R4,0 GET INDEX
SLL R4,1 INTO MONTH TABLE
AH R5,MONTBL(R4) ADD TO DAYS
CVD R5,CVTWA CONVERT TO DECIMAL
UNPK WORK(3),CVTWA+6(2) CONVERT TO EBCDIC
OI WORK+2,C'0' MAKE SURE LAST CHAR IS PRINTABLE
MVC WORK+3(2),=CL2' PUT IN ENDING PAREN
MVC DTOUT(5,R6),WORK MOVE REPLACEMENT TO RECORD
BR R10 RETURN
*
CVTWA DS D CONVERSION WORKAREA
SAVE35 DS 18F ENTRY REGISTER SAVE AREA
WORK DS CL40 WORK AREA
BLANKS DC CL40' ' FOR BLANKING OUT AREAS
MONTBL DC H'0',H'31',H'59',H'90',H'120',H'151'
DC H'181',H'212',H'243',H'273',H'304',H'334'
LTORG
END
/*

```

Figure 265 (Part 4 of 6). Sample RMMRECVL Execution JCL

```

//LINK35 EXEC PGM=HEWL,PARM='XREF,LET,LIST,NCAL',COND=(8,LE)
//SYSPRINT DD SYSOUT=*
//OBJ DD DSN=&XE35OBJ,DISP=(OLD,PASS)
//SYSLMOD DD DSN=&EXIT,DISP=(,PASS),SPACE=(TRK,(10,10,10)),
// UNIT=SYSDA,DCB=(RECFM=U,BLKSIZE=13030)
//SYSLIN DD *
INCLUDE OBJ(E35FILL)
NAME E35FILL
//*
//CLEAN EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE 'SILVER.RMM.RECOVERA.CLIST'
DELETE 'SILVER.RMM.RECOVERA.DATA'
//*
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=*
//EXIT DD DSN=&EXIT,DISP=(OLD,PASS)
//IN1 DD DSN=SILVER.CLIST.CLIST(LOSTTAPE),DISP=SHR
//IN2 DD DSN=DFRMM.EXTRACT.BACKUP(-1),DISP=SHR
//OUT1 DD DSN=&01,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//OUT2 DD DSN=&02,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
/* MAKE SURE FILE WITH ALL FIELDS TO BE SHOWN IS FIRST
//CONCAT DD DSN=*.OUT2,VOL=REF=*.OUT2,DISP=(OLD,PASS)
// DD DSN=*.OUT1,VOL=REF=*.OUT1,DISP=(OLD,PASS)
//OUT3 DD DSN=&03,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//FINAL DD DSN=&TEMP,DISP=(,PASS),UNIT=SYSDA,SPACE=(CYL,(2,1))
//COMMANDS DD DSN=SILVER.RMM.RECOVERA.CLIST,DISP=(,CATLG),
// LRECL=370,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(370,(1,1),RLSE)
//DFSMSG DD SYSOUT=*
//TOOLIN DD *
MODE CONTINUE
** CREATE NEW DATA SETS WITH FIELD IN SAME COLUMNS
* MOVE RACK NUMBER (6,6,CH) FIELD FROM FILE1 TO POSITION 5
COPY FROM(IN1) TO(OUT1) USING(CPY1)
* MOVE RACK NUMBER (9,6,CH) FIELD FROM FILE2 TO POSITION 5
* MOVE OTHER INFORMATION (9,500,CH) FIELDS FROM FILE2 TO POSITION 11
COPY FROM(IN2) TO(OUT2) USING(CPY2)
* SELECT ONLY THE RECORDS WITH DUPLICATE FIELDS
SELECT FROM(CONCAT) TO(OUT3) ON(5,6,CH) ALLDUPS
* KEEP ONLY THE FIRST OF THE DUPLICATE RECORDS
SELECT FROM(OUT3) TO(FINAL) ON(5,6,CH) FIRST
* BUILD ADDVOLUME COMMANDS USING OUTREC AND E35
COPY FROM(FINAL) TO(COMMANDS) USING(CMDT)

```

Figure 265 (Part 5 of 6). Sample RMMRECVL Execution JCL

```

//CPY1CNTL DD *
  OUTREC FIELDS=(1,4,6,6)
//CPY2CNTL DD *
  INCLUDE COND=(5,1,CH,EQ,C'V')
  OUTREC FIELDS=(1,4,361,6,9,500)
//CMDTCNTL DD *
* BUILD ADDVOLUME COMMANDS USING OUTREC AND E35
  OUTREC FIELDS=(1,4,C' AV ',11,6,
    C' STATUS(',320,8,C')',
    C' RACK(',363,8,C')',
    C' UNIT(',355,8,C')',
    C' LABEL(',335,3,C')',
    C' DENSITY(',131,4,C')',
*   E35 IS USED TO CHANGE '*****' TO '*'
    C' USE(',353,2,4X,C')',
*   E35 IS USED TO TEST PASSED FLAGS AND FILL IN USE FIELD:
*     FIRST FLAG = 'Y' -> 'MVS'
*     SECOND FLAG = 'Y' -> 'VM'
    C' DESCRIPTION('',443,30,C'')',
    C' ACCOUNT('',403,40,C'')',
    C' EXPDT(',127,4,C'/',121,2,124,2,C')',
*   AMERICAN DATE FORMAT USED ('MM/DD/YYYY')
*   E35 IS USED TO CHANGE PASSED DATE OF 'YYYY/MMDD' TO
*   REQUIRED JULIAN DATE FORMAT OF 'YYYY/DDD'
    C' OWNERACCESS(',351,1,5X,C')',
*   E35 IS USED TO TEST PASSED FLAG AND FILL IN OWNERACCESS FIELD:
*     'R' -> 'READ'
*     'U' -> 'UPDATE'
*     'A' -> 'ALTER'
    C' SECLEVEL(',278,8,C')',
*   E35 IS USED TO CHANGE ' ' TO 'NONE'
    C' OWNER(',254,8,C')',
    C' ASDATE(',244,4,C'/',238,2,241,2,C')',
*   AMERICAN DATE FORMAT USED ('MM/DD/YYYY')
*   E35 IS USED TO CHANGE PASSED DATE OF 'MM/DD/YYYY' TO
*   REQUIRED JULIAN DATE FORMAT OF 'YYYY/DDD'
    C' ASTIME(',248,2,250,2,252,2,C')',
    C' RELEASEACTION(',339,7,X,
*   E35 IS USED TO CHANGED PASSED VALUE OF 'OWNER' TO 'RETURN' AND
*   LEAVE PASSED VALUE OF 'SCRATCH' AS IS IN RELEASEACTION FIELD
      347,4,21X,C')',
*   E35 IS USED TO TEST PASSED FLAGS AND FILL IN RELEASEACTION FIELD:
*     FIRST FLAG = 'Y' -> 'REPLACE'
*     SECOND FLAG = 'Y' -> 'INIT'
*     THIRD FLAG = 'Y' -> 'ERASE'
*     FOURTH FLAG = 'Y' -> 'NOTIFY'
    370:X) PAD OUT RECORDS TO 370 BYTES

*   E35 IS USED TO BUILD COMPLEX PARTS OF COMMAND AS INDICATED ABOVE
MODS E35=(E35FILL,1000,EXIT)

```

Figure 265 (Part 6 of 6). Sample RMMRECVL Execution JCL

D.3.4 SMF Reports

The jobs presented below can be used to create reports from SMF data.

D.3.4.1 Listing SMF DFSMSrmm Volume Records: Job RMMSMF

Figure 266 shows a job that lists all DFSMSrmm SMF volume records in readable format.

```
//RMMSMF JOB ,
//          MSGLEVEL=(1,1),MSGCLASS=H,
//          NOTIFY=&SYSUID,USER=&SYSUID,
//          CLASS=0,TIME=5,REGION=4M
//*****
/* INPUT: RAWSMF DD STATEMENT - RAW SMF DATA
/* OUTPUT: VREPT DD STATEMENT
/* ALSO COPY RAW SMF DATA TO 'SILVER.RMM.SMFDATA' TO LOOK FOR MORE INFO
//*****
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=*      ICETOOL MESSAGES
//DFSMSG DD SYSOUT=*      DFSORT MESSAGES
//TOOLIN DD *              CONTROL STATEMENTS
* FIND THE RMM SMF AUDIT 'VOLUME' RECORDS
  COPY FROM(RAWSMF) TO(RMMV) USING(SMFV)
* DISPLAY VARIOUS FIELDS FROM THE SMF HEADER AND VOLUME SECTION
  DISPLAY FROM(RMMV) LIST(VREPT) -
    TITLE('RMM SMF AUDIT RECORDS') DATE TIME PAGE -
    BLANK -
* SMF HEADER FIELDS
  HEADER(' TIME') ON(8,3,HEX) -
  HEADER(' DATE') ON(11,4,PD) -
  HEADER(' SYS') ON(15,4,CH) -
  HEADER(' USER') ON(35,8,CH) -
  HEADER(' ACT') ON(43,1,CH) -
* VOLUME SECTION FIELDS
  HEADER(' VOLUME') ON(46,6,CH) -
  HEADER(' CREATE') ON(104,4,PD) -
  HEADER(' LASTCH') ON(128,4,PD) -
  HEADER(' USER') ON(136,16,CH) -
  HEADER(' LASTUSCH') ON(152,4,PD)
//RAWSMF DD DSN=ACCT.SJFEMVSA.D921213.T230004,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921214.T060007,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921214.T120005,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921214.T180004,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921214.T230004,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921217.T230004,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921218.T060007,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921218.T120005,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921218.T180005,DISP=SHR
//          DD DSN=ACCT.SJFEMVSA.D921218.T230004,DISP=SHR
//RMMV DD DSN=&&TEMPV,REFDD=*.RAWSMF,DISP=(,PASS)
```

Figure 266 (Part 1 of 2). Sample RMMSMF Execution JCL

```

//SMFVCNTL DD *
  INCLUDE COND=(6,1,BI,EQ,X'FC',          - RMM SMF AUDIT RECORDS -
                AND,44,1,CH,EQ,C'V')      - VOLUME TYPE RECORDS -
  OPTION VLSHRT                           - IGNORE SHORT RECORDS -
//VREPT DD SYSOUT=*
/*
//CLEANUP EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
  DELETE SILVER.RMM.SMFDATA
//S10F1 EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSUT1 DD DSN=*.STEP1.RMMV,DISP=(OLD,PASS),LRECL=32756
//SYSUT2 DD DSN=SILVER.RMM.SMFDATA,DISP=(,CATLG,CATLG),
//      RECFM=VB,LRECL=32756,BLKSIZE=32760,DSORG=PS,
//      SPACE=(CYL,(100,100),RLSE),UNIT=SYSDA
//SYSUT3 DD UNIT=SYSDA,SPACE=(CYL,(100,100))
//SYSUT4 DD UNIT=SYSDA,SPACE=(CYL,(100,100))
//SYSIN DD DUMMY

```

Figure 266 (Part 2 of 2). Sample RMMSMF Execution JCL

D.3.4.2 List Number of SMF DFSMSrmm Records: Job SMFPRNT

Figure 267 shows a job that gives a count of each SMF record found.

```

//SMFLOC JOB ,
//      MSGLEVEL=(1,1),MSGCLASS=H,
//      NOTIFY=&SYSUID,USER=&SYSUID,
//      CLASS=0,TIME=5,REGION=4000K
//*****
/* INPUT : RAWSMF DD STATEMENT - RAW SMF DATA
/* OUTPUT: VREPT DD STATEMENT
//*****
/*
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=* ICETOOL MESSAGES
//DFSMSG DD SYSOUT=* DFSORT MESSAGES
//TOOLIN DD * CONTROL STATEMENTS
  OCCUR FROM(RAWSMF) LIST(VREPT) -
    TITLE('RMM SMF AUDIT RECORDS') DATE TIME PAGE -
    BLANK -
    ON(6,1,BI) ON(VALCNT) ON(6,1,HEX)
* LOOK AT 6,1 AND FIND ALL OCCURANCES AND TOTAL OCCURANCES ON(VALCNT)
* AND SHOW IN HEX (FINDS ALL SMF RECORDS)
//RAWSMF DD DSN=ACCT.SJFEMVSA.D921213.T230004,DISP=SHR
//VREPT DD SYSOUT=*

```

Figure 267. Sample SMFPRNT Execution JCL

D.3.5 BARCODE Utility Reports

The jobs presented below can be used to simplify inventory operations using a barcode reader.

D.3.5.1 Create ADDVOLUME Commands: Job BARCMDS

Figure 268 shows a job that creates ADDVOLUME commands from a list of barcode scanned tapes.

```
//BARCMDS JOB ,
//      MSGLEVEL=(1,1),MSGCLASS=H,
//      NOTIFY=&SYSUID,USER=&SYSUID,
//      CLASS=0,TIME=4,REGION=4000K
//*****
//*      INPUT:
//*      BARCODE DD STATEMENT: LIST OF BARCODE SCANNED TAPES
//*      OUTPUT:
//*      RMMCMD DD STATEMENT: LIST OF RMM COMMANDS
//*****
//CLEAN EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DELETE 'SILVER.RMMBAR.CLIST'
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=* ICETOOL MESSAGES
//DFSMSG DD SYSOUT=* DFSORT MESSAGES
//TOOLIN DD * CONTROL STATEMENTS
COPY FROM(BARCODE) TO(RMMCMD) USING(TEMP)
//BARCODE DD DSN=SILVER.RMM.BARCODES,DISP=SHR
//RMMCMD DD DSN=SILVER.RMMBAR.CLIST,DISP=(,CATLG,DELETE),
// LRECL=256,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(256,(1,1),RLSE)
//TEMPCNTL DD *
INCLUDE COND=(5,3,CH,EQ,C'IBM')          BARCODE SCANNED TAPE LIST
OUTREC FIELDS=(1,4,C'RMM AV ',9,6,C' DATE(99365)') RMM COMMAND
OPTION VLSHRT
```

Figure 268. Sample BARCMDS Execution JCL

D.3.5.2 Audit Using Barcode Scanner: Job BARCOMP

Figure 269 on page 666 shows a job that audits tapes using barcode scanner output.

```

//BARCOMP JOB ,
//          MSGLEVEL=(1,1),MSGCLASS=H,
//          NOTIFY=&SYSUID,USER=&SYSUID,
//          CLASS=A,TIME=5,REGION=4000K
//*****
//*
//* COMPARISON WITH RMM EXTRACT FILE
//*   LISTS WHAT IS IN BOTH THE LIBRARY AND THE RMM EXTRACT FILE
//*       WHAT IS ONLY IN THE LIBRARY
//*       WHAT IS ONLY IN THE RMM EXTRACT FILE
//*
//*****
//CLEAN EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
    DELETE 'DFRMM.REPT.BARCODE.MATCH.LIBRARY'
    DELETE 'DFRMM.REPT.BARCODE.LIBRARY.ONLY'
    DELETE 'DFRMM.REPT.BARCODE.RMMDBASE.ONLY'
//STEP1 EXEC PGM=ICETOOL
//TOOLMSG DD SYSOUT=* ICETOOL MESSAGES
//DFSMSG DD SYSOUT=* DFSORT MESSAGES
//BARCODE DD DSN=SILVER.RMM.BARCODES,DISP=SHR
//EXTRACT DD DSN=DFRMM.MASTER.EXTRACT,DISP=SHR
//BARCTEMP DD DSN=&01,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//EXTRTEMP DD DSN=&02,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//EXCEPT DD DSN=&EXCEPT,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//COMP1 DD DSN=&COMP1,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//COMPL DD DSN=&COMPL,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//COMPR DD DSN=&COMPR,DISP=(,PASS),UNIT=SYSDA,
// SPACE=(CYL,(2,1))
//* MAKE SURE FILE WITH ALL FIELDS TO BE SHOWN IS FIRST
//*CONCAT DD STATEMENT HAS VOLUMES THAT WERE BARCODE SCANNED (IN LIBRARY)
//*       AND ARE IN THE RMM DATABASE
//CONCAT DD DSN=*.EXTRTEMP,VOL=REF=*.EXTRTEMP,DISP=(OLD,PASS)
//       DD DSN=*.BARCTEMP,VOL=REF=*.BARCTEMP,DISP=(OLD,PASS)

```

Figure 269 (Part 1 of 3). Sample BARCOMP Execution JCL

```

/*CONCATL DD STATEMENT HAS VOLUMES THAT WERE BARCODE SCANNED (IN LIBRARY)
/*      BUT ARE NOT IN THE RMM DATABASE
/*      EXCEPT HAS RECORDS THAT ARE IN BARCODE BUT NOT IN RMM DBASE
/*      AND IN RMM DBASE BUT NOT IN BARCODE
/*      BARCTEMP HAS RECORDS THAT ARE WERE BARCODE SCANNED
//CONCATL DD DSN=*.BARCTEMP,VOL=REF=*.BARCTEMP,DISP=(OLD,PASS)
//      DD DSN=*.EXCEPT,VOL=REF=*.EXCEPT,DISP=(OLD,PASS)
/*CONCATR DD STATEMENT HAS VOLUMES THAT WERE IN THE RMM DATABASE BUT
/*      WERE NOT BARCODE SCANNED (NOT IN LIBRARY)
/*      EXCEPT HAS RECORDS THAT ARE IN BARCODE BUT NOT IN RMM DBASE
/*      AND IN RMM DBASE BUT NOT IN BARCODE
/*      EXTRTEMP HAS RECORDS THAT ARE IN THE RMM DATABASE
//CONCATR DD DSN=*.EXTRTEMP,VOL=REF=*.EXTRTEMP,DISP=(OLD,PASS)
//      DD DSN=*.EXCEPT,VOL=REF=*.EXCEPT,DISP=(OLD,PASS)
//COMPSAME DD DSN=DFRMM.REPT.BARCODE.MATCH.LIBRARY,DISP=(,CATLG),
// LRECL=256,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(256,(1,1),RLSE)
//INLIBR DD DSN=DFRMM.REPT.BARCODE.LIBRARY.ONLY,DISP=(,CATLG),
// LRECL=256,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(256,(1,1),RLSE)
//INRMM DD DSN=DFRMM.REPT.BARCODE.RMMDBASE.ONLY,DISP=(,CATLG),
// LRECL=256,RECFM=VB,DSORG=PS,AVGREC=K,SPACE=(256,(1,1),RLSE)
//TOOLIN DD *
MODE CONTINUE
** CREATE NEW DATA SETS WITH FIELD IN SAME COLUMNS
COPY FROM(BARCODE) TO(BARCTEMP) USING(BARC)
COPY FROM(EXTRACT) TO(EXTRTEMP) USING(EXTR)
*
* REPORT OF RACK #S THAT ARE IN LIBRARY AND IN RMM DBASE
*
* SELECT ONLY THE RECORDS WITH DUPLICATE FIELDS
SELECT FROM(CONCAT) TO(COMP1) ON(5,6,CH) ALLDUPS
* KEEP ONLY THE FIRST OF THE DUPLICATE RECORDS
SELECT FROM(COMP1) TO(COMPSAME) ON(5,6,CH) FIRST
*
* REPORT OF RACK #S THAT ARE IN LIBRARY BUT NOT IN RMM DBASE
*
* SELECT ONLY THE RECORDS THAT ARE NOT DUPLICATE
*      EXCEPT HAS RECORDS THAT ARE IN BARCODE BUT NOT IN RMM DBASE
*      AND IN RMM DBASE BUT NOT IN BARCODE
SELECT FROM(CONCAT) TO(EXCEPT) ON(5,6,CH) NODUPS
* SELECT RECORDS THAT ARE IN BARCODE BUT NOT IN RMM DBASE
SELECT FROM(CONCATL) TO(COMPL) ON(5,6,CH) ALLDUPS
* KEEP ONLY THE FIRST OF THE DUPLICATE RECORDS
SELECT FROM(COMPL) TO(INLIBR) ON(5,6,CH) FIRST
*
* REPORT OF RACK #S THAT ARE IN RMM DBASE BUT NOT IN LIBRARY
*
* SELECT ONLY THE RECORDS THAT ARE NOT DUPLICATE
*      EXCEPT HAS RECORDS THAT ARE IN BARCODE BUT NOT IN RMM DBASE
*      AND IN RMM DBASE BUT NOT IN BARCODE

```

Figure 269 (Part 2 of 3). Sample BARCOMP Execution JCL

```

*SELECT FROM(CONCAT) TO(EXCEPT) ON(5,6,CH) NODUPS (ALREADY DONE ABOVE)
* SELECT RECORDS THAT ARE IN RMM DBASE BUT NOT IN BARCODE
  SELECT FROM(CONCATR) TO(COMPR) ON(5,6,CH) ALLDUPS
* KEEP ONLY THE FIRST OF THE DUPLICATE RECORDS
  SELECT FROM(COMPR) TO(INRMM) ON(5,6,CH) FIRST
//BARCNTL DD *
*      MOVE 9,6 CH FIELD FROM BARCODE FILE TO POSITION 5 TO ALIGN COLS
  INCLUDE COND=(5,3,CH,EQ,C' IBM')
  OUTREC FIELDS=(1,4,9,6)
//EXTRCNTL DD *
*      MOVE 361,6 CH FIELD FROM EXTRACT FILE TO POSITION 5 TO ALIGN COLS
  INCLUDE COND=(5,1,CH,EQ,C' V')   USE VOL REC SO GET ONLY IN USE RACKS
  OUTREC FIELDS=(1,4,366,6)        RACK IS IN COLUMN 366 IN VOL REC

```

Figure 269 (Part 3 of 3). Sample BARCOMP Execution JCL

D.4 Sample VRS Conversion from TLMS RMF

This section presents a sample VRS conversion from TLMS RMF data.

Data in Figure 270 on page 669 is converted to the DFSMSrmm commands shown in Figure 271 on page 669. Actually, DFSMSrmm CDS records are produced, rather than the TSO subcommands, but you can use the EDGCVT program VRSCMDS DD to generate the ADDVRS commands so that you can easily check what is generated for DFSMSrmm.

This example assumes that the following TLMSLOCN statements are used to map the TLMS locations to DFSMSrmm equivalents:

```

TLMSLOCN TLMSID=DC RMMLOC=SHELF
TLMSLOCN TLMSID=TL RMMLOC=SHELF
TLMSLOCN TLMSID=CL RMMLOC=SHELF
TLMSLOCN TLMSID=D2 RMMLOC=D2
TLMSLOCN TLMSID=FS RMMLOC=FS
TLMSLOCN TLMSID=OS RMMLOC=OS

```

Q-IND	JOB-NAME																		
	QUALIFIER																		
	ADD DATE			I---- AUTHORIZATION ----I															
	DATA CENTER LOCATION			1ST OFF-SITE			2ND OFF-SITE			3RD OFF-SITE			4TH OFF-SITE			5TH OFF-SITE			
	T	ID	NUM	VER	T	ID	NUM	T	ID	NUM	T	ID	NUM	T	ID	NUM	T	ID	NUM
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	DEFAULT										13/03/89			STSG002			00022000		
	1 DC																		
F	ACSS.DUMP.LOADDATA.D0001										27/07/89			SDAC005					
	2 DC 0090																		
Q	ACSS.DUMP.RAWDATA.										22/06/89			SDAC005					
	2 DC 0090																		
P	ADB2.ARCHLOG										01/06/89			SDAC005					
	6 DC										5 FS 0001			5 TL 0013					
Q	ADB2.IMAGCOPY.										18/05/89			SDAC005					
	6 DC										5 FS 0001			5 TL 0020					
Q	PCSS.DFDSS.										30/01/90			SDAC018					
	6 DC										4 FS 0002			1 TL					
Q	STSG.VTAM.BACKUP.										27/06/90			STSG001					
	6 DC										4 FS 0002			4 D2 0005			1 TL		

Figure 270. Sample Retention Selection Report for Migration

ADDVRS DSN('**') WHILECATALOG LOC(HOME) COUNT(99999)	-
OWNER(STSG002)	
ADDVRS DSN('ACSS.DUMP.LOADDATA.D0001') DAYS LOC(HOME)	-
COUNT(90) OWNER(SDAC005)	
ADDVRS DSN('ACSS.DUMP.RAWDATA.**') DAYS LOC(HOME)	-
COUNT(90) OWNER(SDAC005)	
ADDVRS DSN('ADB2.ARCHLOG*.*') DAYS LOC(FS)	-
COUNT(14) OWNER(SDAC005) STORENUMBER(1)	
ADDVRS DSN('ADB2.IMAGCOPY.*') DAYS LOC(FS)	-
COUNT(21) OWNER(SDAC005) STORENUMBER(1)	
ADDVRS DSN('PCSS.DFDSS') GDG WHILECATALOG CYCLES LOC(FS)	-
COUNT(99999) OWNER(SDAC018) STORENUMBER(2)	
ADDVRS DSN('STSG.VTAM.BACKUP.*') WHILECATALOG CYCLES	-
LOC(FS) COUNT(99999) OWNER(STSG001)	-
STORENUMBER(2) NEXTVRS(VRSSTSG1)	
ADDVRS NAME(VRSSTSG1) OWNER(STSG001)	-
LOC(D2) STORENUMBER(5)	

Figure 271. Sample Converted VRS Definitions

Appendix E. Modification Required for EPIC and DFSMSrmm Parallel Run

Figure 272 shows the modifications required in DFSMSrmm for EPIC to run in parallel with DFSMSrmm.

```
//STSGMALO JOB  ,'AVIS.M RMM OCE LNK',CLASS=S,USER=STSGMA,
//  NOTIFY=STSGMA,MSGCLASS=X,COND=(4,LT)
/*JOBPARM  R=SS,L=20
/** THIS JOB PROVIDES AND INSTALLS A BYPASS TO ENABLE RMM AND EPIC
/** TO RUN IN PARALLEL.
/** UPDATED FOR EPIC 3.2
/** UPDATED FOR UW90355,UW32712,UW32713,UW32922
/**
/** WHY THE BYPASS IS REQUIRED
/** Without this bypass, if you have EPIC
/** installed on the system at the same time as you have RMM started
/** with EDGSSSI in IEFSSNxx, or have started RMM in a mode other
/** than MANUAL, tape usage will be affected.
/** EPIC does not need to be active to result in these problems;
/** if the TSID entry is in IEFSSNxx that counts as installed.
/** Tape usage is affected by:
/** - In warning and protect modes, RMM issues no messages about
/** incorrect tapes mounted and cannot validate tape usage.
/** This is because EPIC bypasses RMMs validation.
/** - A loop in EOVS processing can occur
/**
/** If you do not install the bypass, you must IPL with EPIC
/** removed from the system in order to have fully functional RMM
/** tape processing. With the bypass installed you can more easily
/** move from EPIC to RMM.
/**
/** INSTRUCTIONS:
/** It is not advisable to install this bypass using SMP/E.
/** Although the ZAP can be installed under SMP/E, the link-edit
/** required for IFG0194A MUST be performed outside of SMP/E or the
/** incorrect link-edit will be done.
/**
/** 1) Use IEBCOPY or ISPF/PDF 3.3 to save a copy of your current
/** LPALIB data set. Specify this data set name in the LPASAVE
/** DD in the LK job step.
/** If you rerun this job, you should not need to repeat the
/** saving of the original LPALIB.
/** 2) If you have DFSMSrmm V1.1 installed you should update the
/** ZAP5E and ZAP4A job steps to use the alternative, commented-out
/** ZAP. The current ZAPs are correct for all other levels
/** of DFSMSrmm and DFRMM R2.
/** OFFSETS MAY NEED TO BE UPDATED FOR LATEST RMM MAINTENANCE.
/** IF THE ZAPS DO NOT FIT, REVIEW THE COMMENTS IN THE AMASZAP
/** CONTROL STATEMENTS. IF YOU STILL NEED HELP, CALL SUPPORT CENTER
```

Figure 272 (Part 1 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

/** 3) Update the SYSLMOD DD in LK job step to
/**    identify the LPALIB you are installing the bypass on.
/** 4) Update the SYSLIB DD in AMASPZAP job steps to
/**    identify the LPALIB you are installing the bypass on.
/** 5) Update the SYSLIB DD in AMBLIST job step to
/**    identify the LPALIB you are installing the bypass on.
/** 6) Tailor the job card and submit the job
/** 7) Use the AMASPZAP DUMPT and AMBLIST output to verify that
/**    the bypass is implemented correctly.
/** 8) Ensure that the SYSLIB DD statement in the EPIC startup procedure
/**    identifies the LPALIB updated by this install job.
/** 9) Should the ZAP steps not complete with return code zero,
/**    review the comments in the control statements for AMASPZAP.
/**
/** VALIDATION OF BYPASS
/** If you wish to see trace messages for the front-end to verify
/** that it works correctly, see the assembler source for the
/** front-end and remove the * comment from the WTO macro calls.
/** If the validation fails you may want to activate the tracing
/** to identify the problem.
/** To activate RMM and EPIC tracing you can do the following:
/** 1.Active RMM trace, with operator command.
/**    F DFRMM,DEBUG=8000
/**    to turn off trace
/**    F DFRMM,NODEBUG
/** 2.Active EPIC messages using parmlib option
/**    OPTION MSG05=YES,MSG28=YES,MSG33=YESNO,MSG47=YES,MSG51=YES
/**    to turn off messages revert to your existing options.
/**
/** 1) IPL with CLPA using the new LPALIB created by this install job.
/** 2) Start RMM in WARNING or PROTECT mode.
/**    REPLY 'RETRY' to message EDG0103D if it is issued.
/**    Start EPIC.
/** 3) Submit a job that creates a new tape data set on a scratch
/**    tape. Select a tape which is not scratch status in RMM, which
/**    can be safely used for output. Mount the tape.
/**    RMM should issue a reject message.
/**    In WARNING mode, RMM will allow the tape to be used and
/**    will update the RMM CDS with information about the new tape
/**    data set. Use RMM LV volser command to list the volume details.
/**    In PROTECT mode, RMM does not allow the volume to be used.
/**    In any case, EPIC should have recorded the latest details for the
/**    new tape file created.
/**    This verifies that the change to IFG0194A was successful.
/** 4) Submit a job that creates a new tape data set on a SL tape,
/**    that is known to RMM.
/**    Mount an acceptable tape. No error messages should be issued.
/**    The job should end as normal.
/**    Verify that both RMM and EPIC have details of the data set and
/**    volume you used. Use the RMM LV command or the RMM ISPF dialog.
/**    This verifies that the change to IGC0005E was successful.

```

Figure 272 (Part 2 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run


```

/*
//ASM      EXEC PGM=IEV90,REGION=5M,TIME=5,
//          PARM=(' DECK,NOOBJECT,NORLD,BATCH,XREF(SHORT)',
//          'LINECOUNT(55)')      LINECOUNT=55 IS DISTRIBUTION DEFAULT
//SYSPRINT DD SYSOUT=(*,FE4A)
//SYSUT1   DD UNIT=SYSDA,SPACE=(CYL,(3,1))
//SYSPUNCH DD UNIT=VIO,SPACE=(TRK,(1,1)),BLKSIZE=800,DISP=(,PASS),
//          DSN=&&FE4A
//SYSLIB   DD DCB=BLKSIZE=32000,DISP=SHR,DSN=SYS1.MACLIB
//          DD DISP=SHR,DSN=SYS1.MODGEN
//SYSIN    DD *
194A      TITLE 'A FRONT END TO IFG0194A FOR RMM TO COEXIST WITH EPIC'
IFG0194A  CSECT
IFG0554A  EQU   IFG0194A
IFG0204A  EQU   IFG0194A
          ENTRY IFG0554A
          ENTRY IFG0204A
          BASR 3,0          DUMP BASE REG.
          USING *,3        DECLARE ADRESSABILITY
          USING FORCORE,4  DECLARE ADRESSABILITY
          LA 1,WT0Z9        POINT AT WTO FOR Z9 CALLED
          L 11,=V(IFG019Z9) LOAD Z9 ADDRESS
          CLC TZ9,6(6)      IFG019Z9 CALLED?
          BE WTOTCALL       YES, DO PROCESSING
          L 11,=V(EDG0194A) LOAD 4A ADDRESS
          CLC T4#,6(6)      IFG0194# CALLED?
          BE CALL4A         YES, DO PROCESSING
          LA 1,WTOUNKN      POINT AT WTO FOR UNKOWN CALLED
          CLC =C'4A',6(6)   IFG0194A CALLED?
          BNE WTOTCALL      NO, THEN SELECT IFG0194A
          LA 1,WT04ANC      POINT AT IFG0194A, R14 - 12
          C 14,CON12        R14 = 12
          BNE WTODCALL      NO, THEN SELECT IFG0194A
          SPACE 2
          L 15,DXUCBADR     GET THE UCB
          CLI UCBTBYT3-UCBCMSEG(15),UCB3TAPE    CALLED FOR TAPE
          BNE NOTTAPE       NO, THEN SKIP
          L 11,=V(IFG019Z9) LOAD Z9 ADDRESS
          LA 1,WT04AZ9      POINT AT WTO FOR Z9 TO BE CALLED
          MVC 6(2,6),TZ9    SET WTG TABLE
          B WTOTCALL        NO, THEN SKIP
          SPACE 2
CALL4A    LA 1,WT04#        POINT AT WTO FOR Z9 CALLING 4A
          MVC 6(2,6),T4A    SET WTG TABLE
          B WTOTCALL        NO, THEN SKIP
          SPACE 2

```

Figure 272 (Part 3 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

NOTTAPE LA 1,WT04ANT          POINT AT WTO FOR 4A CALLED
        B  WTODCALL          CONTINUE PROCESSING
        SPACE 2
WTOTCALL DS 0H                CONTINUATION ADDRESS FOR TAPE CALLS
*      WTO MF=(E,(1))          ISSUE SELECTED WTO
***    REMOVE ASTERISKS FROM WTO STATEMENT TO ENABLE TRACING
        LR 15,11              COPY SELECTED MODULE ADDR
        BR 15                  AND CONTINUE
        SPACE 2
WTODCALL DS 0H                CONTINUATION ADDRESS FOR DASD CALLS
****   WTO MF=(E,(1))          ISSUE SELECTED WTO
***    REMOVE ASTERISKS FROM WTO STATEMENT TO ENABLE TRACING
        LR 15,11              COPY SELECTED MODULE ADDR
        BR 15                  AND CONTINUE
        SPACE 3
        LTORG ,
        SPACE 2
        PRINT NOGEN
CON12   DC F'12'
WT0Z9   WTO MF=L,ROUTCDE=2,DESC=4, X
        ' IFG0194A FRONT-END DETECTED CALL TO IFG019Z9'
WT0UNKN WTO MF=L,ROUTCDE=2,DESC=4, X
        ' IFG0194A FRONT-END DETECTED CALL TO UNKNOWN'
WT04#   WTO ' IFG0194A FRONT-END DETECTED CALL TO IFG0194A BY IFG019Z9', X
        MF=L,ROUTCDE=2,DESC=4
WT04ANT WTO MF=L,ROUTCDE=2,DESC=4, X
        ' IFG0194A FRONT-END DETECTED CALL TO IFG0194A, NOT TAPE'
WT04ANC WTO MF=L,ROUTCDE=2,DESC=4, X
        ' IFG0194A FRONT-END DETECTED CALL TO IFG0194A, R14 - 12'
WT04AZ9 WTO ' IFG0194A FRONT-END DETECTED CALL TO IFG0194A, Z9 SELECTED',
        ,MF=L,ROUTCDE=2,DESC=4
        SPACE 2
TZ9     DC C'Z9'
T4#     DC X'F4C0'
T4A     DC C'4A'
        SPACE 2
        IECDSECS (MAIN),EXPAND=YES
        IEFUCBOB
        END
/*

```

Figure 272 (Part 4 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

/*
//LK      EXEC PGM=LINKEDIT,REGION=2048K,
//          PARM='MAP,LIST,RENT,LET,NCAL'
//SYSPRINT DD  SYSOUT=(*,,LK01)
//SYSUT1   DD  UNIT=SYSDA,SPACE=(CYL,(2,2))
//LPASAVE  DD  DISP=SHR,DSN=STSGMA.SAVECOPY.LPALIB
//FE4A     DD  DISP=(OLD,DELETE),DSN=&&FE4A
//SYSLMOD  DD  DISP=SHR,VOL=SER=8E5SY1,UNIT=3380,DSN=SYS1.LPALIB
//SYSLIN   DD  *
            INCLUDE FE4A
            INCLUDE LPASAVE(IFG0194A)
            CHANGE IFG0194A(EDG0194A)
            INCLUDE LPASAVE(IFG0194A)
            ALIAS IFG0194F
            ALIAS IFG0194J
            ALIAS IFG0194K
            ALIAS IFG0195G
            ALIAS IFG0204A
            ALIAS IFG0204J
            ALIAS IFG0234J
            ALIAS IFG0554A
            ALIAS IFG0554C
            ALIAS IFG0554F
            ALIAS IFG0554J
            ALIAS IFG0554K
            ALIAS IGG0190A
            ALIAS IGG0550P
            ENTRY IFG0194A
            NAME IFG0194A(R)
            INCLUDE LPASAVE(IGC0005E)
            EXPAND IFG05522(24)
            ALIAS IFG0551F
            ALIAS IFG0551L
            ALIAS IFG0551N
            ALIAS IFG0552X
            ALIAS IFG0553F
            ALIAS IFG0554P
            ALIAS IFG0554T
            ALIAS IGC0003A
            ALIAS IGG0550B
            ALIAS IGG0550D
            ALIAS IGG0550F
            ALIAS IGG0550H
            ENTRY IGC0005E
            NAME IGC0005E(R)
/*

```

Figure 272 (Part 5 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

/*
/*
//ZAP4A EXEC PGM=AMASPZAP
//SYSPRINT DD SYSOUT=(*,,ZAPM)
//SYSLIB DD DISP=SHR,DSN=SYS1.LPALIB,UNIT=3380,VOL=SER=8E5SY1
//SYSIN DD *
DUMPT IFG0194A IFG019Z9
NAME IFG0194A IFG019Z9
* The X'F4C1' to be changed is the first entry in the table
* created by the XCTLTABL MACRO.
* The MACRO starts with a LTORG which means that it will start
* with the double word aligned string
* CL8' IFG019Z9'
* followed by a VCON and other data including the string C'4F'.
* The next data is double word aligned, so that the C'4F' may be
* followed by some bytes of garbage, possible X'0000'.
* The target double word aligned string starts
* DC CL3'019',CL2'4A',VL3(IFG0194A)
* and it is the A of the C12'4A' which must be changed to X'C0'
* Following ZAP is for JDZ11B4 DFSMSrmm 1.2 (UW32712)
* and JDZ11C0 DFSMSrmm 1.3 (UW90355)
* and HDP3RM2 DFRMM R2 (UW32922)
VER 02A0 C9C6C7F0F1F9E9F9 =CL8' IFG019Z9'
VER 02AC E9F9 =CL2' Z9'
VER 02AE F4C6 =C'4F'
VER 02B8 F0F1F9 IHB0017A DC 0D'0',CL3'019'
VER 02BB F4C1 IDZ94A DC CL2'4A'
VER 02C0 F0F1F9 DC CL3'019'
VER 02C3 F4D1 IDZ94J DC CL2'4J'
REP 02BB F4C0 Change IDZ94A to X'F4C0'
* Following ZAP is for JDZ1150 DFSMSrmm 1.1 (UW32713)
* VER 0258 C9C6C7F0F1F9E9F9 =CL8' IFG019Z9'
* VER 0264 E9F9 =CL2' Z9'
* VER 0266 F4C6 =C'4F'
* VER 0268 F0F1F9 IHB0017A DC 0D'0',CL3'019'
* VER 026B F4C1 IDZ94A DC CL2'4A'
* VER 0270 F0F1F9 DC CL3'019'
* VER 0273 F4D1 IDZ94J DC CL2'4J'
* REP 026B F4C0 Change IDZ94A to X'F4C0'
DUMPT IFG0194A IFG019Z9
/*

```

Figure 272 (Part 6 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

/*
//ZAP5E EXEC PGM=AMASPZAP
//SYSPRINT DD SYSOUT=(*,ZAPM)
//SYSLIB DD DISP=SHR,DSN=SYS1.LPALIB,UNIT=3380,VOL=SER=8E5SY1
//SYSIN DD *
DUMPT IGC0005E IFG055Z2
NAME IGC0005E IFG055Z2
* The patch area will be the 24 bytes of hex zeros at the end of
* the module. To reach it, use the fact that register 3 contains
* the address of the 2nd half-word in the module.
* Parts of the patch code vary, other than due to the relative
* position of the area itself. For example,
* the address of the VL3(IFG0551T) loaded by the ICM instruction.
* This can be located by the double-word aligned string
* CL5'0551T'
* which is followed by the VL3(IFG0551T).
* The CL5'0551T' will be before another double word aligned string
* which starts with X'FF' and is followed by the FMID of the product.
* The area containing the 0551T string is generated by the
* XCTLTABL MACRO which should be towards the end of the original
* module.
* Following ZAP is for JDZ11B4 DFSMSrmm 1.2 (UW32712)
* and JDZ11C0 DFSMSrmm 1.3 (UW90355)
* and HDP3RM2 DFRMM R2 (UW32922)
VER 0002 47F0,3138 BRANCH ROUND ID
VER 02A8 0000,0000,0000,0000 CHECK OFFSET OF PATCH AREA
VER 02B0 0000,0000,0000,0000 CREATED BY LINKAGE EDITOR
VER 02B8 0000,0000,0000,0000 EXPAND OPTION
VER 0238 C9C6C7F0F5F5E9F2 =CL8' IFG055Z2'
VER 0244 E9F2 =CL2' Z2'
VER 0250 F0F5F5F1E3 DC CL5'0551T', AL3(IFG0551T)
VER 0258 C9C6C7F0F5F5E9F2 =CL8' IFG055Z2'
REP 0002 47F0,32A6 BRANCH TO PATCH AREA
REP 02A8 D501,6006,3242 CLC 6(2,6),C' Z2'
REP 02AE 4780,3138 BE PAST EYECATCHER
REP 02B2 1FFF SLR 15,15
REP 02B4 BFF7,3253 ICM 15,7,ADDRESS OF IFG0551T
REP 02B8 07FF BR 15

```

Figure 272 (Part 7 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

```

* Following ZAP is for JDZ1150 DFSMSrmm 1.1 (UW32713)
* VER 0002 47F0,3138          BRANCH ROUND ID
* VER 0258 0000,0000,0000,0000 CHECK OFFSET OF PATCH AREA
* VER 0260 0000,0000,0000,0000 CREATED BY LINKAGE EDITOR
* VER 0268 0000,0000,0000,0000 EXPAND OPTION
* VER 01F0 C9C6C7F0F5F5E9F2   =CL8' IFG055Z2'
* VER 01FC E9F2                =CL2' Z2'
* VER 0200 F0F5F5F1E3          DC CL5'0551T', AL3(IFG0551T)
* VER 0208 C9C6C7F0F5F5E9F2   =CL8' IFG055Z2'
* REP 0002 47F0,3256          BRANCH TO PATCH AREA
* REP 0258 D501,6006,31FA      CLC 6(2,6),C' Z2'
* REP 025E 4780,3138          BE PAST EYECATCHER
* REP 0262 1FFF               SLR 15,15
* REP 0264 BFF7,3203          ICM 15,7,ADDRESS OF IFG0551T
* REP 0268 07FF               BR 15
DUMPT IGC0005E IFG055Z2
/*
/**
//MAPMDL EXEC PGM=AMBLIST,PARM='LINECNT=48'
//SYSPRINT DD SYSOUT=(*,,MAPM)
//SYSLIB DD DISP=SHR,DSN=SYS1.LPALIB,UNIT=3380,VOL=SER=8E5SY1
//SYSIN DD *
LISTLOAD OUTPUT=XREF,MEMBER=(IFG0194A)
LISTLOAD OUTPUT=XREF,MEMBER=(IGC0005E)
/*

```

Figure 272 (Part 8 of 8). JCL to Modify DFSMSrmm for EPIC Parallel Run

Appendix F. Special Notices

This publication is written for people who are planning to convert from a vendor tape management system to DFSMSrmm. Use it to decide whether to convert to DFSMSrmm, and then use it to help plan and carry out that conversion. We have designed this book to help you with all aspects of the conversion, from the early planning stage through customization and product implementation of DFSMSrmm. We included working examples to use both during and after conversion. The information in this publication is not intended as the specification of any programming interfaces that are provided by DFSMSrmm. See the PUBLICATIONS section of the IBM Programming Announcement for DFSMS/MVS V1R4 for more information about what publications are considered to be product documentation.

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Appendix G. Related Publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this redbook.

G.1 International Technical Support Organization Publications

For information on ordering these ITSO publications, see "How to get IBM Redbooks" on page 683.

- *DFSMS/MVS V1R4 Technical Guide*, SG24-4892
- *IBM Magstar 3494 Tape Libraries: A Practical Guide*, SG24-4632

G.2 IBM Redbooks collections

Redbooks are also available on the following CD-ROMs. Click the CD-ROMs button at <http://www.redbooks.ibm.com/> for information about all the CD-ROMs offered, updates and formats.

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G.3 Other Publications

These publications are also relevant as further information sources:

- *DFSMS/MVS General Information*, GC26-4900
- *DFSMS/MVS Planning for Installation*, SC26-4919
- *MVS/ESA System Management Facilities*, GC28-1457
- *DFSMSrmm Guide and Reference*, SC26-4931
- *DFSMSrmm Implementation and Customization Guide*, SC26-4932
- *IBM 3494 Tape Library Dataserver Introduction and Planning*, GA32-0279
- *OS/390 MVS System Messages, Vol 2*, GC28-1785
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- *OS/390 MVS Initialization and Tuning Reference*, SC28-1751
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- *OS/390 Security Server (RACF) Security Administrator's Guide*, SC28-1915
- *OS/390 Security Server (RACF) Command Language Reference*, SC28-1918
- *DFSMS/MVS DFSMSHsm Storage Administrators Guide*, SH21-1076

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List of Abbreviations

AMS	access method services	MTL	manual tape library
APAR	authorized program analysis report	NL	nonlabeled
APF	authorized program facility	OAM	Object Access Method
ATL	automated tape library	OCDS	offline control data set
BLP	bypass label processing	PDS	partitioned data set
BTLS	Basic Tape Library Support	OPC/ESA	Operations Manager for MVS/ESA
CDS	control data set	PPT	program properties table
CST	cartridge system tape	PTF	program temporary fix
DSNB	data set name block	RACF	Resource Access Control Facility
ECCST	enhanced capacity cartridge system tape	RDS	retention data set
EDM	external data manager	RMF	retention master file
EOV	end of volume	RMM	Removable Media Manager
GDG	generation data group	SAF	security authorization facility
GRS	global resource serialization	SMF	system management facility
HLQ	high-level qualifier	SMS	Storage Management Subsystem
IBM	International Business Machines Corporation	SSD-SP	Storage Systems Division Software Products
ICF	integrated catalog facility	TCDB	tape configuration database
ISMF	Interactive Storage Management Facility	TMC	tape management catalog
ISPF	Interactive System Productivity Facility	TRS	tape retention system
ITSO	International Technical Support Organization	VMF	volume master file
IVP	installation verification procedure	VMS	vault management system
KSDS	key-sequenced data set	VNL	volume not in library
LCS	library control system	VOLCAT	volume catalog
LPA	link pack area	VOLSER	volume serial number
LSR	local shared resources	VPD	vault pattern description
		VPDD	vault pattern description data set
		VRS	vital record specification
		WTOR	write to operator with reply

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